

Split-Zero: escaping fibres and separated-zeta positivity

Proofs of 22–23 September 2026

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This edition contains the complete thirty-four derivations in the order listed below. Its finite algebra, analytic explicit formulas and exact comparison maps retain their own hypotheses. The source and reading guide follows the proofs.

The escaping inverse fibre, its infinitesimal boundary, and its trace form

This derivation retains the original polynomial map, its target coordinates, and its heat constants. It computes an actual dual-number fibre at every double collision, the larger boundary at a triple collision, the exact maps under which infinitesimals vanish, and the signatures of the associated finite trace forms. It does not identify these finite trace forms with the classical Weil distribution. No off-critical zero of the Riemann zeta function is asserted.

Sources actually read

The complete original LaTeX files `tex/satellites/26_incompressible_fibre_heat.tex` and `tex/satellites/23_source_mechanism_transfer.tex` in the local Zeta-Function-Foundation repository were read for this calculation. The received source formulas are `eq:ifh-original`, `eq:ifh-det`, `eq:ifh-chart`, `eq:ifh-target-chart`, `eq:ifh-target-flow`, `eq:ifh-triple`, `eq:retained-node-map`, and `eq:retained-arithmetic-extension`. Their original-map provenance is Tao's displayed map and explanation and Speyer's marked-factor discussion, as cited in those programme sources. The full-support definition used below is reproduced in FSR1–FSR7 of full-support reconstruction proof; its first 70 lines were read. The results below supply their own algebraic proofs.

The exact read coverage was lines 1–480 of the first file, SHA256 72058517553FF3A2C35699844FA95BBA9EC695E774E0E3D4CF33CF7B84DA51DD, and lines 1–666 of the second file, SHA256 D8A65C46B78A5ACE521A164AABC99F246908058E36B60C35BCE938D51C020898. These are local programme source versions, not author-source archives.

The forward heat time called tau in the source is denoted h here, solely to keep it distinct from the global absorber τ_L . The displacement from a specified collision is $d = h - h_c$. All original spatial and target coordinates are retained.

1. Original map and projective inverse chart

Over $\mathbb{C}$, write

$$\begin{aligned} F_1 &= (1 + xy)^3 w + y^2(1 + xy)(4 + 3xy), \\ F_2 &= y + 3x(1 + xy)^2 w + 3xy^2(4 + 3xy), \\ F_3 &= 2x - 3x^2 y - x^3 w. \end{aligned} \tag{EFI1}$$

The target is (a, b, c) . Define

$$P(r) = cr^3 - 2r^2 + br - 2a, \quad \alpha = \frac{P'(r)}{2}, \quad H(r, \alpha, c) = \left(\frac{1}{\alpha}, r - \alpha, 5\alpha^2 - 3r\alpha - c\alpha^3 \right). \tag{EFI2}$$

Direct substitution before imposing $P = 0$ gives

$$F \circ H = (r^2 + r\alpha - cr^3, 4r + 2\alpha - 3cr^2, c). \tag{EFI3}$$

Consequently $F_2 = b$ is exactly $2\alpha = P'(r)$, and then $2(F_1 - a) = P(r)$. Conversely, from a finite point with $x \neq 0$, recover $\alpha = 1/x$ and $r = y + 1/x$; solving $F_3 = c$ recovers every coordinate of EFI2. This proves that simple roots of P are in bijection with all inverse points with $x \neq 0$. No multiple root supplies a finite point in this chart. On the excluded hyperplane,

$$F(0, y, w) = (w + 4y^2, y, 0). \tag{EFI4}$$

Thus a separate finite point $(0, b, a - 4b^2)$ exists exactly when $c = 0$.

Put projective coordinates $[X : Y : W : T]$ on the compactification of the source, with affine coordinates $(x, y, w) = (X/T, Y/T, W/T)$. Multiplying the homogeneous representative of EFI2 by α gives the everywhere defined map

$$\overline{H}(r, \alpha, c) = [1 : \alpha(r - \alpha) : 5\alpha^3 - 3r\alpha^2 - c\alpha^4 : \alpha]. \tag{EFI5}$$

It agrees with the actual inverse point when $\alpha \neq 0$. Every point with $\alpha = 0$ maps to

$$p_\infty = [1 : 0 : 0 : 0]. \tag{EFI6}$$

This equality of projective points does not identify their infinitesimal neighbourhoods. Those neighbourhoods are computed below.

2. The original two escaping trajectories give exact dual numbers

On the original target line $b = c = 0$, the entire root cover is

$$a = -r^2, \quad \alpha = -2r, \quad (x, y, w) = \left(-\frac{1}{2r}, 3r, 26r^2\right) \quad (r \neq 0). \quad (\text{EFI7})$$

These equalities follow by substituting $c = b = 0$ in EFI2. In particular, the projective graph is

$$(r, a) \mapsto ([1 : -6r^2 : -52r^3 : -2r], a = -r^2). \quad (\text{EFI8})$$

In the projective chart $X = 1$, its coordinate $T/X = -2r$ recovers r . Hence this map identifies the graph closure with the affine r -line; its completed local ring at the escaping point is $\mathbb{C}[[r]]$. The base parameter map is exactly

$$\mathbb{C}[[a]] \longrightarrow \mathbb{C}[[r]], \quad a \longmapsto -r^2. \quad (\text{EFI9})$$

The scheme fibre over $a = 0$ is therefore

$$\mathbb{C}[[r]]/(a) = \mathbb{C}[r]/(r^2) \xrightarrow[r \mapsto \epsilon]{r \mapsto \epsilon} D := \mathbb{C}[\epsilon]/(\epsilon^2). \quad (\text{EFI10})$$

The inverse is $\epsilon \mapsto r$. The class r is nonzero because the two residue classes $1, r$ are linearly independent. Thus dual numbers arise from the special fibre of the actual inverse map; they have not been introduced by a resemblance of notation.

The additional finite point of EFI4 is $(0, 0, a)$ and survives at $a = 0$. It is a different component of the compactified inverse fibre. The length-two point EFI10 is supported at infinity, while this extra point is supported at the finite origin. Keeping both preserves total fibre length three on this line.

The original forward heat orbit has

$$a = -\frac{1}{4} + 2h, \quad r^2 = \frac{1}{4} - 2h, \quad d = h - \frac{1}{8}, \quad r^2 = -2d. \quad (\text{EFI11})$$

For $h < 1/8$, taking $r = \mp\sqrt{1 - 8h}/2$ in EFI7 recovers exactly

$$\gamma_{\pm}(h) = \left(\pm(1 - 8h)^{-1/2}, \mp\frac{3}{2}(1 - 8h)^{1/2}, \frac{13}{2}(1 - 8h)\right). \quad (\text{EFI12})$$

Thus the dual-number boundary is attached to those same escaping trajectories with the same time constant.

The exact cancellation and the retained tangent

For a holomorphic germ $f(r) = \sum_{n \geq 0} f_n r^n$ on the graph, the sheet involution is $\sigma(r) = -r$. The trace is

$$\text{Tr}(f) = f(r) + f(-r) = 2 \sum_{n \geq 0} f_{2n} r^{2n}. \quad (\text{EFI13})$$

This is a map to $\mathbb{C}[[a]]$ under EFI9. Its kernel is exactly $r\mathbb{C}[[a]]$, and the trace divided by two has the even-section inclusion as a right inverse. This proves exactly what the sheet trace forgets.

The antisymmetric functional satisfies

$$\frac{f(r) - f(-r)}{2r} \longrightarrow f_1 = f'(0) \quad \text{as } r \longrightarrow 0. \quad (\text{EFI14})$$

The equality follows directly from its power series, which equals $f_1 + f_3 r^2 + f_5 r^4 + \dots$. On a germ pulled back from $a = -r^2$, the numerator is identically zero. On the projective coordinate $T/X = -2r$, the limit is -2 . Therefore the retained limit is a nonzero tangent functional at p_{∞} , killed by projection to the target. Its algebra map is precisely EFI10.

This limit does not assign a finite value to the pole $x = -1/(2r)$. Indeed the two sheet sums and the quadratic moment are

$$x(r) + x(-r) = 0, \quad x(r) - x(-r) = -\frac{1}{r}, \quad x(r)^2 + x(-r)^2 = \frac{1}{2r^2} = -\frac{1}{2a}. \quad (\text{EFI15})$$

Thus a first-moment cancellation coexists with a divergent second moment. The latter is the explicit retained defect in any proposed assertion that all observables cancel.

3. Every double collision in the given heat family

Fix a double root r_0 at a target (a_0, b_0, c_0) . Put

$$q_0 = 3c_0r_0 - 2 \neq 0, \quad t = r - r_0, \quad a = a_0 + 2d, \quad b = b_0 + 6c_0d, \quad c = c_0. \quad (\text{EFI16})$$

The inequality $q_0 \neq 0$ says exactly that the root has multiplicity two rather than three, since $P''(r_0) = 2q_0$. Taylor expansion of this cubic is an exact identity, with no omitted remainder:

$$P = q_0t^2 + c_0t^3 + 2q_0d + 6c_0dt. \quad (\text{EFI17})$$

In the completed local ring at $(d, t) = (0, 0)$, the factor $2q_0 + 6c_0t$ is a unit. Hence EFI17 is equivalent to

$$d = -\frac{t^2(q_0 + c_0t)}{2q_0 + 6c_0t}, \quad \alpha = q_0t + \frac{3}{2}c_0t^2 + 3c_0d. \quad (\text{EFI18})$$

The completed root ring is $\mathbb{C}[[t]]$ with exactly this base map. The coefficient of t in α is $q_0 \neq 0$. A formal series with a nonzero linear coefficient has a unique inverse series: its coefficient of degree one is determined by division by q_0 , and at every later degree the new unknown enters multiplied by the same nonzero q_0 . This recursive argument proves $\mathbb{C}[[\alpha]] = \mathbb{C}[[t]]$.

Since $T/X = \alpha$ is a coordinate of the projective graph, the graph completion is this same ring. In EFI18 the coefficient multiplying t^2 is a unit, so the special fibre is

$$\mathbb{C}[[t]]/(d) = \mathbb{C}[t]/(t^2). \quad (\text{EFI19})$$

The projective tangent has

$$T/X = q_0t, \quad Y/X = r_0q_0t, \quad W/X = 0 \quad \text{modulo } t^2. \quad (\text{EFI20})$$

This retains the original constants and the actual tangent direction. For $c_0 = 0$, one has $q_0 = -2$, and EFI18 gives exactly $d = -t^2/2$, including the quadratic calculation above.

4. Triple collision: root fibre and projective graph fibre differ by exact maps

Now fix $c \neq 0$ and the triple target

$$r_* = \frac{2}{3c}, \quad a_* = \frac{4}{27c^2}, \quad b_* = \frac{4}{3c}. \quad (\text{EFI21})$$

Along its heat orbit $a = a_* + 2d, b = b_* + 6cd$, put $z = r - r_*$. The exact equations are

$$P = c(z^3 + 6dz), \quad \alpha = \frac{3c}{2}z^2 + 3cd. \quad (\text{EFI22})$$

Let $R = \mathbb{C}[d]$ and

$$B = R[z]/(z^3 + 6dz). \quad (\text{EFI23})$$

It is free of rank three over R , with ordered basis $1, z, z^2$, by division by the displayed monic cubic. Its completed local ring is $\hat{B} = \mathbb{C}[[d, z]]/(z^3 + 6dz)$. Its special root fibre is

$$B_0 = B/(d) = \mathbb{C}[z]/(z^3). \quad (\text{EFI24})$$

This is a length-three infinitesimal, rather than a length-two one. It has the surjection $B_0 \rightarrow D, z \mapsto \epsilon$, with kernel (z^2) . It also contains the subalgebra $\mathbb{C}[z^2] \cong D$, with $\epsilon \mapsto z^2$. There is no retraction of this latter inclusion: any homomorphism $B_0 \rightarrow D$ sends z to a nilpotent multiple of ϵ , whose square is zero, whereas a retraction would have to send z^2 to $\epsilon \neq 0$.

The graph closure itself

In the projective chart $X = 1$, write

$$\eta = Y/X = \alpha(r_* + z - \alpha), \quad \omega = W/X = 5\alpha^3 - 3(r_* + z)\alpha^2 - c\alpha^4. \quad (\text{EFI25})$$

The projective graph algebra is the R -subalgebra $A = R[\alpha, \eta, \omega] \subset B$. Define

$$u = z^2 = \frac{2}{3c}(\alpha - 3cd), \quad v = dz = -\frac{\eta - r_*\alpha + \alpha^2}{6c}. \quad (\text{EFI26})$$

The second equality follows because $\alpha z = (3c/2)z^3 + 3cdz = -6cdz$. Thus $u, v \in A$. Conversely $\alpha = (3c/2)u + 3cd$, $\eta = r_*\alpha - \alpha^2 - 6cv$, and

$$\omega = 5\alpha^3 - 3r_*\alpha^2 + 18c\alpha v - c\alpha^4. \quad (\text{EFI27})$$

All projective coordinates therefore belong to $R[u, v]$, proving $A = R[u, v]$ inside B .

Products in this ring satisfy exactly

$$u^2 = -6du, \quad uv = -6dv, \quad v^2 = d^2u. \quad (\text{EFI28})$$

For example $u^2 = z^4 = -6dz^2$, and the other identities follow in the same manner. Every word in u, v reduces, by these identities, to an R -linear combination of $1, u, v$. These three elements are linearly independent in B : a relation would have coefficients $A_0 + A_1z^2 + A_2dz = 0$, and the basis $1, z, z^2$ forces $A_0 = A_1 = dA_2 = 0$; the polynomial ring R has no d -torsion. Hence

$$A \cong R[u, v]/(u^2 + 6du, uv + 6dv, v^2 - d^2u), \quad A = R \oplus Ru \oplus Rv. \quad (\text{EFI29})$$

This proves that the displayed relations are complete.

For $d \neq 0$, $z = v/d$, so $A[d^{-1}] = B[d^{-1}]$, which is the original three-point inverse graph. The algebra B is finite over $R \subset A$, hence is finite over A . The image of its map to the affine projective-coordinate chart is closed, and its coordinate algebra is exactly the subalgebra A . Equivalently, any polynomial vanishing on the graph with $d \neq 0$ is zero in $A[d^{-1}]$; since A is R -free, it was already zero in A . These two facts prove that EFI29 is the scheme-theoretic graph closure.

Its completion is obtained by replacing R with $\mathbb{C}[[d]]$ in EFI29. Its special fibre is

$$A_0 = A/(d) = \mathbb{C}[u, v]/(u^2, uv, v^2). \quad (\text{EFI30})$$

It has two independent square-zero directions. The total graph is flat of rank three over the d -line, as the proved basis shows. This is the exact scheme in which all three escaping branches converge at p_∞ .

The inclusion $A \hookrightarrow B$ induces

$$j_0 : A_0 \longrightarrow B_0, \quad u \longmapsto z^2, \quad v \longmapsto 0. \quad (\text{EFI31})$$

Its kernel is $\mathbb{C}v$, its image is $\mathbb{C} \oplus \mathbb{C}z^2 \cong D$, and its cokernel as a vector space is $\mathbb{C}z$. In particular, the vanishing class v is nonzero in the projective special fibre; its vanishing names this specific morphism, not the absence of an infinitesimal.

The failure of injectivity after specializing has a complete exact sequence. From the two R -bases,

$$0 \longrightarrow A \longrightarrow B \longrightarrow R/(d) \longrightarrow 0, \quad (\text{EFI32})$$

where the last map takes the coefficient of z modulo d . Tensoring the elementary resolution $0 \rightarrow R \xrightarrow{d} R \rightarrow R/(d) \rightarrow 0$ proves

$$0 \longrightarrow \mathbb{C}v \longrightarrow A_0 \xrightarrow{j_0} B_0 \longrightarrow \mathbb{C}z \longrightarrow 0. \quad (\text{EFI33})$$

The first map can be checked without invoking derived notation: the omitted generator $z \in B/A$ is killed by d , and $dz = v \in A$ is its connecting class. This is the precise retained defect of specialization.

Each original branch and its poles

The central root has $z = 0$, $\alpha = 3cd$. Each outer root has $z^2 = -6d$, $\alpha = -6cd$. After the common parameter substitution $d = -q^2/6$, the three roots are $z = 0, q, -q$, and

$$\alpha_{\text{central}} = -\frac{c}{2}q^2, \quad \alpha_{\text{outer}} = cq^2. \quad (\text{EFI34})$$

All projective coordinates in EFI25 converge to EFI6, with T/X of order two in q . The outer branch distinction first enters αz in order three. The exact sum and sum of squares of the original x -coordinates are

$$\sum x_i = \frac{1}{3cd} + 2\left(-\frac{1}{6cd}\right) = 0, \quad \sum x_i^2 = \frac{1}{6c^2d^2}. \quad (\text{EFI35})$$

The second expression is a nonzero pole, so cancellation of the sum does not cancel the family of observables.

5. Exact positivity of the finite trace forms

These forms are attached to the root and graph algebras just proved. They are not identified here with the Weil form. Their value is that they calculate exactly what trace-vanishing of an infinitesimal can and cannot imply.

For real a , let $B_a = \mathbb{C}[r]/(r^2 + a)$, with the conjugation that conjugates coefficients and fixes r . Define

$$\langle f, g \rangle_a = \text{Tr}_{B_a/\mathbb{C}}(M_{\bar{f}g}), \quad (\text{EFI36})$$

where M_b means multiplication by b . In basis $1, r$, $\text{Tr}(1) = 2$, $\text{Tr}(r) = 0$, and $\text{Tr}(r^2) = -2a$. Thus the Gram matrix is

$$G_2(a) = \begin{pmatrix} 2 & 0 \\ 0 & -2a \end{pmatrix}. \quad (\text{EFI37})$$

It is positive definite for $a < 0$, has inertia $(1, 1, 0)$ for $a > 0$, and has radical $\mathbb{C}r$ at $a = 0$. Here inertia lists positive, negative, and zero dimensions. In particular, a nonzero dual-number direction can be trace-null at the collision and become negative on one side of that collision.

For real d , use $B_d = \mathbb{C}[z]/(z^3 + 6dz)$ with coefficient conjugation fixing z . Multiplication in basis $1, z, z^2$ gives

$$\text{Tr}(1) = 3, \quad \text{Tr}(z) = 0, \quad \text{Tr}(z^2) = -12d, \quad \text{Tr}(z^3) = 0, \quad \text{Tr}(z^4) = 72d^2. \quad (\text{EFI38})$$

For example the matrix of M_z has columns $(0, 1, 0)^T$, $(0, 0, 1)^T$, $(0, -6d, 0)^T$, and squaring it proves the second moment; the last two follow from $z^3 = -6dz$. The Gram matrix is

$$G_3(d) = \begin{pmatrix} 3 & 0 & -12d \\ 0 & -12d & 0 \\ -12d & 0 & 72d^2 \end{pmatrix}, \quad \det G_3(d) = -864d^3. \quad (\text{EFI39})$$

For $d \neq 0$, the $1, z^2$ block has positive first entry and determinant $72d^2 > 0$, so completing the square makes both of its directions positive. The remaining entry is $-12d$. Hence the inertia is $(3, 0, 0)$ when $d < 0$, $(2, 1, 0)$ when $d > 0$, and $(1, 0, 2)$ at $d = 0$. At the latter point the radical is the entire nilpotent ideal $(z) \subset B_0$.

For the graph algebra A_d , in the retained basis $1, u, v$, the same computation using EFI28 gives

$$G_{\text{graph}}(d) = \begin{pmatrix} 3 & -12d & 0 \\ -12d & 72d^2 & 0 \\ 0 & 0 & -12d^3 \end{pmatrix}, \quad \det G_{\text{graph}}(d) = -864d^5. \quad (\text{EFI40})$$

To verify that this is the trace intrinsic to A_d , rather than an unproved transport of trace, multiply $1, u, v$ using EFI28. The diagonal entries of M_u are $0, -6d, -6d$, and those of M_v are all zero. Thus $\text{Tr}(u) = -12d$ and $\text{Tr}(v) = 0$, from which every entry follows. The inertias are exactly those of EFI39, while at $d = 0$ the radical is the two-dimensional ideal (u, v) .

The trace forms before and after the graph inclusion are related by the exact basis map $(1, u, v) \mapsto (1, z^2, dz)$. Thus the change of determinant is the factor d^2 . The quotient of length one in EFI32 is recorded by two additional orders of discriminant vanishing, not by cancellation of the sign on $d > 0$.

The full original cubic has discriminant $c^4 \det G_3(d)$, namely $-864c^4d^3$. The factor c^4 is retained because the displayed polynomial is $c(z^3 + 6dz)$, and each of the three root differences is unchanged while a cubic discriminant has leading-coefficient power four. This agrees with its exact coefficient heat discriminant.

6. The zero-prime globalization and the infinitesimal fibre

Let L be a nontrivial bounded distributive lattice and R a nonzero commutative unital ring. The programme semiring is

$$G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\}, \quad (r, \lambda) + (s, \mu) = (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu). \quad (\text{EFI41})$$

Its global additive identity is $\tau_L = (0, 0_L)$, whereas its supported zero is $e_L = (0, 1_L) \neq \tau_L$. Write $Z_L = \{(0, \lambda)\}$. Multiplying by e_L proves $e_L G_L(R) = Z_L$.

The amplitude projection

$$\rho_R : G_L(R) \longrightarrow R, \quad (r, \lambda) \longmapsto r \quad (\text{EFI42})$$

preserves addition, multiplication, and both the global zero and unit. The ideal Z_L is prime exactly when R is a domain. Indeed membership of xy in Z_L is exactly $\rho_R(x)\rho_R(y) = 0$, and every ring element occurs as a supported amplitude. This proves both implications, including the need for the domain hypothesis. For $R = \mathbb{Z}$, the supported-zero prime is therefore the established prime of the original programme; this is not a new claim.

A ring homomorphism $f : R \rightarrow R'$ induces the semiring map

$$G_L(f) : G_L(R) \longrightarrow G_L(R'), \quad (r, \lambda) \longmapsto (f(r), \lambda). \quad (\text{EFI43})$$

It is defined even when a nonzero r maps to zero: its top support remains top, so the image is e_L , not τ_L . The operation identities follow by applying f to amplitudes and leaving the lattice operations unchanged.

In particular EFI10 induces

$$G_L(\mathbb{C}[[r]]) \longrightarrow G_L(D), \quad \hat{r} \longmapsto \hat{\epsilon} =: \eta, \quad \eta^2 = e_L \neq \tau_L. \quad (\text{EFI44})$$

Here $\hat{r} = (r, 1_L)$. The image infinitesimal η is not itself e_L , because its amplitude ϵ is nonzero. Consequently Z_L is not prime in $G_L(D)$: $\eta^2 \in Z_L$ but $\eta \notin Z_L$. The same calculation holds for the two nilpotent directions of $G_L(A_0)$ and the nonzero nilpotents of $G_L(B_0)$. The change is forced by the actual amplitude ring; it cannot be avoided by relabelling e_L as τ_L .

The prime map remains completely explicit. For a ring prime $\mathfrak{p} \subset R$, set

$$Q_{\mathfrak{p}} = Z_L \cup \{(r, 1_L) : r \in \mathfrak{p}\}. \quad (\text{EFI45})$$

The product test through ρ_R proves this is prime. In the dual-number ring every prime contains ϵ , because $\epsilon^2 = 0$; the quotient by (ϵ) is the field $\mathbb{C}$. Thus its unique ring prime is (ϵ) , and the arithmetic prime at the collision is

$$Q_{(\epsilon)} \subset G_L(D), \quad G_L(f)^{-1}(Q_{(\epsilon)}) = Q_{(r)} \subset G_L(\mathbb{C}[[r]]). \quad (\text{EFI46})$$

The contraction follows from $f^{-1}((\epsilon)) = (r)$. The pure support primes Z_a , for lattice prime ideals $\mathfrak{a} \subset L$, contract to the identical support primes: a top-supported amplitude always has top support under EFI43, and a proper lattice ideal does not contain 1_L . This proves the full prime-contraction description for this map.

In the scalar support case $L = \{0, 1\}$, the domain family has

$$(\tau) \subsetneq (e) \subsetneq Q_{(r)}. \quad (\text{EFI47})$$

The infinitesimal fibre instead has the two prime ideals $(\tau) \subsetneq Q_{(\epsilon)}$. Its missing intermediate prime has not been silently deleted: (e) remains an ideal, and EFI44 is its explicit failure of primality. The spectrum map contracts these two primes to (τ) and $Q_{(r)}$, respectively.

Why the pole still needs the compactified boundary

Let $R = \mathbb{C}[[r]]$ and $K = \mathbb{C}((r))$. The inclusion $G_L(R) \hookrightarrow G_L(K)$ is injective because amplitudes and supports are individually preserved. The finite inverse expression $x = -1/(2r)$ belongs to the latter amplitude ring, not the former. If an element $X \in G_L(R)$ extended it while satisfying $-2rX = \hat{1}$, applying EFI42 would give $-2r\rho_R(X) = 1$ in R , impossible by its constant term. At the special fibre, $\eta X = \hat{1}$ is also impossible, since amplitudes would assert that the nilpotent ϵ is invertible. This proves the precise nonextension; EFI8–EFI10 supply the compactified object and its nonzero infinitesimal in its place.

7. Support-valued retention of the calculated trace signatures

Assume L is finite. Define the contracted meet algebra

$$C_L = \mathbb{C}[L, \wedge]/([0_L]). \quad (\text{EFI48})$$

For every $a \neq 0_L$, evaluation $\chi_a([\lambda]) = 1$ when $a \leq \lambda$, and zero otherwise, is multiplicative. In any linear extension of the finite order on $L \setminus \{0_L\}$, their incidence matrix is triangular with ones on the diagonal. Hence the joint evaluation is an algebra isomorphism $C_L \cong \mathbb{C}^{L \setminus \{0_L\}}$. Let E_a denote its coordinate idempotents. Thus $E_a E_b = \delta_{ab} E_a$ and $\sum_a E_a = 1$.

For any one of the finite algebras V above, extend it to $C_L \otimes V$ and give the coefficient algebra the conjugation fixing each E_a . For $x = \sum_a E_a \otimes x_a$ and $y = \sum_a E_a \otimes y_a$, define the support-valued trace form

$$\mathcal{T}_L(x, y) = \sum_a E_a \operatorname{Tr}_V(M_{\overline{x_a y_a}}). \quad (\text{EFI49})$$

This is exactly the trace of the multiplication endomorphism over C_L , because the product decomposition makes its matrix blockwise the multiplication matrix over V . Each χ_a sends EFI49 to the already calculated finite trace form on V , while $v \mapsto E_a \otimes v$ provides its supported section.

For positive real weights $w_a > 0$, applying $\ell(\sum t_a E_a) = \sum w_a t_a$ gives a Hermitian form which is a direct sum of positively weighted copies of the original form. For example, for $a > 0$ in the quadratic family and any nonzero support idempotent E_j ,

$$\ell \mathcal{T}_L(E_j \otimes r, E_j \otimes r) = -2w_j a < 0. \quad (\text{EFI50})$$

For $d > 0$, the triple root direction $E_j \otimes z$ has value $-12w_j d < 0$, and the graph direction $E_j \otimes v$ has value $-12w_j d^3 < 0$. At the collision these nonzero infinitesimal directions are radical directions in every supported component. The support extension therefore retains both the vanishing at the collision and the computed sign on either side. It does not itself prove that the corrected Weil form equals this trace extension.

8. Exact result available for the arithmetic continuation

The actual inverse map supplies a finite cover whose double special fibre is dual numbers, and whose triple projective special fibre has two independent square-zero directions. The sheet trace, graph-to-root specialization, and target projection each have different computed kernels: EFI13, EFI31–EFI33, and EFI14. Their kernels are retained objects, with explicit connecting maps. The original pole observable has the nonzero second moments EFI15 and EFI35. The trace signatures are EFI37, EFI39, EFI40, and their support-valued forms are EFI49.

The exact arithmetic interface already present in the read source is the finite scalar extension $s \mapsto -r^2/2$ of its specified arithmetic quotient, with multiplication matrix $\begin{pmatrix} 0 & -2S \\ I & 0 \end{pmatrix}$. Its square is $-2 \operatorname{diag}(S, S)$. These identities transport the two-sheet geometry while preserving the original spectral coordinate. They do not identify an arbitrary value of the collision parameter with a zero of that arithmetic operator. A corrected Weil calculation must carry its actual test-function map and trace through this interface; no positivity conclusion about that different form has been substituted for this calculation.

9. Exact bridge to the previously observed deformation

The first 135 lines of the programme's OBSERVED_COTANGENT_FROBENIUS_DERIVATION.md were additionally read. Its original coefficients in OCF1–OCF3 are $a_{\text{obs}} > 0$, $\varepsilon_{\text{obs}} > 0$, $r_{\text{obs}} \in \mathbb{C}$,

$$\lambda(s) = a_{\text{obs}} s + \varepsilon_{\text{obs}} r_{\text{obs}} = a_{\text{obs}}(s - s_*), \quad s_* = -\frac{\varepsilon_{\text{obs}} r_{\text{obs}}}{a_{\text{obs}}}, \quad A_{\text{obs}} = \mathbb{C}[s, z_{\text{obs}}]/(z_{\text{obs}}^2 - \lambda(s)z_{\text{obs}}). \quad (\text{EFI51})$$

The subscripts distinguish the observed coefficients from the original map's target a and inverse-root coordinate r . They do not alter any coefficient.

Retain the escaping cover from EFI7 as $C_{\text{esc}} = \mathbb{C}[a, r]/(r^2 + a)$ over $\mathbb{C}[a]$. Define the base map

$$\mathbb{C}[a] \longrightarrow \mathbb{C}[s], \quad a \longmapsto -\frac{\lambda(s)^2}{4}. \quad (\text{EFI52})$$

Its pullback is

$$C_{\text{esc}} \otimes_{\mathbb{C}[a]} \mathbb{C}[s] = \mathbb{C}[s, r]/\left(r^2 - \frac{\lambda(s)^2}{4}\right). \quad (\text{EFI53})$$

There is the exact $\mathbb{C}[s]$ -algebra isomorphism

$$\begin{aligned} \Psi : A_{\text{obs}} &\longrightarrow C_{\text{esc}} \otimes_{\mathbb{C}[a]} \mathbb{C}[s], & z_{\text{obs}} &\longmapsto r + \frac{\lambda}{2}, \\ \Psi^{-1} : C_{\text{esc}} \otimes_{\mathbb{C}[a]} \mathbb{C}[s] &\longrightarrow A_{\text{obs}}, & r &\longmapsto z_{\text{obs}} - \frac{\lambda}{2}. \end{aligned} \quad (\text{EFI54})$$

For proof,

$$\left(z_{\text{obs}} - \frac{\lambda}{2}\right)^2 - \frac{\lambda^2}{4} = z_{\text{obs}}^2 - \lambda z_{\text{obs}}. \quad (\text{EFI55})$$

This verifies both quotient relations, and both compositions fix their generators. Thus the previously observed deformation is the specified base change of the actual escaping cover. This is an isomorphism of these full families, rather than an identification only of their special fibres. The base map is quadratic and is part of the result.

Under EFI54 the sheet involution $r \mapsto -r$ becomes exactly

$$z_{\text{obs}} \mapsto \lambda - z_{\text{obs}}. \quad (\text{EFI56})$$

The two observed factors $z_{\text{obs}} = 0, \lambda$ correspond to $r = -\lambda/2, +\lambda/2$. Consequently the original source coordinates along these sheets, for $\lambda \neq 0$, are respectively

$$\left(\frac{1}{\lambda}, -\frac{3}{2}\lambda, \frac{13}{2}\lambda^2\right), \quad \left(-\frac{1}{\lambda}, \frac{3}{2}\lambda, \frac{13}{2}\lambda^2\right). \quad (\text{EFI57})$$

Substitution into EFI1 gives $(a, b, c) = (-\lambda^2/4, 0, 0)$, as already proved by EFI3. The additional finite point is $(0, 0, -\lambda^2/4)$. Hence the isomorphism carries the actual inverse branches and their remaining finite point, not just the root equation.

The infinitesimal deformation that vanishes to first order

Put $\delta = s - s_*$ and $\kappa = a_{\text{obs}} > 0$, so that $\lambda = \kappa\delta$. Over the first-order base $T_1 = \mathbb{C}[\delta]/(\delta^2)$, EFI54 gives

$$A_{\text{obs}} \otimes_{\mathbb{C}[s]} T_1 \xrightarrow[\sim]{r=z_{\text{obs}}-\kappa\delta/2} T_1[r]/(r^2) = T_1 \otimes_{\mathbb{C}} \mathbb{C}[r]/(r^2). \quad (\text{EFI58})$$

The relation follows because the full centered equation is

$$r^2 = \frac{\kappa^2\delta^2}{4}. \quad (\text{EFI59})$$

Thus the first-order deformation is a product. This is the exact sense in which the deformation class vanishes at first order. The class r itself is nonzero, since $1, r$ are a free T_1 -basis. The map and its inverse in EFI58 reduce to the identity on the central dual-number fibre. Its first-order triviality has therefore not been obtained by removing that fibre.

At second order let $T_2 = \mathbb{C}[\delta]/(\delta^3)$. Then the family is not isomorphic over T_2 to $T_2[\epsilon]/(\epsilon^2)$. To prove this assertion completely, every element of the centered family has a unique form $A + Br$, with $A, B \in T_2$. If it is the image of ϵ under such an isomorphism, its reduction modulo δ must generate the nilpotent ideal of $\mathbb{C}[r]/(r^2)$. Therefore B has nonzero constant term and is a unit. The equation $(A + Br)^2 = 0$ gives, in the free basis $1, r$,

$$2AB = 0, \quad A^2 + B^2 \frac{\kappa^2\delta^2}{4} = 0. \quad (\text{EFI60})$$

Since $2B$ is a unit, the first equation forces $A = 0$. The second then asserts that the nonzero class δ^2 in T_2 is zero after multiplication by the unit $B^2\kappa^2/4$, a contradiction. Thus the first-order triviality has an explicitly nontrivial second-order continuation.

This supplies a precise distinction between a vanishing tangent deformation and a vanishing family. The original escape parameter is $a = -\kappa^2\delta^2/4$: its first derivative is zero at the collision, and its second derivative there is exactly $-\kappa^2/2$.

Trace form and discriminant transported through the isomorphism

Along the real displacement line $s = s_* + \delta$, $\delta \in \mathbb{R}$, the original coefficient $\lambda = \kappa\delta$ is real, even when s_* is not real. Coefficient conjugation fixes this real parameter and fixes z_{obs} . The multiplication trace in basis $1, z_{\text{obs}}$ is

$$G_{\text{obs}}(\lambda) = \begin{pmatrix} 2 & \lambda \\ \lambda & \lambda^2 \end{pmatrix}. \quad (\text{EFI61})$$

Indeed $\text{Tr}(1) = 2$, $\text{Tr}(z_{\text{obs}}) = \lambda$, and $\text{Tr}(z_{\text{obs}}^2) = \lambda^2$, by multiplication in the free basis. For the centered basis $1, r$, the basis-change matrix from its coordinates to the original basis is

$$Q = \begin{pmatrix} 1 & -\lambda/2 \\ 0 & 1 \end{pmatrix}, \quad Q^* G_{\text{obs}} Q = \begin{pmatrix} 2 & 0 \\ 0 & \lambda^2/2 \end{pmatrix}. \quad (\text{EFI62})$$

Direct multiplication verifies this equality. It is precisely the pullback of EFI37 through $a = -\lambda^2/4$: $-2a = \lambda^2/2$. In particular, the observed real-displacement family is positive definite away from the collision and positive semidefinite at the collision, with radical $\mathbb{C}r$ there. Its real base image lies on $a \leq 0$; this is why it does not enter the negative-sign side $a > 0$ of EFI37.

The discriminants satisfy, with every coefficient retained,

$$\text{disc}(r^2 + a) = -4a, \quad (-4a)|_{a=-\lambda^2/4} = \lambda^2 = a_{\text{obs}}^2(s - s_*)^2 = \det G_{\text{obs}}. \quad (\text{EFI63})$$

The original quadratic in EFI2 is $-2(r^2 + a)$, whose polynomial discriminant is $-16a$; its pullback is $4\lambda^2$. This extra factor four belongs to the polynomial's retained leading coefficient. It is not the trace discriminant in the monic algebra basis $1, r$.

The map EFI54 also induces the isomorphism of programme semirings $G_L(\Psi)$ by EFI43, preserving every support label, τ_L , and e_L individually. At $\delta = 0$ it recovers $\hat{r}^2 = e_L \neq \tau_L$. Thus the actual bridge carries the infinitesimal, its zero-prime contraction, and its trace form; no identification of τ_L with supported zero is used.

10. Differential bridge and the primes of the full families

Retain every coordinate of EFI51–EFI55, with $\kappa = a_{\text{obs}}$. Differentiating those exact polynomial maps gives

$$dr = dz_{\text{obs}} - \frac{\kappa}{2} ds, \quad da = -\frac{\kappa\lambda}{2} ds, \quad 2r dr + da = (2z_{\text{obs}} - \lambda) dz_{\text{obs}} - \kappa z_{\text{obs}} ds. \quad (\text{EFI64})$$

For proof substitute $2r = 2z_{\text{obs}} - \lambda$ in the first two equalities: the coefficient of ds is $-\kappa(2z_{\text{obs}} - \lambda)/2 - \kappa\lambda/2 = -\kappa z_{\text{obs}}$. Thus EFI64 is exactly the differential of $r^2 + a = z_{\text{obs}}^2 - \lambda z_{\text{obs}}$. It induces the map between the differential-module presentations

$$\frac{C_{\text{esc}} da \oplus C_{\text{esc}} dr}{C_{\text{esc}}(da + 2r dr)} \longrightarrow \frac{A_{\text{obs}} ds \oplus A_{\text{obs}} dz_{\text{obs}}}{A_{\text{obs}}((2z_{\text{obs}} - \lambda)dz_{\text{obs}} - \kappa z_{\text{obs}} ds)} \quad (\text{EFI65})$$

with scalars extended by EFI52–EFI54. These are the universal modules of differentials: a derivation on a polynomial ring is determined by its values on the two generators, and descends to the quotient exactly when it annihilates the differential of the displayed relation. This proves the presentations and the induced map without assuming smoothness at a collision.

The tangent-deformation comparison can also be proved directly. Over $\mathbb{C}[\eta]/(\eta^2)$, replacing r by $r + \eta h(r)$ changes the first-order coefficient of a relation $r^2 + \eta g(r)$ by $2rh(r)$; multiplying its defining relation by the unit $1 + \eta k(r)$ changes it by $k(r)r^2$. The retained class of that coefficient is therefore

$$[g] \in \mathbb{C}[r]/(r^2, 2r) = \mathbb{C}. \quad (\text{EFI66})$$

For the original escaping unfolding $r^2 + a$, the displacement $a = \eta$ gives $[g] = [1] \neq 0$. For the observed unfolding $z_{\text{obs}}^2 - \kappa\delta z_{\text{obs}}$, the displacement $\delta = \eta$ gives $[g] = [-\kappa z_{\text{obs}}] = 0$. The explicit centering in EFI58 proves actual first-order triviality, not merely the vanishing of an invariant. Equivalently, $da/d\delta = -\kappa^2\delta/2$ vanishes at $\delta = 0$. EFI60 proves exactly what survives at second order.

Primality of supported zero must be checked on the total family as well as on each fibre. The original escape algebra $C_{\text{esc}} \cong \mathbb{C}[r]$ is a domain, so $Z_L = (e_L)$ is prime in $G_L(C_{\text{esc}})$. In the observed total family, $z_{\text{obs}}(z_{\text{obs}} - \lambda) = 0$ with both factors nonzero. Their nonvanishing follows from the free basis $1, z_{\text{obs}}$ and the nonzero polynomial $\lambda = \kappa(s - s_*)$. Hence (e_L) is not prime in $G_L(A_{\text{obs}})$.

There is a precise splitting map between the prime sets. The two ideals $\mathfrak{p}_0 = (z_{\text{obs}})$ and $\mathfrak{p}_1 = (z_{\text{obs}} - \lambda)$ are prime, since their quotients are each $\mathbb{C}[s]$. Their intersection is zero: an element $A(s) + B(s)z_{\text{obs}}$ vanishes on both branches only when $A = 0$ and $B\lambda = 0$, which force $A = B = 0$. They are the only minimal primes, since any prime must contain one factor of $z_{\text{obs}}(z_{\text{obs}} - \lambda)$. Under the composite map $\iota : C_{\text{esc}} \rightarrow A_{\text{obs}}$, $r \mapsto z_{\text{obs}} - \lambda/2$, obtained from base change and Ψ^{-1} in EFI54, r restricts to $-\kappa(s - s_*)/2$ and $+\kappa(s - s_*)/2$ on these branches. Both substitutions are injective maps $\mathbb{C}[r] \rightarrow \mathbb{C}[s]$. Consequently

$$Q_{\mathfrak{p}_0} \cap Q_{\mathfrak{p}_1} = Z_L, \quad G_L(\iota)^{-1}(Q_{\mathfrak{p}_i}) = Z_L \quad (i = 0, 1), \quad (\text{EFI67})$$

with Z_L on the right denoting the zero-amplitude ideal in $G_L(C_{\text{esc}})$. Thus the original supported-zero prime has two arithmetic prime points over it, whose intersection is the retained nonprime zero-amplitude ideal. At the collision both lie below $Q_{(z_{\text{obs}}, \lambda)}$, whose contraction is $Q_{(r)}$. The last contraction follows by evaluating r at zero and retaining the constant coefficient of any polynomial in r .

The triple total families have the same domain issue already before specialization. In B and A respectively,

$$z(z^2 + 6d) = 0, \quad u(u + 6d) = 0, \quad (\text{EFI68})$$

with both factors nonzero. For B this follows from its free basis $1, z, z^2$; for A it follows from its free basis $1, u, v$. Thus their supported-zero ideals are not prime either. Their individual arithmetic primes and their contractions are always the inverse images of the ring primes under EFI42, as proved in EFI45. This statement retains the established prime (e) over domain amplitudes such as $\mathbb{Z}$ and $\mathbb{C}[r]$, and also retains the exact zero-divisor products created by the actual base change and collision.

Exact algebra checks

The companion script `check_escaping_fibre_infinitesimal.py` passes 23 exact polynomial, rational, matrix, and discriminant identities. Its recorded output is `ESCAPING_FIBRE_EXACT_CHECKS.json`. These checks include the original determinant, original inverse composition, both observed branches in the original map, graph-algebra coordinates and products, the two trace matrices, their determinants, and EFI54–EFI63. The ring-theoretic completeness, exact-sequence, and nontrivial second-order arguments are proved in the text rather than inferred from finite sampling.

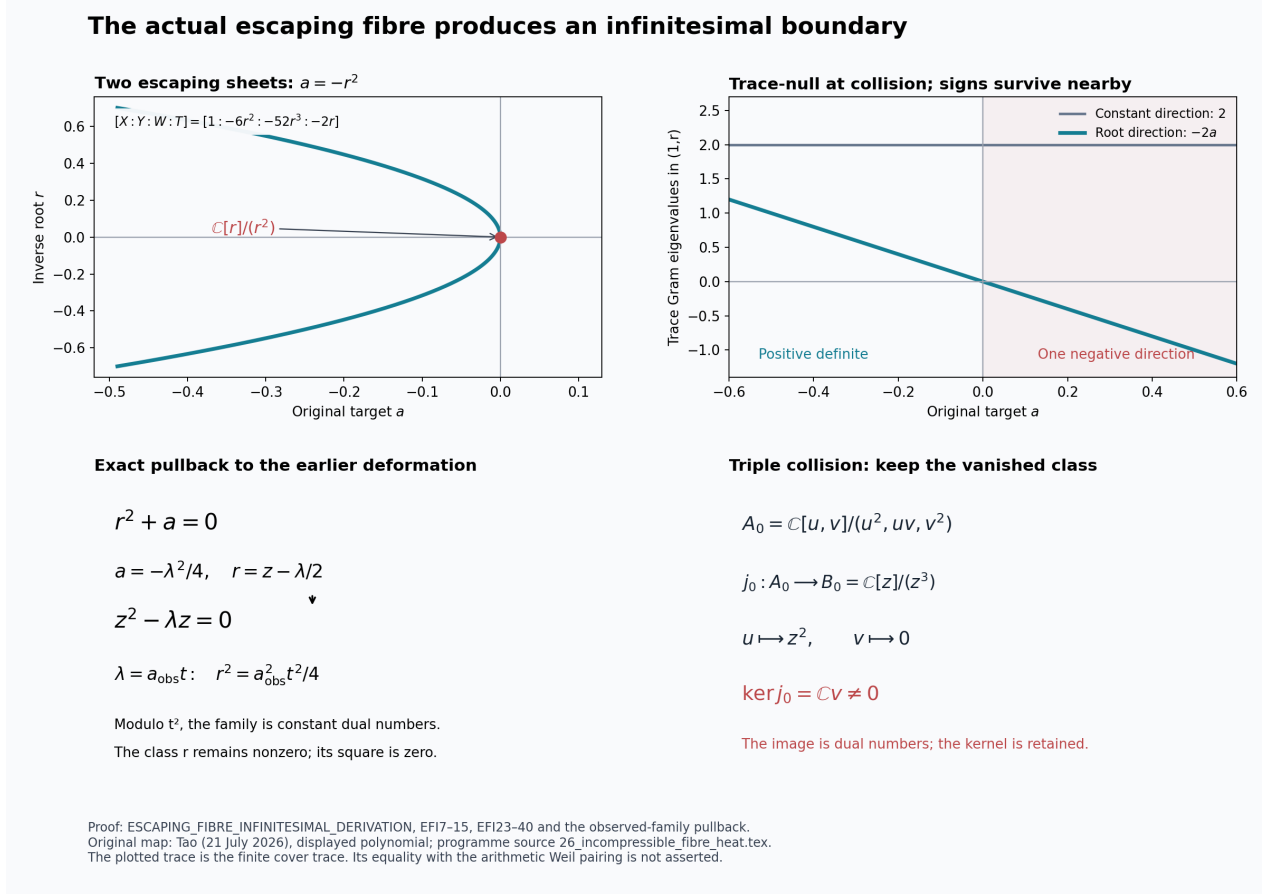


Figure 1: The exact cover $a=-r^2$ and finite trace eigenvalues; the observed-family pullback retains the collision generator. The triple graph map retains its kernel. Proof: EFI7–EFI15, EFI23–EFI40, EFI51–EFI68.

Infinitesimal vanishing, retained currents, and support-valued positivity

22 September 2026. This note computes the several different vanishing statements at the observed collision, the complete signs that survive, and the exact class of corrections capable of changing them. The calculations retain the original observation, metric, Hermitian summand, amplitude multiplier, and full support algebra. No off-line zero of the Riemann zeta function is asserted to exist.

The incoming public programme proofs are *The observed infinitesimal collision and the full arithmetic operator*, OC1–OC17; *Observed support at the shifted nilpotent collision*, OSP1–OSP42; and *Cotangent and Frobenius calculations for the observed collision*, OCF1–OCF23 and OIF1–OIF36. The additional complete local dependencies are full-support reconstruction proof, FSR1–FSR36; finite Weil packet proof, WP1–WP57; and analytic Weil packet proof, WA1–WA30. Their complete sources accompany this derivation; no public link is asserted for an unpublished dependency. The cited proofs retain their original human citations. This is a further finite derivation and an application of the already proved analytic identity WA24, not a replacement for its analytic proof.

1. Original operators and the collision

Keep the original source data and metric from OC1–OC3:

$$\begin{aligned} M &= C + R, \quad C = C^\dagger, \quad R = \epsilon f e^\dagger, \quad e^\dagger e = f^\dagger f = 1, \quad e^\dagger f = 0, \quad \epsilon > 0, \\ J &= G_N^{-1} \Lambda^* Q_B^{1/2}, \quad Q_B = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad J^\dagger J = I_B, \quad C_B = J^\dagger C J = C_B^*. \end{aligned} \quad (\text{ISP1})$$

Here the original arithmetic quotient, cutoff, and observation remain those of OC1: $q = (k+1)^2$, $k \geq 17$, $k \equiv 1 \pmod{4}$, $q-1 \leq N \leq 2q$, and the original surjection $\Lambda : E \rightarrow B$. The star is the adjoint in the Euclidean coordinates supplied by the isometry J ; it does not change the original metric.

Write

$$u = J^\dagger e, \quad v = J^\dagger f, \quad a = u^* u, \quad r = u^* v, \quad d = v^* v, \quad \Delta = ad - |r|^2 > 0.$$

The measured area statement in OC3 supplies this strict inequality. Define

$$g = u/\sqrt{a}, \quad h = \frac{v - (r/a)u}{\sqrt{\Delta/a}}, \quad b = \epsilon\sqrt{\Delta} > 0, \quad P = gg^* + hh^*. \quad (\text{ISP2})$$

The numerator defining h has inner product zero with u , because $u^* v - (r/a)u^* u = 0$. Its squared norm is $d - |r|^2/a = \Delta/a$. Thus g, h are an orthonormal pair and P is their orthogonal projector.

The exact observed family is

$$F_B(s) = (\epsilon v + su)u^*, \quad M_B(s) = C_B + F_B(s), \quad \lambda(s) = as + \epsilon r. \quad (\text{ISP3})$$

Direct multiplication gives $F_B(s)^2 = \lambda(s)F_B(s)$. On the original orthogonal plane $S = \text{im } P$, and on its orthogonal complement H , respectively, it is

$$F_B(s)|_S = \begin{pmatrix} \lambda(s) & 0 \\ b & 0 \end{pmatrix}, \quad F_B(s)|_H = 0. \quad (\text{ISP4})$$

At the collision parameter

$$s_* = -\epsilon r/a, \quad N = F_B(s_*) = \epsilon(v - (r/a)u)u^* = bhg^*, \quad N^2 = 0, \quad Ng = bh \neq 0. \quad (\text{ISP5})$$

The map $\mathbb{C}[\eta]/(\eta^2) \rightarrow \text{End}(B)$, $1 \mapsto I_B$, $\eta \mapsto N$, is injective. Indeed a relation $cI_B + \ell N = 0$ applied to h gives $c = 0$, and then applied to g gives $\ell b = 0$, hence $\ell = 0$. Its image is exactly the generated algebra. The supported version sends its unit to P and has the same proof on S .

2. Exactly what vanishes

For the isolated nilpotent operator, for every scalar t ,

$$\text{tr}(N^j) = 0 \quad (j \geq 1), \quad \det(I_B - tN) = 1, \quad (I_B - tN)^{-1} = I_B + tN. \quad (\text{ISP6})$$

In the basis g, h , N is strictly triangular, and is zero on H . This proves the trace and determinant assertions; multiplication proves the inverse. A marked matrix element retains the exact term

$$x^*((I_B - tN)^{-1} - I_B)y = tb(x^*h)(g^*y). \quad (\text{ISP7})$$

For $x = h, y = g$, this term is tb , which is nonzero when $t \neq 0$. Vanishing of the isolated trace series therefore has an explicit kernel and a nonzero receiver.

The family algebra itself is

$$\mathcal{A} = \mathbb{C}[t, z]/(z(z - at)), \quad t = s - s_*.$$

The invertible change $\xi = z - at/2$, with inverse $z = \xi + at/2$, gives

$$z(z - at) = \xi^2 - a^2t^2/4. \quad (\text{ISP8})$$

Over $\mathbb{C}[t]/(t^2)$, this is an isomorphism with the constant dual-number family, and it reduces to the identity on the fibre $t = 0$. Thus its first-order deformation class vanishes. Over $\mathbb{C}[t]/(t^3)$, the trace discriminant in the basis $1, z$ equals $a^2t^2 \neq 0$; the constant dual-number family's trace discriminant is zero. To compute it, multiplication by z has trace at , and multiplication by $z^2 = atz$ has trace a^2t^2 . The trace matrix is therefore $\begin{pmatrix} 2 & at \\ at & a^2t^2 \end{pmatrix}$, with determinant a^2t^2 . An invertible basis change multiplies its determinant by a unit square. Hence it cannot turn this nonzero discriminant into zero. The first nontrivial order is exactly two.

The original operator representation and its metric retain additional information. Work now over $\mathbb{C}[t]/(t^2)$, and put

$$K = \frac{a}{2b}gh^*, \quad T = I_B + tK, \quad T^{-1} = I_B - tK.$$

Since $[K, N] = (a/2)(gg^* - hh^*)$, direct multiplication proves

$$F_B(s_* + t) = TNT^{-1} + \frac{at}{2}P,$$

$$T^{-1}M_B(s_* + t)T = C_B + N + t\left([C_B, K] + \frac{a}{2}P\right). \quad (\text{ISP9})$$

Here C_B remains the original operator on all of B ; it need not preserve S . These formulas follow by expanding products and using $t^2 = 0$. If the formal parameter is given the real involution $t^* = t$, then

$$T^*T = I_B + t(K + K^*). \quad (\text{ISP10})$$

The coefficient $K + K^*$ is nonzero, because $(K + K^*)h = (a/(2b))g$. Thus this coordinate trivialization is not unitary for the original metric. Equations ISP9–ISP10 retain the exact operator and metric changes attached to the vanishing deformation class.

There is also a choice-dependent Frobenius vanishing. Over a torsionfree integral collision ring $D_p = K_p[\eta]/\eta^2$ with the coefficient Frobenius specified in OIF33, every lift is

$$\Phi_c(\eta) = pc\eta, \quad c \in K_p. \quad (\text{ISP11})$$

To verify this, write an image as $A + B\eta$. Its square is zero, so the coefficient domain forces $A = 0$. Its reduction must equal $\eta^p = 0$, forcing $B \in pK_p$; the converse is immediate. The corrected lift has $c = 0$, so kills η , whereas the earlier multiplicative lift has $c = 1$ and sends it to $p\eta \neq 0$. The same underlying nilpotent algebra therefore does not determine this Frobenius vanishing without the specified lift.

3. The full arithmetic determinant does not vanish infinitesimally

Keep C_B in the characteristic polynomial, and let $n = \dim B$. Rank-one determinant expansion gives

$$\det(\zeta I_B - C_B - N) = \det(\zeta I_B - C_B) - b g^* \text{adj}(\zeta I_B - C_B) h. \quad (\text{ISP12})$$

For a proof, expand the determinant column by column. Every term containing at least two columns from bhg^* vanishes because those columns are proportional to h . The terms containing exactly one such column sum to the displayed adjugate contraction, with the minus sign from the subtracted perturbation. This proof is a polynomial identity and does not require an inverse.

In particular, the isolated identity $\det(I - tN) = 1$ does not remove the second term of ISP12. On the plane, the concrete Hermitian choice $C_B|_S = \begin{pmatrix} 0 & c \\ c & 0 \end{pmatrix}$, $c \in \mathbb{R} \setminus \{0\}$, gives

$$\det(\zeta I_S - C_B|_S - N|_S) = \zeta^2 - c(c + b), \quad \det(\zeta I_S - C_B|_S) = \zeta^2 - c^2. \quad (\text{ISP13})$$

This example disproves an inference from nilpotence alone; it is not substituted for the native C_B .

For the actual family, define

$$p_*(\zeta) = \det(\zeta I_B - M_B(s_*)), \quad q_*(\zeta) = u^* \operatorname{adj}(\zeta I_B - M_B(s_*))u.$$

The same polynomial expansion proves, exactly for every t ,

$$p_{s_*+t}(\zeta) = p_*(\zeta) - tq_*(\zeta). \quad (\text{ISP14})$$

Moreover q_* has degree $n-1$ and leading coefficient $u^*u = a > 0$: the adjugate of $\zeta I_B - M_B(s_*)$ has leading coefficient $\zeta^{n-1}I_B$. Consequently the full characteristic polynomial has a nonzero first derivative in the original parameter, even though the abstract algebra deformation class in ISP8 vanishes. Equivalently, $\operatorname{tr} M_B(s_* + t) = \operatorname{tr} M_B(s_*) + at$. These statements concern the actual original family.

4. The signed current survives with both signs

Use exactly the programme convention

$$W_B(s) = i(M_B(s) - M_B(s)^*).$$

The retained Hermitian term cancels in this particular difference because $C_B = C_B^*$. It remains in ISP12–ISP14. Direct conjugate transposition of ISP4 gives

$$W_B(s)|_S = \begin{pmatrix} -2\operatorname{Im} \lambda(s) & -ib \\ ib & 0 \end{pmatrix}, \quad W_B(s)|_H = 0. \quad (\text{ISP15})$$

At the infinitesimal collision,

$$\begin{aligned} W_* &:= W_B(s_*) = i(N - N^*), & W_*^2 &= b^2 P, \\ W_* \frac{g + ih}{\sqrt{2}} &= b \frac{g + ih}{\sqrt{2}}, & W_* \frac{g - ih}{\sqrt{2}} &= -b \frac{g - ih}{\sqrt{2}}. \end{aligned} \quad (\text{ISP16})$$

These follow by multiplying the two-by-two matrix, or by using $Ng = bh$, $Nh = 0$, $N^*h = bg$, $N^*g = 0$. Its inertia on B is $(1, 1, n-2)$. For $y = c_g g + c_h h + y_H$,

$$y^* W_* y = 2b \operatorname{Im}(\overline{c_g} c_h). \quad (\text{ISP17})$$

Thus the two unit eigenvectors in ISP16 have values $+b$ and $-b$, not zero.

The current correction from the original parameter $s = 0$ is exactly

$$W_* - W_B(0) = 2\epsilon \operatorname{Im}(r) g g^*. \quad (\text{ISP18})$$

For every complex parameter s , the determinant of its active matrix in ISP15 is $-b^2 < 0$. Consequently that rank-one displacement cannot make the current positive semidefinite. Along the real displacement $s = s_* + t$, $t \in \mathbb{R}$, the full current is exactly constant, $W_B(s_* + t) = W_*$, because $\lambda(s_* + t) = at$ is real.

5. All Hermitian corrections on the current plane

Write an arbitrary Hermitian correction on S as

$$D = \begin{pmatrix} \alpha & c \\ \bar{c} & \delta \end{pmatrix}, \quad \alpha, \delta \in \mathbb{R}, \quad c \in \mathbb{C}.$$

Then the complete set of corrections giving a nonnegative form is

$$\boxed{W_* + D \succeq 0 \iff \alpha \geq 0, \delta \geq 0, \alpha\delta \geq |c - ib|^2.} \quad (\text{ISP19})$$

For necessity evaluate on g, h , obtaining the diagonal inequalities, and use the nonnegative determinant. For sufficiency, when $\alpha > 0$, complete the square:

$$\alpha \left| x + \frac{c - ib}{\alpha} y \right|^2 + \left(\delta - \frac{|c - ib|^2}{\alpha} \right) |y|^2 \geq 0.$$

When $\alpha = 0$, the determinant inequality forces $c - ib = 0$, and the form is $\delta|y|^2$. This proves every boundary case as well as the positive-definite case.

In particular, an orthogonal-support correction tP , $t \in \mathbb{R}$, gives positivity exactly for $t \geq b$. A correction supported only on gg^* , or only on hh^* , never suffices. The equality $e^2 = e$ by itself supplies no scalar coefficient t ; ISP19 states the exact correction needed by the operator it would act on.

The analogous complete classification for a larger support space is as follows. Let X, Z be finite-dimensional Hermitian spaces and let

$$\mathcal{Q} = \begin{pmatrix} H & A \\ A^* & D \end{pmatrix} \quad \text{on } X \oplus Z, \quad H = H^*, \quad D = D^*.$$

Let D^+ be the operator equal to the reciprocal of each nonzero eigenvalue of D and zero on its kernel. Then

$$\boxed{\mathcal{Q} \succeq 0 \iff D \succeq 0, \quad A(\ker D) = 0, \quad H - AD^+A^* \succeq 0.} \quad (\text{ISP20})$$

To prove necessity, first restrict to Z . For $z_0 \in \ker D$, the value on (x, tz_0) is $x^*Hx + 2\operatorname{Re}(tx^*Az_0)$. Its nonnegativity for every complex t forces $Az_0 = 0$. Thus A^*x belongs to $(\ker D)^\perp = \operatorname{im} D$. For all x, z , exact multiplication now gives

$$\langle (x, z), \mathcal{Q}(x, z) \rangle = x^*(H - AD^+A^*)x + (z + D^+A^*x)^*D(z + D^+A^*x). \quad (\text{ISP21})$$

Set $z = -D^+A^*x$ for the last necessary inequality. Conversely ISP21 proves positivity from the three displayed conditions. This proof constructs the minimizing support coordinate and includes singular D .

For a fixed Hermitian correction R to the old block, replace H in ISP20 by $H + R$; the result classifies every such corrected extension. Adding independent support coordinates and mixed blocks while leaving H unchanged cannot remove an old negative value: it is still the value on $(x, 0)$. Indeed minimizing over the added positive coordinates subtracts AD^+A^* , exactly as ISP21 records.

6. Full support and every positive scalar trace

Retain the finite bounded distributive lattice L , the actual semiring $G_L(R)$, and its supported zero e_L , with $(e_L) = Z_L$ prime when R is a domain. This is prior programme mathematics, not a new result here. The contracted multiplicative algebra is the proved product

$$\Gamma_L = \mathbb{C}[(G_L(R), \times)]/([\tau_L]) \cong C_L \times D_R, \quad C_L = \mathbb{C}[(L, \wedge)]/([0_L]). \quad (\text{ISP22})$$

In this algebra $[e_L]$ is the identity of the C_L factor. It is a multiplicative idempotent, not the square-zero operator N of ISP5. The latter comes from the separate, explicitly specified deformation/observation receiver. The exact role of the former in the following coefficient extension is multiplication by the support identity.

FSR11–FSR14 give

$$C_L = \bigoplus_{a \in L \setminus \{0\}} \mathbb{C}E_a, \quad E_a E_b = \delta_{ab} E_a, \quad E_a^* = E_a, \quad \sum_a E_a = 1_{C_L}. \quad (\text{ISP23})$$

For completeness, the characters are $\eta_a([\lambda]) = 1_{a \leq \lambda}$; their incidence matrix is triangular with diagonal one, hence invertible. Its inverse gives $E_a = \sum_{\lambda \leq a} \mu(\lambda, a)[\lambda]$. Applying every character proves the stated multiplication and unit identities.

Every positive complex-linear scalar functional on C_L is uniquely

$$\ell\left(\sum_a c_a E_a\right) = \sum_a t_a c_a, \quad t_a \geq 0. \quad (\text{ISP24})$$

Indeed $E_a = E_a^* E_a$ forces $\ell(E_a) \geq 0$; conversely the displayed coefficients give $\ell(v^*v) = \sum_a t_a |v_a|^2 \geq 0$. It is faithful on positive elements precisely when every $t_a > 0$.

Extend the actual current coefficientwise to $B_L = C_L \otimes B$, retaining $1 \otimes C_B$ in its full arithmetic operator. Its support-valued form is

$$\mathcal{W}_L(x, y) = \sum_a E_a x_a^* W_* y_a, \quad x = \sum_a E_a \otimes x_a. \quad (\text{ISP25})$$

For every a with $t_a > 0$, the exact vector

$$x_a^- = E_a \otimes (g - ih)/\sqrt{2} \quad \text{has} \quad \mathcal{W}_L(x_a^-, x_a^-) = -bE_a, \quad \ell\mathcal{W}_L(x_a^-, x_a^-) = -bt_a < 0. \quad (\text{ISP26})$$

Thus every nonzero positive functional retains a negative direction for this form. If exactly r of the weights are positive and $d_L = |L| - 1$, its scalar inertia is

$$(r, r, (n-2)r + n(d_L - r)). \quad (\text{ISP27})$$

This follows by using the two eigenvectors in ISP16 and a basis of H in every sector; the sectors of weight zero contribute their entire n -dimensional space to the radical.

The same proof applied to the full finite Weil pairing FSR28, whose scalar inertia is $(2, 2, 4m - 4)$, gives

$$\text{inertia}(\ell\mathbb{B}_L) = (2r, 2r, (4m-4)r + 4m(d_L - r)). \quad (\text{ISP28})$$

In particular, the regular trace has all weights one. Boolean branch traces select only join-irreducible sectors and put zero weights on the remaining mixed sectors; zero observation there does not prove zero full form. For $L = \mathbb{B}^2$, the mixed projector $w = [1] - [a] - [b]$ obeys $w^2 = w \neq 0$, both branch traces kill it, and FSR34–FSR35 give the exact value $-2mw$ on the specified mixed negative packet.

7. Exact maps from the collision current to a quartet packet

This section constructs a finite comparison without identifying it with the native period observation. Let $\rho = 1/2 + \delta + i\gamma$, $0 < \delta < 1/2$, $\gamma > 2$, and use the four distinct points

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = (\rho, 1 - \bar{\rho}, \bar{\rho}, 1 - \rho).$$

For $m \geq 1$, let $d(s) = \prod_i (s - \alpha_i)$, $h(s) = d(s)^m$, $E_h = \mathbb{C}[s]/(h)$, and let e_i be the full primary idempotents. Let $U = M_{j_h(v)}$ be the original amplitude unit, with its full Taylor coefficients and nonzero values $v(\alpha_i)$. Let $a_h = \text{ev } U : E_h \rightarrow \mathbb{C}^4$. The finite pairing is

$$B_h(x, y) = m(\overline{(a_h x)_2}(a_h y)_1 + \overline{(a_h x)_1}(a_h y)_2 + \overline{(a_h x)_4}(a_h y)_3 + \overline{(a_h x)_3}(a_h y)_4). \quad (\text{ISP29})$$

These algebraic data can be defined for arbitrary such points and an arbitrary specified unit; they do not assert a zeta zero.

Define the typed injective linear map

$$\mathcal{I} : S \longrightarrow E_h, \quad c_g g + c_h h \longmapsto \sqrt{b/m} U^{-1} (c_g e_1 - i c_h e_2). \quad (\text{ISP30})$$

Applying a_h gives $\sqrt{b/m}(c_g, -i c_h, 0, 0)$, so injectivity follows. Substitution in ISP29 gives the exact isometry

$$B_h(\mathcal{I}x, \mathcal{I}y) = x^* W_* y \quad (x, y \in S). \quad (\text{ISP31})$$

Explicitly the right side is $-ib\overline{x_g}y_h + ib\overline{x_h}y_g$, and the left side has the same two coefficients. This is an isometry of Hermitian forms, whose signs are indefinite; it is not asserted to preserve the original positive norm.

There is also a nilpotent operator on the complete packet, including all its jets:

$$\mathcal{N}_h(x) = -ib(a_h x)_1 U^{-1} e_2. \quad (\text{ISP32})$$

Since $(a_h U^{-1} e_2)_1 = 0$, it has square zero. It is nonzero on $U^{-1} e_1$, and direct substitution proves

$$\mathcal{N}_h \mathcal{I} = \mathcal{I} N|_S. \quad (\text{ISP33})$$

This comparison preserves the constant b , its complex phase, and the original U . It sends the radical $J_h = (d)/(d^m)$ to zero by the formula for a_h ; every radical jet remains in the domain and in U . Extending ISP30–ISP33 by $C_L \otimes -$ gives the identical maps in every support sector. The construction is explicit and depends on the chosen quartet; equality with the original period receiver has not been asserted or assumed.

8. Interpolation removes endpoint terms without changing any packet jet

Since none of the four points is 0 or 1, $h(0)h(1) \neq 0$. For a polynomial q , set

$$\boxed{\mathcal{R}_h q(s) = q(s) - h(s) \left((1-s) \frac{q(0)}{h(0)} + s \frac{q(1)}{h(1)} \right)}. \quad (\text{ISP34})$$

This is a linear polynomial map. Substitution at the two endpoints gives

$$\mathcal{R}_h q(0) = \mathcal{R}_h q(1) = 0, \quad \mathcal{R}_h q \equiv q \pmod{h}. \quad (\text{ISP35})$$

The latter congruence preserves the derivatives of every order $0, \dots, m-1$ at every α_i , because h has order m there. Conversely divisibility by h is exactly the kernel of the complete quartet jet map, by successive division by the four distinct linear factors. Therefore every original packet class, including every nilpotent jet and the entire action of U , is retained by this map.

Let $s_h : E_h \rightarrow \mathbb{C}[s]$ select the unique representative of degree below $4m$, as supplied by monic polynomial division. The composite

$$\tilde{s}_h = \mathcal{R}_h s_h : E_h \longrightarrow s(s-1)\mathbb{C}[s] \quad (\text{ISP36})$$

is a linear section of polynomial reduction modulo h , with degree at most $4m+1$. Its image has dimension $4m$, and every class has exactly the stated lift. In particular this endpoint condition does not constrain any quartet jet.

The same statement holds for arbitrary specified finite endpoint orders. For $K \geq 1$, put $b_K(s) = s^K(s-1)^K$. It is a unit in E_h , because each $b_K(\alpha_i) \neq 0$. An explicit inverse is obtained by the finite Taylor reciprocal in each factor $\mathbb{C}[t_i]/(t_i^m)$, followed by the primary idempotent inverse of the jet map. Consequently

$$\tilde{s}_{h,K}(x) = b_K s_h([b_K]^{-1}x) \quad (\text{ISP37})$$

is a linear section of reduction modulo h whose values vanish to order at least K at both 0 and 1. Multiplication followed by reduction proves the section identity exactly. This constructs the connecting map for every finite endpoint jet constraint.

9. Consequence for the explicit formula and zero-prime corrections

The following application concerns the programme's counterfactual zeta packet used in WP1–WP4 and WA1–WA2: its four specified points have exact zero order m for $g = 2\xi$, so $v = g/h$ is entire and has nonzero values there. This is the established test of an off-line packet; it is not an existence claim.

For two polynomials q_1, q_2 , use exactly

$$q^\#(s) = \overline{q(1-\bar{s})}, \quad A_{q_1, q_2}(s) = q_1^\#(s)q_2(s)v(s)^2. \quad (\text{ISP38})$$

Let $\tilde{q}_i = \mathcal{R}_h q_i$. Equations ISP35 and WP21 prove that the original full primary amplitude jets of $\tilde{q}_i v$ and $q_i v$ agree. Furthermore

$$A_{\tilde{q}_1, \tilde{q}_2}(0) = A_{\tilde{q}_1, \tilde{q}_2}(1) = 0. \quad (\text{ISP39})$$

Indeed the factor $\tilde{q}_2$ vanishes at both endpoints. This conclusion also follows by evaluating the reflected first factor; no positivity inference is used.

Take the full-jet negative class

$$x_- = U^{-1}(e_1 - e_2), \quad \tilde{q}_- = \tilde{s}_h(x_-). \quad (\text{ISP40})$$

Its four amplitude values are exactly $(1, -1, 0, 0)$, hence

$$B_h([\tilde{q}_-], [\tilde{q}_-]) = -2m. \quad (\text{ISP41})$$

Its Fourier-coordinate polynomial is $\tilde{P}_-(z) = \tilde{q}_-(1/2 + iz)$; define $F_-(z) = \tilde{P}_-(z)v(1/2 + iz)$,

$$K_-(t) = |F_-(t)|^2, \quad k_-(u) = \frac{1}{2\pi} \int_{\mathbb{R}} K_-(t) e^{itu} dt.$$

WA9–WA16 apply to every polynomial, so the increased degree in ISP36 changes their constants but not the proved admissibility, exponential strip decay, or absolute convergence. Substitution into the full analytic identity WA24, using ISP39, gives

$$\boxed{-2m = \frac{1}{2\pi} \int_{\mathbb{R}} |F_-(t)|^2 \left(\operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi \right) dt - \sum_{n \geq 2} \frac{\Lambda_{\text{ar}}(n)}{\sqrt{n}} (k_-(\log n) + k_-(-\log n)).} \quad (\text{ISP42})$$

This is the exact endpoint-free arithmetic expression for that packet, with every sign and transform convention retained. It does not prove that this packet exists for zeta, and does not prove nonnegativity of the arithmetic expression.

More generally let $\mathcal{E}(q)$ be the vector of any fixed finite list of derivatives at 0, 1, and let $D = D^*$ be any matrix on that vector space. Every correction

$$\mathcal{D}(q_1, q_2) = \mathcal{E}(q_1)^* D \mathcal{E}(q_2) \quad (\text{ISP43})$$

vanishes on the image of ISP37 for sufficiently large K . Yet that image surjects onto every packet jet and contains the negative class ISP40. The same conclusion holds if $\mathcal{E}$ takes the endpoint jets of qv : multiplication by the entire function v preserves the required vanishing orders. Thus no correction factoring solely through finitely many endpoint evaluations or endpoint derivatives makes this packet form positive. This is a classification of that class of corrections, not an assumption that all effects of the zero prime belong to it.

In full support, choose any sector E_a and lift the polynomial as $E_a \tilde{q}_-$. Its exact support-valued form is $-2mE_a$, while all the specified endpoint jets are zero. A positive scalar trace with weight $t_a > 0$ reads $-2mt_a < 0$, by ISP24. Mixed sectors obey the same calculation, including sectors omitted by every separate Boolean branch.

The proved zero-prime structure therefore changes the spectrum and enlarges the retained coefficient and observation spaces, as FSR1–FSR22 specify. A corrected analytic formula must supply its actual distribution or operator on these spaces. The results above already settle three concrete possibilities: nilpotence alone does not kill the observed current; coefficientwise full-support extension does not change its surviving signs; and a finite endpoint correction cannot erase the explicit negative quartet test. The spaces and maps exposed by these statements are ISP20–ISP21, ISP23–ISP28, and ISP34–ISP43. They retain the objects on which a further arithmetic correction must act.

10. The endpoint pairing also has a computable sign

For this quartet polynomial,

$$h(0) = h(1) = (|\rho|^2 |1 - \rho|^2)^m > 0, \quad v(0) = v(1) = c_0 := 1/h(0) > 0.$$

The first equality follows by grouping the conjugate factors and using the reflection permutation of the four roots. The second uses the exact convention $g(0) = g(1) = 1$ of WA1. Hence for endpoint vectors $e(q) = (q(0), q(1))$, the pole pairing itself is

$$\begin{aligned} A_{q_1, q_2}(0) + A_{q_1, q_2}(1) &= c_0^2 (\overline{q_1(1)} q_2(0) + \overline{q_1(0)} q_2(1)) \\ &= e(q_1)^* c_0^2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} e(q_2). \end{aligned} \quad (\text{ISP44})$$

Thus the pole contribution by itself has one positive and one negative endpoint direction. Evaluation onto $\mathbb{C}^2$ is surjective by the polynomial $(1-s)c + sd$, so both directions occur. This computes the actual contribution already present in WA24; it does not assign an unproved coefficient to the additional prime (e_L). The section ISP36 kills both of these directions while preserving every packet jet.

11. Exact finite verification

The reproducible script `check_infinisimal_support_positivity.py` passed 38 exact symbolic checks, recorded in `INFINITESIMAL_SUPPORT_POSITIVITY_CHECKS.json`. The checks include the nilpotent/current identities, both signed eigenvectors, the first-order conjugation with a Hermitian matrix coupling the current plane to its complement, the full characteristic polynomial, the current-to-packet isometry and intertwiner, every retained jet in a multiplicity-two quartet example, endpoint vanishing through order two in the higher-order section, the singular-block square-completion identity, and the eight exact module/metric identities of ISP46–ISP53. The calculations use exact rational and symbolic coefficients. The finite sample points are not asserted to be zeta zeros. These checks support the displayed algebraic calculations; they do not replace the general proofs above or assert a new analytic positivity theorem.

12. The positive escaping trace and the original observed metric

The escaping-family isomorphism EFI52–EFI55 uses the exact base map $a_{\text{esc}} = -\lambda^2/4$ and coordinate $r_{\text{esc}} = z_{\text{obs}} - \lambda/2$. Along the real displacement $s - s_*$, the observed coefficient $\lambda = a(s - s_*)$ is real. Its algebra

multiplication trace in the basis 1, z_{obs} is

$$G_{\text{tr}}(\lambda) = \begin{pmatrix} 2 & \lambda \\ \lambda & \lambda^2 \end{pmatrix}. \quad (\text{ISP45})$$

Indeed $z_{\text{obs}}^2 = \lambda z_{\text{obs}}$, multiplication by z_{obs} has trace λ , and multiplication by its square has trace λ^2 , while the identity has trace two. The centered coordinate $r_{\text{esc}} = z_{\text{obs}} - \lambda/2$ gives the congruent matrix $\text{diag}(2, \lambda^2/2)$. Thus this form is positive definite for real $\lambda \neq 0$, and has one-dimensional radical at zero. The following calculation gives its exact relationship to the original metric and current.

Write $z = z_{\text{obs}}$ for the retained observed generator. Retain $S = \text{span}\{g, h\}$ with its original orthonormal frame, and $b = \epsilon\sqrt{\Delta} > 0$. The supported receiver is

$$\rho_\lambda : \mathbb{C}[z]/(z^2 - \lambda z) \longrightarrow \text{End}(S), \quad 1 \mapsto I_S, \quad z \mapsto F_\lambda = \begin{pmatrix} \lambda & 0 \\ b & 0 \end{pmatrix}. \quad (\text{ISP46})$$

It is a homomorphism because $F_\lambda^2 = \lambda F_\lambda$. It is injective: a linear relation $cI_S + dF_\lambda = 0$ applied to h gives $c = 0$, and applied to g then gives $db = 0$. The cyclic module map from its regular representation to S is

$$T_\lambda : \mathbb{C}[z]/(z^2 - \lambda z) \longrightarrow S, \quad 1 \mapsto g, \quad z \mapsto \lambda g + bh, \quad [T_\lambda]_{(1,z),(g,h)} = \begin{pmatrix} 1 & \lambda \\ 0 & b \end{pmatrix}. \quad (\text{ISP47})$$

Its determinant is $b \neq 0$. Direct multiplication proves $F_\lambda T_\lambda = T_\lambda M_z$, where $M_z = \begin{pmatrix} 0 & 0 \\ 1 & \lambda \end{pmatrix}$. Thus regular multiplication and the supported receiver have the same ordinary traces.

The algebra involution used in ISP45 fixes z when λ is real. Its image is not fixed by the original matrix adjoint:

$$\rho_\lambda(z^*) - \rho_\lambda(z)^* = F_\lambda - F_\lambda^* = -iW_*, \quad W_* = \begin{pmatrix} 0 & -ib \\ ib & 0 \end{pmatrix}. \quad (\text{ISP48})$$

The equality follows by conjugate transposition. It is nonzero for every real λ , including zero. Thus the exact map relating the forms is a faithful algebra receiver with this specified adjoint defect.

There are three retained Hermitian forms on the same two-dimensional coordinate space. The algebra multiplication trace gives ISP45. Pulling back the original vector metric through ISP47 gives

$$G_{\text{vec}} = T_\lambda^* T_\lambda = \begin{pmatrix} 1 & \lambda \\ \lambda & \lambda^2 + b^2 \end{pmatrix}. \quad (\text{ISP49})$$

Pulling back the matrix Hilbert–Schmidt metric through ISP46 gives

$$G_{\text{HS}} = \begin{pmatrix} \text{Tr}(I_S) & \text{Tr}(F_\lambda) \\ \text{Tr}(F_\lambda^*) & \text{Tr}(F_\lambda^* F_\lambda) \end{pmatrix} = \begin{pmatrix} 2 & \lambda \\ \lambda & \lambda^2 + b^2 \end{pmatrix}. \quad (\text{ISP50})$$

Every entry follows by multiplying the displayed matrix F_λ . In particular

$$G_{\text{HS}} - G_{\text{tr}} = \begin{pmatrix} 0 & 0 \\ 0 & b^2 \end{pmatrix}, \quad \|\rho_0(z)\|_{\text{HS}}^2 = b^2 > 0. \quad (\text{ISP51})$$

The trace-null generator at collision has a nonzero, exactly measured operator norm. The supported unit in ISP46 has rank two. Using instead the full unital receiver $1 \mapsto I_B$ changes the upper-left entry of ISP50 to $\dim B$; it does not change its other three entries.

The trace pairing makes multiplication by the real generator self-adjoint, as matrix multiplication verifies:

$$G_{\text{tr}} M_z = M_z^* G_{\text{tr}}.$$

For the positive matrix metric the exact defect is instead

$$G_{\text{HS}} M_z - M_z^* G_{\text{HS}} = \begin{pmatrix} 0 & -b^2 \\ b^2 & 0 \end{pmatrix}. \quad (\text{ISP52})$$

This is obtained by subtracting the two products, with no limiting argument. Thus the extra positive term in ISP51 has a specified effect on multiplication adjoints; it cannot be added while asserting unchanged adjoint compatibility.

Finally, transporting the trace form itself to the original vector plane gives

$$Q_\lambda = (T_\lambda^{-1})^* G_{\text{tr}} T_\lambda^{-1} = \begin{pmatrix} 2 & -\lambda/b \\ -\lambda/b & \lambda^2/b^2 \end{pmatrix}, \quad \det Q_\lambda = \lambda^2/b^2. \quad (\text{ISP53})$$

The inverse $T_\lambda^{-1} = \begin{pmatrix} 1 & -\lambda/b \\ 0 & 1/b \end{pmatrix}$ proves the formula by multiplication. This positive metric for $\lambda \neq 0$ makes F_λ self-adjoint; at the collision it becomes degenerate and kills h . There cannot be a positive definite metric making the nonzero $N = bhg^*$ self-adjoint: that would imply $\langle Nx, Nx \rangle = \langle x, N^2 x \rangle = 0$ for every x , forcing $N = 0$. The explicitly constructed degenerate metric ISP53 is the retained limiting object, while the original positive metric and the current W_* remain unchanged.

The Hermitian summand C_B of the full arithmetic operator remains in ISP12–ISP14. The present algebra comparison concerns F_λ , and makes no commutation assertion about C_B or identification of the trace metric with the classical Weil form.

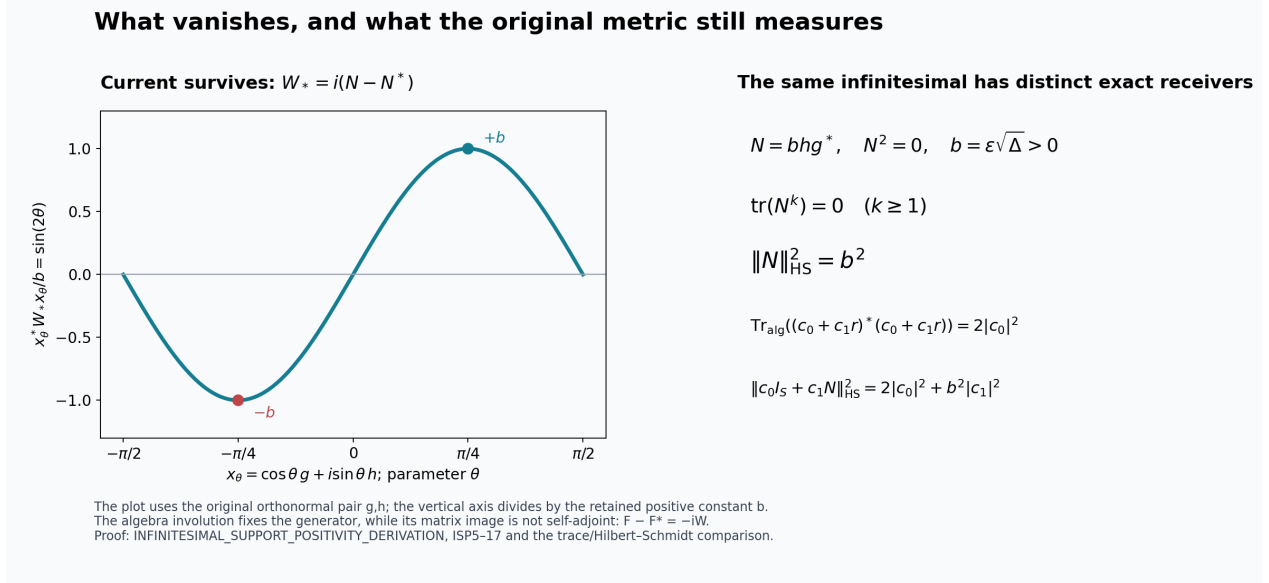


Figure 2: The same nonzero infinitesimal has zero algebraic trace and nonzero matrix norm; its current has both signs in the original metric. Proof: ISP5–ISP6, ISP18–ISP20, ISP45–ISP53.

The logarithmic arithmetic distribution of the separated-support zeta sheet

This calculation uses the actual interpolation in the programme's *Secondary Note on Split-Zero Globalization*, canonical source PUBUNIT-B1A260A5E731EA568C353AA4, equations under `def:hurwitz-specialization`, `thm:universal-taylor-deformation`, `thm:arithmetic-kernels-riemann`, and `cor:first-log-jet-explicit`. It also uses *Split-Zero Support Repair*, some `considerations.tex`, sections “Conservative ideal and zeta side” and “Shifted zeta deformation”. These are prior programme results. The supported-zero prime is also a prior result of Paper 1, attributed by the user to Gemini Pro 2.5. The calculation below derives the signed arithmetic distribution that replaces the prime-power distribution for this particular separated sheet. It does not assign an arbitrary finite norm to the supported-zero prime.

NL1. The original scalar maps and the actual analytic family

Let $R = \mathbb{Z}$, and let $G(R) = R \sqcup \{\tau\}$. Addition and multiplication on supported elements are their ring operations; $\tau + x = x$ and $\tau x = \tau$. Write $e = 0_R$, so $e \neq \tau$, $e^2 = e$, and $er = e$ for every supported r . The two maps

$$p : G(R) \rightarrow R, \quad p(r) = r, \quad p(\tau) = 0, \quad \chi : G(R) \rightarrow \mathbb{B}, \quad \chi(r) = 1, \quad \chi(\tau) = 0 \quad (\text{NL1})$$

preserve both operations and units, directly by these laws. The prime ideal $(e) = \{\tau, e\}$ is $p^{-1}(0)$; its complement consists of nonzero integers and is multiplicatively closed. This proves its primality without identifying its generator with the semiring zero.

The separated residue map $G(\mathbb{Z}) \rightarrow G(\mathbb{Z}/n\mathbb{Z})$, obtained by reducing the supported amplitude and keeping τ , has target cardinality $n + 1$. This holds also for $n = 1$: the zero ring has a supported zero and the distinct τ . Composing with amplitude gives $\mathbb{Z}/n\mathbb{Z}$, of cardinality n . Thus the two defined sums are

$$F_0(s) = \sum_{n \geq 1} n^{-s} = \zeta(s), \quad F_1(s) = \sum_{n \geq 1} (n + 1)^{-s} = \zeta(s) - 1. \quad (\text{NL2})$$

Their interpolation is

$$F_t(s) = \sum_{n \geq 1} (n + t)^{-s} = \zeta(s, 1 + t), \quad t \geq 0, \quad \Re s > 1. \quad (\text{NL3})$$

Here t is a real shift parameter, not the semiring element τ . The equality with Hurwitz zeta is the defining series with its index translated; compare the original TeX formulas at DLMF 25.11.1 and 25.11.3.

The prime (e) cannot be inserted into the old ideal Euler product with a positive finite multiplicative norm extending the integer ideal norm. Indeed $(e)I_{(n)} = (e)$; a multiplicative norm h would satisfy $h((e)) = nh((e))$, so $h((e)) = 0$ for $n = 2$. This equality specifies the obstruction. The separated residue sum (NL3) is a different, already defined analytic receiver of support. Its logarithmic distribution is now calculated in full.

NL2. A convergent expansion with every multiplicative interaction retained

Put

$$b = 1 + t, \quad r_n = \frac{n + t}{1 + t} \quad (n \geq 2), \quad H_t(s) = \sum_{n \geq 2} r_n^{-s}. \quad (\text{NL4})$$

Every $r_n > 1$, and

$$F_t(s) = b^{-s}(1 + H_t(s)). \quad (\text{NL5})$$

There exists $\sigma_0 > 1$ with $H_t(\sigma_0) < 1$. To prove this, fix any $\sigma_1 > 1$. The terms $r_n^{-\sigma}$ tend to zero as $\sigma \rightarrow \infty$, and for $\sigma \geq \sigma_1$ they are dominated by the summable sequence $r_n^{-\sigma_1}$. Dominated convergence gives $H_t(\sigma) \rightarrow 0$.

On every closed half-plane $\Re s \geq \sigma_0$ with $H_t(\sigma_0) < 1$, the series

$$\log F_t(s) = -s \log b + \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} \sum_{n_1, \dots, n_k \geq 2} (r_{n_1} \cdots r_{n_k})^{-s} \quad (\text{NL6})$$

uses the holomorphic logarithm of $1 + H_t$ defined by its power series. Its absolute sum is at most $\sum_{k \geq 1} H_t(\sigma_0)^k/k < \infty$. Differentiation is locally uniformly justified: for a slightly smaller abscissa still exceeding 1 and with $H_t < 1$,

$$J_t(\sigma) = \sum_{n \geq 2} (\log r_n) r_n^{-\sigma} < \infty,$$

and the absolute sum of the differentiated length terms of length k , after the factor $1/k$, is $J_t(\sigma)H_t(\sigma)^{k-1}$. Summing this geometric bound proves

$$-\frac{F'_t(s)}{F_t(s)} = \log b + \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} \sum_{n_1, \dots, n_k \geq 2} \log(r_{n_1} \cdots r_{n_k}) (r_{n_1} \cdots r_{n_k})^{-s}. \quad (\text{NL7})$$

No factor indexed by an ordinary prime has been assumed.

Define the signed measure on $[0, \infty)$

$$\nu_t = (\log b)\delta_0 + \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} \sum_{n_1, \dots, n_k \geq 2} \ell(n_1, \dots, n_k) \delta_{\ell(n_1, \dots, n_k)}, \quad \ell(n_1, \dots, n_k) = \sum_i \log r_{n_i}. \quad (\text{NL8})$$

It is locally finite: if $\ell \leq M$, then $k \leq M/\log r_2$, and every $r_{n_i} \leq e^M$, leaving finitely many tuples. Equal lengths are added with their exact signed multiplicities. The same bound proving (NL7) gives $\int e^{-\sigma\ell} |d\nu_t| < \infty$ in the stated right half-plane. Its Laplace transform is exactly $-F'_t/F_t$.

NL3. An explicit negative arithmetic atom at the separated endpoint

At $t = 1$, $b = 2$, $r_n = (n+1)/2$, and

$$H_1(4) = \sum_{m \geq 3} (2/m)^4 \leq \frac{16}{81} + \int_3^\infty 16x^{-4} dx = \frac{32}{81} < 1. \quad (\text{NL9})$$

Thus (NL7) holds, in particular, for $\Re s \geq 4$, with the convergence estimate just proved. There is a positive atom

$$\nu_1(\{\log(3/2)\}) = \log(3/2). \quad (\text{NL10})$$

Only the one-term tuple $n_1 = 2$ has that length: a tuple of at least two terms has product at least $(3/2)^2$.

There is also the exact negative atom

$$\boxed{\nu_1(\{\log(9/4)\}) = -\log(3/2)}. \quad (\text{NL11})$$

For a one-term tuple, $(n+1)/2 = 9/4$ would give $n = 7/2$, impossible. A tuple of at least three terms has product at least $27/8 > 9/4$. For two terms both factors are at least $3/2$, so equality forces both to be $3/2$, that is, the single ordered tuple $(2, 2)$. Its coefficient in (NL8) is $-\frac{1}{2} \log(9/4) = -\log(3/2)$. This proves (NL11) with no truncation.

Consequently the logarithmic arithmetic distribution for the actual separated sheet is signed. The original von Mangoldt distribution, $\sum_{n \geq 2} \Lambda(n) \delta_{\log n}$, cannot be reused for this sheet. The negative atom alone is not a negative value of the full Weil form: the gamma, divisor, reflection-defect, and boundary terms of a complete contour identity must also be retained. It is an exact arithmetic coefficient that such an identity must contain.

NL4. The first infinitesimal is nonzero and contains mixed prime data

For $|t| < 1$, the binomial series gives, locally uniformly for $\Re s > 1$,

$$F_t(s) = \zeta(s) - t s \zeta(s+1) + O(t^2). \quad (\text{NL12})$$

For completeness, fix a compact set of s and a radius $r < 1$. The Taylor remainder in $(1+t/n)^{-s}$ is bounded by a constant times $|t|^2/n^2$, uniformly on that compact set; after multiplication by n^{-s} it is summable. This proves the asserted termwise expansion and its holomorphic derivatives.

The Euler product of ζ , used only at $t = 0$, gives

$$Q(s) = \frac{\zeta(s+1)}{\zeta(s)} = \sum_{n \geq 1} \alpha(n) n^{-s}, \quad \alpha(1) = 1, \quad \alpha(n) = (-1)^{\omega(n)} \frac{\prod_{p|n} (p-1)}{n}. \quad (\text{NL13})$$

Indeed the local series is

$$\frac{1-p^{-s}}{1-p^{-s-1}} = 1 - \sum_{j \geq 1} \frac{p-1}{p^j} p^{-js}.$$

Its absolute product converges for $\Re s > 1$; multiplying the series proves (NL13). Thus

$$\partial_t \log F_t(s)|_{t=0} = -sQ(s), \quad \partial_t \left(-\frac{F'_t}{F_t} \right) \Big|_{t=0} = Q(s) + sQ'(s) = \sum_{n \geq 1} \alpha(n)(1 - s \log n)n^{-s}. \quad (\text{NL14})$$

The derivative series is absolutely convergent on compact subsets of $\Re s > 1$, since $|\alpha(n)| \leq 1$ and $\sum (1 + \log n)n^{-\sigma} < \infty$. For real $s > 1$, $sQ(s) > 0$, so the first deformation of $\log F_t$ does not vanish. This infinitesimal is not the first-order-trivial node deformation from the observed collision.

Equation (NL14) is equivalently the Laplace transform of the locally finite distribution

$$\nu_0 = \sum_{n \geq 1} \alpha(n) (\delta_{\log n} - (\log n) \delta'_{\log n}). \quad (\text{NL15})$$

Here $\langle \delta'_\ell, f \rangle = -f'(\ell)$, so its Laplace transform is $se^{-s\ell}$. This fixes the derivative sign in (NL15).

The dot also denotes the actual distributional derivative of (NL8), not only a matching transform. On a fixed compact length interval and for $0 \leq t \leq 1$, the inequalities $r_2(t) \geq 3/2$ and $r_n(t) \geq (n+1)/2$ give a uniform finite bound on the tuples which can enter. Differentiating their smooth moving atoms therefore defines the derivative on every compactly supported smooth test. For weighted Laplace tests, $|\partial_t \log r_n| \leq 1$. A tuple of length k has length derivative bounded by k ; differentiating its coefficient and its exponential introduces at most one further factor k , together with its original length. On a sufficiently far right half-plane, uniformly in a neighborhood of zero, the sums are bounded by geometric series of the forms $\sum k B^k$ and $\sum k D B^{k-1}$, with $B < 1$ and $D < \infty$ as above. Thus Laplace transformation commutes with differentiation. At zero all tuple lengths are logs of positive integers. The resulting distribution has the form $\sum_n (a_n \delta_{\log n} + b_n \delta'_{\log n})$. These coefficients are uniquely determined by its Laplace transform: multiply the difference of two such transforms by the exponential of its least remaining length and let the real argument tend to infinity; the leading polynomial of degree at most one must be zero, and induction removes every length. The exponentially weighted convergence just proved bounds the remaining tail in each step. Comparing with (NL14) proves exactly $a_n = \alpha(n)$, $b_n = -(\log n)\alpha(n)$, as in (NL15).

For distinct primes p, q ,

$$\alpha(pq) = \frac{(p-1)(q-1)}{pq} \neq 0. \quad (\text{NL16})$$

These are supported at products of distinct primes, whereas the original von Mangoldt coefficients vanish there. The deformation therefore has explicitly nonzero mixed-prime terms already at first order; they cannot be removed by keeping only a modified weight at each prime power.

NL5. Full finite support and exact specialization

For a finite bounded distributive support lattice L , the actual set is

$$G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\}. \quad (\text{NL17})$$

Let $c = |L| - 1$. Its separated residue over $\mathbb{Z}/n\mathbb{Z}$ has $n + c$ elements: n supported top elements and $|L| - 1$ lower support elements. Hence its scalar cardinality receiver is

$$F_c(s) = \sum_{n \geq 1} (n + c)^{-s} = \zeta(s) - \sum_{n=1}^c n^{-s}. \quad (\text{NL18})$$

This proves exactly how (NL3) receives the full support count. It does not identify different support labels with one another in G_L . The cardinality map forgets that distinction; the full meet-algebra coefficients and their primitive idempotents remain separate in the programme's full-support reconstruction. Formula (NL8) applies with $t = c$ to the count receiver. All logarithmic product tuples, including their collisions in the length variable, are retained.

NL6. Consequence for the positivity calculation

There are three exact facts to carry forward. First, the supported-zero prime remains a nontrivial prime of $G(\mathbb{Z})$; no positive finite multiplicative ideal norm extending the old norm can give it an ordinary new Euler factor. Second, the programme already supplies a different analytic receiver, the separated quotient-size interpolation (NL3), whose logarithmic derivative is (NL7), with the explicit signed atom (NL11). Third,

its infinitesimal variation is (NL14), not zero. A corrected explicit formula for this receiver must use these arithmetic terms and its own completed divisor. The exact escaping-fibre trace and observed-operator current are calculated in the accompanying derivations; equality with a Weil form requires the actual map, not an identification of their names.

The two primitive-defect source files in folders 3 and 4 are byte-identical (SHA256 2f3289f6a04105d3f15fe5f7c36dde286515e61f465caf1582f9808e19f3d2f7). Their negative augmentation retains the kernel of a scalar cancellation. That same instruction is followed here: the full signed length distribution (NL8) and mixed-prime distribution (NL15) are retained rather than discarded when an ordinary Euler presentation fails.

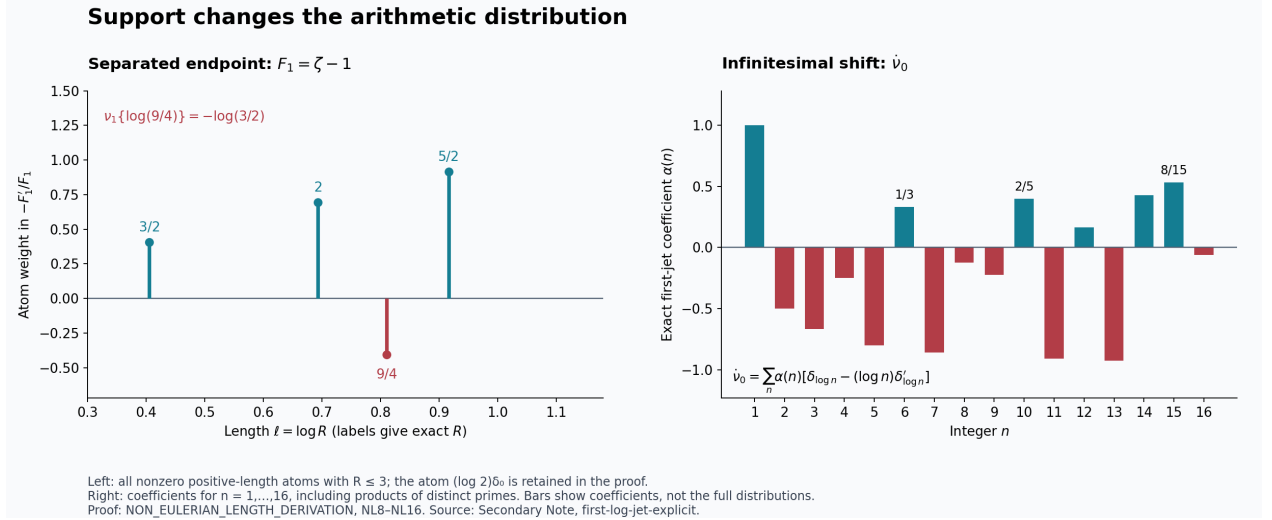


Figure 3: All nonzero positive-length atoms through ratio 3, and the exact first arithmetic coefficients; the zero-length mass is stated separately. Proof: NL7–NL16.

The actual separated zeta sheet, its explicit formula, and its positivity defect

This calculation uses the quotient-size interpolation in the supplied original LaTeX. It retains the supported arithmetic zero e , the unsupported element τ , the distinct residue quotients, the completed-function endpoint residues, every gamma pole crossed by the contour, and the functional-equation defect. The final counterexample concerns this particular shifted zeta sheet. It is not a counterexample to the classical Riemann hypothesis.

Source coverage for this derivation: `globalization nte/2/split_zero_secondary_note.tex`, lines 1111–1555, read in full; `globalization nte/4/split_zero_defect_preprint (1).tex`, lines 500–854, read in full; the standing definitions, quotient sheets, and shifted-deformation sections of `globalization nte/5/some_considerations.tex` were read in the returned source text. The whole-file read of the latter was truncated in an unrelated middle section and is not recorded as complete. The relevant named results are `thm:universal-taylor-deformation`, `cor:jets`, `thm:arithmetic-kernels-riemann`, `cor:first-log-jet-explicit`, `prop:hurwitz-standard-package`, and the defect preprint’s “Endpoint residues” and “Invariant/anti-invariant decomposition.” The secondary note lists The Clankers; `some_considerations.tex` lists “The Clankers / prepared in continuation”; the defect preprint has a blank author field. The source-reading ledger maintained by the root task records canonical source IDs. This file supplies the additional proofs below; it does not claim those older formulas as new.

SW1. The typed quotient maps and the interpolation they actually give

Let $S = G(\mathbb{Z}) = \mathbb{Z} \sqcup \{\tau\}$, with $e = 0_{\mathbb{Z}} \neq \tau$. For $n \geq 1$, the two successive surjections are

$$S \xrightarrow{\pi_n} G(\mathbb{Z}/n\mathbb{Z}) \xrightarrow{p_n} \mathbb{Z}/n\mathbb{Z}, \quad \pi_n(\tau) = \tau_n, \quad \pi_n(a) = a \bmod n, \quad p_n(\tau_n) = 0. \quad (\text{SW1})$$

The first target has $n + 1$ elements, and the second has n . Their zero fibres in the source are respectively $\{\tau\}$ and $\{\tau\} \cup n\mathbb{Z}$. These statements follow directly from the definitions, including at $n = 1$, when the first quotient has two elements.

The two Dirichlet sums, and the supplied interpolation between their numerical weights, are

$$F_t(s) = \sum_{n \geq 1} (n + t)^{-s} = \zeta(s, 1 + t), \quad F_0 = \zeta, \quad F_1 = \zeta - 1, \quad 0 \leq t \leq 1. \quad (\text{SW2})$$

Initially these equalities hold for $\Re s > 1$. The identity at $t = 1$ is a change of index in an absolutely convergent series. The intermediate values are numerical interpolation of the cardinality weights; they are not asserted to count a nonintegral number of elements in a quotient.

In particular, SW2 is not the separate prime-weight Euler product $\prod_p (1 - (p + 1)^{-s})^{-1}$. Its distinction from that product is proved with the actual quotient maps in TN1–TN9 of prime-norm comparison proof. For example the weights satisfy $(m + 1)(n + 1) - (mn + 1) = m + n$; retaining the labels gives the multiplicative vector $(1, n)$, while summing its entries gives the nonmultiplicative number $n + 1$.

SW2. Analytic continuation and estimates used below

Write $a = 1 + t \in [1, 2]$. For any integer $M \geq 1$, Euler summation with $2M$ derivatives gives

$$F_t(s) = \frac{a^{1-s}}{s-1} + \frac{a^{-s}}{2} + \sum_{j=1}^M \frac{B_{2j}}{(2j)!} (s)_{2j-1} a^{1-s-2j} - \frac{(s)_{2M}}{(2M)!} \int_0^\infty \tilde{B}_{2M}(x) (x+a)^{-s-2M} dx, \quad \Re s > 1 - 2M. \quad (\text{SW3})$$

Here $\tilde{B}_k(x) = B_k(\{x\})$, and $(s)_j = s(s+1) \cdots (s+j-1)$. To check the formula, apply integration by parts on every interval $[n, n+1]$, using $B'_k = kB_{k-1}$, $B_k(1) = B_k(0)$ for $k \neq 1$, and $B_1(1) - B_1(0) = 1$. Summing the first boundary jumps gives the values $(n+a)^{-s}$. The remaining boundaries at the lower end are the displayed half-value and even Bernoulli terms; the upper boundaries tend to zero for $\Re s > 1$. The derivative of order j of $(x+a)^{-s}$ is $(-1)^j (s)_j (x+a)^{-s-j}$, giving the sign of the remainder. This proves SW3 first for $\Re s > 1$. Since $\tilde{B}_{2M}$ is bounded, its integral is absolutely and locally uniformly convergent for $\Re s > 1 - 2M$, so it proves the asserted continuation there.

Letting M increase proves that F_t is meromorphic on $\mathbb{C}$ with its sole pole at 1, simple of residue 1. On each fixed vertical strip, away from that pole, SW3 bounds $F_t(s)$ by a fixed polynomial in $1 + |\Im s|$, uniformly for $t \in [0, 1]$. Indeed choose $2M$ larger than one minus the left edge; the integral of $(x + a)^{-\Re s - 2M}$ is bounded uniformly on that strip, and $(s)_{2M}$ is a polynomial. Derivatives in s satisfy the same kind of bound by Cauchy's formula in a slightly larger strip. These estimates, rather than a numerical approximation to a zero sum, justify every infinite vertical integral below.

SW3. The first infinitesimal is explicit and does not vanish

Termwise differentiation, dominated uniformly on compact subsets of $\Re s > 1$, gives

$$\partial_t F_t(s) = -s F_t(s+1), \quad \partial_t^j F_t(s) = (-1)^j (s)_j F_t(s+j). \quad (\text{SW4})$$

Analytic continuation gives these meromorphic identities everywhere. In the dual-number ring $\mathbb{C}[\epsilon]/(\epsilon^2)$, the precise specialization is

$$F_\epsilon(s) = \zeta(s) - \epsilon s \zeta(s+1), \quad \log F_\epsilon = \log \zeta - \epsilon Q, \quad Q(s) = s \frac{\zeta(s+1)}{\zeta(s)}. \quad (\text{SW5})$$

The logarithm equality is local on any open set where $\zeta \neq 0, \infty$; its differentiated form is a global meromorphic identity. In particular

$$\partial_t \left. \frac{F'_t}{F_t} \right|_{t=0} = -Q'(s), \quad \partial_t \left(-\frac{F'_t}{F_t} \right) \Big|_{t=0} = Q'(s). \quad (\text{SW6})$$

The nilpotence $\epsilon^2 = 0$ does not make the coefficient of ϵ zero: at $s = 2$, that coefficient in F_ϵ is $-2\zeta(3) \neq 0$.

For $\Re s > 1$, multiplying the absolutely convergent series of $1/\zeta(s)$ and $\zeta(s+1)$ yields

$$Q(s) = s \sum_{n \geq 1} \alpha(n) n^{-s}, \quad \alpha(1) = 1, \quad \alpha(n) = (-1)^{\omega(n)} \frac{\prod_{p|n} (p-1)}{n}. \quad (\text{SW7})$$

The local factor is $(1 - p^{-s})/(1 - p^{-s-1}) = 1 - \sum_{k \geq 1} (p-1)p^{-k} p^{-ks}$, which proves the displayed coefficients. Their series is absolutely convergent on $\Re s > 1$, also after one logarithmic differentiation. Thus

$$Q'(s) = \sum_{n \geq 1} \alpha(n) (1 - s \log n) n^{-s}. \quad (\text{SW8})$$

This first jet includes all squarefree-prime support combinations in $\alpha(n)$; it is not a single new prime-power term.

SW4. Endpoint residues, logarithmic orders, and the full reflection defect

Use exactly the supplied completion

$$\Lambda_t(s) = \pi^{-s/2} \Gamma(s/2) F_t(s), \quad \ell_t(s) = \Lambda'_t(s) / \Lambda_t(s). \quad (\text{SW9})$$

SW3 at zero, or the first Bernoulli polynomial, gives $F_t(0) = -1/2 - t$. Hence

$$\text{Res}_0 \Lambda_t = -1 - 2t, \quad \text{Res}_1 \Lambda_t = 1, \quad \text{PP}_{0,1} \Lambda_t = \frac{1+t}{s(s-1)} - t \frac{2s-1}{s(s-1)}. \quad (\text{SW10})$$

Both poles are simple for $0 \leq t \leq 1$. Consequently

$$\text{Res}_0 \ell_t = \text{Res}_1 \ell_t = -1. \quad (\text{SW11})$$

For proof, a nonzero principal coefficient gives $\Lambda_t = (s-b)^{-1} u_t(s)$, with $u_t(b) \neq 0$; its logarithmic derivative is $-1/(s-b) + u'_t/u_t$. Thus the weight of the endpoint in a logarithmic explicit formula is its order, not the coefficient $-1 - 2t$ of the original completed function.

At every negative even integer $-2j$, $j \geq 1$, the precise order is

$$\text{ord}_{-2j} \Lambda_t = \text{ord}_{-2j} F_t - 1. \quad (\text{SW12})$$

This formula also retains exceptional parameter values at which a Hurwitz zero cancels a gamma pole. At $t = 1$, $F_1(-2j) = -1$, so all these completed poles are present and simple. In particular multiplying only by $s(s-1)$ does not make Λ_1 entire.

Define the additive and logarithmic reflection defects by

$$\Delta_t(s) = \Lambda_t(s) - \Lambda_t(1-s), \quad d_t(s) = \ell_t(s) + \ell_t(1-s) = \frac{d}{ds} \log \frac{\Lambda_t(s)}{\Lambda_t(1-s)}. \quad (\text{SW13})$$

These are explicit meromorphic objects. Their signs follow by the chain rule. At $t = 0$, the classical functional equation gives both defects zero. SW5–SW6 give the complete infinitesimal logarithmic defect

$$\dot{d}_0(s) = -Q'(s) - Q'(1-s). \quad (\text{SW14})$$

Its endpoint residues vanish because Q is regular at both endpoints: $Q(0) = -2$, from $s\zeta(s+1) \rightarrow 1$, and $Q(1) = 0$. This vanishing is only a statement about these two residues. It does not set SW14 or the moving zero divisor to zero.

SW5. Exact contour identity before invoking any positivity

For entire functions f, g , set

$$f^\#(s) = \overline{f(1-\bar{s})}, \quad A(s) = f^\#(s)g(s), \quad K(u) = A(1/2 + iu), \quad k(v) = \frac{1}{2\pi} \int_{\mathbb{R}} K(u) e^{iuv} du. \quad (\text{SW15})$$

In the formulas below the entire tests have Gaussian decay, up to a polynomial factor, in each fixed vertical strip. We construct such tests explicitly in SW9. The more general finite-rectangle residue identity requires no decay hypothesis.

Fix $c > 1$ and let $a_c = 1 - c$. First take a finite rectangle with these two real edges and imaginary edges $\pm T$, avoiding nonremovable poles on its boundary. The residue theorem gives

$$\frac{1}{2\pi i} \oint \ell_t(s) A(s) ds = \sum_{F_t(\rho)=0 \text{ inside}} m_\rho A(\rho) - A(1) - \sum_{j \geq 0: -2j \text{ inside}} A(-2j). \quad (\text{SW16})$$

Zeros and gamma poles are written separately, so exact cancellations in SW12 occur by addition of their integer orders. All quantities in SW16 are finite.

If $\ell_t A$, $\ell_t(s)A(1-s)$, and $\ell_t(1-s)A(1-s)$ have only finitely many nonremovable poles in the strip and their horizontal integrals tend to zero, then taking $T \rightarrow \infty$ gives

$$\begin{aligned} Z_{t,c}(A) - A(1) - \sum_{j \geq 0: -2j > 1-c} A(-2j) &= \frac{1}{2\pi i} \int_{\Re s=c} \ell_t(s)(A(s) + A(1-s)) ds \\ &\quad - \frac{1}{2\pi i} \int_{\Re s=c} d_t(s)A(1-s) ds. \end{aligned} \quad (\text{SW17})$$

Here $Z_{t,c}(A)$ is the finite sum of the zero terms not annihilated by A , counted with multiplicity. The actual tests in SW9 satisfy every property in this sentence by a displayed cancellation identity, so SW17 is used below without a missing analytic assumption. For the sign, the left upward integral becomes the right upward integral of $\ell_t(1-s)A(1-s)$; the positive contour subtracts it. Substituting $\ell_t(1-s) = d_t(s) - \ell_t(s)$ proves SW17.

SW6. The non-Eulerian arithmetic term and the completed formula

Put $a = 1 + t$, $r_n = (n+t)/a$ for $n \geq 2$, and choose c so large that $\sum_{n \geq 2} r_n^{-c} < 1$. Write $r_{\mathbf{n}} = \prod_{i=1}^k r_{n_i}$. Absolute convergence of the logarithm series gives

$$-\frac{F'_t}{F_t}(s) = \log a + \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} \sum_{n_1, \dots, n_k \geq 2} \log r_{\mathbf{n}} r_{\mathbf{n}}^{-s}. \quad (\text{SW18})$$

To verify differentiation and absolute convergence, take a slightly smaller real part on which the sum is still less than 1. Its positive margin bounds the differentiated geometric series. Explicitly, if $B(c) = \sum r_n^{-c} < 1$ and $D(c) = \sum (\log r_n) r_n^{-c} < \infty$, the sum of the absolute differentiated terms is $\sum_{k \geq 1} D(c) B(c)^{k-1} = D(c)/(1-B(c))$. The general-length derivation and its signed atoms are developed independently in non-Eulerian length proof.

Define the absolutely convergent functional

$$\begin{aligned} \mathcal{L}_t(k) &= 2 \log(a) k(0) \\ &+ \sum_{k_0 \geq 1} \frac{(-1)^{k_0+1}}{k_0} \sum_{n_1, \dots, n_{k_0} \geq 2} \frac{\log r_{\mathbf{n}}}{\sqrt{r_{\mathbf{n}}}} [k(\log r_{\mathbf{n}}) + k(-\log r_{\mathbf{n}})]. \end{aligned} \quad (\text{SW19})$$

Gaussian strip decay implies $|k(v)| \leq C_R e^{-R|v|}$ for every fixed $R > 0$: shift the vertical contour defining k by R in the appropriate direction. Choose $R = c - 1/2$, or a slightly larger value, and the absolute majorant just proved for SW18 proves convergence of SW19. All exchanges of its sum and integral are therefore justified.

Let

$$w_\infty(u) = \Re \psi(1/4 + iu/2) - \log \pi, \quad \mathcal{D}_{t,c}(A) = \frac{1}{2\pi i} \int_{\Re s=c} d_t(s) A(1-s) ds. \quad (\text{SW20})$$

Then SW17 is exactly

$$\begin{aligned} Z_{t,c}(A) &= A(1) + \sum_{j \geq 0: -2j > 1-c} A(-2j) \\ &+ \frac{1}{2\pi} \int_{\mathbb{R}} K(u) w_\infty(u) du - \mathcal{L}_t(k) - \mathcal{D}_{t,c}(A). \end{aligned} \quad (\text{SW21})$$

For completeness, $\ell_t = -\frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2) + F'_t/F_t$. The gamma part can be moved from c to $1/2$ without crossing a gamma pole, and reflection of the integration variable gives w_∞ . The digamma series on $\Re(s/2) > 0$ bounds it polynomially, so Gaussian decay kills the horizontal edges. A term r^{-s} in SW18 integrates to $r^{-1/2} [k(-\log r) + k(\log r)]$, by shifting its entire integrand to the critical line. The constant gives $2 \log(a) k(0)$. Finally the minus sign of $\mathcal{D}$ is the sign derived in SW17. This proves every sign and endpoint contribution in SW21.

At $t = 0$, $d_0 = 0$, and SW18 reduces to the usual logarithmic prime-power series. For tests annihilating the classical trivial zeros, $A(-2j) = 0$ for $j \geq 1$. SW21 then has exactly the two endpoints, archimedean term, and prime term in WA24. At $t = 1$, neither the negative-even terms nor $\mathcal{D}_{1,c}$ can be dropped merely because the original endpoint principal part has been computed.

SW7. Local variation of the original zero packet

Let ρ be an actual zero of ζ , of multiplicity m , and take a small circle containing no other zero and neither endpoint. Its continuation as a zero cluster of F_t is counted by

$$T_t(A) = \frac{1}{2\pi i} \oint \frac{F'_t(s)}{F_t(s)} A(s) ds. \quad (\text{SW22})$$

The compact boundary is zero-free at $t = 0$, so uniform continuity makes it zero-free for all sufficiently small t . The argument principle proves that this is the sum of A over the moving cluster, counted with multiplicities. Differentiating on that compact contour and integrating an exact derivative gives

$$\dot{T}_0(A) = -\frac{1}{2\pi i} \oint Q'(s) A(s) ds = \text{Res}_{s=\rho} (Q(s) A'(s)). \quad (\text{SW23})$$

The integration-by-parts equality follows from the zero contour integral of $(QA)'$. It does not require the zero to be simple. If it is simple, the result is

$$\dot{\rho}(0) = \frac{\rho \zeta(\rho+1)}{\zeta'(\rho)}, \quad \dot{T}_0(A) = \dot{\rho}(0) A'(\rho). \quad (\text{SW24})$$

For a nontrivial zero, the numerator is nonzero because $\rho \neq 0$ and $\Re(\rho+1) > 1$. Thus a simple nontrivial zero has a nonzero first displacement in this interpolation. For a multiple zero, SW23 retains the full residue of order up to m , rather than assuming a differentiable labeling of its individual branches.

The arithmetic first variation on the right-hand side is explicit as well. With the conventions of SW15, the integral of $-Q'(s)(A(s) + A(1-s))$ on a right line is

$$\sum_{n \geq 1} \frac{\alpha(n)}{\sqrt{n}} \left[\left(\frac{\log n}{2} - 1 \right) (k(-\log n) + k(\log n)) + \log n (k'(-\log n) - k'(\log n)) \right]. \quad (\text{SW25})$$

To verify it, differentiation of the Fourier integral gives $\frac{1}{2\pi} \int (1/2 + iu) K(u) e^{iuv} du = \frac{1}{2} k(v) + k'(v)$. Reflection replaces the second derivative term by its negative. Insert SW7 and differentiate termwise. Absolute convergence follows from the same exponential contour bounds for k, k' . The reflection correction SW14 remains alongside SW25; omitting it is not a deformation of the same contour identity.

SW8. The exact symmetric lift and the missing Hermitian property

Let

$$G_t(s) = (s-1)F_t(s), \quad H_t(s) = G_t(s)G_t(1-s) = -s(s-1)F_t(s)F_t(1-s). \quad (\text{SW26})$$

Both G_t and H_t are entire by SW3. They have real conjugation symmetry, and $H_t(1-s) = H_t(s)$. The actual map is the multiplicative norm for the involution on entire functions,

$$N : \mathcal{O}(\mathbb{C}) \longrightarrow \mathcal{O}(\mathbb{C})^{s \mapsto 1-s}, \quad N(G)(s) = G(s)G(1-s). \quad (\text{SW27})$$

It satisfies $N(G_1G_2) = N(G_1)N(G_2)$, and its divisor is the original divisor plus its reflected divisor, with multiplicities. It is not asserted to be additive or to preserve the zero count without doubling. On the critical line $H_t(s) = |G_t(s)|^2 \geq 0$.

The zero pairing of H_t , on finite packets closed under $\rho \mapsto 1-\bar{\rho}$, is Hermitian. Proof: applying complex conjugation and this permutation to $\sum m_\rho f(1-\bar{\rho})g(\rho)$ exchanges f and g . By contrast, a divisor lacking that permutation symmetry need not define a Hermitian form. The next calculation proves the failure, and a negative direction in the symmetric lift, for the specific endpoint $t=1$.

SW9. A fully explicit negative test for the separated endpoint

The exact special values are

$$F_1(-20) = -1, \quad F_1(-19) = \frac{174611}{6600} - 1 = \frac{168011}{6600} > 0. \quad (\text{SW28})$$

They follow from $\zeta(-n) = -B_{n+1}/(n+1)$, $B_{21} = 0$, and $B_{20} = -174611/330$, also obtainable from SW3. Continuity therefore gives at least one real zero in $(-20, -19)$. Define α to be the least such real zero. This is well-defined: the endpoint values exclude neighborhoods of the two endpoints, and a nonzero holomorphic function has only finitely many zeros in the remaining compact interval. Write $m \geq 1$ for its exact multiplicity, and put $\beta = 1 - \alpha \in (20, 21)$.

Since $F_1(\sigma) = \sum_{n \geq 2} n^{-\sigma} > 0$ for real $\sigma > 1$, β is not a zero of F_1 . The divisor of H_1 , however, has both α and β , each with multiplicity m . Define

$$h(s) = (s-\alpha)^m(s-\beta)^m, \quad W(s) = e^{(s-1/2)^2} \frac{H_1(s)}{h(s)}. \quad (\text{SW29})$$

The quotient is entire, $W(\alpha), W(\beta) \neq 0$, and $W^\# = W$. Indeed reflection exchanges the two factors of h with total sign $(-1)^{2m} = 1$, and all the coefficients are real. SW3 proves that W and every fixed derivative are bounded by a polynomial times $e^{-(\Im s)^2}$ on each fixed vertical strip; away from a bounded region the denominator is a polynomial bounded below by a positive constant times $|\Im s|^{2m}$, and inside that region removability gives boundedness.

For arbitrary prescribed $z_\alpha, z_\beta \in \mathbb{C}$, take the degree-at-most-one polynomial

$$q(s) = \frac{z_\alpha}{W(\alpha)} \frac{s-\beta}{\alpha-\beta} + \frac{z_\beta}{W(\beta)} \frac{s-\alpha}{\beta-\alpha}, \quad f = qW, \quad A = f^\# f. \quad (\text{SW30})$$

Then $f(\alpha) = z_\alpha$ and $f(\beta) = z_\beta$. At every other zero of H_1 , W vanishes at least to the full multiplicity. Therefore A annihilates all of those zero terms. Take the strip $-21 < \Re s < 22$. Its native and symmetric zero functionals are exactly

$$Z_{1,22}(A) = m\bar{z}_\beta z_\alpha, \quad Z_{H_1}(A) = m(\bar{z}_\beta z_\alpha + \bar{z}_\alpha z_\beta). \quad (\text{SW31})$$

No other zero contributes, including a multiple zero or a zero with large imaginary part, because of the preceding vanishing orders.

For $(z_\alpha, z_\beta) = (1, -1)$, this gives

$$\boxed{Z_{1,22}(f^\# f) = -m < 0, \quad Z_{H_1}(f^\# f) = -2m < 0.} \quad (\text{SW32})$$

For $(z_\alpha, z_\beta) = (1, i)$, it gives

$$\boxed{Z_{1,22}(f^\# f) = -im,} \quad (\text{SW33})$$

which is not real. Thus the original shifted divisor pairing is not Hermitian, and the exact symmetric lift has a genuine negative direction. Nonnegativity of H_1 as a function on the critical line, proved in SW8, does not imply nonnegativity of its zero pairing.

Here are the promised contour checks for the actual tests. Since

$$\frac{H'_1}{H_1}A = q^\# q e^{2(s-1/2)^2} \frac{H'_1 H_1}{h^2}, \quad (\text{SW34})$$

the only possible nonremovable poles are at α, β . Every other zero is canceled as an identity of meromorphic functions. SW3 and its differentiated estimate bound the numerator polynomially on each strip, so the horizontal edges tend to zero with Gaussian speed. Likewise, using $H_1 = G_1 G_1(1-s)$, each occurrence of F'_1/F_1 or its reflected counterpart multiplied by A cancels a factor of F_1 in H_1^2 . The same assertion holds for $A(1-s)$, since its denominator is the same reflection-invariant h^2 . The remaining digamma factors have finitely many poles in this strip and at most polynomial growth on high horizontal edges, by the digamma recurrence and its convergent right-half-plane series. Consequently SW17–SW21 apply with no unproved horizontal zero-avoidance estimate.

At $t = 1, c = 22$, the absolute logarithm expansion required in SW18 holds. For instance

$$\sum_{n \geq 2} \left(\frac{2}{n+1} \right)^{22} \leq \left(\frac{2}{3} \right)^{22} + \int_3^\infty \left(\frac{2}{x} \right)^{22} dx = \left(\frac{2}{3} \right)^{22} \left(1 + \frac{3}{21} \right) < 1. \quad (\text{SW35})$$

Thus the corrected explicit formula evaluated on the negative test is the exact equality

$$-m = A(1) + \sum_{j=0}^{10} A(-2j) + \frac{1}{2\pi} \int_{\mathbb{R}} |f(1/2 + iu)|^2 w_\infty(u) du - \mathcal{L}_1(k) - \mathcal{D}_{1,22}(A). \quad (\text{SW36})$$

Every term is defined in SW15 and SW19–SW20, and every series and integral converges by the estimates above. This is a concrete positivity analysis of the actual non-Eulerian separated sheet, including its retained completion and reflection terms. It does not insert an arbitrary Euler factor for the supported-zero prime.

The tested divisor is the full shifted divisor in the strip $-21 < \Re s < 22$, and its selected real zero lies outside the classical critical strip. Thus SW32 does not establish nonpositivity for a separately defined pairing restricted to $0 < \Re s < 1$. Nor does it rule out a different boundary subtraction or a selected positivity cone. Such an operation must specify its map and retain the component it removes. The exact maps for this detected component are given next.

SW10. The retained two-point object and its positive and negative projections

Keep the actual function W of SW29 and define the vector space

$$V = W\mathbb{C}[s], \quad E : V \longrightarrow \mathbb{C}^2, \quad E(f) = (f(\alpha), f(\beta)). \quad (\text{SW37})$$

Let $j : \mathbb{C}^2 \rightarrow V$ be the interpolation formula SW30. Direct evaluation proves $Ej = \text{id}$. Since $W(\alpha), W(\beta) \neq 0$, a polynomial multiple qW lies in $\ker E$ precisely when q is divisible by both distinct linear factors. Hence

$$K := \ker E = W(s - \alpha)(s - \beta)\mathbb{C}[s], \quad V = j(\mathbb{C}^2) \oplus K. \quad (\text{SW38})$$

The decomposition is explicit: $f = jE(f) + (f - jE(f))$, and its second term lies in K . An element of the intersection has evaluation both equal to its coefficient vector and zero, proving directness.

For $z = E(f)$ and $w = E(g)$, the native and reflected pairings factor through these maps as

$$B_{\text{nat}}(f, g) = m \overline{z_\beta} w_\alpha, \quad B_{\text{sym}}(f, g) = m(\overline{z_\beta} w_\alpha + \overline{z_\alpha} w_\beta). \quad (\text{SW39})$$

The same cancellation used in SW31 proves this for arbitrary $f, g \in V$. Both forms vanish when either input lies in K . The descended matrix of B_{sym} is $m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which is invertible, so its full radical is exactly K . Also $B_{\text{sym}} = B_{\text{nat}} + B_{\text{nat}}^*$, where $B^*(f, g) = \overline{B(g, f)}$; this is the exact passage to the Hermitian part, including its factor of two.

Define

$$\Pi_+ = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \Pi_- = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad P_\pm = j\Pi_\pm E, \quad P_K = 1 - jE. \quad (\text{SW40})$$

Multiplication gives $\Pi_\pm^2 = \Pi_\pm$, $\Pi_+\Pi_- = 0$, and $\Pi_+ + \Pi_- = 1$. Using $Ej = 1$ proves that P_+, P_-, P_K are mutually annihilating idempotents summing to 1_V . Their images are respectively $j\mathbb{C}(1, 1)$, $j\mathbb{C}(1, -1)$, and K . These three summands are orthogonal for B_{sym} . To verify all signs and the orthogonality, write

$$c_+(f) = \frac{f(\alpha) + f(\beta)}{2}, \quad c_-(f) = \frac{f(\alpha) - f(\beta)}{2}.$$

Substitution in SW39 gives

$$B_{\text{sym}}(f, g) = 2\overline{mc_+(f)}c_+(g) - 2\overline{mc_-(f)}c_-(g). \quad (\text{SW41})$$

Thus the nonnegative cone is exactly $\{|c_-| \leq |c_+|\}$, with the radical directions retained, and $j\mathbb{C}(1, 1) \oplus K$ is a maximal nonnegative linear subspace. Indeed any vector outside it has $c_- \neq 0$; subtracting its positive component, already in this subspace, gives a negative vector by SW41, so it cannot be adjoined while preserving nonnegativity.

A particular boundary subtraction can now be defined, rather than left implicit:

$$B_+(f, g) := B_{\text{sym}}(f, g) + 2\overline{mc_-(f)}c_-(g) = B_{\text{sym}}(P_+f, P_+g) = 2\overline{mc_+(f)}c_+(g). \quad (\text{SW42})$$

It is positive semidefinite, and its removed negative direction is exactly the recorded image of P_- . Equations SW37–SW42 identify the defect object, its quotient, section, radical, orthogonal summands, and one exact corrected form. They do not claim that this particular subtraction is the arithmetic or prismatic choice required elsewhere in the programme; they provide the complete finite object such a comparison would have to map.

SW11. Consequences for the programme’s vanishing-infinitesimal question

The exact distinctions now have proved maps. The quotient-size construction SW1 maps to the interpolation SW2; its first-order base change is SW5; its completed divisor is measured by SW16; its failure of reflection is the meromorphic function SW13; and its reflected Hermitian lift is the norm map SW27. The first-order endpoint residue variation of the logarithmic derivative is zero, while the first-order function, logarithmic derivative, and moving zero cluster are respectively SW5, SW6, and SW23 and need not vanish. SW24 proves nonvanishing of the displacement at every simple nontrivial classical zero.

The previously derived prime-weight formula TN21 concerns a different function, a holomorphic-unit multiple of classical ζ in $\Re s > 0$. Its unchanged zero divisor and corrected archimedean term cannot be transferred to $F_1 = \zeta - 1$. The actual map from this interpolation to a symmetric divisor is SW27, which adds the reflected divisor and exposes SW32. A prismatic or deformation comparison must carry these explicit objects and the reflection defect through its maps; nilpotence of a parameter or vanishing of an endpoint residue does not erase them.

Reconstruction over the full support lattice

This calculation starts from the programme's actual support-amplitude semiring. It proves the precise comparison with the earlier scalar calculations, identifies information lost by all separate Boolean branch maps after linearization, and constructs a support-valued extension of the finite Weil packet form. It does not assert that a zeta zero off the critical line exists, or that positivity for the classical Weil distribution has been proved.

The source definition and the existing spectrum theorem are in *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, and Frobenius-Perfect Quantization*, The Clankers, June 2026, Version 11. The file read for this derivation is `split_support_geometry_arithmetic_curve_v11910.tex`, SHA256 8443cc0401b18d373939d992f0a2f5384ee2fa1130f059753d5f3972eec5052d, Definition `def:lattice-split`, Theorems `thm:lattice-normal-form`, `thm:birkhoff-support-coordinates`, `thm:lattice-prime-classification`, `thm:spectral-ordinal-sum`, and `thm:support-base-change`, source lines1180–1871. Those results are prior programme mathematics. The different file `split_support_geometry_arithmetic_curve_v11.tex`, SHA256 5fa6a55aa9d1a2e57599e206087d3d2466b25dd99fa1a90d5f86367aafab1b4d, has the explicit branch-spectrum theorem at lines1925–1985. The editions are not treated as byte-identical.

1. The base and its split comparison maps

Let R be a nonzero commutative unital ring and L a nontrivial bounded distributive lattice. Write its bounds as 0_L and 1_L . Set

$$S_L = G_L(R) = \{z_\lambda = (0, \lambda) : \lambda \in L\} \cup \{\widehat{r} = (r, 1_L) : r \in R\}, \quad z_{1_L} = \widehat{0} = e_L, \quad z_{0_L} = \tau_L. \quad (\text{FSR1})$$

The overlap in this union consists of the one element e_L . The operations are

$$(r, \lambda) + (s, \mu) = (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu). \quad (\text{FSR2})$$

In particular, $\widehat{r} + z_\lambda = \widehat{r}$ and $\widehat{r}z_\lambda = z_\lambda$, including $r=0$. Closure follows because every nonzero amplitude has top support. The semiring laws follow from those of R and the distributive lattice law. Its zero is τ_L and its unit is $\widehat{1}$.

Put $S = G_L(R)$, with $B = \{0, 1\}$. The endpoint inclusion and each Boolean branch give maps

$$\iota : S \longrightarrow S_L, \quad \tau \mapsto \tau_L, \quad r \mapsto \widehat{r}, \quad \beta_\epsilon : S_L \longrightarrow S, \quad z_\lambda \mapsto z_{\epsilon(\lambda)}, \quad \widehat{r} \mapsto r, \quad \beta_\epsilon \iota = \text{id}_S. \quad (\text{FSR3})$$

Here $\epsilon : L \rightarrow B$ preserves the bounds, joins, and meets. Both maps preserve every operation in FSR2: on the zero fibre this is the lattice-map identity; on two supported amplitudes it is the identity of R ; on a mixed pair it follows from the two identities after FSR2. Thus ι is injective and every β_ϵ is surjective. The kernel congruence of β_ϵ identifies exactly the zero-fibre elements with the same ϵ value, and no two distinct nonzero amplitudes. Its inverse image of the actual scalar zero is

$$\beta_\epsilon^{-1}(\tau) = \{z_\lambda : \epsilon(\lambda) = 0\}. \quad (\text{FSR4})$$

Its inverse image of the supported scalar zero is the complementary set of zero-fibre elements. This is the exact distinction between a zero ideal and a congruence in this comparison.

The amplitude projection $p_L : S_L \rightarrow R$ has zero fibre $Z_L = \{z_\lambda\}$. Every homomorphism to a ring kills Z_L : the equality $e_L + z_\lambda = e_L$ implies this by additive cancellation in the target. The restriction to the supported copy then gives a unique ring homomorphism $R \rightarrow A$. This proves the universal property of p_L , without identifying e_L with τ_L in S_L .

For a domain R , the supported zero is a prime element in the divisibility sense:

$$e_L S_L = Z_L, \quad e_L \mid xy \iff p_L(x)p_L(y) = 0 \iff e_L \mid x \text{ or } e_L \mid y. \quad (\text{FSR5})$$

The first equality follows from $e_L z_\lambda = z_\lambda$ and $e_L \widehat{r} = e_L$. The next equivalence follows from the first equality and the multiplication law. The last uses precisely the absence of zero divisors in R . The ideal Z_L is proper because $\widehat{1}$ is outside it. The element e_L is not a unit, since every product with it has amplitude zero. It is reducible, since $e_L = e_L e_L$ with two nonunits. Its primality and its multiplicative idempotence are simultaneous facts.

For completeness the entire semiring-prime comparison can be obtained directly. If an ideal contains a supported amplitude, multiplication by e_L puts e_L in it, hence puts all of Z_L in it. Its supported

amplitudes form a ring ideal J . An ideal containing no supported amplitude is Z_a for a proper lattice ideal a : joins follow from addition and downward closure follows from $z_\mu z_\lambda = z_\mu$ when $\mu \leq \lambda$. Thus all ideals are Z_a and

$$Q_J = Z_L \cup \{\hat{r} : r \in J\}. \quad (\text{FSR6})$$

Testing FSR2 shows that Z_a is prime exactly when a is a prime lattice ideal, and Q_J is prime exactly when J is a ring prime. Every support prime is strictly below every arithmetic prime. For $R=Z$ this includes

$$Z_a \subsetneq Z_L = (e_L) = Q_{(0)} \subsetneq Q_{(p)}. \quad (\text{FSR7})$$

The basic opens are $D(z_\lambda) = \{Z_a : \lambda \notin a\}$ and $D(\widehat{r}) = \text{Spec_lat}(L) \cup D_R(r)$. These formulas prove the topological ordinal sum, including its topology. Contraction along β_ϵ sends the scalar prime $\{\tau\}$ to $Z_{(\epsilon^{-1})}$ and Q_q to Q_q . Contraction along ι sends every Z_a to $\{\tau\}$ and Q_q to Q_q . Consequently the scalar spectrum is a retract with an explicitly chosen section; it is not the entire support spectrum.

2. Joint reconstruction before linearization

Now assume L finite and let $J(L)$ be its nonzero join-irreducible elements. For $j \in J(L)$ put $\theta_j(\lambda) = 1$ if $j \leq \lambda$ and 0 otherwise. A join-irreducible is join-prime: if $j \leq \lambda \vee \mu$, distributivity gives $j = (j \wedge \lambda) \vee (j \wedge \mu)$, so one term equals j . Therefore θ_j is a bounded lattice homomorphism. Every nonzero element is the join of the join-irreducibles below it, by induction in the finite partially ordered set. Hence these characters jointly distinguish elements of L .

Conversely, a Boolean branch has a prime filter whose least element j is the meet of its elements. If $j = x \vee y$ with neither x nor y equal to j , the primality of the filter contradicts the minimality of j . Thus j is join-irreducible, and the branch is θ_j . This proves the entire branch list.

The product of all branch maps is the injective semiring map

$$B : S_L \longrightarrow \prod_{j \in J(L)} S, \quad B(x) = (\beta_j(x))_j. \quad (\text{FSR8})$$

Its image is exactly the following set. All coordinate amplitudes have a common value r . If $r \neq 0$, all coordinates are the same supported element r . If $r = 0$, each coordinate is τ or e , and the set of coordinates equal to e is an order ideal in $J(L)$. Indeed it is $\{j : j \leq \lambda\}$ for the source z_λ . Conversely, an order ideal A gives $\lambda = \text{join } A$; join-primality shows that $\{j : j \leq \lambda\} = A$. This proves both surjectivity onto the displayed image and injectivity.

For $L = B^d$ the poset $J(L)$ is an antichain of d atoms. Every subset is an order ideal, so FSR8 becomes an isomorphism

$$G_{B^d}(R) \cong \underbrace{G(R) \times_R \cdots \times_R G(R)}_{d \text{ factors}}, \quad (\text{FSR9})$$

where the maps to R are amplitude projections. This is different from the product of independent amplitude coordinates: the common-amplitude condition is part of the isomorphism, not an omitted restriction.

For a general finite distributive L , FSR8 is the subsemiring of that fibre product whose zero-amplitude masks satisfy the stated order-ideal relations. The additive and multiplicative laws remain FSR2. All earlier scalar identities pull back along each β_j . Conversely, if two S_L -valued expressions have equal images under every β_j , FSR8 proves equality in S_L . This last conclusion concerns semiring-valued expressions; it does not yet concern their free additive envelopes.

3. The exact extra kernel created by linearization

For a commutative coefficient ring k define the contracted meet algebra

$$C_L = k[(L, \wedge)] / ([0_L]). \quad (\text{FSR10})$$

It is free as a k -module on $u_\lambda = [\lambda]$ for $\lambda \neq 0_L$, with $u_0 = 0$, unit u_1 , and $u_\lambda u_\mu = u_{(\lambda \wedge \mu)}$. It retains multiplication from support and introduces its own additive linear combinations. In particular $u_{(\lambda \vee \mu)}$ is not defined to be $u_\lambda + u_\mu$.

For every nonzero $a \in L$, whether join-irreducible or not, define

$$\eta_a(u_\lambda) = \mathbf{1}_{\{a \leq \lambda\}}. \quad (\text{FSR11})$$

This is a unital k -algebra homomorphism: $a \leq \lambda \wedge \mu$ holds exactly when both inequalities hold. With any linear extension of the order, its matrix has entries $1_{a \leq \lambda}$, is triangular with diagonal 1, and is invertible over the integers. Consequently

$$\eta : C_L \xrightarrow{\sim} \prod_{a \in L \setminus \{0\}} k. \quad (\text{FSR12})$$

No division in k is used.

An explicit inverse uses the Möbius function of the finite order. Define $\mu(a, a) = 1$ and, for $a < b$, $\mu(a, b) = -\sum_{a \leq c < b} \mu(a, c)$. The upper triangular incidence matrix $Z_{ab} = 1_{a \leq b}$ has inverse with these entries: the recurrence proves $MZ = I$, and finite triangular matrices then also satisfy $ZM = I$. Define

$$E_a = \sum_{\lambda \leq a} \mu(\lambda, a) u_\lambda. \quad (\text{FSR13})$$

Then $\eta_b(E_a) = \sum_{b \leq \lambda \leq a} \mu(\lambda, a) = \delta_{ab}$. The term $\lambda = 0$ contributes zero. Thus

$$E_a E_b = \delta_{ab} E_a, \quad \sum_{a \neq 0} E_a = u_1, \quad u_\lambda = \sum_{0 < a \leq \lambda} E_a. \quad (\text{FSR14})$$

These identities follow by applying the isomorphism FSR12, so hold over every coefficient ring.

The linearized Boolean branch θ_j is precisely η_j . Write

$$K_L = \bigcap_{j \in J(L)} \ker \eta_j = \bigoplus_{a \in L \setminus (\{0\} \cup J(L))} k E_a. \quad (\text{FSR15})$$

This proves the full kernel, not only its dimension. Over a field its dimension is $|L| - 1 - |J(L)|$. It is a direct product of copies of k , with multiplicatively idempotent coordinate projectors. It is zero exactly when every nonzero lattice element is join-irreducible, equivalently when L is a chain. To verify the last equivalence, a chain has that property. If L is not a chain, choose incomparable a, b ; then $a \vee b$ is nonzero and join-reducible.

For $L = B^2 = \{0, a, b, 1\}$, with $a \wedge b = 0$ and $a \vee b = 1$,

$$E_a = u_a, \quad E_b = u_b, \quad w = E_1 = u_1 - u_a - u_b, \quad w^2 = w \neq 0, \quad \eta_a(w) = \eta_b(w) = 0. \quad (\text{FSR16})$$

Indeed the three terms are distinct basis vectors, proving nonzero, and direct expansion gives $(u_1 - u_a - u_b)^2 = u_1 + u_a + u_b - 2u_a - 2u_b = u_1 - u_a - u_b$. Thus the semiring branch maps in FSR8 distinguish every support element, but their separately linearized maps omit w . This is a proved failure of linearized reconstruction from separate branches. Its retained object is K_L with inclusion, coordinate projectors, and quotient given in FSR15.

The nonzero a that are not join-irreducible give genuine mixed observables. For a with join-irreducible decomposition $a = \text{join}_{j \leq a} j$,

$$\eta_a(u_\lambda) = \prod_{j \leq a} \theta_j(\lambda). \quad (\text{FSR17})$$

Both sides are 1 exactly when every $j \leq a$ lies below λ , equivalently when $a \leq \lambda$. Thus a joint product of branch tests restores this missing coordinate. Separate branch expectations do not determine expectations of these products. FSR12 gives the complete recovery map using all these joint tests.

4. Contracted multiplicative coefficients and the scalar retract

Let $\Gamma_L = k[(S_L, \text{multiplication})]/([\tau_L])$. Put $E = [e_L]$ and $F = 1 - E$. Multiplication by E sends $[z_\lambda]$ to itself and $[\widehat{r}]$ to $[e_L]$. Thus $E \Gamma_L$ is C_L , embedded by $u_\lambda \mapsto [z_\lambda]$. The complementary factor $F \Gamma_L$ has basis

$$v_r = [\widehat{r}] - [e_L], \quad r \in R \setminus \{0\}, \quad (\text{FSR18})$$

and $v_0 = 0$. Their multiplication is $v_r v_s = v_{rs}$, with zero when $rs = 0$. Indeed expansion gives $[\widehat{rs}] - [e_L] - [e_L] + [e_L]$. The two sets $\{[z_\lambda] : \lambda \neq 0\}$ and $\{v_r : r \neq 0\}$ form a basis: each original nonzero supported basis element is $v_r + [e_L]$, and comparison of the distinct $[\widehat{r}]$ coefficients proves linear independence. Hence

$$\Gamma_L \cong C_L \times D_R, \quad D_R = k[(R, \times)]/([0_R]). \quad (\text{FSR19})$$

The formulas prove this for rings with zero divisors as well as domains. For $R=\mathbb{Z}$ the second factor is

$$D_{\mathbb{Z}} \cong k[t, X_p : p \text{ rational prime}]/(t^2 - 1), \quad v_{-1} \mapsto t, \quad v_p \mapsto X_p. \quad (\text{FSR20})$$

Unique prime factorization of every nonzero integer proves a bijection of the displayed monomial bases and compatibility with multiplication.

For scalar support the same construction is $\text{Gamma_B}=k \times D_R$. Under FSR19 each branch and the scalar inclusion become

$$\Gamma(\beta_j) : (c, d) \mapsto (\eta_j(c), d), \quad \Gamma(\iota) : (a, d) \mapsto (a u_1, d). \quad (\text{FSR21})$$

These formulas follow on z_lambda and $\widehat{r}$, which generate the algebras. The composite is the identity. The intersection of all branch kernels is exactly $K_L \times 0$. Thus the earlier arithmetic prime-label algebra D_R is preserved as an actual direct factor, and the missing support coefficient object is explicitly identified. This is stronger than saying that earlier work might survive.

The evaluation map $\text{Gamma_L} \rightarrow R$ when $k=\mathbb{Z}$ sends $[\widehat{r}]$ to r and every $[z_lambda]$ to 0. For $R=\mathbb{Z}$, FSR19–20 identify its kernel as

$$C_L \times (t + 1, X_p - p \text{ for every prime } p) \subset C_L \times D_{\mathbb{Z}}. \quad (\text{FSR22})$$

The quotient of D_Z by the displayed ideal is $\mathbb{Z}$: every polynomial reduces to its integer evaluation, and that evaluation is the inverse of the resulting inclusion of $\mathbb{Z}$. This proves equality of the kernel ideal. In particular all of C_L , including K_L , is killed by arithmetic addition. It remains present before that quotient. The inclusion and quotient in FSR19 and FSR22 give its exact relation to the arithmetic factor.

5. Extension of complexes, operators, and all primary jets

Take $k=\mathbb{C}$ for this section. Let $(V^{\bullet}, d)$ be any specified complex of complex vector spaces, with its original differentials. Its full support extension is

$$V_L^{\bullet} = C_L \otimes_{\mathbb{C}} V^{\bullet} = \bigoplus_{a \neq 0} E_a \otimes V^{\bullet}, \quad d_L(E_a \otimes v) = E_a \otimes dv. \quad (\text{FSR23})$$

This is a defined construction, not an identification with a prismatic complex. Decomposition FSR14 proves directly

$$\ker d_L = \bigoplus_a E_a \otimes \ker d, \quad \text{im } d_L = \bigoplus_a E_a \otimes \text{im } d, \quad H^q(V_L) = \bigoplus_a E_a \otimes H^q(V). \quad (\text{FSR24})$$

The first two equalities follow by equating the coefficients of the independent E_a . The third is their quotient. Thus a nonzero cohomology class of a specified old complex remains nonzero in every individual support sector. For every linear chain map T , the map $1 \otimes T$ preserves all sectors. A homotopy $dH + Hd = T - T'$ extends to $1 \otimes H$ and satisfies the identical formula. The maps $\text{eta_j} \otimes \text{id}$ are chain maps, with joint kernel $K_L \otimes V^{\bullet}$. The scalar inclusion $v \mapsto u_1 \otimes v$ is split by every $\text{eta_j} \otimes \text{id}$. These formulas specify exactly which part of this statement is functorial and which prismatic comparison still requires identifying the actual input complex.

For the finite packet take the four distinct points

$$\alpha_1 = \rho, \quad \alpha_2 = 1 - \bar{\rho}, \quad \alpha_3 = \bar{\rho}, \quad \alpha_4 = 1 - \rho,$$

where $\rho = 1/2 + \delta + i\gamma$, $0 < \delta < 1/2$ and $\gamma > 2$. Put $d(s) = \prod_{i=1}^4 (s - \alpha_i)$, $h = d^m$ with $m \geq 1$, and $A_h = \mathbb{C}[s]/(h)$. These parameters define a finite algebra for every such ρ ; they do not assert that ρ is a zeta zero. Pairwise coprimality gives the isomorphism to the four rings $\mathbb{C}[t_i]/(t_i^m)$ by Taylor coefficients. To see its inverse explicitly, take $h_i = h/(s - \alpha_i)^m$ and multiply h_i by the degree- $(m-1)$ Taylor polynomial of $1/h_i$ at α_i . The resulting class e_i is 1 modulo $(s - \alpha_i)^m$ and 0 modulo each other primary power. Then

$$\sum_{i=1}^4 \sum_{b=0}^{m-1} c_{i,b} e_i (s - \alpha_i)^b \quad (\text{FSR25})$$

is the inverse image of the four Taylor polynomials. Injectivity follows because a polynomial vanishing to order m at all four points is divisible by h . This proves the complete primary decomposition and retains every jet.

Evaluation $\text{ev} : A_h \rightarrow \mathbb{C}^4$ is surjective with kernel $J_h = (d)/(d^m)$, of dimension $4m - 4$. Surjectivity follows from the e_i . The kernel statement follows by divisibility by the four distinct linear factors. Choose any

specified unit $u \in A_h$ and write $U = M_u$. Its four constant Taylor coefficients are nonzero; conversely that condition makes it a unit by the finite inverse of each truncated Taylor series. Then $a_h = \text{ev } U$ is onto with kernel J_h . The old programme choice $u = j_h(g/h)$, wherever that entire quotient defines a unit, is retained exactly by this construction; it is not substituted by 1.

Extend A_h, J_h, U and a_h coefficientwise as in FSR23. If $u_{i,b}$ denotes its Taylor coefficient, the full operator remains

$$U_L((x_{a,i,b})_{a,i,b})_{a,i,b} = \sum_{c=0}^b u_{i,c} x_{a,i,b-c}. \quad (\text{FSR26})$$

Thus no amplitude derivative, multiplicity, or support sector is omitted. Multiplication by s still acts on each basis element by α_i times that element plus the next jet, with the last next jet zero. The kernel of the extended a_h is $C_L \otimes J_h$, while the joint kernel of all branch maps on the entire algebra is $K_L \otimes A_h$. These are distinct explicitly defined subspaces.

6. The complete support-valued finite Weil pairing

Let σ exchange 1 with 2 and 3 with 4. On $\mathbb{C}^4$ set

$$b(c, c') = m \sum_{i=1}^4 \overline{c_{\sigma i}} c'_i, \quad B_h(x, y) = b(a_h x, a_h y). \quad (\text{FSR27})$$

Complex conjugation and reindexing by σ prove Hermitian symmetry. The matrix of b is the sum of two blocks $m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which is nonsingular and has one positive and one negative direction per block. Since a_h is onto with kernel J_h , the radical of B_h is J_h and its inertia is $(2, 2, 4m - 4)$. A full diagonal basis is

$$\frac{U^{-1}(e_1 + e_2)}{\sqrt{2m}}, \quad \frac{U^{-1}(e_1 - e_2)}{\sqrt{2m}}, \quad \frac{U^{-1}(e_3 + e_4)}{\sqrt{2m}}, \quad \frac{U^{-1}(e_3 - e_4)}{\sqrt{2m}},$$

together with $U^{-1}e_i(s - \alpha_i)^b$ for $1 \leq i \leq 4$ and $1 \leq b < m$. Direct evaluation gives squared values $+1, -1$ and 0 as stated. This proves the finite result independently of any analytic zeta assertion.

Give C_L the involution conjugating coefficients and fixing each E_a . For $x = \sum_a E_a \otimes x_a$ and $y = \sum_a E_a \otimes y_a$ define

$$\mathbb{B}_L(x, y) = \sum_{a \neq 0} E_a B_h(x_a, y_a). \quad (\text{FSR28})$$

The definition is unambiguous by FSR14. It is C_L -Hermitian and sesquilinear by the scalar identities and orthogonality of the E_a . Its radical is exactly $C_L \otimes J_h$: if its values against all y vanish, choosing y supported on one sector shows that the corresponding x_a lies in the scalar radical. For every branch j ,

$$\eta_j(\mathbb{B}_L(x, y)) = B_h((\eta_j \otimes \text{id})x, (\eta_j \otimes \text{id})y). \quad (\text{FSR29})$$

This is the proved comparison with the earlier scalar packet form.

The intrinsic positive cone of the finite product algebra C_L is

$$C_L^+ = \left\{ \sum_a c_a E_a : c_a \in \mathbb{R}_{\geq 0} \right\}. \quad (\text{FSR30})$$

It equals $\{v^*v : v \in C_L\}$, since each coordinate is an absolute square, and conversely each nonnegative coordinate has its nonnegative real square root. The regular-representation trace is $\text{tr}_{\text{reg}}(\sum_a c_a E_a) = \sum_a c_a$: multiplication by E_a is the projection onto its one-dimensional coordinate. Consequently $\text{tr}_{\text{reg}} \mathbb{B}_L$ has inertia

$$(2(|L| - 1), 2(|L| - 1), (4m - 4)(|L| - 1)). \quad (\text{FSR31})$$

This follows from the direct sum of the explicit diagonal bases after FSR27. No claim identifies this finite trace with the adelic trace without the corresponding analytic map.

Retaining only the sum of Boolean-branch traces means using $\sum_{j \in J(L)} \eta_j$. The radical of that form is

$$\left(\bigoplus_{j \in J(L)} E_j \otimes J_h \right) \oplus (K_L \otimes A_h), \quad (\text{FSR32})$$

and its inertia is

$$(2|J(L)|, 2|J(L)|, (4m-4)|J(L)| + 4m(|L| - 1 - |J(L)|)). \quad (\text{FSR33})$$

Each observed sector has exactly the scalar radical and each unobserved sector has its entire $4m$ -dimensional space in the radical; this proves both formulas.

For B^2 let w be FSR16 and choose the fully specified element

$$x = w \otimes U^{-1}(e_1 - e_2). \quad (\text{FSR34})$$

Both branch images of x vanish, but

$$\mathbb{B}_{B^2}(x, x) = -2m w \neq 0. \quad (\text{FSR35})$$

The equality follows from FSR27 and $w^2 = w$. Thus zero results from all separate branch observations do not make the full support-valued form zero or positive. Replacing the minus by a plus in FSR34 gives $+2mw$, so the omitted space carries both signs in this finite packet model. This is a concrete comparison result about the defined form. It is not a counterexample to RH: the arbitrary ρ in FSR25 has not been asserted to be a zeta zero.

Finally the scalar inclusion $x \mapsto u_1 \otimes x$ obeys

$$\mathbb{B}_L(u_1 \otimes x, u_1 \otimes y) = u_1 B_h(x, y). \quad (\text{FSR36})$$

It therefore preserves the old calculation as a diagonal subspace. Full support adds exact sectors and additional observations; it does not by itself repair a negative scalar value. This specifies the part of the old Weil construction that survives and the additional calculation required for an actual full-support adelic realization.

The localized Weil packet: values, jets, and the observation multiplier

This paper computes the finite Hermitian form specified by the zero side of the Weil pairing on the programme's entire functions. It keeps the polynomial coordinate change, all primary jets, and the actual multiplier $U = M_{j_h(v_h)}$. The form has a four-dimensional nondegenerate quotient and a fully described nilpotent radical. Every jet remains present in the source algebra and in the multiplier, although the zero-side pairing evaluates only the zeroth jets.

The definitions of the quartet, E_h , the Taylor remainder j_h , the unit U , and the primary basis are those of the programme's OPG1 and OPG10–OPG11 in *Full observation proofs*. They are restated and proved to the extent needed here. The starting suggestion is Proposition 3 in the supplied *Split-Zero programme: orientation and new angles*, 22 September 2026. The formula there changes from the Fourier coordinate to the s -coordinate without changing the polynomial's name. Equations (WP5)–(WP11) below give the exact coordinate maps.

No analytic explicit-formula identity, Fourier inversion assertion, or positivity theorem is assumed in the finite calculation. Section 10 specifies the precise zero-side identity proved here and its analytic scope.

1. The complete-order quartet and the two polynomial coordinates

Retain the programme's hypotheses

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad m \in \mathbb{Z}_{\geq 1}. \quad (\text{WP1})$$

Order its four distinct points as

$$\alpha_1 = \rho, \quad \alpha_2 = 1 - \bar{\rho}, \quad \alpha_3 = \bar{\rho}, \quad \alpha_4 = 1 - \rho, \quad \mathcal{R} = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}. \quad (\text{WP2})$$

The real parts distinguish points with the same imaginary part because $\delta > 0$, and the imaginary parts distinguish the upper and lower pairs because $\gamma > 0$. Let $g = 2\xi$ have a zero of exact order m at each $\alpha \in \mathcal{R}$. Put

$$d(s) = \prod_{\alpha \in \mathcal{R}} (s - \alpha), \quad h(s) = d(s)^m, \quad n = 4m, \quad v_h(s) = \frac{g(s)}{h(s)}. \quad (\text{WP3})$$

Here s is the original zeta coordinate. At each α , write $g(s) = (s - \alpha)^m b_\alpha(s)$, where b_α is holomorphic and $b_\alpha(\alpha) \neq 0$. If $h_\alpha(s) = h(s)/(s - \alpha)^m$, then $h_\alpha(\alpha) \neq 0$, and

$$v_h(s) = \frac{b_\alpha(s)}{h_\alpha(s)} \quad \text{near } \alpha, \quad v_h(\alpha) = \frac{g^{(m)}(\alpha)}{m! h_\alpha(\alpha)} \neq 0. \quad (\text{WP4})$$

Thus all the apparent singularities of g/h are removable and v_h is entire. At every zero $\rho' \notin \mathcal{R}$ of g , it vanishes because $h(\rho') \neq 0$.

Define the Fourier coordinate and its inverse by

$$\theta(s) = \frac{s - 1/2}{i}, \quad \iota(z) = \frac{1}{2} + iz, \quad \theta\iota = \text{id}_{\mathbb{C}}, \quad \iota\theta = \text{id}_{\mathbb{C}}. \quad (\text{WP5})$$

For $P \in \mathbb{C}[z]$, the associated s -polynomial is

$$q_P(s) = P(\theta(s)), \quad F_P(z) = P(z)v_h(\iota(z)). \quad (\text{WP6})$$

In particular $F_P(\theta(s)) = q_P(s)v_h(s)$. The two polynomials P and q_P have the same degree, but their coefficients and variables differ by (WP5). A factor $P(1 - \bar{\alpha})$ is therefore not the value obtained from F_P unless P has first been replaced by q_P .

Put $\beta_\alpha = \theta(\alpha)$. In the order (WP2),

$$(\beta_{\alpha_1}, \beta_{\alpha_2}, \beta_{\alpha_3}, \beta_{\alpha_4}) = (\gamma - i\delta, \gamma + i\delta, -\gamma - i\delta, -\gamma + i\delta). \quad (\text{WP7})$$

Let $\sigma\alpha = 1 - \bar{\alpha}$. This involution interchanges α_1, α_2 and α_3, α_4 . Direct substitution gives

$$\overline{\theta(\alpha)} = \theta(\sigma\alpha), \quad \iota(\overline{\beta_\alpha}) = \sigma\alpha. \quad (\text{WP8})$$

Define

$$H(z) = \prod_{\alpha \in \mathcal{R}} (z - \beta_\alpha)^m = i^{-n} h(\iota(z)), \quad E_z = \mathbb{C}[z]/(H), \quad E_h = \mathbb{C}[s]/(h). \quad (\text{WP9})$$

Here $n = 4m$, so $i^{-n} = 1$; the scalar is displayed to record how the defining polynomials transform. The maps

$$\mathcal{C} : E_z \longrightarrow E_h, \quad [P] \longmapsto [P \circ \theta], \quad \mathcal{C}^{-1} : E_h \longrightarrow E_z, \quad [q] \longmapsto [q \circ \iota] \quad (\text{WP10})$$

are inverse algebra isomorphisms. Indeed substitution sends H to $i^{-n}h$ and h to $i^n H$, so the maps descend to the quotients. Their two composites fix every polynomial by (WP5). They also retain the multiplication operators exactly:

$$\mathcal{C} M_z = M_{\theta(s)} \mathcal{C}, \quad \mathcal{C} M_{\iota(z)} = M_s \mathcal{C}. \quad (\text{WP11})$$

2. All the jets and the entire-function remainder

For each $\alpha \in \mathcal{R}$, introduce an independent nilpotent coordinate t_α and the algebra

$$A_\alpha = \mathbb{C}[t_\alpha]/(t_\alpha^m). \quad (\text{WP12})$$

There is an algebra isomorphism

$$\mathcal{J} : E_h \longrightarrow \prod_{\alpha \in \mathcal{R}} A_\alpha, \quad [q] \longmapsto \left(\sum_{a=0}^{m-1} \frac{q^{(a)}(\alpha)}{a!} t_\alpha^a \right)_{\alpha \in \mathcal{R}}. \quad (\text{WP13})$$

Every derivative in this formula is part of the specified map.

Here is a direct proof including its inverse. Define

$$T_\alpha(s) = \sum_{a=0}^{m-1} \frac{(1/h_\alpha)^{(a)}(\alpha)}{a!} (s - \alpha)^a, \quad e_\alpha = [h_\alpha T_\alpha] \in E_h. \quad (\text{WP14})$$

Modulo $(s - \alpha)^m$, the product $h_\alpha T_\alpha$ equals 1, by multiplication of the Taylor expansion and its truncated inverse. Modulo $(s - \eta)^m$ for $\eta \neq \alpha$, it equals 0, because h_α is divisible by that factor. Thus $\mathcal{J}(e_\alpha)$ is 1 in the α -factor and 0 elsewhere. The inverse of (WP13) is

$$\mathcal{J}^{-1} \left(\left(\sum_{a=0}^{m-1} c_{\alpha,a} t_\alpha^a \right)_\alpha \right) = \sum_{\alpha \in \mathcal{R}} \sum_{a=0}^{m-1} c_{\alpha,a} e_\alpha (s - \alpha)^a. \quad (\text{WP15})$$

This proves surjectivity. For injectivity, a polynomial in the kernel has a zero of order at least m at each of the four distinct points. Successive division by their linear factors shows it is divisible by their product h . The kernel is therefore zero in E_h . Taylor multiplication modulo t_α^m proves multiplicativity. Equations (WP13)–(WP15) also prove

$$e_\alpha e_\eta = 0 \ (\alpha \neq \eta), \quad e_\alpha^2 = e_\alpha, \quad \sum_{\alpha} e_\alpha = 1, \quad (\text{WP16})$$

and give the basis $e_\alpha (s - \alpha)^a$, $0 \leq a < m$, of E_h .

For an entire function f , define its finite Taylor remainder by

$$j_h(f) = \mathcal{J}^{-1} \left(\left(\sum_{a=0}^{m-1} \frac{f^{(a)}(\alpha)}{a!} t_\alpha^a \right)_\alpha \right) \in E_h. \quad (\text{WP17})$$

This is an algebra homomorphism: addition and scalar multiplication follow coefficient by coefficient, and the Leibniz rule gives multiplication of the truncated Taylor series. It agrees with the quotient map on polynomials. The class $j_h(f)$ has a unique representative of degree less than n , by division by the monic polynomial h .

In the programme notation $j_h v_h$, the symbol j_h denotes this map applied to v_h . It is not a second scalar function multiplied by v_h . Put

$$u_h = j_h(v_h), \quad U = M_{u_h} : E_h \longrightarrow E_h. \quad (\text{WP18})$$

In the α -factor of (WP13), set

$$v_{\alpha,a} = \frac{v_h^{(a)}(\alpha)}{a!}, \quad u_\alpha(t_\alpha) = \sum_{a=0}^{m-1} v_{\alpha,a} t_\alpha^a. \quad (\text{WP19})$$

All constants $v_{\alpha,0}$ are nonzero by (WP4). Write $u_\alpha = v_{\alpha,0}(1 + b_\alpha)$, with $b_\alpha \in (t_\alpha)$. Since $b_\alpha^m = 0$, its inverse is

$$u_\alpha^{-1} = v_{\alpha,0}^{-1} \sum_{r=0}^{m-1} (-b_\alpha)^r. \quad (\text{WP20})$$

Multiplication of this finite series by u_α gives $1 - (-b_\alpha)^m = 1$. Hence u_h is a unit and U an automorphism. In full jet coordinates its formula is

$$(Ux)_{\alpha,a} = \sum_{b=0}^a v_{\alpha,b} x_{\alpha,a-b} \quad (0 \leq a < m). \quad (\text{WP21})$$

It therefore keeps every derivative of the amplitude through order $m-1$. The determinant in the primary basis is $\prod_{\alpha \in \mathcal{R}} v_h(\alpha)^m \neq 0$, because each block in (WP21) is triangular with that constant diagonal.

3. The zeroth-jet quotient and its exact multiplier map

Let $V_{\mathcal{R}} = \mathbb{C}^{\mathcal{R}}$, with coordinatewise multiplication, and define

$$e : E_h \longrightarrow V_{\mathcal{R}}, \quad e([q]) = (q(\alpha))_{\alpha \in \mathcal{R}}. \quad (\text{WP22})$$

This is a surjective algebra homomorphism: it is the constant-term map in each factor of (WP13), and (WP15) lifts any tuple by $\sum c_\alpha e_\alpha$. Its kernel is

$$J = \ker e = (d)/(d^m) \subset E_h, \quad \dim_{\mathbb{C}} J = 4m - 4. \quad (\text{WP23})$$

Indeed a polynomial has value zero at every α precisely when it is divisible by d . In the primary coordinates, J is $\prod_{\alpha} (t_\alpha)$; it has basis $e_\alpha(s - \alpha)^a$, $1 \leq a < m$, and the asserted dimension follows. Every element of J has its m -th power zero. Conversely, a nilpotent element must have value zero in each complex coordinate, since a complex number with a vanishing positive power is zero. Thus J is exactly the nilradical of E_h .

Write

$$V_h : V_{\mathcal{R}} \longrightarrow V_{\mathcal{R}}, \quad (c_\alpha)_\alpha \longmapsto (v_h(\alpha)c_\alpha)_\alpha, \quad a_h = eU : E_h \longrightarrow V_{\mathcal{R}}. \quad (\text{WP24})$$

Then

$$eU = V_h e, \quad \ker a_h = J, \quad a_h \text{ is surjective.} \quad (\text{WP25})$$

The identity follows either from evaluation of a product or from the $a=0$ case of (WP21). The map V_h is invertible because its four diagonal entries are nonzero, so the kernel and surjectivity assertions follow from (WP22)–(WP23). Thus the exact comparison with the programme's multiplier is the commuting diagram of typed maps

$$\begin{array}{ccc} E_h & \xrightarrow{U} & E_h \\ e \downarrow & & \downarrow e \\ V_{\mathcal{R}} & \xrightarrow{V_h} & V_{\mathcal{R}} \end{array} \quad (\text{WP26})$$

The left and right vertical maps retain the same kernel J ; the horizontal map on the upper row retains all the additional coefficients (WP21).

The entire-function family is a further, distinct vector space

$$\mathcal{F}_h = \{F_P : P \in \mathbb{C}[z]\}. \quad (\text{WP26a})$$

The map $P \mapsto F_P$ is a linear isomorphism onto this family. Surjectivity is its definition. For injectivity, v_h is nonzero at a packet root, hence nonzero on a neighborhood of that root. If F_P vanishes identically, P vanishes on the corresponding open set in the z -coordinate. A nonzero polynomial has only finitely many roots, so $P = 0$.

The full-jet map from this entire-function family is

$$\begin{aligned} \mathcal{T}_h : \mathcal{F}_h &\longrightarrow E_h, & \mathcal{T}_h(F) &= j_h(F \circ \theta), \\ \mathcal{T}_h(F_P) &= U\mathcal{C}[P], & (\mathcal{J}\mathcal{T}_h(F))_{\alpha,a} &= i^{-a} \frac{F^{(a)}(\beta_\alpha)}{a!}. \end{aligned} \quad (\text{WP26b})$$

The first identity on the second line follows from (WP6), (WP17), and (WP18). The second follows by differentiating the affine change of variable θ exactly a times; its derivative is $1/i$. Thus the factor i^{-a} is retained in every jet coordinate.

The map $\mathcal{T}_h$ is onto and its kernel is

$$\ker \mathcal{T}_h = \{F_{HQ} : Q \in \mathbb{C}[z]\}. \quad (\text{WP26c})$$

Indeed U and $\mathcal{C}$ are isomorphisms, so $\mathcal{T}_h(F_P) = 0$ exactly when P is divisible by H . There is a separate surjection

$$e\mathcal{T}_h : \mathcal{F}_h \longrightarrow V_{\mathcal{R}}, \quad F \longmapsto (F(\beta_\alpha))_\alpha, \quad \ker(e\mathcal{T}_h) = \{F_{D_z Q} : Q \in \mathbb{C}[z]\}, \quad D_z(z) = \prod_\alpha (z - \beta_\alpha). \quad (\text{WP26d})$$

The coordinate formula follows from (WP26b) at $a = 0$. It is onto by (WP25). Since $v_h(\alpha) \neq 0$, its value at F_P is zero exactly when P vanishes at all the distinct β_α , which is equivalent to divisibility by D_z . Thus the entire functions themselves are not identified with their finite jets or with their values: (WP26b)–(WP26d) give the actual surjections and both exact kernels.

4. The finite Hermitian form and its radical

Use the convention that a Hermitian form is conjugate-linear in its first argument and linear in its second. On $V_{\mathcal{R}}$, define

$$b_{\mathcal{R}}(c, c') = m \sum_{\alpha \in \mathcal{R}} \overline{c_{\sigma\alpha}} c'_\alpha. \quad (\text{WP27})$$

It is Hermitian: the conjugate of $b_{\mathcal{R}}(c', c)$, after replacing the summation index α by $\sigma\alpha$, is exactly (WP27). It is nondegenerate. If it vanishes against every c' , selecting c' with a single nonzero coordinate at α gives $m\overline{c_{\sigma\alpha}} = 0$, hence every coordinate of c is zero.

Define the packet form on the full algebra by the pullback

$$B_h(x, y) = b_{\mathcal{R}}(a_h x, a_h y) = m \sum_{\alpha \in \mathcal{R}} \overline{v_h(\sigma\alpha) x(\sigma\alpha)} v_h(\alpha) y(\alpha), \quad x, y \in E_h. \quad (\text{WP28})$$

Evaluation is independent of polynomial representatives by (WP22), so the formula is well defined. It is Hermitian because (WP27) is. Let $\text{rad } B_h$ mean the set of x with $B_h(x, y) = 0$ for every $y \in E_h$.

Theorem WP1 (exact radical and quotient). There is an exact sequence

$$0 \longrightarrow J \longrightarrow E_h \xrightarrow{a_h} V_{\mathcal{R}} \longrightarrow 0, \quad \text{rad } B_h = J. \quad (\text{WP29})$$

The induced map $\bar{a}_h : E_h/J \rightarrow V_{\mathcal{R}}$ is an algebra-multiplier comparison and a linear isometry from the descended Hermitian form to $b_{\mathcal{R}}$. More precisely, $e : E_h/J \rightarrow V_{\mathcal{R}}$ is an algebra isomorphism, and $\bar{a}_h = V_h \bar{e}$ is a linear isomorphism; $\bar{a}_h$ need not preserve the unit or multiplication.

Proof. Exactness is (WP25). If $x \in J$, then $a_h x = 0$, so (WP28) vanishes for every y . Conversely, if (WP28) vanishes for every y , surjectivity of a_h implies that $b_{\mathcal{R}}(a_h x, c') = 0$ for every c' . Nondegeneracy of (WP27) gives $a_h x = 0$, hence $x \in J$. The quotient map is an isomorphism by exactness. Formula (WP28) proves the isometry identity. The distinction between $\bar{e}$ and $\bar{a}_h$ follows from (WP24): multiplying all coordinates by a fixed nonconstant unit is a linear automorphism, and is not in general an algebra homomorphism. $\square$

This theorem does not remove the nilpotent directions from the original packet. It identifies the precise quotient seen by this particular Hermitian form, and identifies every omitted direction as an element of the specified ideal J .

5. Complete inertia and explicit negative representatives

In the order (WP2), write $c = (a, b, c_3, c_4)$. Then

$$b_{\mathcal{R}}(c, c') = m(\bar{b}a' + \bar{a}b' + \bar{c}_4 c'_3 + \bar{c}_3 c'_4). \quad (\text{WP30})$$

The matrix is the direct sum of two blocks $m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. In each pair the two vectors $(1, 1)/\sqrt{2m}$ and $(1, -1)/\sqrt{2m}$ have squared values $+1$ and -1 , and their mutual pairing is zero. They form a basis of that two-dimensional pair. Consequently

$$\text{inertia}(b_{\mathcal{R}}) = (2, 2, 0), \quad \text{inertia}(B_h) = (2, 2, 4m - 4), \quad (\text{WP31})$$

where inertia lists positive, negative, and radical dimensions in that order. This is inertia under congruence of Hermitian forms, rather than an assertion that an arbitrary coefficient-frame matrix of B_h has eigenvalues $\pm m$.

For a complete diagonal basis of the original packet, use

$$x_1^\pm = \frac{1}{\sqrt{2m}} U^{-1}(e_{\alpha_1} \pm e_{\alpha_2}), \quad x_2^\pm = \frac{1}{\sqrt{2m}} U^{-1}(e_{\alpha_3} \pm e_{\alpha_4}), \quad (\text{WP32})$$

together with

$$x_{\alpha,a} = U^{-1}(e_\alpha(s - \alpha)^a), \quad \alpha \in \mathcal{R}, \quad 1 \leq a < m. \quad (\text{WP33})$$

These form a basis because the original primary vectors do, the two changes of basis at order zero are invertible, and U is invertible. The last $4m - 4$ vectors lie in J . Formula (WP28) gives $B_h(x_j^+, x_j^+) = 1$, $B_h(x_j^-, x_j^-) = -1$, all pairings between distinct displayed positive or negative vectors zero, and all pairings with (WP33) zero. This proves (WP31) on the entire vector space, including all primary multiplicities.

These representatives are explicit finite Taylor expressions. For example

$$U^{-1}e_\alpha = e_\alpha \sum_{a=0}^{m-1} \frac{(1/v_h)^{(a)}(\alpha)}{a!} (s - \alpha)^a. \quad (\text{WP34})$$

The reciprocal is taken only in a neighborhood of α , where it is holomorphic by (WP4). Taylor multiplication proves (WP34), as also follows from (WP20). Formula (WP34) prescribes the full jet of a representative whose amplitude has constant primary jet 1 at one root and zero at the others.

There are also degree-three representatives, without imposing any condition on their higher amplitude jets. Define the Lagrange polynomials

$$\ell_\alpha(s) = \prod_{\eta \in \mathcal{R} \setminus \{\alpha\}} \frac{s - \eta}{\alpha - \eta}. \quad (\text{WP35})$$

Their denominators are nonzero, and $\ell_\alpha(\eta) = \delta_{\alpha\eta}$. Put

$$q_-(s) = \frac{\ell_{\alpha_1}(s)}{v_h(\alpha_1)} - \frac{\ell_{\alpha_2}(s)}{v_h(\alpha_2)}, \quad P_-(z) = q_-(1/2 + iz). \quad (\text{WP36})$$

Then $a_h[q_-] = (1, -1, 0, 0)$, so

$$B_h([q_-], [q_-]) = -2m. \quad (\text{WP37})$$

Dividing either polynomial in (WP36) by $\sqrt{2m}$ gives value -1 ; dividing by $\sqrt{m}$ gives value -2 . The second negative direction is obtained with α_3, α_4 . This formula includes the exact Fourier-coordinate representative P_- .

For $m > 1$, the class $[q_-]$ in (WP36) and the full-jet representative $U^{-1}(e_{\alpha_1} - e_{\alpha_2})$ need not agree. Their difference lies in J , because their images under a_h agree and $\ker a_h = J$. Thus the two explicit choices have the same pairings with every packet class, with their difference retained as a specified nilpotent class.

6. The surviving jet filtration and multiplication operators

The radical has the finite filtration

$$E_h = J^0 \supset J^1 \supset \dots \supset J^{m-1} \supset J^m = 0, \quad J^r = (d^r)/(d^m). \quad (\text{WP38})$$

In primary coordinates, $J^r = \prod_\alpha (t_\alpha^r)$. To verify this from d , write $d(s) = (s - \alpha)c_\alpha(s)$, with $c_\alpha(\alpha) \neq 0$. The factor c_α is a unit in A_α , so the ideal generated by d^r in that primary factor is exactly (t_α^r) . The isomorphism (WP13) then proves (WP38) for every r .

For $0 \leq r < m$, the exact layer map is

$$\lambda_r : J^r/J^{r+1} \longrightarrow V_{\mathcal{R}}, \quad [x] \longmapsto \left(\frac{x^{(r)}(\alpha)}{r!} \right)_\alpha. \quad (\text{WP39})$$

For $r = 0$, this means the map e on E_h/J . A polynomial representative of an element of J^r has all derivatives of orders less than r zero at each root. Changing it by a multiple of h changes none of the derivatives in (WP39), since $r < m$. Changing its class by J^{r+1} also leaves those coefficients unchanged. Formula (WP13)

shows that the kernel is exactly J^{r+1} , and that the elements $e_\alpha(s - \alpha)^r$ map to the coordinate basis. Hence (WP39) is an isomorphism and every layer has dimension 4.

All these layers are preserved by U , its inverse, and $A_h = M_s$. Their induced operators are

$$\lambda_r \operatorname{gr}_r(U) = V_h \lambda_r, \quad \lambda_r \operatorname{gr}_r(A_h) = \operatorname{diag}(\alpha) \lambda_r. \quad (\text{WP40})$$

For the first equality, use (WP21): the coefficient of order r in a product with an element whose lower coefficients are zero is $v_{\alpha,0} x_{\alpha,r}$. For the second, multiply its Taylor series by $\alpha + t_\alpha$; the extra term only enters the next layer. Before passage to a layer, the complete formulas remain (WP21) and

$$A_h(e_\alpha(s - \alpha)^a) = \alpha e_\alpha(s - \alpha)^a + e_\alpha(s - \alpha)^{a+1}, \quad (\text{WP41})$$

where the last term is zero for $a = m - 1$. Thus the original Jordan chains and the amplitude derivatives survive in the source and its filtered maps. The form (WP28) reads only the layer $r = 0$; this follows from its evaluated formula and does not assert that higher derivatives are unavailable to the observation map.

7. The exact adjoint relation for every polynomial multiplier

For $q \in \mathbb{C}[s]$, define

$$q^\#(s) = \overline{q(1 - \bar{s})}. \quad (\text{WP42})$$

This is a polynomial: if $q(s) = \sum_k c_k s^k$, then $q^\#(s) = \sum_k \bar{c}_k (1 - s)^k$. It is an antilinear, multiplicative involution. Since σ permutes $\mathcal{R}$,

$$h^\#(s) = \prod_{\alpha \in \mathcal{R}} (1 - s - \bar{\alpha})^m = (-1)^{4m} \prod_{\alpha \in \mathcal{R}} (s - \sigma\alpha)^m = h(s). \quad (\text{WP43})$$

It therefore descends to an antilinear algebra involution on E_h . Its action on all the jets is

$$(x^\#)_{\alpha,a} = (-1)^a \overline{x_{\sigma\alpha,a}}. \quad (\text{WP44})$$

Indeed, substitute $s = \alpha + t$ into (WP42), expand q at $\sigma\alpha$, and conjugate its expansion at the increment $-\bar{t}$. This gives the displayed coefficient, including its sign.

For the original completed zeta function, its functional identities are $g(1 - s) = g(s)$ and $g(\bar{s}) = \overline{g(s)}$. They imply $g^\# = g$. The quartet polynomial has exactly the same two identities: conjugation permutes its four factors, and replacing s by $1 - s$ gives the sign $(-1)^{4m} = 1$ and the permutation $\alpha \mapsto 1 - \alpha$. Therefore

$$v_h^\# = v_h, \quad v_h(\sigma\alpha) = \overline{v_h(\alpha)}, \quad (j_h v_h)^\# = j_h v_h. \quad (\text{WP44a})$$

The first identity follows by division off the finite zero set of h and then by continuity at the removable points (WP4). The second is its evaluation, and the third follows from the signed jet formula (WP44). No scalar factor accompanies these identities for the monic degree- $4m$ polynomial in (WP3).

Theorem WP2 (multiplier adjoints). For every $q, x, y \in E_h$,

$$B_h(M_q x, y) = B_h(x, M_{q^\#} y). \quad (\text{WP45})$$

In particular, if $Y_h = (A_h - \frac{1}{2}I)/i = M_{\theta(s)}$, then

$$B_h(A_h x, y) = B_h(x, (I - A_h)y), \quad B_h(Y_h x, y) = B_h(x, Y_h y). \quad (\text{WP46})$$

Proof. At each root α , $q^\#(\alpha) = \overline{q(\sigma\alpha)}$. Inserting a factor $q(\sigma\alpha)$ in the conjugated first entry of (WP28) therefore has exactly the same effect as inserting $q^\#(\alpha)$ in its second entry. This proves (WP45) term by term. For the coordinate polynomial, $s^\# = 1 - s$. For $\theta(s) = -i(s - 1/2)$, antilinearity and this equality give $\theta^\#(s) = i((1 - s) - 1/2) = -i(s - 1/2) = \theta(s)$. These substitutions prove both assertions in (WP46). $\square$

The operator Y_h is consequently self-adjoint with respect to the specified Hermitian form. Its induced eigenvalues on E_h/J are exactly (WP7), since (WP40) gives the diagonal multiplier $(\alpha - 1/2)/i$. The nonreal values $\gamma \pm i\delta$, $-\gamma \pm i\delta$ are paired by the two hyperbolic blocks (WP30). There is no claim that self-adjointness for this indefinite form makes those values real. Equations (WP40), (WP44), and (WP46) give the exact relation between the full primary operator and its value quotient.

In particular the original amplitude unit is itself self-adjoint for this form, by (WP44a) and (WP45). For polynomial inputs, the entire function

$$\mathcal{A}_{P,Q}(s) = q_P^\#(s)q_Q(s)v_h(s)^2 \quad (\text{WP46a})$$

has the packet values

$$\mathcal{A}_{P,Q}(\alpha) = \overline{q_P(\sigma\alpha)v_h(\sigma\alpha)}q_Q(\alpha)v_h(\alpha). \quad (\text{WP46b})$$

This follows from $q_P^\#(\alpha) = \overline{q_P(\sigma\alpha)}$ and (WP44a). On the complement of the zeros of g , the exact logarithmic-derivative identity is

$$\frac{\xi'(s)}{\xi(s)}\mathcal{A}_{P,Q}(s) = q_P^\#(s)q_Q(s)\frac{g'(s)g(s)}{h(s)^2}. \quad (\text{WP46c})$$

Here $g = 2\xi$, so $\xi'/\xi = g'/g$; multiplication by $(g/h)^2$ proves the formula with no additional factor of two. At a zero of g outside the packet, the right side is holomorphic because h does not vanish there. At $\alpha \in \mathcal{R}$, writing $g = (s - \alpha)^m b_\alpha$ and $h = (s - \alpha)^m h_\alpha$ shows that g'/h^2 has the Laurent term $mv_h(\alpha)^2/(s - \alpha)$. Thus its residue after multiplication by $q_P^\#q_Q$ is exactly $m\mathcal{A}_{P,Q}(\alpha)$. These are local meromorphic identities; no contour estimate or prime-side identity is used to prove them.

8. The original period observation and what each map reads

The map $a_h = eU$ applies the same full-jet unit as the original observation and then takes values. To make the relation to the period stage explicit, let $\Pi : E_h \rightarrow \mathbb{C}^n$ be the programme's OPG4 linear map, defined on the unique degree-less-than- n representative by its prescribed convergent period integrals. Retain its actual contours, exponential weight, and basis; none is changed here. Put

$$\mathbf{b}_{\alpha,a} = \Pi(e_\alpha(s - \alpha)^a). \quad (\text{WP47})$$

For every $x \in E_h$, linearity and (WP21) give the identity of vectors in $\mathbb{C}^n$

$$\Pi Ux = \sum_{\alpha \in \mathcal{R}} \sum_{a=0}^{m-1} \mathbf{b}_{\alpha,a} \sum_{b=0}^a v_{\alpha,b} x_{\alpha,a-b}. \quad (\text{WP48})$$

This proof uses a finite sum of the original primary classes; it does not assign a period integral to an arbitrary entire function. The value map, with the same coefficients before projection, is

$$a_h x = (v_{\alpha,0} x_{\alpha,0})_\alpha. \quad (\text{WP49})$$

Thus (WP48) and (WP49) are two exact maps out of the same source after the same multiplication. In particular, for $x = j_h(f)$, the ring-homomorphism property (WP17) gives

$$Uj_h(f) = j_h(v_h f), \quad \Pi Uj_h(f) = \Pi j_h(v_h f), \quad a_h j_h(f) = (v_h(\alpha)f(\alpha))_\alpha. \quad (\text{WP50})$$

The middle formula evaluates Π on a finite Taylor remainder, exactly as in OPG11. It is not the integral of the unreduced entire function unless a separate equality proves that assertion.

There is also a precise test for whether any proposed observation $T : E_h \rightarrow W$, to a complex vector space W , factors through the value map a_h . The factorization exists exactly when $T(J) = 0$; when it exists its only possible formula is

$$\overline{T}(c) = T \left(U^{-1} \sum_{\alpha \in \mathcal{R}} c_\alpha e_\alpha \right). \quad (\text{WP51})$$

To prove this, the expression in parentheses has a_h -image c by (WP15) and (WP24). If $T(J) = 0$, any two lifts with that image differ by (WP29) in J , so their T -images agree, and (WP51) gives a well-defined linear factorization. Surjectivity of a_h forces uniqueness. Conversely a factorization through a_h kills its kernel J . This computes the exact space of factorizing maps as the injective image

$$\text{Hom}_{\mathbb{C}}(V_{\mathcal{R}}, W) \xrightarrow{a_h^*} \text{Hom}_{\mathbb{C}}(E_h, W), \quad \overline{T} \mapsto \overline{T}a_h, \quad \text{im}(a_h^*) = \{T : T(J) = 0\}. \quad (\text{WP52})$$

It does not presume that the period map, or the later cyclic tensor observation, lies in that subspace.

9. Why the off-line and on-line cases have different packet algebras

The hypotheses (WP1) guarantee four distinct roots. Substituting $\delta = 0$ into a formula which assumes four distinct roots merges α_1 with α_2 and α_3 with α_4 . If one retains the product of four factors literally, their orders double. Exact zero order m would then no longer justify dividing g by that new polynomial. The on-line packet must instead be defined from its distinct roots and their actual orders.

For completeness, the general finite value calculation is as follows. Let $\mathcal{A}$ be a finite set of distinct complex points stable under $\sigma\alpha = 1 - \bar{\alpha}$. Assign positive integer orders m_α , with $m_{\sigma\alpha} = m_\alpha$, and suppose an entire g has these exact orders there. Set

$$h_{\mathcal{A}}(s) = \prod_{\alpha \in \mathcal{A}} (s - \alpha)^{m_\alpha}, \quad v_{\mathcal{A}} = g/h_{\mathcal{A}}. \quad (\text{WP53})$$

With the corresponding jet algebra, remainder multiplier, and amplitude-value map, define

$$B_{\mathcal{A}}(x, y) = \sum_{\alpha \in \mathcal{A}} m_\alpha \overline{v_{\mathcal{A}}(\sigma\alpha)} x(\sigma\alpha) v_{\mathcal{A}}(\alpha) y(\alpha). \quad (\text{WP54})$$

Let f be the number of fixed points of σ in $\mathcal{A}$, and let p be the number of its two-element orbits. Then

$$\text{inertia}(B_{\mathcal{A}}) = \left(f + p, p, \sum_{\alpha \in \mathcal{A}} (m_\alpha - 1) \right). \quad (\text{WP55})$$

To prove this directly, use the local rings $\mathbb{C}[t_\alpha]/t_\alpha^{m_\alpha}$. The interpolation proof (WP14)–(WP15) applies with the individual order in each factor, because the other factors are still units at that root. Evaluation has kernel $\prod_{\alpha} (t_\alpha)$ of the stated dimension, and amplitude multiplication is invertible because its constants are nonzero. On its value quotient, a fixed point contributes the positive form $m_\alpha |c_\alpha|^2$. A two-element orbit contributes the block $m_\alpha \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, with one positive and one negative direction. Distinct orbits are orthogonal since their coordinates never occur together in (WP54). This proves the radical and all three entries of (WP55).

In particular, a packet supported entirely on the critical line has $p = 0$, and (WP54) is positive semidefinite. Its radical still contains all higher jet directions. A distinct off-line quartet has $f = 0, p = 2$, giving (WP31). Both assertions concern the respective actual packet algebras and the exact orders in (WP53).

10. The exact zero-side localization and analytic scope

Index the distinct nontrivial zeros of ξ by ρ' , and let $m_{\rho'}$ be their multiplicities. For the entire functions (WP6), form the discrete zero pairing

$$\mathcal{W}_0(F_P, F_Q) = \sum_{\rho'} m_{\rho'} \overline{F_P(\theta(\rho'))} F_Q(\theta(\rho')). \quad (\text{WP56})$$

This particular sum has only four potentially nonzero summands. Indeed, when $\rho' \notin \mathcal{R}$, (WP4) gives $F_Q(\theta(\rho')) = q_Q(\rho') v_h(\rho') = 0$. Thus no issue of summation order remains in (WP56). At the quartet, the multiplicities equal m , and (WP8) gives the exact identity

$$\begin{aligned} \mathcal{W}_0(F_P, F_Q) &= m \sum_{\alpha \in \mathcal{R}} \overline{q_P(\sigma\alpha)} v_h(\sigma\alpha) q_Q(\alpha) v_h(\alpha) \\ &= B_h(\mathcal{C}[P], \mathcal{C}[Q]). \end{aligned} \quad (\text{WP57})$$

The bracketed arguments on the second line are classes in E_z , mapped into E_h by (WP10). If a polynomial is changed by a multiple of H , its amplitude has all its first m jets zero at each Fourier root. Therefore (WP57) descends to E_z exactly as claimed. In fact the form already factors through the smaller value quotient $\mathbb{C}[z]/\prod_{\alpha} (z - \beta_\alpha)$; the explicit kernel of the map from E_z to that quotient is $\mathcal{C}^{-1}J$.

For $m > 1$, the multiplicity in (WP56) multiplies a value. It does not differentiate F_P or F_Q . This is why the correct inertia on the full $4m$ -dimensional packet is (WP31), and why the higher jet directions appear as its precisely computed radical. Their maps remain (WP21), (WP38)–(WP44), and (WP48).

The proof above does not identify (WP56) with pole, Gamma, and prime terms for this test family. Such an identity involves its analytic test class and convergence statements; none follows from the finite quotient alone. The finite algebra supplies a complete target for that calculation: the Hermitian form (WP28), with the explicit negative representatives (WP32) or (WP36), the complete radical (WP23), and the exact relation to the native amplitude multiplier (WP26). No assertion here establishes positivity of the prime-side expression or a conclusion about the existence of an off-line zero of ξ .

Exact residue-duality continuation

RD1–RD31b proves the complete Jacobian map from perfect residue duality to the retained regular trace, identifies the actual packet radical with cotangent cohomology, and computes the supported endpoint compensation. The existing formulas and all their coordinates remain unchanged.

The explicit formula on the original entire-function packet

This note proves the analytic identity needed to connect the finite packet to the arithmetic side of Weil’s explicit formula. It uses the programme’s actual multiplier, retains both polynomial coordinates, and proves convergence of every term. The result is an identity, not a proof of positivity or of the Riemann hypothesis.

The classical source for the formula and the Fourier/Mellin conventions is Alain Connes and Caterina Consani, *Weil positivity and Trace formula, the archimedean place*, arXiv:2006.13771v1, original author source `weil-compo.tex`, Appendix A, “Fourier versus Mellin transforms” (`appenmellinapp`), and Appendix B, “Explicit formula” (`appendix2`), source lines 2011–2070. In particular `bombieriexplicit`, `bombieriexplicit1`, and `burnolexPLICIT1` fix the pole terms, prime terms, and archimedean sign. That source states its formula for compactly supported functions. The proof below establishes the required extension directly for the present functions; it does not infer admissibility from a numerical residual.

WA1. Objects, involutions, and the exact integrand

Let

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (\text{WA1})$$

Use the entire continuation of this classical function, with $g(1-s) = g(s)$, $\overline{g(\bar{s})} = g(s)$, and $g(0) = g(1) = 1$. Retain the programme’s counterfactual quartet

$$\rho = \tfrac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \tfrac{1}{2}, \quad \gamma > 2, \quad \mathcal{R} = (\rho, 1 - \bar{\rho}, \bar{\rho}, 1 - \rho).$$

Each of these four distinct points is a zero of exact order $m \geq 1$ of g . This is the original counterfactual input, not an assertion that such a zero has been found. Put

$$h(s) = \prod_{\alpha \in \mathcal{R}} (s - \alpha)^m, \quad v(s) = g(s)/h(s), \quad \theta(s) = (s - \tfrac{1}{2})/i, \quad \iota(z) = \tfrac{1}{2} + iz. \quad (\text{WA2})$$

All apparent singularities of v are removable. At each packet point,

$$v(\alpha) = \frac{g^{(m)}(\alpha)}{m! \prod_{\eta \in \mathcal{R} \setminus \{\alpha\}} (\alpha - \eta)^m} \neq 0.$$

For polynomials $P, Q \in \mathbb{C}[z]$, define

$$q_P = P \circ \theta, \quad q_Q = Q \circ \theta, \quad F_P(z) = P(z)v(\iota(z)), \quad F_Q(z) = Q(z)v(\iota(z)).$$

For an entire function a in the s -coordinate, write $a^\#(s) = \overline{a(1 - \bar{s})}$. Conjugation followed by coefficient conjugation makes this entire, and the operation respects products. Reflection and conjugation permute the four factors of h ; the reflection contributes the sign $(-1)^{4m} = 1$. Thus $h^\# = h$ and $v^\# = v$. The exact pairing integrand is

$$A(s) = q_P^\#(s)q_Q(s)v(s)^2. \quad (\text{WA3})$$

It is entire. On the critical line,

$$K(t) := A(\tfrac{1}{2} + it) = \overline{F_P(t)}F_Q(t), \quad t \in \mathbb{R}. \quad (\text{WA4})$$

At a packet point it is

$$A(\alpha) = \overline{q_P(1 - \bar{\alpha})v(1 - \bar{\alpha})}q_Q(\alpha)v(\alpha). \quad (\text{WA5})$$

These identities follow by substitution in the displayed involution. They do not replace q_P by P .

WA2. Uniform decay in every fixed vertical strip

Here is a decay proof that needs no zero-free horizontal line or estimate of ζ'/ζ inside the critical strip.

For $\Re s > 1$, integration of the absolutely convergent series for ζ gives

$$\zeta(s) = s \int_1^\infty [x]x^{-s-1} dx = \frac{s}{s-1} - s \int_1^\infty \{x\}x^{-s-1} dx. \quad (\text{WA6})$$

For the first equality, write $[x] = \sum_{n \geq 1} 1_{[n, \infty)}(x)$, exchange the absolutely convergent sum and integral, and integrate each term. The last integral is holomorphic for $\Re s > 0$, by uniform domination on compact subsets there. It therefore gives the meromorphic continuation to this half-plane and the bound

$$|\zeta(s)| \leq |s/(s-1)| + |s|/(\Re s). \quad (\text{WA7})$$

In particular it is bounded by a constant times $1 + |\Im s|$ when the real part ranges in a fixed compact interval contained in $(0, \infty)$ and $|\Im s| \geq 2$.

For $z = \sigma + it$, $\sigma > 0$, rotate the Euler integral for $\Gamma(z)$ through the angle $\vartheta \operatorname{sgn}(t)$, where $0 < \vartheta < \pi/2$. The integral on the small circular arc tends to zero because $\sigma > 0$; the large arc tends to zero because the real part of its ray is positive. Cauchy's theorem gives

$$\Gamma(z) = e^{i\vartheta \operatorname{sgn}(t)z} \int_0^\infty e^{-re^{i\vartheta \operatorname{sgn}(t)}} r^{z-1} dr, \quad |\Gamma(\sigma + it)| \leq e^{-\vartheta|t|} \frac{\Gamma(\sigma)}{(\cos \vartheta)^\sigma}. \quad (\text{WA8})$$

The bound is uniform for σ in any positive compact interval. Apply it with $\vartheta = \pi/3$ to $\Gamma(s/2)$. Equations WA1 and WA7 show, uniformly in every fixed interval $1/2 \leq \Re s \leq B$,

$$|g(s)| \leq C_B(1 + |\Im s|)^3 e^{-\pi|\Im s|/6} \quad (|\Im s| \geq 2). \quad (\text{WA9})$$

Reflection $g(s) = g(1 - s)$ gives the same type of estimate on every fixed vertical strip, including negative real parts. The constant may depend on that strip. Applying Cauchy's derivative formula on circles of radius $1/4$, inside a slightly larger strip, gives the same estimate with an adjusted constant for every fixed derivative of g . The exponential changes on such a circle by at most a constant factor.

For $|\Im s|$ larger than twice the largest modulus of a packet point plus the strip width, every factor of h has modulus bounded below by a fixed positive multiple of $1 + |\Im s|$. Division by this polynomial and multiplication by the finite polynomials $q_P^\#, q_Q$ therefore give

$$|A^{(j)}(s)| \leq C_{B,P,Q,j}(1 + |\Im s|)^{D_j} e^{-\pi|\Im s|/3} \quad (\text{WA10})$$

on each fixed vertical strip, for each fixed derivative order j . On its bounded remaining part, entireness supplies a finite maximum. The same argument gives exponential strip decay of F_P, F_Q and every fixed derivative, with exponent $\pi/6$. These are estimates on the actual functions, not assertions of uniform constants as the packet or the polynomials vary.

WA3. The inverse transform, convolution, and prime-tail convergence

Define

$$f_P(u) = \frac{1}{2\pi} \int_{\mathbb{R}} F_P(t) e^{itu} dt, \quad f_Q(u) = \frac{1}{2\pi} \int_{\mathbb{R}} F_Q(t) e^{itu} dt, \quad k(u) = \frac{1}{2\pi} \int_{\mathbb{R}} K(t) e^{itu} dt. \quad (\text{WA11})$$

Exponential decay and its derivative estimates show that these inverse transforms are Schwartz functions. More explicitly, differentiate under the integral for derivatives in u , and integrate by parts any number of times for powers of u . All boundary terms vanish by WA10, and all resulting integrals converge absolutely. Fourier inversion on Schwartz functions gives $\widehat{f}_P = F_P$, $\widehat{f}_Q = F_Q$ with the convention $\widehat{f}(t) = \int f(u) e^{-iut} du$.

For any real $R > 0$, shift the t -contour in the last integral of WA11 to $\Im t = R$ for $u \geq 0$, and to $\Im t = -R$ for $u < 0$. The vertical connecting pieces tend to zero by WA10; multiplication by e^{itu} is bounded on each connecting segment for fixed u . Thus

$$|k(u)| \leq C_R e^{-R|u|}, \quad C_R = \frac{1}{2\pi} \max_{\epsilon=\pm 1} \int_{\mathbb{R}} |A(\tfrac{1}{2} - \epsilon R + it)| dt < \infty. \quad (\text{WA12})$$

The same proof applies to f_P, f_Q , and to their derivatives after inserting powers of the integration variable. Fubini's theorem on Schwartz functions gives the exact convolution

$$k = \widetilde{f}_P * f_Q, \quad \widetilde{f}_P(u) = \overline{f_P(-u)}. \quad (\text{WA13})$$

Its Fourier transform is $\overline{F_P(t)} F_Q(t) = K(t)$, which proves the equality by Fourier inversion.

Let $\Lambda_{\text{ar}}(n) = \log \ell$ when $n = \ell^r$ for a prime ℓ and $r \geq 1$, and zero otherwise. This is the arithmetic von Mangoldt function, not the observation map. Since $0 \leq \Lambda_{\text{ar}}(n) \leq \log n$, taking any $R > 1/2$ in WA12 proves absolute convergence of

$$\sum_{n \geq 2} \frac{\Lambda_{\text{ar}}(n)}{\sqrt{n}} (k(\log n) + k(-\log n)). \quad (\text{WA14})$$

There is also the explicit bound for every integer $N \geq 2$:

$$\sum_{n > N} \frac{\Lambda_{\text{ar}}(n)}{\sqrt{n}} (|k(\log n)| + |k(-\log n)|) \leq 2C_R \sum_{n > N} \frac{\log n}{n^{R+1/2}}. \quad (\text{WA15})$$

For example, with $R = 2$, the right side is at most

$$2C_2N^{-3/2} \left(\frac{2}{3} \log N + \frac{4}{9} \right). \quad (\text{WA16})$$

Indeed $(\log x)x^{-5/2}$ is decreasing for $x \geq 2$, so its sum beyond N is bounded by its integral from N to infinity; integration by parts gives the displayed expression. A numerical error certificate would additionally require a certified numerical bound for C_2 ; the existence proof alone is not such a certificate.

WA4. Cancellation of every unwanted zero

Set

$$L(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s) = \frac{g(s)}{s(s-1)}, \quad \ell(s) = L'(s)/L(s). \quad (\text{WA17})$$

This meromorphic completed function has simple poles at $0, 1$, is invariant under $s \mapsto 1-s$, and otherwise has exactly the nontrivial zeta zeros with their multiplicities. These assertions also follow from WA1 and the stated entire continuation of g . As a meromorphic identity,

$$\ell(s)A(s) = q_P^\#(s)q_Q(s) \frac{g'(s)g(s)}{h(s)^2} - \frac{A(s)}{s} - \frac{A(s)}{s-1}. \quad (\text{WA18})$$

Away from the roots of g , this is the logarithmic derivative identity and cancellation of one factor of g . Both sides then have the same meromorphic continuation.

At a zero outside $\mathcal{R}$, the first term in WA18 is holomorphic: its denominator does not vanish there. At $\alpha \in \mathcal{R}$, the logarithmic derivative of g has residue m ; multiplication by the entire function A gives residue $mA(\alpha)$, also when that residue is zero. At $0, 1$, the residues are respectively $-A(0), -A(1)$. These are the only possible poles. This proves exact cancellation at every other zero, regardless of its location or multiplicity.

Choose a rectangle with vertical sides $\Re s = 3/2$ and $\Re s = -1/2$, and horizontal sides $\Im s = \pm T$, for $T > \gamma + 1$. There are no poles on its boundary after the removable singularities in WA18 have been filled. Equations WA9–WA10 bound its horizontal integrals by a polynomial in T times $e^{-\pi T/3}$. They therefore tend to zero for all sufficiently large T , without selecting a sequence avoiding other zeta zeros. The residue theorem and $\ell(1-s) = -\ell(s)$ give

$$m \sum_{\alpha \in \mathcal{R}} A(\alpha) - A(0) - A(1) = \frac{1}{2\pi i} \int_{\Re s = 3/2} \ell(s) (A(s) + A(1-s)) ds. \quad (\text{WA19})$$

The integral is oriented upwards. To verify the sign, the positively oriented left edge is downward. Writing it as minus an upward integral and substituting $w = 1-s$ changes $\ell(1-w)$ to $-\ell(w)$; its total contribution is therefore the plus $A(1-s)$ term in WA19.

WA5. The arithmetic integral and its exact signs

On $\Re s = 3/2$, the absolutely convergent Euler product gives

$$\ell(s) = -\frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2) - \sum_{n \geq 2} \Lambda_{\text{ar}}(n) n^{-s}, \quad \psi = \Gamma'/\Gamma. \quad (\text{WA20})$$

For completeness, absolute convergence follows from $\sum_{n \geq 2} (\log n) n^{-3/2} < \infty$. Expanding each Euler factor as a geometric series gives ζ ; expanding its logarithm and differentiating on any strictly smaller closed half-plane gives the last series with the displayed coefficients. Uniform convergence justifies that differentiation and shows the product is nonzero there.

The gamma factor in WA20 is holomorphic between real parts $1/2$ and $3/2$. Its growth is at most polynomial, which is sufficient here. One direct bound uses

$$\psi(z) = -\gamma_E + \sum_{j=0}^{\infty} \left(\frac{1}{j+1} - \frac{1}{j+z} \right), \quad \Re z > 0. \quad (\text{WA21})$$

This expansion follows by logarithmic differentiation of Euler's gamma product, uniformly on compact subsets of this half-plane. For $\Re z \geq a > 0$, its summand has modulus at most $|z-1|/((j+1)(j+a))$, so the series gives $|\psi(z)| \leq C_a(1+|z|)$. Gamma conjugation gives $\psi(\bar{z}) = \overline{\psi(z)}$. Shift the gamma integral in WA19 to

$\Re s = 1/2$; the horizontal pieces vanish by this bound and WA10. Then substitute $s = 1/2 + it$, and in the term with $A(1 - s)$ replace t by $-t$. The result is

$$\frac{1}{2\pi} \int_{\mathbb{R}} K(t) \left(\Re \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi \right) dt. \quad (\text{WA22})$$

All these integrals are absolutely convergent.

For the Dirichlet series part, WA10 and its absolute convergence on $\Re s = 3/2$ permit termwise integration by absolute domination. For each fixed n , shift the entire integrand $n^{-s}(A(s) + A(1 - s))$ to the critical line. WA11 then gives

$$\frac{1}{2\pi i} \int_{\Re s=3/2} n^{-s} (A(s) + A(1 - s)) ds = \frac{k(-\log n) + k(\log n)}{\sqrt{n}}. \quad (\text{WA23})$$

The first sign comes from $n^{-it} = e^{-it \log n}$; reflection changes it for the second term. Thus the prime contribution is exactly minus WA14. Its convergence can be justified either before the shift by the absolutely convergent Dirichlet series or afterwards by WA12–WA15.

Combining WA19, WA22 and WA23 proves the complete identity

$$\boxed{\begin{aligned} m \sum_{\alpha \in \mathcal{R}} \bar{q}_P(1 - \bar{\alpha}) v(1 - \bar{\alpha}) q_Q(\alpha) v(\alpha) &= A(0) + A(1) \\ &+ \frac{1}{2\pi} \int_{\mathbb{R}} \overline{F_P(t)} F_Q(t) \left(\Re \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi \right) dt \\ &- \sum_{n \geq 2} \frac{\Lambda_{\text{ar}}(n)}{\sqrt{n}} (k(\log n) + k(-\log n)). \end{aligned}} \quad (\text{WA24})$$

It holds for every pair of polynomials and every packet satisfying the retained original hypotheses. No assumption of Weil positivity enters its proof.

WA6. Exact comparison with the published multiplicative convention

Define a function on $\mathbb{R}_{>0}$ by

$$b_A(x) = x^{-1/2} k(-\log x). \quad (\text{WA25})$$

It has decay sufficient for all the following integrals by WA12. Substituting $x = e^u$ in its Mellin transform gives

$$\int_0^\infty b_A(x) x^{s-1} dx = \int_{\mathbb{R}} k(-u) e^{(s-1/2)u} du = A(s). \quad (\text{WA26})$$

On the critical line this is Fourier inversion, and on the whole plane it follows by the same absolutely convergent transform and analytic continuation. Its involution satisfies

$$b_A^\sharp(x) = x^{-1} b_A(x^{-1}) = x^{-1/2} k(\log x). \quad (\text{WA27})$$

Consequently the two pole integrals in `bombieriexplicit` are precisely $A(1), A(0)$; for the second integral use the substitution $y = x^{-1}$. At a prime power $x = \ell^r$, its two finite-place terms are exactly $\ell^{-r/2}(k(-r \log \ell) + k(r \log \ell))$. The opposite of the archimedean distribution in `burnolexplicit1` is the integral in WA24. Equations WA25–WA27 prove the complete change of conventions, including its inverse $k(u) = e^{-u/2} b_A(e^{-u})$. The function is not asserted compactly supported; WA19–WA24 provide the required analytic extension.

WA7. Consequence for the programme's finite form, with the radical retained

Let $E_h = \mathbb{C}[s]/(h)$, let $U = M_{j_h(v)}$ be the literal finite Taylor-remainder multiplier, and let $e : E_h \rightarrow \mathbb{C}^{\mathcal{R}}$ evaluate the zeroth jets. The finite proofs WP13–WP25 give its complete jet formula and establish

$$a_h = eU, \quad a_h([q]) = (q(\alpha) v(\alpha))_\alpha, \quad \ker a_h = (d)/(d^m), \quad d = \prod_{\alpha \in \mathcal{R}} (s - \alpha). \quad (\text{WA28})$$

The elementary proof of the kernel is that all $v(\alpha)$ are nonzero and a polynomial vanishes at the four distinct points exactly when divisible by d . Surjectivity follows from interpolation at those points. The actual multiplier retains the higher jets, as specified in WP21; WA28 takes their stated quotient.

In the displayed order of $\mathcal{R}$, the left side of WA24 is the pullback under a_h of

$$b(c, d) = m(\bar{c}_2 d_1 + \bar{c}_1 d_2 + \bar{c}_4 d_3 + \bar{c}_3 d_4). \quad (\text{WA29})$$

Each pair has matrix $m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, with eigenvectors $(1, 1)$, $(1, -1)$ and eigenvalues $m, -m$. Hence this form has inertia $(2, 2)$, and its pullback has inertia $(2, 2, 4m - 4)$ and radical exactly WA28. The degree-three polynomial

$$q_-(s) = \frac{\prod_{\eta \neq \alpha_1} (s - \eta)}{v(\alpha_1) \prod_{\eta \neq \alpha_1} (\alpha_1 - \eta)} - \frac{\prod_{\eta \neq \alpha_2} (s - \eta)}{v(\alpha_2) \prod_{\eta \neq \alpha_2} (\alpha_2 - \eta)} \quad (\text{WA30})$$

has amplitude values $(1, -1, 0, 0)$. Taking $P_- = q_- \circ \iota$, and $P = Q = P_-$ in WA24, proves that its full arithmetic right side is exactly $-2m$. This is the arithmetic identity associated with the assumed off-line packet. It is not a constructed off-line zero and does not establish that the arithmetic right side is nonnegative for zeta.

The analytic identity closes the admissibility question for these noncompact test functions. The remaining RH question is an actual sign statement for the right side of WA24 on this programme family. Neither positive source norms nor the nonzero divided-power classes alone prove that sign. Their comparison must supply a proved map and an inequality before any RH conclusion can be drawn.

Tau residue norms and the exact change in the Weil formula

The original semiring is $S = G(\mathbb{Z}) = \{\tau\} \sqcup \mathbb{Z}$, with the original integer operations, $\tau + x = x$, and $\tau x = \tau$. Write $e = 0_{\mathbb{Z}}$, retaining $e \neq \tau$. This calculation starts from its residue and support maps. It does not replace a chosen residue norm by a different one without recording the map.

The originating programme's `globalization_note.tex`, `ideal/congruence/zeta` sections, proves the original arithmetic ideal norm and its zeta function. Its prime spectrum is two-dimensional. The companion `TAU_PRIME_SPECTRUM_DERIVATION.md` checks that classification, its localizations and the coefficient maps in full. Here the calculation continues from the different residue constructions to their exact prime distributions and Weil forms.

TN1. Three numerical constructions, with their quotient maps

For $n \geq 1$, reduction of supported integers defines the support-preserving semiring map

$$\pi_n : S \longrightarrow G(\mathbb{Z}/n\mathbb{Z}), \quad \tau \longmapsto \tau_n, \quad a \longmapsto (a \bmod n)^{\bullet}. \quad (\text{TN1})$$

It is onto and preserves both operations by their definitions. Its equivalence classes keep τ separate from every integer. Its global-zero fibre is only $\{\tau\}$. Arithmetic projection gives a second surjection

$$G(\mathbb{Z}/n\mathbb{Z}) \longrightarrow \mathbb{Z}/n\mathbb{Z}, \quad \tau_n \longmapsto 0, \quad b^{\bullet} \longmapsto b. \quad (\text{TN2})$$

The composite has zero fibre $I_n = \{\tau\} \cup n\mathbb{Z}$. The two target cardinalities are $n + 1$ and n , respectively. At $n = 1$ the first target is the two-element Boolean semiring and the second is the one-element ring; this case is retained.

The usual ideal product is $I_m I_n = I_{mn}$. Every finite sum of products of multiples of m, n is a multiple of mn , and every multiple of mn is itself such a product; the adjoined identity τ causes no further integer values. Hence the second cardinality is multiplicative, whereas the first has the exact defect

$$(m + 1)(n + 1) - (mn + 1) = m + n. \quad (\text{TN3})$$

Consequently the two actual ideal sums on I_n , for $\Re s > 1$, are

$$Z_{\text{ar}}(s) = \sum_{n \geq 1} n^{-s} = \zeta(s), \quad Z_{\text{card}}(s) = \sum_{n \geq 1} (n + 1)^{-s} = \zeta(s) - 1. \quad (\text{TN4})$$

The second equality is an index change in an absolutely convergent series. It does not have the prime Euler product obtained by replacing every p by $p + 1$.

There is also a precisely defined multiplicative prime weight

$$N_{\text{pr}}(n) = \prod_{p|n} (p + 1)^{v_p(n)}, \quad N_{\text{pr}}(1) = 1. \quad (\text{TN5})$$

This agrees with the first residue cardinality at prime integers, but not at general integers or at 1. Unique factorization gives its separate Euler product

$$E_{\text{pr}}(s) = \sum_{n \geq 1} N_{\text{pr}}(n)^{-s} = \prod_p \frac{1}{1 - (p + 1)^{-s}}, \quad \Re s > 1. \quad (\text{TN6})$$

Absolute convergence follows from $N_{\text{pr}}(n) \geq n$; expanding finite products and then passing to the absolutely convergent limit proves the identity. Thus all three functions in TN4 and TN6 are defined, and none is substituted for another.

TN2. The exact mixed-support reason for TN3

For commutative rings R, T , let $\chi_R : G(R) \rightarrow \mathbb{B}$ send τ to 0 and every supported element to 1; similarly for T . There is a semiring isomorphism

$$G(R \times T) \xrightarrow{\sim} G(R) \times_{\mathbb{B}} G(T), \quad \tau \longmapsto (\tau_R, \tau_T), \quad (r, t)^{\bullet} \longmapsto (r^{\bullet}, t^{\bullet}). \quad (\text{TN7})$$

The fibre product contains exactly the displayed two types, because equality of support means either both entries are absent or both are supported. The map is a bijection. Addition and multiplication on two

supported entries are the ring product operations; entries involving the absent pair obey the external identity and absorber laws. These checks prove preservation of both operations, both identities, and the inverse laws. They also prove the pullback universal property: a pair of semiring maps with equal support takes values in exactly this subset, and therefore factors uniquely through it.

If m, n are coprime, choose integers u, v with $um + vn = 1$. The usual reduction map $\mathbb{Z}/mn \rightarrow \mathbb{Z}/m \times \mathbb{Z}/n$ is injective because divisibility by both coprime integers implies divisibility by their product; it is surjective because $bvn + cum$ has residues b, c . Applying TN7 gives

$$G(\mathbb{Z}/mn) \cong G(\mathbb{Z}/m) \times_{\mathbb{B}} G(\mathbb{Z}/n). \quad (\text{TN8})$$

Its embedding into the full product omits precisely

$$(\mathbb{Z}/m)^{\bullet} \times \{\tau_n\}, \quad \{\tau_m\} \times (\mathbb{Z}/n)^{\bullet}. \quad (\text{TN9})$$

These disjoint mixed-support subsets have sizes m, n . TN3 is therefore the count of explicitly identified subsets. No supported cancellation is renamed as absence. In support-separated counts the vector is $(1, n)$; fibre-product multiplication is componentwise, $(1, m)(1, n) = (1, mn)$. Forgetting the labels by summing coordinates gives $n + 1$, and TN3 measures exactly the failure of that sum map to be multiplicative.

TN3. Exact analytic comparison for the prime weight

On $\Re s > 0$, use the branch

$$\log(1 - x^{-s}) = - \sum_{j \geq 1} \frac{x^{-js}}{j}, \quad x \geq 2,$$

and define

$$L_{\tau}(s) = \sum_p (\log(1 - p^{-s}) - \log(1 - (p+1)^{-s})), \quad R_{\tau}(s) = e^{L_{\tau}(s)}. \quad (\text{TN10})$$

This definition has no branch ambiguity because $|x^{-s}| < 1$. To prove convergence, differentiate the summand with respect to the real variable x :

$$\log(1 - p^{-s}) - \log(1 - (p+1)^{-s}) = -s \int_p^{p+1} \frac{x^{-s-1}}{1 - x^{-s}} dx. \quad (\text{TN11})$$

Fix $a > 0$, and put $d_a = 1 - 2^{-a} > 0$. On $\Re s \geq a$, the absolute value of the integrand without s is at most x^{-a-1}/d_a . The prime intervals are disjoint subsets of $[2, \infty)$, so

$$|L_{\tau}(s)| \leq \frac{|s|2^{-a}}{ad_a}. \quad (\text{TN12})$$

The same domination proves local uniform convergence of the series on the open half-plane. It therefore defines a holomorphic function, and its exponential is holomorphic and nowhere zero. On $\Re s > 1$, the absolutely convergent Euler products give

$$E_{\text{pr}}(s) = \zeta(s)R_{\tau}(s). \quad (\text{TN13})$$

This supplies a meromorphic continuation of the particular function TN6 to $\Re s > 0$. Its zero and pole divisor there equals that of ζ , with multiplicities, because R_{τ} is a holomorphic unit. This is a statement about TN6 with TN5, not about TN4's function $\zeta - 1$, and not about an isomorphism of spectra or trace spaces.

The retained logarithmic-derivative correction is

$$r_{\tau}(s) = \frac{R'_{\tau}(s)}{R_{\tau}(s)} = \sum_p \left(\frac{\log p}{p^s - 1} - \frac{\log(p+1)}{(p+1)^s - 1} \right). \quad (\text{TN14})$$

One may differentiate locally uniformly in TN10 by holomorphy and Cauchy's formula on slightly larger compact sets. A direct formula, including a vertical bound, is

$$r_{\tau}(s) = \sum_p \int_p^{p+1} x^{-s-1} \left(\frac{s \log x}{(1 - x^{-s})^2} - \frac{1}{1 - x^{-s}} \right) dx, \quad (\text{TN15})$$

$$|r_{\tau}(s)| \leq \frac{2^{-a}}{ad_a} + \frac{|s|2^{-a} \left(\frac{\log 2}{a} + \frac{1}{a^2} \right)}{d_a^2}, \quad \Re s \geq a. \quad (\text{TN16})$$

For TN15, differentiate $(\log x)/(x^s - 1)$ in x and integrate its negative derivative from p to $p+1$. The estimates use $\int_2^{\infty} x^{-a-1} dx = 2^{-a}/a$ and $\int_2^{\infty} (\log x)x^{-a-1} dx = 2^{-a}(\log 2/a + 1/a^2)$. They prove absolute local uniform convergence, and show that r_{τ} has at most linear vertical growth in every fixed strip inside this half-plane. All quantities in TN10–TN16 are retained; equality of a zero divisor does not remove this nonzero correction.

TN4. The corresponding prime distributions

Use the programme's actual entire test functions from WEIL_PACKET_ANALYTIC_DERIVATION.md:

$$A(s) = q_P^\#(s)q_Q(s)v(s)^2, \quad K(t) = A(\tfrac{1}{2} + it) = \overline{F_P(t)}F_Q(t), \quad k(u) = \frac{1}{2\pi} \int_{\mathbb{R}} K(t)e^{itu}dt. \quad (\text{TN17})$$

Its proof WA2–WA3 establishes exponential decay of A in every fixed vertical strip and $|k(u)| \leq C_R e^{-R|u|}$ for every fixed $R > 0$. Define the two explicit prime functionals

$$\begin{aligned} \mathcal{P}(k) &= \sum_p \sum_{j \geq 1} (\log p) p^{-j/2} (k(j \log p) + k(-j \log p)), \\ \mathcal{P}_\tau(k) &= \sum_p \sum_{j \geq 1} \log(p+1) (p+1)^{-j/2} (k(j \log(p+1)) + k(-j \log(p+1))). \end{aligned} \quad (\text{TN18})$$

Both sums converge absolutely: for $R = 2$, each is dominated by a constant times $\sum_p (\log(p+1))(p+1)^{-5/2}/(1 - (p+1)^{-5/2})$, or the analogous expression with p ; each is bounded by a convergent sum over all integers.

The exact difference is

$$\boxed{\mathcal{P}(k) - \mathcal{P}_\tau(k) = \frac{1}{2\pi} \int_{\mathbb{R}} K(t) 2\Re r_\tau(\tfrac{1}{2} + it) dt.} \quad (\text{TN19})$$

To prove it, start with

$$\frac{1}{2\pi i} \int_{\Re s = 3/2} r_\tau(s) (A(s) + A(1-s)) ds.$$

On this line, expand both terms in TN14 as absolutely convergent geometric series and integrate term by term. Fourier inversion and an entire contour shift give exactly the difference in TN18, with the factors $p^{-j/2}$ and $(p+1)^{-j/2}$. Alternatively shift the integral itself to $\Re s = 1/2$. No poles are crossed because r_τ is holomorphic for positive real part; its vertical bound TN16 and the strip decay of A make the horizontal integrals tend to zero. Reflect $t \mapsto -t$ in the second summand. Since TN10 has real coefficients under conjugation, $r_\tau(\bar{s}) = r_\tau(s)$. This gives TN19, including its sign and factor 2.

TN5. Weil's condition with the full tau correction

Let

$$w_\infty(t) = \Re \psi(\tfrac{1}{4} + \tfrac{it}{2}) - \log \pi, \quad w_{\infty, \tau}(t) = w_\infty(t) - 2\Re r_\tau(\tfrac{1}{2} + it). \quad (\text{TN20})$$

The complete analytic identity WA24 and TN19 give

$$\boxed{B_h([q_P], [q_Q]) = A(0) + A(1) + \frac{1}{2\pi} \int_{\mathbb{R}} \overline{F_P(t)} F_Q(t) w_{\infty, \tau}(t) dt - \mathcal{P}_\tau(k).} \quad (\text{TN21})$$

Here B_h is the literal full-jet packet form with its exact radical, as constructed in WP28–WP29. To check the sign directly, TN19 says $\mathcal{P} = \mathcal{P}_\tau + D$, where D is the integral of $2\Re r_\tau$. Substitution into the original term $-\mathcal{P}$ subtracts this integral, giving TN20. Thus altering the prime norms alone while keeping the previous infinite-place term does not produce the same form; TN20 is its required exact correction.

For $P = Q$, the right side of TN21 is real. Its nonnegativity is exactly the original packet's Weil positivity condition, since TN21 is an equality of forms. This calculation does not prove that condition. It supplies a full new expression of the condition through the separately defined prime-residue weight, preserving the support cardinality defect and the correcting distribution. In particular the degree-three negative representative WP36 of an assumed off-line packet still gives the exact value $-2m$ in TN21.

TN6. What this does and does not identify

There are exact maps from the original tau semiring to two different residue targets, an exact support fibre product explaining the failure of scalar count multiplicativity, and a complete analytic comparison from one specified prime-residue Euler product to its Weil functional. These statements use the original addition and keep its supported zero. They do not replace the two-dimensional semiring spectrum by the one-dimensional arithmetic spectrum, or identify either with Connes's adèle-class orbit space.

The further Connes calculation must retain the supported coordinate-zero hyperplanes, their stabilizers and transverse additive fields, and the new absence strata. Those are the objects entering Section VI,

equations (5)–(15), of Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, arXiv:math/9811068v1. The independent endpoint and adelic comparison continues in TAU_CONNES_ENDPOINT_DERIVATION.md and TAU_PRIME_SPECTRUM_DERIVATION.md. No equality of their Hilbert-space traces is assumed from the scalar function identity TN13.

The original three-point fibre, the prime-zero endpoints, and the heat sign change

This calculation answers the specific comparison between the target $(-1/4, 0, 0)$ of the original polynomial map and the pole pairing at $s = 0, 1$. It proves an isomorphism preserving the specified involution and trace, retains the third inverse point and the infinitesimal at infinity, and computes the exact change to the completed zeta function when its endpoint factor is moved. Positivity proved here concerns these finite trace forms. The complete zeta-zero form is retained in the completion calculation.

The original polynomial and three points are Tao, displayed formula (1) and the following equalities. Earlier programme proofs are the complete inverse chart, `eq:retained-cubic`, `eq:retained-inverse`, and `eq:retained-heat`, the original heat trajectories, and EFI7–EFI15, including the projective graph and dual-number fibre. The complete arguments needed for this comparison follow.

1. Original coordinates and the entire target line

Write the third source coordinate as W ; this is the identity renaming of the w in the cited source, to distinguish it from quotient generators. The original map is

$$\begin{aligned} F_1 &= (1 + xy)^3 W + y^2(1 + xy)(4 + 3xy), \\ F_2 &= y + 3x(1 + xy)^2 W + 3xy^2(4 + 3xy), \\ F_3 &= 2x - 3x^2y - x^3W. \end{aligned} \tag{HEB1}$$

Its target coordinates remain (a, b, c) . On $b = c = 0$, the complete inverse curve has the scheme decomposition

$$F^{-1}\{(a, 0, 0) : a \in \mathbb{C}\} \simeq \text{Spec } \mathbb{C}[a] \sqcup \text{Spec } \mathbb{C}[r, r^{-1}], \tag{HEB2}$$

with maps to the target respectively $a \mapsto a$ and $a \mapsto -r^2$. The source coordinates on the two components are

$$(x, y, W) = (0, 0, a), \quad (x, y, W) = \left(-\frac{1}{2r}, 3r, 26r^2\right). \tag{HEB3}$$

Proof. Put $K = 2 - 3xy - x^2W$. Then $F_3 = xK$ and $K + x(3y + xW) = 2$. In the coordinate ring defined by $F_1 - a, F_2$, the ideals (x) and (K) are comaximal. Their intersection is their product: if $v \in (x) \cap (K)$, multiply v by a representation $1 = j_x + j_K$ with $j_x \in (x)$ and $j_K \in (K)$; each summand is in (xK) . The converse inclusion is immediate. The Chinese remainder map therefore splits the quotient by F_3 into these two components, with no lost scheme structure.

On $x = 0$, $F_2 = y$ and $F_1 = W + 4y^2$, giving the first component. On $K = 0$, x is a unit with inverse $(3y + xW)/2$. Set $\alpha = 1/x$ and $r = y + 1/x$. Solving $F_3 = 0$ gives $W = 5\alpha^2 - 3r\alpha$. Direct substitution gives

$$F_2 = 4r + 2\alpha, \quad F_1 = r^2 + r\alpha. \tag{HEB4}$$

Consequently $\alpha = -2r$, $a = -r^2$, and (HEB3) follows. Conversely (HEB3) satisfies all three target equations and recovers $r = y + 1/x$. These mutually inverse coordinate substitutions prove (HEB2). Explicit component idempotents are $K/2$ for the finite component and $(3xy + x^2W)/2$ for the escaping component. Their sum is one and their product is zero. $\square$

At $a = -1/4$, the two roots are $r = -1/2, +1/2$. The complete fibre is precisely

component coordinate	(x, y, W)	
finite	$(0, 0, -1/4)$	
$r = -1/2$	$(1, -3/2, 13/2)$	
$r = +1/2$	$(-1, 3/2, 13/2)$	(HEB5)

Thus the number $1/4$ in the root relation is exactly the negative of the original target coordinate, rather than a new choice of scale.

This target is not the only noninjective one. Every $a \neq 0$ has the separate finite point and two distinct complex points in (HEB3). For example at $a = +1/4$ these are

$$(0, 0, 1/4), \quad (i, 3i/2, -13/2), \quad (-i, -3i/2, -13/2). \tag{HEB6}$$

Direct substitution or (HEB3) proves all three images are $(1/4, 0, 0)$. The two nonreal points cannot be dropped when discussing complex injectivity.

2. Compactification and the involution

In source projective coordinates $[X : Y : Z : T]$, whose affine coordinates are $(x, y, W) = (X/T, Y/T, Z/T)$, the escaping graph extends to

$$r \longmapsto ([1 : -6r^2 : -52r^3 : -2r], a = -r^2). \quad (\text{HEB7})$$

The displayed graph is closed in $\mathbb{P}^3 \times \mathbb{A}^1$: it is the scheme defined by $Y = 6aX$, $Z = -26aT$, and $T^2 + 4aX^2 = 0$. These equations have no projective point with $X = 0$, because they would successively force $Y = T = Z = 0$. In the chart $X = 1$ its coordinate ring is $\mathbb{C}[a, T]/(T^2 + 4a)$, and the mutually inverse substitutions $r = -T/2$, $a = -r^2$ identify this ring with $\mathbb{C}[r]$. Substitution recovers exactly all four coordinates in (HEB7), proving the asserted scheme equality. The coordinate $T/X = -2r$ is its inverse coordinate, so the graph is exactly the affine r -line. Its fibre at $a = 0$ is the length-two algebra $\mathbb{C}[r]/(r^2)$ supported at $[1 : 0 : 0 : 0]$. The finite component remains $[0 : 0 : a : 1]$, disjoint from (HEB7). Hence the compactified inverse graph over the target line is finite flat of rank three, with coordinate algebra

$$\mathcal{C} = \mathbb{C}[a] \times \mathbb{C}[a, r]/(r^2 + a). \quad (\text{HEB8})$$

Freeness of the second factor in the basis $1, r$ follows by division by its monic relation. The original affine inverse curve (HEB2) is obtained by removing the point $r = 0$ from the second component, not by quotienting away its infinitesimal.

The original source involution and its target map are

$$R(x, y, W) = (-x, -y, W), \quad F \circ R = (F_1, -F_2, -F_3). \quad (\text{HEB9})$$

Every term of (HEB1) verifies this equality: xy, y^2, x^2W remain fixed and the indicated odd factors change sign. Thus R preserves our target line, fixes the finite component, and sends $r = y + 1/x$ to $-r$. For real a , compose this algebraic reflection with coefficient conjugation to obtain

$$r^\# = -r, \quad z^\# = \bar{z} \quad (z \in \mathbb{C}). \quad (\text{HEB10})$$

The relation $r^2 + a$ is stable and $\#^2 = 1$, so this is a well-defined conjugate-linear ring involution. It is not ordinary coefficient conjugation with r fixed; (HEB9) is the exact map connecting the two.

3. An isometry with the actual endpoint pairing

For a compactly supported smooth complex test f on $\mathbb{R}$, retain

$$\widehat{f}(z) = \int_{\mathbb{R}} f(v) e^{-ivz} dv, \quad M_f(s) = \widehat{f}((s - 1/2)/i). \quad (\text{HEB11})$$

Set $A = M_f(0) = \widehat{f}(i/2)$ and $B = M_f(1) = \widehat{f}(-i/2)$. The endpoint contribution is

$$W_0(f, g) = \overline{A}_f B_g + \overline{B}_f A_g. \quad (\text{HEB12})$$

Use uppercase A, B here to avoid confusion with the original target a, b, c .

At the exact target $a = -1/4$, there are mutually inverse algebra maps

$$\begin{aligned} \mathbb{C}[s]/(s(s-1)) &\longrightarrow \mathbb{C}[r]/(r^2 - 1/4), & s &\longmapsto 1/2 + r, \\ \mathbb{C}[r]/(r^2 - 1/4) &\longrightarrow \mathbb{C}[s]/(s(s-1)), & r &\longmapsto s - 1/2. \end{aligned} \quad (\text{HEB13})$$

They respect the relations by expansion and are inverse on their generators. Under (HEB13), $s^\# = 1 - s$. Evaluation at $r = -1/2, +1/2$ maps the second algebra isomorphically to $\mathbb{C}^2$, with inverse

$$(A, B) \longmapsto p_{A,B} = (A + B)/2 + (B - A)r. \quad (\text{HEB14})$$

Both evaluations and the coefficient formula prove the inverse assertion. The involution acts on endpoint values as $(A, B)^\# = (\overline{B}, \overline{A})$.

For $E_a = \mathbb{C}[r]/(r^2 + a)$, define the regular trace by the trace of its multiplication operator. In the basis $(1, r)$,

$$L_r = \begin{pmatrix} 0 & -a \\ 1 & 0 \end{pmatrix}, \quad \text{Tr}_{E_a}(c + dr) = 2c. \quad (\text{HEB15})$$

Multiplying $(\bar{c} - \bar{d}r)(e + kr)$ gives scalar coefficient $\bar{c}e + a\bar{d}k$, and hence

$$q_a(c + dr, e + kr) := \text{Tr}_{E_a}((c + dr)^\#(e + kr)) = 2\bar{c}e + 2a\bar{d}k. \quad (\text{HEB16})$$

In particular substitution of (HEB14) gives

$$q_{-1/4}(p_{A,B}, p_{A,B}) = \frac{1}{2}|A + B|^2 - \frac{1}{2}|A - B|^2 = 2\text{Re}(A\bar{B}). \quad (\text{HEB17})$$

Polarization, or the same multiplication with two inputs, proves equality with (HEB12) in both slots. These are isomorphisms of algebras with involution and isometries of the trace pairings. They do not identify the separate third component with either endpoint. On the whole algebra (HEB8) the form is

$$q_a^{\text{full}}(v, c + dr) = |v|^2 + 2|c|^2 + 2a|d|^2. \quad (\text{HEB18})$$

4. The original heat clock crosses the sign boundary

Retain the original forward heat parameter h , with its original speed and initial target:

$$a(h) = -1/4 + 2h, \quad b = c = 0, \quad r^2 = 1/4 - 2h. \quad (\text{HEB19})$$

The original cubic on this line is $P = -2r^2 - 2a$; it satisfies $\partial_h P = -4 = \partial_r^2 P$. Thus (HEB19) is the original coefficient heat flow. The exact pulled-back form in the free basis $(1, r)$ is

$$G_h = \begin{pmatrix} 2 & 0 \\ 0 & 4h - 1/2 \end{pmatrix}, \quad q_h(r, r) = 4h - 1/2. \quad (\text{HEB20})$$

It has one positive and one negative direction for $h < 1/8$; at $h = 1/8$ it has one positive direction and radical $\mathbb{C}r$; for $h > 1/8$ it is positive definite. This follows directly from the two real diagonal entries. In the full rank-three algebra (HEB18), the finite point adds a positive direction in every case.

At the collision $r \neq 0$ as an algebra element and $r^2 = 0$: division by the monic polynomial gives independent classes $1, r$. It is the trace pairing of this nonzero infinitesimal that vanishes. Under the supported embedding into $G(E_0)$ its square is the supported zero e , distinct from the global unsupported zero τ . No positivity conclusion follows from identifying either of these zeros with the other.

The two spectral coordinates obtained from (HEB13) throughout the family are

$$s_\pm(h) = 1/2 \pm \sqrt{1/4 - 2h}. \quad (\text{HEB21})$$

At $h = 0$ they are the actual endpoints $0, 1$. At $h = 1/8$ they coincide at $1/2$ with multiplicity two. At $h > 1/8$ they lie on the critical line, at $1/2 \pm i\sqrt{2h - 1/4}$. In the original source, however, the corresponding x, y have become nonreal; the original real trajectories escape to infinity at the collision. Formula (HEB7) gives the continuation as a complex algebraic graph, not a finite continuation in $\mathbb{R}^3$.

There is also an operator version with the metric retained. Put $S = I/2 + L_r$ and $D = (S - I/2)/i = -iL_r$. Direct multiplication gives

$$L_r^* G_h = -G_h L_r, \quad D^* G_h = G_h D. \quad (\text{HEB22})$$

For $h > 1/8$, G_h is positive definite, so if $Dv = \lambda v$ with $v \neq 0$, the equality of the two pairings gives $\lambda = \bar{\lambda}$. Thus S has real part $1/2$ on every eigenvalue. At the collision, D is a nonzero nilpotent; it cannot be self-adjoint for any positive definite metric, because that would give $\|Dv\|^2 = \langle v, D^2 v \rangle = 0$ for every v . Its specified semidefinite trace metric retains exactly the radical already computed. This is a complete finite-dimensional positivity and spectral calculation.

5. Moving this factor in the completed zeta function

Define

$$q_h(s) = s(s - 1) + 2h, \quad q_0 = s(s - 1), \quad \Lambda(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad 2\xi(s) = q_0(s) \Lambda(s). \quad (\text{HEB23})$$

The entire endpoint test M_f has a unique class modulo q_h : for two distinct roots use their two values, and at a double root use its value and first derivative. Existence follows by degree-one interpolation; uniqueness follows because a polynomial of degree at most one with the two required vanishing conditions is zero. The involution is the same $s^\# = 1 - s$ with conjugate coefficients. Its trace pairing is

$$P_h(f, g) = \sum_{q_h(\rho)=0}^{\text{mult}} \overline{M_f(1 - \bar{\rho})} M_g(\rho). \quad (\text{HEB24})$$

This equals (HEB16) for those interpolating classes: multiplication matrices at distinct roots are diagonalized by evaluation, and at a double root their trace is twice their scalar coefficient. Therefore $P_0 = W_0$, $P_{1/8}(f, f) = 2|M_f(1/2)|^2$, and

$$P_h(f, f) = |M_f(s_+(h))|^2 + |M_f(s_-(h))|^2 \geq 0 \quad (h > 1/8). \quad (\text{HEB25})$$

An exact realization of this moving denominator is

$$\Lambda_h(s) = \frac{2\xi(s)}{q_h(s)} = \Lambda(s) \frac{q_0(s)}{q_h(s)}. \quad (\text{HEB26})$$

As identities of meromorphic functions and signed divisors,

$$\frac{\Lambda'_h}{\Lambda_h} = \frac{\Lambda'}{\Lambda} + \frac{q'_0}{q_0} - \frac{q'_h}{q_h}, \quad \text{div } \Lambda_h = \text{div } \xi - \text{div } q_h. \quad (\text{HEB27})$$

Logarithmic differentiation proves the first identity; factoring at each point proves the second, including cancellations when a zero of q_h coincides with a zero of ξ . One must keep those cancellations; at such a parameter, the factor trace (HEB24) is not a list of uncanceled poles of Λ_h .

Here is the exact effect on Weil's formula. Write its original decomposition as

$$W(f, g) = P_0(f, g) + \mathcal{A}(f, g) - \mathcal{P}(f, g), \quad (\text{HEB28})$$

where $\mathcal{A}$ is the original archimedean term and $\mathcal{P}$ the original rational-prime term. Define the archimedean term for the modified completion by

$$\mathcal{A}_h = \mathcal{A} + P_0 - P_h. \quad (\text{HEB29})$$

Then exactly

$$W = P_h + \mathcal{A}_h - \mathcal{P}. \quad (\text{HEB30})$$

This is both an algebraic identity of forms and the residue identity supplied by the rational logarithmic correction in (HEB27). Indeed q'_j/q_j has residue equal to the root multiplicity at every root of q_j , so pairing these finite divisors with $M_f^\# M_g$ gives $P_0 - P_h$. The remaining zeta factor and its prime logarithmic derivative are unchanged. Thus the sign change of the endpoint receiver can be made exact while preserving the complete original zero form W , but it is accompanied by the equally exact correction (HEB29). Positivity of (HEB25) alone is not positivity of the complete form (HEB30).

For clarity, Λ_h in (HEB26) is a rationally modified completion of the original ζ . It is not the Hurwitz family $\zeta(s, 1+t)$ and it is not the de Bruijn–Newman heat evolution of ξ . Each has its own proved map and parameter. Replacing any of them by (HEB26) without its correction would change the calculation.

6. The integral coefficient before quadratic base change

Before choosing the heat clock, the integral algebra is

$$E_{\mathbb{Z}} = \mathbb{Z}[u, r]/(r^2 - u), \quad \sigma(r) = -r, \quad \sigma(u) = u. \quad (\text{HEB31})$$

The geometric substitutions are $u = -a$ and, over $\mathbb{Z}[1/2]$, $u = 1/4 - 2h$. The ramified coefficient family used for the reflected analytic pair is the exact base change $u \mapsto d^2$, $r \mapsto w$. The image has $u \geq 0$ when d is real. The original heat line instead has access to both signs of u . These are different maps from the same coefficient space; neither map is omitted.

For the integral Frobenius lift $\Phi_p(u) = u^p$, $\Phi_p(r) = r^p$, coefficient integers fixed, the relation is preserved and the reduction modulo p is Frobenius. The ring is torsionfree, so $\delta_p(x) = (\Phi_p(x) - x^p)/p$ is defined. Direct computation on the generator at $p = 2$ gives

$$(\sigma\Phi_2 - \Phi_2\sigma)(r) = 2u, \quad (\sigma\delta_2 - \delta_2\sigma)(r) = u, \quad q_u(r, r) = -2u. \quad (\text{HEB32})$$

Thus the divided arithmetic defect is the same coefficient whose sign controls the real reflected form, through explicit integral and complex realizations. In the collision quotient $u = 0$, r survives with $r^2 = 0$. There is no claim that a p -adic coefficient has an intrinsic real sign, or that quotienting by u chooses the positive real half-line. The complete quotient, conormal and prism maps are calculated in WRF; (HEB31)–(HEB32) give the exact common source needed here.

The accompanying `check_endpoint_heat_bridge.py` verifies the original three images, the inverse chart, source reflection, matrix isometry, and rational logarithmic correction using exact rational symbolic arithmetic. Its finite verification scope is recorded in `ENDPOINT_HEAT_BRIDGE_CHECKS.json`. All arguments above are supplied independently of that check.

The earlier first-order vanishing in the ramified observed family also has an exact clock map. Under $u = d^2$ and $u = 1/4 - 2h$,

$$h = \frac{1}{8} - \frac{d^2}{2}, \quad r^2 = d^2, \quad q(r, r) = -2d^2. \quad (\text{HEB32a})$$

Thus modulo d^2 the algebra is the constant dual-number algebra and this paired value vanishes, while its coefficient of d^2 is exactly -2 . The first derivative of the displayed heat clock in d is zero at zero and its second derivative is -1 . This is why first-order vanishing in d does not itself choose the positive side of the heat clock: real d maps to $h \leq 1/8$. The original real heat parameter and this quadratic pullback are both retained through the displayed polynomial map.

7. An admissible test through the vanishing infinitesimal and positive quarter

The sign comparison can be evaluated on an actual compactly supported smooth test. Let

$$\psi(v) = C \begin{cases} \exp(-1/(1-v^2)), & |v| < 1, \\ 0, & |v| \geq 1, \end{cases} \quad \int_{\mathbb{R}} \psi(v) dv = 1, \quad f = \psi'. \quad (\text{HEB33})$$

The constant C is the reciprocal of the positive finite integral of the displayed function. The function is even, nonnegative, smooth and compactly supported: each one-sided derivative at $|v| = 1$ is an exponential of a negative reciprocal times a polynomial in reciprocal powers, hence tends to zero. This last assertion follows from $t^{-m}e^{-1/t} \rightarrow 0$ for every nonnegative integer m as $t \downarrow 0$, proved by the exponential series bound $e^{1/t} \geq t^{-(m+1)}/(m+1)!$. Thus f is an admissible real odd test in (HEB11).

Define the entire function $B(t) = \int \psi(v)e^{-tv} dv$. Entirety and differentiation under the integral follow uniformly on every compact set of t from $|v| \leq 1$. Integration by parts, with zero boundary values, gives

$$M_f(s) = (s - 1/2)B(s - 1/2), \quad B(-t) = B(t), \quad B(0) = 1. \quad (\text{HEB34})$$

The first equality uses the exact Fourier convention (HEB11); the second follows by $v \mapsto -v$. In particular $M_f(1/2) = 0$ and $M'_f(1/2) = 1$. Hence its class at the collision is precisely the nonzero class r in $\mathbb{C}[r]/r^2$. Its value vanishes, its first derivative survives, and its reflected regular trace pairing is zero, by (HEB16).

For every real a , introduce the entire coefficient function

$$b(a) = \sum_{k=0}^{\infty} \frac{(-a)^k}{(2k)!} \int \psi(v)v^{2k} dv. \quad (\text{HEB35})$$

The moments have absolute value at most one, so the series converges uniformly on bounded subsets of a . The even series of B and the relation $r^2 = -a$ give exactly $[M_f] = b(a)r$ in E_a , including its double-root class at zero. Equivalently $b(a) = \int \psi(v) \cos(\sqrt{a}v) dv$ for $a \geq 0$ and $b(a) = \int \psi(v) \cosh(\sqrt{-a}v) dv$ for $a \leq 0$. The exponential series and uniform convergence on $|v| \leq 1$ prove both formulas, with no choice of complex square root needed.

Put $b_0 = b(-1/4) = \int \psi(v) \cosh(v/2) dv$ and $b_1 = b(1/4) = \int \psi(v) \cos(v/2) dv$. Both are strictly positive: the integrands are nonnegative, and $\cos(v/2) \geq \cos(1/2) > 0$ on the support. Formula (HEB16) consequently gives the exact values and the actual compensation

$$\begin{aligned} P_0(f, f) &= -\frac{1}{2}b_0^2 < 0, \\ P_{1/8}(f, f) &= 0, \\ P_{1/4}(f, f) &= \frac{1}{2}b_1^2 > 0, \\ (\mathcal{A}_{1/4} - \mathcal{A})(f, f) &= -\frac{1}{2}(b_0^2 + b_1^2) < 0. \end{aligned} \quad (\text{HEB36})$$

The fourth equality follows from (HEB29), and cancels exactly the gain $P_{1/4}(f, f) - P_0(f, f)$. All values belong to the same admissible test f , without replacing it by an arbitrary endpoint vector or a polynomial outside the test space. This proves both the actual sign crossing and its complete explicit-formula compensation. It does not assign a sign to the remaining total $W(f, f)$: that total is unchanged by (HEB30).

The zero-prime endpoint form through a nilpotent collision

23 September 2026. This note starts with the specified endpoint contribution $W_0(a, b) = 2 \operatorname{Re}(a\bar{b})$. It constructs the exact double-root family carrying that form, computes its sign on every real fibre, proves what a nilpotent thickening can retain, and constructs the positive correction by reflection. The calculation concerns this endpoint contribution. It does not identify the reflected endpoint contribution with the full corrected Weil distribution.

The original observed-family coordinates used in Section 8 are those of the complete escaping-to-observed comparison, EFI51–EFI63. Their operator/metric distinctions are proved in the original metric comparison, ISP45–ISP53, which was read for this calculation. Every finite algebra and form used below is defined and calculated here.

1. The endpoint form and its negative direction

Let $V = \mathbb{C}^2$, with coordinates $x = (a, b)$, coordinatewise multiplication, and unit $(1, 1)$. Define the antilinear algebra involution and linear trace

$$x^\# = (\bar{b}, \bar{a}), \quad \operatorname{tr}(a, b) = a + b. \quad (\text{PZS1})$$

Applying the involution twice fixes each coordinate; it preserves products because multiplication is coordinatewise. Its trace-Hermitian form is

$$W_0(x, y) = \operatorname{tr}(x^\# y) = \bar{b} a' + \bar{a} b', \quad y = (a', b'). \quad (\text{PZS2})$$

It is conjugate-linear in its first entry, linear in its second, and Hermitian by direct conjugation and reordering of the two summands. Its matrix in the coordinate basis is

$$J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad W_0(x, y) = x^* J y. \quad (\text{PZS3})$$

In particular

$$W_0(x, x) = 2 \operatorname{Re}(a\bar{b}) = \frac{1}{2}(|a + b|^2 - |a - b|^2). \quad (\text{PZS4})$$

Expansion of the absolute squares proves the last identity. The vectors $(1, 1)/\sqrt{2}$ and $(1, -1)/\sqrt{2}$ are orthonormal eigenvectors of J with eigenvalues $+1, -1$; hence the form has inertia $(1, 1, 0)$, in the order positive, negative, radical dimensions. This keeps the whole negative direction, rather than only its numerical value.

2. A flat algebra family with the required involution

For each real number u , define

$$A_u = \mathbb{C}[w]/(w^2 - u), \quad (c + dw)^\# = \bar{c} - \bar{d}w. \quad (\text{PZS5})$$

The involution is well-defined because $(w^2 - u)^\# = w^2 - u$ for real u . It is multiplicative, antilinear, and has square the identity. The complete family is

$$\mathcal{A} = \mathbb{C}[u, w]/(w^2 - u), \quad \mathbb{C}[u] \longrightarrow \mathcal{A}, \quad u \longmapsto w^2. \quad (\text{PZS6})$$

Monic division gives the free $\mathbb{C}[u]$ -basis $1, w$; therefore the family is flat of rank two, including at $u = 0$. Its fibre is the stated quotient A_u . The family has the coefficient-conjugating involution fixing u and sending w to $-w$; its real fibres inherit PZS5.

In basis $1, w$, multiplication by $c + dw$ has matrix

$$M_{c+dw} = \begin{pmatrix} c & ud \\ d & c \end{pmatrix}. \quad (\text{PZS7})$$

Its regular trace is $2c$. Thus for $p = c + dw$, $q = c' + d'w$,

$$\begin{aligned} q_u(p, q) &:= \operatorname{Tr}_{A_u/\mathbb{C}}(M_{p^\# q}) \\ &= 2\bar{c}c' - 2u\bar{d}d', \end{aligned} \quad [q_u]_{(1, w)} = \begin{pmatrix} 2 & 0 \\ 0 & -2u \end{pmatrix}. \quad (\text{PZS8})$$

To verify this, expand $(\bar{c} - \bar{d}w)(c' + d'w)$, whose constant coefficient is $\bar{c}c' - u\bar{d}d'$, and apply PZS7. The displayed matrix also proves Hermitian symmetry.

3. The original endpoints occur at the exact fibre $u = 1/4$

Evaluation at the two roots gives an algebra map

$$E : A_{1/4} \longrightarrow V, \quad p \longmapsto (p(-1/2), p(1/2)). \quad (\text{PZS9})$$

Its explicit inverse is

$$E^{-1}(a, b) = \frac{a+b}{2} + (b-a)w. \quad (\text{PZS10})$$

Substitution verifies both endpoint values. Conversely, for $p = c + dw$, these values are $(c - d/2, c + d/2)$, and PZS10 recovers c, d . This proves the algebra isomorphism, since evaluation is multiplicative.

One has the commuting identities

$$E(p^\#) = E(p)^\#, \quad \text{Tr}_{A_{1/4}}(M_p) = \text{tr}(E(p)), \quad q_{1/4}(p, q) = W_0(E(p), E(q)). \quad (\text{PZS11})$$

The first follows because negating the real roots interchanges them; the second follows by summing $c - d/2$ and $c + d/2$; the third follows from the first two and multiplicativity. It also follows directly by substituting PZS10 into PZS8:

$$q_{1/4}(E^{-1}x, E^{-1}x) = \frac{1}{2}|a+b|^2 - \frac{1}{2}|b-a|^2 = W_0(x, x). \quad (\text{PZS12})$$

Thus the constant, variable, involution, and endpoint ordering are all fixed by exact maps.

4. The sign changes through the actual collision

The matrix PZS8 gives the full list

$$\text{inertia}(q_u) = \begin{cases} (1, 1, 0), & u > 0, \\ (1, 0, 1), & u = 0, \\ (2, 0, 0), & u < 0. \end{cases} \quad (\text{PZS13})$$

For $u \neq 0$, the two complex roots of $w^2 - u$ are distinct, each with multiplicity one. For $u = 0$, the fibre is

$$A_0 = \mathbb{C}[w]/w^2, \quad w \neq 0, \quad \text{rad } q_0 = \mathbb{C}w. \quad (\text{PZS14})$$

Here $w \neq 0$ follows from the free basis, and the radical follows from the diagonal matrix. The collision has length two; no multiplicity is removed when its two points merge.

For $u > 0$, put $r = \sqrt{u} > 0$. Evaluation at $-r, r$ identifies A_u with two coordinate factors, and the involution interchanges their conjugates. Its trace form is therefore the same hyperbolic endpoint form as PZS2. For $u < 0$, put $r = \sqrt{-u} > 0$. Evaluation at $-ir, ir$ instead has

$$(p^\#)(\pm ir) = \overline{p(\pm ir)}, \quad q_u(p, p) = |p(-ir)|^2 + |p(ir)|^2. \quad (\text{PZS15})$$

Indeed $(p^\#)(z) = \overline{p(-\bar{z})}$, and each imaginary root satisfies $-\bar{z} = z$. The sum of the values still equals the regular trace. Evaluation is invertible because the roots are distinct. This proves directly why the involution and sign change together while the total length remains two.

There is an explicit continuing negative vector: the polynomial $p = -2w$ corresponds at $u = 1/4$ to $(a, b) = (1, -1)$, and

$$q_u(-2w, -2w) = -8u. \quad (\text{PZS16})$$

It is negative for $u > 0$, zero for $u = 0$, and positive for $u < 0$. The fixed free module $\mathbb{C}[u] \oplus \mathbb{C}[u]w$ supplies this precise continuation through the collision. A sign change is therefore possible in the real family. The nilpotent fibre itself makes this direction radical; positivity of that direction occurs after continuing to the other side.

The two signs are not connected by an involution-preserving isomorphism of the endpoint algebras. For $u > 0$, the two nontrivial primitive idempotents $(1 \pm w/\sqrt{u})/2$ are exchanged by $\#$. For $u < 0$, the two primitive idempotents associated with the imaginary roots are each fixed by $\#$, because evaluation has componentwise conjugation by PZS15. An algebra isomorphism permutes the two primitive idempotents, so cannot change an exchanged pair into a fixed pair while preserving the involution. There are nevertheless explicit algebra isomorphisms: for $u > 0, v < 0$,

$$A_u \longrightarrow A_v, \quad w \longmapsto i\sqrt{u/(-v)}w', \quad (\text{PZS17})$$

with inverse multiplication by the reciprocal coefficient. Its square is u because $(w')^2 = v$. Its precise involution defect on w is

$$\phi(w^\#) - \phi(w)^\# = -2i\sqrt{u/(-v)}w' \neq 0. \quad (\text{PZS18})$$

This proves both the relationship and the information not preserved by that isomorphism.

5. A nilpotent thickening of the original endpoints keeps their negative residue

Let R be a commutative real algebra with a nilpotent ideal I and a fixed isomorphism $R/I \cong \mathbb{R}$. Extend conjugation to $R_{\mathbb{C}} = R \otimes_{\mathbb{R}} \mathbb{C}$ by fixing R . Let $\tilde{V} = R_{\mathbb{C}}^2$ and let $\tilde{G}$ be any Hermitian matrix whose reduction modulo I is J of PZS3. This includes every matrix deformation of the original endpoint pairing over this nilpotent base.

For every lift $\tilde{x}$ of $(1, -1)$, reduction gives

$$\tilde{x}^* \tilde{G} \tilde{x} \pmod{I} = -2. \quad (\text{PZS19})$$

This follows by applying the quotient homomorphism to the entire matrix product. On the other hand every element of the Hermitian-square cone

$$\Sigma_R = \left\{ \sum_{j=1}^n z_j^* z_j : n \geq 0, z_j \in R_{\mathbb{C}} \right\} \subset R \quad (\text{PZS20})$$

has nonnegative residue, because its residue is $\sum_j |\pi(z_j)|^2$, where $\pi : R_{\mathbb{C}} \rightarrow \mathbb{C}$ is the quotient homomorphism. Therefore the value PZS19 is outside Σ_R , for every possible nilpotent correction. No deformation of the pairing reducing to the original endpoint form can make all its values Hermitian-square-positive.

This result also has a scalar receiver: compose the deformed pairing with $R \rightarrow \mathbb{R}$. The resulting scalar form is exactly W_0 and retains value -2 on every such lift. Thus the negative direction cannot be removed by adding terms that vanish under this residue map. Crossing the family from $u = 1/4$ to $u < 0$ in Section 4 changes the residue parameter; it is not a nilpotent thickening with residue $u = 1/4$.

6. Positive functionals at the collision and their exact radical

On the involutive algebra A_0 , call a complex-linear functional ℓ positive when $\ell(p^\# p)$ is a nonnegative real number for every p . Then all such functionals are exactly

$$\ell(c + dw) = tc, \quad t \in \mathbb{R}_{\geq 0}. \quad (\text{PZS21})$$

Here is a direct proof. Positivity at $p = 1$ gives $t = \ell(1) \geq 0$. For $p = 1 + dw$, the product is $p^\# p = 1 + (d - \bar{d})w$. With $d = iy$, $y \in \mathbb{R}$, positivity requires $t + 2iy\ell(w)$ to be real and nonnegative for every real y . Its slope must be real because the whole expression is real, and must be zero because otherwise it takes negative values for one sign of sufficiently large y . Hence $\ell(w) = 0$. Linearity proves the asserted form. Conversely $\ell(p^\# p) = t|c|^2 \geq 0$, proving sufficiency.

For $t > 0$, the induced form $\ell(p^\# q)$ has radical exactly $\mathbb{C}w$; for $t = 0$, its radical is all of A_0 . Thus every positive functional kills the nonzero infinitesimal direction, with the explicit augmentation map

$$\epsilon : A_0 \rightarrow \mathbb{C}, \quad c + dw \mapsto c, \quad \ker \epsilon = \mathbb{C}w, \quad \ell = t\epsilon. \quad (\text{PZS22})$$

The scalar section $c \mapsto c$ splits this map and leaves its square-zero kernel specified.

More generally, let B be a positive semidefinite Hermitian form on the regular vector space of A_0 , compatible with the specified multiplication adjoints:

$$B(M_a x, y) = B(x, M_{a^\#} y). \quad (\text{PZS23})$$

Then $B(w, w) = B(1, -w^2) = 0$. A zero-norm vector of a positive semidefinite form lies in its radical: expanding $B(x + tz, x + tz) \geq 0$ with $B(z, z) = 0$ and varying complex t forces $B(x, z) = 0$. Apply this to $z = w$. Thus compatible positivity cannot give a positive norm to the nilpotent regular generator. A positive norm on that generator requires a changed adjoint relation, whose concrete original-observation example is ISP51–ISP52.

7. An explicit positive correction on the fixed endpoint space

The linear involution $J(a, b) = (b, a)$ is also an algebra automorphism of the endpoint algebra. Define

$$W_+(x, y) = W_0(x, Jy). \quad (\text{PZS24})$$

Using $J^2 = I$ in PZS3 gives

$$W_+(x, y) = x^* y, \quad W_+(x, x) = |a|^2 + |b|^2. \quad (\text{PZS25})$$

Thus it is positive definite. Its exact difference from the original form is the positive rank-one form

$$W_+(x, y) - W_0(x, y) = \overline{(a - b)}(a' - b'), \quad W_+(x, x) - W_0(x, x) = |a - b|^2. \quad (\text{PZS26})$$

Expansion proves these formulas. The correction has kernel $\{(a, a) : a \in \mathbb{C}\}$ and takes value 4 on $(1, -1)$, changing the old value -2 to $+2$. This is an explicit way to flip the negative direction while retaining its removed contribution as a known form.

The same change can be expressed as a changed algebra involution. Put

$$x^\star = J(x^\#) = (\bar{a}, \bar{b}). \quad (\text{PZS27})$$

Then $\text{tr}(x^\star y) = W_+(x, y)$. The original involution $\#$ and the new $\star$ are related by the exact automorphism J , but are not equal. On $A_{1/4}$, the automorphism is the sheet map $w \mapsto -w$. Under PZS27 the generator satisfies $w^\star = w$; under the original involution it satisfies $w^\# = -w$. This records the data altered by the correction.

The endpoint multiplication operator corresponding to the coordinate s is

$$S = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \quad (\text{PZS28})$$

Its adjoint for W_0 is

$$S^{\dagger_0} = JS^\star J = I - S, \quad W_0(Sx, y) = W_0(x, (I - S)y), \quad (\text{PZS29})$$

whereas its adjoint for W_+ is $S^{\dagger_+} = S$. These follow by the displayed matrix products. Also $JSJ = I - S$, proving the exact operator intertwining with the sheet switch.

Consequently the centered Fourier coordinate $Y = (S - I/2)/i$ is self-adjoint for W_0 , but skew-adjoint for W_+ :

$$Y^{\dagger_0} = Y, \quad Y^{\dagger_+} = -Y. \quad (\text{PZS30})$$

The sign in the first assertion follows because adjoints conjugate $1/i$ and replace $S - I/2$ by its negative. The second uses the ordinary adjoint of the real diagonal matrix $S - I/2$. Thus positivity of the corrected form does not retain the old adjoint identity for this operator.

Let $\mathcal{P} = \mathbb{C}[s]$, let $\text{ev}_{0,1}(f) = (f(0), f(1))$, and define the reflection automorphism $(Tf)(s) = f(1 - s)$. Then

$$\text{ev}_{0,1} T = J \text{ev}_{0,1}, \quad T^2 = I, \quad TM_s = (I - M_s)T. \quad (\text{PZS31})$$

Each identity follows by evaluation or direct substitution. Therefore

$$W_0(\text{ev}_{0,1} f, \text{ev}_{0,1}(Tg)) = \overline{f(0)}g(0) + \overline{f(1)}g(1). \quad (\text{PZS32})$$

This proves the positive correction as an exact reflected-test map for the endpoint contribution alone. Applying T to a test in a full analytic explicit formula also changes its other terms; PZS32 does not assert their invariance or positivity.

8. The observed quadratic pullback selects a side and an involution

Retain the observed family

$$B_\lambda = \mathbb{C}[z]/(z^2 - \lambda z), \quad \lambda = a_{\text{obs}}(s - s_*), \quad a_{\text{obs}} > 0. \quad (\text{PZS33})$$

For a real displacement $s - s_*$, $\lambda \in \mathbb{R}$. The exact coordinate map is

$$w = z - \lambda/2, \quad u = \lambda^2/4, \quad w^2 - u = z^2 - \lambda z. \quad (\text{PZS34})$$

It identifies B_λ with $A_{\lambda^2/4}$. The inverse is $z = w + \lambda/2$. Under the endpoint involution $w^\# = -w$, one has

$$z^\# = \lambda - z, \quad [q_\lambda^-]_{(1,w)} = \begin{pmatrix} 2 & 0 \\ 0 & -\lambda^2/2 \end{pmatrix}, \quad [q_\lambda^-]_{(1,z)} = \begin{pmatrix} 2 & \lambda \\ \lambda & 0 \end{pmatrix}. \quad (\text{PZS35})$$

For the second matrix, trace of z is λ and $z^\# z = (\lambda - z)z = 0$. Its determinant is $-\lambda^2$. Thus along this original real observed path, the endpoint form is indefinite for every $\lambda \neq 0$, and becomes radical at $\lambda = 0$. The base map $u = \lambda^2/4$ stays on $u \geq 0$ and cannot reach the positive side $u < 0$ in PZS13.

The positive escaping trace previously calculated in EFI61–EFI62 and ISP45 instead uses the coefficient-conjugating involution $w^\star = w$, equivalently $z^\star = z$. With that involution its matrices are

$$[q_\lambda^+]_{(1,w)} = \begin{pmatrix} 2 & 0 \\ 0 & \lambda^2/2 \end{pmatrix}, \quad [q_\lambda^+]_{(1,z)} = \begin{pmatrix} 2 & \lambda \\ \lambda & \lambda^2 \end{pmatrix}. \quad (\text{PZS36})$$

This is the same algebra and the same regular trace, with the different stated involution. Their connecting automorphism is

$$\sigma(w) = -w, \quad \sigma(z) = \lambda - z, \quad \star = \sigma \circ \#, \quad q_\lambda^+(p, q) = q_\lambda^-(p, \sigma q). \quad (\text{PZS37})$$

To prove the form identity, regular trace is invariant under σ : conjugation by an algebra automorphism sends a multiplication matrix to a similar matrix. Hence $\text{Tr}(p^\# \sigma q) = \text{Tr}(\sigma(p^\#) q) = \text{Tr}(p^\star q)$. One can also verify it using

$$[\sigma]_{(1,z)} = \begin{pmatrix} 1 & \lambda \\ 0 & -1 \end{pmatrix}, \quad [q_\lambda^+] - [q_\lambda^-] = \begin{pmatrix} 0 & 0 \\ 0 & \lambda^2 \end{pmatrix}. \quad (\text{PZS38})$$

Thus the positive escaping trace has an exact relationship to the original negative endpoint form: it performs the fundamental-symmetry correction on the same fibres. This is not evidence that the original endpoint involution already had the positive trace.

At $\lambda = 0$, both trace forms have matrix $\text{diag}(2, 0)$, although their involutions still differ on the nonzero nilpotent generator. For $\lambda \neq 0$, their signs differ exactly as PZS35–PZS36 calculate. This specifies the coincidence at the collision and the data that remain distinct there.

There is also a real obstruction to reaching the positive side through this fixed observed base map, valid over any real nilpotent parameter algebra with residue field $\mathbb{R}$: the residue of $\lambda^2/4$ is nonnegative. To access $u < 0$, one must change the real base path, or change the involution as in PZS37. Both constructions are explicitly available above, and neither is identified with a nilpotent correction leaving the original endpoint form unchanged.

9. What the calculation establishes for the proposed sign flip

The free family PZS6 and its original involution provide an actual passage from a negative direction, through a nonzero trace-null dual-number direction, to a positive direction, as PZS13–PZS16 prove. The original observed quadratic base map restricts this family to its nonnegative- u side. Its previously positive trace uses the alternative involution, connected by PZS37. On the fixed endpoint space, PZS24–PZS32 construct the positive correction and identify exactly its rank-one contribution and altered multiplication adjoint. A nilpotent thickening that reduces to the original endpoint pairing retains its negative residue by PZS19; at the collision, every positive functional has the explicit radical and augmentation in PZS21–PZS23.

These are complete calculations of the proposed endpoint mechanism. They retain the nilpotent, the full multiplicity, the sign-changing continuation, and the correction maps. The analytic effect of applying the same reflection or continuation to every other term of the corrected Weil formula is a separate calculation on those actual terms; no endpoint-only identity above asserts that effect.

10. Exact verification

The reproducible script `check_prime_zero_sign_deformation.py` passed 16 exact checks, recorded in `PRIME_ZERO_SIGN_DEFORMATION_CHECKS.json`. They verify the endpoint interpolation and its inverse, the involution and trace isometries, the continuing negative vector, the rank-one fundamental-symmetry correction, both endpoint multiplication adjoints, the two observed-coordinate trace forms and their sheet-switch relation, and preservation of a negative residue under a specified matrix deformation with nilpotent parameters. These checks use symbolic arithmetic. They support the calculations but do not replace the proofs for arbitrary parameters and nilpotent bases, and do not assert positivity of the full Weil formula.

The ES quarter spectrum, its tetrahedral metric, and the original heat trace

This note starts with the actual four raw ES cells and their quarter-inversion operator. It constructs a form-preserving linear isomorphism with two positive scalar directions and the reflected trace of the original quadratic heat family. It proves the transported symmetry actions, the precise multiplication maps, and the surviving collision algebra. The ES form is not presumed to be the complete Weil form.

The original source read for this calculation is `ES-Fable-C123/es_fable_c123_preprint.tex`: the definition `eq:Hermitian-circle-coefficients`; Corollaries `cor:raw-blade-cells-quarter-three-plus-one` and the following raw-cell projector corollary, including `eq:four-cell-explicit-quarter-images`, `eq:raw-negative-cell-projector`, `eq:raw-positive-cell-projector`; Theorem `thm:Fable-D3-raw-cell-action`; Corollary `cor:Fable-three-character-tetrahedron`; Theorem `thm:Fable-forced-regular-four-simplex`; and `eq:quarter-inversion-matrix` through `eq:quarter-negative-eigenline`. These are actual passages of the source, rather than identifications inferred from a four-element count. The receiving programme proof read in full is the complete heat endpoint sign bridge proof, HEB1–HEB32. The original heat equations also have the public proof *The original heat trajectories*, `eq:ifh-target-flow` and the ensuing trajectories, while EFI7–EFI15 give their projective graph and collision algebra. All finite-dimensional claims made below have their proofs here.

1. The source operator and its complete intertwiner

Let

$$V = M_2(\mathbb{R}), \quad x = x_{11}E_{11} + x_{12}E_{12} + x_{21}E_{21} + x_{22}E_{22}, \quad (\text{ESQ1})$$

with the displayed ordering of the four basis vectors. We retain both the vector space and its original matrix multiplication. The symbol $I_2 = E_{11} + E_{22}$ denotes the matrix identity, whereas I_V denotes the identity linear operator on the four-dimensional vector space.

The quarter-inversion operator pulled back from the ES source is

$$A_0 = \text{diag}(1/4, 1/4, -1/4, 1/4). \quad (\text{ESQ2})$$

Its two invariant spaces and its negative projector are

$$U_+ = \text{span}_{\mathbb{R}}\{E_{11}, E_{12}, E_{22}\}, \quad U_- = \mathbb{R}E_{21}, \quad P_-x = x_{21}E_{21} = \left(\frac{1}{2}I_V - 2A_0\right)x. \quad (\text{ESQ3})$$

These formulas follow by evaluating the diagonal operator on all four basis vectors. In particular $P_-^2 = P_-$, its kernel is U_+ , and its image is U_- .

Here is the full source intertwiner, including its inverse. In Hermitian coefficient coordinates write

$$H(\alpha, \beta_0, \beta_1, \gamma) = \begin{pmatrix} \alpha & \beta_0 + i\beta_1 \\ \beta_0 - i\beta_1 & \gamma \end{pmatrix}. \quad (\text{ESQ4})$$

The coefficient vector of $\mathcal{T}_{\square}x$ is

$$\begin{aligned} \alpha &= x_{11} + 2x_{21} + x_{22}, & \beta_0 &= (x_{11} - x_{22})/2, \\ \beta_1 &= x_{12}, & \gamma &= (x_{11} + x_{22})/4 - x_{21}/2. \end{aligned} \quad (\text{ESQ5})$$

This is exactly the linear extension of the four source images. Its inverse is

$$x_{11} = \alpha/4 + \gamma + \beta_0, \quad x_{22} = \alpha/4 + \gamma - \beta_0, \quad x_{12} = \beta_1, \quad x_{21} = \alpha/4 - \gamma. \quad (\text{ESQ6})$$

Substitution in either direction returns all four starting coefficients, proving that it is an isomorphism of real vector spaces. The source coefficient inversion is

$$\mathcal{I}_- = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1/4 & 0 & 0 \\ 0 & 0 & 1/4 & 0 \\ 1/16 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{ESQ7})$$

Applying this matrix to the four columns of (ESQ5) multiplies the E_{21} column by $-1/4$ and each other column by $1/4$. Since the four columns form a basis, this proves the entire operator identity

$$\mathcal{T}_{\square}A_0 = \mathcal{I}_-\mathcal{T}_{\square}. \quad (\text{ESQ8})$$

The negative line is therefore the specific raw E_{21} line, not an unspecified member of a quartet. In the source channel order $(++)$, $(+-)$, $(-+)$, $(--)$, it is the $(-, +)$ channel. The source specializations $\mathcal{M}(53) = 2E_{21}$ and $\mathcal{M}(109) = E_{12}$ retain their different eigenspaces under (ESQ8).

2. The tetrahedral metric and the rank-one heat motion

Retain the source's forced metric

$$G = \hat{G}_F = \begin{pmatrix} 1/16 & 0 & 0 & -7/144 \\ 0 & 2/9 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -7/144 & 0 & 0 & 1/16 \end{pmatrix}. \quad (\text{ESQ9})$$

On U_+ , the sum and difference of E_{11}, E_{22} have respective eigenvalues $1/72, 1/9$, and E_{12} has eigenvalue $2/9$. The orthogonal fourth direction E_{21} has eigenvalue 1. These four positive numbers prove that G is positive definite. Its tetrahedral origin is proved again in Section 7.

Use the original forward heat parameter with its original constants,

$$a(h) = -\frac{1}{4} + 2h, \quad h \in \mathbb{R}. \quad (\text{ESQ10})$$

Define the actual deformation of the source operator by

$$A_h = A_0 + 2hP_- = \frac{1}{4}(I_V - P_-) + a(h)P_-. \quad (\text{ESQ11})$$

It is G -self-adjoint: the decomposition $U_+ \oplus U_-$ is G -orthogonal, and the operator is scalar on each summand. Equivalently $A_h^\top G = GA_h$, as is also immediate from the displayed matrices. Consequently

$$B_h(x, y) = x^\top GA_h y \quad (\text{ESQ12})$$

is a symmetric bilinear form. Its full matrix and quadratic expression are

$$GA_h = \begin{pmatrix} 1/64 & 0 & 0 & -7/576 \\ 0 & 1/18 & 0 & 0 \\ 0 & 0 & a(h) & 0 \\ -7/576 & 0 & 0 & 1/64 \end{pmatrix}, \quad (\text{ESQ13})$$

$$B_h(x, x) = \frac{x_{11}^2 + x_{22}^2}{64} - \frac{7}{288}x_{11}x_{22} + \frac{x_{12}^2}{18} + a(h)x_{21}^2. \quad (\text{ESQ14})$$

Matrix multiplication proves every coefficient. The unchanged positive eigenvalues of this matrix are $1/288, 1/36, 1/18$, and its remaining eigenvalue is $a(h)$. Thus it has signature $(3, 1)$ for $h < 1/8$, has three positive directions and radical $\mathbb{R}E_{21}$ for $h = 1/8$, and is positive definite for $h > 1/8$. At $h = 1/4$, $A_h = I_V/4$ and $B_h = G/4$. The exact difference formula is

$$B_{h_2}(x, y) - B_{h_1}(x, y) = 2(h_2 - h_1)x_{21}y_{21}. \quad (\text{ESQ15})$$

This follows because $GP_- = P_-$. It proves that the entire changing part has rank one and is increasing as a quadratic form when heat time increases.

For the two marked source specializations this gives the exact values

$$B_h(\mathcal{M}(53), \mathcal{M}(53)) = 4a(h) = -1 + 8h, \quad B_h(\mathcal{M}(109), \mathcal{M}(109)) = 1/18. \quad (\text{ESQ15a})$$

They follow from $\mathcal{M}(53) = 2E_{21}$, $\mathcal{M}(109) = E_{12}$, and (ESQ13), retaining the factor two on the first source vector.

The original intertwiner also transports the entire heat family, not only its initial value. Define

$$\mathcal{I}_h = \mathcal{I}_- + 2h \left(\frac{1}{2}I - 2\mathcal{I}_- \right) = hI + (1 - 4h)\mathcal{I}_-. \quad (\text{ESQ16})$$

Substitution of (ESQ3) in (ESQ8) proves $\mathcal{T}_\square A_h = \mathcal{I}_h \mathcal{T}_\square$. Transport the metric by $G_T = (\mathcal{T}_\square^{-1})^\top G \mathcal{T}_\square^{-1}$. In the coordinates of (ESQ4) its quadratic expression is

$$G_T(y, y) = \frac{(\alpha + 4\gamma)^2}{576} + \frac{2}{9}(\beta_0^2 + \beta_1^2) + (\alpha/4 - \gamma)^2. \quad (\text{ESQ17})$$

This is obtained by inserting (ESQ6) in (ESQ9). It proves the metric transport exactly; no identification with the Lorentz form $-\det H$ is being substituted for it. Likewise the transported heat form is

$$B_{T,h}(y, y) = \frac{(\alpha + 4\gamma)^2}{2304} + \frac{\beta_0^2 + \beta_1^2}{18} + a(h)(\alpha/4 - \gamma)^2. \quad (\text{ESQ18})$$

3. The exact trace receiver and the form-preserving isomorphism

For real a , let

$$E_a = \mathbb{R}[r]/(r^2 + a). \quad (\text{ESQ19})$$

Every element has a unique expression $c + dr$, by division by the monic quadratic. The real algebra involution is $(c + dr)^\# = c - dr$. It preserves the relation and squares to the identity. The regular trace is the trace of multiplication on the basis $(1, r)$. Since

$$L_r = \begin{pmatrix} 0 & -a \\ 1 & 0 \end{pmatrix}, \quad \text{Tr}_{E_a}(c + dr) = 2c, \quad (\text{ESQ20})$$

the full reflected trace pairing is

$$\frac{1}{2} \text{Tr}_{E_a}((c + dr)^\#(e + fr)) = ce + adf. \quad (\text{ESQ21})$$

Indeed the product has scalar coefficient $ce + adf$, and (ESQ20) gives twice it. At complex coefficients the involution also conjugates c, d , and the same proof gives $\bar{c}e + a\bar{d}f$.

Define the real vector space

$$W_a = \mathbb{R}^2 \oplus E_a \quad (\text{ESQ22})$$

with symmetric form

$$Q_a((z_1, z_2, p), (t_1, t_2, q)) = z_1 t_1 + z_2 t_2 + \frac{1}{2} \text{Tr}_{E_a}(p^\# q). \quad (\text{ESQ23})$$

The map proposed for the raw ES cells is exactly

$$\begin{aligned} \Theta_h : V &\longrightarrow W_{a(h)}, \\ z_1 &= \frac{x_{11} + x_{22}}{24}, \quad z_2 = \frac{x_{11} - x_{22}}{6\sqrt{2}}, \\ p &= c + dr = \frac{x_{12}}{3\sqrt{2}} + x_{21}r. \end{aligned} \quad (\text{ESQ24})$$

It is a vector-space isomorphism for every h , including the collision. Its inverse is

$$x_{11} = 12z_1 + 3\sqrt{2}z_2, \quad x_{22} = 12z_1 - 3\sqrt{2}z_2, \quad x_{12} = 3\sqrt{2}c, \quad x_{21} = d. \quad (\text{ESQ25})$$

These formulas are mutually inverse by substitution, without dividing by a . They prove in particular that no vector is dropped when $a = 0$.

The exact isometry identity is

$$B_h(x, y) = Q_{a(h)}(\Theta_h x, \Theta_h y). \quad (\text{ESQ26})$$

For the diagonal terms, (ESQ21) gives $c^2 + ad^2 = x_{12}^2/18 + ax_{21}^2$. The remaining two squares are

$$\frac{(x_{11} + x_{22})^2}{576} + \frac{(x_{11} - x_{22})^2}{72} = \frac{x_{11}^2 + x_{22}^2}{64} - \frac{7}{288} x_{11} x_{22}. \quad (\text{ESQ27})$$

Expansion proves the equality with (ESQ14). For two arguments the identical expansion with products proves (ESQ26), or it follows by real polarization. Complex-linear extension gives the corresponding Hermitian identity, with all squared terms replaced by absolute squares and first coefficients conjugated.

The scalar factor one half in (ESQ23) is essential: HEB16 uses the full regular trace. More precisely, if the whole HEB rank-three form is

$$q_a^{\text{full}}(v, p) = |v|^2 + \text{Tr}_{E_a}(p^\# p),$$

then the explicit map

$$(z_1, z_2, p) \longmapsto (z_1, (z_2, p/\sqrt{2})) \quad (\text{ESQ28})$$

is an isometry from Q_a to one additional positive scalar form plus q_a^{full} . Its inverse multiplies the last coordinate by $\sqrt{2}$. This gives the exact relation with HEB's whole finite receiver while retaining the additional positive dimension.

The raw involution $4A_0 = I_V - 2P_-$ is also intertwined:

$$\Theta_h(4A_0 x) = (z_1, z_2, p^\#) \quad \text{over real coefficients.} \quad (\text{ESQ29})$$

Over complex coefficients the matching conjugate-linear raw involution is $(I_V - 2P_-)\bar{x}$. Each formula follows because exactly x_{21} changes sign, with coefficient conjugation added in the latter case. The heat operator itself becomes the diagonal map

$$(z_1, z_2, c, d) \mapsto (z_1/4, z_2/4, c/4, a(h)d). \quad (\text{ESQ30})$$

Under the same coordinates the fixed positive metric is $4|z_1|^2 + 4|z_2|^2 + 4|c|^2 + |d|^2$. This is obtained from G by (ESQ25), and together with (ESQ30) reproduces Q_a .

The trace-factor projection and its kernel are completely explicit:

$$0 \longrightarrow \text{span}\{E_{11}, E_{22}\} \longrightarrow V \xrightarrow{\pi_h} E_{a(h)} \longrightarrow 0, \quad \pi_h(x) = x_{12}/(3\sqrt{2}) + x_{21}r. \quad (\text{ESQ31})$$

It is onto with linear section $c + dr \mapsto 3\sqrt{2}cE_{12} + dE_{21}$, and its kernel follows from the independence of $1, r$. The kernel is orthogonal to that section for B_h , and carries the two positive squares (ESQ27). All assertions hold at the collision as well.

4. The endpoint algebra and the changing spectral pair

The exact all-parameter polynomial map to the HEB endpoint family is

$$\mathbb{R}[s]/(s(s-1)+2h) \xrightarrow{\sim} E_{a(h)}, \quad s \mapsto \frac{1}{2} + r, \quad \text{inverse: } r \mapsto s - \frac{1}{2}. \quad (\text{ESQ32})$$

The two relations correspond because $(s-1/2)^2 + a(h) = s(s-1) + 2h$. Substitution proves that the displayed maps are inverse unital algebra maps. The involution corresponds to $s^\# = 1-s$ over the reals, with conjugate coefficients in the complex version. Thus this is the actual HEB polynomial receiver, with the original heat speed 2 retained.

At $h = 0$, evaluation of $p = c + dr$ at $r = -1/2, +1/2$ gives

$$A = c - d/2, \quad B = c + d/2, \quad c = (A+B)/2, \quad d = B-A. \quad (\text{ESQ33})$$

The inverse formulas prove the algebra identification with the two endpoints. For two inputs the half-trace term becomes

$$\frac{1}{2} \text{Tr}(p^\# q) = \frac{1}{2} (\bar{A}_p B_q + \bar{B}_p A_q). \quad (\text{ESQ34})$$

To check the constants directly, substitute (ESQ33) into $\bar{c}e - \bar{d}f/4$ from (ESQ21); the diagonal endpoint terms cancel and the two cross terms have coefficient one half. Therefore the ES form is the two positive squares plus **one half** the HEB endpoint pairing, through the explicit invertible coordinate map, not through an assumed equality of names.

For all heat times the complex roots are

$$s_\pm(h) = \frac{1}{2} \pm \sqrt{\frac{1}{4} - 2h}. \quad (\text{ESQ35})$$

For $h > 1/8$, writing $t = \sqrt{2h - 1/4}$ gives $s_\pm = 1/2 \pm it$. The two evaluations of p are $c \pm itd$, and their squared absolute values have sum $2|c|^2 + 2a(h)|d|^2$. Thus the half-trace is half the sum of these two squares, exactly as in HEB25. At $h = 1/8$, evaluation alone sees the scalar coefficient c , while the quotient algebra still retains the independent derivative coefficient d . The relation is $r^2 = 0$, not $r = 0$.

This algebra and trace comparison does not replace the full Weil identity. HEB28–HEB30 retain its other terms when the endpoint factor is moved. No positivity claim for those other terms, or for all zeta zeros, is implied by (ESQ26) or (ESQ35).

5. Matrix multiplication and the exact collision algebra

There is no nonzero multiplicative ring map from $M_2(\mathbb{R})$ into a commutative ring. To prove this without invoking simplicity, let f be such a map. Commutativity of the target and the products $E_{12}E_{21} = E_{11}$, $E_{21}E_{12} = E_{22}$ give $f(E_{11}) = f(E_{22}) = e$. Since $E_{11}^2 = E_{11}$, one has $e^2 = e$; since $E_{11}E_{22} = 0$, one also has $e^2 = 0$. Therefore $e = 0$, and $f(I_2) = 0$. For every x , multiplicativity gives $f(x) = f(I_2x) = 0$. This proves the assertion and, in particular, excludes a unital algebra map from the full raw matrix algebra to $\mathbb{R} \times \mathbb{R} \times E_a$. Consequently (ESQ24), (ESQ28), and (ESQ31) have their stated linear meanings; their nonzero values could not obey that multiplication law.

There is nevertheless an exact algebra map in the opposite direction for the whole heat family:

$$\rho_a : E_a \hookrightarrow M_2(\mathbb{R}), \quad \rho_a(c + dr) = \begin{pmatrix} c & -ad \\ d & c \end{pmatrix} = cI_2 + dK_a, \quad K_a = E_{21} - aE_{12}. \quad (\text{ESQ36})$$

Since $K_a^2 = -aI_2$, the map respects the relation. Products of two displayed matrices have diagonal coefficient $ce - adf$ and lower-left coefficient $cf + de$, precisely the coefficients of $(c + dr)(e + fr)$. It is unital, and its diagonal and lower-left entries recover c, d , proving injectivity. This construction is already defined over $\mathbb{Z}[a]$; it uses no square roots and no division by the parameter.

Put $J = \text{diag}(1, -1)$. Direct multiplication gives

$$JK_a J^{-1} = -K_a, \quad \rho_a(p^\#) = \overline{J\rho_a(p)} J^{-1} \quad (\text{ESQ37})$$

in the complex version, with the bar omitted over the reals. The ordinary matrix trace of $\rho_a(p)$ is $2c$, hence exactly the regular trace (ESQ20). It follows, using the proved multiplicativity, that

$$\frac{1}{2} \text{tr} \left(\overline{J\rho_a(p)} J^{-1} \rho_a(q) \right) = \frac{1}{2} \text{Tr}_{E_a}(p^\# q). \quad (\text{ESQ38})$$

Thus the multiplication and reflected trace have an exact matrix realization, even though the linear isometry of Section 3 is not an algebra map.

The algebra involution in (ESQ37) is different from the raw linear reflection $\Omega_\square = 4A_0$, which changes the sign of E_{21} alone. The latter is not multiplicative on the entire matrix algebra: $\Omega_\square(E_{12}E_{21}) = E_{11}$, whereas $\Omega_\square(E_{12})\Omega_\square(E_{21}) = -E_{11}$. Their exact defect on the algebra embedding is

$$\Omega_\square \rho_a(c + dr) - \rho_a((c + dr)^\#) = -2adE_{12} \quad \text{over real coefficients.} \quad (\text{ESQ38a})$$

Indeed the left upper-right entry is $-ad$, and the subtracted entry is $+ad$; every other entry cancels. At complex coefficients, apply coefficient conjugation before $\Omega_\square$, and the corresponding right side is $-2adE_{12}$. At the collision this entire defect vanishes, so the raw reflection restricts there to the actual algebra involution of the collision corner. Away from the collision the reflected trace remains exactly represented by (ESQ37)–(ESQ38), with the precise difference from the raw reflection given here.

Its relation with the original ES metric is also calculable. Substituting $x_{11} = x_{22} = c$, $x_{12} = -ad$, $x_{21} = d$ into (ESQ14), and polarizing, gives

$$B_h(\rho_a(c + dr), \rho_a(e + fr)) = \frac{ce}{144} + \left(a + \frac{a^2}{18} \right) df, \quad a = a(h). \quad (\text{ESQ39})$$

The complex formula conjugates the first coefficients. This retains the exact difference between the algebra embedding and the form-preserving linear map; it does not identify their images without calculating the coefficients.

At the collision $a = 0$, $K_0 = E_{21}$, so (ESQ36) identifies the image with the actual unital matrix subalgebra

$$\mathbb{R}[I_2, E_{21}] = \{cI_2 + dE_{21} : c, d \in \mathbb{R}\} \xrightarrow{\sim} \mathbb{R}[r]/(r^2), \quad I_2 \mapsto 1, \quad E_{21} \mapsto r. \quad (\text{ESQ40})$$

Its inverse is ρ_0 . The multiplication law on both sides is $(c, d)(e, f) = (ce, cf + de)$; this proves every part of the algebra isomorphism and proves that the nilpotent remains nonzero. Complexification gives exactly $\mathbb{C}[I_2, E_{21}] \simeq \mathbb{C}[r]/(r^2)$. On this corner the ES collision form is $\bar{c}e/144$, whereas the half-trace form under (ESQ40) is $\bar{c}e$. The difference is the explicit scalar factor $1/144$. The global linear isometry (ESQ24) instead sends $cI_2 + dE_{21}$ to $(c/12, 0, dr)$, which confirms that its unit and nilpotent coordinates are distributed differently.

6. The native tetrahedral action and its trace-coordinate transport

The actual two ES generators are

$$R = \begin{pmatrix} 1/4 & \sqrt{3}/2 & 0 & 3/4 \\ -\sqrt{3}/4 & -1/2 & 0 & \sqrt{3}/4 \\ 0 & 0 & 1 & 0 \\ 3/4 & -\sqrt{3}/2 & 0 & 1/4 \end{pmatrix}, \quad J_\square = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{ESQ41})$$

Their third columns fix E_{21} , and their other columns preserve U_+ . They consequently commute with P_- and A_h for every h . For a full verification of their group and metric properties, put $(z_1, z_2, c, d) = \Theta_h x$ in the scalar basis of its target. Substitution of (ESQ41) into (ESQ24) gives

$$R : (z_1, z_2, c, d) \mapsto \left(z_1, -\frac{1}{2}z_2 + \frac{\sqrt{3}}{2}c, -\frac{\sqrt{3}}{2}z_2 - \frac{1}{2}c, d \right), \quad (\text{ESQ42})$$

$$J_\square : (z_1, z_2, c, d) \mapsto (z_1, -z_2, c, d). \quad (\text{ESQ43})$$

The two-by-two rotation in (ESQ42) has square obtained by changing both off-diagonal signs and has cube the identity. Conjugating it by the reflection of z_2 changes exactly those two signs, giving its inverse. Thus $R^3 = I$, $J_\square^2 = I$, and $J_\square R J_\square = R^{-1}$. The six maps $I, R, R^2, J_\square, R J_\square, R^2 J_\square$ are distinct: the first three restrict to distinct rotations of the (z_2, c) -plane with determinant 1, and the last three have determinant minus one and distinct restrictions. They are therefore precisely a D_3 action.

The same displayed rotation and reflection preserve $z_2^2 + c^2$, fixing z_1, d . They preserve both the positive metric in (ESQ30) and the trace-coordinate heat form $z_1^2 + z_2^2 + c^2 + ad^2$. Pulling back by the proved isomorphism gives

$$R^\top G R = G, \quad J_\square^\top G J_\square = G, \quad B_h(Rx, Ry) = B_h(x, y), \quad B_h(J_\square x, J_\square y) = B_h(x, y). \quad (\text{ESQ44})$$

This proves the full native symmetry of the deformation, not only preservation of its unique negative line.

These are linear symmetries of the metric and operator. They do not assert automorphisms of the full matrix algebra: for example the displayed matrix gives $(RE_{12})^2 = 3I_2/4$, although $E_{12}^2 = 0$. Nor does R preserve the integral raw-cell lattice, since its E_{12} column has coefficients $\sqrt{3}/2$. There is nevertheless an exact algebra-compatible restriction at the collision. Both generators fix $I_2 = E_{11} + E_{22}$, as summing their first and fourth columns proves, and both fix E_{21} . They therefore fix $\mathbb{R}[I_2, E_{21}]$ pointwise. The collision algebra isomorphism (ESQ40) is consequently equivariant for this native D_3 with the trivial action on $\mathbb{R}[r]/(r^2)$.

It also proves an exact qualification about the trace projection. The decomposition $\mathbb{R}^2 \oplus E_a$ selected by (ESQ24) is not preserved factor by factor under the native rotation: the scalar coefficient c of E_a mixes with z_2 . In fact the vector with coordinates $(0, 1, 0, 0)$ lies in $\ker \pi_h$, whereas its image under R has trace scalar coordinate $-\sqrt{3}/2$. Therefore no linear action on E_a alone can make this particular projection π_h equivariant under R . The full four-dimensional map Θ_h is equivariant with the explicit actions (ESQ42)–(ESQ43). This supplies the exact symmetry map instead of omitting the mixing.

7. The genuine regular tetrahedron and its full vertex symmetry

The three positive source rays and their forced fourth ray are

$$\begin{aligned} r_0 &= 2E_{11} + 2E_{12} + 2E_{22}, \\ r_+ &= (2 + \sqrt{3})E_{11} - E_{12} + (2 - \sqrt{3})E_{22}, \\ r_- &= (2 - \sqrt{3})E_{11} - E_{12} + (2 + \sqrt{3})E_{22}, \\ r_\infty &= -(r_0 + r_+ + r_-) = -6(E_{11} + E_{22}). \end{aligned} \quad (\text{ESQ45})$$

The source's original $3 + 1$ tetrahedron is $\Delta_F = \text{conv}\{E_{21}, r_0, r_+, r_-\}$, as stated in `cor:Table-D3-tetrahedral-realization`, which was read directly. It is distinct from the centered regular tetrahedron $\text{conv}\{r_\infty, r_0, r_+, r_-\}$ entirely inside U_+ . In the first tetrahedron, the three edges from E_{21} have squared G -length 2, while the opposite-face edges have squared length $8/3$, by orthogonality to E_{21} and the pairings proved below. Thus every vertex isometry of this first tetrahedron fixes E_{21} and permutes the other three vertices, giving exactly the native D_3 . No exchange with an apex is a further isometry of that tetrahedron. The second tetrahedron has the full geometric S_4 calculated below.

Their positive Euclidean coordinates $2(z_1, z_2, c)$ under (ESQ24) are respectively

$$\begin{aligned} v_0 &= (1/3, 0, 2\sqrt{2}/3), \\ v_+ &= (1/3, \sqrt{2/3}, -\sqrt{2}/3), \\ v_- &= (1/3, -\sqrt{2/3}, -\sqrt{2}/3), \\ v_\infty &= (-1, 0, 0). \end{aligned} \quad (\text{ESQ46})$$

Their norms are 1, and the dot product of any two different vectors is $-1/3$, by multiplication of these coordinates. For example the two products involving $v_0, v_\pm$ equal $1/9 - 4/9 = -1/3$, while $v_+ \cdot v_- =$

$1/9 - 2/3 + 2/9 = -1/3$; products with v_∞ are immediate. Since $G|_{U_+}$ is exactly the Euclidean form on these doubled coordinates by (ESQ30), this proves

$$\langle r_a, r_b \rangle_G = \begin{cases} 1 & a = b, \\ -1/3 & a \neq b, \end{cases} \quad a, b \in \{\infty, 0, +, -\}. \quad (\text{ESQ47})$$

The first three rays form a basis of U_+ : their coordinate matrix in (E_{11}, E_{12}, E_{22}) has determinant $-24\sqrt{3}$, obtained by subtracting its second column from its third and expanding. Therefore specifying their Gram matrix uniquely specifies the bilinear form. Since (ESQ9) has just been shown to give those entries, this proves the source metric's uniqueness as well as positivity and the regular-tetrahedron assertion.

Every permutation π of these four rays extends uniquely to a G -orthogonal linear operator S_π on U_+ . To see existence, the only linear dependence of the four rays is their sum zero: rank three gives a one-dimensional relation space, and this is a nonzero relation. A permutation preserves that relation, so induces an invertible map on their three-dimensional span. It preserves all the Gram entries (ESQ47), hence the form. Extend it to V by fixing E_{21} . The maps obey $S_\pi S_\rho = S_{\pi\rho}$ on the spanning vectors and are distinct on them, giving a faithful geometric S_4 action. They preserve G and commute with every A_h , since they preserve the two eigenspaces in (ESQ3).

The resulting map on every vector also has the finite formula

$$S_\pi = P_- + \frac{3}{4} \sum_{a \in \{\infty, 0, +, -\}} r_{\pi(a)} r_a^\top G. \quad (\text{ESQ47a})$$

It fixes E_{21} because the rays are orthogonal to that vector. On r_b , (ESQ47) makes the sum equal to $(3/4)(r_{\pi(b)} - (1/3) \sum_{a \neq b} r_{\pi(a)}) = r_{\pi(b)}$, using the zero sum of the rays. These spanning-vector computations prove the stated formula and all its prescribed values.

The native generators act on these rays by

$$R : r_0 \mapsto r_+ \mapsto r_- \mapsto r_0, \quad J_\square : r_0 \mapsto r_0, \quad r_+ \leftrightarrow r_-, \quad r_\infty \mapsto r_\infty. \quad (\text{ESQ48})$$

These follow either by substitution in (ESQ41) or by (ESQ42) on (ESQ46). Thus the native D_3 is exactly the stabilizer of r_∞ in this geometric tetrahedral S_4 .

For a fully explicit additional generator, put

$$K = \begin{pmatrix} 1/3 & -8/3 & 0 & -2/3 \\ -1/6 & 1/3 & 0 & -1/6 \\ 0 & 0 & 1 & 0 \\ -2/3 & -8/3 & 0 & 1/3 \end{pmatrix}. \quad (\text{ESQ49})$$

Let $v = r_0 - r_\infty = (8, 2, 0, 8)^\top$. From (ESQ47), $v^\top G v = 8/3$, and direct multiplication gives

$$K = I_V - \frac{3}{4} v v^\top G. \quad (\text{ESQ50})$$

For any x , this subtracts $2\langle v, x \rangle_G v / \langle v, v \rangle_G$; expansion of its inner product on two vectors cancels the two cross terms with the final quadratic term, proving it is a G -isometry. The formula sends r_0 to r_∞ , r_∞ to r_0 , and fixes $r_\pm, E_{21}$, since their pairings with v vanish. Hence K is the transposition extending the native stabilizer to the full geometric S_4 . Conjugating this transposition by the native three-cycle supplies the transpositions of r_∞ with each other vertex, and these generate every permutation.

There is a useful exact orthonormal four-frame that makes this action a permutation representation. For each tetrahedral vertex put

$$w_a = \frac{1}{2} E_{21} + \frac{\sqrt{3}}{2} r_a, \quad a \in \{\infty, 0, +, -\}, \quad (\text{ESQ51})$$

and let W be the four-column matrix of these vectors in that order. By (ESQ47) and orthogonality to E_{21} , their pairings are 1 on the diagonal and $1/4 + (3/4)(-1/3) = 0$ off it. Therefore

$$W^\top G W = I_4, \quad W^{-1} E_{21} = \frac{1}{2} \mathbf{1}, \quad \mathbf{1} = (1, 1, 1, 1)^\top. \quad (\text{ESQ52})$$

The second identity follows also from $\sum_a w_a = 2E_{21}$. Since each w_a has coefficient $1/2$ on E_{21} , its orthogonal projector in these coordinates is $\mathbf{1}\mathbf{1}^\top/4$. It follows that

$$W^{-1} A_h W = \frac{1}{4} I_4 + \frac{2h - 1/2}{4} \mathbf{1}\mathbf{1}^\top, \quad (\text{ESQ53})$$

and

$$B_h(Wy, Wy) = \frac{1}{4} \sum_{j=1}^4 y_j^2 + \frac{2h - 1/2}{4} \left(\sum_{j=1}^4 y_j \right)^2. \quad (\text{ESQ54})$$

Thus the changing line is the uniform line in this genuine tetrahedral four-frame; the orthogonal sum-zero three-space retains eigenvalue $1/4$. Permuting these four frame vectors is exactly S_π , so (ESQ53) is invariant under all four-vertex permutations. These frame vectors are not the four raw cells: (ESQ51) supplies the exact change of basis between the two presentations.

8. Permuting the four raw cells transports both operator and metric

Let P be a permutation matrix of the original raw basis (ESQ1), rather than a permutation of the rays or frame vectors of Section 7. The transported data are

$$A_h^{(P)} = PA_hP^{-1}, \quad G^{(P)} = P^{-\top}GP^{-1}, \quad B_h^{(P)} = P^{-\top}(GA_h)P^{-1}. \quad (\text{ESQ55})$$

Multiplying the first two formulas gives the third, and substitution yields

$$B_h^{(P)}(Px, Py) = B_h(x, y), \quad \Theta_h^{(P)} = \Theta_h P^{-1}. \quad (\text{ESQ56})$$

Hence there is an exact transported isometry for every raw permutation, with the negative line sent to $P\mathbb{R}E_{21}$. This proves the correspondence without pretending the fixed coordinate metric is invariant under every such permutation.

Indeed the only raw permutation matrices preserving the displayed G are the identity and the swap $E_{11} \leftrightarrow E_{22}$. A metric-preserving raw permutation must preserve its diagonal entries. The values 1 and $2/9$ occur only at E_{21} and E_{12} , so those two positions must be fixed. The remaining two positions have equal diagonal entries $1/16$, giving only their identity or exchange; the off-diagonal entries in (ESQ9) are preserved by both. This proves the assertion exactly. In particular a raw permutation moving the negative cell cannot preserve this fixed metric, even though it always has the transported realization (ESQ55).

There is a stronger fixed-metric check away from the scalar time. If $h \neq 1/4$, a diagonal operator that assigns its changing eigenvalue $a(h)$ to one of the raw diagonal cells E_{11}, E_{22} while retaining G fails to be G -self-adjoint: the nonzero metric entry $G_{14} = -7/144$ connecting those cells forces their assigned eigenvalues to agree, whereas $a(h) \neq 1/4$. Formula (ESQ55) avoids that change of pairing by transporting the whole metric. At the exceptional scalar time $h = 1/4$, $A_h = I_V/4$, so every linear map commutes with that operator; preservation of the fixed G still remains a separate condition.

9. The regular four-simplex and the scalar time

The original source completes the positive tetrahedron to five vectors in the full space:

$$s_* = E_{21}, \quad s_a = \frac{\sqrt{15}}{4}r_a - \frac{1}{4}E_{21} \quad (a \in \{\infty, 0, +, -\}). \quad (\text{ESQ57})$$

Using (ESQ47), each has norm one. Distinct s_a, s_b have pairing $(15/16)(-1/3) + 1/16 = -1/4$, and $\langle s_*, s_a \rangle = -1/4$. Also

$$\sum_{\alpha \in \{*, \infty, 0, +, -\}} s_\alpha = 0, \quad \frac{1}{4} \sum_{a \neq *} s_a = -\frac{1}{4}E_{21}. \quad (\text{ESQ58})$$

These follow by summing (ESQ57) and using $\sum r_a = 0$. They prove the regular four-simplex and its exact opposite-face centroid. The source intertwiner sends that centroid to

$$\mathcal{T}_\square(-E_{21}/4) = \mathcal{I}_- \mathcal{T}_\square(E_{21}) = -\frac{1}{8}(4, 0, 0, -1). \quad (\text{ESQ59})$$

The first equality is (ESQ8) on the negative eigenvector; the final coordinate follows from (ESQ5).

Every permutation of the five simplex vectors gives a unique G -orthogonal map of V . For proof, they span V : s_* gives E_{21} , and the other vectors then give the spanning positive rays. Their sole linear relation is the sum zero, so a permutation induces an invertible map, and their common Gram entries prove isometry on every vector. This realizes S_5 as the simplex's vertex-symmetry group.

For $h \neq 1/4$, the two eigenvalues in (ESQ11) differ, so an operator commuting with A_h must preserve its one-dimensional eigenspace $\mathbb{R}E_{21}$. Among the five simplex vertices only s_* lies on that line: each s_a has a

nonzero positive component. Thus the simplex permutations commuting with A_h are precisely the stabilizer of s_* , namely the geometric S_4 in Section 7. At $h = 1/4$, the operator is scalar and every simplex permutation commutes with it and preserves $B_h = G/4$. This is the exact symmetry enlargement at that heat time; it does not relabel the originally marked source cell.

10. The nilpotent with its support retained

The collision algebra isomorphism (ESQ40) preserves its nonzero nilpotent, its unit and its supported zero. For a fixed bounded distributive support lattice L , any of the proved ring maps f between these commutative algebras induces

$$G_L(f) : (r, \lambda) \mapsto (f(r), \lambda). \quad (\text{ESQ60})$$

This is well defined because a nonzero amplitude can only have top support and a zero image with that same top support is allowed. The ring homomorphism laws preserve amplitudes, and unchanged joins and meets preserve the support operations, proving it is the corresponding support-semiring map. In particular the image of E_{21} in the commutative collision corner is the nonzero supported amplitude r ; its square is the supported zero e , not the unsupported element τ . The radical of the collision form, the nilpotent in the algebra, and absence in the support semiring therefore have exact connecting maps and retain their different definitions.

11. The actual moving source tetrahedron requires its own metric transport

The fixed raw-cell metric G and the coefficient deformation A_h above do not by themselves identify the metric altitude of the moving three-point source fibre. Here is that spatial comparison with its full map. The original source's affine map and metric were read directly in `eq:Fable-collision-to-spatial-affine-map`, `eq:Fable-collision-pullback-metric`, `eq:Fable-source-face-and-axis`, and `eq:Fable-source-signed-quarter-altitude`. In this section (x, y, z) are the three original polynomial-source coordinates; this z is the coordinate denoted W in HEB3. It is not one of the four raw-cell coefficients.

For $a < 0$, let $t = \sqrt{-a} > 0$. The exact three real points from HEB3 and their target are

$$p_0(a) = (0, 0, a), \quad p_r(a) = \left(-\frac{1}{2r}, 3r, -26a\right) \quad (r = \pm t), \quad q_a = (a, 0, 0). \quad (\text{ESQ61})$$

Their centroid, plane, and distinguished displacement are

$$c_a = (0, 0, -17a), \quad H_a = \{(x, y, z) : y - 6ax = 0\}, \quad q_a - c_a = av_F, \quad v_F = (1, 0, 17). \quad (\text{ESQ62})$$

The centroid is the sum of the three displayed points divided by three: the first two coordinate sums vanish and the third is $a - 52a = -51a$. For an escaping point, $y - 6ax = 3r + 3a/r = 3(r^2 + a)/r = 0$, and it is also zero at $p_0(a)$. The two difference vectors from $p_0(a)$ to $p_t(a), p_{-t}(a)$ are independent because their first two coordinates are negatives and their common third coordinate is $-27a \neq 0$; hence this really is the plane of the face. Finally the difference of q_a, c_a is the displayed multiple of v_F .

The original affine map at $a = -1/4$ has linear part

$$L_0 = \begin{pmatrix} 68/9 & 136/27 & -4/9 \\ 1 & 2/3 & 0 \\ 0 & -2\sqrt{3}/3 & 0 \end{pmatrix}, \quad b_0 = (17/9, 3/2, 0)^\top, \quad (\text{ESQ63})$$

with source metric $g_0 = L_0^\top L_0$. Direct multiplication gives $L_0 v_F = (0, 1, 0)$ and

$$g_0 v_F = (1, 2/3, 0)^\top. \quad (\text{ESQ64})$$

Therefore the fixed g_0 -normal covector of this axis is $x + (2/3)y$. It is proportional to the moving face covector $y - 6ax$ exactly when $a = -1/4$: comparing the y -coefficients forces the proportionality constant to be $2/3$, and then the x -coefficients require $1 = -4a$. Consequently the scalar displacement a in (ESQ62) cannot be called a signed normal altitude in the fixed original metric for arbitrary heat time.

For the entire real region $a < 0$, the exact replacement is the explicitly moving affine map

$$\mathcal{A}_a(x) = L_a x + b_0, \quad L_a = \begin{pmatrix} -17/(9a) & 17/(54a^2) & 1/(9a) \\ 1 & -1/(6a) & 0 \\ 0 & -\sqrt{3}/(3\sqrt{-a}) & 0 \end{pmatrix}. \quad (\text{ESQ65})$$

It agrees with (ESQ63) when $a = -1/4$, by substituting that value in all its entries. Its determinant is $-\sqrt{3}/(27a\sqrt{-a}) \neq 0$, by expansion along the third row, so

$$g_a = L_a^\top L_a \quad (\text{ESQ66})$$

is a positive definite real metric: $v^\top g_a v = \|L_a v\|^2 > 0$ for every nonzero v .

The four images, with their signs retained, are

$$\begin{aligned} \mathcal{A}_a(p_0(a)) &= (2, 3/2, 0), \\ \mathcal{A}_a(p_t(a)) &= (-1, 3/2, -\sqrt{3}), \\ \mathcal{A}_a(p_{-t}(a)) &= (-1, 3/2, \sqrt{3}), \\ \mathcal{A}_a(q_a) &= (0, 3/2 + a, 0). \end{aligned} \quad (\text{ESQ67})$$

For the first and fourth these follow by direct multiplication. For p_r , the first coordinate simplifies to $17/(18ar) + 17r/(18a^2) - 1 = -1$ because $r^2 + a = 0$; the second differs from $3/2$ by $-1/(2r) - r/(2a) = 0$; and the third is $-\sqrt{3}r/t$. These prove both escaping images, including the displayed ordering. Thus the three points give a fixed equilateral face in these transported coordinates.

Finally $L_a v_F = (0, 1, 0)$ for every $a < 0$, and the second row gives

$$\langle v_F, v_F \rangle_{g_a} = 1, \quad \langle v_F, v \rangle_{g_a} = x - \frac{y}{6a} \quad (v = (x, y, z)). \quad (\text{ESQ68})$$

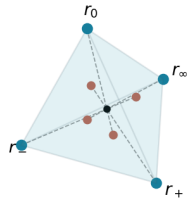
The latter covector vanishes exactly on the plane H_a . Hence v_F is its chosen unit normal in g_a , and (ESQ62) has signed normal altitude exactly a for this transported metric. The map L_a diverges as $a \rightarrow 0^-$, for example through its $17/(54a^2)$ entry. No nonsingular continuation of this specified real spatial transport at the collision is claimed. The fixed four-dimensional coefficient form and the algebraic collision receiver of Sections 2–5 remain defined there by their own displayed maps.

The calculation realizes the original ES quarter 3 + 1 operator with the original heat coefficient and a specified reflected trace. It retains the native tetrahedral symmetry, its action on trace coordinates, the exact algebra embedding and collision isomorphism, and the separate moving spatial metric required by the original three-point geometry. It proves finite forms, operators, and their maps; it makes no identification of the ES form with an unspecified global Weil pairing.

The source edition is available as the 488-page ES reader and its original TeX source and build archive. The TeX entry is `reader/erdos_straus_project_reader.tex`; the source identity and exact reading locations appear in the source guide.

The exact regular tetrahedron in the positive three-space

The four-dimensional map retains the extra direction



$$w_j = \frac{1}{2}E_{21} + \frac{\sqrt{3}}{2}r_j$$

$$\langle w_i, w_j \rangle_G = \delta_{ij}, \quad E_{21} = \frac{1}{2} \sum_j w_j$$

$$W^{-1}A_h W = \frac{1}{4}I_4 + \frac{2h-1/2}{4} \mathbf{1}\mathbf{1}^\top$$

The sum-zero space has eigenvalue $1/4$.

The uniform line has eigenvalue $-1/4 + 2h$.

Face centres: $f_j = -r_j/3$. The line RE_{21} is orthogonal to this space.

Figure 4: Exact tetrahedral vertices and face centres, with the four-dimensional orthonormal frame. ESQ45–ESQ54 prove the coordinates and the uniform changing direction; the orthogonal E21 direction is retained outside the displayed three-space.

The quarter heat family in the actual ES free-shell operator

23 September 2026. This note transports the quarter family through the spectral projectors of the actual raw ES operator into its free-shell carrier. It retains the original forced metric, the full raw matrix product, the two ordered sheets, all three rotations, every orbit label, the ambient simplex kernel, and both directed blades. It constructs the bounded operator on the full orbit Hilbert space, its exact quadratic algebra receiver and collision pairing, and the full Figure 10 support-corner and modular transport. Every finite orbit compression is compatible with these maps. No equality with a Weil distribution is asserted.

The original source read is `ES-Fable-C123/es_fable_c123_preprint.tex`. The precise inputs are `eq:raw-negative-cell-projector`, `eq:raw-positive-cell-projector`, `eq:pulled-back-quarter-cell-signs`, `thm:quarter-three-plus-one`, `eq:origin-resolved-support`, `eq:origin-involution`, `eq:star-kneser-D3-cover-hypotheses`, `thm:general-shell-123-linking-decomposition`, `cor:general-shell-simplex-Lorentz-pullback`, and `thm:general-shell-control-determinant-volume`. The corresponding public reading edition is *Erdős–Straus Project Reader*, pinned edition 20709bee. The mathematical reading for this derivation used the original TeX, not extracted PDF text. All finite and infinite operator claims below are proved here from the explicit source matrices and labels.

1. The original arithmetic labels and the actual shell spaces

For the ES source, let p be a prime, let a_{ar} be a positive integer, and retain

$$R = 4a_{\text{ar}} - p > 2, \quad \gcd(R, pa_{\text{ar}}) = 1, \quad N = pa_{\text{ar}}. \quad (\text{ESHL1})$$

These are `eq:star-kneser-D3-cover-hypotheses`. They imply that p is odd and $p \nmid a_{\text{ar}}$: for $p = 2$, R and pa_{ar} are both even; for odd $p \mid a_{\text{ar}}$, the same prime divides R . Define exactly the origin-resolved set

$$\mathcal{X}_{R,p} = \{(j, u) : j \in \{0, 1, 2\}, u \mid a_{\text{ar}}^2, \quad ua_{\text{ar}}^{-1} = -p^{1-j} \text{ in } (\mathbb{Z}/R\mathbb{Z})^\times\}. \quad (\text{ESHL2})$$

All inverses in this formula exist by ESHL1. Its involution is

$$\iota(j, u) = (2 - j, a_{\text{ar}}^2/u). \quad (\text{ESHL3})$$

It preserves the defining congruence because taking its multiplicative inverse turns $-p^{1-j}$ into $-p^{j-1}$. Applying it twice gives the identity. A fixed point would have $j = 1, u = a_{\text{ar}}$; the congruence would then give $1 = -1 \pmod{R}$, impossible for $R > 2$. Thus all its orbits have size two.

Let $\mathcal{O} = \mathcal{X}_{R,p}/\langle \iota \rangle$ and $q = |\mathcal{O}|$. The two divisors $d(j, u) = p^j u$ in any orbit have product N^2 . They are unequal: uniqueness of the representation $p^j u$, using $p \nmid a_{\text{ar}}$, would otherwise give a fixed point of ι . Thus each orbit has a unique low member $x_{\mathcal{O}}^-$ with divisor below N and high member $x_{\mathcal{O}}^+$ with divisor above N . This proves the original ordering used by the source.

The complete free-cover labels are (O, ϵ, m) , where $\epsilon \in \{0, 1\}$ names the low/high sheet and $m \in \mathbb{Z}/3\mathbb{Z}$ names the rotation. The original bijection and its unitary linearization are

$$\theta_{123}(m, x_{\mathcal{O}}^\epsilon) = (O, \epsilon, m), \quad \Theta_{123} : \mathbb{C}[\tilde{\mathcal{X}}_{R,p}] \longrightarrow \mathcal{H}_q := \mathbb{C}[\mathcal{O}] \otimes \mathbb{C}^2 \otimes \mathbb{C}^3, \quad e_{(m, x_{\mathcal{O}}^\epsilon)} \mapsto e_O \otimes f_\epsilon \otimes g_m. \quad (\text{ESHL4})$$

They are bijections because the three labels recover the original point. The map is unitary for the stated orthonormal label bases. Hence $\dim \mathcal{H}_q = 6q$, retaining every label.

Let $P_2 = I_3 - \mathbf{1}\mathbf{1}^*/3$ be the source triangle support projector. The original support unitary $U_2 : \text{Ran } P_2 \rightarrow \mathbb{C}^2$ satisfies $U_2^* U_2 = P_2$ and $U_2 U_2^* = I_2$, as proved in `eq:simplex-123-unitary-identities`. Write $\eta_\epsilon = U_2^* f_\epsilon$. These are an orthonormal basis of the original triangle support, with their original coordinate choice retained. The actual ambient simplex space and its support are

$$\mathcal{K}_q = \mathbb{C}[\mathcal{O}] \otimes \mathbb{C}^3 \otimes \mathbb{C}^3, \quad \mathcal{H}_q^{\text{simp}} = \mathbb{C}[\mathcal{O}] \otimes \text{Ran } P_2 \otimes \mathbb{C}^3. \quad (\text{ESHL5})$$

The isometry $V_q = I \otimes U_2^* \otimes I_3$ identifies $\mathcal{H}_q$ with $\mathcal{H}_q^{\text{simp}}$. Its range projector is $P_q^{\text{simp}} = I \otimes P_2 \otimes I_3$. The retained complementary space is

$$\mathcal{Z}_q = \mathbb{C}[\mathcal{O}] \otimes \mathbb{C}\mathbf{1} \otimes \mathbb{C}^3, \quad \mathcal{K}_q = \mathcal{H}_q^{\text{simp}} \oplus \mathcal{Z}_q, \quad \dim \mathcal{Z}_q = 3q. \quad (\text{ESHL6})$$

This ambient kernel is different from the source map Π_{123} , which sends all six labels in an orbit to one support label and has kernel dimension $5q$. Explicitly that latter kernel is the subspace whose six coefficients sum to zero on every orbit. Neither of these kernels is substituted for the other.

2. The raw quarter operator and its spectral algebra

Give the raw four-cell vector space $\mathcal{V}_{\text{raw}} = M_2(\mathbb{C})$ its Hilbert–Schmidt inner product, with orthonormal basis $E_{11}, E_{12}, E_{21}, E_{22}$. The original forced ES metric is retained by the explicit isometry ESHL43–ESHL44 below. The source’s pulled-back quarter reflection acts by

$$\mathfrak{Q}_{\square}(E_{11}) = E_{11}, \quad \mathfrak{Q}_{\square}(E_{12}) = E_{12}, \quad \mathfrak{Q}_{\square}(E_{21}) = -E_{21}, \quad \mathfrak{Q}_{\square}(E_{22}) = E_{22}. \quad (\text{ESHL7})$$

Consequently its orthogonal spectral projectors are

$$P_{-}(M) = M_{21}E_{21}, \quad P_{+} = I_{\mathcal{V}_{\text{raw}}} - P_{-}. \quad (\text{ESHL8})$$

They satisfy $P_{\pm}^2 = P_{\pm} = P_{\pm}^*$, $P_{+}P_{-} = 0$, with ranks three and one. Define the exact raw quarter operator and its heat family by

$$A_0 = \frac{1}{4}P_{+} - \frac{1}{4}P_{-}, \quad A_h = \frac{1}{4}P_{+} + a(h)P_{-}, \quad a(h) = -\frac{1}{4} + 2h. \quad (\text{ESHL9})$$

The original coefficient inversion matrix $\mathcal{I}_{-}$ is transported to this raw presentation by the source map $\mathcal{T}_{\square}$, since `eq:pulled-back-quarter-reflection` states $\mathfrak{Q}_{\square} = \mathcal{T}_{\square}^{-1}(4\mathcal{I}_{-})\mathcal{T}_{\square}$. Thus A_0 is the actual pulled-back quarter operator.

The fixed spectral algebra is

$$\mathcal{D} = \mathbb{C}[A_0] = \{cP_{+} + dP_{-} : c, d \in \mathbb{C}\} \cong \mathbb{C}^2. \quad (\text{ESHL10})$$

Indeed $P_{+} = 2A_0 + I/2$, $P_{-} = -2A_0 + I/2$, so both projectors belong to the generated algebra. Their orthogonality proves multiplication is coordinatewise and the involution conjugates c, d . The representation is faithful because both projectors have nonzero range.

The original forced metric in `eq:Fable-forced-full-metric`, also used in ESQ9, has matrix

$$G = \begin{pmatrix} 1/16 & 0 & 0 & -7/144 \\ 0 & 2/9 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -7/144 & 0 & 0 & 1/16 \end{pmatrix}. \quad (\text{ESHL43})$$

Let $v_{+} = (E_{11} + E_{22})/\sqrt{2}$, $v_{-} = (E_{11} - E_{22})/\sqrt{2}$, and let $\Pi_{+}, \Pi_{-}, \Pi_{12}, \Pi_{21}$ be the Hilbert–Schmidt projections on these two unit vectors and on E_{12}, E_{21} , respectively. Multiplying ESHL43 by these four vectors gives the respective eigenvalues $1/72, 1/9, 2/9, 1$. Hence the fully specified positive square root is

$$T = G^{1/2} = \frac{1}{6\sqrt{2}}\Pi_{+} + \frac{1}{3}\Pi_{-} + \frac{\sqrt{2}}{3}\Pi_{12} + \Pi_{21}, \quad T^*T = G, \quad TP_{\pm} = P_{\pm}T, \quad TA_h = A_hT. \quad (\text{ESHL44})$$

The four orthogonal projections sum to the identity. Squaring their positive coefficients proves $T^*T = G$, and replacing the coefficients by their reciprocals defines the inverse. Each projection preserves the decomposition into $\text{Ran } P_{+}$ and $\text{Ran } P_{-}$, proving both commutation identities. Thus $T : (\mathcal{V}_{\text{raw}}, \langle x, Gy \rangle_{\text{HS}}) \rightarrow (\mathcal{V}_{\text{raw}}, \langle x, y \rangle_{\text{HS}})$ is an isometry, and the original heat form is exactly $\langle x, GA_h y \rangle_{\text{HS}} = \langle Tx, A_h Ty \rangle_{\text{HS}}$. Conjugation by T fixes every member of the spectral algebra $\mathcal{D}$. This is a vector-metric transport, and does not change the raw matrix product: for example $T(I_2) = I_2/(6\sqrt{2})$, so T is not a unital algebra map. The separate map preserving that product is ESHL45.

The spectral algebra $\mathcal{D}$ consists of endomorphisms of the raw four-cell vector space. Its negative projector is the two-sided corner operation $P_{-}(X) = E_{22}XE_{11}$, as direct entry multiplication proves. It is not a left multiplication operator: P_{-} has rank one, whereas left multiplication by a nonzero rank-one matrix on $M_2(\mathbb{C})$ has rank two, since each of two columns independently ranges over a fixed one-dimensional image. Section 3 uses the spectral algebra as its domain; ESHL45–ESHL47 construct the full raw matrix algebra map and intertwine this corner operation exactly.

3. The faithful spectral lift into the actual ordered sheets

In $\mathcal{H}_q$, define the original low and high projectors

$$Q_{<} = I_{\mathbb{C}[\mathcal{O}]} \otimes E_{11}^{(2)} \otimes I_3, \quad Q_{>} = I_{\mathbb{C}[\mathcal{O}]} \otimes E_{22}^{(2)} \otimes I_3. \quad (\text{ESHL11})$$

They are exactly **eq:general-shell-ordered-sheet-projectors** under ESHL4. Their ranges are orthogonal, they sum to the identity on $\mathcal{H}_q$, and each has dimension $3q$. For $q \geq 1$, define

$$\boxed{\Psi_q : \mathcal{D} \longrightarrow B(\mathcal{H}_q), \quad cP_+ + dP_- \longmapsto cQ_< + dQ_>} \quad (\text{ESHL12})$$

This is a faithful unital $*$ -homomorphism. Orthogonality gives $(cQ_< + dQ_>)(c'Q_< + d'Q_>) = cc'Q_< + dd'Q_>$, matching the source product. Adjoints conjugate the same two coefficients. The source unit maps to $Q_< + Q_> = I$. If an image vanishes, applying it to any low and any high basis vector gives $c = d = 0$. This proves faithfulness and also the isometry $\|\Psi_q(cP_+ + dP_-)\| = \max(|c|, |d|)$.

The map retains the two spectral labels, but changes their representation multiplicities. Its exact trace comparison is

$$\text{Tr}_{\text{raw}}(cP_+ + dP_-) = 3c + d, \quad \text{Tr}_{\mathcal{H}_q} \Psi_q(cP_+ + dP_-) = 3q(c + d). \quad (\text{ESHL13})$$

Thus this is not a trace-preserving or Hilbert-space unitary identification of the four-cell space with the shell. Both multiplicity lists remain explicit. When $q = 0$, the shell space and all its operators are zero; the resulting map is zero and is not faithful. The nonzero-shell assertion has therefore not been extended to an empty arithmetic shell.

Define the fixed positive weight and its inverse on the support by

$$W = 16Q_< + 4Q_>, \quad W^{-1} = \frac{1}{16}Q_< + \frac{1}{4}Q_>. \quad (\text{ESHL14})$$

Both are verified by multiplication. The weight commutes with $\Psi_q(\mathcal{D})$, and its positive square root is $4Q_< + 2Q_>$. The lifted heat carrier is

$$\boxed{H_h := W\Psi_q(A_h) = 4Q_< + t(h)Q_>, \quad t(h) = -1 + 8h.} \quad (\text{ESHL15})$$

At zero it equals the actual ES carrier

$$H_0 = I_{\mathbb{C}[\mathcal{O}]} \otimes \text{diag}(4, -1) \otimes I_3 = H_{R,p}^{\text{sh}}, \quad (\text{ESHL16})$$

in the source's unitary presentation **eq:general-shell-restricted-Lorentz-carrier**. The map $A \mapsto W\Psi_q(A)$ is a specified positive linear map on self-adjoint elements, since it also equals $W^{1/2}\Psi_q(A)W^{1/2}$; it is not asserted to be an algebra homomorphism. The faithful homomorphism is Ψ_q itself.

The original simplex presentation is $\tilde{H}_h = V_q H_h V_q^*$ on $\mathcal{K}_q$, with zero action on $\mathcal{Z}_q$. This gives the exact commuting relation $\tilde{H}_h V_q = V_q H_h$, and the support-corner homomorphism $D \mapsto V_q \Psi_q(D) V_q^*$ has unit P_q^{simp} . It is unital as a map to that corner, not to the whole ambient algebra with identity $I_{\mathcal{K}_q}$.

4. Positivity, the sharp gap, and every kernel

For $x = x_< + x_>$ in the orthogonal sheet decomposition,

$$\langle x, H_h x \rangle = 4\|x_<\|^2 + t(h)\|x_>\|^2. \quad (\text{ESHL17})$$

Thus for a nonempty finite shell

$$\text{inertia}(H_h) = \begin{cases} (3q, 3q, 0), & h < 1/8, \\ (3q, 0, 3q), & h = 1/8, \\ (6q, 0, 0), & h > 1/8. \end{cases} \quad (\text{ESHL18})$$

In particular $H_h \geq 0$ exactly for $h \geq 1/8$. For $h > 1/8$, the sharp lower and upper bounds are

$$\boxed{\min(4, -1 + 8h)I \preceq H_h \preceq \max(4, -1 + 8h)I.} \quad (\text{ESHL19})$$

Both constants are attained on the appropriate sheet vectors in ESHL17. The lower bound is independent of q . At the critical time, the kernel on the shell is exactly $\text{Ran } Q_>$.

On the full simplex ambient space, the kernel is $\mathcal{Z}_q$ for $h \neq 1/8$, and is

$$\ker \tilde{H}_{1/8} = \mathcal{Z}_q \oplus V_q \text{Ran } Q_>, \quad \dim \ker \tilde{H}_{1/8} = 6q. \quad (\text{ESHL20})$$

The original ambient kernel has not disappeared when the support form becomes positive. The sharp lower bound on the ambient space is the support inequality $\tilde{H}_h \succeq \min(4, t(h))P_q^{\text{simp}}$ for $h > 1/8$; strict positivity is a statement on $\mathcal{H}_q^{\text{simp}}$.

There are also two distinct times at which a single operator loses its ability to recover the fixed spectral labels. At $h = 1/4$, $A_h = I/4$, so $\mathbb{C}[A_h] = \mathbb{C}I$ although the fixed algebra $\mathcal{D}$ and its $P_{\pm}$ remain. At $h = 5/8$, $H_h = 4I$, so its generated algebra is scalar although $Q_{<}, Q_{>}$ remain present. These follow by equating their two coefficients. The lift uses $\mathcal{D} = \mathbb{C}[A_0]$ and the retained labelled projectors throughout; it does not reconstruct them from the possibly scalar single operator at these times.

5. The actual mixed blades remain nonzero

Retain exactly the directed blades of `eq:general-shell-low-high-blade-transport` and `eq:general-shell-high-low-blade-transport`:

$$D = I \otimes 2E_{21}^{(2)} \otimes I_3, \quad U = I \otimes 2E_{12}^{(2)} \otimes I_3 = D^*. \quad (\text{ESHL21})$$

Here D is the original low-to-high sheet map and U its adjoint; these symbols do not denote the amplitude multiplier from a different packet calculation. Matrix-unit multiplication proves

$$D^2 = U^2 = 0, \quad UD = 4Q_{<}, \quad DU = 4Q_{>}, \quad \|D\| = \|U\| = 2. \quad (\text{ESHL22})$$

Their norms follow because each takes a unit vector in its source sheet to a vector of norm two and kills the other sheet. They are nonzero for $q \geq 1$, at every heat time. Their precise source and target spaces are

$$D : \text{Ran } Q_{<} \rightarrow \text{Ran } Q_{>}, \quad U : \text{Ran } Q_{>} \rightarrow \text{Ran } Q_{<}. \quad (\text{ESHL23})$$

The carrier consequently retains the original directed expression

$$H_h = UD + t(h)Q_{>}. \quad (\text{ESHL24})$$

At $h = 0$, this is exactly `eq:general-shell-directed-Gram-support-carrier`. At $h = 1/8$, the high sheet becomes the carrier's kernel, but D still takes the positive sheet onto that kernel and U takes it back. The commutators are

$$[H_h, D] = (t(h) - 4)D, \quad [H_h, U] = (4 - t(h))U. \quad (\text{ESHL25})$$

Both follow by multiplying the two diagonal coefficients with the two matrix units. Thus the mixed maps have not been discarded by diagonalizing the form.

Their generated algebra is exactly

$$I_{\mathbb{C}[\mathcal{O}]} \otimes M_2(\mathbb{C}) \otimes I_3 : \quad E_{11} \mapsto Q_{<}, \quad E_{22} \mapsto Q_{>}, \quad E_{21} \mapsto D/2, \quad E_{12} \mapsto U/2. \quad (\text{ESHL26})$$

The four matrix-unit laws hold by ESHL22 and the sheet actions; independence follows by applying a proposed relation to one low and one high vector. This is an actual copy of the raw matrix algebra $M_2(\mathbb{C})$, with its diagonal subalgebra containing $\Psi_q(\mathcal{D})$. The faithful algebra isomorphism onto this image is R_q in ESHL45. Section 2 identifies the spectral projector P_- with a two-sided corner operation on that algebra; it does not exclude the matrix algebra identification.

There is a faithful dual-number receiver $\mathbb{C}[\eta]/\eta^2 \rightarrow B(\mathcal{H}_q)$, $1 \mapsto I$, $\eta \mapsto D$. Its multiplicativity follows from $D^2 = 0$; applying $cI + dD = 0$ first to a high vector and then to a low vector proves injectivity. These nilpotents are present in the mixed operator algebra at all times. The diagonal homomorphism Ψ_q does not itself create a nilpotent in its reduced domain $\mathbb{C}^2$.

The original rotation acts only on the third factor and commutes with D, U, H_h . The original reflection swaps the two sheet factors and reverses rotation labels, giving

$$\mathbf{SDS}^{-1} = U, \quad \mathbf{SQ}_{<}\mathbf{S}^{-1} = Q_{>}, \quad \mathbf{SH}_h\mathbf{S}^{-1} = t(h)Q_{<} + 4Q_{>}. \quad (\text{ESHL27})$$

These follow directly from the source label action in `eq:general-shell-123-label-action`. The directed carrier is invariant under that reflection only when $t(h) = 4$. Its exact reflected partner, rather than an unproved invariance, is retained in ESHL27.

6. Exact finite traces, lengths, and determinant ratios

For finite $q \geq 1$, the two eigenvalues and their multiplicities give

$$\text{Tr } H_h = 3q(4+t), \quad \|H_h\|_{\text{HS}}^2 = 3q(16+t^2), \quad \det H_h = 4^{3q}t^{3q}, \quad t = -1 + 8h. \quad (\text{ESHL28})$$

The Hilbert–Schmidt formula is the trace of the square because H_h is Hermitian. The determinant is the product of all eigenvalues, including its sign. For $t \neq 0$, its absolute determinant volume and the exact ratio to the original carrier are

$$V_h^{\det} = |4t|^{3q}, \quad \frac{V_h^{\det}}{V_0^{\det}} = |t|^{3q}, \quad \det(H_h H_0^{-1}) = (-t)^{3q} = (1-8h)^{3q}. \quad (\text{ESHL29})$$

The last formula follows also from $H_h H_0^{-1} = Q_{<} - tQ_{>}$. At the critical time the determinant is zero; no logarithm of that zero is treated as finite.

The exact traceless part is

$$K_h = H_h - \frac{4+t}{2}I = \frac{4-t}{2}(Q_{<} - Q_{>}), \quad \|K_h\|_{\text{HS}}^2 = \frac{3q}{2}(4-t)^2. \quad (\text{ESHL30})$$

For the constant-generator paths $\exp(-i\sigma H_h)$, $\exp(-i\sigma K_h)$, $0 \leq \sigma \leq 1$, the Hilbert–Schmidt speed is the norm of the generator: multiplying the derivative by the inverse unitary removes the unitary factor and leaves the same Hilbert–Schmidt norm. Thus their squared lengths equal the two squared norms in ESHL28 and ESHL30. At $h = 0$ they are $51q$ and $75q/2$, exactly the ES source constants.

All per-orbit quantities have exact finite formulas, with the full factor q retained:

$$\frac{\text{Tr } H_h}{q} = 3(4+t), \quad \frac{\|H_h\|_{\text{HS}}^2}{q} = 3(16+t^2), \quad \frac{\log V_h^{\det}}{q} = 3 \log |4t| \quad (t \neq 0). \quad (\text{ESHL31})$$

For $t \neq 0$, these imply

$$\log V_h^{\det} = \frac{\log |4t|}{16+t^2} \|H_h\|_{\text{HS}}^2. \quad (\text{ESHL32})$$

For $t \neq 0, 4$, they also imply

$$\log V_h^{\det} = \frac{2 \log |4t|}{(4-t)^2} \|K_h\|_{\text{HS}}^2. \quad (\text{ESHL33})$$

At $t = -1$, these coefficients are $2 \log 2/17$ and $4 \log 2/25$, reproducing the original determinant-length identities. At $t = 4$, $K_h = 0$ while $V_h^{\det} = 16^{3q}$, so the second quotient is not defined; the original exact formulas still apply. This handles the exceptional scalar carrier explicitly.

7. The actual infinite orbit-space operator

Let $\mathcal{O}_\infty$ be any nonempty countable set of the retained labelled complement orbits, with shell labels included when orbits from different shells are combined. The empty case is the zero space already specified after ESHL13. Form the actual Hilbert completion

$$\mathcal{H}_\infty = \ell^2(\mathcal{O}_\infty) \otimes \mathbb{C}^2 \otimes \mathbb{C}^3. \quad (\text{ESHL34})$$

This definition also covers a finite orbit set. If it is infinite, the following statements concern that same labelled Hilbert space; no assertion that every arithmetic shell is occupied is being introduced. The projectors $Q_{<}, Q_{>}$, blades D, U , weight W , and Ψ_∞ are defined by the identical tensor formulas ESHL11–ESHL15 and ESHL21.

For example, if $x = (x_{O,<,m}, x_{O,>,m})$, then

$$(H_h x)_{O,<,m} = 4x_{O,<,m}, \quad (H_h x)_{O,>,m} = t(h)x_{O,>,m}. \quad (\text{ESHL35})$$

The square sum of these coordinates is bounded by $\max(16, t(h)^2) \|x\|^2$, proving that H_h is everywhere defined and bounded. Choosing a single sheet basis vector proves

$$\|H_h\| = \max(4, |t(h)|), \quad \|H_h - H_{h'}\| = 8|h - h'|. \quad (\text{ESHL36})$$

Its adjoint is itself by the real diagonal coefficients. Its quadratic form is the convergent sum of ESHL17 over all labels, so the positivity criterion and sharp uniform lower bound ESHL19 remain valid without a

dimension-dependent error. In particular, for every fixed $h > 1/8$, H_h is boundedly invertible on the entire orbit support, with

$$H_h^{-1} = \frac{1}{4}Q_{<} + \frac{1}{t(h)}Q_{>}, \quad \|H_h^{-1}\| = \frac{1}{\min(4, t(h))}. \quad (\text{ESHL37})$$

This is an explicit inverse, verified by multiplication. At $h = 1/8$, its kernel is the whole high-sheet Hilbert subspace and its range is the low-sheet subspace. The blades remain bounded with norm two and retain every identity ESHL22–ESHL27.

The infinite ambient simplex space is $\mathcal{K}_\infty = \ell^2(\mathcal{O}_\infty) \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$. Its orthogonal decomposition is the completed version of ESHL5–ESHL6, and $V_\infty = I \otimes \mathbf{U}_2^* \otimes I_3$ gives the same isometry and support projector. The zero action on the entire complementary Hilbert space is retained, including when H_h is uniformly positive on its support.

For completeness, every local matrix unit also remains an actual bounded operator. For basis labels $\alpha = (O, \epsilon, m)$, $\beta = (O', \epsilon', m')$, define

$$E_{\alpha\beta}x = e_\alpha \langle e_\beta, x \rangle. \quad (\text{ESHL38})$$

Its norm is one, its adjoint is $E_{\beta\alpha}$, and $E_{\alpha\beta}E_{\gamma\delta} = \delta_{\beta\gamma}E_{\alpha\delta}$, all by direct application to a vector. Thus the mixed maps, including maps between different orbit labels, are present in $B(\mathcal{H}_\infty)$. The particular global blades are the strong sums $D = 2 \sum_{O, m} E_{(O, >, m), (O, <, m)}$ and its adjoint, whose orthogonal source and target coordinates give norm two.

8. Exact compatibility with all finite compressions

For a finite subset $F \subset \mathcal{O}_\infty$, take all two-sheet and three-rotation labels over its orbits. Let $j_F : \mathcal{H}_F \rightarrow \mathcal{H}_\infty$ be inclusion by zero outside F , and $P_F = j_F j_F^*$. It is an isometry. The tensor formulas give

$$H_h j_F = j_F H_{h,F}, \quad j_F^* H_h j_F = H_{h,F}, \quad P_F H_h = H_h P_F, \quad \Psi_\infty(A) j_F = j_F \Psi_F(A). \quad (\text{ESHL39})$$

The same equations hold for D, U, W and the original rotation and reflection operators because none changes the orbit label. Thus these are the actual finite lifts and their exact inclusions, not new finite approximations with altered sheets or rotations.

For nested finite orbit sets F_n exhausting $\mathcal{O}_\infty$, the square-summable tail identity gives $\|(I - P_{F_n})x\| \rightarrow 0$ for every x . Therefore

$$P_{F_n} H_h P_{F_n} \longrightarrow H_h \quad \text{strongly on } \mathcal{H}_\infty. \quad (\text{ESHL40})$$

Indeed the difference on x has norm at most $\|H_h\| \|(I - P_{F_n})x\|$, since the projections commute with H_h . Positivity and its uniform bound already hold on every finite compression and on the limiting operator by ESHL17; there is no hidden loss of the lower bound in this limit. If every F_n has a nonempty complement, this convergence is not in operator norm: a unit tail vector in a sheet attaining the maximum eigenvalue magnitude has difference norm $\max(4, |t(h)|)$. The exact topology of convergence is thereby specified.

9. Infinite traces and the retained finite-volume data

When $\mathcal{O}_\infty$ is infinite, H_h is never Hilbert–Schmidt: its value on each of infinitely many low-sheet basis vectors has norm four, so the sum of their squared norms diverges. It is not trace class for the same reason, since $|H_h|$ has eigenvalue four infinitely often. For $h \geq 1/8$, its extended positive trace is $+\infty$. For $h < 1/8$, its positive and negative parts both have infinite trace, so an ordinary signed trace would be the undefined difference $\infty - \infty$. None of these facts changes its bounded operator norm or the positive lower bound at $h > 1/8$.

The ordinary finite-dimensional determinant has no direct infinite-dimensional value here. More precisely, for $h \neq 0$, the relative operator is

$$H_h H_0^{-1} = I - 8h Q_{>}. \quad (\text{ESHL41})$$

Its difference from the identity is not compact: it takes an infinite orthonormal sequence in the high sheet to an orthogonal sequence of fixed nonzero norm. Hence that difference is not trace class, and the usual trace-class Fredholm determinant is unavailable. At $h = 0$ the relative operator is the identity and its determinant is one.

The exact finite determinants and per-orbit data remain ESHL28–ESHL33 on every finite subset. Their absolute values can tend to zero, infinity, or remain one, depending on $|4t(h)|$; none of those sequence behaviours alone defines an ordinary determinant of the infinite operator. On the specified constant-fibre algebra $I_{\ell^2(\mathcal{O}_\infty)} \otimes M_6(\mathbb{C})$, there is instead the exact finite fibre trace $\tau_{\text{fibre}}(I \otimes B) = \text{Tr}_{\mathbb{C}^6} B$. It equals

$\text{Tr}_{\mathcal{H}_F}(j_F^*(I \otimes B)j_F)/|F|$ on every nonempty finite orbit set. Thus, for $t(h) \neq 0$, the explicitly defined fibre determinant is

$$\exp(\tau_{\text{fibre}}(\log |H_h|)) = |4t(h)|^3. \quad (\text{ESHL42})$$

This is a finite-fibre observable with its exact comparison map; it is not presented as an ordinary infinite determinant. The full orbit labels, all local injections, the fixed spectral representation, and the ambient kernel remain in the construction.

The result is a lift of the stated quarter heat family into the actual ES free-shell carrier, with a bounded infinite-orbit extension and a uniform positive support bound after $h = 1/8$. The full raw multiplication and quadratic receiver are constructed in ESHL45–ESHL50, the collision pairing in ESHL51–ESHL53, and the complete Figure 10 transport in ESHL54–ESHL64. These maps do not identify the ES spectral algebra or its infinite orbit trace with an arithmetic Weil pairing.

10. Exact finite verification

The reproducible script `check_es_shell_heat_lift.py` passed **217 exact symbolic checks**, recorded in `ES_SHELL_HEAT_LIFT_CHECKS.json`. They verify the raw spectral projectors and their multiplicities, the original forced metric isometry, the full raw matrix product and corner operation, the quadratic receiver and its original involution, the complete collision pairing, both blade products and nilpotence, all stated trace and determinant formulas, the original control-length constants, and the ambient kernel dimensions for one, two, and three complete orbit blocks. They also check finite inclusions retaining all six labels per orbit; the original displayed segment, triangle, and tetrahedron synthesis matrices; all nine rectangular corners and their 27 typed compositions; all 36 density ratios; the full Hilbert–Schmidt map; inter-orbit transport; and the exact ambient multiplicativity defect. The bounded infinite extension, its sharp lower bound, and strong convergence are proved in ESHL34–ESHL40 and ESHL62–ESHL63; they are not inferred from the finite checks.

11. The full raw matrix product and its exact corner operation

For every nonempty finite or countable orbit set $\mathcal{O}$, use its Hilbert space $\mathcal{H}_{\mathcal{O}} = \ell^2(\mathcal{O}) \otimes \mathbb{C}^2 \otimes \mathbb{C}^3$, where $\ell^2(\mathcal{O}) = \mathbb{C}[\mathcal{O}]$ when the set is finite. Define

$$R_{\mathcal{O}} : M_2(\mathbb{C}) \longrightarrow B(\mathcal{H}_{\mathcal{O}}), \quad R_{\mathcal{O}}(X) = I_{\ell^2(\mathcal{O})} \otimes X \otimes I_3. \quad (\text{ESHL45})$$

The tensor product laws give $R(XY) = R(X)R(Y)$, $R(I_2) = I$, and $R(X^*) = R(X)^*$. If $R(X) = 0$, fix an orbit O and rotation m ; its restriction to the span of $e_O \otimes f_0 \otimes g_m, e_O \otimes f_1 \otimes g_m$ is the matrix X , so $X = 0$. This proves faithfulness. The same restriction, together with the square-sum norm bound over all (O, m) , proves $\|R(X)\| = \|X\|$. The inverse on its image is precisely that two-dimensional restriction for any fixed (O, m) , independent of the choice. Its image is the whole blade algebra ESHL26, since it sends $E_{11}, E_{22}, E_{21}, E_{12}$ to $Q_<, Q_>, D/2, U/2$, respectively.

Entry multiplication in the original raw matrix algebra gives $P_-(X) = E_{22}XE_{11}$. Applying the multiplicative map just proved gives

$$R(P_-X) = Q_>R(X)Q_<, \quad R(A_hX) = \frac{1}{4}R(X) + \left(a(h) - \frac{1}{4}\right)Q_>R(X)Q_<. \quad (\text{ESHL46})$$

Thus the quarter operator acts on the blade algebra by the explicitly displayed linear operation on operators. This is the exact relation between the spectral endomorphism and the matrix multiplication. It is different from left multiplication by the heat carrier H_h : the former fixes the upper-right matrix cell with coefficient $1/4$, while $H_hR(E_{12}) = 4R(E_{12})$. Both actions are retained with their domains and constants. The spectral lift $\Psi(A_h)$ of ESHL12 and the operator-valued map $X \mapsto R(A_hX)$ of ESHL46 also have different domains; ESHL46 is their explicitly calculated underlying raw-cell action, not an asserted equality of those maps.

For finite q , the matrix trace on q identical orbit blocks and three identical rotation blocks gives

$$\text{Tr}(R(X)^*R(Y)) = 3q \text{tr}_2(X^*Y), \quad \text{Tr}(R(TX)^*R(TY)) = 3q \langle X, GY \rangle_{\text{HS}}. \quad (\text{ESHL47})$$

The second equality follows from the first and $T^*T = G$; TX means the four-dimensional linear metric map ESHL44 applied to the raw matrix viewed as a vector. If $\hat{A}_h$ denotes the linear operator in the right side of ESHL46, then $\hat{A}_hR(TY) = R(A_hTY) = R(TA_hY)$. Inserting this in ESHL47 gives $\text{Tr}(R(TX)^*\hat{A}_hR(TY)) = 3q \langle X, GA_hY \rangle_{\text{HS}}$. This proves the original metric and form transport alongside the product-preserving map. For an infinite orbit set, the same formulas with the finite fibre trace τ_{fibre} replace $3q$ by 3; the ordinary traces of these repeated operators need not be finite.

12. The original quadratic algebra, its involution, and the vanishing infinitesimal

For real a , define $E_a = \mathbb{C}[r]/(r^2 + a)$, with coefficient-conjugating involution $(c + dr)^\# = \bar{c} - \bar{d}r$. The basis $(1, r)$ is independent because the relation is monic of degree two. Multiplication by r on this basis has matrix $\begin{pmatrix} 0 & -a \\ 1 & 0 \end{pmatrix}$, so the regular trace is $\text{Tr}_{E_a}(c + dr) = 2c$. The original matrix receiver, proved in ESQ36 and re-established by the following multiplication, is

$$\begin{aligned} \rho_a(c + dr) &= cI_2 + d(E_{21} - aE_{12}), \\ \rho_{a,\mathcal{O}} &:= R_{\mathcal{O}}\rho_a, \quad \rho_{a,\mathcal{O}}(c + dr) = cI + \frac{d}{2}(D - aU), \quad K_{a,\mathcal{O}} := \frac{1}{2}(D - aU), \quad K_{a,\mathcal{O}}^2 = -aI. \end{aligned} \quad (\text{ESHL48})$$

For the last identity, ESHL22 gives $(D - aU)^2 = -a(DU + UD) = -4aI$. Therefore the coefficient product is $(c, d)(e, f) = (ce - adf, cf + de)$, exactly the multiplication in E_a . The unit maps to I . Applying the displayed matrix on one fixed orbit and rotation recovers c from either diagonal entry and d from the lower-left entry; hence the map is faithful even when $a = 0$. The map is an isomorphism onto the explicit commutative subalgebra $\mathbb{C}[I, K_{a,\mathcal{O}}]$, with inverse recovering those two entries. Its constants require no division by a , so the isomorphism includes the collision. This construction is initially defined over $\mathbb{Z}[a]$, before specializing the parameter or extending coefficients.

Let $\mathcal{C}$ be coefficient conjugation in the original orbit-sheet-rotation basis and $J_s = Q_{<} - Q_{>}$. The symbol J_s is the sheet-sign operator and is unrelated to the adjunction map on a Hilbert–Schmidt space in Section 13. Both J_s and $\mathcal{C}$ square to the identity and commute. The matrix-unit actions give $J_s D J_s = -D$, $J_s U J_s = -U$, and coefficient conjugation fixes the matrices D, U . Consequently

$$\rho_{a,\mathcal{O}}(p^\#) = J_s \mathcal{C} \rho_{a,\mathcal{O}}(p) \mathcal{C} J_s. \quad (\text{ESHL49})$$

On the full operator algebra the right side defines a conjugate-linear multiplicative order-two map: the two adjacent factors $\mathcal{C} J_s J_s \mathcal{C}$ cancel in a product. It is not the Hilbert adjoint, which reverses product order. On the commutative image in ESHL48 it realizes exactly the source involution $\#$; commutativity makes both product-order descriptions agree there. These facts prove the type of the involution without changing it to the positive Hilbert adjoint.

In a finite shell its trace and reflected pairing retain their full multiplicity:

$$\begin{aligned} \text{Tr}_{\mathcal{H}_q} \rho_{a,q}(p) &= 3q \text{Tr}_{E_a}(p), \\ \text{Tr}_{\mathcal{H}_q}(\rho_{a,q}(p^\#) \rho_{a,q}(q')) &= 6q(\bar{c}e + a\bar{d}f), \quad p = c + dr, \quad q' = e + fr. \end{aligned} \quad (\text{ESHL50})$$

Indeed each orbit and rotation contains the same two-dimensional regular matrix, and $p^\# q'$ has scalar coefficient $\bar{c}e + a\bar{d}f$. The finite fibre trace on an infinite orbit space gives the same two equations with $q = 1$; this does not turn the repeated operator into a trace-class operator. In the basis $(1, r)$, the full finite reflected Gram matrix is $6q \text{diag}(1, a)$. Its inertia is $(1, 1, 0)$ for $a < 0$, $(1, 0, 1)$ for $a = 0$, and $(2, 0, 0)$ for $a > 0$, because the two real diagonal entries have exactly those signs. These are the two algebra-coordinate multiplicities, alongside the actual shell-vector multiplicities ESHL18.

Set $a = a(h) = -1/4 + 2h$; then $t(h) = 4a(h)$, so the algebra collision and the carrier collision occur at the same original heat time $h = 1/8$. The two maps giving them are the quadratic algebra map ESHL48 and the weighted spectral map $H_h = W\Psi(A_h)$, respectively. They are related by their actual common blades and the formulas already proved; they have not been declared to be the same map. At the collision put

$$N := \rho_{0,\mathcal{O}}(r) = D/2, \quad N \neq 0, \quad N^2 = 0, \quad N^* N = Q_{<}, \quad N N^* = Q_{>}, \quad H_{1/8} N = 0, \quad N H_{1/8} = 4N. \quad (\text{ESHL51})$$

The two products with N^* follow from ESHL22. In particular N is nonzero and maps the low sheet isometrically onto the high sheet. Multiplication by $H_{1/8} = 4Q_{<}$ on the left kills its high-sheet image; multiplication on the right multiplies its low-sheet source by four. This proves both asymmetric mixed products and proves that the vanishing does not erase the infinitesimal.

For every pair of vectors v, w in the complete shell Hilbert space, finite or infinite, the complete carrier pairing is

$$\begin{aligned} \langle Nv, H_h Nw \rangle &= t(h) \langle Q_{<} v, Q_{<} w \rangle, \\ \langle Nv, H_{1/8} Nw \rangle &= 0, \quad \langle x, H_{1/8} Nv \rangle = \langle Nv, H_{1/8} x \rangle = 0 \quad \text{for every } x. \end{aligned} \quad (\text{ESHL52})$$

To prove the first identity, $H_h N = t(h)N$ by its high-sheet image, and $N^* N = Q_{<}$; hence $N^* H_h N = t(h)Q_{<}$. The remaining identities follow either by setting $t = 0$ or by using the self-adjointness of the carrier and $H_{1/8} N = 0$. After the collision, $\langle Nv, H_h Nv \rangle > 0$ holds exactly when $Q_{<} v \neq 0$. Before the collision it is

strictly negative for the same nonzero source. At the collision the original form vanishes on that whole image, whereas its Hilbert norm is still $\|Nv\|^2 = \|Q_{<}v\|^2$. These formulas specify the vanishing observable and the surviving norm in the same coordinates.

There is also an exact regular-trace vanishing. In a finite shell,

$$\mathrm{Tr} N^j = 0 \quad (j \geq 1), \quad \det(I - zN) = 1, \quad \mathrm{Tr}(N^*N) = \mathrm{Tr}(NN^*) = 3q. \quad (\text{ESHL53})$$

For $j = 1$ the matrix diagonal is zero; for $j \geq 2$ the operator is zero. On each two-dimensional block $I - zN$ is triangular with both diagonal entries one, proving the determinant, including every rotation and orbit. The last equalities count the ranks of the two projections. In an infinite orbit set, the fibre trace gives the corresponding values zero and three, while N is not trace class and $I - zN$ has no trace-class Fredholm determinant for $z \neq 0$: its difference from the identity sends an infinite orthonormal low-sheet sequence to an orthogonal sequence of norm $|z|$. Thus neither the surviving mixed product nor the infinite trace domain is hidden by the zero regular trace. The receiver $\eta \mapsto D$ in Section 5 is the same collision algebra after the explicit scalar algebra isomorphism $\eta \mapsto 2r$, whose inverse sends $r \mapsto \eta/2$.

13. The full Figure 10 corner map and its modular transport

This section reads and proves the source result `thm:simplex-123-modular-intertwiner` displayed in Figure `fig:simplex-123-modular-linking`. Its nine-dimensional source is the direct sum $\mathbb{C}^2 \oplus \mathbb{C}^3 \oplus \mathbb{C}^4$, rather than the tensor ambient space $\mathbb{C}^3 \otimes \mathbb{C}^3$ of ESHL5. Their exact support transport will be given below, so neither ambient presentation is silently substituted for the other.

For $n = 1, 2, 3$, retain the original regular-simplex vectors $v_0, \dots, v_n$ in the n -dimensional space V_n : their inner products are one on the diagonal and $-1/n$ off the diagonal. For $n = 1$ the two vectors are exactly $1, -1$. Let $S_n : \mathbb{C}^{n+1} \rightarrow V_n$ send e_i to v_i , and define

$$P_n = I_{n+1} - \frac{1}{n+1} \mathbf{1}\mathbf{1}^*, \quad U_n = \sqrt{\frac{n}{n+1}} S_n P_n, \quad P = P_1 \oplus P_2 \oplus P_3, \quad U = U_1 \oplus U_2 \oplus U_3. \quad (\text{ESHL54})$$

The Gram entries give $S_n^* S_n = (n+1)P_n/n$. Thus $U_n^* U_n = P_n$. Its rank is n , equal to $\dim V_n$; its restriction to $\mathrm{Ran} P_n$ is therefore an onto isometry, proving $U_n U_n^* = I_{V_n}$. It follows that $U^* U = P$, $U U^* = I_{\mathcal{V}}$, where $\mathcal{V} = V_1 \oplus V_2 \oplus V_3$ has dimension six and P has rank six. The full ambient kernel is the direct sum of the three constant lines $\mathbb{C}\mathbf{1}_{n+1}$.

The full map and inverse are

$$\Phi : PM_9(\mathbb{C})P \longrightarrow B(\mathcal{V}), \quad \Phi(X) = UXU^*, \quad \Phi^{-1}(Y) = U^* Y U. \quad (\text{ESHL55})$$

The products of the two maps in either order reduce to $PXP = X$ and $IYI = Y$. For supported X, Y , $\Phi(X)\Phi(Y) = UX(U^*U)YU^* = UXPYU^* = \Phi(XY)$. Taking adjoints gives $\Phi(X)^* = \Phi(X^*)$, and $\Phi(P) = I_{\mathcal{V}}$. This proves a unital $*$ -isomorphism from the support corner with its own unit, including surjectivity and injectivity.

For $a, b, c \in \{1, 2, 3\}$, the rectangular maps and their full typed products are

$$\begin{aligned} \Phi_{ab} : P_a M_{a+1, b+1}(\mathbb{C}) P_b &\longrightarrow B(V_b, V_a), & \Phi_{ab}(X) &= U_a X U_b^*, \\ \Phi_{ab}^{-1}(Y) &= U_a^* Y U_b, & \Phi_{ab}(X) \Phi_{bc}(Y) &= \Phi_{ac}(XY), \\ \Phi_{ab}(X)^* &= \Phi_{ba}(X^*), & \dim_{\mathbb{C}} B(V_b, V_a) &= ab. \end{aligned} \quad (\text{ESHL56})$$

The inverse and adjoint follow from the same support identities; in the product the middle factor is $U_b^* U_b = P_b$, and $XP_b Y = XY$. For orthonormal bases $e_{n,k}$ of the original spaces V_n , the ab operators $|e_{a,i}\rangle\langle e_{b,j}|$ form a basis of that rectangle; their inverse images under the stated inverse form a basis of the source rectangle. The nine dimensions are therefore the full array $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$, summing to 36. Every composable rectangular path, including all six orders visiting the three sectors, is carried to its corresponding product by repeated application of the proved product law.

Retain any original positive density b_n invertible on $\mathrm{Ran} P_n$, and write $b = b_1 \oplus b_2 \oplus b_3$, $d_n = U_n b_n U_n^*$, $d = \oplus d_n$. The symbol b here denotes this density, not the quadratic parameter a of Section 12. The corner inverse is $b^{-1,P}$, with product unit P ; its image is d^{-1} , since $\Phi(b)\Phi(b^{-1,P}) = I$. On the two Hilbert–Schmidt spaces define

$$\Delta_b(X) = bXb^{-1,P}, \quad J_P(X) = X^*, \quad \Delta_d(Y) = dYd^{-1}, \quad J_{\mathcal{V}}(Y) = Y^*. \quad (\text{ESHL57})$$

These J 's are conjugate-linear maps on operator Hilbert spaces, rather than the linear sheet-sign matrix J_s in ESHL49. The map Φ is Hilbert–Schmidt unitary: for supported X, Y , cyclically moving the rectangular factors in a finite trace gives $\text{Tr}_{\mathcal{V}}(\Phi(X)^*\Phi(Y)) = \text{Tr}_{\mathbb{C}^9}(X^*PY) = \text{Tr}_{\mathbb{C}^9}(X^*Y)$. This trace identity can also be obtained by summing the equal matrix entries in the orthonormal support bases $\mathbf{U}^*e_{n,k}$.

Choose an orthonormal eigenbasis of the positive matrix d , with eigenvalues $\lambda_i > 0$. On the matrix unit E_{ij} , Δ_d acts by λ_i/λ_j , a positive real number. The corresponding inverse images under Φ give the same eigenvalues for Δ_b , by the multiplicativity already proved. Therefore Δ_b, Δ_d are positive, invertible, self-adjoint operators on their Hilbert–Schmidt spaces, and

$$\begin{aligned} \Phi\Delta_b &= \Delta_d\Phi, & \Phi J_P &= J_{\mathcal{V}}\Phi, & \Phi(-\log \Delta_b) &= (-\log \Delta_d)\Phi, \\ \Phi_{ab}(b_a X b_b^{-1, P_b}) &= d_a \Phi_{ab}(X) d_b^{-1}. \end{aligned} \tag{ESHL58}$$

The first and fourth identities are the explicit product law with the inverse. The second is the adjoint identity. For the third, define the logarithm on each positive eigenvalue; the two sides send each corresponding matrix unit to the same multiple $-\log(\lambda_i/\lambda_j)$, proving it on a basis. This also proves the equality of all imaginary powers $\Phi\Delta_b^{it} = \Delta_d^{it}\Phi$. The vectors $b^{1/2}$ and $d^{1/2}$ correspond by the same eigenbasis argument, and the antilinear maps $X \mapsto b^{-1/2, P} X^* b^{1/2}$, $Y \mapsto d^{-1/2} Y^* d^{1/2}$ equal $J\Delta^{1/2}$ and are intertwined. These are the full finite Tomita maps, as substitution on $Xb^{1/2}$ gives $X^*b^{1/2}$, and likewise on the target.

The source's specific density is retained without replacing its six eigenvalues:

$$d_1 = (1), \quad d_2 = \text{diag}(2, 3), \quad d_3 = \text{diag}(5, 7, 11), \quad b_n = \mathbf{U}_n^* d_n \mathbf{U}_n. \tag{ESHL59}$$

The modular eigenvalue on $|e_{a,i}\rangle\langle e_{b,j}|$ is exactly $\lambda_{a,i}/\lambda_{b,j}$. In a closed path $a \rightarrow b \rightarrow c \rightarrow a$ with matching intermediate basis labels, the product of these ratios is one, because its three numerators and denominators cancel pairwise; the corresponding logarithmic weights sum to zero. These statements concern the actual nine rectangles and their compositions. For the separate source density $b_n = P_n/n$, each diagonal sector has $\Delta = I$, but the (a, b) rectangle has eigenvalue b/a , obtained from $(1/a)/(1/b)$. Thus vanishing of the three diagonal modular generators does not discard the inter-sector modular data.

To transport Figure 10 into the ordered shell, specify a basis choice rather than identifying its sectors by dimension alone. Use the original orthonormal coordinate bases of V_n , and define the following unitary $C : \mathcal{V} \rightarrow \mathbb{C}^2 \otimes \mathbb{C}^3$:

source vector	$e_{1,1}$	$e_{2,1}$	$e_{2,2}$	$e_{3,1}$	$e_{3,2}$	$e_{3,3}$
image under C	$f_0 \otimes g_0$	$f_0 \otimes g_1$	$f_0 \otimes g_2$	$f_1 \otimes g_0$	$f_1 \otimes g_1$	$f_1 \otimes g_2$

(ESHL60)

The table is a bijection between orthonormal bases, so its linear extension is unitary and its inverse reverses the table. These are coordinate basis vectors, not selected simplex vertices: the latter have nonzero off-diagonal Gram entries and could not be sent to an orthonormal basis by a unitary. It is a chosen coordinate transport. In particular the one- and two-dimensional sectors together occupy the low sheet and the three-dimensional sector occupies the high sheet under this choice. This fact is specified by the table; no arithmetic rule identifying sector number with sheet or rotation has been assumed.

Let $L = CU : \mathbb{C}^9 \rightarrow \mathbb{C}^2 \otimes \mathbb{C}^3$. Then $L^*L = P$, $LL^* = I_6$, and the exact shell-corner map is

$$\Lambda(X) = C\Phi(X)C^* = LXL^*, \quad \Lambda^{-1}(Y) = L^*YL. \tag{ESHL61}$$

The proof of ESHL55 applies verbatim using the two displayed identities, and every rectangle in ESHL56 has image $C_a B(V_b, V_a) C_b^*$, where C_n is the restriction of the table to V_n . Its source and target projections are exactly $C_n C_n^*$. The transported density is CdC^* , and all identities ESHL58 follow by this unitary conjugation. To recover any previously specified shell operator Y , including $Q_<, Q_>, D, U, H_h, K_{a,\odot}$ on one orbit, its unique supported Figure 10 source is L^*YL ; hence the relation is onto and retains every mixed operator.

This also gives a precise bridge between the two nine-dimensional ambient presentations. On one orbit, let $V_{\Delta} = \mathbf{U}_2^* \otimes I_3$ be the actual simplex support isometry from ESHL5. The map $Z = V_{\Delta}L$ obeys $Z^*Z = P$, $ZZ^* = P_2 \otimes I_3$, because both L and V_{Δ} have the initial and final projections already proved. Thus $X \mapsto ZXZ^*$ is an isomorphism of their two support corners, with inverse $Y \mapsto Z^*YZ$. It sends the Figure 10 kernel to zero and has range orthogonal to the tensor-ambient kernel; both three-dimensional kernels are retained as the specified complements. It is a partial isometry of ambient spaces and a unitary on the six-dimensional supports, not an unlabelled identification of the ambient coordinates.

The exact effect of using the whole nine-dimensional matrix algebra can also be retained. Extend the same formula as the linear map $\Lambda^{\text{amb}}(X) = LXL^*$ for every $X \in M_9(\mathbb{C})$. For arbitrary X, Y , direct subtraction

gives

$$\Lambda^{\text{amb}}(XY) - \Lambda^{\text{amb}}(X)\Lambda^{\text{amb}}(Y) = LX(I - P)YL^*. \quad (\text{ESHL64})$$

Indeed the second term is $LXPYL^*$ because $L^*L = P$. Thus ESHL64 records exactly the part of a composition passing through the retained ambient kernel. It vanishes on the support corner, recovering the proved algebra isomorphism, but that corner hypothesis has not been omitted for arbitrary ambient matrices.

For an explicit nonzero instance, choose any unit shell vector e , put $u = L^*e$, and take the unit vector $k = (1, 1, 0, \dots, 0)/\sqrt{2}$ in the segment's constant-line kernel. Then $Lu = e$, $Lk = 0$, and u, k are orthogonal unit vectors. The ambient operators $X = uk^*$, $Y = ku^*$ each have zero image under Λ^{amb} , but $XY = uu^*$ has image $ee^* \neq 0$. This computes a composition through the retained kernel, with its nonzero support return, rather than deleting it.

Finally tensor these exact maps over the retained orbit set. Write

$$\begin{aligned} \mathcal{A}_{\mathcal{O}} &= \ell^2(\mathcal{O}) \otimes \mathbb{C}^9, & P_{\mathcal{O}}^{123} &= I \otimes P, & L_{\mathcal{O}} &= I \otimes L, \\ \Lambda_{\mathcal{O}} : P_{\mathcal{O}}^{123} B(\mathcal{A}_{\mathcal{O}}) P_{\mathcal{O}}^{123} &\longrightarrow B(\mathcal{H}_{\mathcal{O}}), & \Lambda_{\mathcal{O}}(X) &= L_{\mathcal{O}} X L_{\mathcal{O}}^*, & \Lambda_{\mathcal{O}}^{-1}(Y) &= L_{\mathcal{O}}^* Y L_{\mathcal{O}}. \end{aligned} \quad (\text{ESHL62})$$

The operators are bounded, with $L_{\mathcal{O}}^* L_{\mathcal{O}} = P_{\mathcal{O}}^{123}$, $L_{\mathcal{O}} L_{\mathcal{O}}^* = I$, by the square-sum coordinate formula. The multiplication, inverse, and adjoint proofs of ESHL55 now use bounded-operator composition with these same identities and prove a unital $*$ -isomorphism of the full corners. It includes inter-orbit operators: $|O\rangle\langle O'| \otimes X_{ab}$ maps to $|O\rangle\langle O'| \otimes C_a \Phi_{ab}(X_{ab}) C_b^*$. Their products multiply the orbit matrix units and the original rectangles simultaneously. Thus this map retains more than the constant-fibre algebra. Finite complete-orbit inclusions commute with $L_{\mathcal{O}}$ and its adjoint by their coordinate formulas, so their compressions commute with $\Lambda_{\mathcal{O}}$ as well.

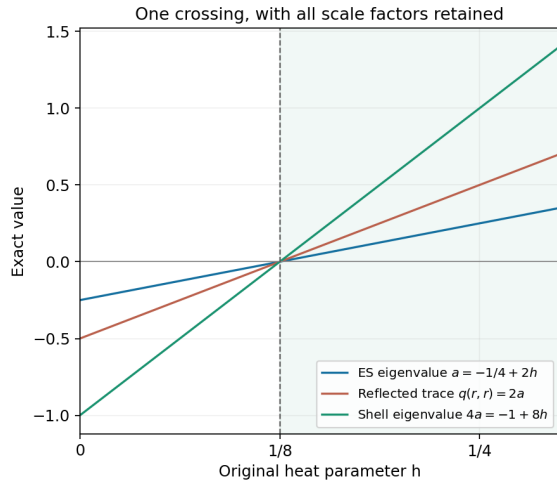
The same map restricts to a unitary between the Hilbert–Schmidt spaces of these support corners. Indeed in the orthonormal support basis $e_{\mathcal{O}} \otimes L^*(f_{\epsilon} \otimes g_m)$, the matrix entries of X equal the entries of $\Lambda_{\mathcal{O}}(X)$ in the original shell basis. Equality of their square sums proves both the Hilbert–Schmidt isometry and its surjectivity. Define the bounded positive support-invertible densities $B_{\mathcal{O}} = I \otimes b$ and $D_{\mathcal{O}} = I \otimes C d C^*$. They correspond under $\Lambda_{\mathcal{O}}$. Their inverses have the same tensor form. Consequently the conjugation operators and adjunction on the complete Hilbert–Schmidt spaces satisfy

$$\begin{aligned} \Lambda_{\mathcal{O}}(B_{\mathcal{O}} X B_{\mathcal{O}}^{-1, P}) &= D_{\mathcal{O}} \Lambda_{\mathcal{O}}(X) D_{\mathcal{O}}^{-1}, \\ \Lambda_{\mathcal{O}}(X^*) &= \Lambda_{\mathcal{O}}(X)^*, \\ \Lambda_{\mathcal{O}}(-\log \Delta_{B_{\mathcal{O}}}) &= (-\log \Delta_{D_{\mathcal{O}}}) \Lambda_{\mathcal{O}}. \end{aligned} \quad (\text{ESHL63})$$

For a complete proof of the operator domains, choose the six finite-fibre density eigenvectors. The complete Hilbert–Schmidt basis is indexed by pairs $(O, i), (O', j)$. Conjugation acts by λ_i/λ_j , independent of the two orbit labels. These ratios lie between $\lambda_{\min}/\lambda_{\max} > 0$ and $\lambda_{\max}/\lambda_{\min} < \infty$. Thus Δ , its inverse, its square root, and its logarithm are all bounded everywhere on the Hilbert–Schmidt space. The logarithm identity follows first on each such basis vector by its identical ratio, then on all square-summable series by boundedness. Adjunction reverses the two labels and conjugates their coefficients, proving its antiunitarity and the displayed intertwining. For the original density ESHL59 the ratio bounds are $1/11, 11$ and the logarithmic norm is $\log 11$, attained on the $1, 11$ matrix unit.

When the orbit set is infinite, $D_{\mathcal{O}}$ is not a trace-class density and $D_{\mathcal{O}}^{1/2}$ is not a Hilbert–Schmidt vector: each of infinitely many orbits contributes the positive sum of its six eigenvalues. ESHL63 is an exact statement about the bounded density-conjugation operators on the complete Hilbert–Schmidt space. It does not introduce a nonexistent finite-trace state vector. On each finite orbit compression the density square root is a Hilbert–Schmidt vector and the finite Tomita computation after ESHL58 applies with the same maps. This states the finite and infinite domains while retaining their exact connecting inclusions.

The source edition is available as the 488-page ES reader and its original TeX source and build archive. The TeX entry is `reader/erdos_straus_project_reader.tex`; the source identity and exact reading locations appear in the source guide.



The same lift on finitely or infinitely many orbits

$$\mathcal{H} = \mathcal{L}^2(\mathcal{O}) \otimes \mathbb{C}^2 \otimes \mathbb{C}^3$$

$$\Psi(P_+) = Q_-, \quad \Psi(P_-) = Q_+$$

$$H_h = 4Q_- + (-1 + 8h)Q_+$$

$$\langle v, H_h v \rangle \geq \min(4, -1 + 8h) \|v\|^2$$

for every $v \in \mathcal{H}$ and $h > 1/8$.

At $h = 1/8$: the negative-sector form vanishes;
the two square-zero blades and their mixed products remain.

Figure 5: The three exact changing coefficients and their common crossing. The quarter coefficient is a , its reflected trace is $2a$, and the shell coefficient is $4a$. ESHL12–ESHL25 and ESHL34–ESHL42 prove the finite and infinite maps, sharp bound, surviving blades and trace scope.

The actual Weil correction has the ES three-plus-one form

This calculation identifies the finite correction in the full explicit formula with the original ES quarter form. It uses actual compactly supported smooth tests, proves a section onto all four evaluation coordinates, and retains the kernel. It does not identify this finite correction with the entire Weil pairing. The needed endpoint conventions and completion identity are proved in the complete heat endpoint sign bridge proof, HEB11–HEB17 and HEB23–HEB36; the exact original ES metric and its isometry are proved in the complete es quarter heat trace derivation proof, ESQ9–ESQ27. Every comparison needed here is calculated below.

1. Four evaluation coordinates of an admissible test

Let $\mathcal{T} = C_c^\infty(\mathbb{R}; \mathbb{C})$ and retain

$$M_f(s) = \int_{\mathbb{R}} f(v) e^{-(s-1/2)v} dv, \quad (s_1, s_2, s_3, s_4) = (0, 1, 1/2 + i/2, 1/2 - i/2). \quad (\text{WEC1})$$

Set $t_j = s_j - 1/2$, $k_j(v) = e^{-t_j v}$, and define

$$L : \mathcal{T} \longrightarrow \mathbb{C}^4, \quad Lf = (A, B, C, D) = (M_f(s_1), M_f(s_2), M_f(s_3), M_f(s_4)). \quad (\text{WEC2})$$

The integral defines an entire function of s , since its derivatives converge uniformly on each compact set and the support in v is compact.

Here is a concrete linear section for every four-tuple. Choose the normalized positive even smooth bump ψ on $(-1, 1)$ in HEB33 and put

$$G_{ij} = \int_{\mathbb{R}} \psi(v) k_i(v) \overline{k_j(v)} dv, \quad S(y)(v) = \psi(v) \sum_{j=1}^4 (G^{-1}y)_j \overline{k_j(v)}. \quad (\text{WEC3})$$

The matrix G is positive definite. Indeed, for $z \in \mathbb{C}^4$, $z^* G z = \int \psi(v) |\sum_i \bar{z}_i k_i(v)|^2 dv$. If this equals zero, the continuous sum vanishes on $(-1, 1)$, where $\psi > 0$. Its derivatives of orders 0, 1, 2, 3 at zero give $\sum_i \bar{z}_i (-t_i)^m = 0$. The four numbers $-t_i$ are distinct, so this Vandermonde matrix is invertible: its determinant is $\prod_{i < j} ((-t_j) - (-t_i)) \neq 0$, obtained by the usual polynomial alternating-factor expansion (both sides have the same degree and leading coefficient). Thus $z = 0$. This proves positive definiteness and existence of the displayed inverse.

The section $S(y)$ is smooth and compactly supported because it is a finite sum of smooth exponentials times ψ . Direct integration gives

$$LS(y) = GG^{-1}y = y. \quad (\text{WEC4})$$

Consequently L is onto, and $\mathcal{T} = \ker L \oplus S(\mathbb{C}^4)$ as complex vector spaces, with projection SL . This proves that no hidden condition on admissible tests prevents the four coordinates used below.

2. The exact endpoint gain and its full-form compensation

At heat time $h = 1/4$, the moving endpoint polynomial is $s(s-1) + 1/2$, whose roots are s_3, s_4 . The two forms on actual tests are therefore

$$\begin{aligned} P_0(f, g) &= \overline{A_f} B_g + \overline{B_f} A_g, \\ P_{1/4}(f, g) &= \overline{C_f} C_g + \overline{D_f} D_g. \end{aligned} \quad (\text{WEC5})$$

This follows from the endpoint involution $s^\# = 1 - s$ together with coefficient conjugation: it exchanges 0, 1 and fixes each critical-line point in the reflected evaluation. Define the finite gain form

$$\mathcal{C} = P_{1/4} - P_0. \quad (\text{WEC6})$$

In the actual four values its quadratic expression is

$$\mathcal{C}(f, f) = |C|^2 + |D|^2 + \frac{1}{2}|A - B|^2 - \frac{1}{2}|A + B|^2. \quad (\text{WEC7})$$

Expansion of the last two squares gives $-\overline{A}B - \overline{B}A$, proving the formula. In coordinates $(C, D, (A-B)/\sqrt{2}, (A+B)/\sqrt{2})$, its matrix is $\text{diag}(1, 1, 1, -1)$. Hence its quotient has signature $(3, 1)$. More precisely its radical on $\mathcal{T}$ is exactly $\ker L$: the inclusion follows from (WEC5), while if $Lf \neq 0$, nondegeneracy of this displayed matrix supplies a vector y with nonzero pairing against Lf , and the admissible test $S(y)$ detects it by (WEC4).

Let $\mathcal{A}$ and $\mathcal{P}$ denote the original archimedean and rational-prime terms. The full Weil form remains

$$W = P_0 + \mathcal{A} - \mathcal{P} = P_{1/4} + (\mathcal{A} - \mathcal{C}) - \mathcal{P}. \quad (\text{WEC8})$$

This is the exact completion correction, not a discarded residual: logarithmic differentiation of $\Lambda_{1/4} = 2\xi/[s(s-1) + 1/2]$ adds $q'_0/q_0 - q'_{1/4}/q_{1/4}$, and the residues at the four simple roots yield $P_0 - P_{1/4} = -\mathcal{C}$. Thus the correction subtracted from the archimedean term is the same nondegenerate $(3, 1)$ quotient just computed; its negative has signature $(1, 3)$.

3. An explicit isometry to the original ES quarter form

Retain the original raw ordering $(E_{11}, E_{12}, E_{21}, E_{22})$ and the forced ES metric and initial quarter operator

$$G_{\text{ES}} = \begin{pmatrix} 1/16 & 0 & 0 & -7/144 \\ 0 & 2/9 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -7/144 & 0 & 0 & 1/16 \end{pmatrix}, \quad A_0 = \text{diag}(1/4, 1/4, -1/4, 1/4). \quad (\text{WEC9})$$

They are the original source matrices used in ESQ9 and ESQ2. Define the linear map $J : \mathbb{C}^4 \rightarrow M_2(\mathbb{C})$ by

$$\begin{aligned} x_{11} &= 12C + 3\sqrt{2}D, & x_{22} &= 12C - 3\sqrt{2}D, \\ x_{12} &= 3(A - B), & x_{21} &= \sqrt{2}(A + B). \end{aligned} \quad (\text{WEC10})$$

It is invertible, with

$$C = \frac{x_{11} + x_{22}}{24}, \quad D = \frac{x_{11} - x_{22}}{6\sqrt{2}}, \quad A = \frac{1}{2} \left(\frac{x_{21}}{\sqrt{2}} + \frac{x_{12}}{3} \right), \quad B = \frac{1}{2} \left(\frac{x_{21}}{\sqrt{2}} - \frac{x_{12}}{3} \right). \quad (\text{WEC11})$$

Substitution in both orders gives each of the four starting coordinates. From the displayed matrix multiplication,

$$x^* G_{\text{ES}} A_0 x = \left| \frac{x_{11} + x_{22}}{24} \right|^2 + \left| \frac{x_{11} - x_{22}}{6\sqrt{2}} \right|^2 + \frac{|x_{12}|^2}{18} - \frac{|x_{21}|^2}{4}. \quad (\text{WEC12})$$

For example the first two squares expand to $(|x_{11}|^2 + |x_{22}|^2)/64 - (7/288) \text{Re}(\overline{x_{11}}x_{22})$, exactly the two diagonal entries and mixed term of $G_{\text{ES}}A_0$. Substituting (WEC10) in (WEC12) gives (WEC7). Polarization, or the same expansion in two slots, therefore proves the isometry

$$\boxed{\mathcal{C}(f, g) = (JLf)^* G_{\text{ES}} A_0 (JLg)}. \quad (\text{WEC13})$$

The map JL is onto, has kernel $\ker L$, and has explicit section SJ^{-1} . This proves an isomorphism of the nondegenerate Hermitian quotient $\mathcal{T}/\ker L$ with the actual complexified ES quarter form, including its coordinates, constants and sign. It is a linear form isomorphism, not an asserted ring isomorphism of a test-function multiplication with raw matrix multiplication.

The form identity links the tetrahedral description to an actual term of the Weil formula: it is the finite endpoint-change correction. It does not identify the remaining $\mathcal{A} - \mathcal{P}$, or the full form W , with this four-dimensional quotient. Equation (WEC8) records exactly where that quotient enters.

4. Actual positive and negative gain tests

The section (WEC3) supplies explicit admissible tests for both signs. For

$$f_- = S(1, 1, 0, 0), \quad f_+ = S(1, -1, 0, 0), \quad (\text{WEC14})$$

one has $P_{1/4}(f_{\pm}, f_{\pm}) = 0$, while $P_0(f_-, f_-) = 2$ and $P_0(f_+, f_+) = -2$. Thus $\mathcal{C}(f_-, f_-) = -2$ and $\mathcal{C}(f_+, f_+) = 2$. These are tests of the endpoint gain; they do not give negative values of the full classical Weil form. The bump derivative in HEB33–HEB36 supplies a single test whose endpoint term passes from negative, through a nonzero nilpotent with zero pairing, to positive, with the exact opposite change in the archimedean term.

5. The actual test involution and its exact ES intertwiner

The involution on tests is $f^\#(v) = \overline{f(-v)}$. Changing variables in (WEC1) gives $M_{f^\#}(s) = \overline{M_f(1 - \bar{s})}$, so its action on the four coordinates is

$$(A, B, C, D)^\# = (\bar{B}, \bar{A}, \bar{C}, \bar{D}). \quad (\text{WEC15})$$

Under the map J in (WEC10), the transported action on raw coordinates is $T_0 x = \text{diag}(1, -1, 1, 1)\bar{x}$: only $A-B$ changes sign. The original ES anti-linear quarter involution is instead $T_{\text{ES}} x = (4A_0)\bar{x} = \text{diag}(1, 1, -1, 1)\bar{x}$. Their exact comparison is

$$T_{\text{ES}} = C_0 T_0, \quad C_0 = \text{diag}(1, -1, -1, 1). \quad (\text{WEC16})$$

On raw matrices C_0 is conjugation by $\text{diag}(1, -1)$, as direct multiplication shows. Both involutions and their relation are thereby retained.

There is also an isometry that intertwines the two anti-linear actions. Put

$$U = \text{diag}(1, i, i, 1), \quad \hat{J} = UJ. \quad (\text{WEC17})$$

The only off-diagonal entries of $G_{\text{ES}} A_0$ connect the first and fourth positions, on which U is the identity. On the other two positions multiplication by i preserves the absolute squares. Hence $U^*(G_{\text{ES}} A_0)U = G_{\text{ES}} A_0$. For the anti-linear comparison the matrix is $U \text{diag}(1, -1, 1, 1)\bar{U}^{-1} = \text{diag}(1, 1, -1, 1)$, by multiplication of its four diagonal entries. Consequently

$$\mathcal{C}(f, g) = (\hat{J}L f)^* G_{\text{ES}} A_0 (\hat{J}L g), \quad \hat{J}L(f^\#) = T_{\text{ES}} \hat{J}L(f). \quad (\text{WEC18})$$

The inverse and section are $J^{-1}U^{-1}$ and $SJ^{-1}U^{-1}$. Explicitly $\hat{J}$ changes just two entries of (WEC10) to $x_{12} = 3i(A - B)$, $x_{21} = i\sqrt{2}(A + B)$. Thus both the Hermitian form and the specified involution are preserved by a proved map. The map U is not a raw-matrix algebra automorphism: it fixes E_{11} , while $U(E_{12})U(E_{21}) = -E_{11}$. No multiplication compatibility beyond its stated linear and involution properties is assumed.

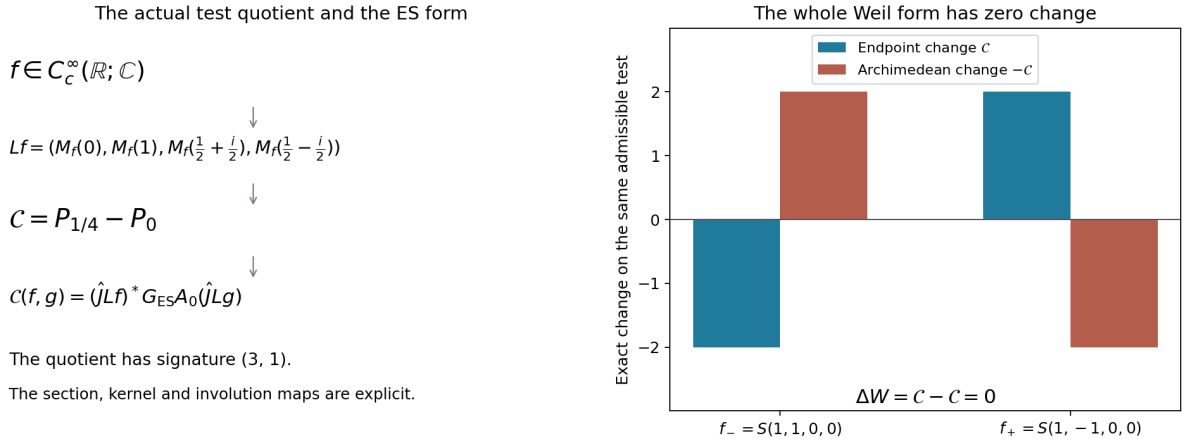


Figure 6: The actual four-value correction, its exact ES isometry, and its cancellation in the full Weil form. WEC1–WEC18 give all maps, the section and the two admissible tests.

The signed eight-state completion and its exact heat receivers

This note compares the original seven-point affine chart with its eight-state signed completion, keeping both original involutions. It calculates the complete trace forms, identifies the missing infinity state, and constructs a target path whose marked-root quotient is exactly the HEB heat family. The whole eight-state form is calculated on that path, including its length-four collision algebra.

The original sources actually read are *The ES–Fable inverse correspondence in the original weighted conductor*, FC36–FC47, and *The finite signed-root completion with all original fibre algebras*, FS1–FS27 and FS30–FS31. The latter pinned TeX has the same bytes as the local source used here. Its reciprocal-coordinate construction, chart transitions, trace and local algebra proofs are retained. The receiving HEB equations are HEB15–HEB20 and HEB31–HEB32 of the complete heat endpoint sign bridge proof, read in full during the quarter comparison. The original heat clock also has the public proof *The original heat trajectories*, eq:ifh-target-flow. All new comparisons and their hypotheses are proved below.

1. The original coordinates and the two completion charts

Write the first source coordinate called a in FS1 as α , solely to distinguish it from the HEB coefficient a_H . This is the identity renaming of that coordinate. The other source coordinates remain (y, z, w) . The original factors are

$$\begin{aligned} b &= i + \alpha y, & c &= -i + 2\alpha y + \alpha^2 z, \\ d &= -iy - \alpha(iz + 2y^2) - \alpha^2 yz, & e &= 2z - 7iy^2 + \alpha w, \\ f &= iw + 3iy^3 - 4yz + \alpha(6iy^2z + wy + 4y^4) + 2\alpha^2 y^3 z. \end{aligned} \quad (\text{ESH1})$$

The target is $\mathbf{U} = (A, B, C, D)$, where

$$P = (\alpha c, \alpha e + bd, \alpha f + be, bf), \quad H_{\mathbf{U}}(U, V) = AU^4 + U^3V + BU^2V^2 + CUV^3 + DV^4. \quad (\text{ESH2})$$

The retained factor identities are $\alpha d + bc = 1$ and $\alpha^3 f - \alpha^2 be + \alpha b^2 d - b^3 c = 1$. They follow by substituting (ESH1), and are the exact identities proved in FS1 and FC44–FC44b. Thus $(\alpha U + bV)(cU^3 + dU^2V + eUV^2 + fV^3) = H_{\mathbf{U}}$, with resultant one. The cubic coefficient is fixed at one; it is not another free parameter.

Use R for the finite marked root called r in FS2, and T for its reciprocal factor called t . The finite completed chart and the infinity chart are

$$F_{\mathbf{U}}(R) = AR^4 + R^3 + BR^2 + CR + D, \quad F_{\mathbf{U}}(R) = 0, \quad T^2 = F'_{\mathbf{U}}(R), \quad (\text{ESH3})$$

$$k_{\mathbf{U}}(v) = A + v + Bv^2 + Cv^3 + Dv^4, \quad k_{\mathbf{U}}(v) = 0, \quad \Theta^2 = -k'_{\mathbf{U}}(v). \quad (\text{ESH4})$$

On their overlap the maps are

$$v = R^{-1}, \quad \Theta = -T/R, \quad R = v^{-1}, \quad T = -\Theta/v. \quad (\text{ESH5})$$

They are inverse by substitution. Differentiating $k(v) = v^4 F(1/v)$ at a root gives $k'(v) = -v^2 F'(1/v)$, which verifies the squared-coordinate relation and its sign. On the original finite factor chart,

$$R = -b/\alpha = -y - i/\alpha, \quad T = 1/\alpha, \quad \alpha^2 F'_{\mathbf{U}}(R) = 1. \quad (\text{ESH6})$$

The finite completion allows $T = 0$; it does not declare $1/\alpha$ to be an invertible coordinate at that boundary.

For reference, the inverse factor formulas on $T \neq 0$ are

$$\begin{aligned} \alpha &= T^{-1}, & b &= -R/T, & c &= AT, \\ d &= (1 + AR)T, & e &= (B + R + AR^2)T, & f &= (C + BR + R^2 + AR^3)T. \end{aligned} \quad (\text{ESH7})$$

They give $\alpha d + bc = 1$ directly. Their four product coefficients are A, B, C , and $-R(C + BR + R^2 + AR^3) = D$, the last by $F(R) = 0$. Their resultant is $T^{-2} F'(R) = 1$. Thus these formulas prove the factor reconstruction with its original reciprocal coordinate.

At $v = 0$, the infinity equations give

$$A = 0, \quad \Theta^2 = -1, \quad \Theta = -i \text{ is included, } \Theta = i \text{ is omitted.} \quad (\text{ESH8})$$

Indeed the infinity factor chart has $b = \Theta^{-1}$; the original source at $\alpha = 0$ has $b = i$ by (ESH1), so its value is $\Theta = -i$. The other sign has $b = -i$ and does not belong to that original chart. This is the precise meaning of the missing eighth state.

The missing infinity point is unramified. The relative derivative matrix of (ESH4) with respect to (v, Θ) has determinant $2\Theta k'(v)$; at $v = 0$ its value is $2\Theta \neq 0$, since $k'(0) = 1$. This matrix is invertible there, proving the vanishing of the relative differential module and the absence of ramification in that chart. On the finite chart the same determinant is $2TF'(R) = 2T^3$, so its ramification support is $T = 0$. The infinity points never belong to that support. These calculations reproduce the relevant part of FS16–FS17 and prove that the omitted infinity sign is not the nilpotent collision.

2. The entire completed fibre at the original seven-point target

At

$$\mathbf{U}_* = (0, 0, -1, 0), \quad H_*(U, V) = UV(U - V)(U + V), \quad F_*(R) = R^3 - R, \quad (\text{ESH9})$$

the projective roots are $\infty, 0, 1, -1$, all simple. At the finite roots the derivative values are

$$F'_*(0) = -1, \quad F'_*(1) = F'_*(-1) = 2. \quad (\text{ESH10})$$

The completed complex fibre algebra is therefore exactly

$$\mathcal{A}_* = \frac{\mathbb{C}[T_0]}{(T_0^2 + 1)} \times \frac{\mathbb{C}[T_+]}{(T_+^2 - 2)} \times \frac{\mathbb{C}[T_-]}{(T_-^2 - 2)} \times \frac{\mathbb{C}[\Theta]}{(\Theta^2 + 1)}. \quad (\text{ESH11})$$

To prove the finite splitting, the idempotents in $\mathbb{C}[R]/(R^3 - R)$ are

$$e_0 = 1 - R^2, \quad e_+ = (R^2 + R)/2, \quad e_- = (R^2 - R)/2. \quad (\text{ESH12})$$

They evaluate as the three separate indicator functions at $0, 1, -1$. Their products, sum and squares are consequently the orthogonal idempotent identities: a polynomial remainder of degree at most two vanishing at the three distinct roots is zero. Applying those idempotents to the relation $T^2 = 3R^2 - 1$ gives the first three factors in (ESH11). The infinity root is the disjoint fourth root of the projective root fibre, and (ESH8) gives its fourth factor. Each quadratic has two distinct roots, so each is reduced and of dimension two. The total dimension is eight and every geometric point is reduced.

The eight points are

$$(0, \pm i), \quad (1, \pm\sqrt{2}), \quad (-1, \pm\sqrt{2}), \quad (\infty, \Theta = \pm i). \quad (\text{ESH13})$$

The first entry marks the projective root; the second is its retained sign coordinate. The source chart includes the first six and $(\infty, -i)$. At the latter, FS13 gives the original point $(\alpha, y, z, w) = (0, 0, i/2, 0)$. At (∞, i) , the factor coordinates are exactly

$$(\alpha, b, c, d, e, f) = (0, -i, i, 0, -i, 0), \quad (\text{ESH14})$$

by (ESH8) and the product coefficients. This is the omitted sign in FC44b, not an additional root or an identification of two labels.

3. Both original involutions, with coefficient conjugation separate

The completed sign deck involution is

$$\Sigma : (R, T) \mapsto (R, -T), \quad (v, \Theta) \mapsto (v, -\Theta). \quad (\text{ESH15})$$

Its formulas agree through (ESH5), preserve the equations, and square to the identity. They are the completed version of the simultaneous factor sign in FC43–FC44. Let κ conjugate coefficients and fix the displayed coordinate generators; it is defined on a fibre over real (A, B, C, D) . The conjugate-deck involution is $\#_\Sigma = \Sigma\kappa$, so it negates T, Θ and fixes the real root variable R , while conjugating scalars.

The other original involution is

$$J(\alpha, y, z, w) = (-\alpha, -y, z, -w). \quad (\text{ESH16})$$

Substituting in (ESH1) sends the factor tuple to $(-\alpha, b, c, -d, e, -f)$, so its target map and completion charts are

$$J_{\text{out}}(A, B, C, D) = (-A, -B, C, -D), \quad (\text{ESH17})$$

$$J : (R, T) \mapsto (-R, -T), \quad (v, \Theta) \mapsto (-v, \Theta). \quad (\text{ESH18})$$

The first formula follows from (ESH6), and the second from (ESH5). To verify the equations on their whole domains, one has $F_{J_{\text{out}}\mathbf{U}}(-R) = -F_{\mathbf{U}}(R)$, $F'_{J_{\text{out}}\mathbf{U}}(-R) = F'_{\mathbf{U}}(R)$, and the identical two identities for k at $-v$. Thus (ESH18) is the full map, including completion points. The target $\mathbf{U}_*$ is fixed, so $\#_J = J\kappa$ is an involution of this fibre. It is different from $\#_\Sigma$.

4. The complete trace forms and their signatures

For the quadratic algebra $K_\mu = \mathbb{C}[T]/(T^2 - \mu)$, with real $\mu \neq 0$, multiplication in the unchanged basis $(1, T)$ gives

$$\text{Tr}_{K_\mu}(c + dT) = 2c. \quad (\text{ESH19})$$

Indeed multiplication by T has matrix $\begin{pmatrix} 0 & \mu \\ 1 & 0 \end{pmatrix}$. Hence the coefficient-conjugation and conjugate-deck Hermitian forms have respective Gram matrices

$$[Q_\kappa] = \text{diag}(2, 2\mu), \quad [Q_{\#_\Sigma}] = \text{diag}(2, -2\mu), \quad Q_\iota(x, y) = \text{Tr}(\iota(x)y). \quad (\text{ESH20})$$

These formulas follow also by multiplying $(\bar{c} + \bar{d}T)(e + fT)$ or $(\bar{c} - \bar{d}T)(e + fT)$ and taking twice the scalar coefficient. They verify the full forms in both arguments.

In the factor order of (ESH11), with each basis $(1, T)$ retained, the whole Gram matrices are

$$[Q_\kappa] = \text{diag}(2, -2, 2, 4, 2, 4, 2, -2), \quad (\text{ESH21})$$

$$[Q_{\#_\Sigma}] = \text{diag}(2, 2, 2, -4, 2, -4, 2, 2). \quad (\text{ESH22})$$

Both happen to have inertia $(6, 2, 0)$, but their negative subspaces belong to different factors. Here and below inertia lists positive, negative and radical dimensions. The complex bilinear trace without an antilinear involution has no ordered signature; (ESH21) is its coefficient-conjugation Hermitian version, or its real coefficient restriction.

The native involution has a different complete matrix. In the ordered basis

$$(e_0, e_0T, e_+, e_-, e_+T, e_-T, e_\infty, e_\infty\Theta)$$

one has

$$[Q_{\#_J}] = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \oplus \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & -4 \\ -4 & 0 \end{pmatrix} \oplus \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}. \quad (\text{ESH23})$$

For proof, (ESH18) fixes the root 0 and negates its sign, interchanges e_+, e_- and negates T , and fixes the infinity coordinate Θ . Thus, for example, $\#_J(e_+T) = -e_-T$, whose pairing with e_-T is $-\text{Tr}(2e_-) = -4$. The mixed products between different root factors vanish, and (ESH20) gives the remaining blocks. Each off-diagonal two-by-two block has one eigenvalue of each sign. Therefore

$$\text{inertia}(Q_{\#_J}) = (5, 3, 0). \quad (\text{ESH24})$$

Neither calculation assumes that one of these finite forms is the global Weil form.

5. Omitting precisely one of the eight states

Let $m = (\infty, i)$ be the missing point. Its idempotent, supported in the infinity factor, is

$$e_m = (0, 0, 0, (1 - i\Theta)/2), \quad e_n = (0, 0, 0, (1 + i\Theta)/2), \quad (\text{ESH25})$$

where $n = (\infty, -i)$ is the retained point. Evaluation at $\Theta = \pm i$ verifies that these are the respective indicator functions. Therefore the exact original seven-state quotient is

$$q : \mathcal{A}_* \longrightarrow \mathcal{A}_7 := \mathcal{A}_*/\mathbb{C}e_m, \quad \mathcal{A}_* \cong \mathcal{A}_7 \times \mathbb{C}. \quad (\text{ESH26})$$

The last factor is evaluation at m . Its inverse inserts that value at the omitted coordinate, while the canonical nonunital section of q inserts zero there. These maps are the full coordinate maps, not a count alone.

Coefficient conjugation in (ESH25) gives the exact stability results

$$\#_\Sigma(e_m) = e_m, \quad \Sigma(e_m) = \kappa(e_m) = e_n, \quad J(e_m) = e_m, \quad \#_J(e_m) = e_n. \quad (\text{ESH27})$$

Thus $\#_\Sigma$ descends to $\mathcal{A}_7$, whereas Σ , κ , and $\#_J$ individually do not. Pure native J does descend. For the descended conjugate-deck trace one has, on arbitrary lifts f, g ,

$$Q_{\#_\Sigma, \tau}(qf, qg) = Q_{\#_\Sigma, *}(f, g) - \overline{f(m)}g(m). \quad (\text{ESH28})$$

This follows from the direct-product trace and the fact that m is fixed by the point action of $\#_\Sigma$. The removed summand is positive. Consequently

$$\text{inertia}(Q_{\#_\Sigma, \tau}) = (5, 2, 0). \quad (\text{ESH29})$$

In particular the two explicit retained vectors e_+T, e_-T still have value -4 , and are mutually orthogonal by (ESH22). Removing the omitted infinity point has not removed either negative direction.

For the native Hermitian form there is instead the restriction to the retained corner $(1 - e_m)\mathcal{A}_*$. Its infinity line $\mathbb{C}e_n$ is radical because $\#_J(e_n) = e_m$, whose product with every retained element is zero. The other six-dimensional block has inertia $(4, 2, 0)$ from (ESH23), so this restricted form has

$$\text{inertia}(Q_{\#_J}|_{(1-e_m)\mathcal{A}_*}) = (4, 2, 1). \quad (\text{ESH30})$$

It is not a trace form of a descended native involution. Indeed $Q_{\#_J}(e_m, e_n) = \text{Tr}(e_n) = 1$, so adding a representative from the killed ideal can change the original pairing. This proves directly why that full form cannot be defined on the quotient by choosing arbitrary lifts.

The exact native-stable quotient generated by this omission is also determined. The smallest $\#_J$ -stable ideal containing $\mathbb{C}e_m$ is

$$\mathcal{I}_{\text{stable}} = \mathbb{C}e_m \oplus \mathbb{C}e_n = e_\infty \mathcal{A}_*. \quad (\text{ESH30a})$$

Every stable ideal containing e_m must contain its image e_n ; conversely their sum is an ideal and is stable by (ESH27). This proves minimality and the universal quotient property: any algebra quotient with an induced native involution and killing e_m factors uniquely through $\mathcal{A}_6 = \mathcal{A}_*/\mathcal{I}_{\text{stable}}$, the product of the three finite root factors. Its native trace form has inertia $(4, 2, 0)$ by the first three blocks of (ESH23). The restricted seven-dimensional form (ESH30) factors through the further projection $\mathcal{A}_7 \rightarrow \mathcal{A}_6$, whose kernel is its radical $\mathbb{C}e_n$. This gives the complete projection and radical maps, rather than an involution assigned to a nonstable quotient.

Restoring the eighth coordinate has a different exact effect. In the infinity point basis (e_m, e_n) , the native trace block is

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (\text{ESH30b})$$

The diagonal entries vanish because the two different point idempotents multiply to zero after applying $\#_J$, and the cross entries are the trace-one values just computed. Its eigenvectors $e_m + e_n$ and $e_m - e_n$ have positive and negative values 2 and -2 . Completing the eighth state therefore restores the full nondegenerate hyperbolic pair, whose restriction to the retained one-dimensional infinity corner was radical. Under $\#_\Sigma$, by contrast, these two point idempotents are separately fixed and form two positive scalar summands, as (ESH27)–(ESH29) show.

There is a literal occurrence of the fraction $1/8$ in the normalized trace, with a different meaning. If $\text{tr}_8 = \text{Tr}/8$, then

$$\text{tr}_8(1) = 1, \quad \text{tr}_8(e_m) = 1/8, \quad \text{tr}_8(e_\infty) = 1/4. \quad (\text{ESH31})$$

These follow because the three idempotents have ranks eight, one and two as multiplication projectors. The value $1/8$ is the normalized mass of that one reduced state. It supplies no parameter evolution by itself. The next sections give actual parameter maps and calculate their full forms.

6. Exact boundary maps to the HEB quadratic forms

Write the HEB algebra and involution as

$$E_{a_H} = \mathbb{C}[r_H]/(r_H^2 + a_H), \quad r_H^\# = -r_H, \quad z^\# = \bar{z} \ (z \in \mathbb{C}). \quad (\text{ESH32})$$

Its full regular trace form in basis $(1, r_H)$ is $\text{diag}(2, 2a_H)$, by the same multiplication argument as (ESH19)–(ESH20). The original heat clock is $a_H(h) = -1/4 + 2h$.

For the infinity algebra $I = \mathbb{C}[\Theta]/(\Theta^2 + 1)$, the exact conjugate-deck isomorphism is

$$\phi_\Sigma : E_{1/4} \xrightarrow{\sim} I, \quad r_H \mapsto \Theta/2, \quad \phi_\Sigma^{-1}(\Theta) = 2r_H. \quad (\text{ESH33})$$

The squared relation follows from $\Theta^2 = -1$, and substitution gives both inverse compositions. Since $\#_\Sigma(\Theta) = -\Theta$, the map preserves the involution. An algebra isomorphism conjugates multiplication matrices in the corresponding bases, so preserves regular trace and hence the full reflected pairing. Equivalently the target matrix $\text{diag}(2, 2)$ pulls back under $r_H = \Theta/2$ to $\text{diag}(2, 1/2)$, exactly the HEB matrix at $a_H = 1/4$. That value corresponds to heat time $h = 1/4$, not the nilpotent time $h = 1/8$.

For the native involution on the same infinity algebra there is a different exact comparison:

$$\phi_J : E_{-1/4} \xrightarrow{\sim} I, \quad r_H \mapsto i\Theta/2, \quad \phi_J^{-1}(\Theta) = -2ir_H. \quad (\text{ESH34})$$

Here $(i\Theta/2)^2 = 1/4$, so the source relation is respected. Native $\#_J$ fixes Θ and conjugates i , hence negates $i\Theta/2$. Both inverse compositions and involution compatibility follow by substitution, and the algebra isomorphism again preserves trace. Its form has inertia $(1, 1, 0)$, corresponding to HEB heat time $h = 0$. The scalar i is part of the map: this is a complex algebra and conjugate-linear-involution comparison, not an identification of the coefficient-fixed real presentations. In particular (ESH33) must not be used as a native-involution isometry.

The factor projection $\mathcal{A}_* \rightarrow I$ and the corner inclusion $I \rightarrow \mathcal{A}_*$, with zero in the three other factors, are explicit algebra maps. The projection is unital; the inclusion sends its unit to e_∞ . The inclusion preserves regular trace and the corresponding involution, while the projection retains only its factor contribution to the total trace. No unital trace-preserving map from the entire rank-eight algebra to this rank-two algebra could exist, since the traces of their units are eight and two. The full trace comparison is the sum of the four factor traces in (ESH11), not the projection alone.

The reduced infinity pair is not the collision algebra E_0 , and their precise morphisms show what is lost. Every unital complex algebra map $I \simeq \mathbb{C}^2 \rightarrow E_0$ is one of the two point evaluations followed by the scalar inclusion. Indeed an idempotent in E_0 has form $c + dr_H$; its square equation gives $c^2 = c$ and $(2c - 1)d = 0$, hence it is 0 or 1. The two complementary idempotents of I must therefore map to 0 and 1. Conversely each choice is a map. Every unital map $E_0 \rightarrow I$ kills r_H , since its image would be nilpotent in the reduced algebra I ; hence it is the reduction map followed by the diagonal scalar inclusion. Thus every composite from E_0 through I kills its nonzero nilpotent. The retained dual-number direction is genuinely present in the other heat receiver constructed next.

7. A target path whose marked-root quotient is exactly the original heat algebra

Define the following explicit path in the unchanged quartic coefficient target:

$$\gamma_+(h) = (0, 0, -1 - 16h, 16h), \quad F_h(R) = R^3 - (1 + 16h)R + 16h = (R - 1)(R^2 + R - 16h). \quad (\text{ESH35})$$

It begins at $\mathbf{U}_*$. This is a target path chosen to pull back the HEB clock; it is not asserted to be a native PDE evolution of the whole quartic. The marked root $R = 1$ is retained for every parameter, and

$$F'_h(1) = 2 - 16h = -8a_H(h). \quad (\text{ESH36})$$

The finite signed algebra over this path is

$$\mathcal{B} = \frac{\mathbb{C}[h, R, T]}{(F_h(R), T^2 - (3R^2 - 1 - 16h))}. \quad (\text{ESH37})$$

It is free of rank six over $\mathbb{C}[h]$: first divide by the monic cubic in R , then by the monic quadratic in T , giving basis $1, R, R^2, T, TR, TR^2$. Evaluation at the marked root gives the surjective algebra map

$$\pi_1 : \mathcal{B} \longrightarrow \mathcal{K} = \mathbb{C}[h, T]/(T^2 - 2 + 16h), \quad R \longmapsto 1, \quad T \longmapsto T, \quad \ker \pi_1 = (R - 1). \quad (\text{ESH38})$$

The kernel assertion follows by taking the quotient of the displayed presentation by $R - 1$; the resulting presentation is exactly $\mathcal{K}$. It supplies the entire scheme-theoretic marked-root quotient even where the root is multiple.

There is a full family isomorphism

$$\mathbb{C}[h, r_H]/(r_H^2 - 1/4 + 2h) \xrightarrow{\sim} \mathcal{K}, \quad r_H \longmapsto \frac{T}{2\sqrt{2}}, \quad \text{inverse: } T \longmapsto 2\sqrt{2}r_H. \quad (\text{ESH39})$$

The relation follows because $T^2/8 = 1/4 - 2h$; the inverse substitution recovers both generators. The scale $2\sqrt{2}$ is retained. For real h , $\#_\Sigma$ fixes the marked root and sends T to $-T$, so the quotient and this isomorphism preserve the HEB involution. The isomorphism preserves the regular trace of the rank-two receiver by conjugating its multiplication matrices. The projection selects that receiver; its relation to the whole trace is given separately below. In particular the receiver's pulled-back full trace form is exactly

$$Q_{\text{heat},h} = 2|c|^2 + 2a_H(h)|d|^2 \quad \text{on } c + dr_H. \quad (\text{ESH40})$$

This is an actual comparison with the HEB threshold $h = 1/8$, arising from a finite marked-root factor, not from the missing infinity sign.

Away from $h = 1/8$, the marked factor is an actual direct summand of the finite fibre. An explicit root idempotent is

$$e_1(R; h) = \frac{R^2 + R - 16h}{2 - 16h}. \quad (\text{ESH41})$$

It has value 1 at $R = 1$ and vanishes on the other quadratic factor. Since the two factors are comaximal when $2 - 16h \neq 0$, these values prove its idempotence in the quotient; alternatively multiply the polynomial identities and reduce by F_h . The inverse to the resulting factor decomposition inserts the rank-two algebra by multiplying its representatives by e_1 . Its regular trace equals the whole-fibre trace on that corner because multiplication is zero on the complementary factors. At $h = 1/8$, the denominator vanishes and that idempotent expression cannot be retained; (ESH38) remains the valid quotient.

The infinity component stays constant along the entire path: $A = 0$, $k'_h(0) = 1$, and $\Theta^2 = -1$ by (ESH8). Thus the completed whole fibre is the finite signed algebra of (ESH37) plus this constant rank-two infinity factor. The omission of $\Theta = i$ is unchanged throughout the path. In the original source its included point is

$$(\alpha, y, z, w) = \left(0, 0, \frac{i(1 + 16h)}{2}, -16h\right), \quad (\text{ESH42})$$

obtained from the original $\alpha = 0$ formula $P = (0, y, 2iz + 7y^2, -4iyz - w - 3y^3)$. Its target is exactly $\gamma_+(h)$. This supplies a concrete check that the old infinity chart and its omitted sign remain different from the moving finite-root collision.

8. The full eight-state trace on the chosen path

For $h \geq 0$, put $s_h = \sqrt{1 + 64h} > 0$. The three finite roots are

$$R_1 = 1, \quad R_+(h) = \frac{-1 + s_h}{2}, \quad R_-(h) = \frac{-1 - s_h}{2}. \quad (\text{ESH43})$$

They are distinct except when $h = 1/8$, where $R_+ = R_1 = 1$ and $R_- = -2$. This follows directly from their formulas; $R_- \leq -1$ never meets the other two. The exact derivative values are

$$\mu_1 = 2 - 16h, \quad \mu_+ = \frac{s_h(s_h - 3)}{2}, \quad \mu_- = \frac{s_h(s_h + 3)}{2} > 0. \quad (\text{ESH44})$$

For proof, differentiate (ESH35). At a root of its quadratic factor the product derivative is $(R - 1)(2R + 1)$, and substitution of (ESH43) gives the last two expressions; the first is (ESH36).

For distinct roots the complete fibre decomposition is

$$\mathcal{A}_h \simeq \mathbb{C}[T_1]/(T_1^2 - \mu_1) \times \mathbb{C}[T_+]/(T_+^2 - \mu_+) \times \mathbb{C}[T_-]/(T_-^2 - \mu_-) \times I. \quad (\text{ESH45})$$

The projections evaluate R at the three displayed roots and use the infinity chart for I . The inverse maps use the three Lagrange idempotents $e_j(R) = \prod_{k \neq j} (R - R_k)/(R_j - R_k)$, multiplying a sign polynomial by its root idempotent. Their denominators are nonzero exactly in the stated distinct-root domain. Evaluations prove their orthogonal-idempotent relations, sum and inverse compositions, as in (ESH12). Thus the entire decomposition and all factor trace comparisons are explicit.

Each real root factor is preserved by $\#_\Sigma$. By (ESH20), a derivative $\mu < 0$ contributes two positive dimensions, while $\mu > 0$ contributes one positive and one negative dimension. The infinity factor always contributes two positive dimensions. For $0 \leq h < 1/8$ the derivative signs in the order (ESH44) are $(+, -, +)$; for $h > 1/8$ they are $(-, +, +)$. Hence

$$\boxed{\text{inertia}(Q_{\#_\Sigma, \mathcal{A}_h}) = (6, 2, 0) \quad (h \geq 0, h \neq 1/8).} \quad (\text{ESH46})$$

The sign gained by the fixed marked-root heat factor is exactly accompanied by the opposite change in the R_+ factor. The R_- factor remains indefinite throughout. An explicit negative vector on that persistent factor is e_-T , with value

$$Q_{\#_\Sigma}(e_-T, e_-T) = -2\mu_- < 0. \quad (\text{ESH47})$$

This follows because its square has scalar coefficient μ_-e_- , the involution negates its sign, and the trace of e_- is 2. Removing the one infinity sign merely removes a positive dimension, so the seven-dimensional quotient has inertia $(5, 2, 0)$ for these noncritical parameters as well.

9. The full collision algebra and its marked-root quotient

At $h_c = 1/8$,

$$F_{h_c}(R) = R^3 - 3R + 2 = (R - 1)^2(R + 2). \quad (\text{ESH48})$$

Near the repeated root write $\epsilon = R - 1$. The other factor $R + 2 = 3 + \epsilon$ is a unit in that local root algebra, so the root relation is exactly $\epsilon^2 = 0$. Differentiating gives $F'_{h_c}(1 + \epsilon) = 6\epsilon + 3\epsilon^2$, hence the signed local algebra is

$$\mathcal{L} = \frac{\mathbb{C}[\epsilon, T]}{(\epsilon^2, T^2 - 6\epsilon)} \xrightarrow{\sim} \mathbb{C}[T]/(T^4), \quad \epsilon \mapsto T^2/6. \quad (\text{ESH49})$$

The inverse sends T to its original sign coordinate. Squaring its relation gives $T^4 = 36\epsilon^2 = 0$, and the two substitutions recover ϵ, T . This proves the isomorphism with its full coefficient 6 retained. Its basis is $1, T, T^2, T^3$, or equivalently $1, \epsilon, T, \epsilon T$, so its dimension is four.

For a fully explicit whole-fibre decomposition at this parameter, the root idempotent for $R = -2$ is

$$e_{-2} = (R - 1)^2/9, \quad e_{\text{dbl}} = 1 - e_{-2}. \quad (\text{ESH50})$$

The first is 1 at -2 and zero in the double-root algebra; therefore it is idempotent modulo (ESH48). These two factors are comaximal because the retained difference is 3, so they give inverse product projections. Since $F'_{h_c}(-2) = 9$, the complete signed fibre is

$$\mathcal{A}_{h_c} \simeq \mathcal{L} \times \mathbb{C}[T_-]/(T_-^2 - 9) \times I. \quad (\text{ESH51})$$

The maps are given by e_{dbl}, e_{-2} , the local substitution (ESH49), and the infinity factor; this retains all eight dimensions.

Under $\#_\Sigma$, ϵ is fixed and T is negated. For any $f \in \mathcal{L}$, multiplication by its nonconstant part is nilpotent: it lies in the nilpotent ideal (T) , and its multiplication operator has fourth power zero. Thus its trace is zero, whereas its constant part acts on four basis vectors. This proves the complete local form

$$\text{Tr}_{\mathcal{L}}(f) = 4f(0), \quad Q_{\#_\Sigma, \mathcal{L}}(f, g) = 4\overline{f(0)}g(0), \quad \text{rad } Q = (T). \quad (\text{ESH52})$$

Its inertia is $(1, 0, 3)$. The simple -2 factor contributes $(1, 1, 0)$, and infinity contributes $(2, 0, 0)$. Therefore

$$\boxed{\text{inertia}(Q_{\#_\Sigma, \mathcal{A}_{h_c}}) = (4, 1, 3)}. \quad (\text{ESH53})$$

In particular the entire collision is still not a positive form: the vector supported on the -2 sign coordinate has value -18 .

The exact HEB receiver at the collision is the further quotient

$$0 \longrightarrow (T^2)/(T^4) \longrightarrow \mathcal{L} \xrightarrow{\bar{\pi}_1} \mathbb{C}[T]/(T^2) \longrightarrow 0, \quad r_H \mapsto T/(2\sqrt{2}). \quad (\text{ESH54})$$

The kernel is a two-dimensional square-zero ideal with basis T^2, T^3 . This follows directly from (ESH49), since the marked-root quotient sets $\epsilon = 0$, equivalently $T^2 = 0$. Both the kernel and quotient are stable under the stated involution. The quotient has trace $2f(0)$, so its trace is related to the full local trace by

$$\text{Tr}_{\mathcal{L}}(f) = 2 \text{Tr}_{E_0}(\bar{\pi}_1 f), \quad Q_{\mathcal{L}}(f, g) = 2Q_{E_0}(\bar{\pi}_1 f, \bar{\pi}_1 g). \quad (\text{ESH55})$$

The factors of two are required by the lengths four and two. The map sends a nonzero nilpotent to the HEB nilpotent and kills the specified further two-dimensional ideal; it does not identify the length-four object with dual numbers.

Deleting only the omitted infinity state from the completed collision gives a seven-dimensional algebra with inertia $(3, 1, 3)$. The original affine source removes more at this parameter: FS15 removes the ramification component $T = 0$, whose open complement is empty on $\mathcal{L}$ because T is nilpotent. Thus its actual three reduced points are the two signs over -2 and the included infinity sign. Their conjugate-deck trace has inertia $(2, 1, 0)$. These assertions follow by deleting the stated direct factors in (ESH51); they distinguish the seven-dimensional completed quotient from the original affine fibre at ramification.

10. The native involution relates two different target paths

The native map in (ESH17) sends the path (ESH35) to

$$\gamma_-(h) = (0, 0, -1 - 16h, -16h), \quad F_h^-(R) = (R + 1)(R^2 - R - 16h). \quad (\text{ESH56})$$

For $h \neq 0$ these are different target points. The exact completion map between their fibres remains

$$(R, T) \mapsto (-R, -T), \quad (v, \Theta) \mapsto (-v, \Theta), \quad (\text{ESH57})$$

as proved in (ESH18). It sends the fixed marked root $+1$ of γ_+ to the fixed marked root -1 of γ_- , whose derivative is also $2 - 16h$. Under the two marked-root isomorphisms (ESH39), it sends the HEB sign generator to its negative; adding coefficient conjugation gives the HEB involution as an antilinear map **between** those two receivers. It is not a self-involution of the whole $\gamma_+(h)$ fibre for nonzero h . At the initial common target it instead exchanges its two different root factors, giving the mixed native form (ESH23). This is why the full-path calculation (ESH46) is specifically for $\#_\Sigma$, without silently substituting the native involution or the global Weil involution.

11. What the two occurrences of one eighth retain

The one-eighth normalized trace value in (ESH31) belongs to one reduced infinity state. The HEB boundary at $h = 1/8$ belongs, under (ESH35)–(ESH39), to a derivative becoming zero at a finite marked root. The explicit maps prove a useful relation to the same eight-state completion, but they locate these events in different factors. The infinity sign pair remains reduced and unramified throughout the path; the finite collision retains the larger length-four algebra (ESH49) and its exact dual-number quotient.

For any fixed support lattice L , the commutative ring quotients and isomorphisms above induce $G_L(f) : (x, \lambda) \mapsto (f(x), \lambda)$. Preservation of the amplitude operations follows from the ring map, and support joins and meets are unchanged, proving these are the actual support-semiring maps. In (ESH54), the retained nonzero amplitude T has supported square e_L in the dual-number quotient, distinct from τ_L . Its nonzero squared amplitude in the length-four source lies in the explicitly killed ideal. The absence of the infinity state is instead the idempotent quotient (ESH26), with the whole kernel $\mathbb{C}e_m$ displayed. The two maps therefore retain different, exactly specified information.

The complete calculation yields no positive whole eight-state trace at the proposed one-eighth boundary: the conjugate-deck inertias are (ESH46) and (ESH53), and the native initial inertia is (ESH24). The positive HEB factor after the boundary is real mathematical data, with the additional indefinite factor and all nilpotent multiplicities retained in the same calculation.

12. The whole trace in a polynomial basis through the collision

The root idempotents in (ESH45) have denominators that vanish at the collision. The following calculation instead uses the free polynomial basis of (ESH37) over the entire parameter line. Introduce the coefficient abbreviations

$$c_h = 1 + 16h, \quad d_h = 16h, \quad K_h = \mathbb{C}[R]/(R^3 - c_h R + d_h), \quad B_h = K_h[T]/(T^2 - 3R^2 + c_h). \quad (\text{ESH58})$$

These are exactly the finite root and signed fibre algebras already used, with the original root and sign coordinates unchanged. The notation also defines free algebras over $\mathbb{C}[h]$ before specializing h . The infinity factor I remains the one in (ESH33), so the whole fibre is $\mathcal{A}_h = B_h \times I$. Use the ordered bases

$$\begin{aligned} \mathcal{E}_R &= (1, R, R^2), & \mathcal{E}_B &= (1, R, R^2, T, TR, TR^2), \\ \mathcal{E}_8 &= ((1, 0), (R, 0), (R^2, 0), (T, 0), (TR, 0), (TR^2, 0), (0, 1), (0, \Theta)). \end{aligned} \quad (\text{ESH59})$$

The entries of $\mathcal{E}_8$ lie in $B_h \times I$; in particular its first entry is the finite-component unit, and its seventh entry is the infinity-component unit. Their sum is the whole unit. This fixed decomposition uses the disjoint finite and infinity root components proved after (ESH41), without splitting the moving finite roots.

Let M_R be multiplication by R in $\mathcal{E}_R$, with each column the image of its corresponding basis vector. The defining monic cubic gives

$$M_R = \begin{pmatrix} 0 & 0 & -d_h \\ 1 & 0 & c_h \\ 0 & 1 & 0 \end{pmatrix}, \quad M_R^3 = c_h M_R - d_h I_3. \quad (\text{ESH60})$$

Thus, writing $p_j = \text{Tr}_{K_h}(R^j) = \text{tr}(M_R^j)$, direct traces of I_3, M_R, M_R^2 and multiplication of (ESH60) by further powers give

$$\begin{aligned} p_0 &= 3, & p_1 &= 0, & p_2 &= 2c_h, & p_j &= c_h p_{j-2} - d_h p_{j-3} \quad (j \geq 3), \\ p_3 &= -3d_h, & p_4 &= 2c_h^2, & p_5 &= -5c_h d_h, & p_6 &= 2c_h^3 + 3d_h^2. \end{aligned} \quad (\text{ESH61})$$

For example the recurrence gives $p_3 = c_h p_1 - d_h p_0 = -3d_h$, then gives each subsequent displayed term by substitution. The calculation is in $\mathbb{C}[h]$, so remains valid on every fibre, including nonreduced ones.

Define the two root-trace matrices in the unchanged basis by

$$\begin{aligned} P_h &= (p_{i+j})_{0 \leq i, j \leq 2} = \begin{pmatrix} 3 & 0 & 2c_h \\ 0 & 2c_h & -3d_h \\ 2c_h & -3d_h & 2c_h^2 \end{pmatrix}, \\ Q_h &= (3p_{i+j+2} - c_h p_{i+j})_{0 \leq i, j \leq 2} = \begin{pmatrix} 3c_h & -9d_h & 4c_h^2 \\ -9d_h & 4c_h^2 & -12c_h d_h \\ 4c_h^2 & -12c_h d_h & 4c_h^3 + 9d_h^2 \end{pmatrix}. \end{aligned} \quad (\text{ESH62})$$

Every entry follows from (ESH61). The second is precisely $\text{Tr}_{K_h}((3R^2 - c_h)R^{i+j})$, retaining the derivative multiplier $3R^2 - c_h = F'_h(R)$.

For clarity, these root traces determine the full signed trace without a reducedness assumption. Multiplication by $a \in K_h$ acts on the two summands $K_h \oplus TK_h$ by the same matrix M_a , so its trace in B_h is $2 \text{Tr}_{K_h}(a)$. Multiplication by Tb , for $b \in K_h$, has block matrix

$$L_{Tb} = \begin{pmatrix} 0 & M_{(3R^2 - c_h)b} \\ M_b & 0 \end{pmatrix} \quad \text{on } K_h \oplus TK_h. \quad (\text{ESH63})$$

Its trace is zero. Even-even products therefore give $2P_h$, mixed products give zero, and odd-odd products give $2Q_h$ for the complex bilinear regular trace. For real h , coefficient conjugation fixes R, T , while $\#_\Sigma$ fixes R and negates T . Consequently their complete Hermitian matrices in $\mathcal{E}_8$ are

$$\begin{aligned} [Q_{\kappa, \mathcal{A}_h}]_{\mathcal{E}_8} &= \text{diag}(2P_h, 2Q_h, 2, -2), \\ [Q_{\#_\Sigma, \mathcal{A}_h}]_{\mathcal{E}_8} &= \text{diag}(2P_h, -2Q_h, 2, 2). \end{aligned} \quad (\text{ESH64})$$

The last two entries are obtained by multiplying in $I = \mathbb{C}[\Theta]/(\Theta^2 + 1)$ with the respective involutions, as in (ESH20). These are polynomial matrix formulas for the whole eight-dimensional family. They do not substitute the native involution $\#_J$, whose distinct target map is still (ESH57).

12.1. The critical ranks and the complete radical

At $h_c = 1/8$, the retained coefficients are $c_{h_c} = 3$ and $d_{h_c} = 2$. Put

$$u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}.$$

Direct substitution in (ESH62) gives

$$\begin{aligned} P_{h_c} &= \begin{pmatrix} 3 & 0 & 6 \\ 0 & 6 & -6 \\ 6 & -6 & 18 \end{pmatrix} = 2uu^\top + vv^\top, \\ Q_{h_c} &= \begin{pmatrix} 9 & -18 & 36 \\ -18 & 36 & -72 \\ 36 & -72 & 144 \end{pmatrix} = 9vv^\top. \end{aligned} \quad (\text{ESH65})$$

For a coefficient vector $z \in \mathbb{C}^3$, $z^* P_{h_c} z = 2|u^\top z|^2 + |v^\top z|^2$. The two rows $u^\top$ and $v^\top$ are independent, since their first entries agree and their second entries differ. Thus P_{h_c} is positive semidefinite of rank two; its kernel is the simultaneous solution of these two linear equations. The second matrix has rank one because $v \neq 0$. The kernels, with all coordinates retained, are

$$\begin{aligned} \ker P_{h_c} &= \mathbb{C}(-2, 1, 1)^\top, \\ \ker Q_{h_c} &= \{(z_0, z_1, z_2)^\top : z_0 - 2z_1 + 4z_2 = 0\} \\ &= \text{span}_{\mathbb{C}}\{(2, 1, 0)^\top, (-2, 1, 1)^\top\}. \end{aligned} \quad (\text{ESH66})$$

The displayed vectors satisfy the equations and are independent; the ranks just proved show that they exhaust the kernels. It follows immediately from the blocks of (ESH64) that the finite conjugate-deck form has inertia $(2, 1, 3)$, and its whole eight-state form has inertia $(4, 1, 3)$. This proves (ESH53) directly in a basis that is defined across the collision.

More explicitly, write any element of $\mathcal{A}_{h_c}$ uniquely as $(p(R) + Tq(R), \eta_0 + \eta_1\Theta)$, with p, q of degree at most two. For two such elements $x = (p + Tq, \eta_0 + \eta_1\Theta)$ and $y = (\tilde{p} + T\tilde{q}, \tilde{\eta}_0 + \tilde{\eta}_1\Theta)$, (ESH64)–(ESH65) give the full pairing

$$Q_{\#_\Sigma}(x, y) = 4\overline{p(1)}\tilde{p}(1) + 2\overline{p(-2)}\tilde{p}(-2) - 18\overline{q(-2)}\tilde{q}(-2) + 2\overline{\eta_0}\tilde{\eta}_0 + 2\overline{\eta_1}\tilde{\eta}_1. \quad (\text{ESH67})$$

Each independent coefficient test in this formula can be realized by a degree-at-most-two polynomial, or by one infinity basis vector. For example the two constant and linear polynomials interpolate arbitrary values at 1, -2 , and a constant polynomial realizes any prescribed $q(-2)$. Hence the radical is exactly the subspace with $p(1) = p(-2) = q(-2) = \eta_0 = \eta_1 = 0$, not the larger cone of vectors whose single diagonal value happens to vanish. Factoring the degree-at-most-two polynomials gives the following complete radical in the original finite coordinates:

$$\mathfrak{N} = \text{rad } Q_{\#_\Sigma, \mathcal{A}_{h_c}} = \text{span}_{\mathbb{C}}\{n_1, n_2, n_3\}, \quad (\text{ESH68})$$

$$n_1 = ((R-1)(R+2), 0), \quad n_2 = (T(R+2), 0), \quad n_3 = (T(R-1)(R+2), 0).$$

Indeed the first polynomial spans the degree-at-most-two polynomials vanishing at both roots, while $R+2$ and $(R-1)(R+2)$ are independent and span the polynomials vanishing at -2 . These vectors are independent in the fixed free basis (ESH59).

12.2. The radical maps to the retained length-four local algebra

The critical product projection (ESH51) sends the three vectors of (ESH68) to zero in the simple -2 and infinity factors. In its length-four factor retain $R = 1 + \epsilon$, $\epsilon^2 = 0$, $T^2 = 6\epsilon$. Their exact images are

$$n_1 \mapsto T^2/2, \quad n_2 \mapsto 3T + T^3/6, \quad n_3 \mapsto T^3/2. \quad (\text{ESH69})$$

For the first image, $(R-1)(R+2) = \epsilon(3+\epsilon) = 3\epsilon = T^2/2$. Multiplication by T , or expansion of $T(3+\epsilon)$, proves the other two. Conversely the local classes T, T^2, T^3 , extended by zero to the other factors, are the images of

$$T \longleftrightarrow n_2/3 - n_3/9, \quad T^2 \longleftrightarrow 2n_1, \quad T^3 \longleftrightarrow 2n_3. \quad (\text{ESH70})$$

Both compositions are checked by substituting (ESH69). Thus the radical is exactly the entire nilpotent ideal (T) of $\mathbb{C}[T]/(T^4)$, extended by zero to the reduced factors. This identifies it as an ideal and as the nilradical of the whole critical algebra, not merely as a vector subspace of zero trace. The connecting maps preserve $\#_\Sigma$: n_1 is fixed, n_2, n_3 are negated, and the respective powers T^2, T, T^3 have those same signs.

The complete multiplication of the three retained generators is

$$n_1^2 = n_1 n_3 = n_2 n_3 = n_3^2 = 0, \quad n_2^2 = 18n_1, \quad n_1 n_2 = 3n_3, \quad (\text{ESH71})$$

$$\mathfrak{N}^2 = \text{span}_{\mathbb{C}}(n_1, n_3), \quad \mathfrak{N}^3 = \mathbb{C}n_3, \quad \mathfrak{N}^4 = 0.$$

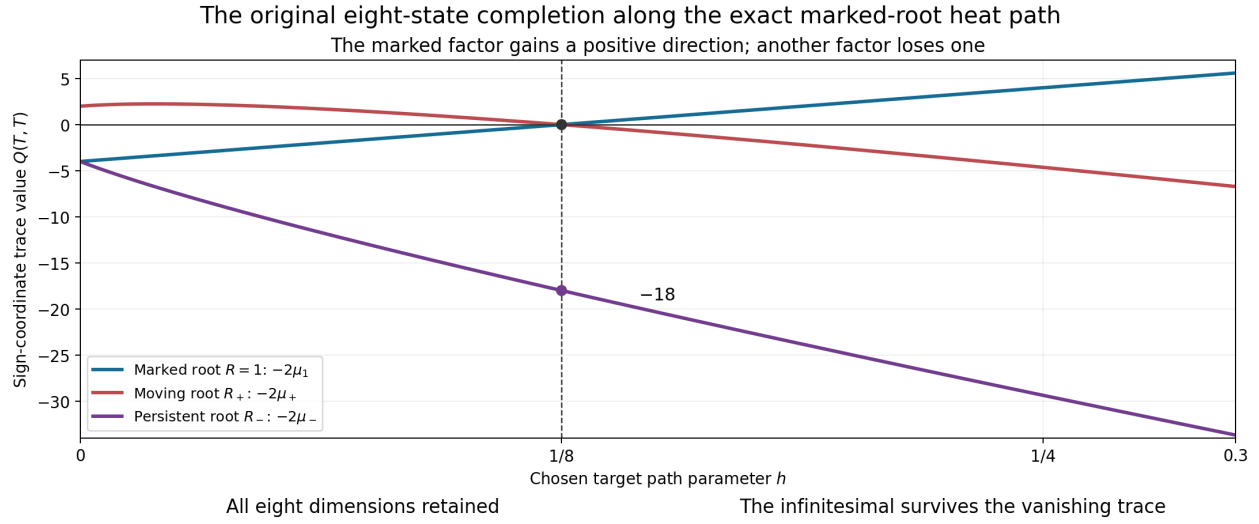
To prove every product, substitute (ESH69), multiply in $\mathbb{C}[T]/(T^4)$, and use the inverse (ESH70). The other components are zero. For instance $(3T + T^3/6)^2 = 9T^2 = 18(T^2/2)$ and $(T^2/2)(3T + T^3/6) = 3T^3/2$. All stated zero products have degree at least four. The displayed nonzero products, together with the independence of the three generators, prove the exact ideal powers and their dimensions. In particular $n_2^3 = 54n_3 \neq 0$ and $n_2^4 = 0$.

Finally, restrict the marked-root map (ESH38) at the collision to this radical. It evaluates $R = 1$ and imposes $T^2 = 0$, so

$$\pi_1(n_1) = 0, \quad \pi_1(n_2) = 3T = 6\sqrt{2}r_H, \quad \pi_1(n_3) = 0, \quad (\text{ESH72})$$

$$0 \longrightarrow \mathfrak{N}^2 \longrightarrow \mathfrak{N} \xrightarrow{\pi_1} \mathbb{C}r_H \longrightarrow 0.$$

Here $\mathbb{C}r_H$ is the nonzero nilpotent ideal in the HEB dual-number receiver, with its original scale $r_H = T/(2\sqrt{2})$. The map is onto since its value on n_2 is nonzero; (ESH68) and these images show that its kernel is precisely $\text{span}(n_1, n_3) = \mathfrak{N}^2$. This is an exact sequence of the corresponding nonunital ideals and of complex vector spaces, with their inherited involutions. It exhibits the full retained infinitesimal, its two-dimensional kernel, and the actual heat quotient in coordinates valid at the collision. No root idempotent with a vanishing denominator enters (ESH58)–(ESH72).



Parameter	Positive	Negative	Radical
$0 \leq h < 1/8$	6	2	0
$h = 1/8$	4	1	3
$h > 1/8$	6	2	0

Involution: $\#_z(T) = -T$, with scalar conjugation.

Infinity stays $\Theta^2 = -1$: two positive states.

The omitted infinity state is unramified.
It is separate from this finite-root collision.

$$F_{1/8}(R) = (R - 1)^2(R + 2)$$

$$\mathcal{L} = \mathbb{C}[T]/(T^4)$$

$$\varepsilon = R - 1 = T^2/6$$

$$\downarrow \text{kernel } (T^2)/(T^4)$$

$$E_0 = \mathbb{C}[T]/(T^2), \quad r_H = T/(2\sqrt{2})$$

$$Q_{\mathcal{L}}(f, g) = 2Q_{E_0}(\tilde{r}_1 f, \tilde{r}_1 g)$$

Figure 7: The full eight-state conjugate-deck trace along the specified target path. ESH35–ESH55 prove the three derivative functions, their sign exchange, the persistent negative factor, and the complete length-four collision with its dual-number quotient.

The supported-zero prime in the full theta and Weil trace distribution

This construction begins with the actual semiring $G(\mathbb{Z})$, includes its supported-zero prime in the full theta sum, and constructs an unconditional trace distribution with every fixed-support endpoint retained. Its arithmetic spectral sector is the rational, unramified, trivial-character sector relevant to the Riemann zeta function. It proves the maps from actual Weil tests to theta functions, from theta functions to their Mellin divisor receiver, and from the full supported object to the arithmetic scalar receiver. The arithmetic projection of carriers, the pushforward of functions, and the projection which forgets a point-function component are different maps and are calculated separately.

The resulting identity is a distributional trace identity on explicitly constructed representations. It is not an assertion of the global operator equality on Connes's adèle-class Hilbert space: the original source explicitly distinguishes its unconditional explicit formula from that further equality. The full supported trace, its scalar arithmetic observation, and its Fourier-closed endpoint enlargement are all given below.

Sources read for this derivation: the complete programme calculations the complete tau prime spectrum derivation proof (TPS1–TPS59), the complete tau connes endpoint derivation proof (TC1–TC41, including TC24a–d), and the complete full support connes derivation proof (FL1–FL58); the original supported-zero prime theorem, labels `thm:prime_ideals_S`, `thm:krull_dim_S` and `eq:maximal_chain`; and Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, arXiv:math/9811068v1. The original author TeX was read at lines 608–822, 1574–1735, 3398–3490, 4217–4416 and 4770–4906. Its Appendix I (36)–(38) supplies the full Poisson formula; Appendix II Theorem 6 is the unconditional explicit formula. Section VI explicitly calls its proposed global operator trace formal. The new maps and all extra support terms below are derived, rather than inferred from the prime spectrum alone.

1. The prime and its actual arithmetic maps

Write

$$S = G(\mathbb{Z}) = \{\tau\} \sqcup \mathbb{Z}^\bullet, \quad e = 0^\bullet \neq \tau, \\ m^\bullet + n^\bullet = (m+n)^\bullet, \quad m^\bullet n^\bullet = (mn)^\bullet, \quad x + \tau = x, \quad x\tau = \tau. \quad (\text{SZW1})$$

The prime element e is a nonunit and is different from the semiring zero τ . Its principal ideal is $P_e = (e) = \{\tau, e\}$. A product belongs to this ideal precisely when one factor is unsupported or the product of its two integer amplitudes is zero. Since the integers are a domain, this forces one factor into P_e . Likewise $P_\tau = \{\tau\}$ is prime because a product is unsupported precisely when a factor is unsupported. All other primes are

$$P_p = \{\tau\} \cup (p\mathbb{Z})^\bullet, \quad P_\tau \subsetneq P_e \subsetneq P_p. \quad (\text{SZW2})$$

Here is a short exhaustive proof of the needed classification. An ideal containing any supported integer contains e , by multiplication by e . Its supported amplitudes are closed under integer multiplication and additive inverses, since multiplication by $(-1)^\bullet$ is allowed. They form an integer ideal, hence are $n\mathbb{Z}$; primeness is exactly primeness of that integer ideal. The remaining ideal is P_τ . This proves (SZW2), including height one for P_e and height two for every P_p .

The two maps

$$\pi : S \rightarrow \mathbb{Z}, \quad \pi(\tau) = 0, \quad \pi(n^\bullet) = n, \quad \chi : S \rightarrow \mathbb{B}, \quad \chi(\tau) = 0, \quad \chi(n^\bullet) = 1 \quad (\text{SZW3})$$

preserve both semiring operations, as direct substitution shows. Their joint map is injective. Contraction under π sends the ordinary generic prime (0) to P_e , and (p) to P_p . Its image is the closed subspace $V(e)$; $D(e) = \{P_\tau\}$. These assertions follow by testing containment of the generating element e in (SZW2).

The diagonal ring maps give semiring maps $S \rightarrow G(\mathbb{A}_\mathbb{Q})$ and $S \rightarrow G(\mathbb{Z}_p)$. If $\mathfrak{n}_v = \ker(\mathbb{A}_\mathbb{Q} \rightarrow \mathbb{Q}_v)$, with $\mathbb{Q}_\infty = \mathbb{R}$, then

$$(S \rightarrow G(\mathbb{A}_\mathbb{Q}))^{-1}(\{\tau\} \cup \mathfrak{n}_v^\bullet) = P_e. \quad (\text{SZW4})$$

Indeed a rational integer has zero image in any local field exactly when it is zero. At the integral prime p , reduction instead gives $S \rightarrow G(\mathbb{F}_p) \rightarrow \mathbb{F}_p$, whose arithmetic zero fibre is P_p . Thus the adelic supported-zero strata retain their separate place labels while contracting to the height-one prime P_e . This is the precise arithmetic connection used in the theta construction; it does not identify e with τ .

2. Full sums, orbital sums, and the supported-zero operator

Use the original self-dual additive measure: Lebesgue measure on $\mathbb{R}$, measure one for $\mathbb{Z}_p$ at each finite place, and hence covolume one for the diagonal $\mathbb{Q}$ in its adèle ring. These are the measures in Connes's Appendix I (34)–(36); the displayed constants are for these fixed measures. Let $\mathcal{S}_{\text{ev}}(\mathbb{R})$ denote the even complex Schwartz functions, with Fourier transform

$$\mathcal{F}\phi(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx. \quad (\text{SZW5})$$

For $x > 0$, put

$$\Theta_e \phi(x) = \sum_{n \in \mathbb{Z}} \phi(nx), \quad P\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad a = \phi(0), \quad b = \int_{\mathbb{R}} \phi. \quad (\text{SZW6})$$

The full supported sum is $\Theta_e \phi = P\phi + a$. It counts the element e exactly once. For every compact interval of positive x , the sums and their derivatives converge absolutely and uniformly: differentiating a term introduces a polynomial power of n , which is dominated by a higher Schwartz decay bound. For $x \geq 1$, the same estimate gives $P\phi(x) = O_N(x^{-N})$ for every $N > 1$. The assertion also holds for all its derivatives and for $\mathcal{F}\phi$.

Poisson summation in the stated Fourier convention gives the exact formulas

$$\begin{aligned} \Theta_e \phi(x) &= x^{-1} \Theta_e(\mathcal{F}\phi)(x^{-1}), \\ P\phi(x) &= x^{-1} P(\mathcal{F}\phi)(x^{-1}) + b/x - a. \end{aligned} \quad (\text{SZW7})$$

For the first formula apply Poisson to $y \mapsto \phi(xy)$, whose Fourier transform is $x^{-1}(\mathcal{F}\phi)(\xi/x)$. Removing precisely the zero-index term on both sides gives the second. Thus the two endpoint coefficients are the value at the supported zero and its Fourier dual, with their full signs retained.

The operator corresponding to multiplication by the supported zero also has an exact domain. On $\mathcal{S}(\mathbb{R}) + \mathbb{C}1$, precomposition with the integer map $x \mapsto nx$ is defined for every $n \in \mathbb{Z}$. For $n = 0$ it is

$$U_e \phi = \phi(0)1, \quad U_e^2 = U_e. \quad (\text{SZW8})$$

It has rank one on this space. It is not an endomorphism of the Schwartz subspace, because the nonzero constant function is not Schwartz. Its Fourier-dual domain is $\mathcal{S}(\mathbb{R}) \oplus \mathbb{C}\delta_0$, on which the exact operator is

$$\mathcal{F}U_e \mathcal{F}^{-1}(\psi + c\delta_0) = \left(\int_{\mathbb{R}} \psi(x) dx + c \right) \delta_0. \quad (\text{SZW8a})$$

Indeed $\mathcal{F}^{-1}(\psi + c\delta_0) = \mathcal{F}^{-1}\psi + c1$, whose value at zero is $\int \psi + c$. Applying U_e and then $\mathcal{F}$ gives the formula since $\mathcal{F}1 = \delta_0$. Its restriction to the Schwartz summand is $\psi \mapsto (\int \psi)\delta_0$. These are different maps with specified domains and targets, not an identification of a point-function with a Dirac distribution.

For the complete two-point carrier, a function is $(\phi, c) \in \mathcal{S}(\mathbb{R}) \oplus \mathbb{C}_\tau$, with value c at τ . Its sum over the rational integer carrier is

$$\Theta_S(\phi, c; x) = \sum_{u \in G(\mathbb{Z})} (\phi, c)(xu) = \Theta_e \phi(x) + c. \quad (\text{SZW9})$$

This converges by (SZW6); the unsupported point is counted once. The contracted convention that an unsupported point contributes no value is the exact invariant subspace $c = 0$, on which (SZW9) is the full supported sum $\Theta_e \phi$, still including e .

This carrier sum is different from evaluating an orbital sum at a fixed point. For $\mathbb{Q}^\times$ acting on $G(\mathbb{A}_{\mathbb{Q}})$, both e and τ are fixed, separately. Hence $\sum_{q \in \mathbb{Q}^\times} F(qe)$ is an infinite repetition of $F(e)$, and converges only when that value is zero; the corresponding statement at τ involves its independent value. The convergence assertion follows because a nonzero constant sequence cannot tend to zero. In contrast, for an idele x , $\sum_{q \in \mathbb{Q}^\times} f(qx)$ samples no zero point and converges by Schwartz decay and the compact support of its finite adelic factors. On the spherical input $f = \phi \otimes 1_{\widehat{\mathbb{Z}}}$ and idele $(x, 1, 1, \dots)$, its contributing rationals are exactly the nonzero integers, so it is precisely $P\phi(x)$. This supplies the actual map from the adelic orbital sum to (SZW6).

3. Full finite support and its Poisson defect

For the actual finite bounded distributive support lattice L , write

$$G_L(A) = \{(a, 1_L) : a \in A\} \cup \{z_\lambda = (0, \lambda) : \lambda \neq 1_L\}, \quad \tau_L = z_{0_L}, \quad e_L = z_{1_L}. \quad (\text{SZW10})$$

Addition uses amplitude addition and support join; multiplication uses amplitude multiplication and support meet. No nonzero amplitude is placed at a nontop support. Let $n = |L|$, and prescribe the function space

$$\mathcal{F}_L = \mathcal{S}_{\text{ev}}(\mathbb{R}) \oplus \bigoplus_{\lambda \neq 1_L} \mathbb{C} \delta_\lambda, \quad F = (\phi, (c_\lambda)), \quad C(F) = \sum_{\lambda \neq 1_L} c_\lambda. \quad (\text{SZW11})$$

This is the global fixed-sector carrier of FL1–FL12. It is not the coordinatewise carrier $Q_F = \prod_{v \in F} G(k_v)$ of TC32–TC41. That latter carrier has positive-dimensional strata $\prod_{v \in E} k_v$, whose integration characters are $\chi_E(j) = \prod_{v \in E} |j_v|_v$, as a change of variables in each coordinate proves. The exact inclusion $G(\prod_{v \in F} k_v) \rightarrow Q_F$ retains only the full-support stratum and the all-unsupported point; restriction of functions to these is the projection of TC41, with kernel the sum of all nonempty proper strata. No such stratum, or its distinct integration character, has been replaced here by a fixed line. The extra weight-one partners constructed in Section 9 are explicitly new Fourier partners of the FL fixed lines.

Here δ_λ is a point-function on the support carrier, not a tempered Dirac distribution. The full sum and the componentwise Fourier map are

$$\begin{aligned} \Theta_L(F; x) &= P\phi(x) + a + C(F), & \mathcal{F}_L^{\text{raw}}(\phi, c) &= (\mathcal{F}\phi, c). \\ \Theta_L(F; x) - x^{-1}\Theta_L(\mathcal{F}_L^{\text{raw}}F; x^{-1}) &= C(F)(1 - x^{-1}). \end{aligned} \quad (\text{SZW12})$$

Every equality follows by inserting (SZW7); thus the extra fixed constants give an exact Poisson defect. They are not extra terms in the punctured sum. With all support labels retained, the defect has coefficients $c_\lambda(1 - x^{-1})$ on the respective nontop labels, so cancellation in the scalar sum $C(F)$ does not remove the labelled defect.

On this space the dilation convention inherited from Connes is $(V(u)\phi)(x) = \phi(x/u)$, $u > 0$; all c_λ are fixed. Its endpoint map is onto:

$$E_L(F) = (a, b, (c_\lambda)_{\lambda \neq 1_L}), \quad B_L = \mathbb{C}(0)_e \oplus \mathbb{C}(1)_{e^\vee} \oplus \bigoplus_{\lambda \neq 1_L} \mathbb{C}(0)_\lambda. \quad (\text{SZW13})$$

The notation $\mathbb{C}(s)$ means the character $u \mapsto u^s$. Evaluation is invariant and $\int \phi(x/u)dx = u \int \phi$, proving these weights. Surjectivity of the original two endpoints follows, for example, by taking one even Schwartz function with nonzero value at zero and an even Schwartz function vanishing there but having nonzero integral, then solving the resulting triangular two-by-two system. The point coefficients give the other coordinate preimages independently. The kernel is exactly $\phi(0) = \int \phi = 0$, with every extra coefficient zero.

4. Mellin regularization retains two different boundary operations

For $\Re s > 1$, the punctured Mellin integral converges absolutely and is

$$Z_\phi(s) = \int_0^\infty P\phi(x)x^s \frac{dx}{x} = 2\zeta(s) \int_0^\infty \phi(x)x^s \frac{dx}{x}. \quad (\text{SZW14})$$

Indeed at zero (SZW7) bounds the integrand by a constant times $x^{\Re s-1} + x^{\Re s}$, and at infinity it decreases arbitrarily fast. Termwise integration is justified by the sum of absolute values; replacing x by nx then gives the displayed Dirichlet series. Evenness accounts for the factor two.

Splitting at one and using (SZW7) proves the complete continuation

$$Z_\phi(s) = \int_1^\infty P\phi(x)x^s \frac{dx}{x} + \int_1^\infty P(\mathcal{F}\phi)(x)x^{1-s} \frac{dx}{x} + \frac{b}{s-1} - \frac{a}{s}. \quad (\text{SZW15})$$

Both integrals are entire: on every compact set of s , every differentiated integrand is dominated by a sufficiently high inverse power of x , times a power of $\log x$. The residues are exactly $-a$ at zero and b at one. Formula (SZW15) has kept the full supported-zero and dual endpoint coefficients.

The full function $\Theta_L(F; x)$ usually has no Mellin convergence strip. Its asymptotic terms are

$$\begin{aligned} \Theta_L(F; x) &= a + C(F) + O_N(x^{-N}) \quad (x \rightarrow \infty), \\ \Theta_L(F; x) &= b/x + C(F) + O_N(x^N) \quad (x \downarrow 0). \end{aligned} \quad (\text{SZW16})$$

The second estimate follows from (SZW7), increasing its Schwartz exponent by one. Define its meromorphic finite part by subtracting these terms on the two half-lines, integrating the remaining rapidly decreasing

functions, and adding the meromorphic continuations of the subtracted elementary integrals. The elementary contributions are

$$\frac{b}{s-1} + \frac{C(F)}{s} - \frac{a+C(F)}{s} = \frac{b}{s-1} - \frac{a}{s}. \quad (\text{SZW17})$$

Here $\int_0^1 x^{s-1} dx = 1/s$ initially for $\Re s > 0$, while $\int_1^\infty x^{s-1} dx = -1/s$ initially for $\Re s < 0$. The equality of their meromorphic continuations does not posit a common domain of ordinary convergence. Thus the full finite part is exactly (SZW15). In particular deleting the small-end term $C(F)/s$ while retaining its large-end counterpart would introduce a spurious pole.

The cancellation in (SZW17) is not the trace of a fixed representation. To see the retained distribution, use $t = \log x$ and Fourier convention $\hat{h}(u) = \int h(t) e^{-iut} dt$. For a constant c , its exponentially damped Fourier transform is

$$\int_{\mathbb{R}} c e^{-\varepsilon|t|} e^{-iut} dt = \frac{c}{\varepsilon + iu} + \frac{c}{\varepsilon - iu} = \frac{2c\varepsilon}{\varepsilon^2 + u^2} \longrightarrow 2\pi c \delta_0(u) \quad (\text{SZW18})$$

in tempered distributions as $\varepsilon \downarrow 0$. For proof, pair the last expression with a Schwartz function g , substitute $u = \varepsilon v$, and apply dominated convergence to $2c \int g(\varepsilon v)/(1 + v^2) dv$. Its limit is $2\pi c g(0)$. The integral $\int (1 + v^2)^{-1} dv = \pi$ follows from the antiderivative $\arctan v$. The fixed line therefore contributes the character distribution $h \mapsto \int h(t) dt$, even though its meromorphic two-tail finite part is zero. These are two explicitly different receiving maps from the same constant function. Each lower support coefficient has its own copy of (SZW18).

5. An exact map from admissible Weil tests to the full theta object

Use precisely the existing HEB11 and WEC1 convention:

$$\mathcal{T} = C_c^\infty(\mathbb{R}; \mathbb{C}), \quad M_f(s) = \int_{\mathbb{R}} f(v) e^{-(s-1/2)v} dv, \quad f^\#(v) = \overline{f(-v)}. \quad (\text{SZW19})$$

Then $M_{f^\#}(s) = \overline{M_f(1 - \bar{s})}$, by changing variables. If the alternative plus-exponent convention is used, its exact relation is $M_f^+(s) = M_f(1 - s) = M_{f(-\cdot)}(s)$; no convention is silently interchanged.

Define a Gaussian dilation map with its complete amplitude:

$$(J_\theta f)(x) = \int_{\mathbb{R}} e^{v/2} f(v) \exp(-\pi e^{2v} x^2) dv. \quad (\text{SZW20})$$

This is even Schwartz. Indeed the support of f bounds e^{2v} above and below by positive constants. Every derivative in x , multiplied by every power of x , is consequently bounded by an integrable multiple of a Gaussian uniformly in v . Differentiation under the integral and the Schwartz estimates follow. Direct evaluation and the Gaussian integral give

$$(J_\theta f)(0) = M_f(0), \quad \int_{\mathbb{R}} J_\theta f = M_f(1), \quad \mathcal{F}(J_\theta f) = J_\theta(f(-\cdot)). \quad (\text{SZW21})$$

For the second equality, the Gaussian has integral e^{-v} . For the third, its Fourier transform is $e^{-v} \exp(-\pi e^{-2v} \xi^2)$; substituting $v \mapsto -v$ gives the formula. Consequently $J_\theta(f^\#) = \mathcal{F} \overline{J_\theta f}$. This is an actual intertwining map between the test involution and the supported-zero Fourier boundary, not only an equality of two pairs of numbers.

The Mellin calculation, first for $\Re s > 1$, is

$$\int_0^\infty (J_\theta f)(x) x^s \frac{dx}{x} = \frac{1}{2} \pi^{-s/2} \Gamma(s/2) M_f(s),$$

$$\boxed{Z_{J_\theta f}(s) = \Lambda(s) M_f(s), \quad \Lambda(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s).} \quad (\text{SZW22})$$

The first integral already converges for $\Re s > 0$. Compactness of the v -support and Gaussian decay justify Fubini, and $x \mapsto e^{-v} x$ gives the factor e^{-sv} . The second formula follows from (SZW14) and continues meromorphically by (SZW15). It retains the full amplitude M_f . Its residues are $-M_f(0)$ and $M_f(1)$, exactly the endpoint map (SZW21).

The two endpoint values range independently over $\mathbb{C}^2$. For an explicit section choose a nonnegative even nonzero bump $\psi \in \mathcal{T}$, put $b_\psi = M_\psi(0) = M_\psi(1) > 0$, and let $\psi_t(v) = \psi(v - t)$, with $t = \log 4$. Then

$$f_{A,B} = \frac{4B - A}{3b_\psi} \psi + \frac{2(A - B)}{3b_\psi} \psi_t, \quad (M_{f_{A,B}}(0), M_{f_{A,B}}(1)) = (A, B). \quad (\text{SZW23})$$

The translated bump has endpoints $(2b_\psi, b_\psi/2)$, which verifies both identities. Thus no universal restriction such as $A = B$, or invariance under an auxiliary finite group, follows merely from admissibility of these Weil tests. Its exact endpoint kernel is $\{f : M_f(0) = M_f(1) = 0\}$, and (SZW23) splits its quotient as a vector space.

6. The complete arithmetic explicit formula, derived with these endpoints

For $h \in \mathcal{T}$, set $H(s) = M_h(s)$, $\widehat{h}(t) = \int h(v) e^{-itv} dv$, and let ρ range over all nontrivial zeros of ζ , with their multiplicities m_ρ . Define

$$\begin{aligned} Z(h) &= \sum_{\rho} m_{\rho} H(\rho), \\ A_{\infty}(h) &= \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{h}(t) (\Re \psi(1/4 + it/2) - \log \pi) dt, \\ P_{\text{fin}}(h) &= \sum_p \sum_{k \geq 1} (\log p) p^{-k/2} \{h(k \log p) + h(-k \log p)\}. \end{aligned} \quad (\text{SZW24})$$

Here $\psi = \Gamma'/\Gamma$ is the digamma function, not the bump used above. The last sum is finite for any particular compactly supported h . The archimedean integral is absolutely convergent since $\widehat{h}$ decreases faster than every inverse power and the digamma factor grows at most logarithmically. The zero sum is absolutely convergent; a proof of the requisite bound is included below.

The exact identity is

$$\boxed{Z(h) = H(0) + H(1) + A_{\infty}(h) - P_{\text{fin}}(h).} \quad (\text{SZW25})$$

The boundary $H(0) + H(1)$ here is obtained from the actual full supported theta (SZW7), (SZW15), and (SZW22). It has not been deleted and then replaced by a numerical Euler factor for e .

Here are the analytic details and signs. For the Gaussian $g(x) = e^{-\pi x^2}$, (SZW15) gives $Z_g = \Lambda$, simple poles at $0, 1$ with residues $-1, 1$, and the functional equation $\Lambda(s) = \Lambda(1 - s)$. The entire function $\xi(s) = \frac{1}{2}s(s-1)\Lambda(s)$ therefore has $\xi(0) = \xi(1) = 1/2$. Its theta integral also gives $\max_{|s| \leq R} |\xi(s)| \leq \exp(CR \log(R+2))$ for some fixed C : bound the rapidly decreasing Gaussian tail by $C_0 e^{-cx^2}$ and each power in the two integrals by x^{R+1} ; its integral is bounded by a constant times $c^{-(R+1)/2} \Gamma((R+1)/2)$. The elementary Stirling bound for the positive real Gamma function gives the claimed estimate. Jensen's formula then implies that the number of zeros in $|s| \leq R$, counted with multiplicity, is $O(R \log(R+2))$, by applying it on radius $2R$ around the nonzero value at zero.

The Euler product is nonzero in $\Re s > 1$. It follows from the functional equation that ξ has no zeros in $\Re s < 0$. For completeness the boundary line $\Re s = 1$ has no zeros either. For $\sigma > 1$, the logarithm of $\zeta(\sigma)^3 |\zeta(\sigma + it)|^4 |\zeta(\sigma + 2it)|$ is a convergent sum of the nonnegative terms $2p^{-k\sigma} (1 + \cos(kt \log p))^2 / k$. The product is therefore at least one. If $t \neq 0$ and $\zeta(1 + it)$ were a zero of order at least one, analyticity there, finiteness at $1 + 2it$, and the simple pole at one would make that product tend to zero as $\sigma \downarrow 1$, a contradiction. The functional equation excludes zeros on $\Re s = 0$; $\xi(0), \xi(1) \neq 0$ handle the two real endpoints. Thus all zeros counted above lie in $0 < \Re s < 1$.

Repeated integration by parts in (SZW19) gives $|H(\sigma + it)| \leq C_{N,I} (1 + |t|)^{-N}$ for every N , uniformly on each fixed bounded real interval I for σ . Together with the zero count, this proves absolute convergence of $Z(h)$.

Set $\ell = \Lambda'/\Lambda$. Integrate $\ell(s)H(s)$ around the rectangle with vertical sides $-1, 2$, oriented positively. The residue theorem gives the zeros minus $H(0) + H(1)$; the residues of ℓ at its two poles are -1 , independently of the pole residues of Λ . A sequence of upper and lower heights tending to infinity can be chosen so that the horizontal integrals tend to zero. One precise justification is the genus-one Hadamard factorization of the order-at-most-one entire function ξ . It gives $\xi'/\xi(s) = B + \sum_{\rho} m_{\rho}((s - \rho)^{-1} + \rho^{-1})$, locally normally away from zeros. The zero bound above implies $\sum m_{\rho}/|\rho|^2 < \infty$. In each interval $[N, N+1]$, exclude intervals of radius N^{-3} around the imaginary parts of zeros with $|\rho| \leq 3N$. Their total length is $O(N^{-2} \log N) < 1$, so a remaining height exists. On that horizontal segment every nearby denominator has modulus at least

N^{-3} , yielding a polynomial bound $O(N^4 \log N)$ for the finite part of the sum. The tail is $O(\log N)$, since the paired summand is bounded by $CN/|\rho|^2$. The same argument applies below the real axis. Subtracting $1/s + 1/(s-1)$ gives the same polynomial bound for ℓ . The arbitrarily high inverse-power decay of H proves the asserted disappearance of the horizontal integrals. This argument uses the classical Hadamard and Jensen theorems of complex analysis, with their hypotheses verified here; it does not assume the Riemann hypothesis or any zero-location assertion beyond the strip proved above.

Since $\ell(1-s) = -\ell(s)$, changing variables on the left edge now gives

$$Z(h) - H(0) - H(1) = \frac{1}{2\pi i} \int_{\Re s=2} \ell(s) \{H(s) + H(1-s)\} ds. \quad (\text{SZW26})$$

Insert $\ell(s) = -\frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2) - \sum_{p,k} (\log p) p^{-ks}$. The prime series converges absolutely on this line. Fourier inversion in each term gives respectively $p^{-k/2} h(-k \log p)$ and $p^{-k/2} h(k \log p)$; this proves its contribution is $-P_{\text{fin}}$. The Gamma part is holomorphic for $\Re s > 0$, so its line can be shifted to $1/2$. Combining the second term after $t \mapsto -t$ gives $-\log \pi + \Re \psi(1/4 + it/2)$, proving its contribution is A_∞ . This proves (SZW25) with the exact signs, constants and domains in (SZW24).

The same identity is Connes's Appendix II Theorem 6 in the unramified trivial-character sector. Its test on C_Q is exactly

$$q_h(j) = |j|^{-1/2} h(-\log |j|), \quad \widehat{q}_h(s) = \int_{C_Q} q_h(j) |j|^s d^* j = H(s), \quad (\text{SZW27})$$

where the norm-one compact group has Haar mass one and the norm quotient has measure du/u . The equality follows by $v = -\log |j|$. The local distribution on the other side is consequently $P_{\text{fin}} - A_\infty$, with all the source principal-value conventions fixed by this equality and by the stated self-dual additive Fourier character. The finite-prime coefficients have remained $(\log p) p^{-k/2}$; none was changed to $\log(p+1)$.

7. An actual summable divisor trace, with the theta image and its cokernel mapped

The zero trace in (SZW24) has an unconditional operator realization that retains every multiplicity. It is a divisor receiver constructed from Λ in (SZW22), not an assumed identification with the entire original adèle-class Hilbert space.

Let $\mathcal{O}$ be the ring of entire functions of s , put $q_0(s) = s(s-1)$, and let $\mathcal{O}(P) = q_0^{-1} \mathcal{O}$ be the meromorphic functions with at most simple poles at $0, 1$. Multiplication gives the injective map

$$\mathcal{O} \xrightarrow{\times \Lambda} \mathcal{O}(P), \quad q_0 \Lambda = 2\xi. \quad (\text{SZW28})$$

Its injectivity follows because Λ is not identically zero. The map $F \mapsto q_0 F$ identifies its cokernel with $\mathcal{O}/(2\xi)$. The other exact quotient is

$$0 \rightarrow \mathcal{O} \rightarrow \mathcal{O}(P) \xrightarrow{(\text{Res}_0, \text{Res}_1)} \mathbb{C}^2 \rightarrow 0. \quad (\text{SZW29})$$

Surjectivity is witnessed by $a/s + b/(s-1)$, and a member with both residues zero is entire. Multiplication by H on this quotient is diagonal with values $H(0), H(1)$. By (SZW22), the actual theta map $f \mapsto Z_{J_\theta} f = \Lambda M_f$ lands in the image of (SZW28); its residue map is $f \mapsto (-M_f(0), M_f(1))$. This proves both the zero-cokernel vanishing of the theta image and the nonvanishing of its independent endpoint data.

At a zero ρ of multiplicity m_ρ , the local cokernel is

$$\mathcal{J}_\rho = \mathbb{C}[z]/(z^{m_\rho}), \quad z = s - \rho. \quad (\text{SZW30})$$

Indeed $2\xi(s) = (s-\rho)^{m_\rho} u_\rho(s)$ with a holomorphic unit $u_\rho(\rho) \neq 0$, and division by that unit gives the stated quotient. The global jet map has the explicit product target

$$\mathcal{O}(P) \longrightarrow \prod_\rho \mathcal{J}_\rho, \quad F \longmapsto ([q_0 F]_{\rho, m_\rho})_\rho. \quad (\text{SZW30a})$$

Its kernel is exactly $\Lambda \mathcal{O}$: vanishing of every required jet makes $q_0 F/(2\xi)$ entire at every zero and elsewhere, proving the reverse inclusion as well as the forward one. Every finitely supported tuple of jets occurs. For one chosen ρ , the entire function $2\xi(s)/(s-\rho)^{m_\rho}$ vanishes with the required orders at every other zero and is a unit at ρ . Multiply it by the Taylor polynomial of the desired jet divided by that unit, of degree below m_ρ , and then divide by q_0 . This gives the prescribed jet and zero jets elsewhere. Finite sums give the general

finite tuple. These are explicit maps into and out of the divisor receiver, preserving the theta amplitude in (SZW22).

In particular, put $\mathcal{C} = \mathcal{O}(P)/\Lambda\mathcal{O}$, let $\iota : \mathcal{C} \hookrightarrow \prod_{\rho} \mathcal{J}_{\rho}$ be the induced injective map, and define

$$\mathcal{C}_{\text{fin}} = \iota^{-1} \left(\bigoplus_{\rho}^{\text{alg}} \mathcal{J}_{\rho} \right). \quad \iota : \mathcal{C}_{\text{fin}} \xrightarrow{\sim} \bigoplus_{\rho}^{\text{alg}} \mathcal{J}_{\rho}. \quad (\text{SZW30b})$$

The superscript denotes the algebraic direct sum, consisting of finite tuples. The preceding interpolation proves this isomorphism. The Hilbert space used next is the completion of this particular subspace with its displayed jet norms. No identification of the entire analytic cokernel or the full product with that Hilbert space is asserted.

Give each $\mathcal{J}_{\rho}$ the Euclidean norm in its full basis $(1, z, \dots, z^{m_{\rho}-1})$, and form their Hilbert direct sum $\mathcal{H}_Z$. Multiplication by $u^{\rho+z}$, $u > 0$, defines a bounded representation: if N_{ρ} is multiplication by z , then $\|N_{\rho}\| \leq 1$, and

$$\|u^{\rho} \exp((\log u)N_{\rho})\| \leq \max(1, u)e^{|\log u|} \quad (\text{SZW31})$$

uniformly in ρ . The bound and density of vectors in finitely many summands prove strong continuity. Integrating against q_h gives multiplication by the full jet of H . This operator is trace class. To verify this directly, if $\text{supp } h \subset [-R, R]$, repeated integration by parts in $H^{(k)}(\rho)/k! = \int (-v)^k h(v) e^{-(\rho-1/2)v} dv/k!$ gives, after summing over all $k \geq 0$, a bound $\sum_k |H^{(k)}(\rho)|/k! \leq C_{N,h}(1 + |\Im \rho|)^{-N}$. The summation is justified because $\sum_k |v|^k/k! = e^{|v|}$, and each fixed-order derivative of the series of absolute derivative bounds remains bounded on the compact interval; the extra falling factorial factors in k are absorbed by this same exponential series. The real part of ρ is in $[0, 1]$, so it introduces only uniform constants. The trace norm of the multiplication matrix is at most $m_{\rho} \sum_k |H^{(k)}(\rho)|/k!$. Summing over zeros converges by the count proved above, taking $N > 2$.

Its trace is therefore the absolutely convergent sum

$$\text{Tr}_{\mathcal{H}_Z}(q_h) = \sum_{\rho} m_{\rho} H(\rho) = Z(h). \quad (\text{SZW32})$$

Every nilpotent jet is retained in (SZW30); the trace of its positive powers is zero because their multiplication matrices have zero diagonal. This is not a quotient identifying such a jet with τ , or even with the supported amplitude zero.

Reflection has the exact maps $\rho \mapsto \rho^{\#} = 1 - \bar{\rho}$ and $p(z) \mapsto \overline{p(-\bar{z})}$ from $\mathcal{J}_{\rho}$ to $\mathcal{J}_{\rho^{\#}}$. The functional equation and reality of ξ give equal multiplicities, so these maps are defined and square to the identity. At the global analytic level the same map is $F^{\#}(s) = \overline{F(1 - \bar{s})}$. The identities $q_0^{\#} = q_0$ and $\Lambda^{\#} = \Lambda$ show that it preserves $\mathcal{O}(P)$ and $\Lambda\mathcal{O}$, descends to $\mathcal{C}$, and intertwines (SZW30a). It preserves $\mathcal{C}_{\text{fin}}$, and its jet map extends to an antilinear isometry J_Z of $\mathcal{H}_Z$, since coefficient conjugation and multiplication of the degree- k coefficient by $(-1)^k$ preserve the displayed Euclidean norms. Its exact dilation covariance is

$$J_Z U_Z(u) = u U_Z(u^{-1}) J_Z, \quad u > 0. \quad (\text{SZW32a})$$

Indeed the reflected multiplier in the $\rho^{\#}$ summand is $u^{\bar{\rho}-z} = u u^{-\rho^{\#}-z}$, proving the identity on every full jet and hence on the Hilbert completion. This is an anti-intertwiner between dilation and reflected dilation; it is not a claim that the dilation generator is self-adjoint for a positive Weil form. All derivative coordinates remain present.

8. The full supported trace identity and its exact scalar observations

For a support label λ , retain its coordinate vector $\mathbf{e}_{\lambda}$ in the vector space $\mathbb{C}[L]$. Define the label-resolved boundary, divisor, and geometric trace distributions by their actual representation traces:

$$\begin{aligned} B_L(h) &= (H(0) + H(1))\mathbf{e}_{1_L} + H(0) \sum_{\lambda \neq 1_L} \mathbf{e}_{\lambda}, \\ Z_L(h) &= Z(h)\mathbf{e}_{1_L}, \\ D_L(h) &= (P_{\text{fin}}(h) - A_{\infty}(h))\mathbf{e}_{1_L} + H(0) \sum_{\lambda \neq 1_L} \mathbf{e}_{\lambda}. \end{aligned} \quad (\text{SZW33})$$

The boundary trace is precisely that of (SZW13) integrated against (SZW27); the zero trace is (SZW32); each lower-coordinate trace is the fixed-character distribution (SZW18). Equivalently, take the finite even

representation B_L and the odd representation $\mathcal{H}_Z$, with their indicated labels. Their integrated operators are trace class, so their label-resolved supertrace is well defined. Equation (SZW25) proves the full identity

$$\boxed{B_L - Z_L = D_L}. \quad (\text{SZW34})$$

Thus all terms come from defined test spaces, the actual full theta, explicit fixed-coordinate representations, and the Mellin divisor receiver. No assertion about an unconstructed global Hilbert trace is used.

Two different scalar maps already give different observations. Let ϵ_{top} pick the coefficient of $\mathbf{e}_{1_L}$, and let ϵ_{all} sum all coefficients. Then

$$\begin{aligned} \epsilon_{\text{top}} D_L &= P_{\text{fin}} - A_{\infty}, \\ \epsilon_{\text{all}} D_L &= P_{\text{fin}} - A_{\infty} + (n-1)H(0), \\ \epsilon_{\text{all}} B_L &= nH(0) + H(1). \end{aligned} \quad (\text{SZW35})$$

The first map forgets all lower fixed traces; the second retains their sum. Their kernels on $\mathbb{C}[L]$ are respectively the span of the nontop coordinate vectors, and the hyperplane of zero coordinate total. These maps must not be called the same arithmetic pushforward.

First take $A = \mathbb{R}$, as in the function space (SZW11). The actual carrier projection $G_L(\mathbb{R}) \rightarrow \mathbb{R}$ identifies all zero points. Its fibre-sum pushforward is

$$(\phi, c) \mapsto \phi + C(F)1_{\{0\}}, \quad \mathcal{F}_L \longrightarrow \mathcal{S}_{\text{ev}}(\mathbb{R}) \oplus \mathbb{C}1_{\{0\}}. \quad (\text{SZW36})$$

The sum is direct because a continuous function on $\mathbb{R}$ supported only at zero is zero. Its kernel consists exactly of $\phi = 0$ and $\sum c_{\lambda} = 0$. The subsequent map $\phi + d1_{\{0\}} \mapsto \phi$ is a separate projection, with kernel the added point-function line. It is this composite which returns the classical regular orbital observation. For $L = \mathbb{B}$, the first pushforward identifies the external coordinate with an independent point correction at supported zero, while the second projection kills that correction. Thus even in the two-point case the full trace and its regular arithmetic observation are connected by two explicit maps.

There is also an adelic version with its own domain $\mathcal{S}(\mathbb{A}_{\mathbb{Q}}) \oplus \mathbb{C}^{L \setminus \{1_L\}}$: it sends (f, c) to $f + C(F)1_{\{0_{\mathbb{A}}\}}$ in $\mathcal{S}(\mathbb{A}_{\mathbb{Q}}) \oplus \mathbb{C}1_{\{0_{\mathbb{A}}\}}$, with exactly the same kernel proof since the adèle group is nondiscrete. The spherical embedding of (SZW11) is explicitly

$$(\phi, c) \mapsto (\phi \otimes 1_{\widehat{\mathbb{Z}}}, c), \quad (\phi, c) \mapsto \phi \otimes 1_{\widehat{\mathbb{Z}}} + C(F)1_{\{0_{\mathbb{A}}\}} \quad (\text{SZW36a})$$

after that adelic pushforward. This states the real-to-adelic map before adding the adelic point correction; the original real function has not silently changed its domain.

More general support maps send point coefficients by finite fibre sums, as in FL19. Separate Boolean branches record the total and the principal-filter sums indexed by join-irreducibles. Their exact kernel is

$$N_L = \{d : \sum_{\lambda} d_{\lambda} = 0, \quad \sum_{\lambda \geq j} d_{\lambda} = 0 \text{ for every join-irreducible } j\}, \quad (\text{SZW37})$$

of dimension $n - |J(L)| - 1$. To verify the dimension, order the join-irreducibles compatibly with their partial order and solve their upper-triangular principal-filter system on the coordinate vectors at those elements; the bottom coordinate then prescribes the total. This gives surjectivity onto $|J(L)| + 1$ moments. The kernel is invariant and trivial under dilation, so it contributes exactly $(n - |J(L)| - 1)H(0)$ to the endpoint trace before it is forgotten. All full filter moments reconstruct the coefficients by finite Möbius inversion, whose inverse follows by multiplying the two triangular incidence matrices. Consequently these lost traces are actual labelled objects, not a claim that equality of all pointwise branch labels gives injectivity on their linear combinations.

9. The reflected endpoint defect and its concrete Fourier closure

For $f, g \in \mathcal{T}$, put $h = f^{\#} * g$, using additive convolution. Fubini gives $M_h(s) = \overline{M_f(1 - \bar{s})} M_g(s)$. If $A_f = M_f(0)$ and $B_f = M_f(1)$, then

$$H(0) = \overline{B_f} A_g, \quad H(1) = \overline{A_f} B_g. \quad (\text{SZW38})$$

The supported-zero pair is therefore exactly $\overline{B_f} A_g + \overline{A_f} B_g$, the hyperbolic Hermitian plane with matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ on (A, B) . It is not a positive form on all admissible tests; (SZW23) supplies arbitrary endpoint pairs, including $(1, -1)$, of value -2 .

The entire unreflected endpoint representation has instead

$$B_L(f, g) = n\overline{B_f}A_g + \overline{A_f}B_g, \quad B_L - B_L^* = (n-1)(H(0) - H(1)). \quad (\text{SZW39})$$

The last equality is an equality of sesquilinear forms, obtained by exchanging the inputs and conjugating. For $(A_f, B_f) = (1, i)$, its diagonal value is $-i(n-1)$, so for $n > 1$ it is not Hermitian. This exact defect is also present in $\epsilon_{\text{all}}\mathbf{D}_L$; (SZW34) therefore retains it on both sides, instead of confusing it with the Hermitian divisor form $Z(f^\# * g)$.

The obstruction defines a concrete missing object. In tempered distributions on $\mathbb{R}$, let

$$B_L^{\mathcal{F}} = (\mathbb{C}1 \oplus \mathbb{C}\delta_0) \otimes \mathbb{C}[L]. \quad (\text{SZW40})$$

For dilation $V(u)\phi(x) = \phi(x/u)$, extend the action to distributions by $\langle V(u)T, \varphi \rangle = u\langle T, \varphi(u\cdot) \rangle$. This agrees with the action on regular distributions by change of variables. It fixes 1 and sends δ_0 to $u\delta_0$, so the two weights are precisely zero and one. Fourier interchanges these two distributions. They are linearly independent: test against a compactly supported smooth function vanishing at zero but having nonzero integral to determine the constant coefficient, then against one nonzero at zero to determine the Dirac coefficient.

There is an injective equivariant map

$$B_L \hookrightarrow B_L^{\mathcal{F}}, \quad a \mapsto a1 \otimes \mathbf{e}_{1_L}, \quad b \mapsto b\delta_0 \otimes \mathbf{e}_{1_L}, \quad c_\lambda \mapsto c_\lambda 1 \otimes \mathbf{e}_\lambda. \\ B_L^{\mathcal{F}}/B_L \cong \bigoplus_{\lambda \neq 1_L} \mathbb{C}(1)_\lambda. \quad (\text{SZW41})$$

The linear independence just proved gives its kernel and quotient. It is the smallest Fourier-stable vector subspace containing the indicated image: the Fourier transform of each missing-label constant is its missing-label Dirac distribution, and after adjoining these the space is Fourier stable. On this space, Fourier followed by coefficient conjugation swaps the two endpoint coordinates at every label. It satisfies the exact reflected covariance $JV(u) = uV(u^{-1})J$. This proves the representation and involution comparison, with no additional group action inferred from the existence of support labels.

Its trace and the required new geometric correction are

$$B_L^{\mathcal{F}}(h) = n(H(0) + H(1)), \\ D_L^{\mathcal{F}}(h) = P_{\text{fin}}(h) - A_\infty(h) + (n-1)(H(0) + H(1)), \quad (\text{SZW42}) \\ B_L^{\mathcal{F}}(h) - Z(h) = D_L^{\mathcal{F}}(h).$$

These follow from (SZW35) by adding the actual quotient trace $(n-1)H(1)$ in (SZW41). The reflected endpoint form has one hyperbolic pair for each independent label; its inertia is $(n, n, 0)$ when all labelled input coordinates are retained. On the diagonal copy of actual scalar tests it is n times (SZW38). Fourier closure therefore makes the missing partners exact; it does not assert positivity of the entire new distribution.

10. The distinct residue-count norm and the comparison required before holonomy

The supported quotient at modulus $m \geq 1$ is $G(\mathbb{Z}/m\mathbb{Z})$, of cardinality $m+1$, while the quotient identifying unsupported and supported zero is $\mathbb{Z}/m\mathbb{Z}$, of cardinality m . Consequently the actual supported residue-count sum is

$$\sum_{m \geq 1} (m+1)^{-s} = \zeta(s) - 1 \quad (\Re s > 1). \quad (\text{SZW43})$$

For coprime moduli the support-preserving Chinese remainder map lands in the fibre product over $\mathbb{B}$; it has size $mn+1$, while the full Cartesian product has size $(m+1)(n+1)$. The difference $m+n$ counts the two mixed-support sectors. Thus $m \mapsto m+1$ is not multiplicative. Extending the prime values $p+1$ multiplicatively would define a different function, and is not used anywhere in (SZW6)–(SZW42). Likewise $P_e^2 = P_e$ prevents any finite positive multiplicative norm other than one at that ideal, and its actual Bourne quotient is the infinite set $\mathbb{Z}$. Neither fact assigns a finite numerical Euler factor to e . The theta endpoint construction instead records its actual fixed operator and Poisson dual.

The complete objects now available for a further holonomy comparison are the full-support endpoint representation (SZW13), the full labelled trace identity (SZW34), its explicit observations (SZW35)–(SZW37), the actual admissible-test lift (SZW20)–(SZW23), the local divisor jets and reflected maps (SZW28)–(SZW32), and the minimal Fourier closure (SZW40)–(SZW42). The raw dilation action fixes every support label, so it

gives identity action on those zero-coordinate lines; this is proved by the carrier multiplication in (SZW10), not assumed from a scalar quotient. A nontrivial action on a separate finite cover must therefore be carried into one of these exact domains by an actual intertwining map before its group average becomes an assertion about this Weil trace. The endpoint map is onto by (SZW23), so invariance under such an average is not forced merely by the test class. These statements specify the maps and the testable defect; they do not discard the finite cover or identify its holonomy with the original dilation action.

11. A nonzero fixed-support trace that the top projection forgets

This example is already present on the original two-point support lattice $L = \mathbb{B}$, with $\tau \neq e$; it requires no extra finite cover. Choose the exact even smooth function

$$\psi(v) = \begin{cases} K \exp\left(-\frac{1}{1-v^2}\right), & |v| < 1, \\ 0, & |v| \geq 1, \end{cases} \quad K = \left(\int_{-1}^1 \exp\left(-\frac{1}{1-v^2}\right) dv\right)^{-1}. \quad (\text{SZW44})$$

The integral defining K is finite and strictly positive. Every derivative of the interior expression tends to zero at $v = \pm 1$: it is the same exponential times a rational function with a finite-order pole there, and an exponential e^{-x} times any fixed polynomial tends to zero as $x \rightarrow +\infty$. Thus $\psi \in \mathcal{T}$, $\psi^\# = \psi$, and $\int \psi = 1$. Set

$$b = \int_{-1}^1 \psi(v) \cosh(v/2) dv > 1, \quad h = \psi^\# * \psi, \quad H = M_h. \quad (\text{SZW45})$$

$$M_\psi(0) = M_\psi(1) = b, \quad H(0) = H(1) = b^2 > 1. \quad (\text{SZW45})$$

Evenness replaces either exponential in the first two Mellin evaluations by its hyperbolic cosine average. Strictness follows from $\cosh(v/2) > 1$ away from zero and $\psi > 0$ on $(-1, 1)$. The convolution identity (SZW38) gives the remaining evaluations. In particular this is an actual compactly supported admissible Weil test. Its actual theta lift is $J_\theta \psi$, with endpoint pair (b, b) , by (SZW20)–(SZW22).

Independently retain the one-dimensional fixed-support function space $E_\tau = \mathbb{C}F_\tau \subset \mathcal{F}_\mathbb{B}$, where $F_\tau = (0, 1)$ is zero on the supported real carrier and one at τ . Its dilation action is the identity, since τ is fixed. The coordinate projection Π_τ onto this line commutes with dilation, descends under the moment map to the lower line in $B_\mathbb{B}$, and acts as zero on the divisor representation $\mathcal{H}_Z$, which has top label. Integrating the trivial action against the exact test (SZW27) gives

$$\text{Tr}_{E_\tau}(q_h) = \int_0^\infty u^{-1/2} h(-\log u) \frac{du}{u} = H(0) = b^2. \quad (\text{SZW46})$$

The change of variable $v = -\log u$ gives the middle integral as $\int h(v) e^{v/2} dv$, proving the sign and weight. The label-resolved supertrace after Π_τ is therefore exactly $b^2 \mathbf{e}_\tau$. It is nonzero; its ϵ_{top} observation is zero and its ϵ_{all} observation is b^2 . For the full unprojected test the lower contribution remains $b^2 \mathbf{e}_\tau$, in addition to the separate top arithmetic term of (SZW33). This identifies an actual kernel element of the top observation and its nonvanishing trace, rather than merely counting possible coordinates.

The same vector also gives a direct comparison with the full theta and Mellin maps:

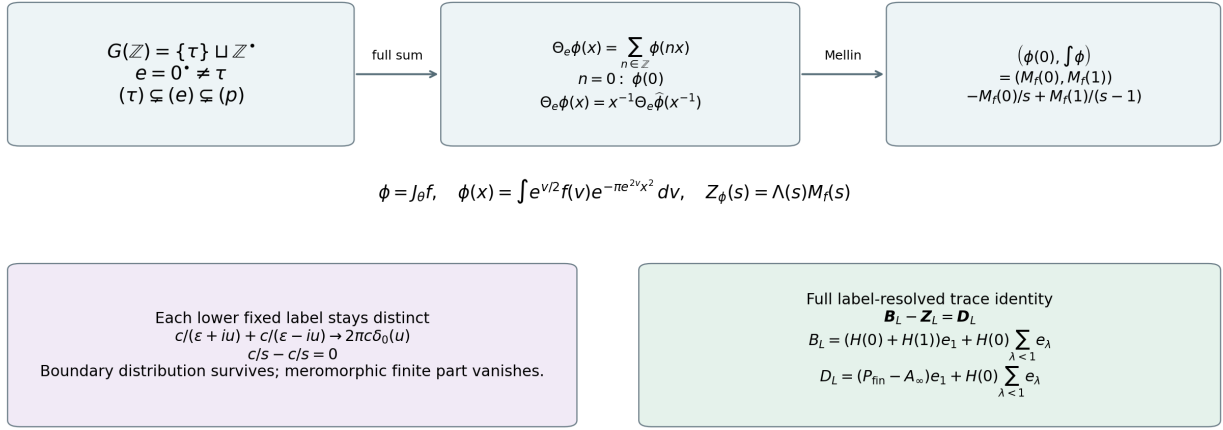
$$\Theta_\mathbb{B}(F_\tau; x) = 1, \quad Z_{F_\tau}^{\text{fp}}(s) = \frac{1}{s} - \frac{1}{s} = 0, \quad \lim_{\varepsilon \downarrow 0} \frac{2\varepsilon}{\varepsilon^2 + t^2} = 2\pi\delta_0(t) \quad \text{in } \mathcal{S}'(\mathbb{R}). \quad (\text{SZW47})$$

The first equality is the actual carrier sum (SZW9). The second is precisely the meromorphic finite-part map (SZW17); the third is the fixed-character boundary map proved in (SZW18). Thus the vanishing under one Mellin observation coexists with the nonzero fixed trace (SZW46), through the exact separate maps already constructed. The surviving object is a trivial dilation representation at its retained support label; it is not a vanishing function, a removed prime, or a zero nilpotent jet.

Exact residue-duality continuation

RD1–RD31b proves the complete Jacobian map from perfect residue duality to the retained regular trace, identifies the actual packet radical with cotangent cohomology, and computes the supported endpoint compensation. The existing formulas and all their coordinates remain unchanged.

Supported zero is retained before taking the Weil trace



SZW1-4: prime and arithmetic maps. SZW6-23: full theta, actual tests and boundary maps.
 SZW33-42: retained labels, scalar observations and minimal Fourier closure.

Figure 8: The actual supported-zero prime, its full theta and Mellin endpoint map, the retained fixed-support boundary distributions, and the full label-resolved trace identity. SZW1–SZW4, SZW6–SZW23 and SZW33–SZW42 prove the displayed maps and constants.

Integral prism continuation

DP1–DP43 constructs the specific bounded prism and proves the original nilradical specialization. EC1–EC48 computes its complete extension class and the exact maps detecting the divided residue. SG1–SG23 gives the supported prime pullbacks, retaining the unsupported point and the original supported-zero prime under coefficient contraction. These calculations retain all earlier trace and endpoint identities.

Actual global heat continuation

The global heat trace retains all zero multiplicities. The arithmetic calculation, HA1–HA26, evaluates the complete supported formula on Cauchy tests and computes its actual derivative. The local response keeps the complete analytic unit; the global response includes mixed prime-product terms. The matrix criterion gives the exact relation of this test family to RH. No sign from an isolated local coefficient replaces the complete pairing.

Exact full-form continuation

The reflection-index theorem, NI1–32, calculates the signed index of every complete Cauchy coefficient tail. The finite-jet and endpoint proof, FJ1–29, preserves the exact finite jets while proving density on the remaining zeros, derives the full exterior index and tail, and calculates the second and third actual heat coefficients. All supported endpoints remain explicit. These results identify the remaining sign problem without asserting its resolution.

One fixed primitive test and the entire prime window

The prime-operator calculation, PM1–41, proves both distinct Fourier support corrections and their exact map into the full Weil formula. The local estimate, PS1–21, gives a positive translation-difference remainder. The fixed-test proof, UP1–35, constructs one compact test whose transform is nonzero at every possible off-critical zero. The complete arithmetic calculation, PW1–22, expresses its entire translated correlation through a fixed-width prime window and an explicit archimedean remainder. Its boundedness is equivalent to RH; that bound remains unresolved. All boundary coordinates and supported-zero labels remain explicit.

Holonomy of the original signed eight-state completion

23 September 2026. This derivation computes continuation of the actual completed ES–Fable cover, including the omitted infinity sign, the two original involutions, the complete finite trace forms, and the surviving collision algebra. It distinguishes a permutation of nearby states from an automorphism extending through a nonreduced fibre, and calculates the residue when such an extension fails. No finite trace form in this note is identified with an arithmetic Weil distribution.

Original sources read: the original finite signed-completion source linked below, FS1–FS31, and *The ES–Fable inverse correspondence*, SM1–SM19, pinned original TeX, source lines 2097–2513, including the explicit paths and sign choices in SM7–SM14. The latter already proves the order-192 monodromy group; that result is not claimed as new here. The completed chart and local algebra source is *The finite signed-root completion*, FS1–FS31, pinned original TeX. Their exact source hashes are, respectively, 10906ba3bd21c06645571560e4c7b0c3948c0ccaf65d8b9e525abf7cd732794c and eaf980a16e7d639e21953e5779e7f76abf18e8bdb80ae923c4160b456c22da6f. The complete group proof needed here is included below. The receiving heat calculation the complete eight-state heat proof, ESH1–ESH72, was read, and every used map and local relation is restated and proved here. Source provenance of the original polynomial is retained in FS’s bibliography: *Erdős–Straus project reader*, source version 69, **explicit-four-time-Jacobian-collision**; the added completion and monodromy are programme derivations.

1. The unchanged cover and its continuation maps

Retain target coordinates $u = (A, B, C, D)$, the fixed cubic coefficient one, and

$$\begin{aligned} H_u(U, V) &= AU^4 + U^3V + BU^2V^2 + CUV^3 + DV^4, \\ F_u(R) &= AR^4 + R^3 + BR^2 + CR + D, \\ k_u(v) &= A + v + Bv^2 + Cv^3 + Dv^4. \end{aligned} \tag{EHM1}$$

The completed finite and infinity charts are

$$F_u(R) = 0, \quad T^2 = F'_u(R), \quad k_u(v) = 0, \quad \Theta^2 = -k'_u(v), \quad v = R^{-1}, \quad \Theta = -T/R. \tag{EHM2}$$

The inverse transition is $R = v^{-1}, T = -\Theta/v$. At a root, differentiation of $k(v) = v^4F(1/v)$ gives $k'(v) = -v^2F'(1/v)$, proving both squared relations and every transition sign. Write $\mathfrak{D}$ for the binary quartic discriminant, with its original value $A^6 \prod_{i < j} (R_j - R_i)^2$ when $A \neq 0$. The completed unramified base is

$$\mathcal{B}^\circ = \{\mathfrak{D} \neq 0\}, \quad \mathcal{B} = \{A\mathfrak{D} \neq 0\} \subset \mathcal{B}^\circ. \tag{EHM3}$$

On $\mathcal{B}^\circ$ each projective root is simple and has two nonzero sign values. On a finite chart the relative Jacobian determinant is $2TF'(R) = 2T^3 \neq 0$. At infinity, $v = 0$ forces $A = 0$ and $\Theta = \pm i$, and the relative determinant is $2\Theta k'(0) = 2\Theta \neq 0$. Thus these eight points have separate local holomorphic inverse charts. Covering a compact path by finitely many such neighborhoods and matching its initial point gives unique continuation of each state. In particular infinity itself is not a branch point of the completed cover.

For a finite root and nonzero T , the original source coordinate α , called a in FS, and the other three source coordinates are

$$\begin{aligned} \alpha &= T^{-1}, \quad y = -R - iT, \\ z &= AT^3 + 2RT + 3iT^2, \\ w &= 7iR^2T + (B - 17R + AR^2)T^2 - 13iT^3 - 2AT^4. \end{aligned} \tag{EHM4}$$

These are FS13 in the unchanged reciprocal coordinate. They invert the original factor equations: $b = -R/T, c = AT, d = (1 + AR)T, e = (B + R + AR^2)T, f = (C + BR + R^2 + AR^3)T$; multiplication gives the four coefficients of EHM1, and the resultant is $T^{-2}F'(R) = 1$. The original source is the completed cover with the ramification divisor and the infinity sign $\Theta = i$ removed. At $A = 0$ its retained sign is $\Theta = -i$, with source point $(0, B, i(7B^2 - C)/2, 11B^3 - 2BC - D)$, as direct substitution in the original polynomial verifies.

For a path entirely in $\mathcal{B}$, order its roots continuously and choose one initial T_j at each. The exact continuation formula is

$$T_j(t) = T_j(0) \exp \left(\frac{1}{2} \int_0^t \left[\frac{A'}{A} + \sum_{k \neq j} \frac{R'_j - R'_k}{R_j - R_k} \right] ds \right). \tag{EHM5}$$

The denominators are nonzero on this path. Differentiating proves that $T_j^2/[A \prod_{k \neq j} (R_j - R_k)]$ is constant with value one. The inverse source sign $\alpha_j = 1/T_j$ has the negative exponent, exactly SM14. EHM2 supplies continuation through infinity where EHM5’s finite coordinates cannot be used.

2. The original order-192 group, with its actual generators

For four finite ordered roots let $\Delta = \prod_{j < k} (R_k - R_j)$. The derivative product is $\prod_j F'(R_j) = A^4 \Delta^2$: each of six unordered pairs contributes a minus sign, whose product is one. Therefore

$$\chi = A^2 \Delta \prod_{j=1}^4 \alpha_j, \quad \chi^2 = 1. \quad (\text{EHM6})$$

This continuous value is constant along any ordered lift. A base loop acts by

$$(j, \epsilon) \mapsto (\pi(j), \sigma_j \epsilon), \quad \prod_j \sigma_j = \text{sgn}(\pi), \quad G = \{(\sigma, \pi) \in \{\pm 1\}^4 \rtimes S_4 : \prod_j \sigma_j = \text{sgn}(\pi)\}. \quad (\text{EHM7})$$

Indeed the final Vandermonde changes by $\text{sgn}(\pi)$, while the product of source signs changes by $\prod_j \sigma_j$, and EHM6 fixes their product. The same signs describe $T_j = 1/\alpha_j$. This gives $|G| = 8 \cdot 24 = 192$ as an upper bound.

The original SM7–SM11 realize the bound without assuming a braid-group presentation. Retain their exact real roots, target, and inverse-source signs:

$$(R_1, R_2, R_3, R_4) = (7, 9, 11, 13), \quad u_{\text{real}} = (-1/40, -59/4, 95, -9009/40), \quad (\text{EHM8})$$

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = (\sqrt{5/6}, -i\sqrt{5/2}, -\sqrt{5/2}, i\sqrt{5/6}).$$

The derivatives are $(6/5, -2/5, 2/5, -6/5)$, proving $\alpha_j^2 F'(R_j) = 1$ in each coordinate. For an adjacent pair, with center $c = (R_j + R_{j+1})/2$, take

$$R_j(\theta) = c - e^{i\theta}, \quad R_{j+1}(\theta) = c + e^{i\theta}, \quad 0 \leq \theta \leq \pi, \quad \beta_j = (j + (j+1) + j - (j+1) -). \quad (\text{EHM9})$$

Keep the other roots and $A = -1/40$ fixed. The root sum stays forty, so the cubic coefficient stays one. The pair disk contains no other root, and the endpoint unordered root set is the initial one, proving this is a base loop. At a fixed root, the moving derivative factor is $(R_k - c)^2 - e^{2i\theta}$, contained in a right-half-plane disk not meeting zero, so its logarithm has zero total change. At a moving root the difference from its partner has argument change π ; each difference from an outside root has a logarithm with zero imaginary endpoint change, because it lies in a left or right half-plane and has real endpoints of the same sign. Thus α_j changes phase by $-\pi/2$. Substituting the four actual signs EHM8 proves exactly the four-cycle EHM9 and fixes every other state.

The squares β_j^2 flip the two adjacent signs. The three vectors $(1, 1, 0, 0), (0, 1, 1, 0), (0, 0, 1, 1)$ are independent over $\mathbb{F}_2$ and span the even-sum subspace, so these squares generate all eight allowed pure sign patterns. The root permutations of the β_j generate S_4 . The generated group therefore has at least $8 \cdot 24$ elements, proving that the upper bound EHM7 is attained. Its product $\beta_1 \beta_2 \beta_3$, applied rightmost first, is the eight-cycle $(1 + 2 + 3 + 4 + 1 - 2 - 3 - 4 -)$. This is the determinant-one signed-permutation group, not the even-sign group called $W(D_4)$: in the latter a signed four-cycle must have positive sign product and order four, while all shorter-cycle combinations have order dividing four or six, so no element has order eight.

The completed cover over $\mathcal{B}^\circ$ has the same monodromy. At a base point with $A \neq 0$, any loop in $\mathcal{B}^\circ$ can be moved slightly to avoid $A = 0$, with its base point fixed. Here is the needed elementary justification. Cover the compact loop by finitely many convex balls contained in $\mathcal{B}^\circ$, subdivide its parameter, and replace each piece by a polygon in these balls. Perturb the finitely many vertices so that their A -coordinates are nonzero, and detour any zero of a segment's linear A -coordinate by a small semicircle in that complex coordinate, within the same ball. The replacements are homotopic to the original paths inside these convex balls and avoid $A = 0$. Thus every completed-loop permutation already occurs over $\mathcal{B}$. Conversely the loops EHM9 remain available, so the group is exactly G . The base is path connected: the discriminant restricted to the complex line through any two allowed endpoints is a nonzero polynomial, and detouring its finitely many zeros connects them. Base-point changes therefore conjugate the permutation representation by an actual bijection of path lifts. That bijection preserves each sign pair, since the two initial opposite signs continue as opposites. The determinant-one subgroup EHM7 is invariant under conjugation by such a signed permutation, so the same description applies in the explicit labels at u_* .

3. Labels and both involutions at the original seven-point target

Use the original target $u_* = (0, 0, -1, 0)$, with $F_*(R) = R^3 - R$. Label all eight completed states by

label	a_+	a_-	b_+	b_-	c_+	c_-	m	n
$(R, T) \text{ or } (\infty, \Theta)$	$(1, \sqrt{2})$	$(1, -\sqrt{2})$	$(0, i)$	$(0, -i)$	$(-1, \sqrt{2})$	$(-1, -\sqrt{2})$	(∞, i)	$(\infty, -i)$

(EHM10)

The missing original state is exactly m . Write $\mathcal{A}_* \cong \mathbb{C}^8$ for the reduced fibre algebra, with point idempotents e_x . Evaluation supplies this isomorphism: the root idempotents at $0, 1, -1$ are $1 - R^2, (R^2 + R)/2, (R^2 - R)/2$, and splitting each nonzero sign quadratic gives its two indicator functions. At infinity, $e_m = (1 - i\Theta)/2$ and $e_n = (1 + i\Theta)/2$, extended by zero elsewhere. These formulas square to themselves, are complementary, and evaluate to the indicated indicator values.

The original deck involution Σ negates T, Θ . Coefficient conjugation κ fixes the coordinate generators. Native J has target map $j_{\text{out}}(A, B, C, D) = (-A, -B, C, -D)$ and chart maps

$$J(R, T) = (-R, -T), \quad J(v, \Theta) = (-v, \Theta). \quad (\text{EHM11})$$

They follow either from EHM4 and $J(\alpha, y, z, w) = (-\alpha, -y, z, -w)$, or by substituting in $F_{j_{\text{out}}u}(-R) = -F_u(R)$ and its derivative. At u_* , the relevant permutations on EHM10 are

$$\begin{aligned} s &= (a_+ a_-)(b_+ b_-)(c_+ c_-)(mn), \\ k &= (b_+ b_-)(mn), \quad j = (a_+ c_-)(a_- c_+)(b_+ b_-), \\ \tau_\Sigma &:= sk = (a_+ a_-)(c_+ c_-), \quad \tau_J := jk = (a_+ c_-)(a_- c_+)(mn). \end{aligned} \quad (\text{EHM12})$$

Here s, j act complex-linearly on the algebra; $\kappa, \#_\Sigma = \Sigma\kappa$, and $\#_J = J\kappa$ conjugate coefficients as well as applying k, τ_Σ, τ_J , respectively. Applying their coordinate formulas to every state verifies EHM12, including the fact that J fixes each infinity sign but $\#_J$ exchanges them.

For any of these conjugate-linear involutions, the complete Hermitian regular-trace pairing is

$$Q_\tau(f, g) = \sum_x \overline{f(\tau x)} g(x), \quad [Q_\tau] = P_\tau, \quad (\text{EHM13})$$

where $P_\tau e_x = e_{\tau x}$. To prove it, multiplication by f on the point basis is diagonal with entries $f(x)$, so its trace is their sum; now multiply $\iota(f)g$. A fixed point of τ contributes one positive scalar form. A transposed pair contributes $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, with one positive and one negative eigenvalue. Hence the full inertias are

$$\text{inertia } Q_k = (6, 2, 0), \quad \text{inertia } Q_{\tau_\Sigma} = (6, 2, 0), \quad \text{inertia } Q_{\tau_J} = (5, 3, 0). \quad (\text{EHM14})$$

In particular e_m is positive for Q_{τ_Σ} , but isotropic for Q_{τ_J} , where it pairs with e_n . The negative native infinity vector is $e_m - e_n$, with value -2 . The omitted state, a negative direction, and a radical direction are therefore explicitly related, but are not interchangeable names.

4. An explicit loop involving the omitted infinity state

The completed coefficient line

$$u(A) = (A, 0, -1, 0), \quad F_A(R) = R(AR^3 + R^2 - 1), \quad \mathfrak{D}(u(A)) = 4 - 27A^2 \quad (\text{EHM15})$$

starts at u_* . The discriminant identity follows either from the Sylvester determinant or from the cubic discriminant of $AR^3 + R^2 - 1$, multiplied by the squared resultant with R , which is one. Put $A_c = 2/(3\sqrt{3})$. For $0 < A < A_c$, the three nonzero roots consist of $R_L < -2/(3A) < R_C < 0 < R_P$: the cubic tends to minus infinity on the left, has positive value $4/(27A^2) - 1$ at $-2/(3A)$, value -1 at zero, and tends to plus infinity on the right. Its derivative has only these two critical points, proving the stated root count and order. As $A \rightarrow 0^+$, $R_C \rightarrow -1, R_P \rightarrow 1$, and $R_L = -1/A + O(A)$, the original infinity branch. At a nonzero root,

$$F'_A(R) = R^2(3AR + 2), \quad \Theta^2 = 3AR + 2. \quad (\text{EHM16})$$

The first follows by differentiating $R(AR^3 + R^2 - 1)$ at a root; the second uses EHM2. For the stated large-root asymptotic, put $w = AR_L$ while $A \neq 0$; the root equation becomes $w^2(w + 1) = A^2$. Its derivative in w at $(w, A) = (-1, 0)$ is one, so its unique analytic branch there satisfies $w = -1 + A^2 + O(A^4)$, obtained by substituting its convergent series. Dividing by the retained nonzero A gives $R_L = -1/A + A + O(A^3)$. EHM16 now gives $\Theta \rightarrow \pm i$. Thus the sign m continues along the large negative root with $T = -R_L\Theta$ positive imaginary, while c_+ continues along R_C with positive real T .

At A_c , the two negative roots collide at $-\sqrt{3}$. The other nonzero root is $\sqrt{3}/2$. The collision is transverse because $\partial_A F_A(R) = R^4 = 9$ and $\partial_R^2 F_A(R) = 2\sqrt{3} \neq 0$ there. Choose a sufficiently small $\epsilon > 0$, approach $A_c - \epsilon$ along the positive real interval, traverse the circle $A = A_c + \epsilon e^{i\theta}$, $\pi \leq \theta \leq 3\pi$, and return along the interval. No other discriminant point lies on or within this small disk. The resulting based-loop permutation is

$$g_\infty = (m \ c_- \ n \ c_+), \quad g_\infty^2 = (mn)(c_+ c_-), \quad (\text{EHM17})$$

fixing $a_{\pm}, b_{\pm}$. Here is the sign computation. In the local coordinate centered at the collision, the two root displacements initially are negative and positive real; their difference makes a positive half-turn as the target makes one turn. The derivative at the negative displacement initially is negative real and at the positive displacement positive real, since $F_{RR} > 0$. Each derivative therefore acquires argument change π ; its square root acquires phase i . Thus positive imaginary T on the large branch ends as negative real T on the other branch, giving $m \mapsto c_-$, whereas positive real T on that branch ends as positive imaginary on the large branch, giving $c_+ \mapsto m$. Negating signs gives the other two arrows. Analytic nonzero factors have logarithms on this small disk and contribute no further winding. The other two root pairs have separate holomorphic nonzero sign branches and are fixed. Returning along the real interval proves the exact labels in EHM17.

This loop leaves $A = 0$; loops entirely in its simple-root locus fix the two infinity sections $\Theta = \pm i$. The missing infinity state is nevertheless in the full monodromy orbit. In particular, two traversals take the retained state n to the omitted state m . On its returning original affine inverse branch, $T = -iR_L(1 + O(A^2))$, so EHM4 gives

$$y = -2R_L + O(A) = 2/A + O(A), \quad \alpha = 1/T \rightarrow 0. \quad (\text{EHM18})$$

Thus the loop has a bounded lift in the completed cover and an escaping lift at the endpoint in the original affine chart. This is the actual relation of the nontrivial holonomy to the seventh/eighth distinction, with its original coordinate divergence.

Let $v_c = e_{c_+} - e_{c_-}$ and $v_{\infty} = e_m - e_n$. Under the point-permutation action $\rho(g)e_x = e_{gx}$,

$$\rho(g_{\infty})v_c = v_{\infty}, \quad \rho(g_{\infty})v_{\infty} = -v_c, \quad Q_{\tau\Sigma}(v_c, v_c) = -2, \quad Q_{\tau\Sigma}(v_{\infty}, v_{\infty}) = 2. \quad (\text{EHM19})$$

All four equations follow by applying EHM17 and EHM12. One negative and one positive direction are exchanged. Both remain in the same eight-dimensional space; the next section calculates exactly what happens to the form.

5. The typed transport of the forms and involutions

The action $\rho(g)e_x = e_{gx}$ is a unital algebra automorphism of $\mathbb{C}^8$, because it permutes the orthogonal primitive idempotents. It preserves the complex bilinear regular trace pairing and the positive coefficient form $\sum_x \overline{f(x)}g(x)$. For an original conjugate-linear involution with point permutation τ , exact substitution gives

$$Q_{\tau}(\rho(g)f, \rho(g)h) = Q_{g^{-1}\tau g}(f, h), \quad \rho(g)\iota_{\tau}\rho(g)^{-1} = \iota_{g\tau g^{-1}}. \quad (\text{EHM20})$$

Thus $\rho(g)$ is an isometry from Q_{τ} to $Q_{g\tau g^{-1}}$. It preserves the original fixed form precisely when $g\tau = \tau g$, since equality of the matrices in EHM13 is then necessary and sufficient. EHM19 does not meet that condition; it is not a sign flip by an isometry of the unchanged indefinite form.

The same fact has a pathwise formulation retaining both native maps. Uniqueness of lifts gives

$$\Sigma M_{\gamma} = M_{\gamma}\Sigma, \quad \kappa M_{\gamma}\kappa = M_{\bar{\gamma}}, \quad JM_{\gamma}J^{-1} = M_{j_{\text{out}}\gamma}. \quad (\text{EHM21})$$

For a path with differing endpoints these are identities between its stated endpoint fibres; at the real, native-fixed base point u_* they are identities on that fibre. Conjugating a lifted path by the respective coordinate map gives a lift of the transformed base path with the corresponding initial point, which proves each equality. Combining the second identity with the first or third gives the corresponding equations for $\#_{\Sigma}$ and $\#_J$. In particular native reflection of $\gamma_+(h) = (0, 0, -1 - 16h, 16h)$ is $\gamma_-(h) = (0, 0, -1 - 16h, -16h)$, generally a different target path. Native reflection is not to be substituted for continuation around a loop of γ_+ .

The seven-state quotient has its exact transported family. For any state x , let $q_x : \mathcal{A}_* \rightarrow \mathcal{A}_*/\mathbb{C}e_x$. Then

$$\bar{\rho}_g : \mathcal{A}_*/\mathbb{C}e_x \xrightarrow{\sim} \mathcal{A}_*/\mathbb{C}e_{gx}, \quad \bar{\rho}_g q_x = q_{gx}\rho(g). \quad (\text{EHM22})$$

This is well defined because $\rho(g)(\mathbb{C}e_x) = \mathbb{C}e_{gx}$; its inverse is induced by g^{-1} . Since G is transitive, the smallest monodromy-stable ideal containing the omitted line $\mathbb{C}e_m$ is all of $\mathcal{A}_*$: it contains every point idempotent and hence their sum one. Consequently there is no nonzero quotient algebra with the full induced monodromy action that kills exactly this one state. The seven-state objects instead form the explicitly transported family EHM22, or a single such quotient over its stabilizer subgroup of order $192/8 = 24$. This proves the obstruction and its replacement object, rather than treating the missing state as unrelated to the cover.

6. Both meridians of the actual chosen heat path

Retain

$$\gamma_+(h) = (0, 0, -1 - 16h, 16h), \quad F_h(R) = (R - 1)(R^2 + R - 16h), \quad \text{Disc } F_h = (2 - 16h)^2(1 + 64h). \quad (\text{EHM23})$$

The factorization follows by expansion. For the discriminant, the quadratic discriminant is $1 + 64h$, and its resultant with $R - 1$ is its value $2 - 16h$ at one; the product of squared pairwise root differences gives their stated product. Thus the path meets the discriminant at $h_c = 1/8$ with multiplicity two and at $h_d = -1/64$ with multiplicity one. On the branch $s(h) = \sqrt{1 + 64h}$ with $s(0) = 1$, its roots and derivatives are

$$R_a = 1, \quad R_b = (-1 + s)/2, \quad R_c = (-1 - s)/2, \quad \mu_a = (9 - s^2)/4, \quad \mu_b = s(s - 3)/2, \quad \mu_c = s(s + 3)/2. \quad (\text{EHM24})$$

The derivatives follow by evaluating $3R^2 - 1 - 16h$, or by differentiating the factorization at each root.

Take the positive meridian around h_c approached along $(0, h_c)$, and the positive meridian around h_d approached along $(h_d, 0)$. In the exact base labels EHM10 their permutations are

$$g_c = (a_+ a_-)(b_+ b_-), \quad g_d = (b_+ c_- b_- c_+). \quad (\text{EHM25})$$

Near h_c , all three root functions are individually holomorphic, μ_a, μ_b have simple zeros, and μ_c has none. The first two square roots therefore change sign once, and the third does not, proving the first formula. Around h_d , let $h = h_d + \epsilon e^{i\theta}$, starting to its right; then $s = 8\sqrt{\epsilon} e^{i\theta/2}$ makes an upper half-circle. For sufficiently small ϵ , the continuous argument of μ_b runs from π to 2π , and that of μ_c from zero to π : in EHM24 the factors $s - 3, s + 3$ remain in their respective left and right half-planes, with real endpoint ratios of the same sign. Consequently $b_+ \mapsto c_-$, $c_+ \mapsto b_+$, and the other two arrows follow by negating signs. This proves the second formula. The infinity states are the constant sections $\Theta = \pm i$ on this path and are fixed by both loops.

These permutations obey $g_c^2 = 1, g_d^4 = 1, g_c g_d g_c = g_d^{-1}$. The four powers of g_d fix $a_{\pm}$, and the four elements $g_c g_d^j$ exchange them, so these eight elements are distinct; the relations reduce every word to one of them. The path's monodromy is thus the group of order eight

$$\langle g_c, g_d \rangle \cong D_8, \quad g_d^2 = (b_+ b_-)(c_+ c_-). \quad (\text{EHM26})$$

The base is the complex h -plane with these two points deleted. To see that these two meridians generate all its loop permutations, surround each puncture by a small disjoint disk and join their boundaries to the base point by nonintersecting arcs. Cutting a large disk containing any given compact loop along those arcs leaves a region without holes, in which the loop reduces to successive boundary traversals of the two punctures. These are the two stated meridians and their inverses. This supplies the elementary topological generation needed for the equality, rather than only a subgroup assertion.

The first loop is an isometry of the fixed $Q_{\tau\Sigma}$, since both permutations in EHM12 and g_c are products of individual pair flips. It sends the negative vector $v_a = e_{a_+} - e_{a_-}$ to $-v_a$ and the positive vector $v_b = e_{b_+} - e_{b_-}$ to $-v_b$; their squared values stay -2 and 2 . The second loop instead sends $v_c \mapsto v_b$ and $v_b \mapsto -v_c$, exchanging values -2 and 2 for the fixed form. Its transported form is exactly EHM20. Therefore neither nontrivial continuation nor a sign change alone proves positivity of the unchanged pairing.

7. The exact local family, the retained radical, and the regular heat-loop extension

Put $t = h - 1/8$, $\epsilon = R - 1$, and

$$\beta(t) = \frac{\sqrt{9 + 64t} - 3}{2}, \quad \beta(\beta + 3) = 16t, \quad \mathcal{L} = \frac{\mathbb{C}\{t\}[\epsilon, T]}{(\epsilon^2 - \beta\epsilon, T^2 + 16t - (6 + 3\beta)\epsilon)}. \quad (\text{EHM27})$$

Here $\mathbb{C}\{t\}$ means convergent power series at zero and the square root has value three. The root branches in this cluster are $\epsilon = 0, \beta$; the third root is separated, so its root factor is invertible in this local algebra. Reducing $F'_h(1 + \epsilon) = 6\epsilon + 3\epsilon^2 - 16t$ by $\epsilon^2 = \beta\epsilon$ proves the second relation. Successive division by the two monic relations proves that $1, \epsilon, T, \epsilon T$ is a free basis. The special fibre is therefore

$$\mathcal{L}_0 = \mathbb{C}[\epsilon, T]/(\epsilon^2, T^2 - 6\epsilon) \cong \mathbb{C}[T]/(T^4), \quad \epsilon = T^2/6. \quad (\text{EHM28})$$

Substitution proves both inverse maps in this isomorphism.

The chosen-loop permutation g_c has the regular local-family realization $\epsilon \mapsto \epsilon, T \mapsto -T$. On the whole eight-state family near h_c , let

$$e_c(h, R) = \frac{(R-1)(R-R_b(h))}{(R_c(h)-1)(R_c(h)-R_b(h))}, \quad R \mapsto R, \quad T \mapsto (2e_c-1)T, \quad \Theta \mapsto \Theta. \quad (\text{EHM29})$$

The denominator stays nonzero near h_c . The numerator evaluates to zero on the cluster and to its denominator at the separated root, so it is the separated-root idempotent. Since $(2e_c-1)^2 = 1$, this map preserves the entire signed relation, flips just the two cluster signs, and fixes the separated pair and infinity states. It is an involution and realizes EHM25. At h_c , $e_c = (R-1)^2/9$. This is a local extension of this loop, not the global sign deck map, which would negate all eight signs.

Retain the original radical generators, extended by zero on the separated and infinity factors:

$$\begin{aligned} n_1 &= (R-1)(R+2) = T^2/2, \\ n_2 &= T(R+2) = 3T + T^3/6, \\ n_3 &= T(R-1)(R+2) = T^3/2. \end{aligned} \quad (\text{EHM30})$$

The right sides follow from $R = 1 + T^2/6$ modulo T^4 . Conversely $T = n_2/3 - n_3/9, T^2 = 2n_1, T^3 = 2n_3$, proving independence and spanning of $\mathfrak{N} = (T)$. Multiplication gives

$$n_2^2 = 18n_1, \quad n_1n_2 = 3n_3, \quad n_1^2 = n_1n_3 = n_2n_3 = n_3^2 = 0, \quad \mathfrak{N}^2 = \langle n_1, n_3 \rangle, \quad \mathfrak{N}^3 = \langle n_3 \rangle, \quad \mathfrak{N}^4 = 0. \quad (\text{EHM31})$$

Each assertion follows by multiplying the displayed powers of T ; the nonzero products and independence prove the stated ideal powers. The specialized loop acts by

$$g_c^* : (n_1, n_2, n_3) \mapsto (n_1, -n_2, -n_3). \quad (\text{EHM32})$$

Here the point map and its pullback are both involutions, so the idempotent-transport convention ρ agrees with the pullback in this case.

Every $f \in \mathcal{L}_0$ has multiplication trace $4f(0)$: its nonconstant part lies in the nilpotent ideal and has a nilpotent multiplication operator, while its constant part acts on four basis vectors. The conjugate-deck form, with $T^\# = -T$, is consequently

$$Q_{\mathcal{L}_0}(f, g) = 4\overline{f(0)}g(0), \quad \text{rad } Q_{\mathcal{L}_0} = \mathfrak{N}. \quad (\text{EHM33})$$

The complete special fibre is $\mathcal{L}_0 \times \mathbb{C}[T_c]/(T_c^2 - 9) \times \mathbb{C}[\Theta]/(\Theta^2 + 1)$. Its latter forms have Gram matrices $\text{diag}(2, -18)$ and $\text{diag}(2, 2)$; direct multiplication proves them as in EHM13. Thus the whole form has inertia $(4, 1, 3)$, and the nonzero negative separated vector remains. The local radical is exactly the kernel of the regular trace pairing, not an assertion that its elements vanish as algebra elements.

The original HEB map is the further quotient setting $R = 1$, hence $T^2 = 0$, with $r_H = T/(2\sqrt{2})$. Its exact sequence on the radical is

$$0 \longrightarrow \mathfrak{N}^2 \longrightarrow \mathfrak{N} \xrightarrow{\pi_1} \mathbb{C}r_H \longrightarrow 0, \quad \pi_1(n_1) = 0, \quad \pi_1(n_2) = 6\sqrt{2}r_H, \quad \pi_1(n_3) = 0. \quad (\text{EHM34})$$

These images follow by evaluation in EHM30, proving the kernel, surjectivity, and constants. The loop g_c negates this surviving infinitesimal and preserves its nonzero algebra class.

8. The transverse order-four collision and its original-coordinate correction

An actual transverse path through the same coefficient point is

$$F_z(R) = ((R-1)^2 - z)(R+2), \quad u(z) = (0, 0, -3-z, 2-2z), \quad x = R-1, \quad x^2 = z, \quad T^2 = 2x(x+3). \quad (\text{EHM35})$$

Expansion proves the target formula, and differentiating on the root relation proves the last equation. Choose the analytic square root $g(x) = \sqrt{2(x+3)}$ with $g(0) = \sqrt{6}$. Then

$$y = T/g(x), \quad y^2 = x, \quad y^4 = z, \quad x = y^2, \quad T = yg(y^2). \quad (\text{EHM36})$$

Both compositions are exact and analytic near the local point because g never vanishes. Positive point continuation is $y \mapsto iy$. Its analytic deck map and coordinate pullback are

$$\alpha : (x, T) \mapsto \left(-x, iT\sqrt{\frac{3-x}{3+x}} \right), \quad \alpha^2 : (x, T) \mapsto (x, -T). \quad (\text{EHM37})$$

The ratio is the specific analytic branch with value one at zero. Substitution in EHM36 proves the equations and the square, without omitting any unit factor. At the special fibre, using $x = T^2/6, x^2 = 0$, its pullback is

$$\alpha_0^*(T) = iT - \frac{i}{18}T^3, \quad \alpha_0^*(n_1) = -n_1, \quad \alpha_0^*(n_2) = in_2 - in_3, \quad \alpha_0^*(n_3) = -in_3. \quad (\text{EHM38})$$

Indeed $\sqrt{(3-x)/(3+x)} = 1 - x/3$ modulo x^2 , giving the first formula. Inserting it in EHM30 gives the remaining three. Its square sends T to $-T$ and its fourth power is the identity. Thus it preserves all products EHM31. The correction $-iT^3/18$ is required for the analytic continuation in the original reciprocal coordinate.

The distinction between point maps and pullbacks is explicit: a point map α has pullback $\alpha^*e_x = e_{\alpha^{-1}x}$. Therefore the forward idempotent transport $\rho(\alpha)e_x = e_{\alpha x}$ of EHM20 is $(\alpha^{-1})^*$. On this special local algebra it sends $T \mapsto -iT + iT^3/18$, with the inverse of the three radical transformations EHM38. Both maps have the same invariants and coinvariants, but opposite eigenvalues $i, -i$ on the corresponding nonreal eigenspaces. This retains the orientation rather than silently changing the convention for a four-cycle.

The relation with the chosen heat path is itself an exact map. Put

$$m(t) = (1 + R_b)/2, \quad d(t) = (1 - R_b)/2, \quad \ell(t) = m(t) - R_c, \quad x = R - m(t). \quad (\text{EHM39})$$

Then $F_h(R) = (x^2 - d(t)^2)(x + \ell(t))$ by its three root factors. Also

$$d(t) = -\frac{16t}{3 + \sqrt{9 + 64t}}, \quad d(t)^2 = \frac{256t^2}{(3 + \sqrt{9 + 64t})^2}. \quad (\text{EHM40})$$

These identities follow by rationalizing $(3 - \sqrt{9 + 64t})/4$. The denominator is an analytic nonzero unit near zero, so one small positive t -loop makes this transverse parameter wind twice. Replacing $x + 3$ in EHM36 by the retained analytic unit $x + \ell(t)$ gives the same exact fourth-root calculation. Its continuation is consequently the square of the transverse order-four map, exactly g_c . This proves the relationship between the two paths and their cycle orders.

At the other collision put $v = h + 1/64, x = R + 1/2$. Direct expansion gives $F_h = (x^2 - 16v)(x - 3/2)$ and $T^2 = 2x(x - 3/2)$ on the root algebra. Choosing $\sqrt{2(x - 3/2)}$ with value $i\sqrt{3}$ gives the exact coordinate $y = T/\sqrt{2(x - 3/2)}$ with $y^4 = 16v$. At the special fibre $x = -T^2/3$, so its positive point continuation has pullback

$$\alpha_{d,0}^*(T) = iT - \frac{2i}{9}T^3, \quad (\alpha_{d,0}^*)^2(T) = -T. \quad (\text{EHM41})$$

This follows by expanding the corresponding ratio square root to first order in x . Its branch has the four-cycle g_d from EHM25. It is a regular specialization at its own collision, which does not imply regular specialization at the different collision h_c .

9. The regular-extension obstruction and its nonzero residue

A based permutation is not automatically a single-valued deck map over the punctured h_c -disk. Such a deck map must commute with the local monodromy g_c : apply it before and after lifting a local loop and use uniqueness. Conversely a commuting permutation extends uniquely over that punctured disk by continuing it in the eight local branches; commutation makes the result independent of the chosen continuation path. Inside the explicitly computed D_8 , the centralizer is

$$C_{D_8}(g_c) = \{1, g_c, g_d^2, g_c g_d^2\}. \quad (\text{EHM42})$$

Indeed $g_c g_d^j = g_d^{-j} g_c$, so commutation requires $g_d^j = g_d^{-j}$, exactly even j ; multiplying by g_c gives the other two. Thus the odd powers of g_d , and their g_c -multiples, already require a slit neighborhood to label their transported maps. They also move some states in the four-point collision cluster to separated reduced states. Any holomorphic extension to the special fibre would preserve its unique nonreduced local factor, so these maps could not extend through the collision even after attempting that identification.

The commuting element g_d^2 has a stronger, explicitly calculable obstruction. It fixes the a -sign and flips only the b -sign in the local cluster. The two local root values of ϵ are zero and β , so its unique generic algebra formula is

$$S_b(\epsilon) = \epsilon, \quad S_b(T) = \left(1 - \frac{2\epsilon}{\beta}\right)T, \quad S_b(\epsilon T) = -\epsilon T. \quad (\text{EHM43})$$

The interpolation multiplier has values one and minus one on the two root branches. Its square is one because $\epsilon^2 = \beta\epsilon$, proving that this is an involutive algebra automorphism on the punctured disk; the last equation

follows from that same relation. This is its only possible continuation there, since the eight reduced points determine the algebra map. The separated c -factor also changes sign under g_d^2 , but this is a bounded, regular operation on that separate factor.

The simple zero of β produces an actual pole in the free basis EHM27. Its residue, meaning the specialization of the operator tS_b at $t = 0$, is

$$\begin{aligned} tS_b(T)|_0 &= -\frac{2t}{\beta}\epsilon T \Big|_0 = -\frac{3}{8}\epsilon T = -\frac{T^3}{16} \neq 0, \\ tS_b(1)|_0 &= tS_b(\epsilon)|_0 = tS_b(\epsilon T)|_0 = 0. \end{aligned} \quad (\text{EHM44})$$

Here $t/\beta = (\beta + 3)/16$ follows from EHM27, and $\epsilon T = T^3/6$ at the collision. Freeness of the four-element basis shows that this nonzero pole cannot be removed by another regular formula agreeing on the punctured disk. Projection onto the analytic cluster summand shows that adjoining the separated and infinity factors cannot cancel it either.

The residue is exactly the derivation

$$\delta = -\frac{T^3}{16} \frac{d}{dT} : \mathbb{C}[T]/(T^4) \longrightarrow \mathbb{C}[T]/(T^4), \quad \ker \delta = \mathbb{C}1 \oplus (T^2), \quad \text{im } \delta = (T^3), \quad \delta^2 = 0. \quad (\text{EHM45})$$

The polynomial derivation preserves the ideal (T^4) , since it sends T^4 to $-T^6/4$; hence it descends and obeys the Leibniz identity. On $1, T, T^2, T^3$, its values are $0, -T^3/16, 0, 0$, exactly EHM44. These four values prove the kernel, image, and square. In the retained radical coordinates this is

$$\delta(n_1) = 0, \quad \delta(n_2) = -\frac{3}{8}n_3, \quad \delta(n_3) = 0, \quad \bar{\delta} : \mathfrak{N}/\mathfrak{N}^2 \xrightarrow{\sim} \mathfrak{N}^3, \quad [n_2] \longmapsto -\frac{3}{8}n_3. \quad (\text{EHM46})$$

The values follow by applying EHM45 to EHM30. Since δ kills $\mathfrak{N}^2$, the quotient map is well defined; both its source and target are one dimensional and its displayed value is nonzero, proving the isomorphism. This residue is an actual nonzero map between the retained infinitesimal layers. Its value depends on the specified original clock $t = h - 1/8$, whose scale is kept in EHM44.

The other nontrivial commuting candidate $g_c g_d^2$ acts on T by the negative of EHM43 and has residue $-\delta$. Consequently exactly

$$\{1, g_c\} \subset \langle g_c, g_d \rangle \quad (\text{EHM47})$$

extends holomorphically over the whole h_c -fiber. These two extensions were constructed in EHM29. All others fail either the punctured-disk commutation condition or the explicit nonzero residue test. This is a complete classification within the actual based heat-path group, not a claim that the different transverse local group has only two extending elements: its four elements were explicitly constructed in EHM37.

10. The action on invariants, coinvariants, and the trace radical

For any finite permutation group K acting on the eight states, let $V = \mathbb{C}^8$, V^K be its invariant vectors, and $V_K = V/W_K$, where $W_K = \text{span}\{\rho(g)v - v : g \in K, v \in V\}$. For an orbit O let $u_O = \sum_{x \in O} e_x$. The exact maps are

$$\mathcal{P}_K v = \sum_O \frac{\sum_{x \in O} v_x}{|O|} u_O, \quad 0 \longrightarrow W_K \longrightarrow V \xrightarrow{\mathcal{P}_K} V^K \longrightarrow 0, \quad V_K \xrightarrow{\sim} V^K, \quad [v] \mapsto \mathcal{P}_K v. \quad (\text{EHM48})$$

On every orbit the differences $e_x - e_y$ span the coefficient-sum-zero subspace: choose one base point in the orbit and subtract it from all the others. Each such difference belongs to W_K because some group element takes one point to the other. Hence W_K is exactly the direct sum of those subspaces, proving the kernel, exactness, and quotient isomorphism. The invariant algebra consists of functions constant on each orbit and is $\mathbb{C}^{\#\text{orbits}}$, with trace weights $|O|$. The map $\mathcal{P}_K$ is linear and is not in general multiplicative: for two different points in a nontrivial orbit, their product is zero, whereas the product of their two images is $u_O/|O|^2 \neq 0$. Thus the linear coinvariant space has not been assigned an unjustified quotient-algebra product. The ideal generated by the differences kills every nontrivial orbit, since multiplying $e_x - e_y$ by e_x gives e_x .

The actual dimensions and restrictions of the unchanged conjugate-deck form are

K	nontrivial state orbits	$\dim V^K = \dim V_K$	$\text{inertia}(Q_{\tau\mathfrak{N}} _{V^K})$
$\langle g_c \rangle$	$\{a_+, a_-\}, \{b_+, b_-\}$	6	(5, 1, 0)
$\langle g_d \rangle$	$\{b_+, c_-, b_-, c_+\}$	5	(4, 1, 0)
$\langle g_c, g_d \rangle$	$\{a_+, a_-\}, \{b_+, b_-, c_+, c_-\}$	4	(4, 0, 0)
G	all eight states	1	(1, 0, 0).

(EHM49)

To prove the signature entries without a dimension inference, every displayed orbit sum has positive value equal to its cardinality, since its orbit is preserved by τ_Σ . For the first row the remaining c_+, c_- block is hyperbolic and the two infinity states are positive. For the second the remaining a_+, a_- block is hyperbolic and infinity is positive. In the last two rows every invariant basis vector is an orbit sum and these sums have disjoint supports, giving mutually orthogonal positive values. These are restrictions to explicitly mapped subspaces, not positivity of the full original form. On the reduced eight-state fibre none of these nontrivial linear coinvariant projections makes the original full form descend by arbitrary lifts: its radical is zero by EHM14, whereas $W_K \neq 0$. A form descends through a projection precisely when its kernel pairs to zero with every vector, which proves this obstruction directly. The projection EHM48 still supplies its specified comparison with the invariant restriction.

At the nonreduced collision, the answer changes in an exactly calculable way. The extending local heat involution has invariants $\langle 1, T^2 \rangle$ and difference space $\langle T, T^3 \rangle$. The latter lies in the trace radical (T) , so the local form EHM33 does descend to its two-dimensional linear coinvariant space, with radical the surviving class of T^2 . Its invariant restriction has inertia $(1, 0, 1)$. On the whole fibre the unchanged separated factor and infinity pair add their signatures, giving the complete invariant/coinvariant form inertia

$$(4, 1, 1) \quad \text{for the extending heat involution at } h_c. \quad (\text{EHM50})$$

By contrast, the transverse order-four map EHM38 has eigenvalues $1, i, -1, -i$ on a suitable basis of $\mathcal{L}_0$. One may use the exact coordinate $y = T/\sqrt{2(3 + T^2/6)}$ from EHM36, in which the basis $1, y, y^2, y^3$ gives these four values; its linear change of basis is invertible since the coefficient of T in y is $1/\sqrt{6} \neq 0$. Thus its invariants are the scalar line and its difference space is the entire nilpotent radical. The local trace descends to the resulting one-dimensional quotient with value $4|c|^2$. On the whole transverse fibre the invariant/coinvariant form is nondegenerate of inertia $(4, 1, 0)$, with the same separated negative direction still present. These conclusions use only the maps that actually specialize; EHM47 excludes assigning a full based D_8 -action to the special fibre.

The residue EHM45 is invisible to this regular trace because its image lies in $\mathfrak{N}^3 \subset \text{rad } Q$, but it is visible in the algebra and its filtration through the isomorphism EHM46. Explicitly $Q(\delta f, g) = Q(f, \delta g) = 0$ for all local f, g , while $\delta(T) = -T^3/16 \neq 0$. These are simultaneous exact identities; zero pairing has not been used to delete the map.

11. Native reflection through both local algebras

Native J maps the positive-path critical point $(0, 0, -3, 2)$ to $(0, 0, -3, -2)$. In the negative-path cluster put $\epsilon_- = R + 1$; its local algebra has $\epsilon_-^2 = 0, T_-^2 = -6\epsilon_-$, while the positive cluster has $T_+^2 = 6\epsilon_+$. The coordinate pullback is

$$J^*(\epsilon_-) = -\epsilon_+, \quad J^*(T_-) = -T_+. \quad (\text{EHM51})$$

Substitution proves both relations and the inverse, which has the same signs. Coefficient conjugation adds the native $\#_J$ map. For the negative-side generators $\nu_1 = (R + 1)(R - 2), \nu_2 = T_-(R - 2), \nu_3 = T_-(R + 1)(R - 2)$, this gives

$$J^*(\nu_1) = n_1, \quad J^*(\nu_2) = n_2, \quad J^*(\nu_3) = -n_3. \quad (\text{EHM52})$$

Every equality follows by replacing R with $-R$ and T_- with $-T_+$, so it preserves the entire nilpotent ideal and its powers, with their original constants.

The positive meridians on the negative target path, in the same initial eight labels, are

$$g_c^- = (c_+ c_-)(b_+ b_-), \quad g_d^- = (b_+ a_- b_- a_+). \quad (\text{EHM53})$$

These follow by applying the native coordinate map and the same root/sign continuation argument used in EHM25. Since coefficient conjugation reverses the orientation of a positive meridian, the correctly typed native identities are $\#_J g_c^+(\#_J)^{-1} = (g_c^-)^{-1}$ and $\#_J g_d^+(\#_J)^{-1} = (g_d^-)^{-1}$; they can also be verified directly from EHM12, EHM25, and EHM53. On the positive path $\#_\Sigma g_c \#_\Sigma^{-1} = g_c$ and $\#_\Sigma g_d \#_\Sigma^{-1} = g_d^{-1}$. At the positive critical local algebra, coefficient conjugation and $\#_\Sigma$ each conjugate α_0^* of EHM38 to its inverse, since they conjugate i to $-i$ and preserve its real coefficients. Its square preserves both involutions. All these local maps preserve the degenerate regular-trace form EHM33 because they fix the constant coefficient, even where they do not commute with the original involution; the whole radical accounts for that exact loss of detection.

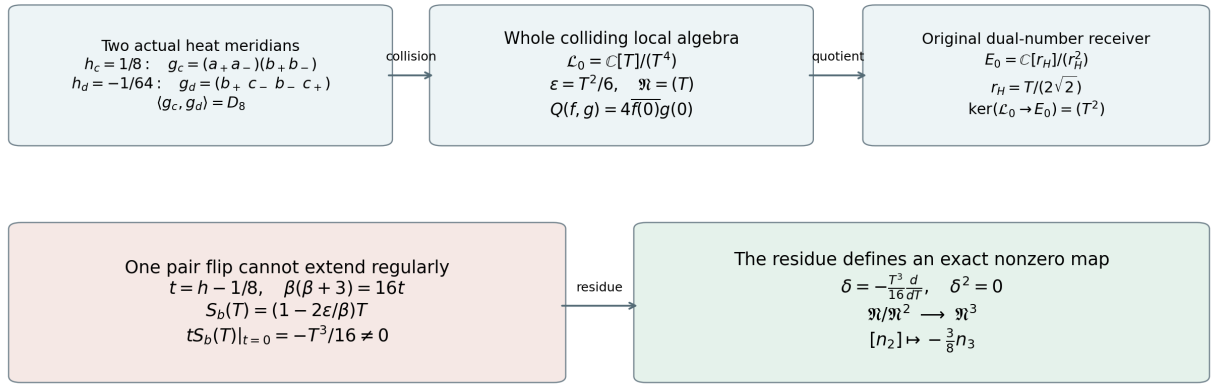
The monodromy receiver for any subsequent programme pairing is now explicit: its fibre is the full eight-state algebra, its transport is $\rho : G \rightarrow \text{Aut}_{\text{C-alg}}(\mathcal{A}_*)$, its two original involutions and their transport are EHM12 and EHM20–EHM21, and its regular collision maps and obstruction residue are EHM29, EHM38, and

EHM43–EHM46. Any proposed scalar pairing transported by these same maps has the directly testable transformation matrix $\rho(g)^*Q\rho(g)$. No equality of that matrix with a new supported-zero Weil formula is assumed here.

12. Exact verification scope

The reproducible script `check_eight_state_holonomy.py` passed **96 exact checks**, recorded in `EIGHT_STATE_HOLONOMY_CHECKS.json`. They reconstruct the 192 signed permutations and the eight-element heat-path group; verify the original involution matrices, all three selected-loop pairing transports, the invariant restriction signatures, the two original discriminants and transverse derivative constants, the order-four local coordinate maps and their original radical products, and the nonzero residue with every basis-monomial Leibniz identity. The analytic path-lifting, chosen logarithm branches, unramified infinity continuation, full local extension classification, and specialization arguments have their complete proofs above; finite symbolic checks are not substituted for those proofs.

Holonomy survives where the infinitesimal trace vanishes



$$n_1 = T^2/2, \quad n_2 = 3T + T^3/6, \quad n_3 = T^3/2; \quad Q(\delta f, g) = 0 \text{ while } \delta(T) \neq 0.$$

EHM23–34 prove the meridians and full local algebra. EHM42–47 prove the pole, residue and exact filtration map.

Figure 9: The two actual heat meridians, the full length-four collision algebra, its dual-number quotient, and the nonzero holonomy residue between infinitesimal layers. EHM23–EHM34 and EHM42–EHM47 give the complete proofs, signs and clock scale.

Exact residue-duality continuation

RD1–RD31b proves the complete Jacobian map from perfect residue duality to the retained regular trace, identifies the actual packet radical with cotangent cohomology, and computes the supported endpoint compensation. The existing formulas and all their coordinates remain unchanged.

Actual signed-cover averaging and the supported-zero endpoint

This calculation uses the original eight-state completion and its actual monodromy. It proves that averaging either of the two retained trace forms over that group produces a positive semidefinite form, calculates the complete defect, and evaluates the change on actual Weil tests. The group action belongs to the auxiliary quartic cover. Its action on an arithmetic global trace is not asserted.

The original cover and its full monodromy are proved in *The ES-Fable inverse correspondence in the original weighted conductor*, SM1–SM19, source lines 2097–2513. The exact fibre, trace and involutions used here are proved in *The original eight-state heat comparison*, ESH9–ESH34. The actual test convention is *The actual Weil correction has the ES three-plus-one form*, WEC1–WEC5. The Gaussian theta map below proves the endpoint comparison directly, retaining supported zero.

1. Exact fibre and point coordinates

At the unchanged target $\mathbf{U}_* = (0, 0, -1, 0)$, the completed algebra is

$$\mathcal{A} = \mathbb{C}[R, T]/(R^3 - R, T^2 - 3R^2 + 1) \times \mathbb{C}[\Theta]/(\Theta^2 + 1). \quad (\text{HWA1})$$

Its eight geometric points, in the order retained throughout, are

$$X = (0+, 0-, 1+, 1-, -1+, -1-, \infty+, \infty-) \quad (\text{HWA2})$$

with coordinates $(0, \pm i), (1, \pm\sqrt{2}), (-1, \pm\sqrt{2}), (\infty, \pm i)$. Evaluation is an algebra isomorphism $\text{ev} : \mathcal{A} \rightarrow \mathbb{C}^X$. Indeed, the root indicators are $e_0 = 1 - R^2, e_1 = (R^2 + R)/2, e_{-1} = (R^2 - R)/2$. Inside a finite root factor with chosen positive sign coordinate t_j , its point indicators are $e_j(1 \pm T/t_j)/2$. The infinity indicators are $(1 \mp i\Theta)/2$. These eight orthogonal idempotents sum to one, evaluate as the eight coordinate vectors, and supply the inverse. Multiplication is diagonal in this basis, so

$$\text{Tr}_{\mathcal{A}}(x) = \sum_{p \in X} x_p. \quad (\text{HWA3})$$

No coordinate weighting is discarded: the coefficient basis and point basis are related by the exact map

$$(c_j, d_j) \mapsto (c_j + t_j d_j, c_j - t_j d_j), \quad (t_0, t_1, t_{-1}, t_\infty) = (i, \sqrt{2}, \sqrt{2}, i). \quad (\text{HWA4})$$

Let Σ negate every sign coordinate and let κ conjugate coefficients. The original native map is $J(R, T) = (-R, -T)$, $J(v, \Theta) = (-v, \Theta)$, where $v = 1/R$ on the reciprocal chart. Combining each with coefficient conjugation gives the following point permutations:

$$q_\Sigma = (1+ \ 1-)(-1+ \ -1-), \quad q_J = (1+ \ -1-)(1- \ -1+)(\infty+ \ \infty-). \quad (\text{HWA5})$$

For example, conjugation at root0 swaps $i, -i$, after which either Σ or J swaps them back. At infinity J fixes Θ , so conjugation still swaps the two points. Formula(HWA5) follows on all eight points. If Q_q denotes the permutation matrix of the involution q , then

$$B_q(x, y) = \text{Tr}(x^{\#_q} y) = x^* Q_q y. \quad (\text{HWA6})$$

Thus B_J has inertia $(5, 3, 0)$ and B_Σ has inertia $(6, 2, 0)$: each swapped pair contributes eigenvalues $1, -1$, and each fixed point contributes 1. The point metric here is the exact sum in the regular trace, not the ES source metric.

2. The actual order-192 group and its average

Label the four root pairs by $j = 0, 1, -1, \infty$. The group proved in SM1–SM19 is

$$G = \{(\sigma, \pi) : \pi \in S_4, \sigma \in \{\pm 1\}^4, \prod_j \sigma_j = \text{sgn } \pi\}, \quad (j, \epsilon) \mapsto (\pi(j), \sigma_j \epsilon). \quad (\text{HWA7})$$

There are exactly $24 \cdot 8 = 192$ elements. For completeness, its appearance in the actual quartic is determined by $a_j^2 H'(r_j) = 1$ and the invariant $A^2 \prod_{j < k} (r_k - r_j) \prod_j a_j \in \{1, -1\}$. Exchanging adjacent roots along semicircles makes the moving derivatives turn through π , so their square roots yield four-cycles $(j + (j+1) + j - (j+1) -)$. Their squares generate all even sign patterns, and their root permutations generate S_4 ; this proves equality with(HWA7), as calculated with exact paths and coordinates in SM7–SM12. The completed cover is unramified

at the infinity pair because its reciprocal equations have determinant $2\Theta k'(0) = 2\Theta \neq 0$. Paths from a nearby point with $A \neq 0$ to $\mathbf{U}_*$ transport its eight separate local charts. Relabelling the roots and their chosen signs conjugates(HWA7) by a signed permutation, which preserves the determinant-one subgroup. We therefore use that same exact group on(HWA2).

Let P_g be its permutation matrix on point functions. Define the explicitly scaled finite conjugation sum

$$\mathcal{E}(Q) = \frac{1}{192} \sum_{g \in G} P_g^* Q P_g. \quad (\text{HWA8})$$

The factor $1/192$ is part of the definition, so the unit vector $\mathbf{1}$, whose original trace square is 8, still has square 8. Let

$$P_{\text{ev}} = \frac{I + P_\Sigma}{2}, \quad P_c = \frac{\mathbf{1}\mathbf{1}^*}{8}. \quad (\text{HWA9})$$

These are orthogonal projections for the point metric, respectively onto sign-even functions and constant functions. Their kernels and ranks follow directly: P_{ev} replaces both values in each pair by their arithmetic mean, so it has rank 4; P_c replaces every value by the mean of all eight, so it has rank 1.

The two exact averages are

$$\boxed{\mathcal{E}(Q_\Sigma) = P_{\text{ev}}, \quad \mathcal{E}(Q_J) = \frac{1}{3}P_{\text{ev}} + \frac{2}{3}P_c.} \quad (\text{HWA10})$$

Here is a direct proof without an assumed irreducibility theorem. The even-sign kernel $K \subset G$ has eight elements. On the sign-even subspace it acts trivially. On the four-dimensional sign-odd subspace it acts by the four distinct sign characters. For two distinct root indices j, k , the average of $\sigma_j \sigma_k$ over K is zero: choose a third root $l \neq j, k$, and multiply each sign pattern by the one flipping j, l , which pairs opposite contributions. Hence averaging a matrix on that subspace over K removes all off-diagonal entries. Averaging then over the root permutations makes its diagonal constant, equal to one quarter of its trace.

Both original matrices commute with Σ , so there are no even-odd blocks. For Q_Σ , the even block is I_4 and the odd block is $\text{diag}(1, -1, -1, 1)$, of trace 0. Its odd average therefore vanishes. For Q_J , the odd block is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad (\text{HWA11})$$

also of trace 0. Its even block is the transposition exchanging roots 1 and -1 . Averaging its conjugates over S_4 averages the six transpositions. The resulting matrix has diagonal $3/6 = 1/2$ and every off-diagonal entry $1/6$, because a given index is fixed by three transpositions and a specified distinct pair is exchanged by one. This is $I_4/3 + \mathbf{1}_4\mathbf{1}_4^*/6$. Transport through the exact even projection gives(HWA10).

For $s_j(x) = x_{j+} + x_{j-}$, the complete quadratic formulas are

$$\begin{aligned} \overline{B}_\Sigma(x, x) &= \frac{1}{2} \sum_j |s_j(x)|^2, \\ \overline{B}_J(x, x) &= \frac{1}{6} \sum_j |s_j(x)|^2 + \frac{1}{12} \left| \sum_j s_j(x) \right|^2. \end{aligned} \quad (\text{HWA12})$$

Thus each has inertia $(4, 0, 4)$. For the native average its eigenvalues on constants, the three-dimensional even sum-zero subspace, and the four-dimensional odd subspace are respectively $1, 1/3, 0$. Their common radical is exactly

$$V_{\text{odd}} = \{x : x_{j-} = -x_{j+} \text{ for all four } j\}. \quad (\text{HWA13})$$

These formulas exhibit positivity from the actual finite group, retaining every root and sign coordinate in the projection and its kernel.

3. Exact defect and multiplication retained by the odd space

Define, on the original eight-dimensional space,

$$D_q = Q_q - \mathcal{E}(Q_q). \quad (\text{HWA14})$$

For Σ , the defect vanishes on even functions and is exactly $\text{diag}(1, -1, -1, 1)$ on odd functions. Its inertia is $(2, 2, 4)$. For J , the even block has eigenvalues $0, 2/3, 2/3, -4/3$, and the odd block is (HWA11), with eigenvalues $1, 1, -1, -1$. To verify the even values, the root transposition fixes the constant line and has eigenvalues $1, 1, -1$ on its sum-zero complement; subtract its average $1/3$ on that complement. Consequently

$$\text{inertia}(D_J) = (4, 3, 1). \quad (\text{HWA15})$$

The exact comparison for arbitrary vectors is

$$B_q(x, y) = \overline{B}_q(x, y) + x^* D_q y. \quad (\text{HWA16})$$

The average cannot be the regular trace paired with an antilinear algebra involution on the original reduced $\mathcal{A}$. Such an involution permutes its eight primitive idempotents and conjugates their coefficients; its trace matrix is an invertible permutation matrix by (HWA3). The matrices in (HWA10) have rank 4. This proves the stated nonidentification and also identifies its exact four-dimensional radical.

There is an explicit linear receiver and an exact multiplication defect. Put

$$E : \mathbb{C}^X \rightarrow \mathbb{C}^4, \quad E(x)_j = (x_{j+} + x_{j-})/2, \quad O(x)_j = (x_{j+} - x_{j-})/2, \quad (\text{HWA17})$$

and let $I : \mathbb{C}^4 \rightarrow \mathbb{C}^X$ duplicate each root value. Then $EI = \text{id}$, $IE = P_{\text{ev}}$, $\ker E = V_{\text{odd}}$, and

$$E(xy) = E(x)E(y) + O(x)O(y). \quad (\text{HWA18})$$

All products on the right are coordinatewise. Expanding $(a_j \pm b_j)(c_j \pm d_j)$ proves the identity in every coordinate. Thus E is a retraction of vector spaces and a bimodule map over the embedded sign-even algebra, while its displayed second term records every product returning from odd to even support. In particular $z = e_{0+} - e_{0-}$ has $E(z) = 0$ but $E(z^2) = e_0 \neq 0$. The kernel is not an ideal, and (HWA17) is not silently promoted to an algebra quotient.

The average forms factor exactly through E :

$$\overline{B}_\Sigma(x, y) = 2 \sum_j \overline{E(x)}_j E(y)_j, \quad \overline{B}_J(x, y) = \frac{2}{3} \sum_j \overline{E(x)}_j E(y)_j + \frac{1}{3} \sum_j \overline{E(x)}_j \sum_j E(y)_j. \quad (\text{HWA19})$$

For any retained support lattice, applying these linear amplitude maps to a supported module keeps its support coordinate unchanged. An amplitude sent to 0 therefore lands at the supported-zero element of that same support, not at the unsupported element τ . Formula (HWA18) remains the precise reason that this amplitude projection is not a semiring homomorphism.

The average of vectors is a third, separately specified map:

$$\frac{1}{192} \sum_g P_g x = P_c x, \quad B_q(P_c x, P_c y) = x^* P_c y. \quad (\text{HWA20})$$

The first identity follows because the group acts transitively on the eight states, so each target coordinate receives each source coordinate equally often. The second uses $Q_q \mathbf{1} = \mathbf{1}$. This rank-one form differs from both rank-four forms (HWA10); all three maps and their different kernels are explicit.

4. Actual supported-zero theta endpoints

Keep $\mathcal{T} = C_c^\infty(\mathbb{R}; \mathbb{C})$ and the programme convention

$$M_f(s) = \int_{\mathbb{R}} f(v) e^{-(s-1/2)v} dv, \quad f^\#(v) = \overline{f(-v)}. \quad (\text{HWA21})$$

The transform of $f^\#$ is $\overline{M_f(1 - \bar{s})}$, by the substitution $v \mapsto -v$. Define the even Schwartz function

$$\phi_f(x) = \int_{\mathbb{R}} e^{v/2} f(v) e^{-\pi e^{2v} x^2} dv. \quad (\text{HWA22})$$

If $\text{supp } f \subset [-R, R]$, every derivative in x of the integrand is a polynomial in x with bounded coefficients times a Gaussian bounded by a constant times $e^{-\pi e^{-2R} x^2}$. Integration proves every Schwartz bound. The Fourier convention $\widehat{\phi}(y) = \int \phi(x) e^{-2\pi i x y} dx$ gives

$$\phi_f(0) = M_f(0), \quad \int \phi_f = M_f(1), \quad \widehat{\phi_f} = \phi_{f(-)}, \quad \widehat{\widehat{\phi_f}} = \phi_{f^\#}. \quad (\text{HWA23})$$

For the integral use $\int e^{-\pi e^{2v} x^2} dx = e^{-v}$. For Fourier use the same Gaussian scaling and substitute $v \mapsto -v$. Absolute integrability from the compact v -support justifies both interchanges.

In $G(\mathbb{Z}) = \{\tau\} \sqcup \mathbb{Z}$, let e denote supported 0. Its principal ideal $\{\tau, e\}$ is prime: a product of two nonzero supported integers is again nonzero supported, so a product in that ideal has at least one factor there. The full supported theta is

$$\Theta_f(x) = \sum_{n \in \mathbb{Z}} \phi_f(nx), \quad x > 0, \quad (\text{HWA24})$$

and retains the $n = 0$ summand $\phi_f(0)$ as the e -term. The external τ is a different label; no integer in this sum represents it. Poisson summation for this Schwartz function gives $\Theta_f(x) = x^{-1} \Theta_{f(-\cdot)}(x^{-1})$. For this particular mixture the identity follows directly by integrating the Gaussian theta identity, with uniform absolute convergence for v in the compact support. Its two endpoint amplitudes are exactly (HWA23), not an arbitrary selected pair of complex numbers. Their regularized Mellin endpoint terms are

$$-\frac{M_f(0)}{s} + \frac{M_f(1)}{s-1}. \quad (\text{HWA25})$$

Indeed split the integral of $\Theta_f(x) - \phi_f(0)$ at 1; use full Poisson on $(0, 1)$, and integrate $x^{s-1}(x^{-1}M_f(1) - M_f(0))$. Initially these elementary integrals converge for $\text{Re } s > 1$, and their displayed meromorphic expressions supply continuation. The remaining transformed integrals over $[1, \infty)$ are entire by rapid decay. Also direct Gaussian integration in the initial half-plane gives

$$\int_0^\infty (\Theta_f(x) - \phi_f(0)) x^{s-1} dx = \pi^{-s/2} \Gamma(s/2) \zeta(s) M_f(s). \quad (\text{HWA26})$$

The two integer signs contribute 2, cancelling the $1/2$ in the Gaussian Mellin integral. Uniform absolute convergence for $\text{Re } s > 1$ proves the exchange. This establishes both the arithmetic amplitude and the exact endpoint residues from the retained supported-zero theta.

For two actual tests the endpoint Hermitian pairing is

$$P_e(f, g) = \overline{M_f(0)} M_g(1) + \overline{M_f(1)} M_g(0). \quad (\text{HWA27})$$

This is the trace of the two-weight endpoint module with its Fourier reflection exchanging the weights. It is the supported-zero endpoint term; it is not all of the integrated Weil distribution. The separate construction SZW retains the remaining finite-prime, archimedean and fixed-support terms.

5. Exact endpoint embedding and the change caused by holonomy averaging

Set $A_f = M_f(0)$, $B_f = M_f(1)$ and define

$$j : \mathcal{T} \rightarrow \mathbb{C}^X, \quad j(f) = (0, 0, A_f/\sqrt{2}, A_f/\sqrt{2}, B_f/\sqrt{2}, B_f/\sqrt{2}, 0, 0). \quad (\text{HWA28})$$

The root assignment is stated in (HWA2): the two endpoint weights are inserted into roots 1 and -1 . It is a linear comparison with scale $1/\sqrt{2}$, not a unital algebra map. By (HWA5), (HWA21),

$$j(f^\#) = \#_J j(f), \quad B_J(j(f), j(g)) = P_e(f, g). \quad (\text{HWA29})$$

Both formulas follow by exchanging A, B and conjugating; the two signs at each root have equal values. Thus this is an exact involution-preserving embedding of the two-dimensional endpoint quotient into the actual completed trace.

The endpoint map $f \mapsto (A_f, B_f)$ is onto. Here is a complete section. Choose a positive smooth compact bump ψ on an open interval. Put $k_0(v) = e^{v/2}$, $k_1(v) = e^{-v/2}$, $G_{ab} = \int \psi k_a k_b$. For $z \neq 0$, the nonzero combination $\bar{z}_0 e^{v/2} + \bar{z}_1 e^{-v/2}$ cannot vanish on an interval: multiplying by $e^{v/2}$ would make $\bar{z}_0 e^v + \bar{z}_1$ identically zero, whose derivative forces $z_0 = z_1 = 0$. Hence G is positive definite. The function $S(A, B) = \psi \sum_b (G^{-1}(A, B)^T)_b k_b$ has precisely those two transforms by multiplication with G . It is smooth and compactly supported, proving surjectivity and $\ker j = \{M_f(0) = M_f(1) = 0\}$.

Substituting (HWA28) in (HWA12) gives the averaged endpoint form

$$\overline{P}_e(f, f) = \frac{1}{2}(|A_f|^2 + |B_f|^2) + \frac{1}{3} \text{Re}(\overline{A_f} B_f). \quad (\text{HWA30})$$

Its matrix is $\begin{pmatrix} 1/2 & 1/6 \\ 1/6 & 1/2 \end{pmatrix}$, with eigenvalues $2/3, 1/3$, so it is strictly positive on the endpoint quotient. The exact change is

$$K_e = \overline{P}_e - P_e, \quad K_e(f, f) = \frac{1}{2}(|A_f|^2 + |B_f|^2) - \frac{5}{3} \operatorname{Re}(\overline{A_f} B_f). \quad (\text{HWA31})$$

In the explicitly related coordinates $u = (A + B)/\sqrt{2}, v = (A - B)/\sqrt{2}$,

$$P_e = |u|^2 - |v|^2, \quad \overline{P}_e = \frac{2}{3}|u|^2 + \frac{1}{3}|v|^2, \quad K_e = -\frac{1}{3}|u|^2 + \frac{4}{3}|v|^2. \quad (\text{HWA32})$$

Thus actual tests with $(A, B) = (1, -1)$ have old value -2 and averaged value $2/3$, while tests with $(A, B) = (1, 1)$ have old value 2 and averaged value $4/3$. The section just proved supplies both tests. The change has both signs on admissible tests, and its radical on $\mathcal{T}$ is exactly $\ker j$: its two-dimensional matrix is invertible and the section detects every nonzero endpoint vector.

The root-pair image of j is not invariant under the full group: a signed root permutation taking root1 to root0 sends a vector with $A \neq 0, B = 0$ outside it. Such a permutation is in (HWA7) after choosing signs with product equal to its permutation sign. This proves a specific failure of invariance of this comparison map, with the group action and endpoint map explicitly retained. It does not assert that every other arithmetic comparison fails.

The full supported-zero construction *The supported-zero prime in the full theta and Weil trace distribution*, SZW24–SZW42 proves the actual integrated formula. Here are its explicit terms and their exact transport. Put $h = f^\# * g$, $H = M_h$, and retain

$$\begin{aligned} \mathcal{Z}(f, g) &= \sum_{\rho} m_{\rho} \overline{M_f(1 - \bar{\rho})} M_g(\rho), \\ \mathcal{A}_{\infty}(f, g) &= \frac{1}{2\pi} \int_{\mathbb{R}} \overline{\widehat{f}(t)} \widehat{g}(t) (\operatorname{Re} \psi(1/4 + it/2) - \log \pi) dt, \\ \mathcal{P}_{\text{fin}}(f, g) &= \sum_{p, k \geq 1} (\log p) p^{-k/2} \{h(k \log p) + h(-k \log p)\}. \end{aligned} \quad (\text{HWA33})$$

The sum in the last line is over primes p and integers $k \geq 1$. It is finite for each compactly supported h . The absolutely convergent zero sum and archimedean integral, with their complete proof and the original minus-exponent convention, are SZW24–SZW32. Substituting $P_e = \overline{P}_e - K_e$ gives the actual equality

$$\boxed{\mathcal{Z} = P_e + \mathcal{A}_{\infty} - \mathcal{P}_{\text{fin}} = \overline{P}_e + \mathcal{A}_{\infty} - \mathcal{P}_{\text{fin}} - K_e.} \quad (\text{HWA34})$$

Formula (HWA31) gives the entire changed term on every admissible test. In particular the negative endpoint -2 becomes $2/3$, while its retained complement changes by $-8/3$. This is an exact calculation inside the derived supported-zero formula.

The full labels can be kept before any scalar projection. Let L be the finite fixed-support lattice of SZW10–SZW13, $\mathbf{e}_1$ its top coordinate, and $\mathbf{s}_{<} = \sum_{\lambda \neq 1} \mathbf{e}_{\lambda}$. Here lower labels are distinct fixed zero points, not the independently different coordinatewise adelic support construction. The original labelled formula, now on $h = f^\# * g$, is

$$\begin{aligned} \mathbf{B} &= P_e \mathbf{e}_1 + H(0) \mathbf{s}_{<}, \\ \mathbf{D} &= (\mathcal{P}_{\text{fin}} - \mathcal{A}_{\infty}) \mathbf{e}_1 + H(0) \mathbf{s}_{<}, \\ \mathbf{B} - \mathcal{Z} \mathbf{e}_1 &= \mathbf{D}. \end{aligned} \quad (\text{HWA35})$$

Keeping these lower fixed distributions exactly and applying the holonomy comparison at the top endpoint gives

$$\begin{aligned} \overline{\mathbf{B}} &= \overline{P}_e \mathbf{e}_1 + H(0) \mathbf{s}_{<}, \\ \overline{\mathbf{D}} &= (\mathcal{P}_{\text{fin}} - \mathcal{A}_{\infty} + K_e) \mathbf{e}_1 + H(0) \mathbf{s}_{<}, \\ \overline{\mathbf{B}} - \mathcal{Z} \mathbf{e}_1 &= \overline{\mathbf{D}}. \end{aligned} \quad (\text{HWA36})$$

This follows by adding precisely $K_e \mathbf{e}_1$ to both sides of (HWA35). No lower support trace was merged with the top endpoint or discarded.

For the minimal Fourier-closed endpoint module proved in SZW40–SZW42, every lower fixed constant acquires its exact weight-one Dirac partner. Its labelled boundary and geometric distributions are

$$\mathbf{B}^{\mathcal{F}} = P_e (\mathbf{e}_1 + \mathbf{s}_{<}), \quad \mathbf{D}^{\mathcal{F}} = (\mathcal{P}_{\text{fin}} - \mathcal{A}_{\infty}) \mathbf{e}_1 + P_e \mathbf{s}_{<}. \quad (\text{HWA37})$$

There is now the defined direct-sum comparison $\bigoplus_{\lambda \in L} j$ from the independent endpoint tests into one completed eight-state vector space for each label. Averaging its native forms label by label and restricting back to the diagonal actual scalar tests gives

$$\begin{aligned}\overline{\mathbf{B}}^{\mathcal{F}} &= \overline{P}_e(\mathbf{e}_1 + \mathbf{s}_{<}), \\ \overline{\mathbf{D}}^{\mathcal{F}} &= (\mathcal{P}_{\text{fin}} - \mathcal{A}_{\infty} + K_e)\mathbf{e}_1 + \overline{P}_e\mathbf{s}_{<}, \\ \overline{\mathbf{B}}^{\mathcal{F}} - \mathcal{Z}\mathbf{e}_1 &= \overline{\mathbf{D}}^{\mathcal{F}}.\end{aligned}\tag{HWA38}$$

Proof: add K_e on every label in(HWA37), using the original identity $\mathbf{B}^{\mathcal{F}} - \mathcal{Z}\mathbf{e}_1 = \mathbf{D}^{\mathcal{F}}$. The direct-sum comparison is a specified linear map, and(HWA29) proves its involution compatibility on each summand. The group is the actual auxiliary-cover group on each receiver; no unproved action on the arithmetic zeros or on the original test space has been introduced. Thus the positive averaged boundary, its full geometric correction, the supported-zero prime, the unsupported coordinate and every mixed-support label are retained in one exact identity.

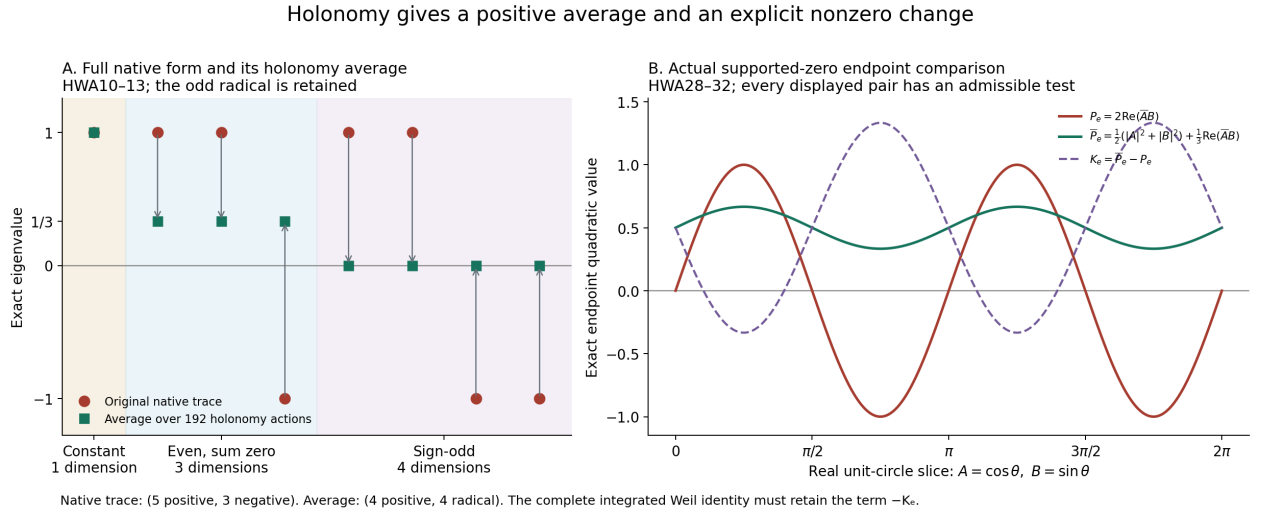


Figure 10: The complete native trace spectrum before and after its order-192 holonomy average, and the exact supported-zero endpoint comparison on actual admissible tests. HWA10–HWA15 prove the left panel; HWA28–HWA32 prove the right panel; HWA34–HWA38 retain its change in the full supported-zero formula.

Exact full-form continuation

The reflection-index theorem, NI1–32, calculates the signed index of every complete Cauchy coefficient tail. The finite-jet and endpoint proof, FJ1–29, preserves the exact finite jets while proving density on the remaining zeros, derives the full exterior index and tail, and calculates the second and third actual heat coefficients. All supported endpoints remain explicit. These results identify the remaining sign problem without asserting its resolution.

Prime holonomy in the supported-zero Weil distribution

The prime-orbit holonomy here is the actual arithmetic Frobenius in the finite abelian covers constructed by Alain Connes and Caterina Consani, *Knots, primes and class field theory*, arXiv:2501.06560v1. The original author source was read at lines 464–695: Definition `coverdef`, Lemma `Gacts`, Proposition `mappingtorus`, Theorem `mappingtorus1`, Fact `artinrec`, Theorem `main`, and Remark `archimedeanplace`. Root subsequently read the entire body and bibliography. Their finite cover and its arithmetic identification are prior results. The formulas below derive its oriented trace, inertia maps, fixed-support endpoints, character-resolved Euler factors, and exact comparison with SZW. They do not identify an unrelated finite-group action with this arithmetic cover.

The full supported-zero object, its prime spectrum, test convention, and trace receiver are those proved in the full supported-zero Weil proof, SZW1–SZW47. In particular

$$S = G(\mathbb{Z}) = \{\tau\} \sqcup \mathbb{Z}^\bullet, \quad e = 0^\bullet \neq \tau, \quad (\tau) \subsetneq (e) \subsetneq (p). \quad (\text{PHW1})$$

Both zero points will remain separate. No expression $\log 0$, positive period for a fixed point, or numerical Euler factor for e is introduced.

1. The finite cover, inertia, and the two orientation coordinates

Put $K = \widehat{\mathbb{Z}}^\times$, and let $\chi : K \twoheadrightarrow G$ be a continuous surjection to a finite abelian group. Use exactly the source’s balanced quotient

$$X^\chi = (Y_{\mathbb{Q}} \times G)/K, \quad (z, g) \sim (zw, \chi(w)g), \quad Y_{\mathbb{Q}} = \mathbb{Q}^\times \backslash \mathbb{A}_{\mathbb{Q}}. \quad (\text{PHW2})$$

The group G acts on the right in the second coordinate. At a prime p , define

$$I_p = \chi(\mathbb{Z}_p^\times), \quad \tilde{p}_p = 1, \quad \tilde{p}_\ell = p \ (\ell \neq p), \quad g_p = \chi(\tilde{p}) \in G, \quad \ell_p = \log p > 0. \quad (\text{PHW3})$$

Here $\tilde{p} \in K$; at every $\ell \neq p$, the rational integer p is an ℓ -adic unit. The class of g_p in G/I_p is independent of a change of the p -component. At an unramified prime $I_p = 1$, this is the source’s $\chi(p) = \text{Frob}_p$, the arithmetic Frobenius. At ramified primes the same quotient class acts on the inertia-invariant fibre.

For completeness the actual orientation map follows directly from (PHW2). Before the finite quotient, the source identifies the inverse image of C_p with $p^{\mathbb{Z}} \backslash (H_p \times \mathbb{R}_+^\times)$, where $H_p = \prod_{\ell \neq p} \mathbb{Z}_\ell^\times$ and the generator is $(h, \lambda) \mapsto (ph, p\lambda)$. Write $t = \log \lambda$. Reducing (h, g) by the compact action gives the natural fibre coordinate $g_0 = \chi(h)^{-1}g$, modulo I_p . Thus the finite mapping torus is exactly

$$\mathfrak{C}_p = ((G/I_p) \times \mathbb{R}) / ((g, t) \sim (g_p^{-1}g, t + \ell_p)). \quad (\text{PHW4})$$

The equality $\chi(ph)^{-1}g = g_p^{-1}\chi(h)^{-1}g$ proves the formula. This proof also gives the ramified version, since the missing p -unit coordinate acts on the fibre by I_p . The positive flow is $[g, t] \mapsto [g, t + a]$. Its first return from $t = 0$ to $t = \ell_p$, expressed again at $t = 0$, is $g \mapsto g_p g$. The opposite traversal is $g \mapsto g_p^{-1}g$.

The map $(g, t) \mapsto (g^{-1}, t)$ turns (PHW4) into the displayed source convention with generator $(g, t) \mapsto (g_p g, t + \ell_p)$. It intertwines right deck multiplication by k with multiplication by k^{-1} . This is the exact coordinate conversion; using the printed generator without this conversion would reverse the fibre representation. The arithmetic Frobenius and its inverse are both retained below.

Let V be a finite-dimensional complex unitary representation ρ of G . Define

$$P_{I_p} = \frac{1}{|I_p|} \sum_{i \in I_p} \rho(i), \quad V_p = V^{I_p} = \text{im } P_{I_p}, \quad T_p = \rho(g_p)|_{V_p}. \quad (\text{PHW5})$$

Direct multiplication of the finite sums shows $P_{I_p}^2 = P_{I_p}$; unitarity gives $P_{I_p}^* = P_{I_p}$. It fixes precisely the invariant vectors. Since G is abelian, T_p is a well-defined unitary operator on V_p , unchanged by replacing g_p by $g_p i$, $i \in I_p$. No assumption about unramified primes is needed in this formula.

2. An actual trace on every prime circle

The coefficient space of sections over (PHW4) is

$$\mathcal{H}_{p,V} = \{v \in L_{\text{loc}}^2(\mathbb{R}, V_p) : v(t + \ell_p) = T_p^{-1}v(t)\}, \quad \|v\|^2 = \int_0^{\ell_p} \|v(t)\|^2 dt. \quad (\text{PHW6})$$

Indeed a function on the natural fibre satisfying $F(gk, t) = \rho(k)^{-1}F(g, t)$ has precisely this boundary condition after (PHW4). The unitary flow on sections is inverse pullback, $U_p(a)v(t) = v(t-a)$; consequently $U_p(\ell_p) = T_p$. For $\nu \in C_c^\infty(\mathbb{R})$, the bounded operator

$$A_{p,V}(\nu) = \int_{\mathbb{R}} \nu(a) U_p(a) da \quad (\text{PHW7})$$

is trace class, and its full trace, including the zero iterate, is

$$\boxed{\text{Tr } A_{p,V}(\nu) = \ell_p \sum_{k \in \mathbb{Z}} \text{Tr}(T_p^k) \nu(k\ell_p).} \quad (\text{PHW8})$$

Here and below $\text{Tr}(T_p^k)$ means the trace of the matrix power; it is not the power of its trace. To prove (PHW8), diagonalize T_p . For an eigenvalue $e^{i\theta}$, an orthonormal Fourier basis is $\ell_p^{-1/2} e^{i(2\pi n - \theta)t/\ell_p}$, $n \in \mathbb{Z}$. The eigenvalues of (PHW7) are $\widehat{\nu}((2\pi n - \theta)/\ell_p)$, with the minus Fourier convention. They are absolutely summable by repeated integration by parts. Poisson summation of this Schwartz function gives

$$\sum_{n \in \mathbb{Z}} \widehat{\nu}((2\pi n - \theta)/\ell_p) = \ell_p \sum_{k \in \mathbb{Z}} e^{ik\theta} \nu(k\ell_p). \quad (\text{PHW9})$$

For example this equality follows by expanding the periodic Dirac distribution $\sum_n e^{-2\pi i n a/\ell_p} = \ell_p \sum_k \delta(a - k\ell_p)$, whose Fourier coefficients on one period are all one, and multiplying by $\nu(a)e^{i\theta a/\ell_p}$. Summing the finitely many eigenlines proves (PHW8), including the term $\ell_p \dim(V_p)\nu(0)$.

The original supported Weil test space remains

$$\mathcal{T} = C_c^\infty(\mathbb{R}; \mathbb{C}), \quad H(s) = M_h(s) = \int h(v) e^{-(s-1/2)v} dv, \quad f^\#(v) = \overline{f(-v)}. \quad (\text{PHW10})$$

Choose once and for all an even smooth cutoff κ , zero for $|a| \leq \frac{1}{4} \log 2$ and one for $|a| \geq \frac{1}{2} \log 2$. Such a function is obtained from $b(x) = e^{-1/x}$ for $x > 0$, zero otherwise, using $b(x)/(b(x) + b(1-x))$ on the intervening interval; its even extension is smooth because it is constant near zero. Define the exact map

$$\nu_h(a) = \kappa(a) e^{-|a|/2} h(-a) \in C_c^\infty(\mathbb{R}). \quad (\text{PHW11})$$

The cutoff is identically one at every nonzero prime iterate $\pm k \log p$, so (PHW8) gives

$$\boxed{P_V(h) = \sum_p \text{Tr } A_{p,V}(\nu_h) = \sum_p \sum_{k \geq 1} (\log p) p^{-k/2} \{ \text{Tr}(T_p^k) h(-k \log p) + \text{Tr}(T_p^{-k}) h(k \log p) \}.} \quad (\text{PHW12})$$

The trace sum in this formula has only finitely many nonzero terms: if $\text{supp } h \subset [-R, R]$, then $k \log p \leq R$. It is independent of every permitted choice of κ . This proves the actual comparison between the original prime-orbit circle and the weighted Weil distribution. The weight $e^{-|a|/2}$ is an explicit test map; its arithmetic contour identity is proved in (PHW23) below. The omitted identity iterate is exactly the separately known functional $\ell_p \dim(V_p)\nu(0)$ of (PHW8).

There is a necessary operator-domain distinction even though (PHW12) is a complete distribution. The bounded direct sum $\bigoplus_p A_{p,V}(\nu)$ exists, with norm at most $\|\nu\|_1$, but for nonzero V and nonzero ν it is not trace class. To prove this, $\widehat{\nu}$ is not identically zero by Fourier inversion, so there is a real interval of positive length on which $|\widehat{\nu}| \geq c > 0$. All but finitely many p have $V_p = V \neq 0$. In any eigenline the frequency grid has spacing $2\pi/\ell_p$, so the number of grid points in that interval is at least its length times $\ell_p/(2\pi) - 1$. The trace norm of the corresponding diagonal block therefore grows at least as a positive constant times $\log p$. The total trace norm diverges. For $\nu = \nu_h$, the finite-prime-cutoff traces nevertheless stabilize once $p > e^R$, by (PHW8). Thus (PHW12) is an exact stabilized trace distribution, not an ordinary trace on the infinite direct sum. The operator family, the finite cutoffs, the stabilized trace, and the failure of the full trace-class norm are all retained.

3. Determinants and the character-resolved arithmetic map

For $\Re s > 1$, define

$$L_V(s) = \prod_p \det(1 - p^{-s} T_p \mid V_p)^{-1}. \quad (\text{PHW13})$$

Since every eigenvalue of T_p has modulus one and $\dim V_p \leq \dim V$, the logarithm

$$\log L_V(s) = \sum_p \sum_{k \geq 1} \frac{\text{Tr}(T_p^k)}{k} p^{-ks}, \quad -\frac{L'_V}{L_V}(s) = \sum_p \sum_{k \geq 1} (\log p) \text{Tr}(T_p^k) p^{-ks} \quad (\text{PHW14})$$

converges absolutely, uniformly on every half-plane $\Re s \geq 1 + \varepsilon$, with its differentiated series. Indeed it is dominated respectively by constant multiples of $\sum_{n \geq 2} n^{-1-\varepsilon}$ and $\sum_{n \geq 2} (\log n) n^{-1-\varepsilon}$. The determinant identity follows by applying $-\log(1-z) = \sum_{k \geq 1} z^k/k$ to each eigenvalue. In particular these formulas specify the branch of the logarithm and prove that L_V has no zero or pole in this half-plane.

Let $\widehat{G}$ denote the character group. The explicit mutually orthogonal idempotents are

$$P_\eta = \frac{1}{|G|} \sum_{g \in G} \overline{\eta(g)} \rho(g), \quad V = \bigoplus_{\eta \in \widehat{G}} V_\eta, \quad V_\eta = \text{im } P_\eta, \quad m_\eta = \dim V_\eta. \quad (\text{PHW15})$$

For a nontrivial character α , the sum $\sum_g \alpha(g)$ is zero: multiplication of its index by an element where $\alpha \neq 1$ multiplies the sum by that unequal value. This proves the orthogonality identities for (PHW15). Simultaneous diagonalization of the commuting unitary matrices proves completeness; every eigenline supplies one of the characters, and (PHW15) projects onto it.

The character $\eta \circ \chi$ factors through a finite unit group. At every prime choose its smallest local congruence exponent through which it factors, and take the product of those prime powers. This gives its conductor q_η and a primitive Dirichlet character η_0 modulo q_η . Set $a_\eta \in \{0, 1\}$ by $\eta(\chi(-1)) = (-1)^{a_\eta}$. At p , the summand V_η lies in V_p precisely when $p \nmid q_\eta$; its eigenvalue is then $\eta_0(p)$. This follows by restricting $\eta \circ \chi$ to the p -unit group and evaluating $\tilde{p}$ at the remaining conductor components. Therefore (PHW13) is exactly

$$L_V(s) = \prod_{\eta \in \widehat{G}} L(s, \eta_0)^{m_\eta}. \quad (\text{PHW16})$$

This includes primes ramified in the extension but unramified for a particular character: they have not been removed merely because they divide the conductor of the whole cover. The trivial character has $q_1 = 1$, $a_1 = 0$, and $L(s, 1) = \zeta(s)$.

There is a fully labelled version before taking a representation trace. In $\mathbb{C}[G]$, put $E_I = |I|^{-1} \sum_{i \in I} i$. Define

$$\mathcal{P}_G(h) = \sum_{p, k \geq 1} (\log p) p^{-k/2} \{E_{I_p} g_p^k h(-k \log p) + E_{I_p} g_p^{-k} h(k \log p)\}. \quad (\text{PHW17})$$

It is a finite sum for every test. Applying $\text{Tr } \rho$ gives (PHW12), because $\rho(E_I) = P_I$. The character Fourier map $x \mapsto (\eta(x))_\eta$ is an isomorphism $\mathbb{C}[G] \rightarrow \mathbb{C}^{\widehat{G}}$, with inverse given by the character idempotents; its orthogonality proof is the same finite-sum calculation as (PHW15). Hence no holonomy character is lost before a projection is specified.

The trivial-character observation is augmentation $\epsilon(\sum c_g g) = \sum c_g$. Its kernel is the span of the elements $g - 1$, and its exact section is $c \mapsto cE_G$. Indeed $\epsilon(E_G) = 1$, $gE_G = E_G$, and every coefficient vector of total zero equals $\sum c_g(g - 1)$. Consequently

$$\epsilon \mathcal{P}_G = P_{\text{fin}}, \quad E_G \mathcal{P}_G = P_{\text{fin}} E_G, \quad \mathcal{P}_G = P_{\text{fin}} E_G + (1 - E_G) \mathcal{P}_G. \quad (\text{PHW18})$$

The second term contains exactly the nontrivial character components; it is an explicitly retained augmentation-ideal distribution. The first is the original arithmetic prime term of SZW24, with its original $\log p$ and $p^{-k/2}$ constants.

4. The fixed-support endpoints of the same cover

At the global supported zero $e \in G(\mathbb{A}_\mathbb{Q})$, every element of K fixes the point. Thus its fibre in (PHW2) is $G/\chi(K) = G/G$, a singleton. Extending the same construction to the full fixed-sector carrier $G_L(\mathbb{A}_\mathbb{Q})$ gives a separate singleton above each z_λ , including e_L and τ_L . The balanced quotient never relates two distinct support labels. Coefficients in V over each such fibre are therefore

$$W = V^G = \text{im } P_G, \quad P_G = \frac{1}{|G|} \sum_{g \in G} \rho(g), \quad V = W \oplus \ker P_G. \quad (\text{PHW19})$$

These maps are actual stabilizer invariants, not a hypothesis of holonomy-invariant test functions. The same averaging proof as (PHW5) shows that P_G is a projection, and (PHW15) identifies its kernel as the sum of all nontrivial character spaces. Let $m_0 = \dim W = m_1$.

The exact Schwartz domain verifies both endpoint weights. Use

$$\mathcal{F}_{L,V} = \{f \in \mathcal{S}(\mathbb{A}_{\mathbb{Q}}) \otimes V : f(aw) = \rho(\chi(w))^{-1} f(a) \ (w \in K)\} \oplus \bigoplus_{\lambda \neq 1_L} W_{\lambda}. \quad (\text{PHW20})$$

Evaluation gives $f(0) \in W$, since zero is fixed by every w . The integral $\int f$ also lies in W , because multiplication by a compact unit preserves the specified self-dual additive Haar measure. Thus

$$\beta_{L,V}(f, c) = \left(f(0), \int f, (c_{\lambda}) \right), \quad B_{L,V} = W(0)_e \oplus W(1)_{e^{\vee}} \oplus \bigoplus_{\lambda \neq 1_L} W(0)_{\lambda}. \quad (\text{PHW21})$$

The map is onto: for $w \in W$, embed an even real Schwartz test as $\phi \otimes 1_{\widehat{\mathbb{Z}}} \otimes w$, use the two moment preimages proved in SZW13, and prescribe the lower coefficients independently. Its kernel is exactly $f(0) = \int f = 0$ and all $c_{\lambda} = 0$. Dilation by an idele fixes the first and lower coordinates and multiplies the integral by its norm, proving the stated weights. This connects the same coefficient system's prime-orbit spaces (PHW6) to its fixed-support quotient without identifying their carriers.

On the original SZW test $q_h(j) = |j|^{-1/2} h(-\log |j|)$, the label-resolved boundary trace is consequently

$$B_{L,V}(h) = m_0 \left((H(0) + H(1)) \mathbf{e}_{1_L} + H(0) \sum_{\lambda \neq 1_L} \mathbf{e}_{\lambda} \right). \quad (\text{PHW22})$$

The coefficient is $\dim V^G$, not $\dim V$. The original two-point case has its separate τ -term $m_0 H(0)$, while the supported-zero/Fourier pair contributes $m_0(H(0) + H(1))$. The Fourier closure of SZW40–42 tensors with W , so it adds exactly $(|L| - 1)m_0$ weight-one partners and no partners in the nontrivial character spaces.

For a finite prime orbit the zero-coordinate set is instead $\{p\}$, giving inertia I_p and the circle (PHW4). Its adelic zero-coordinate prime contracts to $(e) \subset S$, by SZW4, but its other nonzero coordinates and its place label remain present. Thus contraction to the supported-zero prime does not collapse these circles into the global zero point. The source's archimedean orbit has fibre $G/\langle \chi(-1) \rangle$; its invariant coefficient space is the even-parity sum in (PHW15). This is another actual stabilizer map, distinct from both global G -invariants and finite-prime inertia invariants.

5. The convergent arithmetic contour and the full trivial sector

Let $V^{\vee}$ be the dual representation, so the local traces are those of T_p^{-k} . For every $c > 1$, the Euler series already proved in (PHW14) gives

$$\begin{aligned} P_V(h) &= \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} -\frac{L'_V(s)}{L_V(s)} H(s) ds \\ &\quad + \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} -\frac{L'_{V^{\vee}}(s)}{L_{V^{\vee}}(s)} H(1-s) ds. \end{aligned} \quad (\text{PHW23})$$

These are absolutely convergent integrals: the logarithmic derivative is bounded on each indicated vertical line by its absolutely convergent Euler series, and the transforms decrease faster than every power of the height by integration by parts. Termwise integration is therefore valid. To compute the first integral term with $\ell = k \log p$, Fourier inversion gives

$$\frac{1}{2\pi} \int_{\mathbb{R}} e^{-it\ell} H(c+it) dt = h(-\ell) e^{(c-1/2)\ell}.$$

Multiplication by $p^{-kc} = e^{-c\ell}$ gives $p^{-k/2} h(-\ell)$. Replacing $H(s)$ by $H(1-s)$ gives $p^{-k/2} h(\ell)$; the dual representation supplies $\text{Tr}(T_p^{-k})$. This proves (PHW23) with the original signs and weights. This argument uses only the convergent half-plane, not a presumed functional equation for an arbitrary representation.

For the trivial character, both integrals and the fixed-support endpoints are exactly those in SZW. Writing $E_G = |G|^{-1} \sum_g g$ in the group algebra, the exact retained decomposition is

$$\mathcal{P}_G = P_{\text{fin}} E_G + \mathcal{P}_G^{\text{aug}}, \quad \mathcal{P}_G^{\text{aug}} = (1 - E_G) \mathcal{P}_G, \quad \epsilon \mathcal{P}_G^{\text{aug}} = 0. \quad (\text{PHW24})$$

Its proof is (PHW18); its kernel is the full augmentation ideal, not a declaration that the complementary distribution vanishes. The trace receiving P_{fin} is the complete supported-zero identity

$$\begin{aligned}
\mathbf{B}_L &= (H(0) + H(1))\mathbf{e}_{1_L} + H(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda, \\
\mathbf{Z}_L &= Z(h)\mathbf{e}_{1_L}, \\
\mathbf{D}_L &= (P_{\text{fin}} - A_\infty)\mathbf{e}_{1_L} + H(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda, \\
\mathbf{B}_L - \mathbf{Z}_L &= \mathbf{D}_L.
\end{aligned} \tag{PHW25}$$

The proof is SZW20–SZW34: the full theta lift gives the endpoints, the convergent contour gives the arithmetic identity, and its explicit divisor receiver gives $Z(h)$. Tensoring with the invariant coefficient space W multiplies every term by $\dim W$, by the trace of an identity tensor factor. The Fourier closure tensors likewise with W , retaining every support label and its partner. This is the actual trivial-sector map from the holonomy construction into the programme formula. Equations (PHW17), (PHW23), and (PHW24) retain the other character-resolved arithmetic distributions with their full Euler data. The explicit signed-cover quotient which reaches a nontrivial one of these sectors is proved in CBR1–CBR29.

From the signed cover to an arithmetic quadratic cover

This calculation uses Alain Connes and Caterina Consani, Knots, primes and class field theory, arXiv:2501.06560v1, introductory Theorem 1 and the source labels `coverdef`, `mappingtorus1`, `artinrec`, and `main`. The original author TeX, rather than a PDF, was read through its entire body and bibliography. Their theorem supplies the arithmetic cover and its Frobenius monodromy. The calculations below specify the quotient of the programme's signed cover which enters that theorem, and the infinitesimal information its quotient forgets.

The incoming monodromy and collision proofs are EHM1–EHM53. The original order-192 group is the prior programme result SM1–SM19, source lines 2097–2513. Its abelian quotient is computed here; no novelty is claimed for quadratic fields, discriminants, or Gauss sums.

1. The exact abelian quotient

Write the full signed permutation group in its original convention as

$$G = \{(\sigma, \pi) : \sigma \in \{1, -1\}^4, \pi \in S_4, \prod_j \sigma_j = \text{sgn } \pi\}. \quad (\text{CBR1})$$

It acts on the eight states (j, ϵ) , with one pair over each of the four roots, by signed permutations. It has order $8 \cdot 24 = 192$. Let

$$N = \{(\sigma, 1) : \prod_j \sigma_j = 1\}, \quad \chi_{\text{disc}}(\sigma, \pi) = \text{sgn } \pi. \quad (\text{CBR2})$$

Projection to S_4 is onto: choose any sign vector whose product is the required parity. Its kernel is N .

We have the exact calculation

$$[G, G] = \{(\sigma, \pi) : \pi \in A_4, \prod_j \sigma_j = 1\}, \quad G^{\text{ab}} \cong \{1, -1\} \quad (\text{CBR3})$$

with quotient map χ_{disc} . Here is a proof retaining the kernel. Identify N with the even-coordinate-sum subspace of $\mathbb{F}_2^4$. Conjugation by any lift of π permutes its coordinates. Given distinct a, b, c , take $v = e_a + e_b$ and $\pi = (b \ c)$. The commutator of a lift of π with v is $\pi v - v = e_b + e_c$. These pair vectors span all of N , so $N \subseteq [G, G]$. The derived subgroup maps onto $[S_4, S_4] = A_4$: the sign kills commutators, and the commutators of two transpositions sharing one letter are three-cycles; three-cycles generate A_4 . The latter generation follows by splitting every even-length word in transpositions into pairs, and writing two disjoint transpositions as a product of two three-cycles. Thus the derived subgroup contains the entire preimage of A_4 , and containment in the reverse direction follows from the sign homomorphism. This proves (CBR3), including its order 96 and its universal property for homomorphisms from G into abelian groups.

There is a different quotient on the original set of eight points. The subgroup $[G, G]$ is transitive on it: an even permutation moves a chosen root label to any other root label, and an even sign change involving that root and one other root prescribes its sign. Hence

$$\{\text{eight states}\}/[G, G] = \{*\}. \quad (\text{CBR4})$$

Any equivariant map of the eight-point set into a set on which the action factors through an abelian group therefore has singleton image. In contrast, the regular frame torsor of G has the two-point quotient $G/[G, G]$. The double cover below is a quotient of this frame cover, not an unidentified double quotient of the eight individual states. Equations (CBR3) and (CBR4) prove the exact relation and the distinct kernels.

There is also an explicit common finite cover retaining both objects. Let H stabilize the state $(1, +)$, and let $D_G = [G, G]$. Then $|H| = 24$. The restriction of χ_{disc} to H is onto: a transposition of two other root labels, together with a sign change at a third other root, fixes $(1, +)$ and has character -1 . Hence $|H \cap D_G| = 12$, and

$$G/(H \cap D_G) \xrightarrow{\sim} (G/H) \times (G/D_G), \quad g(H \cap D_G) \longmapsto (gH, gD_G). \quad (\text{CBR4a})$$

It is injective by the intersection of the stabilizers. To hit a prescribed pair, first choose the desired gH , then multiply g on the right by an element of H with the required sign; this adjusts the second coordinate without changing the first. This proves surjectivity and equivariance. Projection to the eight-state factor has two-point fibres; projection to the character factor has eight-point fibres. Applied to the frame torsor over the discriminant complement, the same equivariant maps give its associated degree-16 cover and the two covering maps. Thus the failed double quotient of the eight states instead determines a common cover with completely specified maps.

2. The discriminant in the original coefficients

The polynomial in the original affine chart is

$$F(R) = AR^4 + R^3 + BR^2 + CR + D. \quad (\text{CBR5})$$

On $A \neq 0$ with distinct roots $R_1, \dots, R_4$, put $\Delta = \prod_{i < j} (R_i - R_j)$ and $\mathfrak{D} = A^6 \Delta^2$. This discriminant is a polynomial in the coefficients and extends to the binary-quartic chart with a root at infinity. On the frame cover choose the component whose derivative square roots satisfy

$$T_j^2 = F'(R_j), \quad w = A^3 \Delta = A \prod_{j=1}^4 T_j, \quad w^2 = \mathfrak{D}. \quad (\text{CBR6})$$

Indeed $\prod_j F'(R_j) = A^4 \Delta^2$: each unordered root pair contributes a minus sign, and there are six pairs. Thus the ratio of the two candidate square roots in (CBR6) is a constant sign; the component choice sets it to one. Under a signed permutation, Δ changes by $\text{sgn } \pi$, and $\prod T_j$ changes by $\prod \sigma_j$. These agree exactly by (CBR1). Consequently the double cover

$$w^2 = \mathfrak{D} \quad (\text{CBR7})$$

is the character quotient (CBR3). On the discriminant complement its derivative with respect to w is nonzero, so it is an unramified double cover. The reciprocal chart of EHM1–EHM5 retains this cover at $A = 0$; the value of $\mathfrak{D}$ does not require dividing by A .

Along the original infinity path $(A, B, C, D) = (A, 0, -1, 0)$,

$$F(R) = R(AR^3 + R^2 - 1), \quad \mathfrak{D} = 4 - 27A^2. \quad (\text{CBR8})$$

The product discriminant identity gives the cubic discriminant because the resultant with R is -1 , whose square is one. Substituting $a = A, b = 1, c = 0, d = -1$ into the cubic discriminant, or eliminating the common root of the cubic and its derivative, gives $4 - 27A^2$. At the rational parameter $A = 1$, (CBR7) is the field

$$K = \mathbb{Q}[w]/(w^2 + 23). \quad (\text{CBR9})$$

The polynomial is irreducible over $\mathbb{Q}$, since a rational square cannot be negative. This is a specialization of the double-cover equation, not an assertion that the full geometric order-192 monodromy specializes unchanged to an arithmetic Galois group.

3. An explicit map into the class-field cover

Let $\zeta = e^{2\pi i/23}$, and let $\lambda(a)$ be the quadratic character of $\mathbb{F}_{23}^\times$, extended by zero at zero. Its square residues are

$$1, 2, 3, 4, 6, 8, 9, 12, 13, 16, 18. \quad (\text{CBR10})$$

These are the squares of $1, \dots, 11$ modulo 23, so exactly eleven nonzero elements are squares, and -1 is not. Multiplication by a nonsquare exchanges the two cosets of the subgroup of squares, proving $\sum_{a \neq 0} \lambda(a) = 0$. Define

$$\tau_{23} = \sum_{a=1}^{22} \lambda(a) \zeta^a. \quad (\text{CBR11})$$

Conjugation gives $\bar{\tau}_{23} = -\tau_{23}$. Directly,

$$|\tau_{23}|^2 = \sum_{t \neq 0} \lambda(t) \sum_{b \neq 0} \zeta^{(t-1)b} = 22 - \sum_{t \neq 1} \lambda(t) = 23. \quad (\text{CBR12})$$

The inner sum is 22 for $t = 1$ and -1 otherwise, by the geometric sum for the 23rd roots of unity. Therefore $\tau_{23}^2 = -23$, and

$$K \xrightarrow{\sim} \mathbb{Q}(\tau_{23}) \subset \mathbb{Q}(\zeta), \quad w \mapsto \tau_{23} \quad (\text{CBR13})$$

is a field isomorphism onto its image. The polynomial $1 + X + \dots + X^{22}$ is irreducible by Eisenstein after $X \mapsto X + 1$; thus all substitutions $\zeta \mapsto \zeta^r$, $r \in \mathbb{F}_{23}^\times$, are the cyclotomic automorphisms. Reindexing the finite sum gives

$$\sigma_r(\tau_{23}) = \lambda(r) \tau_{23}. \quad (\text{CBR14})$$

This proves the arithmetic character without inferring it solely from the degree of the field.

In the notation of Connes–Consani the required continuous surjection is

$$\chi_{23} : \widehat{\mathbb{Z}}^\times \rightarrow \{1, -1\}, \quad u \mapsto \lambda(u_{23} \bmod 23). \quad (\text{CBR15})$$

It is onto by (CBR10), trivial at every other local unit factor, and factors at 23 through reduction modulo 23. It does not factor through the trivial unit quotient, so its conductor is exactly 23. Equations (CBR13)–(CBR15), followed by the source’s Artin map, identify its field with (CBR9).

For every prime $p \neq 23$, the source theorem now supplies the precise periodic return

$$\text{Mon}(C_p) = \lambda(p), \quad |C_p| = \log p. \quad (\text{CBR16})$$

In particular $p = 2$ has two separate lifted circles, each of length $\log 2$, and $p = 5$ has one circle of length $2 \log 5$. At 23 the quadratic character is ramified. These assertions follow by following multiplication by $+1$ or -1 on the two-element fibre, not by treating the sign as a positive or negative norm.

The integral model at 2 matters. Put $\theta = (1 + w)/2$. Then

$$\theta^2 - \theta + 6 = 0, \quad \mathbb{Z}[w] \subset \mathbb{Z}[\theta], \quad \mathbb{Z}[\theta]/\mathbb{Z}[w] \cong \mathbb{Z}/2. \quad (\text{CBR17})$$

Both modules have basis $(1, w)$ or $(1, \theta)$, and $w = 2\theta - 1$, proving the index. In fact $\mathbb{Z}[\theta]$ is the full ring of integers: an integral $x = a + bw$ has trace $m = 2a \in \mathbb{Z}$ and norm $a^2 + 23b^2 \in \mathbb{Z}$. Hence $23(2b)^2 \in \mathbb{Z}$. If $2b = r/s$ in lowest terms, $s^2 \mid 23$, forcing $s = 1$; write $2b = n \in \mathbb{Z}$. Norm integrality then says $m^2 + 23n^2 \equiv 0 \pmod{4}$, so m, n have the same parity. Thus $x = (m - n)/2 + n\theta \in \mathbb{Z}[\theta]$. Conversely this ring is integral by the displayed monic polynomial. Its two exact reductions are

$$\mathbb{Z}[w]/2 \cong \mathbb{F}_2[\eta]/(\eta^2), \quad \eta = w + 1; \quad \mathbb{Z}[\theta]/2 \cong \mathbb{F}_2 \times \mathbb{F}_2. \quad (\text{CBR18})$$

The second isomorphism is evaluation at $\theta = 0, 1$, since the defining polynomial reduces to $\theta(\theta - 1)$. It is the maximal order which records the unramified split prime 2. The nonzero nilpotent of the first reduction records the nonmaximal order and must not be used to assign ramification to the field cover.

4. The heat collision retains more than this character

On the actual heat path of EHM23,

$$F_h(R) = (R - 1)(R^2 + R - 16h), \quad \mathfrak{D}(h) = (2 - 16h)^2(1 + 64h). \quad (\text{CBR19})$$

For a direct calculation, the quadratic discriminant is $1 + 64h$, and its resultant with $R - 1$ is its value $2 - 16h$ at 1. Put $t = h - 1/8$. The pulled-back double-cover ring over convergent complex germs is

$$B = \mathbb{C}\{t\}[w]/(w^2 - 256t^2(9 + 64t)). \quad (\text{CBR20})$$

Choose the analytic square root $a(t) = \sqrt{9 + 64t}$ with $a(0) = 3$; its convergent binomial series specifies the branch. Define

$$B \hookrightarrow \widetilde{B} = \mathbb{C}\{t\} \oplus \mathbb{C}\{t\}, \quad f(t) + g(t)w \mapsto (f + 16ta(t)g, f - 16ta(t)g). \quad (\text{CBR21})$$

The map respects the relation, is injective, and its image consists exactly of pairs whose difference lies in $t\mathbb{C}\{t\}$. Indeed the inverse expressions are $f = (x + y)/2$ and $g = (x - y)/(32ta(t))$. Thus

$$\widetilde{B}/B \cong \mathbb{C}, \quad (x, y) \mapsto (x - y) \bmod t. \quad (\text{CBR22})$$

The product ring is the integral closure of B in its total quotient ring. To see this, its idempotent $(1, 0)$ is integral, and adjoining it to the diagonal germ ring produces all of $\widetilde{B}$; each convergent germ ring is a discrete valuation ring and hence integrally closed. An element integral over B in either fraction component is integral over the corresponding germ ring after projection, so belongs to that ring. This proves both containments. Its conductor back into B is $t\widetilde{B}$: multiplying by the two component idempotents tests that both coordinates vanish modulo t .

The special fibres and the induced map are

$$B/tB = \mathbb{C}[w]/(w^2) \longrightarrow \widetilde{B}/t\widetilde{B} = \mathbb{C} \oplus \mathbb{C}, \quad c + dw \mapsto (c, c). \quad (\text{CBR23})$$

Its kernel is $\mathbb{C}w$, and its cokernel is the difference line. These facts do not contradict injectivity before taking the fibre: tensoring the exact sequence in (CBR22) with $\mathbb{C}\{t\}/(t)$ gives the connecting isomorphism

from its Tor_1 to that kernel. Explicitly the free resolution with differential multiplication by t computes $\text{Tor}_1(\mathbb{C}, \mathbb{C}) = \mathbb{C}$; exactness or lifting a difference representative yields the nonzero kernel line. The dual-number collision and its two separate normalized points are both retained by (CBR21)–(CBR23).

On a punctured disk at $h = 1/8$ the character cover is trivial: its two sheets are the analytic graphs $w = \pm 16ta(t)$. The zero of the discriminant has order two and its square root has no branch monodromy there. At $h = -1/64$, the factor $1 + 64h$ has a simple zero and $2 - 16h \neq 0$, so the two sheets are exchanged. On the signed-cover group this agrees exactly with

$$\chi_{\text{disc}}(g_c) = +1, \quad \chi_{\text{disc}}(g_d) = -1, \quad \chi_{\text{disc}}(g_d^2) = +1. \quad (\text{CBR24})$$

Indeed g_c changes signs in two root pairs without permuting roots; g_d interchanges two root labels; and its square again has trivial root permutation, using the explicitly based loops EHM25.

The original full collision algebra is the different algebra

$$\mathcal{L}_0 = \mathbb{C}[T]/(T^4), \quad \mathfrak{N} = (T), \quad \mathcal{L}_0/\mathfrak{N}^2 \cong \mathbb{C}[\epsilon]/(\epsilon^2), \quad \epsilon \longleftrightarrow T \bmod T^2. \quad (\text{CBR25})$$

One explicit isomorphism from the double-cover special fibre (CBR23) to this last quotient sends w to $T \bmod T^2$. This is an isomorphism of the displayed complex algebras, with zero kernel and inverse determined by $T \bmod T^2 \mapsto w$. It is not the fibre of the normalization map (CBR21), whose kernel is nonzero. No identification of the two families or of their monodromy actions follows from this special-fibre isomorphism.

The based transport g_d^2 , invisible to (CBR24), has the nonzero operator residue proved in EHM43–EHM46:

$$\delta = -\frac{T^3}{16} \frac{d}{dT}, \quad \delta(T) = -\frac{T^3}{16}, \quad \delta(1) = \delta(T^2) = \delta(T^3) = 0. \quad (\text{CBR26})$$

The derivation is well defined modulo T^4 because it sends T^4 to a multiple of T^6 ; it is nonzero and has square zero. In the original generators

$$n_1 = T^2/2, \quad n_2 = 3T + T^3/6, \quad n_3 = T^3/2, \quad \delta : \mathfrak{N}/\mathfrak{N}^2 \xrightarrow{\sim} \mathfrak{N}^3, \quad [n_2] \mapsto -3n_3/8. \quad (\text{CBR27})$$

Each space is one-dimensional and the displayed coefficient is nonzero. In particular the vanishing character monodromy in (CBR24) coexists with a nonzero map between infinitesimal layers. Neither averaging the quadratic character nor passing to the normalized two-sheet fibre retains this map automatically.

5. The trace map into the supported-zero programme

The oriented periodic trace, including inertia invariants and both traversals, is proved in PHW1–PHW22. For the quadratic character (CBR15), it specializes to the finite distribution on a compactly supported smooth test h

$$P_\lambda(h) = \sum_{p \neq 23} \sum_{k \geq 1} (\log p) p^{-k/2} \lambda(p)^k (h(-k \log p) + h(k \log p)). \quad (\text{CBR28})$$

If the support is in $[-R, R]$, only $k \log p \leq R$ contribute. For $p = 5$, odd iterates carry coefficient -1 and even iterates coefficient $+1$; for $p = 2$, all carry $+1$. These are actual powers of Frobenius, not a choice to replace a negative trace by its absolute value.

For the two-dimensional regular representation of $\{1, -1\}$, the exact character decomposition is $\mathbb{C} \oplus \mathbb{C}_\lambda$. Therefore its periodic distribution is $P_{\text{fin}} + P_\lambda$; at 23 only its trivial line survives the inertia projection. At every separate supported or unsupported global zero, the coefficient space is instead the single trivial line, since the full compact unit group fixes that carrier point. The projector is

$$P_0 = \frac{1}{2}(I + \rho(-1)), \quad \text{im } P_0 = \mathbb{C}, \quad \ker P_0 = \mathbb{C}_\lambda. \quad (\text{CBR29})$$

In the group-ring trace of PHW17, augmentation has exactly this trivial-character observation. It returns the complete supported-zero formula SZW33–SZW42, with the separate fixed-support labels and Fourier partners. The complementary character retains (CBR28). Thus the arithmetic quotient has a defined map to the current Weil calculation while the residue (CBR27) remains additional filtered information in the full signed cover.

The classical RH question still concerns positivity on all admissible tests in that trivial zeta sector. The source paper relates its ambient adèle-class construction to the spectral realization of zeta zeros and to local explicit-formula terms; it does not prove positivity from nontrivial monodromy. Here the exact character, conductor, local prime actions, projection kernels, and surviving infinitesimal residue have been calculated. In particular neither a nontrivial return map nor the positive averaged endpoint deletes the correction retained in HWA33–HWA38.

Exact verification

The reproducible checker `check_class_field_bridge.py` verifies 72 finite-group and polynomial identities, including the full derived subgroup, its eight-state action, both original discriminants, all 22 cyclotomic character actions, the integral reduction at 2, and the residue derivation. The source theorem and the analytic and module arguments have their proofs or exact source locators above.

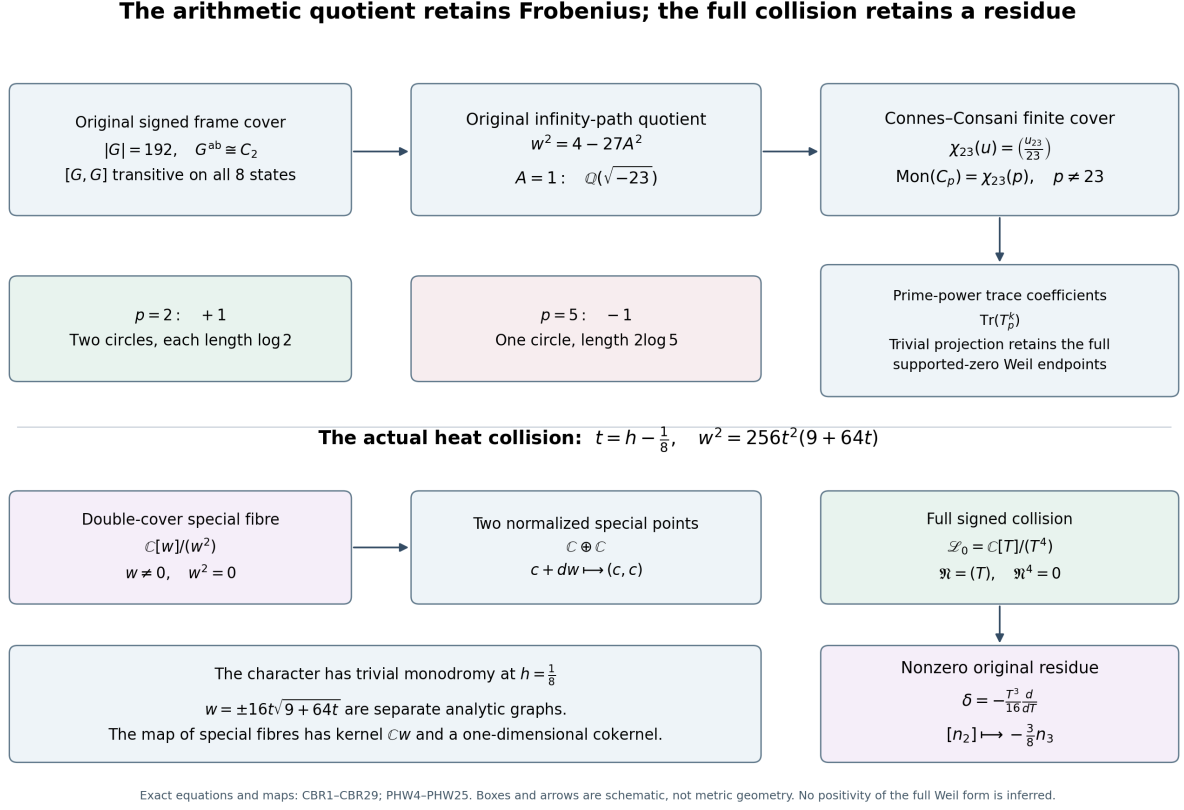


Figure 11: The original frame-cover quotient, its arithmetic specialization and prime returns, the two different special-fibre maps, and the surviving holonomy residue. CBR1-CBR29 and PHW4-PHW25 prove these exact maps. The box arrangement is schematic.

The original supported-zero identity on the heat semimodule

23 September 2026. The calculation is at the unlocalized supported-zero object. It retains the original semilattice and its module fibres. Supported zero is the multiplicative identity of the original singlet algebra, and this fact is used below rather than replaced by a localization argument.

Primary programme source: *An Algebraic Structure Incorporating a $\mathbb{Z}/1\mathbb{Z}$ -Symmetric Element Adjoined to the Integers*, original source, theorem `thm:structure_A` and proof, lines 1146–1260. The construction and operation table are at lines 578–625. The support-semimodule calculation below is proved directly from those operations. The full support reconstruction gives the earlier complete module comparison. The heat source is Rodgers–Tao, arXiv:1801.05914v5, introduction, equations `hoz`, `phidef`, `htdef` and theorem `main`; the bound 0.2 is Platt–Trudgian, arXiv:2004.09765v1, subsection *The de Bruijn–Newman constant*, its displayed corollary. These literature results are cited, not claimed as new proofs here.

1. Supported zero is already invertible on the original singlet algebra

Let R be a nonzero commutative unital ring. In

$$S = G(R) = \{\tau\} \sqcup R^\bullet, \quad e = 0_R^\bullet, \quad (\text{ZH1})$$

use the original operations $a^\bullet + b^\bullet = (a + b)^\bullet$, $a^\bullet b^\bullet = (ab)^\bullet$, $\tau + x = x$ and $\tau x = \tau$. On the original subobject $A = \{\tau, e\}$, their complete tables are

+	τ	e
τ	τ	e
e	e	e

multiplication	τ	e
τ	τ	τ
e	τ	e

They prove

$$0_A = \tau, \quad 1_A = e, \quad e^{-1} = e \text{ in } A. \quad (\text{ZH2})$$

Closure is in the tables; associativity, commutativity and distributivity restrict from S . Hence this is a unital semiring already inside the original set, without any localization. The inclusion $i : A \rightarrow S$ preserves addition, multiplication and additive zero. It takes 1_A to e , whereas $1_S = 1_R^\bullet$. The exact unital retraction in the other direction is

$$r : S \rightarrow A, \quad r(x) = ex, \quad ri = \text{id}_A. \quad (\text{ZH3})$$

Distributivity proves additivity; $(ex)(ey) = e^2xy = exy$ proves multiplicativity; $r(1_S) = e = 1_A$. These identities retain both original units and prove their comparison.

2. The identity acts on the entire semilattice of supported zeros

Let M be any S -semimodule, with its global zero 0_M . Define

$$L_M = eM, \quad p_M : M \rightarrow L_M, \quad m \mapsto em, \quad i_M : L_M \hookrightarrow M. \quad (\text{ZH4})$$

The equations $e^2 = e$, $e + e = e$, $r^\bullet e = e$ show that L_M is a join-semilattice with addition as join and bottom 0_M . Both p_M and i_M are S -linear, where supported scalars act identically on L_M and τ sends it to bottom. Their exact compositions are

$$p_M i_M = \text{id}_{L_M}, \quad i_M p_M = e \cdot (-) : M \rightarrow M, \quad e|_{L_M} = \text{id}_{L_M}. \quad (\text{ZH5})$$

For example $p_M(r^\bullet m) = er^\bullet m = em = r^\bullet p_M(m)$, and the τ equation follows from the semimodule zero law. Additivity is distributivity. The inclusion has the same scalar laws since they are restricted from M . This proves every assertion in ZH4–ZH5. In particular the action of supported zero on L_M is invertible, with inverse itself, while every label of L_M remains present.

The exact adjunction is also explicit. Regard any join-semilattice L with bottom as an S -semimodule $J(L)$, with every supported scalar acting as identity and τ acting as bottom. For an S -linear $f : M \rightarrow J(L)$,

$$f(m) = ef(m) = f(em). \quad (\text{ZH6})$$

Its restriction $g : L_M \rightarrow L$ preserves joins and bottom. Conversely any such g gives the S -linear map gp_M . The two constructions are inverse, since p_M restricts to the identity. Thus

$$\text{Hom}_S(M, J(L)) \simeq \text{Hom}_{V, \perp}(eM, L). \quad (\text{ZH7})$$

This proves the reflection onto the semilattice of supported zeros, with its unit p_M . The natural inclusion i_M is retained as well.

The same functor is also right adjoint to J . Indeed, for an S -linear $h : J(L) \rightarrow M$, one has $h(\lambda) = h(e\lambda) = eh(\lambda)$, so its image is in eM . Its unique factorization through i_M is a map preserving joins and bottom; conversely every such map gives h by inclusion. Therefore

$$\text{Hom}_S(J(L), M) \simeq \text{Hom}_{V, \perp}(L, eM). \quad (\text{ZH7a})$$

The two adjunctions and ZH5 give the exact unit and counit maps on the original zero object. No quotient of the ambient scalar semiring is needed for these identities.

For each $\lambda \in L_M$ the fibre $M_\lambda = \{m : em = \lambda\}$ is an R -module with local zero λ . The group identities are $m + \lambda = (1_S + e)m = m$, $m + (-1_R)^\bullet m = em = \lambda$, and $e((-1_R)^\bullet m) = \lambda$. They give inverses and closure within the fibre. Scalar distributivity and associativity restrict from M . For $\lambda \leq \mu$, the linear transition is $m \mapsto m + \mu$, as direct substitution proves. Thus p_M returns the actual local zero of each separate module fibre, retaining their separate labels.

Every S -linear $F : M \rightarrow N$ obeys

$$p_N F = (F|_{eM}) p_M, \quad F i_M = i_N (F|_{eM}). \quad (\text{ZH8})$$

Both are the identity $F(em) = eF(m)$. If F is invertible, its restriction to eM is invertible, with inverse the restriction of F^{-1} . That restriction can have nontrivial permutations of support labels. The equations do not assert that the full operator or its action on a module fibre is the identity.

3. A specified function space on which the actual heat flow is invertible

Let V consist of even complex measurable functions q on $\mathbb{R}$, up to almost-everywhere equality, with

$$\|q\|_a := \int_{\mathbb{R}} e^{au^2} |q(u)| du < \infty \quad \text{for every real } a > 0. \quad (\text{ZH9})$$

For every real t define

$$U_t q(u) = e^{tu^2} q(u). \quad (\text{ZH10})$$

This is a linear map $V \rightarrow V$: choose $b > \max(a + t, 0)$, then $\|U_t q\|_a \leq \|q\|_b$. Its exact group laws are

$$U_t U_s = U_{t+s}, \quad U_t^{-1} = U_{-t}, \quad U_0 = \text{id}_V. \quad (\text{ZH11})$$

These follow by multiplying the displayed exponential functions. Each inverse is on the same function space V ; no bounded inverse on an unspecified Hilbert space is asserted.

The cosine transform

$$(\mathcal{C}q)(z) = \int_0^\infty q(u) \cos(zu) du \quad (\text{ZH12})$$

is entire. On any compact set $|z| \leq B$, every fixed derivative is bounded by $\int_0^\infty |q(u)| u^k e^{Bu} du$, which is finite because $u^k e^{Bu} \leq C_{a,B,k} e^{au^2}$. Dominated differentiation proves entire dependence. It is injective on V : twice its restriction to real z is the Fourier transform of the even L^1 function q . For completeness Fourier uniqueness follows by multiplying that transform by $e^{-\epsilon x^2}$, applying Fubini with the Gaussian Fourier identity, and obtaining $q * g_\epsilon = 0$; the Gaussian approximate identities converge to q in L^1 , so $q = 0$. All integrals used in this argument are absolutely convergent. Define $F = \mathcal{C}(V)$ with its transported vector-space structure. Then

$$T_t = \mathcal{C} U_t \mathcal{C}^{-1} : F \rightarrow F \quad (\text{ZH13})$$

is an actual linear automorphism with inverse T_{-t} . Differentiating ZH12 gives $\partial_t T_t F = -\partial_z^2 T_t F$.

Let Φ be exactly the Rodgers–Tao kernel

$$\Phi(u) = \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}. \quad (\text{ZH14})$$

The source proves its evenness by Poisson summation. For $u \geq 0$, factoring half the leading exponential gives

$$|\Phi(u)| \leq C e^{9u} e^{-(\pi/2)e^{4u}}, \quad (\text{ZH15})$$

where C is bounded by the convergent sum of $(2\pi^2 n^4 + 3\pi n^2) e^{-(\pi/2)n^2}$. Evenness supplies the negative half-line. This bound proves $\Phi \in V$. Hence the actual family is on this space:

$$H_t = C U_t \Phi = T_t H_0, \quad H_0(z) = \frac{1}{8} \xi(\frac{1}{2} + iz/2). \quad (\text{ZH16})$$

Thus the actual heat evolution is invertible at time 0 and at every other real time on the explicitly defined space F . This is a calculation of the operator, not an inference from whether its current function has real zeros.

4. Carrying that heat group through the unlocalized split-zero construction

For any complex vector space W , its split extension is

$$G(W) = \{\tau_W\} \sqcup W^\bullet, \quad e_W = 0_W^\bullet, \quad (\text{ZH17})$$

with supported vector addition inherited from W , absent additive identity τ_W , $c^\bullet w^\bullet = (cw)^\bullet$, supported scalars fixing τ_W , and τ sending every vector to τ_W . These laws make an $S = G(\mathbb{C})$ -semimodule by direct use of vector-space distributivity, with the cases containing τ giving the global zero.

The exact lifted flow is

$$G(T_t)(w^\bullet) = (T_t w)^\bullet, \quad G(T_t)(\tau_W) = \tau_W, \quad W = F. \quad (\text{ZH18})$$

It is S -linear, has inverse $G(T_{-t})$, and satisfies

$$G(T_0) = \text{id}_{G(F)}, \quad G(T_t)|_{e_{G(F)}} = \text{id}_{\{\tau_F, e_F\}} \quad (t \in \mathbb{R}). \quad (\text{ZH19})$$

The second equation follows because each linear T_t sends 0_F to 0_F , and the first follows from $T_0 = \text{id}$. On this original zero semilattice, supported zero itself acts as the same identity by ZH5. On a general module diagram, ZH8 retains all labels and their induced maps rather than replacing them by this particular two-label example.

Evaluation at z is the S -linear map

$$G(\text{ev}_z) : G(F) \rightarrow G(\mathbb{C}), \quad w^\bullet \mapsto w(z)^\bullet, \quad \tau_F \mapsto \tau. \quad (\text{ZH20})$$

Consequently

$$G(\text{ev}_z)(H_t^\bullet) = e \iff H_t(z) = 0. \quad (\text{ZH21})$$

This is the actual map from the evolving function to the supported zero at which it vanishes. Invertibility of $G(T_t)$ as an evolution operator does not assert invertibility of any evaluation or make ZH21 vacuous.

There is a direct test on the Riemann family itself. Rodgers–Tao proves $\Lambda \geq 0$, so H_{-1} has a nonreal zero. Platt–Trudgian proves $\Lambda \leq 0.2$, so every zero of $H_{0.2}$ is real. Both are related by the actual invertible map $T_{1.2}$, and both induce the identity on the zero semilattice in ZH19. Therefore neither invertibility of the heat evolution nor the identity action of supported zero determines real-rootedness. This uses two actual members of the Riemann heat family.

At $t = 0$, ZH21 still asks exactly whether a nonreal z can evaluate $H_0^\bullet$ to e . The preceding calculations preserve this evaluation map and leave that value question undetermined.

5. The actual infinitesimal residue survives the zero identity

The previously computed complex collision algebra is $C_0 = \mathbb{C}[T]/(T^4)$, with retained derivation

$$D(T) = -T^3/16, \quad D(1) = D(T^2) = D(T^3) = 0. \quad (\text{ZH22})$$

This is the residue in CBR25–CBR29, read in its specified original collision coordinate. It descends from $-T^3\partial_T/16$ because it sends T^4 to $-T^6/4$, in the defining ideal. Its square is zero and its image $\mathbb{C}T^3$ has square zero. Consequently, for each $a \in \mathbb{C}$,

$$V_a = \text{id}_{C_0} + aD, \quad V_a V_b = V_{a+b}, \quad V_a^{-1} = V_{-a} \quad (\text{ZH23})$$

are algebra automorphisms. Indeed the Leibniz rule and the zero product of two D -images prove multiplicativity, and $D^2 = 0$ proves the group law. The value $V_a(T) = T - aT^3/16$ differs from T for $a \neq 0$. These are the automorphisms generated by the retained residue; no equality of the parameter a with heat time t or with a particular meridian is imposed.

Their lifts $G(V_a)$ commute with $p_{G(C_0)}$ and restrict to the identity on the zero semilattice by ZH8. Thus the nonzero infinitesimal action and the invertible identity action of e coexist in the actual collision receiver. Supported-zero identity does not remove the infinitesimal direction.

The exact positive-form consequence for these automorphisms is computable. Fix $a \neq 0$ and $N = aD$. A positive semidefinite Hermitian form Q on C_0 is invariant under V_a precisely when $\text{im } D$ is in its radical. To prove necessity, invariance and $V_a^n = \text{id} + nN$ give, for every integer n ,

$$Q(v + nNv, v + nNv) = Q(v, v). \quad (\text{ZH24})$$

The coefficient of n^2 is $Q(Nv, Nv)$, hence zero. For a positive semidefinite Hermitian form a vector of square zero is radical: expand $Q(x + zy, x + zy) \geq 0$ for arbitrary complex z to force $Q(x, y) = 0$ when $Q(x, x) = 0$. Thus $Q(Nv, w) = 0$ for all v, w . Conversely that condition makes the cross and quadratic terms vanish, proving invariance. Such forms correspond bijectively, by pullback, to positive semidefinite Hermitian forms on $C_0/\mathbb{C}T^3$. The quotient kernel $\mathbb{C}T^3$ is retained as a specified nonzero line; no positive definite invariant form exists on the full four-dimensional space. This is an exact statement about the residue-generated action. The following residue-duality calculation, RD18–RD31b, gives its actual trace comparison and the corresponding full Weil-packet map.

6. The full supported Weil formula and its constant decomposition

The established supported-zero Weil identity, SZW33–SZW35 retains every label:

$$\mathbf{B}_L - \mathbf{Z}_L = \mathbf{D}_L. \quad (\text{ZH25})$$

Its lower fixed-support terms occur explicitly on both $\mathbf{B}_L$ and $\mathbf{D}_L$; its arithmetic divisor receiver is $\mathbf{Z}_L$. They are retained by the identity $e|_{eM}$. In particular, a real-valued function on the labelled zero set can distinguish τ from e ; it need not be an additive semiring character. The complete coordinate comparison is recorded in HZ25–HZ29 of the accompanying heat-specialization calculation.

The pole part of $\Lambda(s)$ is $1/(s-1) - 1/s$. Multiplication by $s(s-1)/2$ makes that term exactly $1/2$, and the Rodgers–Tao factor $1/8$ makes it $1/16$ in H_0 . Here $\Lambda(s)$ denotes the completed meromorphic zeta function, separately from the real de Bruijn–Newman constant. ZH12 represents H_0 as the Fourier transform of the absolutely continuous even measure $(\Phi(u)/2) du$, which has no atom at zero. In the enlarged finite-measure Fourier representation,

$$\frac{1}{2}\Phi(u) du = \frac{1}{16}\delta_0 + \left(\frac{1}{2}\Phi(u) du - \frac{1}{16}\delta_0\right). \quad (\text{ZH26})$$

This is the exact source of the decomposition $H_0 = 1/16 + (H_0 - 1/16)$. The opposite atom in the remainder is retained. Multiplication by e^{tu^2} fixes both atoms, so their cancellation persists. A labelled endpoint contribution in the full supported theta construction can be retained independently; its map to the unlabelled Φ measure has exactly the cancellation in ZH26.

The calculation therefore accepts the original invertibility of supported zero on its singlet algebra and on the entire zero semilattice, constructs the lifted invertible heat group through it, and retains the actual nonzero infinitesimal residue. The supported-zero identity occurs with both real and nonreal zeros in the actual heat family. The undecided arithmetic value is still the complete evaluation and divisor pairing at time0.

Independent verification and illustration

The complete ZH1–ZH26 calculation, including ZH7a, received an independent mathematical review. No mathematical error was found. The constant $1/16$ and $H_0 - 1/16$ individually lie outside the weighted-function cosine image F , by the Riemann–Lebesgue property, while their sum H_0 lies in F . The enlarged measure representation in ZH26 is the stated domain for their separate terms.

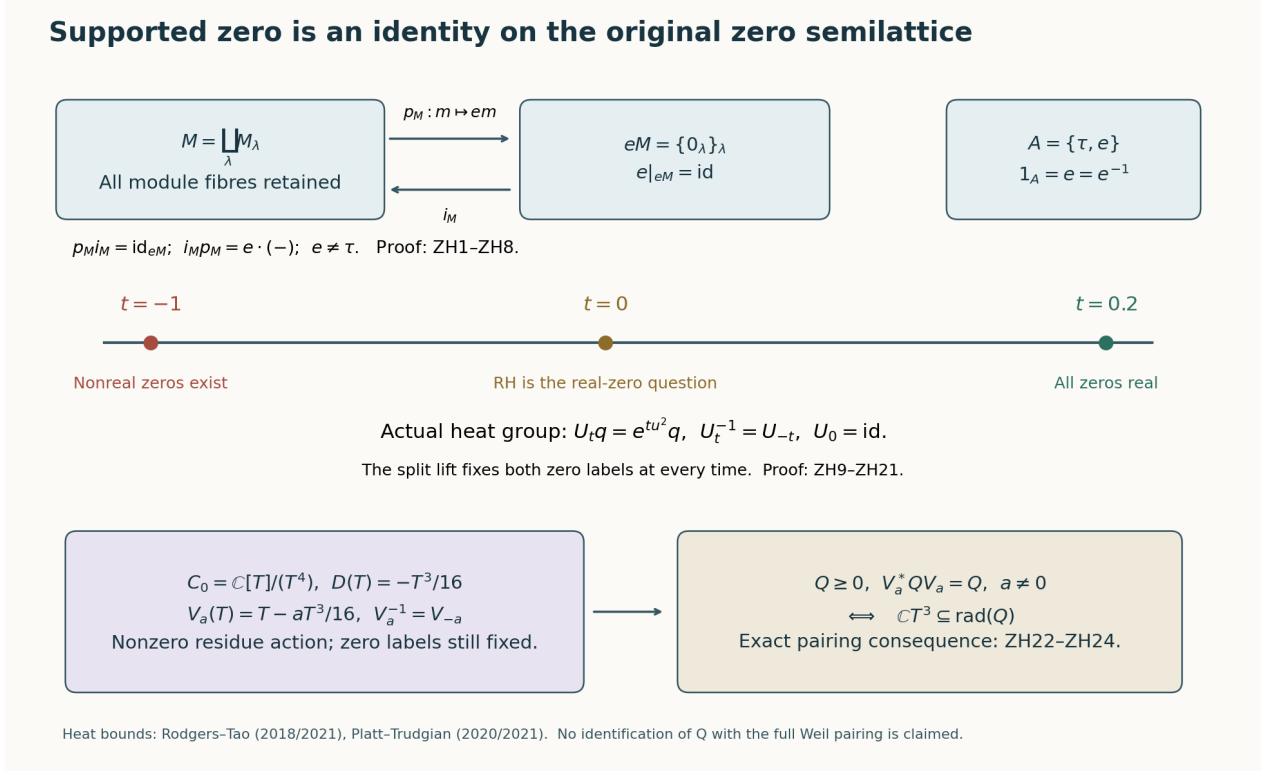


Figure 12: The original zero-semimodule retraction, the actual invertible heat evolution and its two known root regimes, and the exact invariant-form consequence of the retained residue. ZH1–ZH26 prove the displayed maps and statements. Time-marker positions are schematic and are not a numerical scale.

The reproducible figure source is `draw_supported_zero_heat_identity.py`; both its PNG and SVG are retained. The PNG was visually inspected. The two arrows at the top preserve all module fibres and support labels; the lower arrow concerns the residue-generated action, with no identification with an arithmetic Weil pairing assumed.

Actual global heat continuation

The global heat trace retains all zero multiplicities. The arithmetic calculation, HA1–HA26, evaluates the complete supported formula on Cauchy tests and computes its actual derivative. The local response keeps the complete analytic unit; the global response includes mixed prime-product terms. The matrix criterion gives the exact relation of this test family to RH. No sign from an isolated local coefficient replaces the complete pairing.

Residue duality, the collision Jacobian, and the supported Weil trace

23 September 2026.

This calculation gives the exact map from a perfect pairing that detects the collision infinitesimal to the regular trace pairing that annihilates it. It then proves the corresponding identity for the programme's full Weil packets, including their original amplitude multiplier and involution. The trace radical is identified with a specified cotangent cohomology group. No off-critical zeta zero is constructed, and no positivity of the full arithmetic form is asserted.

The incoming collision and meridian are EHM43–EHM46. The incoming Weil multiplier and analytic identity are WP13–WP46b and WA24–WA30. The supported-zero identity used here is SZW33–SZW35. All additional finite algebra identities are proved below.

1. The integral algebras and their complete duality maps

Over the integers put

$$B = \mathbb{Z}[x, T]/(x^2(x+3), T^2 - 3x(x+2)), \quad C = \mathbb{Z}[\epsilon, T]/(\epsilon^2, T^2 - 6\epsilon). \quad (\text{RD1})$$

The original coordinate is $x = R_{\text{original}} - 1$. First reducing powers of T , and then powers of x , proves that the retained bases are

$$\mathcal{B}_B = (1, x, x^2, T, xT, x^2T), \quad \mathcal{B}_C = (1, \epsilon, T, \epsilon T). \quad (\text{RD2})$$

Uniqueness follows by viewing B as the monic quadratic extension of $\mathbb{Z}[x]/(x^2(x+3))$, and C as the monic quadratic extension of $\mathbb{Z}[\epsilon]/(\epsilon^2)$. These are free of ranks six and four. The surjective algebra map

$$q : B \longrightarrow C, \quad q(x) = \epsilon, \quad q(T) = T \quad (\text{RD3})$$

is well defined: the first relation maps to zero and the second to $T^2 - 6\epsilon$. Its kernel is the free subgroup generated by x^2, x^2T , equivalently the ideal (x^2) . This follows directly from RD2.

Define λ_B to be the coefficient of x^2T , and λ_C the coefficient of ϵT . Their bilinear pairing matrices $\lambda(ab)$ in RD2 are

$$G_B = \begin{pmatrix} 0 & G \\ G & 0 \end{pmatrix}, \quad G = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & -3 \\ 1 & -3 & 9 \end{pmatrix}, \quad G_C = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{RD4})$$

For example $x^3 = -3x^2$ and $x^4 = 9x^2$, which give every entry of G . A product with an even total power of T has no T -coefficient, giving the two zero diagonal blocks. The analogous calculations with $\epsilon^2 = 0, T^2 = 6\epsilon$ give G_C . In particular

$$G^{-1} = \begin{pmatrix} 0 & 3 & 1 \\ 3 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}. \quad (\text{RD5})$$

Thus both pairings are perfect over $\mathbb{Z}$, not only over a field. The maps

$$\Theta_B : B \xrightarrow{\sim} B^\vee, \quad a \mapsto (f \mapsto \lambda_B(af)), \quad \Theta_C : C \xrightarrow{\sim} C^\vee \quad (\text{RD6})$$

are isomorphisms of modules over the respective algebras, where $A^\vee = \text{Hom}_{\mathbb{Z}}(A, \mathbb{Z})$ and $(a\ell)(f) = \ell(af)$. This module assertion is immediate from associativity. The inverse matrices in RD4–RD5 prove bijectivity integrally.

For a finite free algebra A , write $\text{tr}_A(f) = \text{Tr}(m_f)$, the regular multiplication trace. If b_i, b_i^* are dual for λ , then the coefficient of b_i in fb_i is $\lambda(b_i^*fb_i)$. Summing the diagonal entries therefore gives

$$\text{tr}_A(f) = \lambda \left(f \sum_i b_i b_i^* \right). \quad (\text{RD7})$$

For B , the dual basis is

$$(3xT + x^2T, 3T + xT, T, 3x + x^2, 3 + x, 1).$$

Its product sum is $6x(x+2)T$. For C , the dual basis is $(\epsilon T, T, \epsilon, 1)$, with product sum $4\epsilon T$. These are exactly the two Jacobian determinants:

$$J_B = \det \begin{pmatrix} 3x(x+2) & -6(x+1) \\ 0 & 2T \end{pmatrix} = 6x(x+2)T, \quad J_C = \det \begin{pmatrix} 2\epsilon & -6 \\ 0 & 2T \end{pmatrix} = 4\epsilon T. \quad (\text{RD8})$$

Columns here are the differentials of the two defining relations, and rows are dx, dT , or $d\epsilon, dT$. Consequently the full trace maps are

$$\mathrm{tr}_B(f) = \lambda_B(J_B f), \quad \mathrm{tr}_C(f) = \lambda_C(J_C f). \quad (\text{RD9})$$

For $f = p_0 + p_1x + p_2x^2 + (q_0 + q_1x + q_2x^2)T$, the first equals $6p_0 - 6p_1 + 18p_2$. The second trace, for $f = a + b\epsilon + cT + d\epsilon T$, equals $4a$. These formulas also follow by direct multiplication matrices and hold after every base change.

The dual of RD3 is particularly informative:

$$q^\vee \lambda_C = (x+3)\lambda_B, \quad q^\vee(c\lambda_C) = (x+3)\tilde{c}\lambda_B. \quad (\text{RD10})$$

Here $\tilde{c}$ is any lift of c . Indeed the coefficient of x^2T in $(x+3)f$ is q_1 , because $x^3 + 3x^2 = 0$; this equals $\lambda_C(qf)$. Independence of the lift follows from $(x+3)x^2 = 0$. Thus RD10 proves the complete dual comparison, not just an equality of dimensions. On traces it gives the exact decomposition

$$J_B = 4x(x+3)T + 2x^2T, \quad q^\vee \mathrm{tr}_C = 4x(x+3)T \lambda_B. \quad (\text{RD11})$$

The two trace functionals evaluate to $4p_0$ and $2(p_0 - 3p_1 + 9p_2)$. This formula holds over $\mathbb{Z}$, including at the prime three. It is not a claim that the integral algebra splits as a direct product.

2. The nonzero residue acts on perfect duality

Base change now to $\mathbb{Z}[1/2]$. Put

$$n_1 = x(x+3), \quad n_3 = Tx(x+3), \quad D_B(x) = 0, \quad D_B(T) = -n_3/8, \quad D_C(\epsilon) = 0, \quad D_C(T) = -3\epsilon T/8. \quad (\text{RD12})$$

The derivations descend: $D_B(x^2(x+3)) = 0$, and the derivative of the second relation is $-T^2n_1/4 = 0$, since $T^2n_1 = 3x^2(x+2)(x+3) = 0$. For C the only further check is $2TD_C(T) = -3\epsilon T^2/4 = 0$. Direct substitution gives $qD_B = D_Cq$. Also $n_1^2 = 0$, $D_B(n_1) = 0$, and $\epsilon^2 = D_C(\epsilon) = 0$, proving $D_B^2 = D_C^2 = 0$.

For either algebra let $D^\vee \ell = -\ell \circ D$. Then

$$D_B^\vee \lambda_B = \frac{n_1}{8} \lambda_B, \quad D_C^\vee \lambda_C = \frac{3\epsilon}{8} \lambda_C. \quad (\text{RD13})$$

For the general element f in RD9, $D_B f = -q_0 n_3/8$: multiplying n_3 by x gives zero. Hence $-\lambda_B(D_B f) = q_0/8$, while $\lambda_B(n_1 f) = q_0$, proving the first identity. The calculation on the four coefficients of C gives the second. For an arbitrary multiplier a , the Leibniz rule proves the full connection formula

$$D^\vee(a\lambda) = (Da + ra)\lambda, \quad r_B = n_1/8, \quad r_C = 3\epsilon/8. \quad (\text{RD14})$$

Indeed $-\lambda(aDf) = -\lambda(D(af)) + \lambda((Da)f)$. This is the dual action on the specified dualizing module, not a Frobenius lift. Compatibility with $q^\vee$ follows both from $qD_B = D_Cq$ and the exact identity $(x+3)n_1 = 3n_1$.

The residue acts nontrivially on the perfect dual generators, but

$$D_B J_B = 0, \quad r_B J_B = 0, \quad D_C J_C = 0, \quad r_C J_C = 0. \quad (\text{RD15})$$

Each assertion follows by inserting RD8 into RD12–RD13 and using $x^2(x+3) = 0$ or $\epsilon^2 = 0$. Thus the two regular trace functionals are fixed by the dual action. RD9 and RD13–RD15 identify the precise map at which the nonzero action disappears.

3. The complete Hermitian calculation on the complex collision

After base change to $\mathbb{C}$, there is the explicit isomorphism

$$C_C \simeq C_0 := \mathbb{C}[T]/(T^4), \quad \epsilon \longmapsto T^2/6. \quad (\text{RD16})$$

Its inverse sends T to T ; $T^4 = 36\epsilon^2 = 0$. This isomorphism does not hold over $\mathbb{Z}$: the natural map $\mathbb{Z}[T]/(T^4) \rightarrow C$ has image generated by $1, T, 6\epsilon, 6\epsilon T$, and cokernel $(\mathbb{Z}/6)^2$ on the ϵ and ϵT coordinates.

Use $f^*(T) = \overline{f(-T)}$, and define $\ell(f) = [T^3]f$. Then $\ell(f^*) = -\overline{\ell(f)}$, so

$$Q_{\text{res}}(f, g) = i\ell(f^*g) \quad (\text{RD17})$$

is Hermitian. Its matrix in $(1, T, T^2, T^3)$ has antidiagonal entries $i, -i, i, -i$, all other entries zero. Each of the pairs $(1, T^3)$ and (T, T^2) has one positive and one negative eigenvalue, of absolute value one. Thus the form is nondegenerate with inertia $(2, 2, 0)$.

The regular trace pairing and its exact comparison are

$$Q_{\text{tr}}(f, g) = \text{Tr}(m_{f^*g}) = 4\overline{f_0}g_0 = Q_{\text{res}}(f, Kg), \quad K = -4iT^3. \quad (\text{RD18})$$

The final equality follows by taking the coefficient of T^3 . The multiplier $-4iT^3$ is fixed by $*$, so K is self-adjoint for Q_{res} . It satisfies

$$K^2 = 0, \quad \ker K = (T), \quad \text{im } K = \mathbb{C}T^3. \quad (\text{RD19})$$

The trace form is positive semidefinite with radical (T) . These properties coexist with the indefinite, perfect residue form. They are related by the displayed map; they are not contradictory positivity claims about a single pairing.

All constants agree with the integral calculation. Under RD16, $\lambda_C = 6\ell$ and $J_C = (2/3)T^3$. Their product is $4\ell(T^3 \cdot)$, exactly RD18.

The retained infinitesimal is $D(T) = -T^3/16$, so $Df = -f_1T^3/16$. The maps $V_a = 1 + aD$, for real a , are algebra automorphisms: $D^2 = 0$ and $(Df)(Dg) = 0$ show multiplicativity and $V_aV_b = V_{a+b}$. Their entire effect on RD17 is

$$Q_{\text{res}}(V_af, V_ag) - Q_{\text{res}}(f, g) = -\frac{ia}{16}(\overline{f_0}g_1 - \overline{f_1}g_0). \quad (\text{RD20})$$

Expanding the two factors proves this equality; the quadratic term is zero because it contains T^6 . For $a \neq 0$ the difference has inertia $(1, 1, 2)$, with nonzero eigenvalues $\pm|a|/16$. Congruence by the invertible V_a leaves the full residue form's inertia $(2, 2, 0)$. The difference of the trace pairings is identically zero, since V_a preserves f_0 .

This is a nonzero signed response to the infinitesimal, together with an exact zero response after RD18. The V_a are the automorphisms generated by the already computed residue. They are not identified with the original finite-order meridian or with a new value of the original heat clock.

4. A complete monic-polynomial identity

Let $h(s)$ be any monic complex polynomial of degree $n \geq 1$, let $E_h = \mathbb{C}[s]/(h)$, and let $\lambda_h(f)$ be the coefficient of s^{n-1} in the remainder of f modulo h . The bilinear form $\lambda_h(fg)$ is perfect: its matrix in $1, s, \dots, s^{n-1}$ is zero when the row and column indices sum to less than $n-1$, and is one when they sum to $n-1$. Reversing the columns gives a triangular matrix with diagonal entries one. This proves invertibility without any hypothesis on repeated roots.

The exact identity for every such polynomial is

$$\text{Tr}_{E_h}(m_f) = \lambda_h(h'f). \quad (\text{RD21})$$

Here is a proof including the repeated-root case. For a polynomial with distinct roots α_j , Lagrange interpolation of the remainder r gives

$$\lambda_h(r) = \sum_j \frac{r(\alpha_j)}{h'(\alpha_j)}.$$

Apply this to the remainder of $h'f$; the result is $\sum_j f(\alpha_j)$. Evaluation is an algebra isomorphism to $\mathbb{C}^n$, so this is also the regular trace. For each of the basis choices $f = 1, s, \dots, s^{n-1}$, both sides of RD21 are polynomial expressions in the n coefficients of the monic h : the multiplication matrices and all remainders are obtained by monic polynomial division. Their difference vanishes when the discriminant is nonzero. A polynomial vanishing on that nonempty open subset of $\mathbb{C}^n$ is identically zero, for instance by the identity theorem successively in each variable. Hence RD21 holds for every monic h , including all multiplicities. Linearity then proves it for all $f \in E_h$.

Suppose now that n is even and $h^\# = h$, where

$$f^\#(s) = \overline{f(1 - \overline{s})}. \quad (\text{RD22})$$

This involution preserves (h) . A representative of degree less than n remains of degree less than n after reflection, and its top coefficient changes by $(-1)^{n-1} = -1$ and conjugation. Thus $\lambda_h(f^\#) = -\overline{\lambda_h(f)}$. It follows exactly as in RD17 that

$$Q_h(f, g) = i\lambda_h(f^\#g) \quad (\text{RD23})$$

is a nondegenerate Hermitian form. Differentiating $h^\# = h$ yields $(h')^\# = -h'$, so

$$K_h = -iM_{h'}, \quad Q_h(f, K_h g) = \text{Tr}(m_{f^\# g}). \quad (\text{RD24})$$

The multiplier $-ih'$ is fixed by $\#$, proving its self-adjointness for Q_h . The factor h' , rather than a freely chosen positive metric, is forced by the trace identity RD21.

5. The actual amplitude and the cotangent group of the Weil packet

Use the programme's $g = 2\xi$, the original s -coordinate, and its finite involution-stable divisor $h = d^m$, where d has four distinct roots. The full quotient $v_h = g/h$ is entire and nonzero at those roots, because m is their exact multiplicity. Let

$$u = j_h(v_h) \in E_h, \quad U = M_u. \quad (\text{RD25})$$

The remainder j_h is defined by simultaneous Taylor interpolation through order $m - 1$ at each of the four roots. The Chinese remainder theorem constructs it uniquely modulo h , and its kernel is the entire functions divisible by h . Since each constant Taylor coefficient of v_h is nonzero, its inverse in each truncated Taylor algebra is the finite geometric-series inverse; therefore u and U are invertible. All derivatives of v_h through the retained orders occur in U . None is replaced by its constant value in RD25.

The original packet pairing WP28 equals

$$B_h(f, g) = \text{Tr}_{E_h}(m_{(uf)^\#(ug)}) = \lambda_h(h' u^\# u f^\# g) = Q_h(Uf, K_h U g). \quad (\text{RD26})$$

To prove the first identity, in the primary factor at α a multiplication matrix is triangular with diagonal its constant coefficient repeated m times. That coefficient here is $\overline{v_h(1 - \bar{\alpha})} f(1 - \bar{\alpha}) v_h(\alpha) g(\alpha)$. Summing over the four factors gives precisely WP28. The next two equalities are RD21 and RD24. In particular all the amplitude factors, the factor m , and the sign in the reflected jets are retained.

The complete cotangent complex of this hypersurface over $\mathbb{C}$ has the presentation

$$L_{E_h/\mathbb{C}} = [E_h \xrightarrow{h'} E_h ds], \quad \text{in degrees } -1, 0. \quad (\text{RD27})$$

One construction is the free differential graded resolution with degree-zero generator s and degree-minus-one exterior generator η , differential $d\eta = h(s)$. Its only underlying two terms are multiplication by h on $\mathbb{C}[s]$; this is injective and has cokernel E_h , so it resolves E_h . Taking its differentials and tensoring with E_h gives $d\eta \mapsto h'ds$, proving RD27. Thus its degree-minus-one cohomology is the explicitly included submodule $\text{Ann}_{E_h}(h')$ of E_h .

The precise comparison is

$$\text{rad } B_h = \ker K_h = \text{Ann}_{E_h}(h') = \frac{(d)}{(d^m)} = H^{-1}(L_{E_h/\mathbb{C}}). \quad (\text{RD28})$$

For the radical equality, RD26 and perfection of λ_h show that the radical in the second argument is $\text{Ann}(h' u^\# u)$. The factor $u^\# u$ is a unit, so this is $\text{Ann}(h')$. Hermitian symmetry gives the same radical in the first argument. Since $h' = md^{m-1}d'$ and d is squarefree in characteristic zero, d' is relatively prime to d ; cancellation in $\mathbb{C}[s]$ proves that d^m divides $h'f$ exactly when d divides f . This proves the middle ideal equality. At $m = 1$ the displayed ideal is zero, as it should be.

Every map in RD28 is specified: a cohomology class represented by $a\eta$ maps to $a \in E_h$; the radical inclusion is this same map. The trace-to-dual map is $\Theta_h \circ M_{h'}$, where $\Theta_h(a)(f) = \lambda_h(af)$. Multiplication by $u^\# u$ gives the amplitude-weighted version, with the same kernel. Its image has dimension four, while its kernel and cokernel both have dimension $4m - 4$. These dimensions follow from RD28 and the perfect $4m$ -dimensional duality, without discarding the kernel.

More precisely $\text{im } K_h = (d^{m-1})/(d^m)$, because md' is a unit modulo d^m . For $m \geq 2$, $2m - 2 \geq m$ shows $K_h^2 = 0$. For $m = 1$, K_h is invertible. In the off-critical quartet, each reflected pair of primary summands is isotropic for Q_h , and they pair perfectly with each other. A dual basis for that pairing gives a Hermitian block $\begin{pmatrix} 0 & I_m \\ I_m & 0 \end{pmatrix}$, with inertia $(m, m, 0)$. The two reflected pairs therefore give inertia $(2m, 2m, 0)$ for the full residue form. This retains the precise difference between the perfect form and its rank-four trace image.

This identifies the surviving nilpotent packet directions with a cotangent cohomology group through the trace's actual Jacobian. It does not identify the whole collision algebra C_0 with an arbitrary actual zeta packet: their specified presentations and multiplicities remain part of the comparison.

The other cotangent cohomology group is also retained:

$$H^0(L_{E_h/\mathbb{C}}) = (E_h/(h')) ds, \quad H^{-1}(L_{E_h/\mathbb{C}}) \times H^0(L_{E_h/\mathbb{C}}) \longrightarrow \mathbb{C}, \quad (a\eta, [b ds]) \longmapsto \lambda_h(ab). \quad (\text{RD28a})$$

This pairing is well defined because $ah' = 0$. It is perfect: multiplication by h' is self-adjoint for the perfect bilinear form $\lambda_h(ab)$, by commutativity. Therefore the orthogonal complement of its image is its kernel. Identifying E_h with its dual by this form identifies that kernel with the dual of its cokernel. This proves both nondegeneracy assertions and supplies the exact duality map. For $m > 1$ these two groups have dimension $4m - 4$; for $m = 1$ both are zero. Thus the radical invisible to the regular trace is still detected by an explicit pairing with the cotangent cokernel.

6. The supported-zero formula keeps its endpoint terms

Retain the finite support semilattice L , its top 1_L , and its separate coordinate vectors $\mathbf{e}_\lambda$. For an admissible test with Mellin transform $A(s)$, write $P(A)$ and $A_\infty(A)$ for exactly the convergent finite-prime and archimedean terms of SZW24, or their proved extension WA24. The full supported formula is

$$\begin{aligned} B_L(A) &= (A(0) + A(1))\mathbf{e}_{1_L} + A(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda, \\ Z_L(A) &= Z(A)\mathbf{e}_{1_L}, \\ D_L(A) &= (P(A) - A_\infty(A))\mathbf{e}_{1_L} + A(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda, \\ B_L(A) - Z_L(A) &= D_L(A). \end{aligned} \quad (\text{RD29})$$

For the entire polynomial packet tests, take $A_{f,g}(s) = f^\#(s)g(s)v_h(s)^2$, with polynomial representatives f, g . The functional equation gives $v_h^\# = v_h$, and WA24 proves the full convergent analytic identity for these tests. All zeros outside the chosen packet contribute zero because v_h vanishes there. Therefore

$$Z_L(A_{f,g}) = Q_h(U[f], K_h U[g]) \mathbf{e}_{1_L}. \quad (\text{RD30})$$

The endpoint values in RD29 still use the actual entire function, not merely its jet class.

Here is their exact behavior on a change in an invisible jet direction. For a polynomial b , replacing g by $g + db$ changes its packet class by $(d)/(d^m)$, and changes A by $\Delta A = f^\# db v_h^2$. Every packet value of ΔA is zero, and the other zeros still give zero. Thus $\Delta Z = 0$. Subtracting the two complete analytic identities gives

$$\Delta(P - A_\infty) = \Delta A(0) + \Delta A(1), \quad \Delta B_L = \Delta D_L. \quad (\text{RD31})$$

The lower-label coordinates on both sides are exactly $\Delta A(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda$. This proves the compensation with every supported-zero term retained. It would be incorrect to infer that the endpoints individually vanish because the packet trace does.

There is an explicit nonzero endpoint witness. In the original quartet write $\rho = 1/2 + \delta + i\gamma$ and put

$$D_0 = d(0) = d(1) = |\rho(1 - \rho)|^2 = (1/4 - \delta^2 + \gamma^2)^2 + 4\delta^2\gamma^2 > 0. \quad (\text{RD31a})$$

The product formula for d proves the equalities. Since $2\xi(0) = 2\xi(1) = 1$, one has $v_h(0) = v_h(1) = D_0^{-m}$. Taking $f = 1, b = 1$ in the preceding perturbation gives

$$\Delta A(0) = \Delta A(1) = D_0^{1-2m}, \quad \Delta(P - A_\infty) = 2D_0^{1-2m}, \quad \Delta Z = 0. \quad (\text{RD31b})$$

Every lower support coordinate changes by the same positive number D_0^{1-2m} . This is an exact endpoint calculation on the programme packet construction; it does not assert existence of an off-critical zero. At $m = 1$ even the finite packet class is unchanged, while these actual entire-test endpoint changes remain nonzero.

For completeness supported zero e remains the identity of $\{\tau, e\}$, with $e^2 = e$; the support projection on an S -semimodule is $p_M(m) = em$, with $p_M i_M = \text{id}_{eM}$. On the split extension of any algebra above, a vanishing linear trace has value e , while the unsupported argument has value τ . The linear maps in RD9 lift by $f^\bullet \mapsto \text{tr}(f)^\bullet$, $\tau \mapsto \tau$. The Hermitian pairings in RD18 and RD26 lift as sesquilinear maps: two supported inputs give the bullet of their pairing, and an unsupported input gives τ . No assertion of complex linearity of a Hermitian first argument is intended. These formulas show precisely how a nonzero algebra element can have supported trace zero without being identified with the unsupported element.

The sign calculation remains exact. In an off-critical quartet the value form is two blocks $m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, so it has two positive and two negative directions, plus the $4m - 4$ radical directions in RD28. The full endpoint expression equals that same value form by RD29–RD30. This analysis does not establish the existence of an off-critical quartet. It proves that modifying only its radical cannot turn its negative value directions positive: for r in RD28, $B_h(f + r, f + r) = B_h(f, f)$. A proof of the desired arithmetic positivity must therefore determine the sign on the retained value quotient; the nonzero infinitesimal and its perfect residue pairing have now been mapped to that exact question.

7. Reproducible verification and figure

The accompanying exact symbolic checker verifies the integral duality matrices, Jacobians, trace maps, quotient dual map, dual Lie weights, and both complex Hermitian defects. The proofs above establish the identities over their stated coefficient rings and for all monic polynomials; finite checks are additional verification. The diagram displays RD18–RD19 and RD26–RD31 with the full cotangent kernel retained.

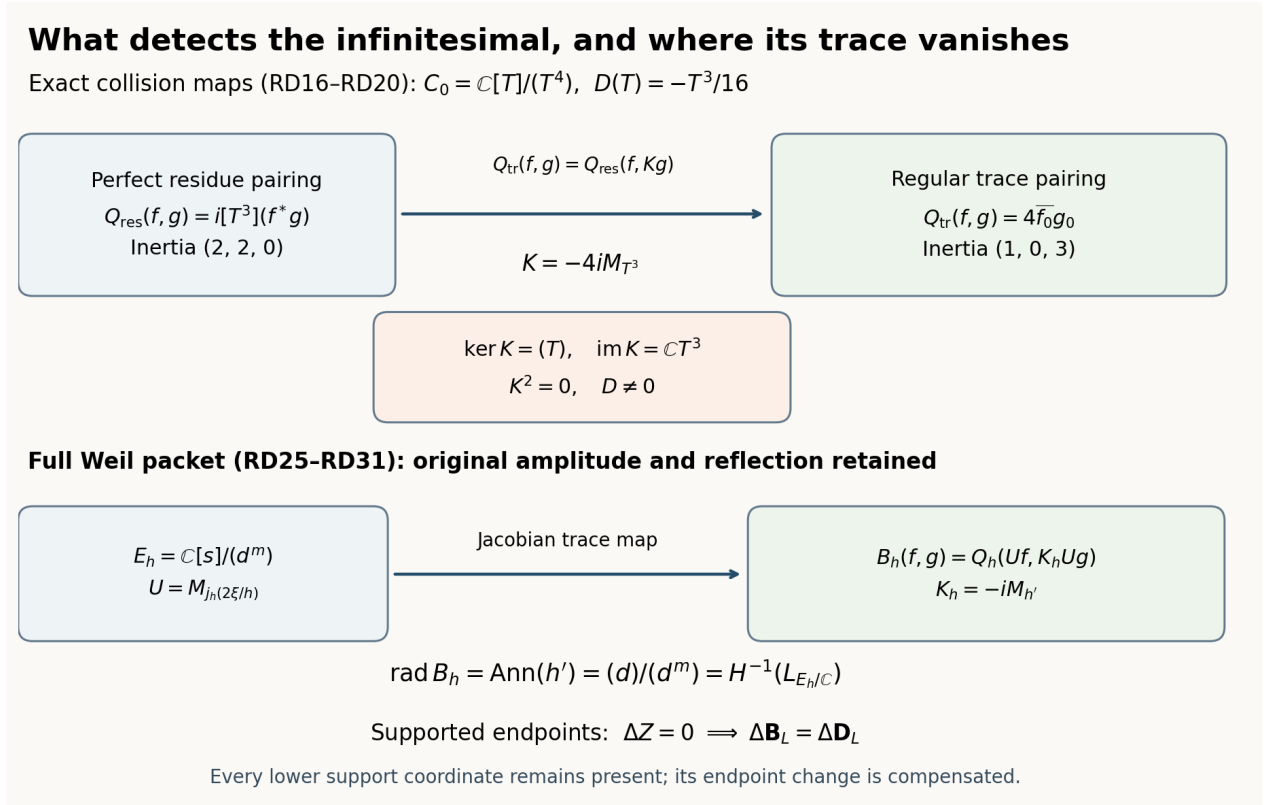


Figure 13: The Jacobian map from perfect residue duality to trace, and the retained supported endpoint terms.

Figure: the upper row is the complex collision, with $K = -4iM_{T^3}$, kernel (T) , and image $\mathbb{C}T^3$. The lower row is the actual amplitude-weighted packet map, whose radical is RD28. The right-hand equations retain the supported-zero boundary from RD29. This is a diagram of proved linear maps and forms; it is not a plot of actual off-critical zeta zeros.

Integral prism continuation

DP1–DP43 constructs the specific bounded prism and proves the original nilradical specialization. EC1–EC48 computes its complete extension class and the exact maps detecting the divided residue. SG1–SG23 gives the supported prime pullbacks, retaining the unsupported point and the original supported-zero prime under coefficient contraction. These calculations retain all earlier trace and endpoint identities.

Actual global heat continuation

The global heat trace retains all zero multiplicities. The arithmetic calculation, HA1–HA26, evaluates the complete supported formula on Cauchy tests and computes its actual derivative. The local response keeps the

complete analytic unit; the global response includes mixed prime-product terms. The matrix criterion gives the exact relation of this test family to RH. No sign from an isolated local coefficient replaces the complete pairing.

The integral extension class, its connecting maps, and the divided residue

Independent bounded derivation, 23 September 2026.

This proof continues NILRADICAL_COTANGENT_DIVIDED_RESIDUE_DERIVATION.md, NC21–NC43. It computes the actual extension class over both $O = \mathbb{Z}_3$ and the original algebra C , gives its connecting maps, and relates its additional C -module data to the original divided residue. The earlier source is unchanged.

EC1. The retained extension and scalar actions

Put

$$C = O[\epsilon, T]/(\epsilon^2, T^2 - 6\epsilon), \quad R = C/(3, \epsilon) = \mathbb{F}_3[T]/T^2. \quad (\text{EC1})$$

The original modules, here denoted N, E , are

$$N = (O/3O)a \oplus (O/9O)b, \quad E = (O/9O)c_1 \oplus (O/9O)c_2 \oplus (O/3O)w. \quad (\text{EC2})$$

Their generator actions are

$$\epsilon a = 0, \quad \epsilon b = -3b, \quad Ta = -3b, \quad Tb = 0; \quad (\text{EC3})$$

$$\epsilon c_1 = \epsilon c_2 = 0, \quad \epsilon w = 3c_2, \quad Tc_1 = c_2, \quad Tc_2 = Tw = 0. \quad (\text{EC4})$$

These actions respect the relations of C . For example $T^2a = T^2b = 0$, while $6\epsilon b = -18b = 0$; on E , T^2 is zero and $6\epsilon w = 18c_2 = 0$. The two generator actions commute on every displayed basis vector, and $\epsilon^2 = 0$.

The exact sequence is

$$0 \longrightarrow N \xrightarrow{\kappa} E \xrightarrow{\psi} R \longrightarrow 0, \quad (\text{EC5})$$

where

$$\kappa(a) = 3c_1, \quad \kappa(b) = w - c_2, \quad \psi(c_1) = 1, \quad \psi(c_2) = \psi(w) = T. \quad (\text{EC6})$$

The action tables prove C -linearity. In particular

$$T\kappa(a) = 3c_2 = \kappa(-3b), \quad \epsilon\kappa(b) = 3c_2 = \kappa(-3b).$$

The c_2 -coordinate of $\kappa(xa + yb) = 0$ first forces $y = 0$ modulo 9, and the c_1 -coordinate then forces $x = 0$ modulo 3. Thus κ is injective. The map ψ is onto, and its kernel consists of c_1 -coefficients divisible by 3 and pairs of c_2, w -coefficients whose sum is zero modulo 3. These are exactly the image of κ .

Write

$$M = N[3] = \mathbb{F}_3a \oplus \mathbb{F}_3(3b), \quad 3N = \mathbb{F}_3(3b). \quad (\text{EC7})$$

On all of N , $\epsilon = -3\text{id}_N$. On M , $\epsilon = 0$, $Ta = -3b$, and $T(3b) = 0$. Consequently

$$R \xrightarrow{\sim} M, \quad 1 \mapsto a, \quad T \mapsto -3b \quad (\text{EC8})$$

is a C -linear isomorphism. This fixes the signs of the later connecting maps.

EC2. The coefficient-3 connecting map

For $r \in R$, choose a lift $h \in E$. Since $3r = 0$, the element $3h$ belongs to $\kappa(N)$. Define

$$\beta : R \longrightarrow N/3N, \quad \beta(r) = [\kappa^{-1}(3h)]. \quad (\text{EC9})$$

If h is changed to $h + \kappa(n_0)$, the element inside brackets changes by $3n_0$. Thus the definition is independent of the lift. Adding lifts and multiplying them by a scalar in C proves that β is C -linear; in particular it is O -linear.

For the chosen lifts c_1, c_2 ,

$$3c_1 = \kappa(a), \quad 3c_2 = \kappa(-3b).$$

Hence

$$\boxed{\beta(1) = \bar{a}, \quad \beta(T) = 0.} \quad (\text{EC10})$$

Here

$$N/3N = \mathbb{F}_3\bar{a} \oplus \mathbb{F}_3\bar{b},$$

and both ϵ, T act as zero: their values on N lie in $3N$. Consequently the formula on T is also required by C -linearity once $\beta(1)$ has been fixed. The exact modules are

$$\ker \beta = \mathbb{F}_3 T, \quad \text{im } \beta = \mathbb{F}_3 \bar{a}, \quad \text{coker } \beta = \mathbb{F}_3 \bar{b}. \quad (\text{EC11})$$

All three are C -modules on which $3, \epsilon, T$ act as zero.

This map is the connecting map in the full coefficient-3 sequence

$$0 \rightarrow N[3] \rightarrow E[3] \rightarrow R \xrightarrow{\beta} N/3N \rightarrow E/3E \rightarrow R \rightarrow 0. \quad (\text{EC12})$$

One can obtain it by applying the kernel-cokernel sequence to multiplication by 3 on EC5. There is also a direct verification of every map. The groups are

$$E[3] = \mathbb{F}_3(3c_1) \oplus \mathbb{F}_3(3c_2) \oplus \mathbb{F}_3 w,$$

$$E/3E = \mathbb{F}_3 \bar{c}_1 \oplus \mathbb{F}_3 \bar{c}_2 \oplus \mathbb{F}_3 \bar{w}.$$

The first injection sends

$$a \mapsto 3c_1, \quad 3b \mapsto -3c_2.$$

The next map sends $w \mapsto T$ and kills $3c_1, 3c_2$, so its image is the kernel of EC10. The map $N/3N \rightarrow E/3E$ sends

$$\bar{a} \mapsto 0, \quad \bar{b} \mapsto \bar{w} - \bar{c}_2.$$

Its kernel is precisely $\text{im } \beta$. Finally $E/3E \rightarrow R$ sends $\bar{c}_1 \mapsto 1, \bar{c}_2, \bar{w} \mapsto T$, and its kernel is the line generated by $\bar{w} - \bar{c}_2$. This proves exactness of EC12 without leaving its maps implicit.

The actions in EC12 are retained. On $E[3]$,

$$T(3c_1) = 3c_2, \quad \epsilon w = 3c_2,$$

and all other actions of ϵ, T on the indicated three generators vanish. In particular ϵ does not annihilate $E[3]$. On $E/3E$, $\epsilon = 0$, $T\bar{c}_1 = \bar{c}_2$, and $T\bar{c}_2 = T\bar{w} = 0$. Each displayed map respects these actions.

EC3. The exact class after forgetting to O

As an O -module, R has the free resolution

$$0 \longrightarrow O^2 \xrightarrow{3} O^2 \longrightarrow R \longrightarrow 0,$$

where the final basis vectors map to $1, T$. Applying $\text{Hom}_O(-, N)$ gives

$$\text{Ext}_O^1(R, N) \simeq (N/3N)^2 \simeq \text{Hom}_{\mathbb{F}_3}(R, N/3N) \simeq \mathbb{F}_3^4. \quad (\text{EC13})$$

Explicitly, an extension is assigned the two values $3h_1, 3h_T$ in N , modulo $3N$, for lifts of $1, T$. Changing the lifts adds multiples of 3. Conversely any pair of representatives $\alpha_1, \alpha_T \in N$ gives the extension

$$(N \oplus Oh_1 \oplus Oh_T)/(3h_1 - \alpha_1, 3h_T - \alpha_T).$$

Injection of N follows by comparing the two free coordinates in a proposed relation. Two pairs differing by multiples of 3 give equivalent extensions by changing their lifts. This proves EC13 as a classification of extensions and identifies its class map with EC9.

Thus the underlying O -class of EC5 is the nonzero homomorphism

$$\varphi(1) = \bar{a}, \quad \varphi(T) = 0. \quad (\text{EC14})$$

It has order 3 in the extension group. The extension is therefore nonsplit over O . Directly, no lift of 1 can be killed by 3: its c_1 -coefficient is a unit modulo 3 and has nonzero triple modulo 9.

The full target in EC13 has dimension four. The forgetful map from C -extensions below does not have this whole group as its image.

EC4. Presentation of the C -extension group

Let $I = (3, \epsilon) \subset C$, so $R = C/I$. The map

$$C^2 \longrightarrow I, \quad (u, v) \longmapsto 3u + \epsilon v$$

has kernel generated by

$$(\epsilon, -3), \quad (0, \epsilon). \quad (\text{EC15})$$

To prove completeness of these syzygies, write $u = u_0 + u_1\epsilon + u_2T + u_3\epsilon T$ and likewise for v . Since C is O -free on $1, \epsilon, T, \epsilon T$, the equation $3u + \epsilon v = 0$ gives

$$u_0 = u_2 = 0, \quad v_0 = -3u_1, \quad v_2 = -3u_3.$$

It follows that

$$(u, v) = (u_1 + u_3T)(\epsilon, -3) + (v_1 + v_3T)(0, \epsilon).$$

This proves EC15 without dividing by 3 in C .

Because C is a free module over itself, the sequence $0 \rightarrow I \rightarrow C \rightarrow R \rightarrow 0$ yields

$$\text{Ext}_C^1(R, N) \simeq \frac{\text{Hom}_C(I, N)}{\{i \mapsto in_0 : n_0 \in N\}}.$$

A homomorphism $F : I \rightarrow N$ is determined by $n = F(3)$, $m = F(\epsilon)$. The syzygies give

$$\epsilon n = 3m, \quad \epsilon m = 0. \quad (\text{EC16})$$

Write $n = ua + yb$, $m = u'a + zb$. Since $\epsilon b = -3b$, the second relation forces $z \in 3O/9O$, and the first then forces $y \in 3O/9O$. Thus both n, m lie in $M = N[3]$, and every pair in M^2 satisfies EC16. The restrictions from C are

$$(3n_0, \epsilon n_0) = (3n_0, -3n_0).$$

They form the one-dimensional subspace generated by $(3b, -3b)$. Consequently

$$\boxed{\text{Ext}_C^1(R, N) = \frac{M \oplus M}{\mathbb{F}_3(3b, -3b)} \simeq \mathbb{F}_3^3.} \quad (\text{EC17})$$

Here is the exact extension associated to a pair (n, m) . Use the homomorphism F just proved and form

$$E_{n,m} = (N \oplus Cu) / \{(-F(i), iu) : i \in I\}. \quad (\text{EC18})$$

Then $N \rightarrow E_{n,m}$ is injective: if $(z, 0) = (-F(i), iu)$, the free Cu -coordinate forces $i = 0$, hence $z = 0$. The quotient is $C/I = R$. The chosen lift u of 1 satisfies

$$3u = n, \quad \epsilon u = m. \quad (\text{EC19})$$

Conversely a lift u in any C -extension defines $F(i) = iu$ and gives an isomorphism from EC18 to that extension, fixing both N and R . Changing u by $n_0 \in N$ changes its pair by $(3n_0, \epsilon n_0)$. This proves the full equivalence relation, and also shows that addition of pairs is the usual addition of extension classes.

For EC5 the lift $u = c_1$ gives

$$(n, m) = (a, 0). \quad (\text{EC20})$$

Thus the actual class has been specified in the extension group, not only tested for nonsplitting.

EC5. The full C -action and the invisible nonzero class

Set

$$E_1 = [(a, 0)], \quad E_2 = [(0, a)], \quad E_3 = [(3b, 0)] = [(0, 3b)]. \quad (\text{EC21})$$

The last equality is exactly the boundary in EC17. These form an $\mathbb{F}_3$ -basis. The C -action, defined by multiplication on the target N , is

$$TE_1 = TE_2 = -E_3, \quad TE_3 = 0, \quad 3E_j = \epsilon E_j = 0. \quad (\text{EC22})$$

For example $T(a, 0) = (-3b, 0)$, and $T(0, a) = (0, -3b)$. These prove every action in EC22.

Under forgetting to O , the two underlying lifts are u, Tu , so the coefficient-3 class is

$$1 \mapsto [n], \quad T \mapsto [Tn] \quad \text{in } N/3N. \quad (\text{EC23})$$

For

$$n = ua + 3vb, \quad m = u'a + 3v'b \quad (u, v, u', v' \in \mathbb{F}_3),$$

one has $Tn = -3ub$, which lies in $3N$. Therefore forgetting sends

$$E_1 \mapsto \varphi, \quad E_2, E_3 \mapsto 0. \quad (\text{EC24})$$

Its exact image is the one-dimensional line $\mathbb{F}_3\varphi$ inside the four-dimensional group EC13; its kernel is $\mathbb{F}_3E_2 \oplus \mathbb{F}_3E_3$, isomorphic to R by

$$1 \mapsto E_2, \quad T \mapsto -E_3.$$

The target C -action here is the one induced by multiplication on N . On $N/3N$, both ϵ, T act as zero, so the target in EC13 is a trivial R -module. This specifies the action and proves C -linearity of the forgetful map. Precomposition by multiplication on R gives a separate action on the full group EC13; the two actions agree on the image of C -extensions.

The exact kernel-image sequence of the forgetful map splits as R -modules by

$$\varphi \mapsto E_1 - E_2,$$

because $T(E_1 - E_2) = 0$. Thus

$$\text{Ext}_C^1(R, N) = RE_2 \oplus \mathbb{F}_3(E_1 - E_2) \simeq R \oplus \mathbb{F}_3. \quad (\text{EC25})$$

This is a splitting of a sequence of extension groups. It is not a splitting of EC5.

The actual class is E_1 , and

$$TE_1 = -E_3 \neq 0. \quad (\text{EC26})$$

The underlying O -class of TE_1 is zero by EC24. The annihilator of E_1 in C is exactly $(3, \epsilon)$: the action factors through R , and E_1, TE_1 are independent over $\mathbb{F}_3$. Hence the cyclic module CE_1 is isomorphic to R . The nonzero T -layer of this class is invisible after forgetting C .

There is a concrete presentation of this invisible class. For $-E_3$, take

$$3u = -3b, \quad \epsilon u = 0.$$

Changing the lift to $u' = u + b$ gives

$$3u' = 0, \quad \epsilon u' = -3b. \quad (\text{EC27})$$

Then u', Tu' define an O -linear section of the quotient, since both are killed by 3. A C -linear section would need a lift of 1 killed by both 3 and ϵ . Any other lift killed by 3 differs from u' by an element of $M = N[3]$, and $\epsilon M = 0$; its ϵ -value is therefore still $-3b \neq 0$. This proves directly that the same extension is split over O and nonsplit over C .

EC6. A canonical second connecting map

The scalar $3 + \epsilon$ annihilates both N and R . Consequently every extension $0 \rightarrow N \xrightarrow{\kappa'} E' \xrightarrow{\psi'} R \rightarrow 0$ has a canonical C -linear map

$$\rho_{E'} : R \longrightarrow N, \quad \kappa' \rho_{E'}(r) = (3 + \epsilon)h \quad (\psi' h = r). \quad (\text{EC28})$$

The right side lies in the image of κ' , since $3 + \epsilon$ annihilates R . It is independent of h , since the same scalar annihilates N . These facts prove existence, uniqueness, and C -linearity. Its image lies in M , since R is killed by 3. For the cocycle of EC19,

$$\rho_{E_{n,m}}(1) = n + m, \quad \rho_{E_{n,m}}(T) = T(n + m). \quad (\text{EC29})$$

The two class data give the exact C -linear isomorphism

$$\boxed{\text{Ext}_C^1(R, N) \xrightarrow{\sim} (M/3N) \oplus M, \quad [(n, m)] \mapsto ([n], n + m).} \quad (\text{EC30})$$

The boundary $(3n_0, -3n_0)$ changes neither coordinate. If both coordinates vanish, $n \in 3N$ and $m = -n$, so the pair is precisely a boundary. Given $([\alpha], z)$, choose $\alpha \in M$ and take $(n, m) = (\alpha, z - \alpha)$; changing α by

an element of $3N$ changes this pair by a boundary. This proves surjectivity and gives a well-defined inverse. Componentwise scalar multiplication proves C -linearity.

The first coordinate of EC30 is the entire possible underlying O -class, interpreted by $1 \mapsto [n], T \mapsto 0$. The second coordinate specifies the whole map ρ , since $\text{Hom}_C(R, N) \simeq M$ by evaluation at 1. Thus this is a classification by actual connecting maps.

For EC5,

$$\rho(1) = a, \quad \rho(T) = -3b. \quad (\text{EC31})$$

In particular $\rho : R \rightarrow M$ is the isomorphism EC8, and its class coordinates in EC30 are $(\bar{a}, a)$. The class $E_1 - E_2$ has the same underlying O -class but $\rho = 0$. Their difference E_2 is nonzero and invisible to forgetting. There are exactly nine C -extension classes over each O -class in the image, because the second summand M has nine elements.

For the actual extension, β is the reduction of ρ modulo $3N$:

$$\beta = \pi\rho, \quad \pi : M \hookrightarrow N \rightarrow N/3N. \quad (\text{EC32})$$

This follows from EC10 and EC31. It is a statement about this actual class; for a general pair (n, m) , the reduction of $\rho(1)$ is $[n + m]$, whereas its coefficient-3 connecting value is $[n]$.

EC7. Divided residues as exact maps on this extension

Retain the original operators from NC36–NC40:

$$A_N = -\frac{\epsilon}{8}\Big|_N, \quad A_H = -\frac{5\epsilon}{8}\Big|_E. \quad (\text{EC33})$$

Their complete basis formulas are

$$\begin{aligned} A_N(a) &= 0, & A_N(b) &= \frac{3}{8}b, \\ A_H(c_1) &= A_H(c_2) = 0, & A_H(w) &= -\frac{15}{8}c_2. \end{aligned} \quad (\text{EC34})$$

Since 8 is a unit in O , these are integral endomorphisms of the retained torsion modules. They are the maps already obtained by division on the free nilradical and cotangent lattices before passing to the quotient. No operation of dividing an arbitrary torsion-module map is being used.

The equality of maps is

$$A_H\kappa = 5\kappa A_N, \quad \psi A_H = 0. \quad (\text{EC35})$$

It follows directly from scalar multiplication and C -linearity of κ, ψ , or by substitution in EC34. Their exact kernels and images are

$$\begin{aligned} \ker A_N &= M, & \text{im } A_N &= 3N; \\ \ker A_H &= (O/9O)c_1 \oplus (O/9O)c_2, & \text{im } A_H &= \mathbb{F}_3(3c_2) = \kappa(3N). \end{aligned} \quad (\text{EC36})$$

Each image has order 3. Both squares and both triples are zero. Their cokernels are

$$\begin{aligned} \text{coker } A_N &= (O/3O)a \oplus (O/3O)b, \\ \text{coker } A_H &= (O/9O)c_1 \oplus (O/3O)c_2 \oplus (O/3O)w. \end{aligned}$$

These follow by quotienting by the two explicitly given image subgroups.

The image has an exact description using the second connecting map:

$$\text{im } A_N = \rho(TR), \quad \text{im } A_H = \kappa\rho(TR). \quad (\text{EC37})$$

Indeed $\rho(T) = -3b$ generates $3N$. The first connecting map β kills TR , by EC11. Thus the additional T -layer in ρ , which is invisible to the coefficient-3 class, gives precisely the order-three image observed by the divided residue.

The factor 5 is retained in EC35. On these order-three images, multiplication by 5 equals multiplication by -1 , since $5 - (-1) = 6$ annihilates them. Therefore EC35 also implies $A_H\kappa = -\kappa A_N$ on these torsion modules. This is an identity in characteristic 3 on the specified finite images. It is not a statement about an order, a sign of a real quadratic form, or real positivity.

EC8. Factoring the divided residue through both connecting maps

Define the C -linear map

$$\theta : E \longrightarrow N, \quad \kappa\theta(h) = 3h. \quad (\text{EC38})$$

Its values and its restriction to the submodule are

$$\theta(c_1) = a, \quad \theta(c_2) = -3b, \quad \theta(w) = 0, \quad \theta\kappa = 3\text{id}_N. \quad (\text{EC39})$$

It exists because $3R = 0$. Its image is M , and its kernel is $E[3]$, either by these formulas or by injectivity of κ . Reduction of θ modulo $3N$ is $\beta\psi$, which is precisely the lift-independent construction EC9.

The definitions of θ, ρ give the identity in the original middle module

$$\epsilon \text{id}_E = \kappa(\rho\psi - \theta). \quad (\text{EC40})$$

Both sides evaluated on h equal $(3 + \epsilon)h - 3h$. Thus

$$F = \frac{\theta - \rho\psi}{8} : E \longrightarrow N$$

is an integral C -linear map with

$$F(c_1) = F(c_2) = 0, \quad F(w) = \frac{3}{8}b.$$

Moreover

$$F\kappa = A_N, \quad A_H = 5\kappa F. \quad (\text{EC41})$$

The first equality follows from $\psi\kappa = 0$, $\theta\kappa = 3$, and $A_N = 3/8$ on N , because $\epsilon = -3$ there. The second follows from EC40. This proves an exact relation among the two connecting maps, the divided nilradical operator, and the divided cotangent operator. It also explains the original factor 5 through actual maps.

The pushout of the actual extension by $A_N : N \rightarrow N$ has zero extension class. This follows algebraically because ϵ kills the extension group EC22, and it also has a concrete split model. Represent the pushout as

$$E_{A_N} = (N \oplus E) / \{(A_N n, -\kappa n) : n \in N\}.$$

For $r \in R$, choose h above it and set

$$s(r) = [-F(h), h]. \quad (\text{EC42})$$

Changing h by κn changes this pair by $(-A_N n, \kappa n)$, which is zero in the pushout. Thus s is well defined and C -linear, and its composite with the quotient to R is the identity. This is a proved splitting of a different, specified pushout extension; EC5 itself remains nonsplit.

The map $(3 + \epsilon)\text{id}_E = \kappa\rho\psi$ has kernel exactly $\kappa(N)$, because ρ is injective, and image $\kappa(M)$. Its image has order 9. This supplies another direct check that ρ records actual middle-module data.

EC9. The complete family over the same O -class

Write a general cocycle as

$$n = ua + 3vb, \quad m = u'a + 3v'b.$$

Changing the lift by a multiple of b changes (v, v') by $(t, -t)$. Thus the three independent coordinates are

$$(u, u', s), \quad s = v + v' \in \mathbb{F}_3. \quad (\text{EC43})$$

Every class has a representative

$$3z = ua, \quad \epsilon z = u'a + 3sb. \quad (\text{EC44})$$

This follows by replacing the earlier lift by $z - vb$. The connecting maps are

$$\beta(1) = u\bar{a}, \quad \beta(T) = 0, \quad \rho(1) = (u + u')a + 3sb. \quad (\text{EC45})$$

These are the exact coordinates of EC30. In particular the actual class is $(1, 0, 0)$.

All middle modules have order 3^5 , since $|N| = 3^3$ and $|R| = 3^2$. Their underlying O -module types are

$$E_{n,m} \simeq \begin{cases} (O/9O)^2 \oplus O/3O, & u \neq 0, \\ O/9O \oplus (O/3O)^3, & u = 0. \end{cases} \quad (\text{EC46})$$

Here is a direct proof. Put $z_0 = z$ for the representative in EC44, and $z_1 = Tz_0 + ub$. Then

$$3z_0 = ua, \quad 3z_1 = T(ua) + 3ub = 0.$$

The lifts z_0, z_1 project to the basis $1, T$ of R , so together with N they generate the whole underlying O -module. When $u = 0$, they each have order 3 and give an O -linear section, yielding the second type. When $u \neq 0$, z_0 has order 9 and includes a in its generated subgroup; the elements z_0, b, z_1 have respective orders 9, 9, 3. The homomorphism from the direct sum of those cyclic groups to the middle module is onto. Both sides have order 3^5 , hence it is an isomorphism. This proves the first type and shows that u', s do not affect it.

On every such C -extension define the specified operator

$$L_{n,m} = -\frac{5\epsilon}{8}. \quad (\text{EC47})$$

It restricts to $5A_N$ on N , induces zero on R , and has square zero. The subgroup $\epsilon E_{n,m}$ is generated over $\mathbb{F}_3$ by $3b$ and m : ϵN is generated by $3b$, $\epsilon z = m$, and $\epsilon Tz = Tm$ is again a multiple of $3b$. There are no further generators because N, z, Tz generate the underlying module. Hence

$$\dim_{\mathbb{F}_3} \text{im } L_{n,m} = \begin{cases} 1, & u' = 0, \\ 2, & u' \neq 0. \end{cases} \quad (\text{EC48})$$

Multiplication by $-5/8$ is a unit, so it does not change this image. The image is always killed by 3, so the unscaled operator $3L_{n,m}$ is zero in every middle module.

For the actual class E_1 , one has $u' = 0$, and EC48 gives the original order-three image. The class $E_1 - E_2$ has the same underlying O -class φ but $u' = -1$, so its operator has image of order 9. These are two explicit extensions with the same underlying O -extension data and different residue images. They show exactly what the additional C -class can change.

EC10. The resulting maps

The original extension is the class $E_1 = [(a, 0)]$ in the fully computed group

$$\text{Ext}_C^1(R, N) \simeq (M/3N) \oplus M \simeq \mathbb{F}_3 \oplus R.$$

Its first coordinate is the coefficient-3 connecting map β ; its second is the isomorphism $\rho : R \rightarrow N[3]$, with values in EC31. Forgetting to O keeps only the first coordinate and lands in the one-dimensional subspace $\mathbb{F}_3\varphi$ of the four-dimensional group EC13.

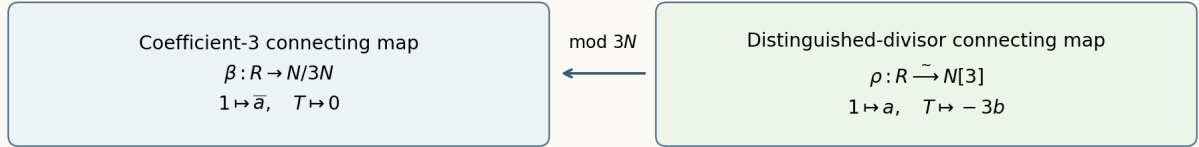
The actual class has the nonzero invisible multiple $TE_1 = -E_3$. The actual divided residue images are $\rho(TR)$ and $\kappa\rho(TR)$, while $\beta(TR) = 0$. Equations EC40–EC42 give their full map-level relation and a concrete splitting of the divided-operator pushout. The numerical factor 5 remains exact; its interpretation as -1 is confined to the stated characteristic-three images.

Exact connecting maps in the figure

Two connecting maps detect different parts of the same extension

$$0 \longrightarrow N \xrightarrow{\kappa} E \xrightarrow{\psi} R \longrightarrow 0, \quad R = \mathbb{F}_3[T]/T^2$$

$$N = (O/3)a \oplus (O/9)b, \quad E = (O/9)c_1 \oplus (O/9)c_2 \oplus (O/3)w$$



$$\beta(TR) = 0, \quad \rho(TR) = 3N = \text{im } A_N \neq 0$$

$$\text{Ext}_C^1(R, N) \cong (N[3]/3N) \oplus N[3] \cong \mathbb{F}_3 \oplus R$$

$$[(n, m)] \mapsto ([n], n + m), \quad E \mapsto (\bar{a}, a)$$

$$\kappa\theta = 3 \text{ id}_E, \quad \kappa\rho\psi = (3 + \varepsilon) \text{ id}_E$$

$$A_H = 5\kappa \frac{\theta - \rho\psi}{8}, \quad A_H\kappa = 5\kappa A_N$$

Full proofs: EC9–EC41. The factor 5 is retained; these finite modules carry no real order.

Figure 14: The two connecting maps, their extension coordinates, and their exact divided-residue factorization. Full proofs EC9–EC41.

The distinguished collision prism and its actual module specializations

Independent derivation, 23 September 2026.

The element $3 + \epsilon$ already occurring in the radical-cotangent extension defines a bounded prism on the retained four-state integral collision algebra. This proof constructs that prism, its exact map from the original unsplit algebra, and the specializations of the original nilradical and cotangent modules. It retains the given Frobenius parameter and original residue coefficients.

The source definitions are Bhargav Bhatt and Peter Scholze, *Prisms and Prismatic Cohomology*, arXiv:1905.08229v4, original author source, labels DefPrismCat, FrobLiftTheta, distinguishedDivide, prismcrit, and PrismMapTaut. The local original-source reading record is given in DP11. Every application to the programme algebras is proved below.

DP1. Original algebras and the retained Frobenius family

Let $O = \mathbb{Z}_3$, fix $c \in O$, and retain

$$B = O[x, T]/(x^2(x+3), T^2 - 3x(x+2)), \quad C = O[\epsilon, T]/(\epsilon^2, T^2 - 6\epsilon). \quad (\text{DP1})$$

Successive monic division gives the free O -bases

$$1, x, x^2, T, xT, x^2T, \quad 1, \epsilon, T, \epsilon T. \quad (\text{DP2})$$

The quotient $q : B \rightarrow C$, $x \mapsto \epsilon, T \mapsto T$, is onto and has kernel

$$J = (x^2) = Ox^2 \oplus Ox^2T. \quad (\text{DP3})$$

Put $n_1 = x(x+3)$, $n_2 = T(x+3)$, $n_3 = Tx(x+3)$. The retained Frobenius maps fix O and have values

$$\begin{aligned} \phi_B(x) &= 0, & \phi_B(T) &= 3cn_3; \\ \phi_C(\epsilon) &= 0, & \phi_C(T) &= 9c\epsilon T. \end{aligned} \quad (\text{DP4})$$

They are algebra homomorphisms: $n_3^2 = 0$ and $(\epsilon T)^2 = 0$, so the quadratic relations map to zero, and the other relations map to zero because x, ϵ do. Modulo 3, both maps are Frobenius. On $B/3$, $x^3 = T^3 = 0$; on $C/3$, $\epsilon^3 = T^3 = 0$; and constants have the usual characteristic-three Frobenius.

Both rings are 3-torsionfree, so

$$\delta_A(f) = \frac{\phi_A(f) - f^3}{3} \quad (A = B, C) \quad (\text{DP5})$$

is a well-defined integral operation. Expansion gives

$$\begin{aligned} \delta_A(f+g) &= \delta_A(f) + \delta_A(g) - fg(f+g), \\ \delta_A(fg) &= f^3\delta_A(g) + g^3\delta_A(f) + 3\delta_A(f)\delta_A(g), \end{aligned}$$

and $\delta_A(0) = \delta_A(1) = 0$, proving the delta-ring identities directly. In particular

$$\begin{aligned} \delta_B(x) &= x^2, & \delta_B(T) &= cn_3 - x(x+2)T, \\ \delta_C(\epsilon) &= 0, & \delta_C(T) &= (3c-2)\epsilon T. \end{aligned} \quad (\text{DP6})$$

The quotient satisfies $q\phi_B = \phi_Cq$, because $q(n_3) = 3\epsilon T$. Applying DP5 and using that C is 3-torsionfree proves $q\delta_B = \delta_Cq$. The generator formulas DP6 also verify this equality directly.

The parameter is transported exactly: if the target family in DP4 had parameter c' , compatibility on T would require $9(c-c')\epsilon T = 0$. The free basis DP2 then gives $c = c'$. Every $c \in O$ gives the displayed compatible pair; no reduction of that parameter is being substituted.

DP2. The distinguished element and all of its torsion

Retain the two corresponding elements

$$a_B = 3 + x, \quad a_C = 3 + \epsilon. \quad (\text{DP7})$$

They satisfy

$$a_C(3 - \epsilon) = 9. \quad (\text{DP8})$$

If $a_C f = 0$, multiplication by $3 - \epsilon$ gives $9f = 0$, so $f = 0$ by DP2. Thus a_C is a nonzerodivisor.

In B , multiplication of a general coefficient polynomial gives

$$(3 + x)(p_0 + p_1x + p_2x^2) = 3p_0 + (p_0 + 3p_1)x + p_1x^2.$$

It vanishes exactly when $p_0 = p_1 = 0$. The same calculation applies to the T -coefficient polynomial. Therefore

$$\text{Ann}_B(a_B) = J.$$

For every $r \geq 1$,

$$\text{Ann}_B(a_B^r) = J. \quad (\text{DP9})$$

Indeed, if $a_B^r f = 0$, applying q gives $a_C^r q(f) = 0$, so $q(f) = 0$ and $f \in J$. Conversely $a_B J = 0$. Hence the quotient q kills exactly all a_B -power torsion, with no additional elements.

The Frobenius and delta values are

$$\begin{aligned} \phi_B(a_B) &= \phi_C(a_C) = 3, \\ \delta_B(a_B) &= -8 - 9x - 2x^2, \quad \delta_C(a_C) = -8 - 9\epsilon. \end{aligned} \quad (\text{DP10})$$

For B , expand $(3 + x)^3$ and use $x^3 = -3x^2$; for C , use $\epsilon^2 = 0$. Both delta values are units, with exact inverses

$$\begin{aligned} \delta_B(a_B)^{-1} &= -\frac{1}{8} + \frac{9}{64}x + \frac{11}{64}x^2, \\ \delta_C(a_C)^{-1} &= -\frac{1}{8} + \frac{9}{64}\epsilon. \end{aligned} \quad (\text{DP11})$$

Multiplying and reducing by $x^3 = -3x^2$, $x^4 = 9x^2$, or by $\epsilon^2 = 0$, proves these formulas. All denominators are powers of 2 and are units in O .

Thus both elements are distinguished, meaning their delta values are units. The pair $(B, (a_B))$ fails the Cartier-divisor condition, since the nonzero ideal J annihilates a_B . The quotient C removes precisely this obstruction, by DP9.

DP3. Complete verification of the bounded-prism axioms

A prism at 3 is a delta-ring A with a Cartier ideal I , derived $(3, I)$ -complete, and satisfying $3 \in I + \phi(I)A$. It is bounded when A/I has bounded 3-power torsion.

Take $I_C = (a_C)$. It is Cartier by DP8. Since $\phi_C(a_C) = 3$,

$$3 \in \phi_C(I_C)C, \quad (\text{DP12})$$

which proves the required ideal containment.

For completeness, retain the exact ideals

$$(3, a_C) = (3, \epsilon), \quad (3, \epsilon)^r = (3^r, 3^{r-1}\epsilon) \quad (r \geq 1). \quad (\text{DP13})$$

The second formula follows by expanding a product of r generators and using $\epsilon^2 = 0$. These powers are cofinal with the powers of (3) . Since C is finite free over the complete ring O , its ordinary completion for this ideal is C .

There is also a direct derived-completeness calculation. Use the Koszul tower

$$K_r = \text{Kos}_C(3^r, \epsilon^r).$$

For $r \geq 2$, one has $\epsilon^r = 0$, and 3^r is regular. Thus the only cohomology groups of K_r are $C/3^r C$ in degrees 0 and -1 . The transition $K_{r+1} \rightarrow K_r$ multiplies the generator for 3^{r+1} by 3 and the generator for ϵ^{r+1} by ϵ . On

degree-zero cohomology the transition is reduction; on degree-minus-one cohomology it includes multiplication by ϵ . Every two-step transition in the latter tower is therefore zero.

The degree-zero tower is surjective, so its higher inverse limit vanishes and its inverse limit is C . The degree-minus-one tower is pro-zero, so both its inverse limit and higher inverse limit vanish. The inverse-limit cohomology sequence for these bounded complexes consequently gives

$$R\varprojlim_r K_r \simeq C$$

in degree zero. This proves derived $(3, \epsilon)$ -completeness, hence derived $(3, a_C)$ -completeness, with the actual transition maps specified.

The a_C -adic topology itself is also cofinal with the 3-adic topology:

$$a_C^2 = 9 + 6\epsilon \in 3C, \quad 9 \in a_C C. \quad (\text{DP14})$$

Indeed $a_C^{2r} C \subset 3^r C$ and $3^{2r} C \subset a_C^r C$. No inversion of 3 or of a_C is involved.

The quotient is exactly

$$D := C/(a_C) \xrightarrow{\sim} (O/9O)[T]/T^2, \quad \epsilon \mapsto -3, \quad T \mapsto T. \quad (\text{DP15})$$

To verify this, impose $\epsilon = -3$. The relation $\epsilon^2 = 0$ becomes $9 = 0$, and $T^2 - 6\epsilon = 0$ becomes $T^2 + 18 = 0$, which is $T^2 = 0$ modulo 9. These substitutions define inverse maps. The quotient is killed by 9, and its element 1 has additive order 9. Hence its full 3-power torsion has exact exponent 9.

It follows from DP8 and DP12–DP15 that

$$\boxed{(C, (3 + \epsilon)) \text{ is a bounded, oriented prism.}} \quad (\text{DP16})$$

This holds for every retained parameter $c \in O$.

DP4. The exact universal quotient and its prism form

The map q is universal among delta-maps from B for which the image of a_B is regular. Precisely, let $f : B \rightarrow A$ be a delta-ring map and assume multiplication by $f(a_B)$ is injective on A . For $j \in J$,

$$f(a_B)f(j) = f(a_B j) = 0,$$

so $f(j) = 0$. There is therefore a unique ring map $\bar{f} : C = B/J \rightarrow A$ with $\bar{f}q = f$. It is a delta-map: for $b \in B$,

$$\bar{f}(\delta_C(qb)) = \bar{f}(q\delta_B b) = f(\delta_B b) = \delta_A(fb).$$

Surjectivity of q proves the required compatibility for every element of C . This establishes the universal property with its exact domain of targets.

For prism targets there is a stronger formulation. A distinguished element d belonging to the Cartier ideal I of a prism (A, I) generates I , and is therefore regular. Here is the argument with its hypotheses retained. The original prism criterion supplies a faithfully flat delta-localization $A \rightarrow A'$ on which $IA' = (e)$, with $e, 3$ in the Jacobson radical. Write $d = eu$ there. The delta product identity gives

$$\delta(d) = e^3\delta(u) + u^3\delta(e) + 3\delta(e)\delta(u).$$

The left side is a unit, and the first and third terms on the right lie in the Jacobson radical. Consequently $u^3\delta(e)$ is a unit, and hence u is a unit. Thus $IA' = (d)$. Faithful flatness descends the ideal equality $I = (d)$; regularity follows after the same cover, where d is a unit times the regular Cartier generator e .

Apply this to a delta-map $f : B \rightarrow A$ with (A, I) a prism and $f(a_B) \in I$. The element $f(a_B)$ is distinguished because the unit in DP10 maps to $\delta_A(f(a_B))$. It therefore generates I and is regular. The factorization just proved gives a unique prism morphism

$$(C, (a_C)) \longrightarrow (A, I). \quad (\text{DP17})$$

Thus DP16 is the initial prism equipped with such a delta-map from the original B carrying a_B into its prism ideal. This is a property of the specified original element and specified Frobenius family. It does not assert a prism structure with ideal (a_B) on B .

DP5. The actual map to the crystalline prism

The same delta-ring C also supports the bounded crystalline prism $(C, (3))$: multiplication by 3 is injective, C is derived 3-complete by its free O -basis, $3 \in (3)$, and $C/3$ has its 3-torsion killed by 3.

The Frobenius ϕ_C is itself a delta-map. Indeed, in the 3-torsionfree ring C , applying ϕ_C to DP5 and using that it fixes 3 gives

$$3\phi_C(\delta_C f) = \phi_C^2(f) - \phi_C(f)^3 = 3\delta_C(\phi_C f).$$

Cancelling 3 proves the assertion. Since $\phi_C(a_C) = 3$, it defines a prism morphism

$$\phi_C : (C, (3 + \epsilon)) \longrightarrow (C, (3)). \quad (\text{DP18})$$

The extended image of the source ideal is exactly (3) .

The identity ring map is a prism morphism in neither direction. One has $a_C \notin 3C$, since its ϵ -coefficient is 1 in the free basis. Also $3 \notin a_C C$, since 3 has nonzero image in the characteristic-nine quotient DP15.

On prism quotients, DP18 induces the exact map

$$D = (O/9O)[T]/T^2 \longrightarrow C/3 = \mathbb{F}_3[\epsilon, T]/(\epsilon^2, T^2), \quad T \mapsto 0, \quad \bar{z} \mapsto z \bmod 3. \quad (\text{DP19})$$

Its image is the constants $\mathbb{F}_3$, and its kernel is $(3, T) \subset D$. These formulas follow immediately from DP4, including its factor $9c$.

DP6. The nilradical quotient is this prism specialization

Retain

$$N_B = On_1 \oplus On_2 \oplus On_3, \quad N_C = O\epsilon \oplus OT \oplus O\epsilon T. \quad (\text{DP20})$$

These are the original nilradicals. For C , the stated ideal is nilpotent and its quotient is the reduced ring O . For B , evaluation at $(0, 0), (-3, 3), (-3, -3)$ has precisely this kernel: a coefficient expression $p(x) + Tq_0(x)$ is killed by all three maps exactly when $p(0) = p(-3) = q_0(-3) = 0$, giving the displayed three generators. Their products satisfy

$$n_1^2 = n_1 n_3 = n_2 n_3 = n_3^2 = 0, \quad n_1 n_2 = 3n_3, \quad n_2^2 = 18n_1,$$

so that kernel is nilpotent. Every nilpotent must vanish in the reduced evaluation target, proving equality with the nilradical.

The actions are

$$xn_1 = xn_3 = 0, \quad xn_2 = n_3, \quad Tn_1 = n_3, \quad Tn_2 = 6n_1, \quad Tn_3 = 0. \quad (\text{DP21})$$

In particular $JN_B = 0$, so N_B is a C -module through q . The map

$$i : N_B \xrightarrow{\sim} N_C, \quad n_1 \mapsto \epsilon, \quad n_2 \mapsto T, \quad n_3 \mapsto \epsilon T \quad (\text{DP22})$$

is C -linear: every generator action in DP21 agrees with multiplication by ϵ, T on its images. Its basis map is bijective.

The actual quotient-induced map q_N is

$$q_N(n_1) = 3\epsilon, \quad q_N(n_2) = 3T + \epsilon T, \quad q_N(n_3) = 3\epsilon T.$$

Thus

$$\boxed{q_N = m_{a_C} \circ i.} \quad (\text{DP23})$$

Multiplication by a_C is injective on N_C , since it is injective on C . Therefore

$$Q_N = \text{coker } q_N = N_C/a_C N_C \simeq N_C \otimes_C^{\mathbf{L}} D \quad (\text{DP24})$$

with the derived tensor concentrated in degree zero. To prove the derived assertion, use the free resolution $0 \rightarrow C \xrightarrow{a_C} C \rightarrow D \rightarrow 0$. Tensoring gives the two-term map $a_C : N_C \rightarrow N_C$, whose kernel is zero; its cokernel is exactly DP24.

There is also an injection $N_C/a_C N_C \rightarrow C/a_C C$. If $a_C f \in N_C$, its constant coefficient is $3f_0$. That coefficient vanishes only when $f_0 = 0$, so $f \in N_C$. Hence $N_C \cap a_C C = a_C N_C$, proving injection. Under DP15 its image is precisely

$$\boxed{Q_N \xrightarrow{\sim} (3, T) \subset D, \quad [\epsilon] \mapsto -3, \quad [T] \mapsto T.} \quad (\text{DP25})$$

The ideal $(3, T)$ is nilpotent, with

$$(3, T)^2 = (3T), \quad (3, T)^3 = 0.$$

Its quotient in D is $\mathbb{F}_3$, which is reduced, so it is exactly $\text{Nil}(D)$. Thus the same Q_N is the original module specialization, the nilradical of the prism quotient, and the kernel of DP19.

In original generators $a = [\epsilon]$, $b = [T]$, its module presentation remains

$$Q_N = (O/3O)a \oplus (O/9O)b, \quad [\epsilon T] = -3b. \quad (\text{DP26})$$

The sign agrees with the embedding $\epsilon \mapsto -3$ in DP25.

DP7. Both cotangent quotients and their exact comparison

Retain the original relative cotangent kernels

$$H_B = H^{-1}(L_{B/O}), \quad H_C = H^{-1}(L_{C/O}).$$

Their free bases are

$$\begin{aligned} u_1 &= (n_1, 0), & u_2 &= (n_3, 0), & u_3 &= (n_2, n_3); \\ v_1 &= (\epsilon, 0), & v_2 &= (\epsilon T, 0), & v_3 &= (3T, \epsilon T). \end{aligned} \quad (\text{DP27})$$

These are kernels of the respective Jacobian matrices

$$\begin{pmatrix} 3x(x+2) & -6(x+1) \\ 0 & 2T \end{pmatrix}, \quad \begin{pmatrix} 2\epsilon & -6 \\ 0 & 2T \end{pmatrix}.$$

For completeness, let $A_0 = O[x]/(x^2(x+3))$. Multiplication by x sends $p_0 + p_1x + p_2x^2$ to $p_0x + (p_1 - 3p_2)x^2$, so its kernel is On_1 . Also $x+2$ is a unit, with inverse $1/2 - x/4 - x^2/4$. Writing $B = A_0 \oplus TA_0$, the equation $T(p + Tq_0) = 0$ forces $p = 0$ and $3x(x+2)q_0 = 0$. Freeness permits cancellation of 3, and the unit inverse gives $q_0 \in On_1$. Thus $\text{Ann}_B(T) = On_3$ and $\text{Ann}_B(T^2) = On_1 \oplus On_3$. In the first Jacobian kernel, the lower equation gives $b = zn_3$, and the upper equation gives $T^2a = 6zn_3$, since $(x+1)n_3 = n_3$. Since $T^2n_2 = 6n_3$, this means $a - zn_2 \in On_1 \oplus On_3$, exactly the three displayed cycles. For the second kernel, coefficient comparison in the free basis of C gives $\text{Ann}_C(T) = O\epsilon T$; the equation $\epsilon a = 3b$ then gives its stated basis. These are the original presentations, with no coordinate change.

The maps previously proved in NC18–NC19 are

$$\begin{aligned} \kappa_B(n_1) &= u_1, & \kappa_B(n_2) &= u_3, & \kappa_B(n_3) &= u_2; \\ \kappa_C(\epsilon) &= 3v_1, & \kappa_C(T) &= v_3 - 2v_2, & \kappa_C(\epsilon T) &= 3v_2. \end{aligned} \quad (\text{DP28})$$

The first is a C -linear isomorphism and the second a C -linear injection. Their linearity follows from the generator actions

$$\epsilon v_1 = \epsilon v_2 = 0, \quad \epsilon v_3 = 3v_2, \quad T v_1 = v_2, \quad T v_2 = 0, \quad T v_3 = 18v_1$$

and DP21; injectivity follows from their independent displayed coefficients. Direct substitution gives

$$q_H(u_1) = 9v_1, \quad q_H(u_2) = 9v_2, \quad q_H(u_3) = 3v_3 - 3v_2,$$

and hence the exact factorization

$$\boxed{q_H = \kappa_C m_{a_C} i \kappa_B^{-1}.} \quad (\text{DP29})$$

Consequently the actual cotangent cokernel is

$$Q_H = H_C/(a_C \kappa_C(N_C)). \quad (\text{DP30})$$

This is connected to the prism specialization of the entire H_C by the surjection

$$Q_H \longrightarrow H_C/a_C H_C.$$

Multiplication by a_C is injective on the O -free module H_C : if $a_C h = 0$, then $9h = (3 - \epsilon)a_C h = 0$. Its induced map therefore identifies

$$a_C H_C / (a_C \kappa_C(N_C)) \simeq H_C / \kappa_C(N_C).$$

The latter is $R = \mathbb{F}_3[T]/T^2$, via $[v_1] \mapsto 1, [v_2] \mapsto T, [v_3] \mapsto 2T$. This yields the exact sequence

$$0 \longrightarrow R \xrightarrow{a_C} Q_H \longrightarrow H_C / a_C H_C \longrightarrow 0. \quad (\text{DP31})$$

The first arrow means: lift a class from $H_C / \kappa_C(N_C)$, multiply its lift by a_C , and take its class in DP30. It is well defined and injective by the preceding cancellation proof.

The second quotient is explicitly

$$H_C / a_C H_C \simeq \mathbb{F}_3\{[v_1], [v_2], [v_3]\} \simeq R \oplus \mathbb{F}_3 \quad (\text{DP32})$$

as C -modules. Indeed $a_C v_1 = 3v_1$, $a_C v_2 = 3v_2$, and $a_C v_3 = 3v_3 + 3v_2$, so the quotient has exactly the three relations $3v_j = 0$. There ϵ acts as zero, $T[v_1] = [v_2]$, and $T[v_2] = T[v_3] = 0$, giving the displayed module decomposition. Its derived specialization is again concentrated in degree zero because a_C is injective on H_C .

Thus DP30 and DP32 have both their actual presentations and their connecting maps. In particular Q_H , of order 3^5 , is not replaced by the order- 3^3 specialization DP32.

DP8. The prism element in the original extension

Use the original notation

$$Q_N = (O/3)a \oplus (O/9)b, \quad Q_H = (O/9)c_1 \oplus (O/9)c_2 \oplus (O/3)w,$$

where $c_1 = [v_1]$, $c_2 = [v_2]$, $w = [v_3 - v_2]$. The original extension maps are

$$\kappa(a) = 3c_1, \quad \kappa(b) = w - c_2, \quad \psi(c_1) = 1, \quad \psi(c_2) = \psi(w) = T. \quad (\text{DP33})$$

The connecting map associated with the annihilator element a_C is

$$\rho : R \rightarrow Q_N[3], \quad \rho(1) = a, \quad \rho(T) = -3b.$$

It is an isomorphism by the original T -action $Ta = -3b$. Direct multiplication gives

$$\boxed{a_C \text{id}_{Q_H} = \kappa \rho \psi}. \quad (\text{DP34})$$

On c_1, c_2, w , the two sides are respectively $3c_1, 3c_2, 3c_2$, since $3w = 0$ and $\epsilon w = 3c_2$. Therefore

$$a_C^2 Q_H = 0, \quad a_C Q_H = \kappa(Q_N[3]), \quad \ker(a_C : Q_H \rightarrow Q_H) = \kappa(Q_N). \quad (\text{DP35})$$

The image has order 9 and identifies with the kernel in DP31. The kernel assertion follows from DP34 and injectivity of κ, ρ .

Retain also the coefficient-3 map

$$\theta : Q_H \rightarrow Q_N, \quad \kappa \theta = 3 \text{id}_{Q_H},$$

whose values are $\theta(c_1) = a, \theta(c_2) = -3b, \theta(w) = 0$. Then the original divided cotangent operator has the exact expression

$$L = -\frac{5\epsilon}{8} = \frac{5(3 - a_C)}{8} = \frac{5}{8} \kappa(\theta - \rho \psi). \quad (\text{DP36})$$

This is an equality of actual integral endomorphisms on Q_H , not division of a torsion map by 3. It relates the coefficient-3 connecting map and the prism-element connecting map with the original coefficient 5 retained.

DP9. The residue descends to a nonzero derivation on the prism quotient

The original integral residues are

$$D_B(x) = 0, \quad D_B(T) = -n_3/8, \quad D_C(\epsilon) = 0, \quad D_C(T) = -3\epsilon T/8. \quad (\text{DP37})$$

They preserve the defining relations by $T^2 n_1 = 0$ and $\epsilon^2 = 0$, respectively. They commute with q . For the Frobenius family DP4,

$$D_B \phi_B = \phi_B D_B = 0, \quad D_C \phi_C = \phi_C D_C = 0.$$

Indeed the images of the residues are the coefficient lines On_3 and $O\epsilon T$, respectively; Frobenius kills these lines, and the residue kills every Frobenius image. This is compatibility with the stated Frobenius, not an assertion of commutation with the nonlinear delta operation.

On the whole algebra C , the divided operator is an integral derivation:

$$\partial = \frac{D_C}{3}, \quad \partial(\epsilon) = 0, \quad \partial(T) = -\epsilon T/8. \quad (\text{DP38})$$

Its action on $T^2 - 6\epsilon$ is $-\epsilon T^2/4 = -3\epsilon^2/2 = 0$, and its action on ϵ^2 is zero, proving that it descends from the polynomial algebra. It fixes a_C , so it induces a derivation on DP15. Its exact formula there is

$$\bar{\partial} : (O/9)[T]/T^2 \longrightarrow (O/9)[T]/T^2, \quad \bar{\partial}(z_0 + z_1 T) = \frac{3}{8} z_1 T. \quad (\text{DP39})$$

It kills constants and is nonzero, since $3T \neq 0$. Its square and its triple are zero. The unscaled $D_C = 3\partial$ consequently induces the zero derivation on the prism quotient.

Under the actual nilradical identification DP25, its restriction is exactly

$$A_N(a) = 0, \quad A_N(b) = \frac{3}{8}b. \quad (\text{DP40})$$

In particular its image is the ideal $(3T) = \text{Nil}(D)^2$, of order 3. It is not multiplication by $3/8$ on the whole ring D , since its value at 1 is zero; multiplication by $3/8$ describes only its restriction to the nilradical, where $3 \cdot 3 = 0$.

The asymmetry with the source B is exact. Dividing $D_B(T)$ by 3 would give

$$-\frac{x^2 T}{24} - \frac{xT}{8},$$

whose $x^2 T$ -coefficient does not belong to O . Thus $D_B/3$ is not an integral algebra derivation on B , although its original restriction to N_B is divisible by 3, as proved in NC36–NC43. The map to C , and then the prism specialization, are the specified morphisms relating these facts.

DP10. The exact ordinary spectra

The nilradical of C is $N_C = (\epsilon, T)$, as proved in DP20, and $C/N_C = O$. Every prime ideal contains the nilradical. Since $O = \mathbb{Z}_3$ has exactly the prime ideals (0) and (3) , the complete ordinary spectrum is

$$\text{Spec}(C) = \{\mathfrak{n}_C, \mathfrak{m}_C\}, \quad \mathfrak{n}_C = (\epsilon, T), \quad \mathfrak{m}_C = (3, \epsilon, T). \quad (\text{DP41})$$

Here $\mathfrak{n}_C \subset \mathfrak{m}_C$; the first point is generic and the second closed. The quotient rings by these ideals are respectively O and $\mathbb{F}_3$, so each displayed ideal is prime and the list is exhaustive.

For the actual prism quotient, DP25 proves

$$\text{Nil}(D) = (3, T) =: \mathfrak{m}_D, \quad D/\mathfrak{m}_D = \mathbb{F}_3. \quad (\text{DP42})$$

Thus $\text{Spec}(D)$ has the single prime $\mathfrak{m}_D$. The quotient map $C \rightarrow D$ pulls it back to $\mathfrak{m}_C$: the three elements $3, \epsilon, T$ map into $(3, T)$, while the residual scalar map is reduction modulo 3. Therefore the ordinary closed immersion has image

$$\text{Spec}(D) \longrightarrow \text{Spec}(C), \quad \mathfrak{m}_D \longmapsto \mathfrak{m}_C, \quad \text{im} = V(a_C) = \{\mathfrak{m}_C\}. \quad (\text{DP43})$$

Indeed the image of a_C in $C/\mathfrak{n}_C = O$ is the nonzero scalar 3, so $\mathfrak{n}_C$ does not contain a_C , whereas $\mathfrak{m}_C$ does. This is specialization at the prime 3 with the complete characteristic-nine nilpotent structure of D retained. In particular $D[1/3] = 0$, but 3 and T are both nonzero in D .

DP11. Exact source-reading and use record

Canonical source ID: PUBUNIT-3372AF0EC9F522F58C6F6CEE.

Authors: Bhargav Bhatt and Peter Scholze. Title: *Prisms and Prismatic Cohomology*. Version: arXiv:1905.08229v4, original author source archive retained locally. The existing machine-readable route sources/prismatic/prisms_index_routes.json was consulted before reading. The actual original source read is

`sources/prismatic/1905.08229v4/prisms.tex`, SHA256 `f155d2051758e2805efad8513690a91d7b18c19461514abdfd3d32f1676cf38b`.

Actual bounded reading for this derivation: source lines 134–180 for introductory prism and boundedness definitions; 484–558 for delta-ring identities, Frobenius lifts, and quotient compatibility; 689–742 for distinguished-element products and division; 936–1084 for Koszul conventions, the exact prism definition `DefPrismCat`, `prismcrit`, rigidity `PrismMapTaut`, and the start of the bounded-prism properties. These original LaTeX portions were read directly. The rest of that paper is not claimed as newly read here.

The received mathematical uses are precise: DP1 applies the delta-ring definition directly to the retained Frobenius family; DP3 checks every prism and boundedness axiom; DP4 applies the distinguished-division argument and the source’s faithfully flat delta-localization construction; DP5 uses the definition of a prism morphism. All specialised identities, torsion kernels, coefficient signs, completeness calculations, and module specializations are proved in this file.

Incoming programme proofs are Integral eight-state collision proof, the original integral algebra and Frobenius calculations; Nilradical and cotangent divided-residue proof, NC1–NC43; and Integral extension-class proof, EC1–EC48. This is a construction of a specific bounded prism and its actual module maps. It does not identify ordinary de Rham or cotangent cohomology with an uncomputed prismatic cohomology group.

Supported-zero spectra of the collision prism

23 September 2026. This proof computes the prime-spectrum maps of the actual coefficient and divisor quotients. Supported zero and the unsupported element remain distinct throughout.

The original semiring conventions and the integer spectrum are proved in TPS1–TPS8. The original collision algebras and nilradical quotient are NC1–NC13. Their spectra and all maps used here are proved below.

SG1. The original supported semiring and all its prime ideals

For a nonzero commutative unital ring A , set

$$G(A) = \{\tau\} \sqcup A^\bullet, \quad e_A = 0_A^\bullet. \quad (\text{SG1})$$

Here the bullet is a label, so $e_A \neq \tau$. Define

$$a^\bullet + b^\bullet = (a + b)^\bullet, \quad a^\bullet b^\bullet = (ab)^\bullet, \quad z + \tau = z, \quad z\tau = \tau. \quad (\text{SG2})$$

These operations form a commutative semiring with additive identity τ and multiplicative identity $1_A^\bullet$. Associativity and distributivity for entirely supported arguments are the corresponding ring identities. If an argument is τ , its additive-identity and multiplicatively absorbing rules give the identities directly. These cases exhaust all arguments.

An ideal contains τ , is closed under addition, and under multiplication by every semiring element. Its intersection with $A^\bullet$ is either empty or $I^\bullet$ for a ring ideal $I \subset A$. Indeed multiplication by e_A produces e_A from any supported member; multiplication by $(-1_A)^\bullet$ supplies supported additive inverses. The other ring-ideal properties are the given closure properties. Conversely these properties prove that every set

$$P_\tau(A) = \{\tau\}, \quad P_I(A) = \{\tau\} \cup I^\bullet \quad (\text{SG3})$$

is a semiring ideal. This classifies all ideals, including the whole semiring when $I = A$.

A proper ideal is prime if membership of a product forces membership of a factor. A product of supported elements is supported, so $P_\tau(A)$ is prime for every A . For $P_I(A)$, products of supported elements give exactly the ring prime-ideal test for I , and products involving τ already satisfy the test. Thus

$$\text{Spec } G(A) = \{P_\tau(A)\} \sqcup \{P_{\mathfrak{p}}(A) : \mathfrak{p} \in \text{Spec } A\}. \quad (\text{SG4})$$

Every arithmetic point $P_{\mathfrak{p}}(A)$ contains e_A . The point $P_\tau(A)$ does not, and is contained strictly in every arithmetic point. The topology has basic opens $D(z) = \{P : z \notin P\}$. For a supported element,

$$D(a^\bullet) = \{P_\tau(A)\} \cup \{P_{\mathfrak{p}}(A) : a \notin \mathfrak{p}\}, \quad D(e_A) = \{P_\tau(A)\}. \quad (\text{SG5})$$

These formulas prove that the arithmetic points form the closed subspace $V(e_A)$, homeomorphic to $\text{Spec } A$.

For a unital ring map $f : A \rightarrow A'$, the map $G(f)$ sends τ to τ and $a^\bullet$ to $f(a)^\bullet$. It preserves both operations and both identities. Its spectrum map is precisely

$$G(f)^* P_\tau(A') = P_\tau(A), \quad G(f)^* P_{\mathfrak{p}'}(A') = P_{f^{-1}\mathfrak{p}'}(A). \quad (\text{SG6})$$

Taking preimages of the basic opens proves continuity. No supported element is mapped to τ .

SG2. The coefficient algebra and its distinguished divisor

Put $O = \mathbb{Z}_3$, the ring of 3-adic integers, and retain

$$B = O[x, T]/(x^2(x+3), T^2 - 3x(x+2)), \quad C = O[\epsilon, T]/(\epsilon^2, T^2 - 6\epsilon). \quad (\text{SG7})$$

The map $q : B \rightarrow C$ sends x to ϵ and T to T ; substituting gives both relations of B . Its images generate C , so it is onto. Successive division by the monic equations gives the free O -basis $1, \epsilon, T, \epsilon T$ of C .

Set

$$\mathfrak{n} = (\epsilon, T) \subset C, \quad \mathfrak{m} = (3, \epsilon, T) \subset C, \quad a_C = 3 + \epsilon. \quad (\text{SG8})$$

The equalities $\epsilon^2 = 0$ and $T^2 = 6\epsilon$ give

$$\mathfrak{n}^2 = (6\epsilon, \epsilon T), \quad \mathfrak{n}^3 = (6\epsilon T), \quad \mathfrak{n}^4 = 0. \quad (\text{SG9})$$

For instance multiplying the two displayed generators of $\mathfrak{n}^2$ by ϵ, T gives exactly the ideal generated by $6\epsilon T$; one further multiplication gives zero. The quotient $C/\mathfrak{n} = O$ is reduced. Therefore $\mathfrak{n}$ is exactly the nilradical. Every prime contains every nilpotent (apply primeness repeatedly to a vanishing power); primes of C thus correspond exactly to primes of O .

The primes of O are (0) and (3) . To see completeness, a nonzero element has the form $3^k u$, with u a unit. A nonzero proper prime containing it must contain 3 , and the ideal (3) is maximal because its quotient is $\mathbb{F}_3$. Thus

$$\text{Spec } C = \{\mathfrak{n} \subsetneq \mathfrak{m}\}. \quad (\text{SG10})$$

Let $D = C/(a_C)$. In this quotient $\epsilon = -3$, so $\epsilon^2 = 0$ imposes 9 and $T^2 = 6\epsilon$ imposes $T^2 = -18 = 0$. Conversely these equations satisfy all the original relations and $3 + \epsilon = 0$. The two substitution maps are inverse, proving

$$D \xrightarrow{\sim} (O/9O)[T]/T^2, \quad \epsilon \mapsto -3, \quad T \mapsto T. \quad (\text{SG11})$$

The ideal $\mathfrak{d} = (3, T) \subset D$ satisfies

$$\mathfrak{d}^2 = (3T), \quad \mathfrak{d}^3 = 0, \quad D/\mathfrak{d} = \mathbb{F}_3. \quad (\text{SG12})$$

Here $3T \neq 0$, because $1, T$ is a free $O/9O$ -basis. These facts prove that $\mathfrak{d}$ is the nilradical and the unique prime of D . The quotient map $\pi : C \rightarrow D$ pulls it back to $\mathfrak{m}$, since its composite to $\mathbb{F}_3$ kills exactly $(3, \epsilon, T)$.

SG3. Every supported point and its exact pullback

Applying SG4 and SG6 gives the complete chains

$$\begin{aligned} \text{Spec } G(C) : \quad P_\tau(C) \subsetneq P_{\mathfrak{n}}(C) \subsetneq P_{\mathfrak{m}}(C), \\ \text{Spec } G(D) : \quad P_\tau(D) \subsetneq P_{\mathfrak{d}}(D). \end{aligned} \quad (\text{SG13})$$

The spectrum map of the actual divisor quotient is

$$G(\pi)^* : P_\tau(D) \mapsto P_\tau(C), \quad P_{\mathfrak{d}}(D) \mapsto P_{\mathfrak{m}}(C). \quad (\text{SG14})$$

Its image is the two-point subspace $\{P_\tau(C), P_{\mathfrak{m}}(C)\}$. It is a homeomorphism onto this subspace: its opens are the empty set, the unsupported singleton, and both points, as follows immediately from SG5 in both spaces. It is not a closed subset of $\text{Spec } G(C)$, since it contains the dense point $P_\tau(C)$ but omits $P_{\mathfrak{n}}(C)$. The ring closed immersion $\text{Spec } D \rightarrow \text{Spec } C$ therefore has this explicitly different supported extension. Its inverse images and topology, rather than an assumed closed-immersion rule for a semiring congruence quotient, specify the map.

The coefficient inclusion $O \rightarrow C$ pulls $\mathfrak{n}$ back to (0) and $\mathfrak{m}$ back to (3) . Hence its supported spectrum map is

$$P_\tau(C) \mapsto P_\tau(O), \quad P_{\mathfrak{n}}(C) \mapsto P_{(0)}(O), \quad P_{\mathfrak{m}}(C) \mapsto P_{(3)}(O). \quad (\text{SG15})$$

This is a homeomorphism of the three-point chains, by their basic opens. The composite $G(O) \rightarrow G(D)$ thus sends its arithmetic point back to the prime over 3 . The pullbacks to the original algebra B are also explicit:

$$q^{-1}\mathfrak{n} = (x, T), \quad q^{-1}\mathfrak{m} = (3, x, T), \quad (\text{SG16})$$

because the two composite quotients are respectively O and $\mathbb{F}_3$. Equations SG6 and SG16 give the corresponding supported prime ideals of $G(B)$, without claiming a classification of every other point of $\text{Spec } G(B)$.

The ideal generated by e_C is exactly $\{\tau, e_C\}$: multiplying a supported element by e_C yields e_C , and addition does not produce another element. It is not prime, since $(\epsilon^\bullet)^2 = e_C$ while $\epsilon^\bullet$ is outside that ideal. Its radical, defined by powers entering the ideal, is exactly

$$\sqrt{(e_C)} = P_{\mathfrak{n}}(C). \quad (\text{SG17})$$

For a supported element, entering (e_C) by a power is exactly nilpotence in C . This proves SG17 and also gives $\sqrt{(e_D)} = P_{\mathfrak{d}}(D)$. Over the original domain O , the same calculation gives $(e_O) = P_{(0)}(O)$, which is itself prime. SG15 is therefore the exact map from the collision's nilpotent prime to the original supported-zero prime; these points are not discarded or identified with P_τ .

SG4. The infinitesimal quotient as an actual ideal

Multiplication by a_C is injective on C , because

$$(3 - \epsilon)a_C = 9 \quad (\text{SG18})$$

and C is free over the domain O . Its restriction to $\mathfrak{n}$ is injective as well. Moreover

$$a_C C \cap \mathfrak{n} = a_C \mathfrak{n}. \quad (\text{SG19})$$

Indeed, writing $f = f_0 + f_1\epsilon + f_2T + f_3\epsilon T$, the constant coefficient of $a_C f$ is $3f_0$. If this belongs to $\mathfrak{n}$, then $f_0 = 0$, hence $f \in \mathfrak{n}$. The other inclusion is immediate. Consequently the natural quotient map is an injection

$$\mathfrak{n}/a_C \mathfrak{n} \hookrightarrow D$$

with image generated by $-3, T, -3T$, namely $\mathfrak{d}$. Thus

$$\mathfrak{n}/a_C \mathfrak{n} \xrightarrow{\sim} \mathfrak{d}, \quad [\epsilon] \mapsto -3, \quad [T] \mapsto T, \quad [\epsilon T] \mapsto -3T. \quad (\text{SG20})$$

The free resolution $0 \rightarrow C \xrightarrow{a_C} C \rightarrow D \rightarrow 0$ shows that $\text{Tor}_1^C(\mathfrak{n}, D)$ is the kernel of multiplication by a_C on $\mathfrak{n}$, hence zero. All higher Tor groups vanish because this resolution has length one. This proves that the same ordinary quotient computes the derived specialization; no discarded higher group is being assumed to vanish.

For the retained Frobenius family, define Φ_C on O to be the identity, on ϵ to be zero, and on T by $9c\epsilon T$, with $c \in O$. The images satisfy both relations. Modulo 3, the original generators have cubes zero, so Φ_C lifts the characteristic-three Frobenius. Since $\Phi_C(a_C) = 3$, it induces the unital ring map

$$\overline{\Phi} : D \longrightarrow C/3C, \quad f_0 + f_1 T \longmapsto \overline{f}_0. \quad (\text{SG21})$$

The target is $\mathbb{F}_3[\epsilon, T]/(\epsilon^2, T^2)$. The displayed formula proves $\ker \overline{\Phi} = \mathfrak{d}$ and $\text{im } \overline{\Phi} = \mathbb{F}_3$. Its additive cokernel is the vector space with basis $\epsilon, T, \epsilon T$. Both quotient rings have exactly the primes specified by their nilradicals, so its supported spectrum map preserves the unsupported point and the sole arithmetic point. The killed ideal is exactly the retained infinitesimal ideal SG20, and is recorded as a kernel rather than erased from the calculation.

Finally the O -linear derivation on C defined by

$$\partial(\epsilon) = 0, \quad \partial(T) = -\epsilon T/8 \quad (\text{SG22})$$

respects the relations: $\partial(\epsilon^2) = 0$ and $\partial(T^2 - 6\epsilon) = -\epsilon T^2/4 = 0$. Here 8 is a unit of O . It kills a_C and descends to the derivation of D

$$\overline{\partial}(f_0 + f_1 T) = \frac{3}{8} f_1 T. \quad (\text{SG23})$$

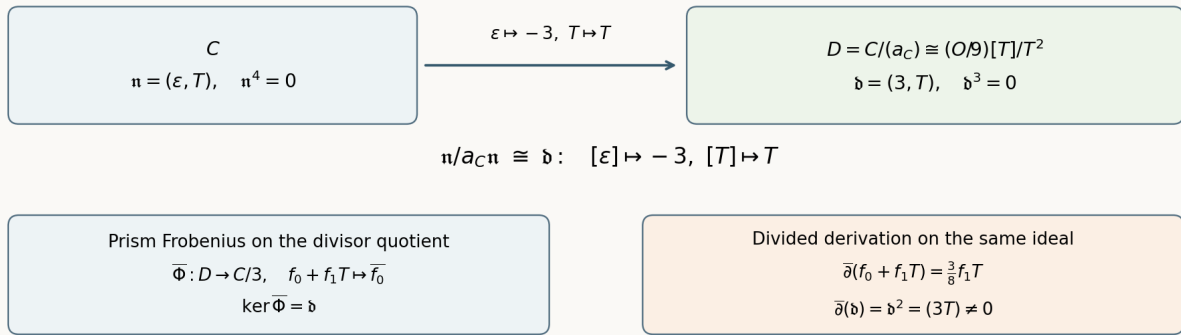
It is nonzero, its square is zero, and its triple is zero: its image is the nonzero ideal generated by $3T$, on which another factor of 3 vanishes. Under SG20 its restriction to $\mathfrak{d}$ is the original divided nilradical operator $A_N(a) = 0, A_N(b) = 3b/8$, with $a = [\epsilon], b = [T]$. Thus the prism Frobenius kills $\mathfrak{d}$, while the divided derivation measures its last nonzero power $\mathfrak{d}^2 = (3T)$. These are exact maps on the same ideal and the same supported spectrum.

The two-element subsemiring $\{\tau, e_A\}$ still has multiplicative identity e_A , and $e_A e_A = e_A$, for each ring here. Hence e_A is its own inverse in that subsemiring. Its inclusion into $G(A)$ preserves both operations and the additive identity, but sends its multiplicative identity to e_A , rather than $1_A^\bullet$. The retraction $z \mapsto e_A z$ is unital when the codomain has identity e_A . These exact maps preserve the intrinsic supported-zero inversion throughout SG1–SG23.

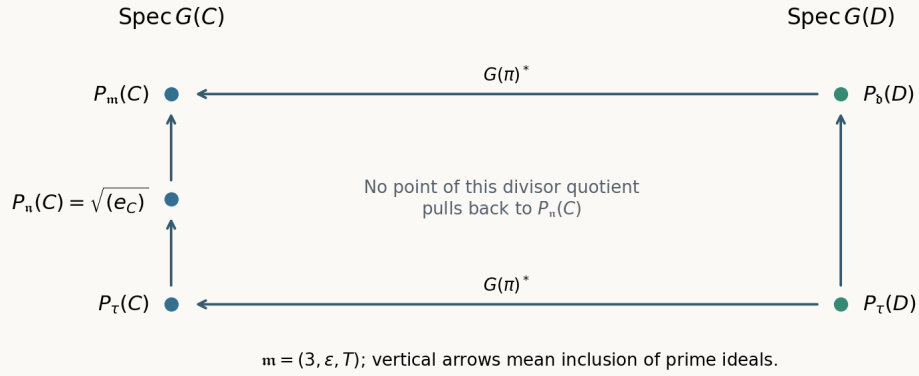
The retained spectrum and operators in the figure

The collision prism retains its infinitesimal and its supported spectrum

$$O = \mathbb{Z}_3, \quad C = O[\varepsilon, T]/(\varepsilon^2, T^2 - 6\varepsilon), \quad a_C = 3 + \varepsilon$$



Prime pullbacks, with the unsupported point retained (SG13-SG17)



Full proofs: SG1–SG23. Positions are schematic; all algebraic labels and maps are exact.

Figure 15: The divisor quotient and prime pullbacks, including the nonclosed supported image. The Frobenius kernel and divided-derivation image are ideals in the same ring. Full proofs SG13–SG23.

The global heat zero trace and its exact infinitesimal variation

This proof establishes an actual holomorphic family of global zero traces for rational tests, and a separately specified regularized variation for rapidly decreasing entire tests. The two domains and their comparison maps are retained. The original heat family, constants and supported endpoint coordinates are used throughout.

1. Uniform growth of the original entire family

Use the original Rodgers–Tao integral and coordinates

$$\begin{aligned} H_t(Z) &= \int_0^\infty e^{tu^2} \Phi(u) \cos(Zu) du, \\ \Phi(u) &= \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \\ g_t(s) &= 16H_t(-2i(s - 1/2)), \quad g_0 = 2\xi_R, \quad \partial_t g_t = \tfrac{1}{4}g_t''. \end{aligned} \tag{AG1}$$

The author source is Rodgers–Tao, arXiv:1801.05914v5, equations `hoz`, `phidef`, `htdef` and its subsequent heat equation. The coordinate verification is also given directly in HA1–HA4 of this edition.

For $u \geq 0$, summing the defining series after retaining a fixed part of its n^2 exponent gives constants $C, c > 0$ with

$$|\Phi(u)| \leq C e^{9u - ce^{4u}}. \tag{AG2}$$

For example split $\pi n^2 e^{4u}$ into two equal parts. One is at least $(\pi/2)e^{4u}$, and the other makes $\sum n^4 e^{-\pi n^2/2}$ converge uniformly. The term with $n^2 e^{5u}$ has the same bound. For each fixed T , the function $Tu^2 - (c/2)e^{4u}$ is bounded above on $[0, \infty)$. Thus on $|t| \leq T, |s| \leq R$, the integral is bounded by a constant depending on T times

$$\int_0^\infty \exp((2R + 10)u - (c/2)e^{4u}) du \leq \exp(C_T(R + 1) \log(R + 2)). \tag{AG3}$$

For the last inequality put $x = e^{4u}$, enlarge the resulting integral from $[1, \infty)$ to $[0, \infty)$, and use the Gamma integral with exponent $(2R + 10)/4$. Its positive-real Stirling bound gives (AG3). Differentiating in s, t adds powers of u , still integrable by (AG2). Consequently g is jointly entire and has order at most one in s , uniformly on bounded time sets in the growth sense of (AG3).

Every real time t_* has $g_{t_*}(0) > 0$. In fact $\Phi(u) > 0$ termwise for $u \geq 0$, since $2\pi n^2 e^{4u} - 3 > 0$, and the integral at $s = 0$ uses $\cosh u > 0$. Continuity therefore supplies a complex time disc about t_* on which $g_t(0)$ stays bounded away from zero. Jensen’s formula on radii R and $2R$, combined with (AG3), gives the uniform zero count

$$N_t(R) := \sum_{|\rho| \leq R} m_{\rho,t} = O(R \log(R + 2)). \tag{AG4}$$

The constant is uniform on a smaller closed time disc. Dyadic summation gives the useful uniform tail estimate

$$\sum_{|\rho| > R} \frac{m_{\rho,t}}{|\rho|^2} = O\left(\frac{\log(R + 2)}{R}\right). \tag{AG5}$$

No common vertical strip for the moving zeros has been assumed.

2. A rational test has a finite-pole global trace

Let $A(s)$ be rational, with $A(s) = O(s^{-2})$ at infinity. Fix a real time t_* at which none of its finitely many poles is a zero of g_{t_*} . Compact continuity supplies a smaller complex time disc D on which all those values and $g_t(0)$ stay nonzero. Define

$$Z_t(A) = \sum_{g_t(\rho)=0} m_{\rho,t} A(\rho), \quad t \in D. \tag{AG6}$$

For each t , the sum is absolutely convergent by (AG4)–(AG5). Its far tail is uniformly bounded by $C_A \log(R + 2)/R$; fixed poles are excluded by the choice of D .

Hadamard factorization for an entire function of order at most one, with the nonzero value at zero, gives

$$L_t(s) := g_t'(s)/g_t(s) = \beta_t + \sum_{\rho} m_{\rho,t} \left(\frac{1}{s - \rho} + \frac{1}{\rho} \right). \tag{AG7}$$

The series is normally convergent away from the zeros. Its normal convergence follows directly for the logarithmic derivative from $\sum m/|\rho|^2 < \infty$, because the paired summand is $O_K(|\rho|^{-2})$ on a fixed compact set K . The exponential factor in the genus-one product supplies the constant β_t .

Let $\mathcal{P}(A)$ be the finite pole set of A . The exact global identity is

$$\boxed{Z_t(A) = - \sum_{p \in \mathcal{P}(A)} \text{Res}_{s=p} (A(s) L_t(s)).} \quad (\text{AG8})$$

To prove it, $\sum_p \text{Res}_p A = 0$, since $A = O(s^{-2})$. Thus the constant β_t and each $1/\rho$ correction in (AG7) give zero after summing residues. For one zero ρ , the rational function $A(s)/(s - \rho)$ has zero residue at infinity, so its residues at the poles of A sum to $-A(\rho)$. Normal convergence on small circles around the finite pole set justifies taking residues term by term. This proves (AG8), including repeated poles of A and repeated zeros of g_t .

The right side of (AG8) is holomorphic in t on D . This proves holomorphic dependence of the actual whole zero trace. In particular its derivative is

$$\dot{Z}_t(A) = -\frac{1}{4} \sum_{p \in \mathcal{P}(A)} \text{Res}_{s=p} \left(A(s) \partial_s \frac{g_t''(s)}{g_t(s)} \right). \quad (\text{AG9})$$

This conclusion does not exchange a derivative with an unrestricted infinite moving-zero sum. The finite residue expression proves the derivative of that sum as a holomorphic function.

3. The Cauchy kernel and the original test domain

For $\Re z, \Re w > 1$, set

$$F_z(s) = (s - z)^{-1}, \quad F^\#(s) = \overline{F(1 - \bar{s})}, \quad A_{z,w} = F_z^\# F_w = -\frac{1}{(s - (1 - \bar{z}))(s - w)}. \quad (\text{AG10})$$

At time zero both poles are nonzeros of $g_0 = 2\xi_R$, by the unconditional critical-strip result proved in SZW25. Formula (AG8) now evaluates both simple residues:

$$Z_t(A_{z,w}) = \frac{L_t(w) - L_t(1 - \bar{z})}{w + \bar{z} - 1} = \frac{L_t(w) + L_t(\bar{z})}{w + \bar{z} - 1}. \quad (\text{AG11})$$

Reflection gives the second equality for complex time. For real time $L_t(\bar{z}) = \overline{L_t(z)}$. For a fixed finite set of tests the time disc is common to all its poles. If $z, w > 1$ are real, the positive integral and reflection make their poles nonzeros at every real time, so the same local argument applies around any such time.

The minus-exponent Mellin convention is the original one:

$$M_h(s) = \int_{\mathbb{R}} h(u) e^{-(s-1/2)u} du. \quad (\text{AG12})$$

For (AG10) its inverse is exactly

$$h_{z,w}(u) = \frac{1}{w + \bar{z} - 1} \begin{cases} e^{-(\bar{z}-1/2)u}, & u \geq 0, \\ e^{(w-1/2)u}, & u \leq 0. \end{cases} \quad (\text{AG13})$$

Direct integration proves (AG12) on $1 - \Re z < \Re s < \Re w$. The values at zeros outside this strip at other times use the rational meromorphic continuation (AG10); this is not an enlargement of the domain of convergence of the integral.

More generally take any rational $A = O(s^{-2})$ whose poles all lie in $\Re s < 0$ or $\Re s > 1$. Its inverse on the critical line has the explicit residue representation

$$h_A(u) = \begin{cases} \sum_{\Re p < 0} \text{Res}_{s=p} \left(e^{(s-1/2)u} A(s) \right), & u \geq 0, \\ - \sum_{\Re p > 1} \text{Res}_{s=p} \left(e^{(s-1/2)u} A(s) \right), & u \leq 0. \end{cases} \quad (\text{AG14})$$

Partial fractions and integration of exponential polynomials verify this directly. At zero the two expressions agree because the finite residues of A sum to zero. There is $d > 1/2$ such that h_A , its first derivative on the

two half-lines and its distributional second derivative have finite weighted norms with weight $e^{d|u|}$; for the second derivative the norm means total variation of a measure. The possible derivative jump contributes one delta atom. Repeated poles produce only polynomial factors multiplying the decreasing exponentials and do not affect this conclusion after decreasing d slightly.

Let $h_{R,\varepsilon}$ be a smooth compact cutoff of h_A on $[-R-1, R+1]$, convolved with a smooth mollifier of radius $\varepsilon \rightarrow 0$. The first two cutoff derivatives can be bounded independently of R . Consequently the weighted norms just stated remain bounded uniformly for $R \geq 1, \varepsilon \leq 1$. Distributional integration by parts twice gives

$$|M_{h_{R,\varepsilon}}(\sigma + iy)| \leq \frac{C}{1+y^2} \quad (0 \leq \sigma \leq 1), \quad |h_{R,\varepsilon}(u)| \leq Ce^{-d|u|}. \quad (\text{AG15})$$

For the latter inequality apply the fundamental theorem of calculus to the absolutely continuous weighted function, using the bounded weighted value and first-derivative norms. The transforms converge pointwise to A on the strip. The zero sum is dominated by $\sum m/(1+|\Im \rho|^2) < \infty$, from SZW25 and (AG4). The prime sum is dominated by $C \sum \Lambda(n)n^{-1/2-d} < \infty$, and the archimedean integrand by $C \log(2+|y|)/(1+y^2)$. Endpoint convergence follows from the weighted L^1 bound. Applying dominated convergence to the complete compact-test formula SZW25 therefore proves its extension to h_A , with all its terms.

In particular, for $A = A_{z,w}$, the exact supported identity remains

$$\begin{aligned} B_L &= (A(0) + A(1))\mathbf{e}_{1_L} + A(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda, \\ Z_L &= Z_0(A)\mathbf{e}_{1_L}, \\ D_L &= (P_{\text{fin}}(h_A) - A_\infty(h_A))\mathbf{e}_{1_L} + A(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda, \\ B_L - Z_L &= D_L. \end{aligned} \quad (\text{AG16})$$

These are the original SZW representations and label maps. Explicit evaluation of each Cauchy term and its actual heat derivative is proved in HA10–HA20.

4. Every finite collision has an exact contour derivative

Let ρ be any actual zero of g_0 , of multiplicity m , and write $g_0(s) = (s - \rho)^m u_\rho(s)$, $u_\rho(\rho) \neq 0$. A small fixed circle containing only this zero has no zeros on its boundary for small complex time. For every test A holomorphic near its closed disc, the argument principle gives the cluster trace and derivative

$$Z_{\rho,t}(A) = \frac{1}{2\pi i} \int_{\partial D_\rho} A(s) L_t(s) ds, \quad \dot{Z}_{\rho,0}(A) = -\frac{m(m-1)}{4} A''(\rho) - \frac{m}{2} \frac{u'_\rho(\rho)}{u_\rho(\rho)} A'(\rho). \quad (\text{AG17})$$

Indeed $g_0''/(4g_0)$ has principal part $m(m-1)/(4(s-\rho)^2) + (m/2)(u'_\rho/u_\rho)(\rho)/(s-\rho)$. Differentiate this principal part in s , multiply by the Taylor expansion of A , and take its residue. Compact nonvanishing on the fixed circle permits the time derivative. No derivative of an individually labelled colliding root is required.

5. An entire-test variation with its exact regularization

For this section let A be entire and assume, for every integer $N \geq 0$ and $j = 0, 1, 2$,

$$\sup_{0 \leq \Re s \leq 1} (1 + |\Im s|)^N |A^{(j)}(s)| < \infty. \quad (\text{AG18})$$

This includes original compactly supported smooth Mellin tests by repeated integration by parts. The zeros in this section are the distinct zeros of g_0 , with multiplicities m_ρ ; they lie strictly inside the critical strip. Put $b = g'_0(0)/g_0(0)$, and for distinct zeros define

$$T_A(\rho, \eta) = \frac{A'(\rho) - A'(\eta)}{\rho - \eta} + \frac{A'(\rho)}{\eta} + \frac{A'(\eta)}{\rho}. \quad (\text{AG19})$$

The following expression is absolutely convergent:

$$\begin{aligned} \mathcal{V}(A) &= -\frac{1}{4} \left[\sum_\rho m_\rho(m_\rho - 1) A''(\rho) + 2b \sum_\rho m_\rho A'(\rho) + 2 \sum_\rho \frac{m_\rho^2 A'(\rho)}{\rho} \right. \\ &\quad \left. + 2 \sum_{\{\rho, \eta\}} m_\rho m_\eta T_A(\rho, \eta) \right]. \end{aligned} \quad (\text{AG20})$$

The last sum is over unordered pairs. To verify convergence, arrange $r = |\rho| \leq R = |\eta|$. For $R < 2r$, either the distance is at least $r/2$, when direct rapid-decrease estimates apply, or the straight segment joining them stays at modulus at least $r/2$. In the second case its real part stays inside the strip and $(A'(\rho) - A'(\eta))/(\rho - \eta) = \int_0^1 A''(\eta + t(\rho - \eta))dt$. Thus $|T_A| \leq C_N(1 + r)^{-N}$. The total multiplicity weight of comparable pairs in a dyadic annulus is $O(r^2 \log^2(2 + r))$, which is summable for large N .

If $R \geq 2r$, retain the exact cancellation

$$T_A(\rho, \eta) = \frac{\rho A'(\rho)}{\eta(\rho - \eta)} + \frac{\eta A'(\eta)}{\rho(\eta - \rho)}, \quad |T_A| \leq \frac{2r|A'(\rho)|}{R^2} + \frac{2|A'(\eta)|}{r}. \quad (\text{AG21})$$

The zero count gives $\sum_{|\eta| \geq R} m_\eta/|\eta|^2 = O(\log(2 + R)/(1 + R))$ and $\sum_{|\rho| \leq R} m_\rho/|\rho| = O(\log^2(2 + R))$. These bounds and (AG18) prove convergence of the disparate-pair sum. The single sums in (AG20) converge as well, using the bound on $\sum m_\rho^2$ in an annulus by the square of its total multiplicity. Finitely many zeros near the origin cause no problem since $g_0(0) \neq 0$.

Here is the precise relation of (AG20) to the original family. For every sufficiently large integer j , select $T_j \in [j, j + 1]$ at distance at least j^{-3} from every $|\Im \rho|$. Such a value exists: only zeros of modulus at most $2j + 4$ matter, and their excluded intervals have total length $O(j^{-2} \log j) < 1$. Let Γ_j be the rectangle with real edges $-1, 2$ and heights $\pm T_j$. For each fixed j , its zero trace is analytic for sufficiently small time. Then

$$\mathcal{V}(A) = \lim_{j \rightarrow \infty} \frac{d}{dt} \bigg|_0 \frac{1}{2\pi i} \int_{\Gamma_j} A(s) L_t(s) ds. \quad (\text{AG22})$$

To prove it, expand the regular part of (AG7) at ρ :

$$\frac{u'_\rho(\rho)}{u_\rho(\rho)} = b + \frac{m_\rho}{\rho} + \sum_{\eta \neq \rho} m_\eta \left(\frac{1}{\rho - \eta} + \frac{1}{\eta} \right). \quad (\text{AG23})$$

The series converges for this fixed zero by genus-one cancellation. Insert it in (AG17), summed over zeros inside Γ_j . Pairs both inside combine to (AG19). The remaining cross-boundary sum is

$$E_T = \sum_{|\Im \rho| < T < |\Im \eta|} m_\rho m_\eta A'(\rho) \left(\frac{1}{\rho - \eta} + \frac{1}{\eta} \right). \quad (\text{AG24})$$

For $|\rho| \leq T/2$ its absolute value is $O(\log T/T)$, by the paired tail bound and summability of $m_\rho |\rho A'(\rho)|$. For $T/2 < |\rho| \leq 2T$, $T \leq |\eta| \leq 4T$, the separated height gives $|\rho - \eta| \gg T^{-3}$; the total is $O_N(T^{5-N} \log^2 T)$. For the same ρ and $|\eta| > 4T$, genus-one cancellation bounds the sum by $O_N(T^{1-N} \log^2 T)$. Hence $E_T \rightarrow 0$. Absolute convergence of (AG20) now proves (AG22) and independence from the chosen separated exhaustion.

Formula (AG22) is a contour-regularized global variation: it is the limit of the derivatives of finite traces, with its order of operations specified. It does not assert that arbitrary entire tests in (AG18) have a holomorphic global moving-zero sum at nearby times. The stronger statement for rational tests is (AG6)–(AG9), proved by their finite pole map. Both receivers retain their exact domains rather than being identified without a convergence argument.

Sources and receiving calculations

The exact original heat source and original author equations are cited at (AG1). Classical Jensen and Hadamard theorems are used with their growth and nonvanishing hypotheses verified in (AG2)–(AG7). The complete original supported formula and its zero-count and prime-contour proofs are SZW19–SZW38. Its human Connes attribution and original source convention are retained. HA applies (AG8) to the complete arithmetic variation; HM applies it to all time derivatives; the Cauchy–Weil criterion applies (AG11) at zero. No assertion of RH or its negation follows from convergence alone.

The actual heat pairing, its supported endpoints, and mixed prime products

This calculation uses the original heat family and the full supported-zero explicit formula. It gives an actual global family of pairings and computes its arithmetic derivative. Positivity of that family at time zero is not assumed. All lower support coordinates remain in the formula.

1. Original family and coordinates

Write $\xi_R(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$. The original Rodgers–Tao conventions are

$$\begin{aligned}\Phi(u) &= \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \\ H_t(Z) &= \int_0^\infty e^{tu^2} \Phi(u) \cos(Zu) du, \\ H_0(Z) &= \frac{1}{8}\xi_R(\tfrac{1}{2} + iZ/2), \quad \partial_t H_t = -\partial_Z^2 H_t.\end{aligned}\tag{HA1}$$

These are the original source equations `phidef`, `htdef`, `hoz`, and the sentence following `htdef` in Rodgers–Tao, arXiv:1801.05914v5, author TeX. Define, retaining the factor 16 and the coordinate factor $-2i$,

$$g_t(s) = 16H_t(-2i(s - \tfrac{1}{2})), \quad g_0(s) = 2\xi_R(s), \quad L_t(s) = \frac{\partial_s g_t(s)}{g_t(s)}.\tag{HA2}$$

The superexponential decrease in (HA1) makes the integral and all its derivatives converge uniformly on compact subsets of $\mathbb{C}_t \times \mathbb{C}_s$. For example, on $|t| \leq T, |s| \leq R$, its integrand is bounded by a constant times $\exp(Tu^2 + 2(R+1)u + 9u - ce^{4u})$, an integrable function; derivatives add only polynomial factors in u . It follows by differentiation that

$$\partial_t g_t = \tfrac{1}{4}\partial_s^2 g_t, \quad \partial_t L_t = \tfrac{1}{4}\partial_s \left(\frac{g_t''}{g_t} \right) = \tfrac{1}{4}(L_t'' + 2L_t L_t').\tag{HA3}$$

The factor $1/4$ follows because $(-2i)^2 = -4$. For every complex t , $g_t(1-s) = g_t(s)$. For real t , $g_t(\bar{s}) = \overline{g_t(s)}$. Thus

$$L_t(1-s) = -L_t(s), \quad L_t(\bar{s}) = \overline{L_t(s)} \quad (t \in \mathbb{R}).\tag{HA4}$$

These identities hold wherever the displayed quotients are defined.

2. The Cauchy tests in the original Mellin convention

Let $z, w \in \mathcal{U} = \{s : \Re s > 1\}$, and put

$$q = w + \bar{z} - 1, \quad F_z(s) = \frac{1}{s-z}, \quad A_{z,w}(s) = F_z^\#(s)F_w(s) = -\frac{1}{(s-(1-\bar{z}))(s-w)},\tag{HA5}$$

where $F^\#(s) = \overline{F(1-\bar{s})}$. In particular, $\Re q > 1$, and both poles lie strictly outside the closed critical strip. The exact original inverse Mellin tests are

$$f_z(u) = -1_{(-\infty, 0]}(u)e^{(z-1/2)u}, \quad h_{z,w} = f_z^\# * f_w,\tag{HA6}$$

with time-domain involution $f^\#(u) = \overline{f(-u)}$. Direct integration gives

$$h_{z,w}(u) = \frac{1}{q} \begin{cases} e^{-(\bar{z}-1/2)u}, & u \geq 0, \\ e^{(w-1/2)u}, & u \leq 0, \end{cases} \quad A_{z,w}(s) = \int_{\mathbb{R}} h_{z,w}(u) e^{-(s-1/2)u} du.\tag{HA7}$$

The second identity initially holds for $1 - \Re z < \Re s < \Re w$, including the closed critical strip; it then gives the meromorphic expression (HA5). The function h is continuous at zero, and its derivative jump is exactly -1 . Its distributional second derivative is the sum of two exponentially decreasing functions and $-\delta_0$. Neither this atom in the test derivative nor the supported-zero point is identified with the unsupported element τ .

The complete extension of the compact-test explicit formula to these tests is proved in the global heat zero-distribution calculation. Here is also the needed convergence argument. Choose d strictly between $1/2$ and $\min(\Re z, \Re w) - 1/2$. Cut h off with a smooth function equal to one on $[-R, R]$ and zero outside

$[-R-1, R+1]$, and convolve with a smooth approximate identity of radius $\varepsilon_R \leq 1$, where $\varepsilon_R \rightarrow 0$ as $R \rightarrow \infty$. The resulting smooth compact tests have uniformly bounded weighted L^1 norms of their value and first derivative and uniformly bounded weighted total variation of the second derivative, with weight $e^{|u|}$. This follows directly from (HA7), its derivative jump, the bounded cutoff derivatives and the compact support of the mollifier. Their transforms therefore obey a uniform bound $C/(1+|y|^2)$ for $0 \leq \Re s \leq 1$, by distributional integration by parts twice. Their pointwise limits give A . The unconditional zero count $N(R) = O(R \log(2+R))$, already proved in SZW25, makes the zero sums uniformly summable. The prime sum is dominated by $C \sum \Lambda(n) n^{-1/2-d}$, and the archimedean integral by $C \log(2+|y|)/(1+y^2)$. Endpoints converge by the same weighted L^1 estimate. Thus dominated convergence applies to every term of SZW25, retaining both endpoints and every support label.

3. Exact global pairing and all arithmetic terms at zero

Zeros are counted with their multiplicities. The rational zero trace is

$$K_t(z, w) = \sum_{g_t(\rho)=0} m_\rho A_{z,w}(\rho). \quad (\text{HA8})$$

For each fixed finite set of z, w there is a complex time disc around zero in which no pole of its tests is a zero of g_t . The full global Hadamard argument in the zero-distribution calculation proves convergence and holomorphic dependence on that disc, and gives

$$K_t(z, w) = \frac{L_t(w) - L_t(1 - \bar{z})}{q} = \frac{L_t(w) + L_t(\bar{z})}{q}. \quad (\text{HA9})$$

For real time the numerator is $L_t(w) + \overline{L_t(z)}$. The proof can be seen directly from the genus-one logarithmic derivative: subtract its values at the two poles. The constant term and each $1/\rho$ correction cancel; the remaining summand divided by q is exactly (HA5) at ρ . Absolute convergence follows from $A(\rho) = O(|\rho|^{-2})$ and the uniform zero count. Multiplicities are retained in this argument.

Introduce three distinct functions on $\Re s > 1$:

$$\begin{aligned} e_0(s) &= \frac{1}{s} + \frac{1}{s-1}, \\ b(s) &= -\frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2), \\ r(s) &= \frac{\zeta'(s)}{\zeta(s)} = -\sum_{n \geq 2} \Lambda(n) n^{-s}. \end{aligned} \quad L_0 = e_0 + b + r. \quad (\text{HA10})$$

The notation $e_0(s)$ is a scalar logarithmic-derivative function, not the programme element e . The endpoint, archimedean and finite-prime values for (HA7) are, separately,

$$\begin{aligned} E(z, w) &= A_{z,w}(0) + A_{z,w}(1) = \frac{1}{w(\bar{z}-1)} + \frac{1}{\bar{z}(w-1)} = \frac{e_0(w) + \overline{e_0(z)}}{q}, \\ G(z, w) &= \frac{b(w) + \overline{b(z)}}{q}, \\ P(z, w) &= \frac{1}{q} \sum_{n \geq 2} \Lambda(n) (n^{-w} + n^{-\bar{z}}) = -\frac{r(w) + \overline{r(z)}}{q}, \\ K_0(z, w) &= E(z, w) + G(z, w) - P(z, w). \end{aligned} \quad (\text{HA11})$$

The endpoint equality follows by common denominators, and the prime equality follows from (HA7) in SZW24. For completeness, the Gamma equality has a direct integral proof. The convergent digamma representation for $\Re a > 0$ is

$$\psi(a) = -\gamma + \int_0^\infty \frac{e^{-x} - e^{-ax}}{1 - e^{-x}} dx. \quad (\text{HA12})$$

It follows from the logarithmic derivative of Euler's product for Gamma: expand $(1 - e^{-x})^{-1}$, integrate finite partial sums, and use the cancellation at zero to obtain $-\gamma + \sum_{j \geq 0} (1/(j+1) - 1/(j+a))$. With the original Fourier convention, the archimedean integral is consequently

$$\begin{aligned} A_\infty(h) &= -(\gamma + \log \pi) h(0) \\ &\quad + \int_0^\infty \frac{e^{-x} h(0) - \frac{1}{2} e^{-x/4} (h(x/2) + h(-x/2))}{1 - e^{-x}} dx. \end{aligned} \quad (\text{HA13})$$

One may first truncate the x integral. Fourier inversion then proves the identity. Passing to the limit is legitimate: at zero the numerator is $O(x)$ by the Lipschitz property of h , and at infinity it decreases exponentially. On the Fourier side, the truncated integrals are bounded by $C \log(2 + |y|)$, uniformly in the cutoffs. To verify that bound, split at $x = 1/(1 + |y|)$ and at 1 and use $|1 - \cos(yx/2)| \leq \min(2, y^2 x^2/8)$. The remaining nonoscillatory difference is integrable independently of y . The bound on $A(1/2 + iy)$ proved above supplies domination. Substituting (HA7) in (HA13) gives (HA11) by (HA12), with every factor $1/2$ retained.

Let L now denote the finite support semilattice of SZW, with top 1_L ; this is distinct from the function $L_t(s)$. In the unchanged vector space with coordinate basis $\mathbf{e}_\lambda$, the full identity is

$$\begin{aligned} \mathbf{B}_L(z, w) &= E(z, w)\mathbf{e}_{1_L} + A_{z,w}(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda, \\ \mathbf{Z}_L(0; z, w) &= K_0(z, w)\mathbf{e}_{1_L}, \\ \mathbf{D}_L(0; z, w) &= (P(z, w) - G(z, w))\mathbf{e}_{1_L} + A_{z,w}(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda, \\ \mathbf{B}_L - \mathbf{Z}_L(0) &= \mathbf{D}_L(0). \end{aligned} \tag{HA14}$$

This is exactly SZW33–34 evaluated on (HA7). In particular, the lower fixed traces have not been dropped. Summing all coordinates and projecting to the top coordinate are different maps with the kernels proved in SZW35–37. The coefficient of each lower coordinate is generally not a Hermitian form by itself. The Hermitian positivity question below concerns the full reflected zero trace K_0 , as in SZW32 and SZW38, with (HA14) giving its exact supported arithmetic realization.

4. The actual derivative and its mixed prime products

Set $a = e_0 + b$, and define $v = a' + a^2$. For $\Re s > 1$, the exact series is

$$\frac{g_0''}{g_0}(s) = v(s) + \sum_{n \geq 2} [\Lambda(n) \log n + (\Lambda * \Lambda)(n) - 2a(s)\Lambda(n)] n^{-s}, \tag{HA15}$$

where $(\Lambda * \Lambda)(n) = \sum_{d|n} \Lambda(d)\Lambda(n/d)$ is ordered Dirichlet convolution. Indeed $g_0''/g_0 = L'_0 + L_0^2$, and inserting (HA10) gives (HA15). Each series, each required derivative and each product converges absolutely and locally uniformly on $\Re s > 1$, because $\Lambda(n) \leq \log n$ and $\sum (\log n)^j n^{-c} < \infty$ for every fixed $c > 1$.

It follows that the genuine time derivative is

$$\begin{aligned} J(s) &:= \partial_t L_t(s)|_{t=0} = \frac{1}{4} v'(s) + \frac{1}{4} \sum_{n \geq 2} C_n(s) n^{-s}, \\ C_n(s) &= -\Lambda(n)(\log n)^2 - (\Lambda * \Lambda)(n) \log n - 2a'(s)\Lambda(n) + 2a(s)\Lambda(n) \log n. \end{aligned} \tag{HA16}$$

This includes the endpoint–Gamma and endpoint–prime cross terms in a , not merely a heat deformation of the old prime weights. Equations (HA3) and (HA9) yield the exact global derivative

$$\boxed{\partial_t K_t(z, w)|_{t=0} = \frac{J(w) + \overline{J(z)}}{w + \bar{z} - 1}.} \tag{HA17}$$

It is a derivative of the actual infinite zero trace, not only a derivative of a finite set of zeros. The holomorphic finite-pole formula (HA9) supplies the justification; no unproved interchange with a moving infinite zero sum is used.

There is a useful exact distinction among arithmetic indices. If $n = p^k$, then

$$\Lambda(n) \log n + (\Lambda * \Lambda)(n) = (2k - 1)(\log p)^2. \tag{HA18}$$

The $k - 1$ ordered proper prime-power factorizations give the convolution. If $n = p^\alpha q^\beta$, where $p \neq q$ and $\alpha, \beta \geq 1$, then

$$\Lambda(n) = 0, \quad (\Lambda * \Lambda)(n) = 2 \log p \log q, \quad \frac{1}{4} C_n(s) = -\frac{1}{2} \log p \log q \log(p^\alpha q^\beta). \tag{HA19}$$

These are the two possible ordered factorizations into prime powers. With three or more distinct prime divisors both functions vanish. Thus the first actual heat derivative already contains products of two distinct

prime contributions. Formula (HA19) does not arise from replacing an ordinary prime by a new numerical norm for supported zero. It is forced by the nonlinear term $2L_0L'_0$ in the exact evolution.

To retain the whole support vector through this calculation, let $\Lambda_t(s) = g_t(s)/(s(s-1))$. At real time close to zero this function has simple poles at zero and one, since $g_0(0) = g_0(1) = 1$ and compact continuity keeps both values nonzero. Its logarithmic derivative is $L_t - e_0$, with residue -1 at each pole. Their residues as poles of Λ_t itself are $-g_t(0)$ and $g_t(1)$, and may vary; these two statements are distinct.

The logarithmic-divisor receiver at time t therefore has unchanged fixed endpoint weights and explicitly computed top coefficient

$$\begin{aligned} Z_L(t) &= K_t \mathbf{e}_{1_L}, \\ D_L(t) &= \left(E - \frac{L_t(w) + L_t(\bar{z})}{q} \right) \mathbf{e}_{1_L} + A_{z,w}(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda, \\ B_L - Z_L(t) &= D_L(t), \\ \dot{D}_L(0) &= -\frac{J(w) + \overline{J(z)}}{q} \mathbf{e}_{1_L}, \quad \dot{B}_L = 0. \end{aligned} \tag{HA20}$$

Here the first two lines are evaluated from the actual meromorphic heat function and its divisor. At zero, (HA11) identifies them with the original arithmetic formula; (HA16) computes their arithmetic first variation. No Euler product for g_t at nonzero time, or new theta-cohomology identification at such time, is asserted. The lower fixed distributions are the same explicit representations used in SZW, and their zero derivative is a calculation for these fixed tests, not a deletion of those distributions.

5. Exact relation to a colliding infinitesimal

Fix any actual zero ρ of g_0 , of multiplicity m , and write $g_0(s) = (s - \rho)^m u(s)$, $u(\rho) \neq 0$. On a small circle enclosing just this cluster, the residue theorem gives its weighted trace. Differentiating that fixed contour using (HA3) gives

$$\partial_t Z_{\rho,t}(A)|_0 = -\frac{m(m-1)}{4} A''(\rho) - \frac{m}{2} \frac{u'(\rho)}{u(\rho)} A'(\rho). \tag{HA21}$$

For proof, $g_0''/(4g_0)$ has principal part $\frac{m(m-1)}{4}(s - \rho)^{-2} + \frac{m}{2}(u'/u)(\rho)(s - \rho)^{-1}$. Its derivative has residue against A equal to the displayed expression. The contour derivative exists without choosing or differentiating individual colliding roots. Formula (HA21) retains the full analytic unit.

For a critical-line zero $\rho = 1 - \bar{\rho}$, take local tests F, G vanishing at ρ and put $A = F^\# G$. Then $A'(\rho) = 0$ and $A''(\rho) = -2F'(\rho)G'(\rho)$. Hence

$$\partial_t Z_{\rho,t}(F^\# G)|_0 = \frac{m(m-1)}{2} \overline{F'(\rho)} G'(\rho). \tag{HA22}$$

This is the exact positive first response on the first vanishing-jet layer of a critical collision. At an exchanged pair $\rho, \rho^\#$, the same calculation gives the cross pairing of the two derivative values; its Gram matrix is a positive multiple of $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, provided both tests vanish at both points and $m \geq 2$. Thus the actual first response preserves the distinction between a critical fixed point and an exchanged pair. If $m = 1$, this particular first-jet response vanishes, while (HA21) still retains the unit-dependent term for arbitrary tests.

For a finite rational test span $F = \sum_j c_j F_{z_j}$, the global quadratic form and derivative are exactly

$$Z_t(F^\# F) = \sum_{i,j} \bar{c}_i K_t(z_i, z_j) c_j, \quad \partial_t Z_t(F^\# F)|_0 = \sum_{i,j} \bar{c}_i \frac{J(z_j) + \overline{J(z_i)}}{z_j + \bar{z}_i - 1} c_j. \tag{HA23}$$

Every zero and every endpoint is retained by (HA8) and (HA14). A sign on (HA22), or on a coefficient of (HA19), does not assign a sign to (HA23). The complete criterion for the latter family at zero is proved in Cauchy–Weil positivity.

6. A positive actual diagonal response and its precise scope

The kernel permits an unconditional sign calculation on the real diagonal. For every real t , $\sigma > 1$, put $v = \sigma - 1/2 > 0$ and give $[0, \infty)$ the probability measure

$$d\mu_{t,v}(u) = \frac{e^{tu^2} \Phi(u) \cosh(2vu) du}{\int_0^\infty e^{ty^2} \Phi(y) \cosh(2vy) dy}. \tag{HA24}$$

The measure is positive: every summand of $\Phi(u)$ is $\pi n^2 e^{5u} (2\pi n^2 e^{4u} - 3) e^{-\pi n^2 e^{4u}} > 0$ for $u \geq 0$. Its density is strictly positive and all moments exist. Differentiation of the actual integral (HA1), with the compact domination already proved, gives

$$L_t(\sigma) = 2 \mathbb{E}_{\mu_{t,v}}[u \tanh(2vu)] > 0, \quad \partial_t L_t(\sigma) = 2 \operatorname{Cov}_{\mu_{t,v}}(u^2, u \tanh(2vu)) > 0. \quad (\text{HA25})$$

For the second identity differentiate a normalized integral. Its strict sign follows from

$$2 \operatorname{Cov}(f, g) = \iint (f(u) - f(y))(g(u) - g(y)) d\mu(u) d\mu(y). \quad (\text{HA26})$$

Both u^2 and $u \tanh(2vu)$ are strictly increasing on $(0, \infty)$. The integrand is positive away from the diagonal, and the positive continuous density gives positive mass to two disjoint intervals. Thus the covariance is strictly positive, not merely nonnegative.

Consequently $K_t(\sigma, \sigma) = 2L_t(\sigma)/(2\sigma - 1) > 0$ and its actual time derivative is strictly positive for every real time. The rational trace identity applies at every such fixed real time: its two real poles are not zeros because the same positive integral makes $g_t(\sigma) > 0$ and reflection gives $g_t(1 - \sigma) > 0$. Its growth and Hadamard argument are local on arbitrary bounded time intervals. This calculation keeps the original measure and all coefficients; it is not positivity imposed on an auxiliary model.

The result identifies precisely what the real diagonal observes. It does not control the off-diagonal matrix entries. In particular, positive values and a positive first derivative on this diagonal also occur at negative heat times. Rodgers–Tao’s original Theorem `main`, together with the defining real-zero threshold stated in their introduction, proves that those negative times do have nonreal zeros in the original Z coordinate. Thus this actual diagonal positivity is strictly weaker than real-rootedness of the actual heat family. The complete matrix criterion retains the missing off-diagonal data.

Sources and exact programme maps

The human heat source is Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, arXiv:1801.05914v5, original author source `fmp-template.tex`, equations `hoz`, `sas`, `phidef`, `htdef`. The original archive is retained intact. The programme uses the minus-exponent Mellin convention of SZW19–SZW38, including its proof of the full explicit formula and every labelled trace. Its human explicit-formula source is Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Appendix II, Theorem 6, as read and applied there. The current finite-pole heat trace, the matrix criterion and the arithmetic calculation have complete separate proofs linked above; none of those links identifies a pending result as an already published edition.

Exact formula diagram

Exact full-form continuation

The reflection-index theorem, NI1–32, calculates the signed index of every complete Cauchy coefficient tail. The finite-jet and endpoint proof, FJ1–29, preserves the exact finite jets while proving density on the remaining zeros, derives the full exterior index and tail, and calculates the second and third actual heat coefficients. All supported endpoints remain explicit. These results identify the remaining sign problem without asserting its resolution.

One fixed primitive test and the entire prime window

The prime-operator calculation, PM1–41, proves both distinct Fourier support corrections and their exact map into the full Weil formula. The local estimate, PS1–21, gives a positive translation-difference remainder. The fixed-test proof, UP1–35, constructs one compact test whose transform is nonzero at every possible off-critical zero. The complete arithmetic calculation, PW1–22, expresses its entire translated correlation through a fixed-width prime window and an explicit archimedean remainder. Its boundedness is equivalent to RH; that bound remains unresolved. All boundary coordinates and supported-zero labels remain explicit.

The actual heat pairing retains every endpoint

Original coordinates: $g_t(s) = 16H_t(-2i(s - 1/2))$, $g_0 = 2\xi_R$, $\text{Re } z, \text{Re } w > 1$.

$\begin{aligned} \partial_t g_t &= \frac{1}{4} g_t'' \\ L_t &= g_t'/g_t \\ \partial_t L_t &= \frac{1}{4} (L_t'' + 2L_t L_t') \end{aligned}$	→	$K_t(z, w) = \sum_p m_p F_z^\#(\rho) F_w(\rho) = \frac{L_t(w) + \bar{L}_t(z)}{w + \bar{z} - 1}$ $F_z(s) = 1/(s - z), \quad F^\#(s) = \overline{F(1 - \bar{s})}$ All zeros and multiplicities; real time in the fixed-pole zero-free interval.
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At time zero: three separately evaluated terms

Supported-zero endpoints $E = \frac{1}{w(z-1)} + \frac{1}{\bar{z}(w-1)}$	Archimedean term $G = \frac{b(w) + \bar{b}(\bar{z})}{w + \bar{z} - 1}$	Finite primes $P = \frac{\sum_{n \geq 2} \Lambda(n)(n^{-w} + n^{-\bar{z}})}{w + \bar{z} - 1}$
--	--	---

$$K_0 = E + G - P, \quad b(s) = -\frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2)$$

$\begin{aligned} \mathbf{B}_L &= E \mathbf{e}_{1_L} + A(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda, \quad \mathbf{Z}_L = K_0 \mathbf{e}_{1_L}, \quad \mathbf{D}_L = (P - G) \mathbf{e}_{1_L} + A(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda \\ \mathbf{B}_L - \mathbf{Z}_L &= \mathbf{D}_L \quad \text{— each lower support coordinate remains present.} \end{aligned}$

First heat response: a product of two distinct prime contributions

$\begin{aligned} p^\alpha, q^\beta, \quad p \neq q \\ \Lambda(p^\alpha) &= \log p \\ \Lambda(q^\beta) &= \log q \end{aligned}$	→	$\begin{aligned} n &= p^\alpha q^\beta: \quad (\Lambda * \Lambda)(n) = 2 \log p \log q \\ [n^{-s}] \dot{L}_0(s) &= -\frac{1}{2} \log p \log q \log(p^\alpha q^\beta) \\ \dot{K}_0(z, w) &= \frac{\dot{L}_0(w) + \bar{\dot{L}}_0(\bar{z})}{w + \bar{z} - 1} \end{aligned}$
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The displayed mixed coefficient is exact. Its sign alone does not determine the sign of the full pairing.

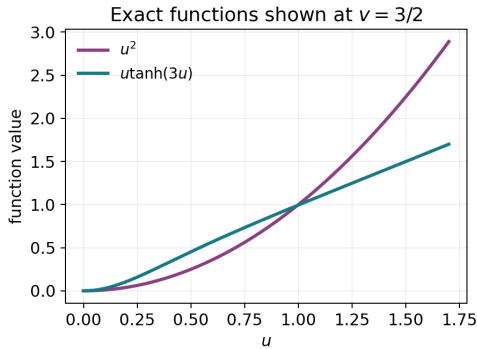
Proofs: HA1–HA20; original heat source: Rodgers–Tao, arXiv:1801.05914v5, hoz / htdef. Full support: SZW33–34.

Figure 16: The actual heat coordinates, complete supported arithmetic identity, and first mixed-prime coefficient. Every constant and support coordinate is retained; full proofs HA1–HA20. Here $A = F_z$ sharp times F_w as defined in HA5.

A positive response of the original heat measure

Every real t , every $\sigma > 1$, and $v = \sigma - 1/2$: full proof HA24–HA26.

$d\mu_{t,v}(u) = \frac{e^{tu^2} \Phi(u) \cosh(2vu) du}{\int_0^\infty e^{ty^2} \Phi(y) \cosh(2vy) dy}$ $\Phi(u) > 0, \quad L_t(\sigma) = 2 \int_{\mathbb{R}} \mu[utanh(2vu)]$
--



Positive real diagonal values occur at negative heat times too; full matrix positivity also tests off-diagonal entries.

$\begin{aligned} \partial_t L_t(\sigma) &= 2 \text{Cov}_\mu(u^2, utanh(2vu)) > 0 \\ K_t(\sigma, \sigma) &> 0, \quad \partial_t K_t(\sigma, \sigma) > 0 \end{aligned}$

Both functions increase strictly.
The double-integral covariance formula proves the sign.

Figure 17: The original positive heat measure and the strict covariance calculation. The two plotted functions are shown at the stated value $v=3/2$; the theorem and exact integral hold for every $v=\sigma-1/2$ with σ greater than one. Full proof HA24–HA26.

Every time derivative of the actual heat logarithmic derivative

The original family is $g_t(s) = 16H_t(-2i(s - 1/2))$, with $\partial_t g_t = \frac{1}{4}\partial_s^2 g_t$. Its coordinate, analytic convergence, and complete supported arithmetic receiver are proved in HA1–HA26. This note computes its time derivatives without replacing any Gamma or endpoint coefficient by a constant.

1. The exact differential-polynomial recurrence

Let $\mathcal{P} = \mathbb{Q}[X_0, X_1, \dots]$ be the polynomial ring in countably many variables; each polynomial uses only finitely many of them. Define commuting spatial and evolutionary derivations by

$$DX_j = X_{j+1}, \quad VX_j = D^j \left(\frac{X_2 + 2X_0X_1}{4} \right). \quad (\text{HM1})$$

Both extend uniquely by the Leibniz rule and linearity. Their commutator is a derivation and is zero on every generator, because $DVX_j = VDX_j$; therefore $DV = VD$. Put

$$B_0 = X_0, \quad B_{k+1} = VB_k = \sum_{j \geq 0} \frac{\partial B_k}{\partial X_j} D^j \left(\frac{X_2 + 2X_0X_1}{4} \right). \quad (\text{HM2})$$

The sum is finite for each k . Where $g_t \neq 0$, set $L_t = g'_t/g_t$. HA3 and the chain rule prove by induction the exact identity

$$\partial_t^k L_t(s) = B_k(L_t(s), L'_t(s), L''_t(s), \dots). \quad (\text{HM3})$$

These are derivatives, not coefficients of a Taylor series; the Taylor coefficient is $B_k/k!$. Each formula is valid as a meromorphic identity in s and locally holomorphically in t away from its zeros.

Give every X_j polynomial degree one. The derivation D preserves this degree. Split $V = V_1 + V_2$, where

$$V_1X_j = \frac{1}{4}X_{j+2}, \quad V_2X_j = \frac{1}{2}D^j(X_0X_1). \quad (\text{HM4})$$

Thus V_1 preserves homogeneous degree and V_2 increases it by one. In particular $\deg B_k \leq k + 1$. Its homogeneous component of degree $k + 1$ is exactly

$$\boxed{B_k^{[k+1]} = \frac{1}{2^k(k+1)} D^k(X_0^{k+1})}. \quad (\text{HM5})$$

Here is the full induction. At $k = 0$ the formula gives X_0 . As with V , the derivation V_2 commutes with D . The only term capable of producing degree $k + 2$ in B_{k+1} is $V_2B_k^{[k+1]}$. Applying the induction hypothesis gives

$$\begin{aligned} V_2B_k^{[k+1]} &= \frac{1}{2^k(k+1)} D^k \left((k+1)X_0^k \frac{X_0X_1}{2} \right) \\ &= \frac{1}{2^{k+1}(k+2)} D^{k+1}(X_0^{k+2}), \end{aligned} \quad (\text{HM6})$$

as required. This polynomial is nonzero; its monomial $X_0^k X_k$ has a positive coefficient. Hence $\deg B_k = k + 1$, not only the upper bound.

For clarity the first two nonconstant polynomials are

$$\begin{aligned} B_1 &= \frac{1}{4}(X_2 + 2X_0X_1), \\ B_2 &= \frac{1}{16}(X_4 + 8X_1X_2 + 4X_0X_3 + 8X_0X_1^2 + 4X_0^2X_2). \end{aligned} \quad (\text{HM7})$$

They follow by one and two direct applications of (HM2). In particular the coefficient of X_1X_2 is eight; no factor is absorbed into a change of heat time.

2. The specified arithmetic expansion

On $\Re s > 1$, retain the exact split

$$\begin{aligned} L_0(s) &= a(s) + r(s), \\ a(s) &= \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2), \\ r(s) &= - \sum_{n \geq 2} \Lambda(n) n^{-s}. \end{aligned} \quad (\text{HM8})$$

The coefficient functions $a, a', \dots$ are retained. The expansion meant here has an explicit rule: in B_k substitute $X_j = a^{(j)} + r^{(j)}$, expand each monomial, and multiply the absolutely convergent Dirichlet series in the $r^{(j)}$ factors by ordered Dirichlet convolution. A term with no r factor remains its original coefficient function. This is a specified representation; no uniqueness claim is made for expansions with arbitrary s -dependent coefficients.

Every factor $r^{(j)}$ has series

$$r^{(j)}(s) = - \sum_{n \geq 2} \Lambda(n) (-\log n)^j n^{-s}. \quad (\text{HM9})$$

Each derivative converges absolutely and locally uniformly on the half-plane. Finite products do too, because their absolute multiple sums are bounded by the product of the convergent single sums. Thus the expansion rule and every spatial derivative used below are justified there.

Since each nonzero coefficient of one factor is supported on prime powers, a product of j such factors is supported on integers with at most j distinct prime factors. By (HM4)–(HM5), the arithmetic coefficient at an integer with more than $k + 1$ distinct prime factors is zero in this specified expansion of $\partial_t^k L_t|_0$.

For an integer with exactly $k + 1$ distinct prime factors, write

$$n = \prod_{i=1}^{k+1} p_i^{\alpha_i}, \quad p_i \neq p_j \ (i \neq j), \quad \alpha_i \geq 1. \quad (\text{HM10})$$

Only the homogeneous component of degree $k + 1$, with every factor chosen from r , can contribute. Any a factor would reduce the number of available prime-power factors. Therefore its coefficient is the coefficient in

$$\frac{1}{2^k(k+1)} \partial_s^k (r(s)^{k+1}). \quad (\text{HM11})$$

To compute it, each of the $k + 1$ ordered factors must receive one of the distinct prime powers in (HM10). There are exactly $(k + 1)!$ such assignments. No exponent can split between two factors, because that would leave a different prime without a factor. Hence the coefficient of r^{k+1} at n is $(-1)^{k+1} (k + 1)! \prod_i \log p_i$. Applying ∂_s^k multiplies it by $(-\log n)^k$. The two signs combine to minus, proving

$$\boxed{[n^{-s}] \partial_t^k L_t(s)|_0 = - \frac{k!}{2^k} (\log n)^k \prod_{i=1}^{k+1} \log p_i.} \quad (\text{HM12})$$

The bracket refers exactly to the expansion rule after (HM8). The displayed coefficient has no remaining $a(s)$ dependence. At lower distinct-prime counts all coefficient functions prescribed by that rule remain present.

For $k = 0$, (HM12) is the original coefficient $-\log p$ at p^α . For $k = 1$ it is $-\frac{1}{2} \log p \log q \log(p^\alpha q^\beta)$, agreeing with HA19. For $k = 2$ it is $-\frac{1}{2} (\log n)^2 \log p \log q \log r$ at three distinct prime factors. Formula (HM12) concerns a time derivative; dividing by $k!$ gives the coefficient in the actual Taylor series in time.

3. The global supported receiver

For the original Cauchy tests at z, w in $\Re s > 1$, the exact global family of HA5–HA20 gives, for every k ,

$$\partial_t^k K_t(z, w)|_0 = \frac{B_k(L_0, L'_0, \dots)(w) + \overline{B_k(L_0, L'_0, \dots)}(z)}{w + \bar{z} - 1}. \quad (\text{HM13})$$

Indeed the finite-pole formula is holomorphic on a time disc for this finite test set, so it may be differentiated any finite number of times; (HM3) supplies the derivatives. All coefficients of B_k are real. The fixed supported boundary and every lower coordinate have zero time derivative for $k \geq 1$. The corresponding geometric-divisor response is the negative of (HM13) in the top coordinate, with no suppressed lower distribution, by HA20.

This constructs actual arithmetic mixed-prime data at every infinitesimal order. The negative coefficient in (HM12) is not the value of a quadratic form: the matrix denominator, conjugate counterpart, remaining coefficients, Gamma factors and endpoints in (HM13) are still present. The complete sign question is the Cauchy–Weil matrix criterion.

Verification and sources

The accompanying `check_heat_mixed_prime_recurrence.py` computes the entire recurrence through order six in the differential polynomial ring, checks its exact degree, and compares every highest-degree component with (HM5). Its monomial counts are 1, 2, 5, 11, 23, 44 and 82. These finite checks supplement the all-order proof (HM4)–(HM6).

The human heat source remains Brad Rodgers and Terence Tao, arXiv:1801.05914v5, original author TeX, equations `hoz`, `htdef` and the subsequent heat equation. The full supported explicit formula is SZW19–SZW38, with its original human sources retained. The logarithmic-derivative equation is the classical Burgers transformation of the heat equation; its exact normalization and every recurrence used here have been proved above. No novelty claim for that transformation is made.

The complete Cauchy–Weil matrix criterion

This note proves an equivalence with the classical Riemann hypothesis. It does not assume or establish positivity of the matrices occurring in that equivalence. The full supported-zero arithmetic evaluation of those matrices is HA10–HA20; every endpoint and lower support coordinate remains there.

1. The original kernel and zero sum

Put $g = 2\xi_R$, $L = g'/g$, and $\mathcal{U} = \{z : \Re z > 1\}$. Define

$$K(z, w) = \frac{L(w) + \overline{L(z)}}{w + \bar{z} - 1}, \quad z, w \in \mathcal{U}. \quad (\text{CK1})$$

The unconditional Euler product and functional equation imply that g is nonzero on $\mathcal{U}$, so this definition requires no RH assumption. The exact global trace identity AG10–AG11 is

$$K(z, w) = \sum_{\rho} m_{\rho} \overline{F_z(\rho^{\#})} F_w(\rho), \quad F_z(s) = \frac{1}{s - z}, \quad \rho^{\#} = 1 - \bar{\rho}. \quad (\text{CK2})$$

The sum converges absolutely and keeps all multiplicities. The reflection and Hadamard proof, with the zero count, is given in AG1–AG11.

A kernel is positive semidefinite on a set when, for every integer $n \geq 1$, every finite tuple of points in that set, and every $c \in \mathbb{C}^n$, its quadratic sum $\sum_{i,j} \bar{c}_i K(z_i, z_j) c_j$ is nonnegative. Under RH each $\rho^{\#} = \rho$, so (CK2) gives

$$\sum_{i,j} \bar{c}_i K(z_i, z_j) c_j = \sum_{\rho} m_{\rho} \left| \sum_j c_j F_{z_j}(\rho) \right|^2 \geq 0. \quad (\text{CK3})$$

This proves one direction. The converse is proved completely below, including the extension of a kernel known only on the observed subdomain.

2. A self-contained Schur extension from a positive kernel

The positive-kernel Hilbert construction and the transfer-function realization used in this section are classical. A primary-source treatment is Joseph A. Ball, Animikh Biswas, Quanlei Fang and Sanne ter Horst, *Multi-variable generalizations of the Schur class: positive kernel characterization and transfer function realization*, arXiv:0705.2042v3, §1, Theorem 1.1(2)–(3) and Proposition 1.2. Their proof discussion of Theorem 1.1, (2) $\Rightarrow$ (3), identifies the equal-Gram construction as the standard lurking-isometry argument. The source orders its Hilbert sum as $\mathcal{H} \oplus \mathbb{C}$, whereas CK5–CK7 use $\mathbb{C} \oplus \mathcal{H}$. The unitary swap $J(\alpha, h) = (h, \alpha)$ sends $V = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ to $JVJ^{-1} = \begin{pmatrix} d & c \\ b & a \end{pmatrix}$. Thus the source’s blocks are $(A, B, C, D) = (d, c, b, a)$, and its transfer formula $D + zC(I - zA)^{-1}B$ is exactly CK7. In the scalar case its kernel $K_f(y, x)$ is CK4’s $S(x, y)$, retaining the conjugate-linear-first convention. The cited theorem is stated on the full disc; the arbitrary-subset extension needed here is proved directly below, rather than asserted to be an additional theorem read in that source.

Let E be any nonempty subset of the unit disc, and let $f : E \rightarrow \mathbb{C}$ satisfy positivity of

$$S(x, y) = \frac{1 - \overline{f(x)}f(y)}{1 - \bar{x}y}. \quad (\text{CK4})$$

We construct a holomorphic function $\tilde{f}$ on the whole unit disc, of modulus at most one, agreeing with f on E . First construct a Hilbert space $\mathcal{H}$ from formal vectors v_x , setting $\langle v_x, v_y \rangle = S(x, y)$, with the inner product conjugate-linear in its first variable. On finite linear combinations this is a nonnegative sesquilinear form. Cauchy–Schwarz for such a form follows from positivity of the quadratic polynomial in a complex scalar; consequently its zero-length vectors form its radical. Quotient by this radical and complete to obtain $\mathcal{H}$.

The identity in (CK4) gives equal Gram matrices for the following two families in $\mathbb{C} \oplus \mathcal{H}$:

$$\begin{pmatrix} 1 \\ xv_x \end{pmatrix} \mapsto \begin{pmatrix} f(x) \\ v_x \end{pmatrix}, \quad 1 + \bar{x}yS(x, y) = \overline{f(x)}f(y) + S(x, y). \quad (\text{CK5})$$

The rule therefore defines a well-defined isometry on their algebraic span and extends to its closure $\mathcal{M}$. Extend it to a contraction V on $\mathbb{C} \oplus \mathcal{H}$ by setting it equal to zero on $\mathcal{M}^{\perp}$. Write its bounded blocks as

$$V = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \|V\| \leq 1. \quad (\text{CK6})$$

Here $a \in \mathbb{C}$, $b : \mathcal{H} \rightarrow \mathbb{C}$, $c : \mathbb{C} \rightarrow \mathcal{H}$ is identified with $c(1)$, and $d : \mathcal{H} \rightarrow \mathcal{H}$. Equation (CK5) says $a + xbv_x = f(x)$ and $c + xdv_x = v_x$. Since $\|d\| \leq 1$, the Neumann series makes $I - zd$ invertible for every $|z| < 1$. Define

$$h(z) = (I - zd)^{-1}c, \quad \tilde{f}(z) = a + zb(I - zd)^{-1}c. \quad (\text{CK7})$$

The series makes both holomorphic and gives $h(x) = v_x$, $\tilde{f}(x) = f(x)$. Moreover $V(1, zh(z)) = (\tilde{f}(z), h(z))$. Contractivity yields

$$|\tilde{f}(z)|^2 + \|h(z)\|^2 \leq 1 + |z|^2\|h(z)\|^2, \quad 1 - |\tilde{f}(z)|^2 \geq (1 - |z|^2)\|h(z)\|^2 \geq 0. \quad (\text{CK8})$$

This proves the required extension without invoking an unproved interpolation theorem. It is the classical Schur transfer construction, with its spaces and maps specified here; no novelty claim for the construction is made.

3. Positive Cauchy matrices exclude every zero off the line

Write $v = s - 1/2$. The half-plane $\Re s > 1/2$ is carried bijectively to the unit disc by

$$x = \frac{v-1}{v+1} = \frac{s-3/2}{s+1/2}, \quad s = \frac{1}{2} + \frac{1+x}{1-x}. \quad (\text{CK9})$$

Let U be any nonempty open subset of $\mathcal{U}$ on which K is positive semidefinite. Its diagonal gives $2\Re L(s)/(2\Re s - 1) \geq 0$, hence $\Re L(s) \geq 0$ on U . In particular $L(s) + 1 \neq 0$. Put

$$f(x) = \frac{L(s) - 1}{L(s) + 1}, \quad x = x(s), \quad s \in U. \quad (\text{CK10})$$

The exact kernel transformation is

$$\frac{1 - \overline{f(x(z))}f(x(w))}{1 - \overline{x(z)}x(w)} = \overline{a(z)}K(z, w)a(w), \quad a(s) = \frac{s + 1/2}{L(s) + 1}. \quad (\text{CK11})$$

Indeed the numerator before division is $2(\overline{L(z)} + L(w))/((\overline{L(z)} + 1)(L(w) + 1))$; the disc denominator is $2(w + \bar{z} - 1)/((\bar{z} + 1/2)(w + 1/2))$. Their quotient gives (CK11) exactly. A diagonal congruence preserves kernel positivity. Section 2 therefore extends f to a Schur function $\tilde{f}$ on the whole unit disc.

The extended function cannot take the value 1 inside the disc. Otherwise the maximum modulus principle makes it identically 1, contradicting (CK10) at any point of U . Thus

$$\tilde{L}(s) = \frac{1 + \tilde{f}(x(s))}{1 - \tilde{f}(x(s))} \quad (\text{CK12})$$

is holomorphic on $\Re s > 1/2$ and equals the original L on U . The holomorphic function $g' - \tilde{L}g$ on this connected half-plane vanishes on U , so the identity theorem makes it zero throughout. If g had a zero ρ of positive multiplicity m there, then g' would have order $m - 1$ at ρ , while $\tilde{L}g$ has order at least m . This contradicts the identity. Hence g has no zeros in $\Re s > 1/2$. The functional equation $g(1 - s) = g(s)$ excludes zeros in $\Re s < 1/2$ as well. Its zeros are exactly the nontrivial zeta zeros, proving RH.

Together with (CK3) this proves the exact equivalence

$$\begin{aligned} \text{RH} &\iff K \text{ is positive semidefinite on } \mathcal{U} \\ &\iff K \text{ is positive semidefinite on } U. \end{aligned}$$

(CK13)

The last phrase means that for each fixed such U , its complete positivity condition is equivalent to RH. It does not replace all finite matrices by a single diagonal inequality.

4. One real point, with every Taylor coefficient retained

Fix a real $\sigma > 1$, write $d = 2\sigma - 1$, and set $\ell_n = L^{(n)}(\sigma)/n!$. All ℓ_n are real. Define the infinite Hermitian coefficient matrix

$$C_{jk} = [X^j Y^k] \frac{L(\sigma + X) + L(\sigma + Y)}{d + X + Y}, \quad j, k \geq 0. \quad (\text{CK14})$$

The bracket denotes the convergent Taylor expansion near $X = Y = 0$, not a truncation of the underlying function. Expanding the denominator geometrically gives the explicit coefficients

$$C_{jk} = \sum_{n=0}^j \ell_n \frac{(-1)^{j+k-n} \binom{j+k-n}{k}}{d^{j+k-n+1}} + \sum_{n=0}^k \ell_n \frac{(-1)^{j+k-n} \binom{j+k-n}{j}}{d^{j+k-n+1}}. \quad (\text{CK15})$$

In particular, ℓ_0 occurs twice, as it must in the numerator of (CK14).

The entries are exactly the original rational-jet pairings

$$C_{jk} = Z_0((Q_j)^\# Q_k), \quad Q_j(s) = \frac{1}{(s - \sigma)^{j+1}}. \quad (\text{CK16})$$

To verify this, $F_{\sigma+Y}(s) = \sum_{k \geq 0} Y^k / (s - \sigma)^{k+1}$ on any sufficiently small parameter disc at each zero. The analogous reflected expansion uses X . Derivatives may be taken in (CK2) locally: its large-zero tail, and every parameter derivative of that tail, are uniformly summable by the $O(|\rho|^{-2})$ bound and the fixed distance of the poles from the critical strip. The finitely many remaining terms are analytic. Coefficient comparison now gives (CK16), with no missing factorial.

We prove the complete single-point criterion:

$$\boxed{\text{RH} \iff (C_{jk})_{0 \leq j, k \leq N} \succeq 0 \text{ for every integer } N \geq 0.} \quad (\text{CK17})$$

Under RH, (CK16) is a positive Gram matrix by (CK3). Conversely suppose every indicated finite section is positive. Choose a small disc $|z - \sigma| < r$ contained in $\mathcal{U}$, on whose product the double Taylor series in (CK14) converges absolutely. For finitely many points z_i in a smaller such disc and coefficients b_i , put $v_j = \sum_i b_i (z_i - \sigma)^j$. Positivity of the finite sections gives $\sum_{j,k=0}^N \bar{v}_j C_{jk} v_k \geq 0$. Absolute convergence allows $N \rightarrow \infty$, yielding $\sum_{i,l} \bar{b}_i K(z_i, z_l) b_l \geq 0$. Thus K is positive on a nonempty open disc, and (CK13) proves RH. This keeps the entire coefficient tower; no finite N is claimed sufficient.

The first two-by-two section retains a useful exact determinant:

$$\begin{pmatrix} C_{00} & C_{01} \\ C_{10} & C_{11} \end{pmatrix} = \begin{pmatrix} 2\ell_0/d & \ell_1/d - 2\ell_0/d^2 \\ \ell_1/d - 2\ell_0/d^2 & -2\ell_1/d^2 + 4\ell_0/d^3 \end{pmatrix}, \quad \det = \frac{4\ell_0^2 - d^2\ell_1^2}{d^4}. \quad (\text{CK18})$$

This follows by expanding the displayed determinant; no derivative of order two enters this particular mixed coefficient. Higher sections do retain higher derivatives through (CK15).

5. The actual heat response of the whole coefficient tower

Let $L_t = g'_t/g_t$ for the original heat family. At the fixed real $\sigma > 1$, the positive integral makes $g_t(\sigma) > 0$ for every real time. Define $C_{jk}(t)$ by the same local Taylor formula as (CK14). For each fixed time and each finite coefficient set the function is holomorphic on a time disc; the proof is AG1–AG11. Therefore

$$\partial_t^k C_{ij}(t)|_0 = [X^i Y^j] \frac{B_k(L_0, L'_0, \dots)(\sigma + X) + B_k(L_0, L'_0, \dots)(\sigma + Y)}{d + X + Y}, \quad (\text{CK19})$$

where the full recurrence and all mixed-prime coefficients are proved in HM1–HM13. Equation (CK19) is the exact map from the arithmetic heat derivative to the complete positivity tower. The rational tests in (CK16) carry the supported endpoints and lower fixed coordinates of SZW through AG16; no coordinate of that identity is removed to define the tower.

HA24–HA26 proves that $C_{00}(t) > 0$ and $\partial_t C_{00}(t) > 0$ for every real time, including negative times when the actual heat function has nonreal zeros. Thus positivity of that one entry is a proved property with a strictly smaller scope than (CK17). The remaining matrix entries and their exact response (CK19) are concrete mathematical data retained by the criterion.

Method and sources

The transfer construction CK4–CK8 is a complete proof of the classical Schur/Pick extension method used here, rather than an appeal to an unstated interpolation hypothesis. Jensen, Hadamard and the original zeta

facts required by CK1–CK3 are proved or invoked with verified hypotheses in AG and SZW. The original human explicit-formula source remains Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Appendix II Theorem 6, as read and applied in SZW19–SZW38. Rodgers–Tao’s original heat coordinates, source locators and archive are retained in AG and HA. No historical novelty claim for equivalent RH criteria or classical transfer methods is made. The exact in-programme contribution is the complete supported arithmetic and original-heat maps into this specified matrix family.

Ball, Joseph A.; Biswas, Animikh; Fang, Quanlei; ter Horst, Sanne. *Multivariable generalizations of the Schur class: positive kernel characterization and transfer function realization*. arXiv:0705.2042v3, 2 November 2007. <https://arxiv.org/abs/0705.2042v3> . Original author TeX inspected, §1, Theorem 1.1, Proposition 1.2 and the proof discussion through (2) $\Rightarrow$ (3).

The inspected treatment attributes the scalar positive-kernel Hilbert-space construction to N. Aronszajn, *Theory of reproducing kernels*, Transactions of the American Mathematical Society 68 (1950), 337–404. It cites J. A. Ball, *Linear systems, operator model theory and scattering: multivariable generalizations*, in *Operator Theory and Its Applications (Winnipeg, MB, 1998)*, Fields Institute Communications 25, American Mathematical Society, 2000, 151–178, for the lurking-isometry terminology. These two original works were not inspected for this repair; this historical attribution is through the inspected Ball–Biswas–Fang–ter Horst source. No historical novelty for these classical methods is claimed.

One fixed primitive test and the entire prime window

The prime-operator calculation, PM1–41, proves both distinct Fourier support corrections and their exact map into the full Weil formula. The local estimate, PS1–21, gives a positive translation-difference remainder. The fixed-test proof, UP1–35, constructs one compact test whose transform is nonzero at every possible off-critical zero. The complete arithmetic calculation, PW1–22, expresses its entire translated correlation through a fixed-width prime window and an explicit archimedean remainder. Its boundedness is equivalent to RH; that bound remains unresolved. All boundary coordinates and supported-zero labels remain explicit.

The exact reflection index of the global Cauchy–Weil pairing

This calculation determines the full negative index of the actual time-zero Cauchy kernel and of its complete coefficient tower at one real pole. The answer is the number of distinct two-point orbits of the actual zero set under reflection in the critical line. That number is allowed to be zero, a finite positive integer, or infinity; the calculation does not assert which alternative occurs. Multiplicity, every local jet, and the full supported arithmetic receiver remain explicit.

The inputs are the complete proofs AG1–AG24, CK1–CK19, and HA1–HA26. All three files were read in full for this derivation. In particular, the convergence below uses the proved actual zero count, and the support calculation uses the full arithmetic identity rather than a replacement of its endpoint terms.

1. The actual zero set, its weights, and reflection

Put

$$g(s) = 2\xi_R(s), \quad L_g(s) = g'(s)/g(s), \quad \mathcal{U} = \{z \in \mathbb{C} : \Re z > 1\}. \quad (\text{NI1})$$

Let $\mathcal{Z}$ be the set of **distinct** zeros of g , and let $m_\rho \geq 1$ be the order of its zero at ρ . The original functional equation and real coefficients give an involution

$$\rho \mapsto \rho^\# = 1 - \bar{\rho}, \quad m_{\rho^\#} = m_\rho. \quad (\text{NI2})$$

The zeros lie in $0 < \Re \rho < 1$, are discrete in $\mathbb{C}$, and have no finite accumulation point. AG4–AG5 gives

$$\sum_{|\rho| \leq R} m_\rho = O(R \log(R+2)), \quad \sum_{|\rho| > R} \frac{m_\rho}{|\rho|^2} = O\left(\frac{\log(R+2)}{R}\right). \quad (\text{NI3})$$

In particular, $\sum_\rho m_\rho/(1+|\rho|^2) < \infty$. Let $\mathcal{Z}_{\text{fix}}$ be the fixed-point set of (NI2), and choose one representative for each two-point orbit to form $\mathcal{P}$. Define

$$\kappa_- := \#\mathcal{P}, \quad \kappa_+ := \#\mathcal{Z}_{\text{fix}} + \#\mathcal{P}, \quad (\text{NI4})$$

where each count is an element of $\{0, 1, 2, \dots, \infty\}$. These are counts of distinct supports, not sums of the multiplicities. A usual off-line quartet with nonzero imaginary part contains two orbits of (NI2), one at each sign of the imaginary part.

Define the Hilbert space, with inner product conjugate-linear in its first argument,

$$\mathcal{H} = \ell^2(\mathcal{Z}, m), \quad \langle a, b \rangle_{\mathcal{H}} = \sum_{\rho \in \mathcal{Z}} m_\rho \bar{a}_\rho b_\rho. \quad (\text{NI5})$$

The reflection operator and the associated Hermitian form are

$$(Ja)_\rho = a_{\rho^\#}, \quad J^2 = I, \quad J^* = J, \quad \|J\| = 1, \quad [a, b] = \langle Ja, b \rangle_{\mathcal{H}}. \quad (\text{NI6})$$

These operator assertions follow by changing the summation variable in (NI5) using (NI2). The zero space, if present, has the same formulas with the evident norm convention. In particular the form is nondegenerate: if $[a, b] = 0$ for all $b \in \mathcal{H}$, then taking $b = Ja$ gives $\|a\|^2 = 0$.

For each fixed point, $\delta_\rho/\sqrt{m_\rho}$ is a unit vector with J -eigenvalue $+1$. For each representative $\rho \in \mathcal{P}$, the two unit vectors

$$u_\rho^+ = \frac{\delta_\rho + \delta_{\rho^\#}}{\sqrt{2m_\rho}}, \quad u_\rho^- = \frac{\delta_\rho - \delta_{\rho^\#}}{\sqrt{2m_\rho}} \quad (\text{NI7})$$

have eigenvalues $+1$ and -1 , respectively. The full collection is an orthonormal basis, since it is the indicated invertible change of basis on each disjoint one- or two-point support. Thus

$$\mathcal{H} = \mathcal{H}_+ \hat{\oplus} \mathcal{H}_-, \quad P_\pm = (I \pm J)/2, \quad [a, a] = \|P_+ a\|^2 - \|P_- a\|^2, \quad \dim \mathcal{H}_\pm = \kappa_\pm. \quad (\text{NI8})$$

Here an infinite dimension means countably infinite Hilbert dimension. In particular, every distinct exchanged pair contributes exactly one negative direction with its original positive multiplicity weight.

2. The exact Cauchy evaluation map

For $z \notin \mathcal{Z}$, put

$$v_z(\rho) = \frac{1}{\rho - z}. \quad (\text{NI9})$$

This is an element of $\mathcal{H}$. Indeed, the terms with $|\rho| \leq 2|z| + 1$ are finite and have nonzero denominators, while on the remaining terms $|\rho - z| \geq |\rho|/2$, so (NI3) applies. On every compact subset of $\mathbb{C} \setminus \mathcal{Z}$, the same bound applies uniformly to the tail, and the finite part has its denominators uniformly separated from zero.

For $z, w \in \mathcal{U}$, AG10–AG11 gives the exact identity

$$\begin{aligned} K_0(z, w) &= \frac{L_g(w) + \overline{L_g(z)}}{w + \bar{z} - 1} = \sum_{\rho} m_{\rho} \overline{v_z(\rho^{\#})} v_w(\rho) \\ &= \langle Jv_z, v_w \rangle_{\mathcal{H}}. \end{aligned} \quad (\text{NI10})$$

The scalar series converges absolutely by Cauchy–Schwarz in (NI5). Its summands are exactly those of the original rational test $F_z^{\#} F_w$, with $F_z(s) = (s - z)^{-1}$ and $F^{\#}(s) = \overline{F(1 - \bar{s})}$. No zero is moved or removed in (NI10).

For a set $E \subset \mathcal{U}$, let $\mathbb{C}^{(E)}$ denote finite coefficient families, and define the linear map

$$V_E : \mathbb{C}^{(E)} \longrightarrow \mathcal{H}, \quad V_E c = \sum_{z \in E} c_z v_z. \quad (\text{NI11})$$

For every finite family,

$$\sum_{z, w \in E} \bar{c}_z K_0(z, w) d_w = [V_E c, V_E d]. \quad (\text{NI12})$$

Thus any repetitions or linear relations in a chosen finite presentation are carried by the kernel of its actual evaluation map; they do not add signed directions.

3. Density, including its convergence and residue proof

Suppose that $E \subset \mathcal{U}$ has an accumulation point inside $\mathcal{U}$. This includes every nonempty open subset and every nondegenerate real interval contained in $(1, \infty)$. We prove

$$\overline{\text{span}\{v_z : z \in E\}}^{\mathcal{H}} = \mathcal{H}. \quad (\text{NI13})$$

Let $h \in \mathcal{H}$ be orthogonal to all those vectors, and form its Cauchy transform

$$\mathcal{C}_h(z) = \sum_{\rho \in \mathcal{Z}} \frac{m_{\rho} \bar{h}_{\rho}}{\rho - z}, \quad z \in \mathbb{C} \setminus \mathcal{Z}. \quad (\text{NI14})$$

The series is absolutely convergent at every such point by Cauchy–Schwarz. More precisely, on a compact set contained in $|z| \leq M$, for $R > 2M + 1$ its tail is bounded uniformly by

$$\sum_{|\rho| > R} \frac{m_{\rho} |h_{\rho}|}{|\rho - z|} \leq 2 \left(\sum_{|\rho| > R} m_{\rho} |h_{\rho}|^2 \right)^{1/2} \left(\sum_{|\rho| > R} \frac{m_{\rho}}{|\rho|^2} \right)^{1/2}. \quad (\text{NI15})$$

This tends to zero. The finite remaining part is holomorphic away from its listed poles. Thus (NI14) converges normally on compact subsets of $\mathbb{C} \setminus \mathcal{Z}$ and defines a holomorphic function there.

This domain is connected. To see this directly, join any two points of the domain by a line segment. The compact segment meets at most finitely many points of the discrete zero set. Small pairwise disjoint discs about those points can be chosen to contain no other zero and neither endpoint. Replacing the portion of the segment in each disc by an arc on its boundary gives a path avoiding the zero set. Orthogonality gives $\mathcal{C}_h = 0$ on E . The identity theorem, first at its accumulation point and then on the connected domain, gives $\mathcal{C}_h = 0$ everywhere in that domain.

Fix $\rho_0 \in \mathcal{Z}$. Choose a disc containing that zero and no other. All terms in (NI14) other than its ρ_0 -term converge normally to a holomorphic function on the disc, by the same tail estimate and the finite denominators. Consequently

$$\text{Res}_{z=\rho_0} \mathcal{C}_h(z) = -m_{\rho_0} \bar{h}_{\rho_0}. \quad (\text{NI16})$$

The transform is zero on the punctured disc, so its residue is zero. Since the multiplicity is positive, $h_{\rho_0} = 0$. This holds at every zero, proving $h = 0$, and the Hilbert orthogonal-complement criterion proves (NI13).

This argument is a density statement for the actual infinite zero set. It does not substitute unrestricted finite interpolation for convergence, and it does not discard a tail of that set.

4. Density of all powers at one real pole

Fix an arbitrary real $\sigma > 1$, and retain every vector

$$q_j(\rho) = \frac{1}{(\rho - \sigma)^{j+1}}, \quad j \geq 0. \quad (\text{NI17})$$

Write $\delta = \sigma - 1 > 0$. Because $|\rho - \sigma| \geq \delta$, $\|q_j\| \leq \delta^{-j} \|q_0\|$. Thus for $|Y| < \delta$ the geometric expansion is an identity in $\mathcal{H}$, with absolute convergence of the vector norm series:

$$v_{\sigma+Y} = \sum_{j=0}^{\infty} Y^j q_j. \quad (\text{NI18})$$

Pointwise the series is the usual geometric identity; convergence in $\mathcal{H}$ and continuity of coordinate evaluations make it the same vector identity. If h is orthogonal to every q_j , taking its inner product with (NI18) gives $C_h(\sigma + Y) = 0$ on a disc. Equations (NI14)–(NI16) therefore imply $h = 0$. We have proved

$$\overline{\text{span}\{q_j : j \geq 0\}}^{\mathcal{H}} = \mathcal{H}. \quad (\text{NI19})$$

Equivalently, $C_h^{(j)}(\sigma) = j! \langle h, q_j \rangle$; (NI18) supplies the derivative justification and the factor $j!$.

The result remains true after deleting **any fixed finite initial set of powers**. More precisely, for every integer $J_0 \geq 0$,

$$\overline{\text{span}\{q_j : j \geq J_0\}}^{\mathcal{H}} = \mathcal{H}. \quad (\text{NI19a})$$

An orthogonal vector makes every Taylor coefficient of C_h at σ of degree at least J_0 vanish. Thus that function agrees near σ with a polynomial of degree less than J_0 , with the zero polynomial understood when $J_0 = 0$. The identity theorem extends this equality to $\mathbb{C} \setminus \mathcal{Z}$. The polynomial has zero residue at every ρ , so (NI16) again forces $h = 0$. No estimate on the polynomial at infinity is needed.

In the original notation of CK14–CK16, put $Q_j(s) = (s - \sigma)^{-j-1}$. Taking the two convergent vector expansions in (NI10) gives

$$C_{jk} = [q_j, q_k] = Z_0((Q_j)^\# Q_k), \quad C_N = (C_{jk})_{0 \leq j, k \leq N}. \quad (\text{NI20})$$

There is no additional factorial or change of signs. In detail, for small independent variables X, Y , use $z = \sigma + \bar{X}$, $w = \sigma + Y$; the first argument of the inner product is conjugate-linear, so the coefficient of $X^j Y^k$ is exactly $[q_j, q_k]$. The numerator and denominator are those of CK14 unchanged.

5. Exact negative and positive indices on dense test spans

For a Hermitian form on a complex vector space, its negative index means the supremum of the dimensions of finite-dimensional subspaces on which the form is negative definite. The positive index is defined with positive definite in place of negative definite. For a finite Hermitian matrix these are the numbers of strictly negative and strictly positive eigenvalues, counted with algebraic multiplicity, by unitary diagonalization.

First, every negative-definite finite-dimensional subspace of $\mathcal{H}$ maps injectively under P_- to $\mathcal{H}_-$. Indeed, a nonzero vector in the kernel would have $[a, a] = \|P_+ a\|^2 \geq 0$, contradicting negative definiteness. Thus its dimension is at most κ_- when that number is finite. Conversely (NI7) supplies a negative orthonormal family of every finite size not exceeding κ_- . The negative index of $\mathcal{H}$ is therefore exactly κ_- , and the identical argument with signs exchanged gives positive index κ_+ .

We prove that either index is retained on **any dense linear subspace** $D \subset \mathcal{H}$. For the negative case, take n different negative unit vectors $u_1, \dots, u_n$ from (NI7). Define $T : \mathbb{C}^n \rightarrow \mathcal{H}$ by $Tx = \sum_i x_i u_i$. Then T is an isometry and $[Tx, Tx] = -\|x\|^2$. By density choose $d_i \in D$ with $\|d_i - u_i\| < 1/(4\sqrt{n})$, and set $Dx = \sum_i x_i d_i$, using the same letter for this finite synthesis map. Its error $E = D - T$ has operator norm less than $1/4$, since $\|Ex\| \leq \sum_i |x_i| \|d_i - u_i\| < \|x\|/4$. Using $\|J\| = 1$, we obtain

$$[Dx, Dx] \leq (-1 + 2\|E\| + \|E\|^2) \|x\|^2 < -\frac{7}{16} \|x\|^2 \quad (x \neq 0). \quad (\text{NI21})$$

Thus the synthesis map is injective and its image is an n -dimensional negative-definite subspace of D . This proves the lower bound for every finite $n \leq \kappa_-$; the upper bound follows from its inclusion in $\mathcal{H}$. For positive unit vectors, the same expansion gives a lower bound $(1 - 2\|E\| - \|E\|^2)\|x\|^2 > 7\|x\|^2/16$, proving the positive statement too.

Finally, the radical of the form restricted to a dense subspace is zero. If $a \in D$ satisfies $[a, b] = 0$ for all $b \in D$, continuity gives that identity for every $b \in \mathcal{H}$. Taking $b = Ja$ then forces $a = 0$. This does not say that each finite matrix cut out of the subspace is nonsingular; a finite restriction can have its own radical.

6. The index theorem for the actual kernel and coefficient tower

Combining (NI12)–(NI13), (NI19)–(NI21) proves the unconditional identity

$$\boxed{\begin{aligned} \sup_{n \geq 1, z_1, \dots, z_n \in E} n_-((K_0(z_i, z_j))_{i,j=1}^n) &= \#\mathcal{P}, \\ \sup_{N \geq 0} n_-(C_N) &= \#\mathcal{P}. \end{aligned}} \quad (\text{NI22})$$

Here E is any subset specified in Section 3, $\sigma > 1$ in the second line is arbitrary, and n_- counts strictly negative eigenvalues. To justify directly the finite matrix passage, the finitely many vectors d_i used in (NI21) involve a common finite union of Cauchy poles, or finitely many powers up to a common maximum N . Their negative Gram form is the pullback of the corresponding finite matrix by their coefficient synthesis map. Its injectivity on the negative subspace proves that matrix has at least n negative eigenvalues. Conversely, any negative eigenspace of a finite matrix maps injectively by its actual evaluation map into a negative-definite subspace of $\mathcal{H}$, using (NI12) or (NI20). This proves both inequalities, including presentations with redundant coefficients.

The same proof gives

$$\sup_{n \geq 1, z_1, \dots, z_n \in E} n_+((K_0(z_i, z_j))) = \sup_{N \geq 0} n_+(C_N) = \kappa_+. \quad (\text{NI23})$$

Using the strengthened density (NI19a), the exact corresponding statement for every fixed tail is

$$\sup_{N \geq J_0} n_-((C_{jk})_{J_0 \leq j, k \leq N}) = \kappa_-, \quad \sup_{N \geq J_0} n_+((C_{jk})_{J_0 \leq j, k \leq N}) = \kappa_+. \quad (\text{NI23a})$$

Indeed, repeat the construction (NI21) with the dense span in (NI19a); the finite list of approximating powers has a common largest index N . This proves both identities with precisely the same upper bound from (NI8). For $\kappa_- < \infty$, some finite matrix attains that entire negative index. In the coefficient tower there is an N_0 such that $n_-(C_N) = \kappa_-$ for all $N \geq N_0$: the lower-bound construction supplies one N_0 , larger spaces retain that negative subspace, and (NI22) supplies the upper bound. For $\kappa_- = \infty$, the integers $n_-(C_N)$ are unbounded. Neither statement supplies a computable universal finite cutoff deciding RH.

The original critical line is exactly the fixed set of (NI2). Consequently (NI22) strengthens CK13 and CK17 from the zero-index equivalence to the exact index:

$$\text{RH} \iff \kappa_- = 0 \iff n_-(C_N) = 0 \text{ for every } N \geq 0. \quad (\text{NI24})$$

All equalities concern the actual zeta zeros. They do not postulate an off-line zero or infer the absence of one. In particular, the positive real diagonal from HA24–HA26 does not obscure an existing negative direction: if $\kappa_- > 0$, it is detected by an off-diagonal finite Gram matrix even when its poles are restricted to any fixed nondegenerate real interval in $(1, \infty)$.

7. Multiplicity and the complete local trace radical

For each actual zero keep its original local algebra and parameter

$$A_\rho = \mathbb{C}[\epsilon_\rho]/(\epsilon_\rho^{m_\rho}), \quad \epsilon_\rho = s - \rho. \quad (\text{NI25})$$

Multiplication by $a_0 + a_1\epsilon_\rho + \dots + a_{m_\rho-1}\epsilon_\rho^{m_\rho-1}$, in the original ordered monomial basis, is triangular with every diagonal entry equal to a_0 . Its multiplication trace is therefore exactly $m_\rho a_0$. Under the original involution, the Taylor coefficients at a reflected zero satisfy

$$\left(\sum_k a_{\rho^\#, k} \epsilon_{\rho^\#}^k \right)^\# \text{ at } \rho = \sum_k (-1)^k \overline{a_{\rho^\#, k}} \epsilon_\rho^k. \quad (\text{NI26})$$

The factor $(-1)^k$ follows from $1 - \overline{\rho + \epsilon_\rho} = \rho^\# - \bar{\epsilon}_\rho$. In particular, the constant-value involution is $a_\rho \mapsto \overline{a_\rho^\#}$.

To retain all jets even for an infinite zero set, define

$$\mathcal{A}_{\text{tr}} = \left\{ a \in \prod_{\rho \in \mathcal{Z}} A_\rho : (a_\rho(0))_\rho \in \mathcal{H} \right\}, \quad \pi(a)_\rho = a_\rho(0), \quad \mathcal{N} = \prod_{\rho \in \mathcal{Z}} (\epsilon_\rho) A_\rho. \quad (\text{NI27})$$

No restriction is imposed on the higher coefficients. This is a complex algebra closed under (NI26): since $m_\rho \geq 1$, every $x \in \mathcal{H}$ has $\sup_\rho |x_\rho| \leq \|x\|$, and hence pointwise multiplication obeys $\|xy\| \leq \|x\|\|y\|$. An identity element is not asserted for this algebra. Its evaluation has the exact split sequence

$$0 \longrightarrow \mathcal{N} \longrightarrow \mathcal{A}_{\text{tr}} \xrightarrow{\pi} \mathcal{H} \longrightarrow 0, \quad \iota(x)_\rho = x_\rho \cdot 1_{A_\rho}, \quad \pi \iota = I. \quad (\text{NI28})$$

Both π and ι preserve addition, scalar multiplication, multiplication, and the indicated involutions. The quotient retains exactly the weighted value space (NI5), while the kernel retains every higher jet. Define on $\mathcal{A}_{\text{tr}}$

$$\mathfrak{T}(a, b) = \sum_\rho \text{Tr}_{A_\rho}(M_{(a \# b)_\rho}) = \sum_\rho m_\rho \overline{a_{\rho^\#}(0)} b_\rho(0) = [\pi a, \pi b]. \quad (\text{NI29})$$

The sum is absolutely convergent by Cauchy–Schwarz. Equation (NI29) is not a claim that the individual unpaired trace sum exists for every a . Its radical is exactly $\mathcal{N}$: containment is immediate, and if the form vanishes against every b , the section in (NI28) and nondegeneracy of (NI6) imply $\pi a = 0$. If multiplicities are unbounded, the product ideal $\mathcal{N}$ need not have a common nilpotence exponent, so it is not called a nilpotent ideal here. Each of its local factors is the specified nilpotent ideal in (NI25).

The exact algebraic description, without any bounded-multiplicity assumption, is

$$\begin{aligned} \sqrt{(0)}_{\mathcal{A}_{\text{tr}}} &= \{a \in \mathcal{N} : \text{there exists } r \geq 1 \text{ such that } a_\rho^r = 0 \text{ for every } \rho\}, \\ \mathcal{N}^r &= \prod_\rho (\epsilon_\rho^r) A_\rho, \quad \bigcap_{r \geq 1} \mathcal{N}^r = 0, \quad \mathcal{A}_{\text{tr}} \cong \varprojlim_r \mathcal{A}_{\text{tr}} / \mathcal{N}^r. \end{aligned} \quad (\text{NI29a})$$

The first equality follows because powers in the product are computed coordinatewise, and any nilpotent has zero constant at every coordinate. For the second equality, multiplication gives one containment. Conversely, for $r \geq 2$, any tuple divisible coordinatewise by ϵ_ρ^r is the product of $r - 1$ copies of the tuple $(\epsilon_\rho)_\rho$ and a tuple in $\mathcal{N}$: shift each original coefficient down by $r - 1$ powers, retaining a factor ϵ_ρ . Every factor lies in $\mathcal{N}$ since its constant coordinates vanish. The case $r = 1$ is its definition. The intersection is zero since each m_ρ is finite. Finally, a compatible inverse-limit element determines each local component when $r \geq m_\rho$, and its constant vector is already the element of $\mathcal{H}$ specified at level $r = 1$. These reconstructed components give an element of $\mathcal{A}_{\text{tr}}$ and the inverse to the displayed map. Thus the trace radical has an exact separated, complete filtration even when it differs from the nilradical. This calculation does not assert any uniform bound, or lack of a uniform bound, for the actual zeta multiplicities.

Every rational Cauchy test and every Q_j has a well-defined Taylor class in each A_ρ . These classes belong to $\mathcal{A}_{\text{tr}}$ by (NI9) and (NI17), and applying π gives exactly v_z and q_j . This proves the full map from the original multiple-zero jet algebras to (NI22), including its kernel. Multiplicity supplies the weight m_ρ ; the local trace radical accounts for why a multiple zero does not by itself add further signed value directions. The retained higher jets can still have nonzero heat responses, as calculated in AG17 and HA21–HA22; (NI29) states their exact time-zero trace behavior only.

8. The complete supported arithmetic receiver

Let $\mathcal{L}$ be the original finite support semilattice with top $1_{\mathcal{L}}$, and let $W_{\mathcal{L}} = \mathbb{C}^{\mathcal{L}}$ have basis $\mathbf{e}_\lambda$. This symbol is distinct from $L_g(s)$. Take F, G in either finite test span of Sections 2 and 4, and put $A = F^\# G$. Its poles lie strictly outside the critical strip and $A = O(s^{-2})$, so the proved extension AG14–AG16 applies. With the unchanged arithmetic functionals P_{fin} and A_∞ , it gives

$$\begin{aligned} B_{\mathcal{L}}(F, G) &= (A(0) + A(1)) \mathbf{e}_{1_{\mathcal{L}}} + A(0) \sum_{\lambda \neq 1_{\mathcal{L}}} \mathbf{e}_\lambda, \\ D_{\mathcal{L}}(F, G) &= (P_{\text{fin}}(h_A) - A_\infty(h_A)) \mathbf{e}_{1_{\mathcal{L}}} + A(0) \sum_{\lambda \neq 1_{\mathcal{L}}} \mathbf{e}_\lambda, \\ B_{\mathcal{L}}(F, G) - D_{\mathcal{L}}(F, G) &= [\text{ev } F, \text{ev } G] \mathbf{e}_{1_{\mathcal{L}}}. \end{aligned} \quad (\text{NI30})$$

Here $\text{ev } F = (F(\rho))_\rho$, and $A(0) = \overline{F(1)}G(0)$, $A(1) = \overline{F(0)}G(1)$. Thus both endpoints and every lower coordinate are explicitly present. Their common lower-coordinate values cancel only in the displayed **difference**, as required by the original full identity.

The precise scalar and vector maps are

$$\begin{aligned} j_{\text{top}} : \mathbb{C} &\longrightarrow W_{\mathcal{L}}, \quad x \longmapsto x\mathbf{e}_{1_{\mathcal{L}}}, \quad \pi_{\text{top}} \left(\sum x_{\lambda} \mathbf{e}_{\lambda} \right) = x_{1_{\mathcal{L}}}, \\ \pi_{\text{top}} j_{\text{top}} &= I, \quad \ker \pi_{\text{top}} = \text{span}\{\mathbf{e}_{\lambda} : \lambda \neq 1_{\mathcal{L}}\}, \\ \Sigma_{\mathcal{L}} \left(\sum x_{\lambda} \mathbf{e}_{\lambda} \right) &= \sum x_{\lambda}, \quad \ker \Sigma_{\mathcal{L}} = \{x : \sum x_{\lambda} = 0\}. \end{aligned} \tag{NI31}$$

Both π_{top} and $\Sigma_{\mathcal{L}}$ recover the scalar form from the difference in (NI30), because that difference lies in the image of j_{top} . They have the displayed kernels, which differ whenever $\mathcal{L}$ has a lower label: for such a label, $\mathbf{e}_{\lambda}$ belongs to $\ker \pi_{\text{top}}$ and not to $\ker \Sigma_{\mathcal{L}}$. For a singleton label set both maps coincide. In the case with lower labels, their agreement on the difference does not identify them on the separate boundary or arithmetic terms. The lower-coordinate pairing $A(0)$ alone is generally not Hermitian and is not assigned a negative index.

For completeness, a top-preserving map of support labels $f : \mathcal{L} \rightarrow \mathcal{M}$ induces the exact linear pushforward $f_* \mathbf{e}_{\lambda} = \mathbf{e}_{f(\lambda)}$. Writing $n_{\mu} = \#\{\lambda \neq 1_{\mathcal{L}} : f(\lambda) = \mu\}$, direct substitution into (NI30) yields

$$\begin{aligned} f_* \mathbf{B}_{\mathcal{L}} &= (A(0) + A(1))\mathbf{e}_{1_{\mathcal{M}}} + A(0) \sum_{\mu} n_{\mu} \mathbf{e}_{\mu}, \\ f_* \mathbf{D}_{\mathcal{L}} &= (P_{\text{fin}}(h_A) - A_{\infty}(h_A))\mathbf{e}_{1_{\mathcal{M}}} + A(0) \sum_{\mu} n_{\mu} \mathbf{e}_{\mu}, \\ f_*(\mathbf{B}_{\mathcal{L}} - \mathbf{D}_{\mathcal{L}}) &= [\text{ev } F, \text{ev } G]\mathbf{e}_{1_{\mathcal{M}}}. \end{aligned} \tag{NI32}$$

Thus even labels that map into the target top contribute with their full counts to both terms, and the exact signed trace is preserved. This is a statement about the indicated linear support map, not an assertion that every such label map has some additional geometric origin. Supported zero and the unsupported element τ remain distinct throughout; no numerical prime norm for supported zero is inserted in these formulas.

Equations (NI22), (NI28)–(NI29), and (NI30)–(NI32) together determine the signed index, the complete local trace kernel, and the full supported arithmetic map. They identify the surviving negative directions as anti-invariant values on actual reflected zero pairs. They do not eliminate such directions by a change of representation or by the vanishing of a nilpotent trace.

Source use and proof scope

The zero-count and entire-function input is AG1–AG8; the actual global kernel is AG10–AG11 and CK1–CK2; the coefficient conventions are CK14–CK16. The complete supported explicit formula and its rational-test extension are AG12–AG16 and HA5–HA14. The original human explicit-formula source remains Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Appendix II, Theorem 6, with original author TeX and all programme conventions retained in SZW19–SZW38. The heat-response reference uses Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, arXiv:1801.05914v5, through the original source locators recorded in AG1 and HA1. No additional source discovery, historical priority claim, or positivity assumption enters this derivation.

Finite local jets, the whole Weil form, and the second heat coefficient

This proof calculates the exact connection between a finite set of original collision jets and the complete supported arithmetic form. It also calculates a negative direction of the first heat Taylor polynomial and the next coefficient of the actual heat trace in that same direction. No off-critical zeta zero is assumed to exist. The indices below count actual zeros, whatever their locations are.

1. The original objects and the finite-jet map

Let $\mathcal{Z}$ be the set of distinct nontrivial zeros of the Riemann zeta function, with their actual multiplicities m_ρ . Write $\rho^\# = 1 - \bar{\rho}$. The functional equation gives $m_{\rho^\#} = m_\rho$. Fix a real $\sigma > 1$ and the vector space and algebra

$$\mathcal{R}_\sigma = \text{span}_{\mathbb{C}}\{(s - \sigma)^{-j-1} : j \geq 0\}. \quad (\text{FJ1})$$

Its elements are proper rational functions with their only possible pole at σ . Products remain in this algebra; the constant function one is not included. The exact reflected form is

$$Q(F, G) = \sum_{\rho \in \mathcal{Z}} m_\rho \overline{F(\rho^\#)} G(\rho). \quad (\text{FJ2})$$

The genus-one growth estimate in AG4–AG5 gives $\sum m_\rho/(1 + |\rho|^2) < \infty$; hence this sum is absolutely convergent. Its arithmetic value is the whole Cauchy coefficient form CK14–CK16, including the endpoints, archimedean term and prime sum in HA10–HA20. The full supported identity is retained in Section 6 below.

Take a finite reflection-stable subset $S \subset \mathcal{Z}$, and positive integers $r_a = r_{a^\#}$ for $a \in S$. These are the number of test coefficients retained at each point. They may equal the actual zero multiplicities, or exceed them when the higher coefficients of fixed tests are needed. Set

$$P(s) = \prod_{a \in S} (s - a)^{r_a}, \quad D = \sum_{a \in S} r_a, \quad W(s) = \frac{P(s)}{(s - \sigma)^D}, \quad \mathcal{J}_S = \bigoplus_{a \in S} \mathbb{C}[v_a]/(v_a^{r_a}). \quad (\text{FJ3})$$

For $S = \emptyset$, use $D = 0, W = 1, \mathcal{J}_S = 0$. The algebra map

$$j_S : \mathcal{R}_\sigma \longrightarrow \mathcal{J}_S, \quad (j_S F)_a = \sum_{k=0}^{r_a-1} \frac{F^{(k)}(a)}{k!} v_a^k \quad (\text{FJ4})$$

is onto. Here is an exact linear section. Given a prescribed jet f , choose the unique polynomial N_f of degree less than D such that

$$N_f(s) \equiv (s - \sigma)^D f_a(s - a) \pmod{(s - a)^{r_a}} \quad (a \in S), \quad \mathfrak{s}(f) = \frac{N_f(s)}{(s - \sigma)^D}. \quad (\text{FJ5})$$

The polynomials $(s - a)^{r_a}$ are pairwise coprime. Repeated Euclidean identities for each pair give the Chinese remainder isomorphism from polynomials modulo P to the displayed finite quotient spaces, proving existence and uniqueness of N_f . Since $a - \sigma \neq 0$, division by the denominator in each quotient proves $j_S \mathfrak{s} = I$. Its numerator degree makes the section proper rational. This is a linear section, not a multiplicative section.

The exact kernel is

$$\ker j_S = W \mathcal{R}_\sigma. \quad (\text{FJ6})$$

One inclusion follows from the numerator factors. For the other, a proper rational F with these vanishing jets has polynomial numerator divisible by P . Thus F/W is rational with no poles outside σ ; moreover it is $O(s^{-1})$, since $W \rightarrow 1$ at infinity. This proves the reverse inclusion without changing the original coordinate.

2. Density while all finite jets are kept exact

Define

$$\mathcal{H}_{\text{ext}} = \ell^2(\mathcal{Z} \setminus S, m), \quad J_{\text{ext}} h(\rho) = h(\rho^\#), \quad T(F) = (j_S F, (F(\rho))_{\rho \notin S}). \quad (\text{FJ7})$$

The inner product on the Hilbert space is conjugate-linear in the first variable and includes the weights m_ρ . The involution J_{ext} is a linear self-adjoint isometry. Equip the finite jet space with the Euclidean norm in its stated coefficient coordinates. The image of T is dense in $\mathcal{J}_S \oplus \mathcal{H}_{\text{ext}}$, and approximation can keep its first coordinate exactly prescribed.

We prove this claim, including the required infinite convergence. If $h \in \mathcal{H}_{\text{ext}}$ is orthogonal to all evaluations of $(s - \sigma)^{-j-1}$, the function

$$C_h(z) = \sum_{\rho \notin S} \frac{m_\rho \overline{h(\rho)}}{\rho - z} \quad (\text{FJ8})$$

converges absolutely and locally uniformly off its discrete pole set. For a compact set $|z| \leq R$, its tail with $|\rho| > 2R$ is bounded by

$$2 \left(\sum_{|\rho| > 2R} m_\rho |h(\rho)|^2 \right)^{1/2} \left(\sum_{|\rho| > 2R} \frac{m_\rho}{|\rho|^2} \right)^{1/2}. \quad (\text{FJ9})$$

On a small disc at σ , termwise Taylor expansion gives $C_h^{(j)}(\sigma) = j! \langle h, ((\rho - \sigma)^{-j-1})_\rho \rangle = 0$. Consequently C_h vanishes there. The complement of a discrete subset of the complex plane is connected: any polygonal path has a compact image meeting only finitely many points of the discrete set and can be diverted in small disjoint discs about those points. The identity theorem now makes C_h zero throughout this complement. Its residue at $a \notin S$ is $-m_a \overline{h(a)}$, so all coordinates of h vanish. This proves density of the rational evaluations in the exterior Hilbert space.

On the remaining zero set, both W and W^{-1} are bounded. Indeed $W(s) \rightarrow 1$ as $|s| \rightarrow \infty$, and only finitely many remaining zeros lie in any fixed disc; none is a zero or pole of W . Multiplication by W is therefore a bounded invertible diagonal operator on $\mathcal{H}_{\text{ext}}$. By (FJ6), evaluations of $\ker j_S$ are also dense there. To approximate an arbitrary pair (f, h) , take $\mathfrak{s}(f)$ from (FJ5), and add elements of $\ker j_S$ whose exterior evaluations tend to $h - \mathfrak{s}(f)|_{\text{ext}}$. Their jets remain exactly f . This proves the assertion. It proves an approximation statement with an exact finite coordinate, not an isomorphism from rational functions onto arbitrary infinite data.

3. Exact inertia of every finite local correction

On the finite jet space define its original trace form

$$B_S(f, g) = \sum_{a \in S} m_a \overline{f_{a\#}} g_{a,0}. \quad (\text{FJ10})$$

All coefficients with index at least one remain present in its radical. The exact jet involution is $(f^\#)_{a,k} = (-1)^k \overline{f_{a\#,k}}$; hence (FJ10) is the regular local trace of $f^\# g$ when $r_a = m_a$. For other r_a , the same expression is the stated constant-coefficient receiver, with the multiplicity of the actual zero retained.

Let H be any Hermitian sesquilinear form on this specified finite space. It defines the precise correction

$$Q_H(F, G) = Q(F, G) + H(j_S F, j_S G) = (B_S + H)(j_S F, j_S G) + \langle J_{\text{ext}} F|_{\text{ext}}, G|_{\text{ext}} \rangle. \quad (\text{FJ11})$$

Write κ_{ext} for the number of distinct two-point orbits of $\rho \mapsto \rho^\#$ in $\mathcal{Z} \setminus S$, with infinity allowed. Then the negative index, defined as the supremum of the dimensions of strictly negative subspaces, is exactly

$$\boxed{\text{ind}_- Q_H = \kappa_{\text{ext}} + \text{ind}_-(B_S + H)}. \quad (\text{FJ12})$$

This statement includes every finite section of the rational coefficient space; it makes no supposition about the existence of off-line zeros.

For the proof, the bounded Hermitian form on $\mathcal{J}_S \oplus \mathcal{H}_{\text{ext}}$ has coefficient operator $A = (B_S + H) \oplus J_{\text{ext}}$. Each fixed reflection point in the exterior contributes one positive direction. Each two-point orbit of multiplicity m , in coordinates divided by $\sqrt{m}$, contributes the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, with one positive and one negative direction. The finite Hermitian matrix $B_S + H$ has its usual orthogonal spectral decomposition. Thus the negative dimension of the full operator is the right side of (FJ12). A negative subspace injects into its negative spectral part: a vector whose negative projection vanishes has nonnegative form value. This gives the upper bound, including for every rational subspace.

For the lower bound, take any finite-dimensional negative spectral subspace with orthonormal basis $u_1, \dots, u_n$. On this subspace the form is bounded above by $-\eta I$ for some $\eta > 0$, since only finitely many negative eigenvalues are involved and the exterior eigenvalue is -1 . By Section 2 choose rational vectors whose images $v_i = T(F_i)$ approach the u_i . For column operators U, V ,

$$\|V^* A V - U^* A U\| \leq \|A\| (2\|U\| \|V - U\| + \|V - U\|^2). \quad (\text{FJ13})$$

This is obtained by expanding $V = U + (V - U)$ and applying the operator norm inequality to its three additional terms. Choose the approximation so the bound is less than $\eta/2$. Their Gram matrix is strictly negative, and the F_i are linearly independent because a dependence would have zero quadratic value. This realizes every such dimension in the rational space, proving (FJ12), also when the index is infinite.

In particular, for a finite set consisting entirely of actual critical-line zeros, every off-line negative direction remains detectable on the exact kernel of all of these jet maps. A local correction supported on these jets cannot remove an exterior negative direction. This is an exact description of which corrections act on which part of the original form. It does not assert that the negative directions exist. If a correction changes the complement as well, its action must be calculated there; (FJ12) concerns precisely the finite-jet corrections (FJ11).

4. A quantitative exterior tail for the actual Cauchy matrix

For fixed poles $z_1, \dots, z_n$ with $\Re z_l > 1$, let K_T be the sum in (FJ2) over $|\Im \rho| \leq T$, evaluated on $F_l = (s - z_l)^{-1}$. Both this cutoff and its complement are reflection-stable. For $T \geq \max(1, 2 \max_l |\Im z_l|)$, each tail vector satisfies $\|(F_l(\rho))_l\| \leq 2\sqrt{n}/|\Im \rho|$, and the same bound holds at $\rho^\#$. Summing the norms of the rank-one matrix contributions gives

$$\|K - K_T\| \leq 4n \sum_{|\Im \rho| > T} \frac{m_\rho}{|\Im \rho|^2} \leq 8n \sum_{|\rho| > T} \frac{m_\rho}{|\rho|^2} = O\left(\frac{n \log(T+2)}{T}\right). \quad (\text{FJ14})$$

For the second inequality use $0 < \Re \rho < 1$, so $|\rho|^2 \leq 1 + |\Im \rho|^2 \leq 2|\Im \rho|^2$. The last estimate is AG5. No numerical value is assigned to its growth constant. The inequality with the actual tail sum is exact, and proves convergence in operator norm. It supplies no sign for the omitted matrix. The same proof applies after deleting any fixed finite set of local clusters, so the complement in HR17 is this actual convergent exterior form.

5. What the first and second heat coefficients actually do

Fix an actual critical-line zero $\rho = 1/2 + ix_0/2$, of order m , in the original family $g_t(s) = 16H_t(-2i(s - 1/2))$, $\partial_t g_t = g_t''/4$. Write

$$g_0(\rho + v) = v^m u(v), \quad a = u'(0)/u(0), \quad F(\rho + v) = f_0 + f_1 v + f_2 v^2 + \dots \quad (\text{FJ15})$$

The reflection $g_0^\# = g_0$ gives $u(v) = (-1)^m \overline{u(-\bar{v})}$, whence $a = -\bar{a}$. All of these are original coefficients; $v = s - \rho = i\xi/2$, with no change of heat time.

For a fixed analytic test A , its local zero trace is a small contour integral of Ag_t'/g_t . Differentiating this finite contour and using the heat equation gives

$$\partial_t Z_{\rho,t}(A)|_0 = -\frac{m(m-1)}{4} A''(\rho) - \frac{m}{2} a A'(\rho). \quad (\text{FJ16})$$

To verify every coefficient, the principal part of g_0''/g_0 is $m(m-1)v^{-2} + 2mav^{-1}$. Its derivative, multiplied by $A/4$, has residue $-m(m-1)A''/4 - maA'/2$. This proves (FJ16) and is the same residue calculation as AG17.

With $A = F^\# G$, one has $A' = \bar{f}_0 g_1 - \bar{f}_1 g_0$ and $A'' = 2\bar{f}_0 g_2 - 2\bar{f}_1 g_1 + 2\bar{f}_2 g_0$. Put $b_m = m(m-1)/2$. The exact first Taylor polynomial of the local trace is therefore the Hermitian matrix on the **test** coefficients (f_0, f_1, f_2)

$$M_\rho^{[1]}(t) = \begin{pmatrix} m & -tma/2 & -tb_m \\ tma/2 & tb_m & 0 \\ -tb_m & 0 & 0 \end{pmatrix}. \quad (\text{FJ17})$$

For $m = 2$, the third test coefficient is still needed: it is not a third basis vector of the two-dimensional original collision algebra. HR7–HR9 gives the exact map of such fixed tests into that algebra, including their higher coefficients. Formula (FJ17) is a Taylor coefficient calculation on fixed tests.

For $m \geq 2$ and $t \neq 0$, the matrix is nonsingular and

$$\det M_\rho^{[1]}(t) = -t^3 b_m^3, \quad (n_+, n_-, n_0) = \begin{cases} (2, 1, 0), & t > 0, \\ (1, 2, 0), & t < 0. \end{cases} \quad (\text{FJ18})$$

Indeed the block on coordinates $(0, 2)$ has determinant $-t^2 b_m^2$, hence one sign of each kind. Its inverse has zero $(0, 0)$ entry, so the Schur complement of this block at coordinate 1 is exactly tb_m , despite the retained coefficient

a. Block Gaussian congruence proves the asserted inertia, and multiplication of the block determinant by its Schur complement gives the determinant. For $m = 1$, $b_m = 0$ and the retained matrix has upper two-by-two determinant $-t^2|a|^2/4$; when $ta \neq 0$ it has one positive and one negative direction and a zero third coordinate, and when $ta = 0$ only its first coordinate is nonzero.

For $m \geq 2$ the particularly direct vector

$$(f_0, f_1, f_2) = \left(\frac{tb_m}{m}, 0, 1 \right) \quad \text{has} \quad f^* M_\rho^{[1]}(t) f = -\frac{t^2 b_m^2}{m}. \quad (\text{FJ19})$$

It is realized by an actual proper rational test depending analytically on t , using the linear section (FJ5) with $S = \{\rho\}$ and $r_\rho = 3$. Its higher Taylor coefficients remain bounded and retained.

We now evaluate the next coefficient of the **actual** local heat trace on this same family. In the original coordinate $v = i\xi/2$, the leading roots are $\xi_\nu = r_\nu \sqrt{t} + O(t)$, where

$$P_m(y) = \sum_{j=0}^{\lfloor m/2 \rfloor} \frac{(-1)^j m!}{j!(m-2j)!} y^{m-2j}. \quad (\text{FJ20})$$

This polynomial is obtained directly by applying $\exp(-t\partial_\xi^2)$ to the leading original term ξ^m . The full-unit derivation and its analytic root expansion are HR1–HR6 and HC1–HC14; only the displayed leading coefficient is used here. Newton's identities, with $p_k = \sum_\nu r_\nu^k$, give

$$p_2 = 2m(m-1), \quad p_4 = 4m(m-1)(2m-3). \quad (\text{FJ21})$$

For verification, the coefficients of y^{m-2} and y^{m-4} are $-m(m-1)$ and $m(m-1)(m-2)(m-3)/2$; the second and fourth Newton identities give these expressions. For $m = 2, 3$, omit the absent fourth coefficient and use the same identity with that coefficient zero; the displayed formula remains exact.

The local Hermitian trace entry on the second test coefficient has leading value $t^2 p_4/16$, since $|(i/2)^2|^2 = 1/16$. The $(0, 2)$ entry has leading value $-tb_m$, the $(0, 0)$ entry is exactly m , and all contributions of the bounded higher test coefficients to the vector (FJ19) are $O(t^3)$. To justify the latter assertion, every higher product has degree at least five in ξ , except a product with the constant coefficient, which has an additional factor t ; odd power sums are analytic in t and start at the next integer power. Uniform convergence of the analytic test expansions on the finitely many roots bounds the remaining tail. Consequently the actual value is

$$\begin{aligned} Z_{\rho,t}(F_t^\# F_t) &= t^2 \left(\frac{p_4}{16} - \frac{b_m^2}{m} \right) + O(t^3) \\ &= \frac{m(m-1)(m-2)}{4} t^2 + O(t^3). \end{aligned} \quad (\text{FJ22})$$

The calculation keeps the negative term from (FJ19) and the positive term that is absent from the first Taylor matrix. For $m = 2$ they cancel at this order; for $m > 2$ the displayed residual coefficient is positive. For sufficiently small positive time the actual local roots are distinct and remain on the critical line, as follows from the real, distinct roots of P_m and their real analytic lifts in the original coordinate. Thus the actual local form is a positive evaluation Gram form, consistent with (FJ22). This is a finite local statement, not a sign assigned to the complement of this cluster.

The simpler restriction $f_0 = 0$ in (FJ17) gives the known positive first-jet coefficient $tb_m \bar{f}_1 g_1$. Equations (FJ18)–(FJ22) identify exactly why that restriction cannot be used as the sign of the complete first Taylor matrix, and why the complete first Taylor matrix cannot be used as the sign of the actual positive-time cluster.

For a double zero the next term can also be calculated exactly for the particular rational section (FJ5), so the cancellation in (FJ22) is not an endpoint of the calculation. Put $\delta = \rho - \sigma$ and take precisely that section with $r_\rho = 3$, $(f_0, f_1, f_2) = (t/2, 0, 1)$. At time zero it is $F_0(\rho + v) = \delta^3 v^2/(\delta + v)^3$, so $f_3(0) = -3/\delta$. The two original arithmetic roots have expansions $v_\pm = \pm i\sqrt{t/2} - (a/2)t + O(t^{3/2})$. To verify the drift directly, substitute $v = bu + du^2$, $t = u^2$ into $g_t = g_0 + (t/4)g_0'' + \dots$: the coefficients of u^2, u^3 , after division by $u(0)$, are $b^2 + 1/2$ and $2bd + ab^3 + 3ab/2$. Thus $b = \pm i/\sqrt{2}$ and $d = -a/2$. All omitted heat terms start at order u^4 .

Substituting the fixed test gives

$$\begin{aligned}
F_t(\rho + v_\pm) &= \frac{\pm i}{\sqrt{2}} \left(-a + \frac{3}{2\delta} \right) t^{3/2} + O(t^2), \\
Z_{\rho,t}(F_t^\# F_t) &= \left| -a + \frac{3}{2\delta} \right|^2 t^3 + O(t^4), \\
\left| -a + \frac{3}{2\delta} \right|^2 &\geq \frac{9(\sigma - 1/2)^2}{4|\rho - \sigma|^4} > 0.
\end{aligned} \tag{FJ22a}$$

For positive time both roots are on the critical line, so the trace is the sum of these two squared moduli. Its analyticity in t removes the possible half-integer remainder. Finally a is purely imaginary and $\Re(1/\delta) = (1/2 - \sigma)/|\delta|^2$, proving the strict bound. This proves that the specified rational test recovers a strictly positive third-order local value at a double collision, while retaining the complete unit derivative and the original pole.

6. Support, compensation and the complete arithmetic receiver

Let L be the retained finite support lattice, and use independent vectors $\mathbf{e}_\lambda$ for its labels. For $A = F^\# G$, the original full formula remains

$$\begin{aligned}
\mathbf{B}_L(A) &= (A(0) + A(1))\mathbf{e}_{1_L} + A(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda, \\
\mathbf{Z}_L(A) &= Q(F, G)\mathbf{e}_{1_L}, \\
\mathbf{D}_L(A) &= (P_{\text{fin}}(A) - A_\infty(A))\mathbf{e}_{1_L} + A(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda, \\
\mathbf{B}_L(A) - \mathbf{Z}_L(A) &= \mathbf{D}_L(A).
\end{aligned} \tag{FJ23}$$

Here the notation for the geometric terms denotes their proved rational-test continuation in AG12–AG16 and HA10–HA14, not an independently assigned extension. This is SZW33–SZW38, with the supported-zero endpoints integrated. Neither supported zero e nor unsupported τ is identified with a vanishing scalar amplitude.

The decomposition (FJ11) is a decomposition of the top coefficient of this formula, while every lower labelled endpoint stays as written. If the new form Q_H is inserted as a new divisor receiver, its exact companion is

$$\mathbf{Z}_{L,H} = \mathbf{Z}_L + H(j_S F, j_S G)\mathbf{e}_{1_L}, \quad \mathbf{D}_{L,H} = \mathbf{D}_L - H(j_S F, j_S G)\mathbf{e}_{1_L}. \tag{FJ24}$$

Substitution proves $\mathbf{B}_L - \mathbf{Z}_{L,H} = \mathbf{D}_{L,H}$. Thus an added local sign correction has an explicitly changed arithmetic counterpart. The original $\mathbf{D}_L$ cannot simultaneously remain equal to that new receiver unless the displayed correction vanishes. The finite-jet and exterior maps in Sections 1–3 identify the spaces where it vanishes and where it acts; no claim of unrelatedness replaces those maps.

Finally the original full heat matrix splits as HR17–HR18:

$$C_{\text{full}}(t) = C_\rho(t) + C_{\text{ext}}(t), \quad \dot{C}_{\text{full}}(0)_{al} = \frac{J_0(z_l) + \overline{J_0(z_a)}}{z_l + \bar{z}_a - 1}, \quad J_0 = \frac{1}{4} \partial_s (L'_0 + L_0^2). \tag{FJ25}$$

AG8–AG11 proves that this is the actual global rational trace. HR7–HR18 gives its exact original local algebra map. Equations (FJ12), (FJ14) and (FJ22) now retain, respectively, the exterior sign count, its convergent matrix tail and the second local coefficient. The sign of the entire arithmetic matrix at time zero remains to be established.

7. Endpoint-annihilating tests still detect all zeros

For every fixed integer $r \geq 1$, impose the exact endpoint conditions $F^{(j)}(0) = F^{(j)}(1) = 0$ for $0 \leq j < r$. Their kernel in $\mathcal{R}_\sigma$ is

$$\mathcal{R}_{\sigma,r}^{\text{end}} = W_r \mathcal{R}_\sigma, \quad W_r(s) = \frac{s^r (s-1)^r}{(s-\sigma)^{2r}}. \tag{FJ26}$$

The divisibility proof of (FJ6) applies verbatim at the two distinct points 0 and 1. Neither is a nontrivial zero. Thus both W_r and its inverse are bounded on the entire actual zero set, by their limit 1 at infinity and

the finitely many nonzero values in a bounded disc. Section 2 proves that evaluations of this exact endpoint kernel are dense in $\mathcal{H} = \ell^2(\mathcal{Z}, m)$. Section 3, with no finite local block, therefore gives

$$\text{ind}_-(Q|_{\mathcal{R}_{\sigma,r}^{\text{end}}}) = \#\{\text{two-point orbits of } \rho \mapsto 1 - \bar{\rho}\}. \quad (\text{FJ27})$$

In particular this is an all-zero detector after every one of the specified endpoint values and derivatives has vanished exactly. For these tests $A = F^\#G$ has $A(0) = A(1) = 0$, and (FJ23) has zero amplitude in every boundary coordinate while retaining each label. The nonzero spectral values are still tested by a dense family. This proves the relation between boundary annihilation and the complete form, without inferring the sign of that form from the annihilation.

Write κ_- for the total number of distinct two-point reflection orbits in $\mathcal{Z}$, allowing infinity. The same finite interpolation and exact-kernel argument proves that $F \mapsto (F(0), F(1), \text{ev } F)$ has dense image in $\mathbb{C}^2 \oplus \mathcal{H}$, with its first two coordinates attained exactly. Therefore for every Hermitian endpoint form H_e ,

$$\text{ind}_-\{Q + H_e(F(0), F(1))\} = \kappa_- + \text{ind}_- H_e. \quad (\text{FJ28})$$

Here notation on the left denotes the polarized sesquilinear form; the right side follows from the bounded direct-sum form and the complete approximation proof (FJ13).

For the actual order-192 holonomy correction proved in HWA, write $A = F(0), B = F(1)$. Its Hermitian matrix and eigenvalues are

$$K_e = \begin{pmatrix} 1/2 & -5/6 \\ -5/6 & 1/2 \end{pmatrix}, \quad \text{spec } K_e = \{-1/3, 4/3\}, \quad \text{ind}_-(Q + K_e) = \kappa_- + 1. \quad (\text{FJ29})$$

Thus the full form obtained by replacing the original endpoint by its positive holonomy average while leaving the original geometric terms fixed has a negative rational-test direction, regardless of RH. To exhibit the direction by its exact construction, prescribe $(A, B) = (1, 1)$, so its correction is $-2/3$. The dense joint map just proved supplies proper rational functions with these exact endpoint values and $\|\text{ev } F\|^2 < 1/3$. Since $|Q(F, F)| \leq \|\text{ev } F\|^2$, each such test has $(Q + K_e)(F, F) < -1/3$.

The two compensation conventions have different signs because they keep different terms fixed. With $B_{L,\text{av}} = B_L + K_e \mathbf{e}_{1L}$, the exact identities are

$$\begin{aligned} B_{L,\text{av}} - D_L &= (Q + K_e) \mathbf{e}_{1L}, \\ D_{L,\text{av}} - D_L + K_e \mathbf{e}_{1L} &= Q \mathbf{e}_{1L}. \end{aligned} \quad (\text{FJ29a})$$

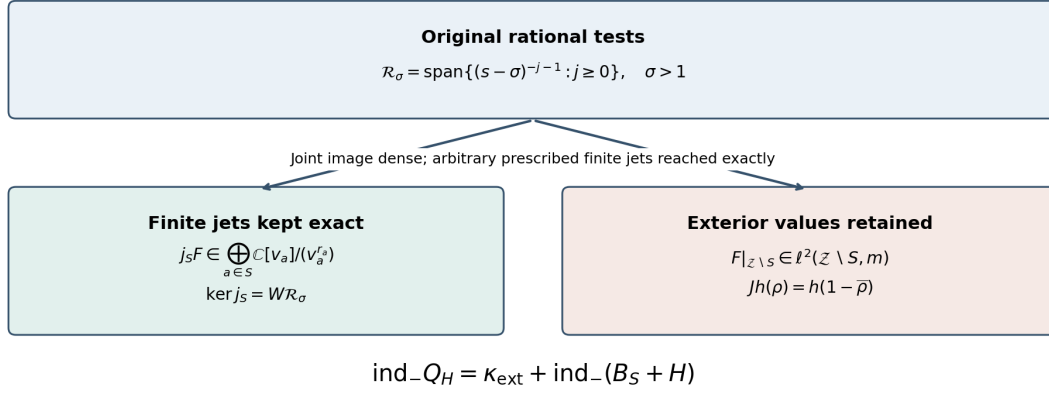
Thus subtracting K_e from the uncorrected averaged scalar form restores Q ; in the averaged-boundary convention the geometric term increases by K_e . By contrast (FJ24) keeps the original boundary fixed while changing the divisor receiver, so its geometric term decreases by that correction. Substitution proves each signed relation. This result describes the corrected and uncorrected receivers separately and proves their connecting maps. It does not assign an RH sign by changing the endpoint.

The endpoint test operator reported in the boundary handoff, $T = \partial_v^2 - 1/4$, has $M_{Tf}(s) = s(s-1)M_f(s)$ on compactly supported smooth tests, by two integrations by parts in the convention $M_f(s) = \int f(v)e^{-(s-1/2)v}dv$. Its multiplier vanishes at the two endpoints and at no nontrivial zero. Formula (FJ26) gives a proper-rational counterpart of this same multiplier, with the exact denominator required for the rational test domain. No surjectivity onto the compact-test domain is inferred from this rational comparison.

Sources and proof locations

The original heat convention is Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, arXiv:1801.05914v5, original author equations `hoz`, `htdef`, `sas`. The entire original author archive and precise reading coverage are retained with AG. The supported explicit formula uses Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, arXiv:math/9811068v1, Appendix II Theorem 6, through the complete SZW proof linked above. The programme proofs AG1–AG24, HA1–HA26, CK1–CK19, and `FIXED_ARITHMETIC_HEAT_FLAGS.tex`, HR1–HR18, are included in the same publication package. The finite Hermite interpolation, Cauchy-transform density, inertia and second-coefficient calculations needed here are proved in full above; no historical novelty claim is made for the classical methods.

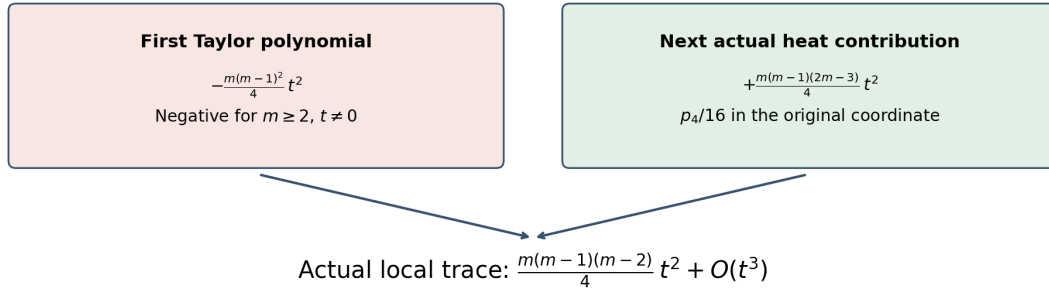
Exact finite jets and every remaining zero



FJ1-FJ13: each exterior reflected pair contributes one negative direction, if it exists.

The same test in the first heat polynomial and the actual trace

$$s - \rho = i\xi/2, \quad b_m = m(m-1)/2, \quad (f_0, f_1, f_2) = (tb_m/m, 0, 1)$$



At $m = 2$ the displayed terms cancel. At $m > 2$ their sum is positive.

FJ15-FJ22: actual local cluster; the full arithmetic complement remains in FJ23-FJ25.

Figure 18: Exact finite-jet and exterior maps, with the retained first and second heat coefficients. FJ1-FJ14 prove the upper diagram; FJ15-FJ22 prove the original-coordinate calculation below. Every endpoint and support coordinate remains in FJ23-FJ29.

One fixed primitive test and the entire prime window

The prime-operator calculation, PM1–41, proves both distinct Fourier support corrections and their exact map into the full Weil formula. The local estimate, PS1–21, gives a positive translation-difference remainder. The fixed-test proof, UP1–35, constructs one compact test whose transform is nonzero at every possible off-critical zero. The complete arithmetic calculation, PW1–22, expresses its entire translated correlation through a fixed-width prime window and an explicit archimedean remainder. Its boundedness is equivalent to RH; that bound remains unresolved. All boundary coordinates and supported-zero labels remain explicit.

The von Mangoldt operator, Fourier support defects, and the full Weil receiver

This derivation uses the original basis and time scale of HCR1–42, complete companion source. It constructs the exact divisibility-projector sum for the von Mangoldt operator, both of its different supported corrections, and its map into the complete Weil formula. The resulting translation form has both signs even after both endpoint values are forced to vanish. No sign of the entire Weil form is inferred from the positive diagonal operator.

1. Objects and the finite arithmetic identity

Let $\mathbb{N} = \{1, 2, \dots\}$, $\mathcal{H} = \ell^2(\mathbb{N})$, and retain the orthonormal basis ξ_n . Let $N\xi_n = n\xi_n$ and $P_d\xi_n = \mathbf{1}_{d|n}\xi_n$. Thus $P_d = S_d S_d^*$ in HCR1–4, with no change to the index or the logarithmic generator $H = \log N$. Write

$$\mu(1) = 1, \quad \mu(n) = \begin{cases} (-1)^r, & n \text{ is a product of } r \text{ distinct primes,} \\ 0, & n \text{ has a squared prime factor,} \end{cases} \quad \Lambda(n) = \begin{cases} \log p, & n = p^k, k \geq 1, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{PM1})$$

In particular $\Lambda(1) = 0$. For an integer $D \geq 1$, put

$$I_D = \{d : 2 \leq d \leq D, \mu(d) \neq 0\}, \quad c_d = -\mu(d) \log d, \quad A_D = \sum_{d \in I_D} c_d P_d, \quad a_D(n) = \sum_{\substack{d \in I_D \\ d|n}} c_d. \quad (\text{PM2})$$

Every coefficient in this displayed sum is nonzero. An empty sum is the zero operator. Each A_D is a bounded self-adjoint operator: it is a finite real linear combination of bounded orthogonal projections, all diagonal in the same basis.

For every $n \geq 1$,

$$\Lambda(n) = - \sum_{d|n} \mu(d) \log d. \quad (\text{PM3})$$

Here and below $\log 1 = 0$. To prove (PM3), let $p_1, \dots, p_r$ be the distinct prime factors of $n > 1$. Only their squarefree products contribute. The coefficient of $\log p_i$ in $\sum_{d|n} \mu(d) \log d$ is

$$\sum_{J \subset \{1, \dots, r\} \setminus \{i\}} (-1)^{1+|J|} = -(1-1)^{r-1}.$$

For $r = 1$ this is -1 ; for $r > 1$ it is zero. This is exactly (PM3). For $n = 1$ the sole divisor contributes zero. Consequently $a_D(n) = \Lambda(n)$ as soon as $D \geq n$. No infinite arithmetic sum is needed at a fixed entry.

2. The precise operator domains and modes of convergence

Define the multiplication operator

$$\mathcal{D}_\Lambda = \left\{ x \in \ell^2(\mathbb{N}) : \sum_n \Lambda(n)^2 |x_n|^2 < \infty \right\}, \quad (Ax)_n = \Lambda(n)x_n. \quad (\text{PM4})$$

It is positive and self-adjoint. Positivity follows by summing the nonnegative quantities $\Lambda(n)|x_n|^2$; that series is finite on $\mathcal{D}_\Lambda$ by Cauchy–Schwarz. To identify the adjoint, test against each basis vector. A vector y belongs to the adjoint domain exactly when the sequence $(\Lambda(n)y_n)_n$ is square summable, and the adjoint is multiplication by that sequence. This is precisely (PM4). For $x \in \mathcal{D}_\Lambda$, its first K coordinates converge to x in both Hilbert norm and the norm of Ax . Thus the finite vectors form a core. On this core, (PM3) proves the exact identity

$$\Lambda(N) = - \sum_{\substack{d \geq 2 \\ \mu(d) \neq 0}} \mu(d) \log(d) P_d, \quad (\text{PM5})$$

where the right side stabilizes on each finite vector and its closure is the operator (PM4).

The convergence assertion cannot be strengthened to strong convergence of $A_D x$ for every $x \in \mathcal{D}_\Lambda$. This has an explicit counterexample. Choose distinct odd primes q_j so large that $\log q_j \geq j^2$, and define

$$x_{2q_j} = 1/j, \quad x_n = 0 \text{ otherwise.} \quad (\text{PM6})$$

Arbitrarily large primes exist: a prime divisor of one plus the product of any finite set of primes is outside that set. Thus this choice is available. The series $\sum 1/j^2$ converges, and $\Lambda(2q_j) = 0$, so $x \in \mathcal{D}_\Lambda$ and $Ax = 0$. At cutoff $D = q_j$, the divisors $2, q_j$ contribute to its $2q_j$ coordinate and $2q_j$ does not. Hence

$$a_{q_j}(2q_j) = \log(2q_j), \quad \|A_{q_j}x\| \geq \log(2q_j)/j \geq j. \quad (\text{PM7})$$

The claimed stronger convergence therefore fails even on the kernel of A .

A useful valid common domain is explicit. Set

$$b(1) = 0, \quad b(n) = \sum_{d|n} |\mu(d)| \log d = 2^{\omega(n)-1} \log \text{rad}(n) \quad (n > 1), \quad \mathcal{D}_b = \left\{ x : \sum b(n)^2 |x_n|^2 < \infty \right\}. \quad (\text{PM8})$$

Here $\omega(n)$ counts its distinct prime factors and $\text{rad}(n)$ is their product. Each factor appears in exactly $2^{\omega(n)-1}$ squarefree divisors, proving the formula for b . The bounds $|a_D(n)| \leq b(n)$ and $\Lambda(n) \leq b(n)$, followed by dominated convergence of the series of squared coordinates, prove $A_D x \rightarrow Ax$ for every $x \in \mathcal{D}_b$. This statement preserves the actual cutoff in (PM2).

3. The weighted trace and its exact factor

Fix $s \in \mathbb{C}$ with $\sigma = \Re s > 1$, using the real logarithm for n^{-s} . The diagonal operator $P_d N^{-s}$ has trace norm

$$\|P_d N^{-s}\|_1 = \sum_{k \geq 1} (dk)^{-\sigma} = d^{-\sigma} \zeta(\sigma), \quad \text{Tr}(P_d N^{-s}) = d^{-s} \zeta(s). \quad (\text{PM9})$$

These are HCR26–28 with $q = a = d$, also proved directly by their displayed sums. Therefore

$$\sum_{d \geq 2, \mu(d) \neq 0} |c_d| \|P_d N^{-s}\|_1 \leq \zeta(\sigma) \sum_{d \geq 2} (\log d) d^{-\sigma} < \infty. \quad (\text{PM10})$$

The scalar majorant converges, for example by integrating $(\log x)x^{-\sigma}$ on $[2, \infty)$. Thus the weighted operator series converges in trace norm. Its diagonal entries stabilize to $\Lambda(n)n^{-s}$, which identifies its limit as AN^{-s} . In particular

$$\begin{aligned} \text{Tr}(AN^{-s}) &= \sum_{n \geq 2} \Lambda(n)n^{-s} = \zeta(s) \sum_{d \geq 2} (-\mu(d) \log d) d^{-s} = -\frac{\zeta'(s)}{\zeta(s)}, \\ \|A_D N^{-s} - AN^{-s}\|_1 &\leq \zeta(\sigma) \sum_{d > D} |\mu(d)| (\log d) d^{-\sigma}. \end{aligned} \quad (\text{PM11})$$

For completeness the last analytic identity follows without differentiating a conditionally convergent series. The elementary divisor identity $\sum_{d|n} \mu(d) = \mathbf{1}_{n=1}$, proved by $(1-1)^{\omega(n)}$, and absolute Dirichlet convolution give $\zeta(s) \sum \mu(d) d^{-s} = 1$ for $\Re s > 1$. The two series and their derivatives converge locally uniformly there, using the same logarithmic majorants as (PM10). Differentiate $\sum \mu(d) d^{-s} = 1/\zeta(s)$, obtaining $\sum (-\mu(d) \log d) d^{-s} = -\zeta'(s)/\zeta(s)^2$, and multiply by $\zeta(s)$. In particular all signs in (PM11) are fixed. This is the function $-r(s)$ of HA10.

4. The original lattice and the two different cutoff corrections

Retain a nontrivial bounded distributive lattice L , with bottom 0_L and top 1_L , and exactly HCR7:

$$\begin{aligned} G_L(\mathbb{C}) &= \{(a, \lambda) : a \neq 0 \implies \lambda = 1_L\}, \\ (a, \lambda) + (b, \eta) &= (a + b, \lambda \vee \eta), \quad (a, \lambda)(b, \eta) = (ab, \lambda \wedge \eta), \\ \tau &= (0, 0_L), \quad e = (0, 1_L), \quad \widehat{c} = (c, 1_L). \end{aligned} \quad (\text{PM12})$$

All matrices below have off-diagonal entries τ , so they are in the row-and-column-finite matrix semiring of HCR7–9. Let

$$(\widehat{P}_d)_{nn} = \begin{cases} 1_S, & d \mid n, \\ \tau, & d \nmid n, \end{cases} \quad (M_d)_{nn} = \begin{cases} \tau, & d \mid n, \\ e, & d \nmid n, \end{cases} \quad R_d = \widehat{1/d} \sum_{j=0}^{d-1} \widehat{E}(j/d). \quad (\text{PM13})$$

The root sum from HCR3, calculated with the actual labels, gives

$$R_d = \widehat{P}_d + M_d, \quad (R_d)_{nn} = (\mathbf{1}_{d|n}, 1_L). \quad (\text{PM14})$$

At a nonmultiple the roots cancel in amplitude, and their sum is e , not τ .

For $D \geq 2$, define the sparse and raw sums using only the nonzero coefficients specified in (PM2):

$$\mathcal{A}_D = \sum_{d \in I_D} \hat{c}_d \hat{P}_d, \quad \mathcal{R}_D = \sum_{d \in I_D} \hat{c}_d R_d, \quad \chi_D(n) = \mathbf{1}_{\exists p \leq D: p \text{ prime}, p|n}. \quad (\text{PM15})$$

Then the full entry formulas are

$$(\mathcal{A}_D)_{nn} = \begin{cases} (a_D(n), 1_L), & \chi_D(n) = 1, \\ \tau, & \chi_D(n) = 0, \end{cases} \quad (\mathcal{R}_D)_{nn} = (a_D(n), 1_L). \quad (\text{PM16})$$

Indeed an active squarefree divisor of n exists exactly when a prime divisor at most D exists. The sparse sum then joins at least one top label, including when its amplitudes subsequently cancel. Every raw summand has top diagonal label, and there is at least one such summand since $2 \in I_D$. This proves (PM16) at every entry. For $D = 1$ both sums are the zero matrix; it is a separate empty-sum case.

The least missing-support correction is

$$(\mathcal{M}_D)_{nn} = \begin{cases} \tau, & \chi_D(n) = 1, \\ e, & \chi_D(n) = 0, \end{cases} \quad \boxed{\mathcal{A}_D + \mathcal{M}_D = \mathcal{R}_D}. \quad (\text{PM17})$$

Every correction C satisfying $\mathcal{A}_D + C = \mathcal{R}_D$ has zero amplitudes, off-diagonal entries τ , top diagonal label when $\chi_D(n) = 0$, and any original label when $\chi_D(n) = 1$. These conditions are necessary by amplitude equality and by the joins in (PM12); they are sufficient by the same calculation. Thus (PM17) is the unique least correction in the entrywise order on zero-amplitude labels, and the entire family of corrections has been specified.

The actual sum of the individual Fourier corrections is usually larger. Put

$$\mathcal{K}_D = \sum_{d \in I_D} \hat{c}_d M_d, \quad Q_D = \prod_{p \leq D, p \text{ prime}} p. \quad (\text{PM18})$$

Each nonzero c_d acts on e as e . Therefore

$$(\mathcal{K}_D)_{nn} = \begin{cases} \tau, & Q_D \mid n, \\ e, & Q_D \nmid n, \end{cases} \quad \boxed{\mathcal{A}_D + \mathcal{K}_D = \mathcal{R}_D}. \quad (\text{PM19})$$

To prove the first formula, all the active d divide n exactly when all primes at most D divide n : the forward implication uses that each of these primes is active, and the reverse uses that an active d is squarefree. The second formula follows by adding (PM14) with its stated coefficients, or directly from (PM16). In particular $\mathcal{M}_D \leq \mathcal{K}_D$, but equality need not hold. For $D = 3$, at $n = 2$ the least correction is τ and the directly summed correction is e . The additional e is absorbed by the already top-labelled entry of $\mathcal{A}_D$; it has not been set equal to τ .

5. All limiting support, including the index one

Every limit in this paragraph is eventual equality at each matrix entry, so it requires no completeness assumption or infinite join in L . The exact limits are

$$\begin{aligned} \mathcal{A}_{nn} &= \begin{cases} \tau, & n = 1, \\ (\Lambda(n), 1_L), & n \geq 2, \end{cases} & \mathcal{R}_{nn} &= (\Lambda(n), 1_L) \quad (n \geq 1), \\ \mathcal{M}_{nn} &= \begin{cases} e, & n = 1, \\ \tau, & n \geq 2, \end{cases} & \mathcal{K} = \mathcal{Z} &= \text{diag}(e, e, \dots). \end{aligned} \quad (\text{PM20})$$

For $n \geq 2$, some prime divisor appears by $D \geq n$, and (PM3) fixes the amplitude. The index 1 has no active divisors. For the last limit choose a prime not dividing the fixed integer n ; once it is included, $Q_D \nmid n$ forever. Existence follows from the elementary infinitude argument already supplied. Thus

$$\boxed{\mathcal{A} + \mathcal{M} = \mathcal{R} = \mathcal{A} + \mathcal{Z}.} \quad (\text{PM21})$$

In particular every integer with at least two distinct prime factors has the entry e in both $\mathcal{A}$ and $\mathcal{R}$, since its zero value of Λ was produced by cancellation. Only the sparse index 1 has τ . The least limiting defect and the directly summed limiting defect are different matrices, even though both complete the same equation.

The stipulated index set is also essential to the full lift. In the complex identity one may insert the zero term $-\mu(1) \log 1 P_1 = 0$. Lifting that newly inserted coefficient as $\widehat{0} = e$ instead gives $e\widehat{P}_1 = Z$, since $P_1 = I$. Thus adding this zero term before lifting changes the sparse lift from $\mathcal{A}$ to $\mathcal{A} + Z = \mathcal{R}$. This is the exact connecting map for the two presentations, not permission to identify their support matrices. The original index set $d \geq 2, \mu(d) \neq 0$ in (PM5) has been retained throughout.

For an arbitrary input with original coordinates $v_n = (x_n, \lambda_n)$, where $x \in \ell^2$ and $x_n \neq 0 \Rightarrow \lambda_n = 1_L$, the finite sums act by

$$(\mathcal{A}_D v)_n = \begin{cases} (a_D(n)x_n, \lambda_n), & \chi_D(n) = 1, \\ \tau, & \chi_D(n) = 0, \end{cases} \quad (\mathcal{R}_D v)_n = (a_D(n)x_n, \lambda_n). \quad (\text{PM22})$$

Their amplitude operators are bounded at each finite D . The limiting amplitude operator has domain (PM4). On that domain the limiting labelled maps act as $(\Lambda(n)x_n, \lambda_n)$ at every $n \geq 2$, with τ at 1 for the sparse map and $(0, \lambda_1)$ at 1 for the raw map. This is an entrywise definition with the domain of the amplitude stated; it does not assert the false Hilbert convergence excluded in (PM7).

The complete coefficient injection of HCR9 sends a matrix entry (a, λ) to $(a, \iota_L(\lambda))$, where $\iota_L : L \hookrightarrow K \otimes_{\mathbb{B}} L$ has its stated retraction. For diagonal masks write χ_D^K and $(1 - \chi_D)^K$ for the separate selected and complementary indicators; the notation is a description of masks, not subtraction in the lattice semiring. The exact images are

	complex amplitude	full support matrix	
$\mathcal{A}_D$	A_D	χ_D^K	
$\mathcal{R}_D$	A_D	I_K	
$\mathcal{M}_D$	0	$(1 - \chi_D)^K$	
$\mathcal{K}_D$	0	$\mathbf{1}_{Q_D \nmid n}^K$	(PM23)

The same map takes the limits to the corresponding masks in (PM20). The retraction recovers every original label, including the λ_n in (PM22). The common-label completion $C(D) = D + Z$ of HCR16 sends $\mathcal{A}_D$ to $\mathcal{R}_D$ and $\mathcal{A}$ to $\mathcal{R}$. Its kernel congruence is equality of amplitudes; the complete injection (PM23) has not taken that quotient.

There is also a supported weighted trace, with its domain expressly fixed. For $\Re s > 1$, multiply the diagonal entries by $(n^{-s}, 1_L)$, take finite basis traces, and then take the absolutely convergent amplitude limit while retaining the eventual label. For $D \geq 2$, sparse finite basis traces are τ at cutoff 1 and top-labelled at every cutoff at least 2. Raw traces are top-labelled at every positive cutoff. Hence both limiting traces are

$$(\text{Tr}(A_D N^{-s}), 1_L), \quad \text{and, as } D \rightarrow \infty, \quad (-\zeta'(s)/\zeta(s), 1_L). \quad (\text{PM24})$$

The correction traces are e , because their zero-amplitude support contains index 1. Equality of the two scalar traces thus does not identify their different support matrices. This construction does not define a trace on arbitrary infinite labelled matrices.

6. The exact map into the complete Weil prime distribution

Use the original SZW19 convention

$$\mathcal{T} = C_c^\infty(\mathbb{R}; \mathbb{C}), \quad M_h(s) = \int_{\mathbb{R}} h(v) e^{-(s-1/2)v} dv, \quad h^\#(v) = \overline{h(-v)}. \quad (\text{PM25})$$

For $h \in \mathcal{T}$, the diagonal operator

$$B_h = N^{-1/2} \{h(\log N) + h(-\log N)\} \quad (\text{PM26})$$

has finite rank: if $\text{supp } h \subset [-R, R]$, its entries vanish for $n > e^R$. Its entry at 1 is $2h(0)$. Since $\Lambda(1) = 0$, the exact trace receiver is

$$\begin{aligned} \text{Tr}(AB_h) &= \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \{h(\log n) + h(-\log n)\} \\ &= \sum_p \sum_{k \geq 1} (\log p) p^{-k/2} \{h(k \log p) + h(-k \log p)\} = P_{\text{fin}}(h). \end{aligned} \quad (\text{PM27})$$

The prime-power expansion uses (PM1), not an altered norm for supported zero. Every sum in (PM27) is finite for a fixed test. At cutoff $D \geq \max(2, \lfloor e^R \rfloor)$, $\text{Tr}(A_D B_h)$ is already equal to (PM27).

To state a full supported test receiver rather than hiding its choices, take $K \geq \max(2, \lfloor e^R \rfloor)$ and the specified finite-basis lift

$$(\widehat{B}_{h,K})_{nn} = \begin{cases} (n^{-1/2}\{h(\log n) + h(-\log n)\}, 1_L), & 1 \leq n \leq K, \\ \tau, & n > K. \end{cases} \quad (\text{PM28})$$

It keeps the zero values inside that chosen finite basis as e . Multiplying by the matrices in (PM20), followed by the finite trace, gives

$$\text{tr}(\mathcal{A}\widehat{B}_{h,K}) = \text{tr}(\mathcal{R}\widehat{B}_{h,K}) = (P_{\text{fin}}(h), 1_L), \quad \text{tr}(\mathcal{M}\widehat{B}_{h,K}) = e. \quad (\text{PM29})$$

The first top label in the sparse trace is at index 2; the raw trace already has it at index 1. These assertions remain true if $h = 0$, because (PM28) deliberately specifies a top-labelled lift of its zero values. This lift therefore is not asserted to be an additive zero-preserving linear map from the complex test space into the ambient split semiring. The amplitude trace is the complex-linear distribution (PM27); the complete matrix injection (PM23) separately retains the support. If a support-adapted lift of B_h is chosen instead, its zero entries have different labels and (PM28)–(PM29) must not be silently reused.

The connection to the entire supported formula is now exact. With $H = M_h$, let $\mathbf{e}_\lambda$ denote the coordinate basis of $\mathbb{C}[L]$ for the finite support lattice used in SZW. Its existing identity is

$$\begin{aligned} B_L(h) &= (H(0) + H(1))\mathbf{e}_{1_L} + H(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda, \\ Z_L(h) &= Z(h)\mathbf{e}_{1_L}, \\ D_L(h) &= (\text{Tr}(AB_h) - A_\infty(h))\mathbf{e}_{1_L} + H(0) \sum_{\lambda \neq 1_L} \mathbf{e}_\lambda, \\ B_L(h) - Z_L(h) &= D_L(h), \end{aligned} \quad (\text{PM30})$$

where

$$Z(h) = \sum_{\rho} m_{\rho} M_h(\rho), \quad A_\infty(h) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{h}(t) \{\Re \psi(1/4 + it/2) - \log \pi\} dt. \quad (\text{PM31})$$

Here the sum ranges over all nontrivial zeta zeros with their multiplicities and $\widehat{h}(t) = \int h(v) e^{-itv} dv$. Formula (PM30) follows by substituting the proved equality (PM27) into SZW24–25 and SZW33–34. Those original formulas retain the theta endpoint traces and prove their convergence. No lower coordinate is replaced by a proposed numerical prime weight. For the noncompact Cauchy convolution in HA7, the same receiver is an absolutely convergent trace: its two terms at $\pm \log n$ give precisely $(n^{-w} + n^{-\bar{z}})/(w + \bar{z} - 1)$, yielding the prime term of HA11 when $\Re z, \Re w > 1$. Thus (PM11), (PM27), and HA11 receive exactly the same arithmetic coefficients with their original constants.

7. Translation is the map relevant to positivity

For $f, g \in \mathcal{T}$, direct substitution in convolution gives

$$(f^\# * g)(v) = \int_{\mathbb{R}} \overline{f(x)} g(x+v) dx = \langle f, U_v g \rangle_{L^2(\mathbb{R})}, \quad (U_v g)(x) = g(x+v), \quad U_v^* = U_{-v}. \quad (\text{PM32})$$

If the difference of the two test supports is contained in $[-R, R]$, all terms outside that interval vanish, and

$$P_{\text{fin}}(f^\# * g) = \left\langle f, \sum_{2 \leq n \leq e^R} \frac{\Lambda(n)}{\sqrt{n}} \{U_{\log n} + U_{-\log n}\} g \right\rangle. \quad (\text{PM33})$$

The finite operator on the right is bounded and self-adjoint; its norm is at most $2 \sum_{n \leq e^R} \Lambda(n)/\sqrt{n}$. The expression stabilizes on each fixed test pair as R grows. An infinite bounded translation operator is not required or asserted. This is the exact receiver from the positive multiplication operator (PM4) to the prime sesquilinear form. The map from finite diagonal coefficients to the sum of translations in (PM33) is linear and respects adjoints, but is not a positive map.

Here is a complete sign test. Write $a = \log 2$, choose

$$0 < 2\delta < \min\{\log 2, \log(3/2)\}, \quad 0 \neq \varphi \in C_c^\infty((-\delta, \delta)), \quad \varphi_a(x) = \varphi(x - a). \quad (\text{PM34})$$

For any nonzero η with this support, put $k = \eta^\# * \eta$ and $g_\pm = \eta \pm \eta_a$. Then

$$g_\pm^\# * g_\pm(v) = 2k(v) \pm k(v - a) \pm k(v + a), \quad k(0) = \|\eta\|_2^2, \quad \text{supp } k \subset [-2\delta, 2\delta]. \quad (\text{PM35})$$

Indeed expand the four convolutions in (PM32); translating both arguments leaves k unchanged and translating one shifts it by $\pm a$. At $v = a$ or $-a$, only one shifted term survives, giving $\pm \|\eta\|_2^2$. At $v = \pm \log n$ with $n \geq 3$, every term vanishes because $a + 2\delta < \log 3$. Consequently

$$\boxed{P_{\text{fin}}(g_{\pm}^{\#} * g_{\pm}) = \pm \frac{2 \log 2}{\sqrt{2}} \|\eta\|_2^2 = \pm \sqrt{2} \log 2 \|\eta\|_2^2.} \quad (\text{PM36})$$

This also proves nonpositivity of the coefficient-to-translation map already on the positive single diagonal coefficient at index 2. Positive diagonal coefficients alone therefore do not imply a positive autocorrelation receiver.

8. The same exact signs survive endpoint annihilation

Keep the test operator and its original constant

$$T = \partial_v^2 - \frac{1}{4}. \quad (\text{PM37})$$

Integration by parts twice, with no boundary term for a compact smooth test, gives

$$M_{Tf}(s) = \left((s - \frac{1}{2})^2 - \frac{1}{4}\right) M_f(s) = s(s-1) M_f(s). \quad (\text{PM38})$$

Thus both endpoint values vanish exactly; at every nontrivial zero the multiplying factor is nonzero. The coefficients are real and the order is even, so $(Tf)^{\#} = T(f^{\#})$. Differentiation of compact convolution gives

$$(Tf)^{\#} * (Tg) = T^2(f^{\#} * g), \quad M_{(Tf)^{\#} * (Tg)}(s) = [s(s-1)]^2 \overline{M_f(1-\bar{s})} M_g(s). \quad (\text{PM39})$$

All support and convergence requirements of (PM27) and (PM30) continue to hold.

For an explicit endpoint-zero version of (PM36), take $\eta = T\varphi$ with φ from (PM34). This η is nonzero. Otherwise $\varphi'' = \varphi/4$ everywhere; solving this equation with its zero value and zero derivative outside the compact support gives the identically zero solution, a contradiction. Its support remains contained in $(-\delta, \delta)$, and T commutes with translation. Hence

$$g_{\pm} = T(\varphi \pm \varphi_a), \quad M_{g_{\pm}}(0) = M_{g_{\pm}}(1) = 0, \quad \boxed{P_{\text{fin}}(g_{\pm}^{\#} * g_{\pm}) = \pm \sqrt{2} \log 2 \|T\varphi\|_2^2.} \quad (\text{PM40})$$

There is no approximation in these signs, and no zero-location assumption in their proof.

For these or any endpoint-zero tests, write $h = g^{\#} * g$. Then $M_h(0) = M_h(1) = 0$ by the convolution formula, so (PM30) is exactly

$$\mathbf{B}_L(h) = 0, \quad \mathbf{D}_L(h) = (P_{\text{fin}}(h) - A_{\infty}(h))\mathbf{e}_{1L}, \quad \mathbf{Z}_L(h) = (A_{\infty}(h) - P_{\text{fin}}(h))\mathbf{e}_{1L}. \quad (\text{PM41})$$

The coordinate spaces and all support maps remain those of (PM30); the lower coefficients here are the calculated zero values of these tests. In particular (PM40) assigns no sign to (PM41) until the archimedean term is included. The exact remaining comparison is between the archimedean form and the translated prime form, with their already fixed constants. The positive operator $\Lambda(N)$, the raw supported zeros, and the endpoint annihilator each have now been connected to that comparison by explicit maps rather than by a positivity assertion.

Source identities and actual reading coverage

- **HCR:** `work/rh_counterfactual_20260913/tau_prime_weil_20260922/cc_totality_20260923/reader_fragments/HURWITZ_CYCLOTOMIC_RECEIVER.tex`, SHA256 `d8a4caa0b66e492e7ffe987ba2aa04beeb201aed3cacc3b30a3b0d4b526e6df38`. Read the complete HCR1–42 and their proofs for this derivation. Operators and full lattice are HCR1–16; the weighted original trace is HCR26–31. Its human operator source is Alain Connes, Caterina Consani and Matilde Marcolli, *Fun with F1*, arXiv:0806.2401v1, original `funBC.tex`, relations `pres`, `idem`, `endo`, `endosigma`, and (c1)–(c4), with exact line ranges recorded in HCR. This derivation has read the HCR proof, not newly reread that author archive; HCR's original-source reading claim is not represented as a new reading here. Equations (PM13)–(PM19) reprove every operator and label identity actually needed.
- **HA:** `HEAT_CAUCHY_ARITHMETIC_DERIVATION.md`, SHA256 `024d3f42723fabf2361cab9400d98d2fe64c57ec114112e6ada6f5430a88f909`. Read HA1–26 for this derivation. HA10–14, pinned public proof supply the exact logarithmic derivative, Cauchy trace and supported coefficients checked in (PM11), (PM30), and the paragraph after (PM31).

- **SZW:** SUPPORTED_ZERO_PRIME_WEIL_DERIVATION.md, SHA256 ff7fbe6cc627720368a7c245052ef8263925ca2d912046dfe45e3be74d4455f4. Read SZW19–26 and SZW33–38, with adjacent text, for this derivation. The complete explicit-formula proof is in SZW24–26 and full support identity SZW33–34, pinned public source. Its human explicit-formula source is Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Appendix II, Theorem 6. The present proof uses the complete programme derivation and does not assert a new reading of that author archive.

The Möbius identity (PM3) is the classical arithmetic inversion identity, proved here with its complete divisor calculation. The new programme calculation is its specified split lift, both distinct support corrections and their limits, its exact full Weil receiver, and the endpoint-zero sign test. No novelty claim about classical Möbius inversion, trace-class diagonal operators, or the classical explicit formula is made.

The actual archimedean form on short-support primitives

This proof gives an explicit rational lower bound, an exact bilinear representation of its positive remainder, and an exact localization on primitive functions retaining all cross terms. It uses the original minus-sign Mellin convention and HA13. No historical novelty claim is made for short-support Weil positivity or for the endpoint-annihilating differential operator.

PS1. Original tests, correlations and the archimedean form

Let $f, g \in C_c^\infty(\mathbb{R}; \mathbb{C})$, and retain the conventions

$$f^\#(u) = \overline{f(-u)}, \quad h_{f,g} = f^\# * g, \quad M_f(s) = \int_{\mathbb{R}} f(u) e^{-(s-1/2)u} du. \quad (\text{PS1})$$

The inner product is $\langle f, g \rangle = \int \overline{f(u)} g(u) du$, conjugate-linear in its first argument. Define the translation $\tau_a f(u) = f(u - a)$. Changing variables in the convolution gives

$$h_{f,g}(a) = \langle \tau_a f, g \rangle, \quad h_{f,g}(-a) = \langle f, \tau_a g \rangle, \quad h_{f,g}(0) = \langle f, g \rangle. \quad (\text{PS2})$$

For the diagonal correlation, $h(-a) = \overline{h(a)}$ and $|h(a)| \leq h(0) = \|f\|^2$ by Cauchy–Schwarz. If the two functions are supported in one interval of length $\ell > 0$, their correlation vanishes outside $[-\ell, \ell]$. This statement does not depend on the center of the interval.

Use precisely the archimedean functional proved in HA13, complete public proof:

$$\begin{aligned} \mathcal{A}(f, g) := A_\infty(h_{f,g}) = & -(\gamma + \log \pi) \langle f, g \rangle \\ & + \int_0^\infty \frac{e^{-x} \langle f, g \rangle - \frac{1}{2} e^{-x/4} (h_{f,g}(x/2) + h_{f,g}(-x/2))}{1 - e^{-x}} dx. \end{aligned} \quad (\text{PS3})$$

Here γ is Euler’s constant. This is a Hermitian sesquilinear form on the stated smooth test space. The cancellation of the numerator at $x = 0$ makes the integral converge; smoothness suffices. No sign convention for an archimedean geometric distribution is substituted for PS3: its sign is the archimedean term appearing positively in the original spectral explicit formula.

PS2. Exact bilinear short-support identity

For $\ell > 0$, set

$$\begin{aligned} C(\ell) = & -\gamma - \log \pi - \log(1 - e^{-2\ell}) \\ & + \int_0^{2\ell} \frac{e^{-x} - e^{-x/4}}{1 - e^{-x}} dx, \quad w(x) = \frac{e^{-x/4}}{1 - e^{-x}}. \end{aligned} \quad (\text{PS4})$$

For f, g supported in one interval of length ℓ , the following is an equality, not merely an inequality:

$$\boxed{\mathcal{A}(f, g) = C(\ell) \langle f, g \rangle + \frac{1}{2} \int_0^{2\ell} w(x) \langle \tau_{x/2} f - f, \tau_{x/2} g - g \rangle dx.} \quad (\text{PS5})$$

To prove it, split PS3 at 2ℓ . Above that point the two correlations vanish, while

$$\int_{2\ell}^\infty \frac{e^{-x}}{1 - e^{-x}} dx = -\log(1 - e^{-2\ell}).$$

Below that point use the unitary translation identity

$$\langle \tau_a f - f, \tau_a g - g \rangle = 2\langle f, g \rangle - h_{f,g}(a) - h_{f,g}(-a).$$

Its substitution separates the remaining constant multiple of $\langle f, g \rangle$, which is exactly the integral in PS4, and leaves the positive-weight integral in PS5. The latter converges at zero: $\|\tau_a f - f\| \leq |a| \|f'\|$, obtained by the fundamental theorem of calculus and Minkowski's inequality. Thus the bilinear integrand is $O(x^2)w(x) = O(x)$ there.

In particular, taking the diagonal gives

$$\mathcal{A}(f, f) = C(\ell) \|f\|^2 + \frac{1}{2} \int_0^{2\ell} w(x) \|\tau_{x/2} f - f\|^2 dx \geq C(\ell) \|f\|^2. \quad (\text{PS6})$$

The remainder is explicitly a Gram integral of translation differences. Consequently the same local lower bound holds as a matrix inequality on every finite family supported in that one common interval. The equality also proves directly that the proposed $C(\ell)$ has the correct tail sign and every factor of two from the original HA13 coordinate.

PS3. The rational lower bound 55/192

For $x > 0$ put $q = e^{-x/4} \in (0, 1)$. The integrand in PS4 is

$$\frac{e^{-x} - e^{-x/4}}{1 - e^{-x}} = -\frac{q(1 + q + q^2)}{1 + q + q^2 + q^3} \geq -\frac{3}{4}. \quad (\text{PS7})$$

The final inequality is equivalent to $q + q^2 + q^3 \leq 3$, which holds term by term. Its limit at zero is $-3/4$, so the integral has no unaccounted singular endpoint. Therefore it is at least $-3\ell/2$.

The elementary inequalities used next can all be retained without decimal approximations. First $\gamma \leq 1$: the decreasing sequence $H_n - \log n$ has limit γ and starts at one. Its decrease follows from $\log(1 + 1/n) \geq 1/(n+1)$, by integration of $1/x$ on $[1, 1 + 1/n]$. Its lower bound follows by comparing the harmonic sum with the same integral, so this passage to the limit is valid. Second $\pi \leq 4$, for example from containment of the unit disc in the square of side two. Third,

$$\log x \geq \frac{2(x-1)}{x+1} \quad (x \geq 1),$$

because the derivative of their difference is $(x-1)^2/[x(x+1)^2] \geq 0$ and the difference is zero at one. At $x = 2$ this yields $\log 2 \geq 2/3$, hence $\log 4 \geq 4/3$.

Finally $1 - e^{-2\ell} \leq 2\ell$ follows by integrating $e^{-x} \leq 1$. Combining all these inequalities in the correct direction gives

$$\begin{aligned} C(\ell) &\geq -1 - \log 4 - \log(2\ell) - \frac{3}{2}\ell \\ &= -1 + \log \frac{1}{8\ell} - \frac{3}{2}\ell \\ &\geq -1 + \log 4 - \frac{3}{64} \geq \frac{1}{3} - \frac{3}{64} = \frac{55}{192} \quad (0 < \ell \leq 1/32). \end{aligned} \quad (\text{PS8})$$

Thus the actual archimedean form satisfies the proved bound

$$\boxed{\mathcal{A}(f, f) \geq \frac{55}{192} \|f\|^2} \quad (\text{PS9})$$

for any smooth f supported in an interval of length at most $1/32$. No endpoint-vanishing condition is required for this archimedean assertion. A smooth function supported in a zero-length interval is zero and satisfies it trivially.

PS4. Endpoint-annihilating primitives and the whole Weil form

Retain the original differential operator

$$T = \frac{d^2}{du^2} - \frac{1}{4}, \quad f = Tg, \quad g \in C_c^\infty(\mathbb{R}). \quad (\text{PS10})$$

Two integrations by parts, with no boundary terms, give

$$M_{Tg}(s) = ((s - 1/2)^2 - 1/4)M_g(s) = s(s - 1)M_g(s). \quad (\text{PS11})$$

Thus both endpoint values vanish exactly, while T does not enlarge the support. Its squared norm, with complex-valued functions permitted, is exactly

$$\begin{aligned} \|Tg\|^2 &= \|g''\|^2 - \frac{1}{2}\Re\langle g'', g \rangle + \frac{1}{16}\|g\|^2 \\ &= \|g''\|^2 + \frac{1}{2}\|g'\|^2 + \frac{1}{16}\|g\|^2. \end{aligned} \quad (\text{PS12})$$

For a compact test f write its original reflected Weil form as

$$\mathcal{Q}(f, g) = \sum_{\rho} m_{\rho} \overline{M_f(\rho^{\#})} M_g(\rho) = M_{f\#*g}(0) + M_{f\#*g}(1) + \mathcal{A}(f, g) - P_{\text{fin}}(f^{\#} * g). \quad (\text{PS13})$$

The complete underlying formula is SZW24–26 and SZW33–38. The sum converges by the original strip zero count and rapid Mellin decay; the displayed identity is SZW25 with its original constants. Its two endpoint values for $f = Tg$ are zero by PS11 and the convolution identity. If g is supported in an interval of length $\ell \leq 1/32$, the correlation of Tg is supported in $[-\ell, \ell]$. Since $\ell < \log 2$, every prime term $h(\pm \log n)$ with $n \geq 2$ vanishes exactly. Consequently the whole spectral form in PS13, with endpoints retained and evaluated, equals the archimedean form and satisfies

$$\boxed{\mathcal{Q}(Tg, Tg) \geq \frac{55}{192} \left(\|g''\|^2 + \frac{1}{2}\|g'\|^2 + \frac{1}{16}\|g\|^2 \right)}. \quad (\text{PS14})$$

The inequality is strict for $g \neq 0$, since its right side is strictly positive. This proves the proposed primitive estimate with the exact positive sign in the original full spectral form. In the support-resolved SZW identity the endpoint amplitude in every boundary coordinate is zero for these tests; the labels themselves remain present. No support element is identified with a scalar zero by this computation.

For precision, every compact smooth test with both endpoint moments zero does have a unique compact smooth primitive in this sense. An explicit inverse, preserving the containing support interval, is

$$g(u) = - \int_{\mathbb{R}} e^{-|u-v|/2} f(v) dv. \quad (\text{PS15})$$

The kernel $-e^{-|u|/2}$ has first derivative jump one and satisfies $TK = \delta_0$ in distributions. Thus $Tg = f$, and convolution with smooth compact f gives a smooth function. If $\text{supp } f \subset [a, b]$, then for $u > b$ its value is $-e^{-u/2}M_f(0) = 0$, and for $u < a$ it is $-e^{u/2}M_f(1) = 0$. Hence $\text{supp } g \subset [a, b]$. The difference of two compact primitives solves the homogeneous equation and is a linear combination of $e^{u/2}$ and $e^{-u/2}$; compact support forces both coefficients to vanish. This proves the stated exact inverse and uniqueness, not only existence of some endpoint-null images of T .

PS5. Exact bilinear localization of the primitive

Let g now be an arbitrary compact smooth function. Choose a finite smooth partition of unity χ_j on a neighborhood of its support, with each χ_j supported in an interval I_j of length at most $1/32$. Such a partition follows by covering the compact support by finitely many shorter open intervals, choosing nonnegative bumps positive on a smaller cover, and dividing by their positive sum on a neighborhood of the support; an auxiliary cutoff inside that neighborhood makes each resulting function compact smooth. Put

$$g_j = \chi_j g, \quad f_j = Tg_j. \quad (\text{PS16})$$

These formulas retain the derivatives of the partition:

$$f_j = \chi_j Tg + 2\chi_j' g' + \chi_j'' g, \quad g = \sum_j g_j, \quad Tg = \sum_j f_j. \quad (\text{PS17})$$

The last equality uses $\sum_j \chi_j = 1$, $\sum_j \chi_j' = \sum_j \chi_j'' = 0$ near $\text{supp } g$; outside that support all the derivatives of g vanish. It is therefore an equality of the original tests, not a replacement by the functions $\chi_j Tg$ alone.

Each piece has both endpoint moments zero and the stated short support. For the exact finite Hermitian matrix

$$q_{jk} = \mathcal{Q}(f_j, f_k),$$

sesquilinearity and the finite convolution expansion give

$$\begin{aligned} \mathcal{Q}(Tg, Tg) &= \sum_{j,k} q_{jk} = \sum_j q_{jj} + 2\Re \sum_{j < k} q_{jk}, \\ q_{jk} &= \mathcal{A}(f_j, f_k) - P_{\text{fin}}(f_j^\# * f_k), \\ q_{jj} &\geq \frac{55}{192} \left(\|g_j''\|^2 + \frac{1}{2} \|g_j'\|^2 + \frac{1}{16} \|g_j\|^2 \right). \end{aligned} \quad (\text{PS18})$$

The endpoint terms vanish also for each cross pair, since both primitive factors have their two endpoint moments zero. The prime term vanishes for each diagonal, but is retained in every off-diagonal entry. The exact support relation is

$$\text{supp}(f_j^\# * f_k) \subset I_k - I_j. \quad (\text{PS19})$$

Thus distinct windows can have nonzero prime correlations even though each window separately has none. PS18 does not infer a sign for the entire matrix from the positive diagonal terms.

The bilinear primitive and convolution maps also retain their exact coefficients:

$$\langle Tg_j, Tg_k \rangle = \langle g_j'', g_k'' \rangle + \frac{1}{2} \langle g_j', g_k' \rangle + \frac{1}{16} \langle g_j, g_k \rangle, \quad (\text{PS20})$$

$$(Tg_j)^\# * (Tg_k) = T^2(g_j^\# * g_k), \quad T^2 = D^4 - \frac{1}{2}D^2 + \frac{1}{16}. \quad (\text{PS21})$$

For PS20 integrate the two cross terms by parts separately. For PS21 use that T has real even coefficients, commutes with reflection, and may act on either factor of a smooth compact convolution. Its Mellin multiplier is correspondingly $s^2(s-1)^2$. Equations PS17–PS21 are the exact bilinear localization; no partition-derivative or inter-window term is removed.

When several pieces are supported in the same one interval of length $\ell \leq 1/32$, PS5 applies to their whole Gram matrix, not just its diagonal. For separated windows the exact entries remain those in PS18–PS19; they are not covered by a common short-interval lower bound. This specifies precisely the local estimate and the remaining arithmetic interactions.

PS6. Original-source reading and comparison

The original author source used for the operator comparison is: Alain Connes and Caterina Consani, *Weil positivity and Trace formula: the archimedean place*, arXiv:2006.13771v1, `sources/weil/2006.13771v1/weil-compo.tex`.

Original file SHA256: B01D353B0423B6FEDEE373C3C33FE3678EEA733F62049810750F6C64EF20F3FC. Intact 2006.13771v1.tar archive SHA256: 771A004B70FB0C36CAE11351BD29CEC50B3DA13896BB1EF5015BBDD68BEADB6D.

Actual reading coverage for this derivation:

- lines 95–139: original explicit-formula convention, endpoint vanishing, the stated local positivity and its historical attribution, and the introductory theorem;
- lines 663–842: Section `sectsupport`, including `boaskac`, `vanishing1`, `vanishing2`, `thmqkey1`, `Qprime`, `remQprime` and `qeasy`, with their proofs.

No PDF was read. The author archive remains intact. These locators record bounded reading, not an exhaustive reading of the paper. Its local operator method and small-support positivity are prior mathematical work; the elementary bound PS8 is derived directly from the programme’s proved HA13 formula and does not claim to replace those stronger results.

The coordinate and sign map to the source operator is explicit. Under $u = \log \rho$, its $Q_{CC} = -(\rho \partial_\rho)^2 + 1/4$ becomes $-D^2 + 1/4 = -T$. Thus applying our T to a primitive corresponds to applying their Q_{CC} to the negative of that primitive. On a convolution square the two minus signs cancel, giving exactly PS21. The source’s support-preserving endpoint-vanishing lemma therefore has the same receiving condition, with this sign retained. PS15 is an independent additive-coordinate inverse verifying that comparison directly.

The symbol τ in that author’s archimedean distribution is a source-specific distribution; it is not identified with the programme’s unsupported element. The present proof uses HA13, PS1 and PS3 to fix its arithmetic sign and all constants.

The concluding bound is an established statement about the short-support diagonal pieces and the common-window matrices. The exact cross-window form in PS18 remains in the global calculation. Nothing here assigns its sign or proves RH from the local estimate alone.

One fixed primitive test and the full translation criterion

This proof constructs one specified compactly supported smooth test, chosen independently of every zeta zero. Its Mellin transform is nonzero at every possible off-critical zero. The full reflected Weil pairing of that test with its translates is bounded on the positive half-line exactly when RH holds. The proof does not establish that boundedness.

The original minus-exponent Mellin convention, actual zero multiset, boundary filter, and global Laplace argument are those of BPZ1–BPZ23. That entire TeX source was read for this derivation. The explicit construction here makes its detecting test independent of a selected or hypothetical zero. The method is an application of the classical Weil-positivity mechanism, with no assertion of historical priority for a bounded-translation criterion.

1. A fixed infinite convolution

The bump used here is classical. Juan Arias de Reyna, *An infinitely differentiable function with compact support: Definition and properties*, Rev. Real Acad. Ciencias Madrid **76** (1982), 21–38, author’s English translation arXiv:1702.05442v1, Theorem 1 and equations (1), (4), (9), constructs the function φ used in the following exact comparison. His Fourier convention is $\widehat{\varphi}(\zeta) = \int \varphi(t)e^{-2\pi i t \zeta} dt$. In the unchanged coordinates of this proof, $b_r(v) = r^{-1}\varphi(v/r)$ and $G_r(z) = \widehat{\varphi}(rz/(2\pi i))$. Indeed, substituting $v = rt$ in the integral gives that transform, and its factor indexed by $h \geq 0$ is $\sinh(rz2^{-h-1})/(rz2^{-h-1})$, the factor indexed by $j = h + 1$ in (UP3). Fourier uniqueness proves the density identity. The factor r^{-1} , the support $[-r, r]$, and total mass one are retained. The complete direct construction below is kept for the reader; it is not a new invention of this bump. Arias de Reyna credits Jessen and Wintner (1935) for its Fourier-transform construction; that original article was not inspected for this citation revision.

Throughout the construction retain

$$r = \frac{1}{64}, \quad a_j = r2^{-j} \quad (j \geq 1), \quad \sum_{j \geq 1} a_j = r, \quad \sum_{j \geq 1} a_j^2 = \frac{r^2}{3}. \quad (\text{UP1})$$

Let $u_j(v) = (2a_j)^{-1}\mathbf{1}_{[-a_j, a_j]}(v)$, and let $b_{r,N} = u_1 * \cdots * u_N$ be its finite convolution density. This is a nonnegative probability density, is real and even, and is supported in $[-r_N, r_N]$, where $r_N = r(1 - 2^{-N})$. These assertions follow successively from the integral formula for convolution: its integral is the product of the integrals, convolution adds support intervals, and reflection preserves the convolution of even functions.

Use the Fourier convention $\widehat{b}(y) = \int_{\mathbb{R}} b(v)e^{-iyv} dv$. Direct integration on each interval gives

$$\widehat{b}_{r,N}(y) = \prod_{j=1}^N \frac{\sin(a_j y)}{a_j y}, \quad G_{r,N}(z) = \int_{\mathbb{R}} b_{r,N}(v)e^{-zv} dv = \prod_{j=1}^N \frac{\sinh(a_j z)}{a_j z}, \quad (\text{UP2})$$

with each quotient assigned its removable value 1 at zero. In particular $\widehat{b}_{r,N}(y) = G_{r,N}(iy)$.

The entire functions in (UP2) converge uniformly on compact subsets of $\mathbb{C}$ to an entire function

$$G_r(z) = \prod_{j=1}^{\infty} \frac{\sinh(a_j z)}{a_j z}, \quad G_r(0) = 1. \quad (\text{UP3})$$

Here is the required product estimate. The power series for $\sinh w/w$ gives

$$\left| \frac{\sinh w}{w} - 1 \right| \leq \frac{|w|^2 e^{|w|}}{6}. \quad (\text{UP4})$$

For the coefficient comparison, $(2k+1)! \geq 6(2k-2)!$ for $k \geq 1$, so the tail is bounded by $|w|^2/6$ times a subseries of $e^{|w|}$. On $|z| \leq R$, (UP4) and (UP1) bound the sum of the deviations of the factors from 1 by $R^2 e^{rR/2} r^2/18$. The tail of this convergent sum is uniform on that disc. The estimate $\prod(1+|d_j|) \leq \exp(\sum |d_j|)$ then shows that the product partial sums are uniformly Cauchy. Their locally uniform limit is entire, proving (UP3).

For real y , each factor in $\widehat{b}_{r,N}$ has modulus at most 1. For every fixed integer $M \geq 1$, every $N \geq M$, and $|y| \geq 1$, keeping its first M factors gives the explicit bound

$$|\widehat{b}_{r,N}(y)| \leq r^{-M} 2^{M(M+1)/2} |y|^{-M}. \quad (\text{UP5})$$

For $|y| \leq 1$ the bound 1 holds. The same two bounds hold for $G_r(iy)$ by taking the limit. Thus the limiting Fourier function is rapidly decreasing, and every polynomial multiple of it is integrable.

Define

$$b_r(v) = \frac{1}{2\pi} \int_{\mathbb{R}} G_r(iy) e^{iyv} dy. \quad (\text{UP6})$$

For every integer $k \geq 0$, (UP5), with $M > k + 1$, permits differentiating (UP6) under the integral k times and gives

$$b_r^{(k)}(v) = \frac{1}{2\pi} \int_{\mathbb{R}} (iy)^k G_r(iy) e^{iyv} dy. \quad (\text{UP7})$$

These derivatives are continuous by dominated convergence. Moreover, for N sufficiently large, Fourier inversion applies to the integrable Fourier transform of the actual convolution density $b_{r,N}$. It gives its continuous representative and the corresponding derivative formula through any fixed order k . The uniform domination (UP5) and pointwise convergence imply

$$\sup_{v \in \mathbb{R}} |b_{r,N}^{(k)}(v) - b_r^{(k)}(v)| \leq \frac{1}{2\pi} \int_{\mathbb{R}} |y|^k |G_{r,N}(iy) - G_r(iy)| dy \rightarrow 0. \quad (\text{UP8})$$

For $k = 0$, every approximating representative is nonnegative and zero outside $[-r, r]$; its continuous representative has these properties everywhere since it has them almost everywhere. The uniform limit therefore has the same properties. Uniform convergence on $[-r, r]$ gives $\int b_r = \lim_N \int b_{r,N} = 1$. Reflection and real-valuedness also pass to the limit. We have proved, with its density and limiting sense specified,

$$b_r \in C_c^\infty(\mathbb{R}, \mathbb{R}), \quad b_r \geq 0, \quad b_r(-v) = b_r(v), \quad \int b_r(v) dv = 1, \quad \text{supp } b_r \subset [-r, r]. \quad (\text{UP9})$$

This is the fixed infinite convolution meant in (UP1). No random choice of a bump or dependence on a zero is involved.

Uniform convergence of the densities on their common compact support permits taking the limit in the integral in (UP2), uniformly for z in compact subsets. Consequently

$$G_r(z) = \int_{\mathbb{R}} b_r(v) e^{-zv} dv, \quad |G_r(z)| \leq e^{r|\Re z|}. \quad (\text{UP10})$$

The function is real on the real axis, has real Taylor coefficients, and satisfies $G_r(-z) = G_r(z)$.

2. The exact zero set of the product

The product zero multiplicities in this section also occur in Arias de Reyna, cited above, equation (9). The coordinate $\zeta = rz/(2\pi i)$ carries his nonzero integer zeros to the exact lattice in (UP11), without changing their orders. The proof below verifies the assertion directly for the present product.

There are no zeros of (UP3) apart from the zeros of its individual factors. For proof, at a point where none of the factors vanishes, choose a small disc on which a finite initial product has no zeros and the tail deviations from 1 have modulus at most $1/2$. Their sum converges uniformly by (UP4). The analytic logarithms of the tail factors, given by the power series for $\log(1+w)$, then have a uniformly absolutely convergent sum on a smaller disc, since $|\log(1+w)| \leq 2|w|$. The tail product equals the exponential of that sum, and is nonzero. At a point where factors vanish, the same reasoning applies after removing the finitely many vanishing factors. Only finitely many can vanish there because $a_j z \rightarrow 0$.

The zeros of a single factor are the simple zeros $z = i\pi n/a_j$, with $n \in \mathbb{Z} \setminus \{0\}$. Their union and the order at each point are therefore

$$\begin{aligned} \{z : G_r(z) = 0\} &= \left\{ \frac{2\pi i k}{r} : k \in \mathbb{Z} \setminus \{0\} \right\} = \{128\pi i k : k \in \mathbb{Z} \setminus \{0\}\}, \\ \text{ord}_{2\pi i k/r} G_r &= 1 + \nu_2(|k|). \end{aligned} \quad (\text{UP11})$$

Indeed the j -th factor vanishes at $2\pi i k/r$ precisely when 2^{j-1} divides k , giving the displayed number of factors, each with order one. In particular, every zero of G_r is on the imaginary axis. The construction does not claim that a critical-line zeta zero cannot lie on that explicitly stated lattice.

3. The original primitive boundary filter

Retain the BPZ operators and coordinate conventions:

$$T = \frac{d^2}{dv^2} - \frac{1}{4}, \quad f_r = T b_r, \quad M_f(s) = \int_{\mathbb{R}} f(v) e^{-(s-1/2)v} dv, \quad f^\#(v) = \overline{f(-v)}. \quad (\text{UP12})$$

The test f_r is real and even and belongs to C_c^∞ , with support contained in $[-r, r]$. Integration by parts twice has no boundary terms and proves

$$M_{f_r}(s) = \left((s - \frac{1}{2})^2 - \frac{1}{4}\right) G_r(s - \frac{1}{2}) = s(s-1) G_r(s - \frac{1}{2}). \quad (\text{UP13})$$

Thus its two endpoint moments vanish, and it is nonzero:

$$M_{f_r}(0) = M_{f_r}(1) = 0, \quad M_{f_r}(\frac{1}{2}) = \int f_r = -\frac{1}{4}. \quad (\text{UP14})$$

The factor T and the original coordinate $s - \frac{1}{2}$ have not been rescaled.

Let ρ be any actual nontrivial zero of ξ_R . BPZ4–BPZ5 gives $0 \leq \Re \rho \leq 1$ and $\rho \notin \{0, 1\}$; its argument does not need a zero-free assertion on the two vertical boundary lines. Equations (UP11)–(UP13) give the exact detecting statement

$$\Re \rho \neq \frac{1}{2} \implies M_{f_r}(\rho) \neq 0. \quad (\text{UP15})$$

The implication follows because then $\rho - 1/2$ is not purely imaginary and the factor $\rho(\rho - 1)$ is nonzero. Formula (UP15) applies simultaneously to all actual off-critical zeros, without selecting one during the construction of b_r .

4. The actual infinite translation pairing

Let ρ range over the distinct zeros of ξ_R , with their original multiplicities m_ρ . For compact smooth tests use exactly the pairing

$$B(f, g) = \sum_{\rho} m_{\rho} \overline{M_f(1 - \bar{\rho})} M_g(\rho), \quad Q(f) = B(f, f), \quad f_a(v) = f(v - a). \quad (\text{UP16})$$

The polynomial zero count and compact-test decay proved in BPZ5–BPZ6 ensure absolute convergence. Here is the estimate specialized to the present test. For every integer $M \geq 0$, integration by parts in the Fourier factor of the Mellin integral gives

$$\sup_{0 \leq \sigma \leq 1} |M_{f_r}(\sigma + i\gamma)| \leq C_M (1 + |\gamma|)^{-M}. \quad (\text{UP17})$$

All derivatives of $f_r(v) e^{-(\sigma-1/2)v}$ have uniformly bounded integrals for $0 \leq \sigma \leq 1$, because of (UP9) and its fixed compact support. These bounds prove (UP17) for $|\gamma| \geq 1$; the original integral covers the bounded interval. Combining this with BPZ5, by dyadic height intervals, proves for every integer $k \geq 0$

$$\sum_{\rho} m_{\rho} |M_{f_r}(\rho)|^2 (1 + |\Im \rho|)^k < \infty. \quad (\text{UP18})$$

Since f_r is real and even, its transform has the exact identities

$$M_{f_r}(1 - s) = M_{f_r}(s), \quad M_{f_r}(\bar{s}) = \overline{M_{f_r}(s)}, \quad \overline{M_{f_r}(1 - \bar{\rho})} = M_{f_r}(\rho). \quad (\text{UP19})$$

Translation in the original variable gives $M_{(f_r)_a}(s) = e^{-(s-1/2)a} M_{f_r}(s)$ for every real a . Therefore, with $c_\rho = \rho - 1/2$,

$$K_r(a) := B((f_r)_a, f_r) = \sum_{\rho} m_{\rho} M_{f_r}(\rho)^2 e^{c_\rho a}, \quad q_r = K_r(0) = Q(f_r). \quad (\text{UP20})$$

The weight off the critical line is the displayed complex square, not an absolute square. Every off-critical weight is nonzero by (UP15). The multiplicity is retained in the coefficient at each distinct exponent.

Since $|\Re c_\rho| \leq 1/2$, equations (UP18) and (UP20) justify all real derivatives locally uniformly:

$$K_r^{(k)}(a) = \sum_{\rho} m_{\rho} M_{f_r}(\rho)^2 c_\rho^k e^{c_\rho a}, \quad K_r \in C^\infty(\mathbb{R}), \quad |K_r(a)| \leq C e^{|a|/2}. \quad (\text{UP21})$$

Changing ρ to $1 - \rho$, respectively $\bar{\rho}$, is legitimate in these absolutely convergent sums and proves

$$K_r(-a) = K_r(a), \quad K_r(a) \in \mathbb{R} \quad (a \in \mathbb{R}). \quad (\text{UP22})$$

No holomorphic extension of this real-time series in the variable a is asserted. Its proof of smoothness and the growth bound are the real-variable statements in (UP21).

5. The global Laplace transform and every possible pole

For $\Re w > 1/2$, (UP18), the strip bound, and absolute integrability permit interchanging the integral and full zero sum. They give

$$\mathcal{K}_r(w) := \int_0^\infty e^{-wa} K_r(a) da = \sum_\rho \frac{m_\rho M_{f_r}(\rho)^2}{w - c_\rho}. \quad (\text{UP23})$$

For example, the integral of the absolute values of all summands is bounded by $(\Re w - 1/2)^{-1} \sum_\rho m_\rho |M_{f_r}(\rho)|^2$, which is finite. The series on the right is a meromorphic function on the entire w -plane. On a compact set $|w| \leq R$ avoiding its listed exponent points, only finitely many zeros have bounded height; their nonzero denominators have a positive minimum. For all sufficiently large $|\Im \rho|$, the denominator has modulus at least $|\Im \rho|/2$, so (UP18) gives uniform absolute convergence of the remaining tail. The same argument applies on a disc after removing any one listed term.

It follows that its residue at every possible exponent point is exactly

$$\text{Res}_{w=c_\rho} \mathcal{K}_r(w) = m_\rho M_{f_r}(\rho)^2. \quad (\text{UP24})$$

The right side is nonzero at every off-critical zero. Distinct actual zeros have distinct c_ρ , so no sum of other zeros can cancel that residue. A point with zero weight is a removable singularity, as specified by (UP24).

Suppose K_r is bounded on $[0, \infty)$. Its Laplace integral in (UP23) then defines a holomorphic function on $\Re w > 0$, because on every compact subset of that half-plane each derivative in w is bounded by an integrable multiple of $a^k e^{-\delta a}$, for some $\delta > 0$. It equals the meromorphic series for $\Re w > 1/2$. The half-plane with the discrete exponent points removed is connected: a segment joining any two points has a compact neighborhood meeting only finitely many such points, and small arcs detour around them. The identity theorem consequently extends their equality throughout that punctured half-plane.

If an actual zero lies off the critical line, reflection provides one with $\Re \rho > 1/2$. Its exponent c_ρ lies in that half-plane and has the nonzero residue (UP24). A holomorphic Laplace integral there cannot agree on a punctured disc with a meromorphic function having that residue. This contradiction proves

$$K_r \text{ bounded on } [0, \infty) \implies \text{RH}. \quad (\text{UP25})$$

This proof uses the whole infinite zero sum and its normally convergent meromorphic continuation. It does not truncate the actual spectrum.

Conversely, under RH each $c_\rho = i\gamma_\rho$ is purely imaginary, and (UP19) gives $M_{f_r}(\rho) \in \mathbb{R}$. Equations (UP18) and (UP20) then give

$$q_r = \sum_\rho m_\rho M_{f_r}(\rho)^2 \geq 0, \quad |K_r(a)| \leq q_r \quad (a \in \mathbb{R}). \quad (\text{UP26})$$

This is the exact triangle inequality for that absolutely convergent sum, with its actual weights. In particular, no independent sign assumption on q_r was used to prove (UP25).

6. A single fixed test gives the complete criterion

Combining (UP25) and (UP26) proves

$$\begin{aligned} \text{RH} &\iff K_r \text{ is bounded on } [0, \infty) \\ &\iff |K_r(a)| \leq q_r \text{ for every } a \geq 0, \quad r = \frac{1}{64}. \end{aligned}$$

(UP27)

All quantities here arise from the one test fixed in (UP1)–(UP14). The last displayed inequality implies boundedness directly: at $a = 0$ it also forces $q_r \geq 0$, and thereafter supplies the finite constant q_r .

The criterion has an exact two-translate formulation. BPZ13–BPZ14, or direct cancellation of the exponentials in (UP16), gives

$$Q((f_r)_a) = q_r, \quad Q((f_r)_a + f_r) = 2q_r + 2K_r(a), \quad Q((f_r)_a - f_r) = 2q_r - 2K_r(a). \quad (\text{UP28})$$

Here Hermitian symmetry of B , the reality (UP22), and its conjugate linearity in the first slot give the cross terms exactly. All three tests retain both zero endpoint moments because translation multiplies those moments by exponentials. Consequently (UP27) is also exactly

$$\text{RH} \iff Q((f_r)_a + f_r) \geq 0 \text{ and } Q((f_r)_a - f_r) \geq 0 \text{ for every } a \geq 0. \quad (\text{UP29})$$

If an off-critical zero exists, the proof of (UP25) makes K_r unbounded. One can then choose $a \geq 0$ with $|K_r(a)| > |q_r|$. The minus sign when $K_r(a) > 0$, and the plus sign when $K_r(a) < 0$, gives the actual compact primitive test with value $2q_r - 2|K_r(a)| < 0$. This witness requires only a translate and a sign of the same fixed test; it does not alter the bump in response to the zero.

Continuity also proves an exact countable version: the inequality in (UP27), or both inequalities in (UP29), can be required for every rational $a \geq 0$, because every real $a \geq 0$ is a limit of such rationals. No corresponding criterion on one fixed, equally spaced translation lattice is asserted here. Sampling on a fixed lattice identifies distinct imaginary frequencies modulo its period; (UP24), which separates their continuous Laplace poles, does not justify discarding that distinction.

7. The convolution test and the full supported receiver

Define the actual convolution

$$h_r = f_r^\# * f_r = f_r * f_r, \quad h_r \in C_c^\infty(\mathbb{R}, \mathbb{R}), \quad h_r(-v) = h_r(v), \quad \text{supp } h_r \subset [-2r, 2r]. \quad (\text{UP30})$$

Fubini's theorem on compact supports gives $M_{h_r} = M_{f_r}^2$. The convolution in the cross pairing of (UP20) is exactly

$$((f_r)_a)^\# * f_r(v) = h_r(v + a), \quad M_{h_r(\cdot + a)}(s) = e^{(s-1/2)a} M_{f_r}(s)^2. \quad (\text{UP31})$$

For proof, $((f_r)_a)^\#(v) = f_r(v + a)$ because f_r is real and even, and a change of variable in the convolution yields the first identity. Both endpoint values of the transform in (UP31) are zero by (UP14). Thus the original full explicit formula on this test reads

$$K_r(a) = A_\infty(h_r(\cdot + a)) - P_{\text{fin}}(h_r(\cdot + a)), \quad (\text{UP32})$$

with the original archimedean and finite-prime terms of BPZ19 and SZW; no endpoint contribution is omitted. The same statement applies to each of the two primitive tests in (UP28), using its own full convolution.

For the finite support semilattice $\mathcal{L}$, write $\mathbf{e}_\lambda$ for the retained coordinate basis of the supported explicit formula. For the test (UP31), its exact vector values are

$$\begin{aligned} \mathbf{B}_{\mathcal{L}} &= 0, \\ \mathbf{Z}_{\mathcal{L}} &= K_r(a) \mathbf{e}_{1_{\mathcal{L}}}, \\ \mathbf{D}_{\mathcal{L}} &= (P_{\text{fin}} - A_\infty)(h_r(\cdot + a)) \mathbf{e}_{1_{\mathcal{L}}}, \\ \mathbf{B}_{\mathcal{L}} - \mathbf{Z}_{\mathcal{L}} &= \mathbf{D}_{\mathcal{L}}. \end{aligned} \quad (\text{UP33})$$

Indeed, both top endpoint coefficients and every lower endpoint coefficient are zero by the evaluated moments in (UP14) and (UP31). These zeros are values of the existing coordinate functionals. They do not identify the supported zero element with external absence.

That distinction is retained directly by the synchronized carriers from BPZ20–BPZ22. For this carrier construction retain the original nontrivial bounded distributive support lattice $\mathcal{L}$, with bottom $0_{\mathcal{L}}$, top $1_{\mathcal{L}}$, and $0_{\mathcal{L}} \neq 1_{\mathcal{L}}$. For a complex vector space V , set

$$\mathcal{G}_{\mathcal{L}}(V) = \{(0, \lambda) : \lambda \in \mathcal{L}\} \cup (V \times \{1_{\mathcal{L}}\}), \quad A^\uparrow(v, \lambda) = (Av, \lambda) \quad (\text{UP34})$$

for each complex linear map A . With the original join addition and meet scalar action, this lift is additive and scalar compatible by direct substitution. Translation, T , and the endpoint map $E(f) = (M_f(0), M_f(1))$ are all such linear maps. Every nonzero primitive test with label $1_{\mathcal{L}}$ maps under $E^\uparrow$ to $(0, 1_{\mathcal{L}})$, the supported zero in the endpoint carrier. The unsupported input $(0, 0_{\mathcal{L}})$ retains its distinct label. Their entire lower-label families are preserved by (UP34).

Likewise the lifted pairing is exactly

$$B^{\mathcal{L}}((u, \lambda), (v, \mu)) = (B(u, v), \lambda \wedge \mu). \quad (\text{UP35})$$

It lies in the synchronized carrier: a nonzero amplitude requires both input amplitudes nonzero, hence both input labels top. Its additivity and conjugate scalar compatibility follow from those of B and distributivity of meet over join. Thus (UP20) and (UP28) retain top support even when their amplitudes vanish. The negative amplitude obtained after (UP29) has this same support label; an endpoint value equal to supported zero does not change its sign. These formulas keep all the original label maps while applying the single fixed primitive test.

Reading and scope

The complete programme source read was `BOUNDARY_PRIMITIVE_ALL_ZERO_RECEIVER.tex`, equations BPZ1–BPZ23 and all intervening proofs, at the path linked above. Its original human input is Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, arXiv:1801.05914v5, with source equations `phidef`, `htdef`, `hoz`, and `sas`, and the original Weil explicit-formula convention is retained through the cited programme sources. This note does not represent a fresh read of those author papers. For the citation revision, Arias de Reyna’s original author TeX `09-Function.tex`, arXiv:1702.05442v1, was read at lines 1–720, including Theorem 1 and equations (1)–(9), their proof, Theorem 4’s derivative formula and the bibliography. The bump and its product zero set are classical results reproduced here. The programme application is their fixed-test nonvanishing at every possible off-critical zero and the full BPZ Laplace argument in the stated coordinates. No arithmetic bound on $K_r(a)$ for all translations has been supplied by this proof.

A fixed primitive test and the complete translated prime window

The boundary filter reaches the whole zero set through a single explicitly constructed test. This calculation gives its entire arithmetic correlation, including the exact archimedean term, the prime contribution and the original support labels. It does not assume the inequality equivalent to RH. Every parameter below is fixed before any zero is selected.

1. Original test and full pairing

Use the test space, involution and Mellin convention

$$\mathcal{T} = C_c^\infty(\mathbb{R}; \mathbb{C}), \quad f^\#(v) = \overline{f(-v)}, \quad M_f(s) = \int_{\mathbb{R}} f(v) e^{-(s-1/2)v} dv, \quad T = \partial_v^2 - \frac{1}{4}. \quad (\text{PW1})$$

The full pairing is conjugate linear in its first slot:

$$B(f, g) = \sum_{\rho} m_{\rho} \overline{M_f(1 - \bar{\rho})} M_g(\rho) = Z(f^\# * g). \quad (\text{PW2})$$

The sum is over the distinct actual nontrivial zeta zeros with their actual multiplicities. Smooth compact tests have Mellin transforms decreasing faster than every inverse power of height, uniformly in the closed critical strip, by repeated integration by parts. The unconditional count of zeros therefore makes (PW2) absolutely convergent. The complete supported explicit formula, with its original coordinate maps, is SZW24–26 and SZW33–38. Its original Gamma integral is HA12–14. Both complete dependencies accompany this derivation.

Fix

$$r = \frac{1}{64}, \quad a_j = r 2^{-j} \ (j \geq 1), \quad b_r = \underset{j \geq 1}{*} \frac{\mathbf{1}_{[-a_j, a_j]}}{2a_j}, \quad G_r(z) = \prod_{j \geq 1} \frac{\sinh(a_j z)}{a_j z}, \quad (\text{PW3})$$

with each quotient interpreted as one at zero. The infinite convolution is the probability law of the absolutely convergent sum of independent uniform variables on the displayed intervals. Its support lies in $[-r, r]$. The complete construction, smoothness and nonvanishing proof is given in the universal primitive translation derivation. Here are the analytic details needed by the present calculation. For each integer N , its Fourier transform is bounded by $\prod_{j=1}^N \min(1, (a_j |y|)^{-1})$, since every remaining factor has modulus at most one. This decreases as $O_N(|y|^{-N})$. Fourier inversion and differentiation therefore give a smooth density b_r ; its being a probability density, even and nonnegative, follows from the original law. Its Laplace transform is the locally uniformly convergent product G_r , since on compact sets each factor differs from one by $O(a_j^2)$. Away from the individual factor zeros the infinite product is nonzero: its tail has a convergent logarithm by the same estimate. All factor zeros have real part zero. Thus $G_r(z) \neq 0$ whenever $\Re z \neq 0$.

Set

$$f_r = T b_r, \quad k_r = b_r^\# * b_r, \quad h_r = f_r^\# * f_r = T^2 k_r, \quad f_{r,a}(v) = f_r(v - a). \quad (\text{PW4})$$

These are real even smooth functions except for the translated test; $k_r \geq 0$, $\int k_r = 1$, and k_r, h_r are supported in $[-2r, 2r]$. The assertions about convolution follow by differentiating under its compact integral. Twice integrating by parts gives

$$M_{f_r}(s) = s(s-1)G_r(s-1/2), \quad M_{f_r}(0) = M_{f_r}(1) = 0. \quad (\text{PW5})$$

The transform is nonzero at every off-critical nontrivial zero: neither $s(s-1)$ nor $G_r(s-1/2)$ vanishes there. The test f_r is nonzero, since a compactly supported solution of $b'' = b/4$ is zero, whereas $\int b_r = 1$.

2. Exact correlation, with every zero retained

Define

$$K_r(a) = B(f_{r,a}, f_r), \quad q_r = K_r(0), \quad a \in \mathbb{R}. \quad (\text{PW6})$$

Translation in the first slot gives $f_{r,a}^\# * f_r(v) = h_r(v + a)$. Reflection and real evenness of f_r give the absolutely convergent expression

$$K_r(a) = \sum_{\rho} m_{\rho} M_{f_r}(\rho)^2 e^{(\rho-1/2)a}. \quad (\text{PW7})$$

For a in a fixed compact real interval the extra exponential is bounded uniformly in the zero set, so the sum is locally uniformly convergent. Complex conjugation and the symmetries of the zeros show that K_r is real

and even; the arithmetic formula below also proves this directly. Translation of both inputs cancels the Mellin exponentials, so

$$B(f_{r,a}, f_{r,a}) = q_r, \quad B(f_r \pm f_{r,a}, f_r \pm f_{r,a}) = 2q_r \pm 2K_r(a). \quad (\text{PW8})$$

The actual short-support estimate proved in the short-support derivation applies because $2r = 1/32 < \log 2$. It gives

$$q_r \geq \frac{55}{192} \|f_r\|_2^2 = \frac{55}{192} (\|b_r''\|_2^2 + \frac{1}{2} \|b_r'\|_2^2 + \frac{1}{16} \|b_r\|_2^2) > 0. \quad (\text{PW9})$$

That proof derives the bound from the full Gamma integral, including its positive translation-difference remainder. The finite-prime terms vanish on this single bump because their first nonzero argument is $\log 2$, outside $[-2r, 2r]$; both endpoint terms vanish by (PW5). Thus (PW9) is a bound for the original full form, not a discarded-term estimate.

For completeness the single-test all-zero criterion has a short full proof. If RH holds, each term in (PW7) is a nonnegative weight times a phase, since $M_{f_r}(1/2 + i\gamma)$ is real. Absolute summability implies $|K_r(a)| \leq q_r$. Conversely suppose K_r is bounded on $[0, \infty)$. For $\Re w > 1/2$, dominated termwise integration gives

$$\int_0^\infty e^{-wa} K_r(a) da = \sum_\rho \frac{m_\rho M_{f_r}(\rho)^2}{w - (\rho - 1/2)}. \quad (\text{PW10})$$

The right side converges normally off its discrete poles: at large height the denominators are bounded below by half the height, and the Mellin values already decrease rapidly. It is meromorphic on the whole complex plane. Every off-critical zero in $\Re \rho > 1/2$ contributes a nonzero residue by (PW5). The assumed boundedness makes the left side holomorphic on $\Re w > 0$. The punctured half-plane is connected, since compact paths can be detoured around its locally finite poles. The identity theorem extends their equality from $\Re w > 1/2$, contradicting any such nonzero residue. Reflection then excludes off-critical zeros on the left too. This proves

$$\boxed{\text{RH} \iff K_r \text{ bounded on } [0, \infty) \iff |K_r(a)| \leq q_r \text{ for every } a \geq 0.} \quad (\text{PW11})$$

Multiplicity is included in the nonzero residues, and no finite zero list is substituted. An eventual bound on $(2r, \infty)$ is equivalent, since K_r is continuous on the remaining compact interval. This argument uses the incoming programme BPZ13–17 Laplace mechanism, whose complete original source is retained as BPZ; the fixed product (PW3) makes its test independent of a selected hypothetical zero. No claim that translation criteria are historically new is made.

3. The entire arithmetic expression, including the origin

For every real a , HA13 applied to $h_r(v + a)$ gives

$$\begin{aligned} K_r(a) = & -(\gamma + \log \pi) h_r(a) \\ & + \int_0^\infty \frac{e^{-x} h_r(a) - \frac{1}{2} e^{-x/4} \{h_r(a + x/2) + h_r(a - x/2)\}}{1 - e^{-x}} dx \\ & - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \{h_r(a + \log n) + h_r(a - \log n)\}. \end{aligned} \quad (\text{PW12})$$

The endpoint values of the translated convolution are zero by (PW5); multiplication by a translation exponential does not change this. Near $x = 0$, the numerator in (PW12) is $O(x)$, by Taylor expansion of the smooth function at a ; the denominator is comparable to x . At infinity the numerator decreases exponentially. Every prime sum here is finite, by compact support. These facts justify the displayed integral without a regularization convention. The factor $1/2$, the shifts $x/2$, and the coefficient $-\log \pi$ are unchanged.

For $a > 2r$, one has $h_r(a) = 0$, $h_r(a + x/2) = 0$ for $x \geq 0$, and $h_r(a + \log n) = 0$. In the remaining integral use $x = 2v$. Its Jacobian cancels the factor $1/2$ exactly. Consequently

$$\boxed{K_r(a) = - \int_{a-2r}^{a+2r} \frac{e^{-v/2}}{1 - e^{-2v}} h_r(a - v) dv - \sum_{\substack{n \geq 2 \\ |\log n - a| \leq 2r}} \frac{\Lambda(n)}{\sqrt{n}} h_r(a - \log n).} \quad (\text{PW13})$$

Thus only the fixed multiplicative interval

$$e^{a-1/32} \leq n \leq e^{a+1/32} \quad (\text{PW14})$$

enters the finite-prime part. This is the original logarithmic coordinate; the width has not been scaled away.

4. Positive archimedean remainder and exact derivative of the prime window

Put $w(v) = e^{-v/2}/(1 - e^{-2v})$ for $v > 0$. Its locally uniformly differentiable expansion is $w(v) = \sum_{j \geq 0} e^{-(2j+1/2)v}$. On the compact integration interval in (PW13), its derivatives of every fixed order converge uniformly. Since k_r and all its derivatives vanish at its support endpoints, four integrations by parts give

$$R_r(a) := \int_{a-2r}^{a+2r} w(v) h_r(a-v) dv = \int_{a-2r}^{a+2r} (T_v^2 w)(v) k_r(a-v) dv, \quad a > 2r. \quad (\text{PW15})$$

For $\lambda_j = 2j + 1/2$, the exact eigenvalue is $(\lambda_j^2 - 1/4)^2 = 4j^2(2j+1)^2$. The $j = 0$ term is annihilated. The rest is strictly positive, giving

$$R_r(a) = \sum_{j \geq 1} 4j^2(2j+1)^2 G_r(2j+1/2)^2 e^{-(2j+1/2)a} > 0, \quad (\text{PW16})$$

$$W(v) := T_v^2 w(v) = \frac{4e^{-5v/2} \{9 + 55e^{-2v} + 31e^{-4v} + e^{-6v}\}}{(1 - e^{-2v})^5}.$$

To verify the rational expression, apply $(q\partial_q)^k$ to $(1-q)^{-1}$ for $k = 2, 3, 4$, and combine $16 \sum j^4 q^j + 16 \sum j^3 q^j + 4 \sum j^2 q^j$, with $q = e^{-2v}$. The numerator is $4q(9 + 55q + 31q^2 + q^3)$, multiplied by $e^{-v/2}$. For the first identity integrate each exponential against $k_r(a-v)$: its value is $e^{-\lambda_j a} G_r(\lambda_j)^2$. Termwise integration follows either from positive monotone convergence or the bound $G_r(\lambda_j) \leq e^{r\lambda_j}$, which leaves a convergent polynomial times $e^{-\lambda_j(a-2r)}$.

The function W is strictly decreasing, since every term in its positive exponential series is. Hence the probability density k_r gives the explicit bounds

$$0 < W(a+2r) \leq R_r(a) \leq W(a-2r), \quad a > 2r. \quad (\text{PW17})$$

In particular $R_r(a) = O_r(e^{-5a/2})$ as $a \rightarrow \infty$, with the coefficient and convergent remainder explicitly specified in (PW16). The archimedean cross term itself is $-R_r(a)$, with a strict negative sign. This sign results from the actual formula, although the diagonal (PW9) is positive.

Define the smooth nonnegative prime-window function

$$Y_r(a) = \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} k_r(a - \log n), \quad a \in \mathbb{R}. \quad (\text{PW18})$$

On each compact interval of a , only finitely many indices occur, so all derivatives can be taken termwise. Because $h_r = T^2 k_r$, (PW13) becomes

$$K_r(a) = - \left(Y_r^{(4)}(a) - \frac{1}{2} Y_r''(a) + \frac{1}{16} Y_r(a) \right) - R_r(a), \quad a > 1/32. \quad (\text{PW19})$$

The operator acting on Y_r has the same coefficients as the endpoint filter squared; no new normalization has been introduced. Nonnegativity of Y_r does not assign a sign to this fourth-order expression. The exact map from positive diagonal prime coefficients to this signed translation form is proved in PM25–41, including endpoint-zero tests of both signs.

Combining the proved implications (PW11) with the exact arithmetic equality (PW19) yields

$$\text{RH} \iff \left| Y_r^{(4)}(a) - \frac{1}{2} Y_r''(a) + \frac{1}{16} Y_r(a) + R_r(a) \right| \leq q_r \quad \text{for every } a > 1/32. \quad (\text{PW20})$$

This is a proved equivalence, not a proof of the inequality on its right. The known positive constant q_r , the strictly positive explicit term R_r , and the complete remaining prime expression have each been computed. The prime inequality in (PW20) remains unresolved.

5. Full supported-zero receiver

Let L be the same finite bounded distributive support lattice as SZW, and let $\mathbf{e}_\lambda$ be its unchanged coordinate basis. Applying the complete formula to $h_r(\cdot + a)$, before any coordinate projection, gives

$$\begin{aligned} B_L(a) &= 0, \\ Z_L(a) &= K_r(a) \mathbf{e}_{1_L}, \\ D_L(a) &= -K_r(a) \mathbf{e}_{1_L}, \quad B_L(a) - Z_L(a) = D_L(a). \end{aligned} \quad (\text{PW21})$$

Every lower coefficient is the evaluated endpoint value $M_{h_r(\cdot+a)}(0) = 0$; the coordinate spaces and their projections remain present. In the original semimodule lift $G_L(V)$, a linear map A acts by $(v, \lambda) \mapsto (Av, \lambda)$. Therefore the endpoint map sends $(f_{r,a}, 1_L)$ to $(0, 1_L) = e$, not to $\tau = (0, 0_L)$. The lifted pairing is

$$B^L((f, \lambda), (g, \mu)) = (B(f, g), \lambda \wedge \mu). \quad (\text{PW22})$$

Additivity and sesquilinearity follow directly from distributivity of meet over join and the corresponding properties of B . The complete coefficient injection $(c, \lambda) \mapsto (c, \iota_L(\lambda))$ is injective because ι_L has the original lattice retraction. It preserves all signs of nonzero amplitudes and all zero labels. Thus an endpoint-zero translated test still carries supported zero, and its value in (PW19) retains its top label even at an amplitude zero.

In the prime operator calculation, the sparse Mobius sum and the Fourier sum also retain distinct support corrections: PM20–24 proves that their least difference is supported zero only at index one, whereas the directly summed Fourier defect is the full supported-zero diagonal. Both map to the same finite prime distribution in (PW13) through the explicitly stated trace receiver; their equality after that receiver does not identify the original matrices. This supplies the exact relation between the boundary filter, every possible off-line zero, the actual prime window and the programme’s retained support data.

Exact support and arithmetic maps

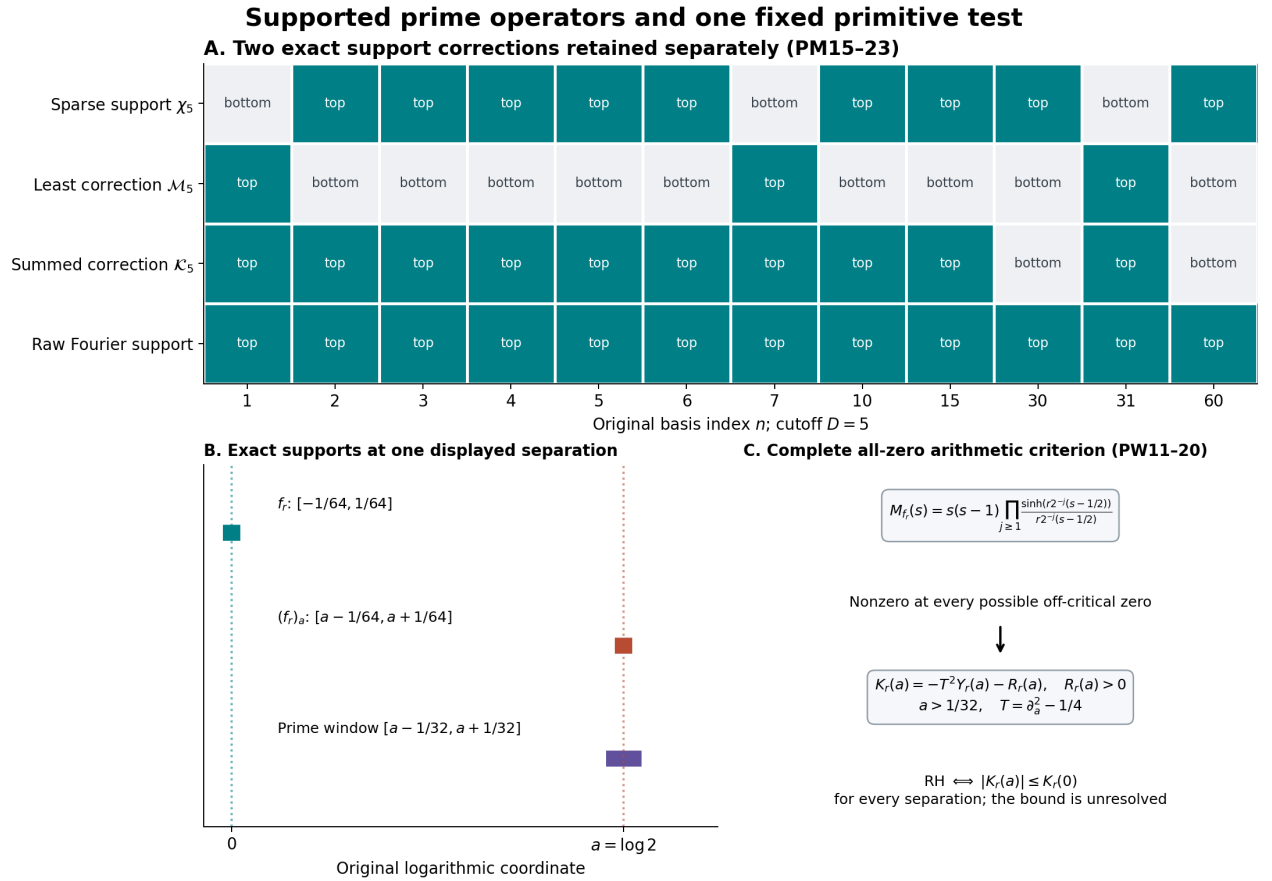


Figure 19: Exact support masks at cutoff five, the original test-support intervals and their prime window, and the complete criterion. The top panel shows all labels at the displayed indices, including the difference between the least correction and the directly summed Fourier correction. The middle support calculation uses the displayed separation $a = \log 2$; the criterion concerns every separation. Proofs: PM15–23 and PW3–20. Both zero endpoint values retain supported-zero labels by PW21–22. The reproducible source draws the proved masks and interval coordinates; it does not sample a hypothetical zero.

Proof sources and scope

The analytic explicit formula and full support vector are the complete SZW source cited above; the Gamma integral is HA13. Their original human source includes Alain Connes, *Trace formula in noncommutative*

geometry and the zeros of the Riemann zeta function, Appendix II, Theorem 6. The incoming BPZ proof supplies the translation/Laplace method with all multiplicities; its full TeX is included unchanged. The present fixed convolution, its complete analytic construction, the short-support estimate, and PM's prime-operator comparison have standalone proofs in this collection. These statements specify their exact mathematical contribution without claiming priority over the extensive classical literature on Weil criteria or smoothed prime formulas.

Sources, exact proof locations, and reading scope

This addition develops the established tau/split-zero base, the original escaping inverse fibre, and the actual quotient-size interpolation. Every new algebraic and analytic calculation appears in full in the accompanying complete proof files. The exact operator input retains its earlier public definitions and proofs; those are linked below. This is not a historical novelty claim or a proof of the Riemann hypothesis.

Earlier programme proofs

- The supported-zero prime and chain are prior programme results. Paper 1 is *An Algebraic Structure Incorporating a $\mathbb{Z}/1\mathbb{Z}$ -Symmetric Element Adjoined to the Integers: Construction, Analysis, and Generalizations*, original `11.tex`, complete public source, labels `thm:prime_ideals_S`, `thm:krull_dim_S`, `eq:maximal_chain`. The source was read completely, 1–1833. The user attributes this result to Gemini Pro 2.5; this is received attribution, not an inferred author field. The present proof of the full-support statement is FSR1–FSR7.
- *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, and Frobenius-Perfect Quantization*, The Clankers, June 2026, version `split_support_geometry_arithmetic_curve_v11910.tex`, SHA256 8443cc0401b18d373939d992f0a2f5384ee2fa1130f059753d5f3972eec5052d. Root reading covers 1–1910; used definitions and proofs are 1180–1871. This is the received full-lattice foundation. The complete receiving proof, including all needed definitions, is FSR1–FSR36, not an assertion that the entire source was read.
- The actual escaping map and heat constants are in Source mechanism transfer, labels `eq:retained-node-map`, `eq:retained-arithmetic-extension`, and Incompressible fibre heat, labels `eq:ifh-original`, `eq:ifh-chart`, `eq:ifh-target-flow`, `eq:ifh-triple`. Both local source versions were read completely; EFI records their exact hashes. The new derivation EFI1–EFI68 proves every received polynomial identity it uses.
- The original metric and observation are in Observed current plane, OCP1 and OCP5–OCP7. Read coverage is 35–169. The published exact incoming family and support maps are Observed deformation and current, DF1–DF16, and Support spectrum and terminal mass, ST1–ST20, both read completely and byte-matched to their public sources.
- The subsequent complete incoming proofs are Observed collision, OC1–OC17; Observed support propagation, OSP1–OSP51; and Observed cotangent and Frobenius, OCF1–OCF23 and OIF1–OIF36. ISP and EFI give the new exact trace/current and escaping-family comparisons.

Supplied original LaTeX on separated zeta interpolation

The source-reading ledger is keyed to canonical IDs from the existing disk index. Its search results were used for routing; they were not counted as reading.

Source	Exact version and root reading	Result used here
<i>Secondary Note on Split-Zero Globalization</i> , The Clankers; PUBUNIT-B1A260A5E731EA568C353AA4	<code>split_zero_secondary_note.tex</code> , SHA256 8e1aa2f934c49fc21e7a3d946ac6a465bc1a68896ec184afcb3d989cc9f8b364; complete 1–1997	<code>def:hurwitz-specialization</code> , <code>thm:universal-taylor-deformation</code> , <code>thm:arithmetic-kernels-riemann</code> , <code>cor:first-log-jet-explicit</code> ; NL1–NL18, SW1–SW42 give the complete receiving proofs.
<i>Split-Zero Support Repair</i> , source author field “The Clankers / prepared in continuation”; PUBUNIT-C4735C0E0B05FF5D811C3F89	<code>some_considerations.tex</code> , SHA256 3f98c8eab35f87996a594d0afa7d77ba561cbc28b632fa33e16d2d6228416702; 1473–1739	Conservative ideal/zeta side and shifted deformation. The multiplicative ideal-norm hypothesis is retained explicitly in NL.
Split-zero defect preprint, blank source author field; PUBUNIT-6C3FFDFC49FDF75FE8EB43E3	<code>split_zero_defect_preprint(1).tex</code> , SHA256 1d5f15f9999b402356d6ba46286f2473ee6c857f55e091c836c905ca15b4695e; 480–854	Endpoint residues and the invariant/anti-invariant decomposition. SW distinguishes function residues from logarithmic-derivative multiplicities and derives the reflection defect.

Source	Exact version and root reading	Result used here
Primitive defect preprint; PUBUNIT- 2B18D475F69385B03F1E2B30	<code>primitive_defect_preprint.tex</code> , SHA256 2f3289f6a04105d3f15fe5f7c36dde286515e61f465caf1582f9808e19f3d2f7; 1–385; the two supplied copies are identical	Retained augmentation kernel; not used as a positivity theorem.
Primitive support defects; PUBUNIT- E4607F9536B56699EB520E23	<code>primitive_support_defects_preprint.tex</code> , SHA256 f12f2c1c36effbf89740e308733d8434989e69cc088a0fc1c686281a157b42c5; 1–440	Source comparison for signed support defects.
Split-zero moonshine repair; PUBUNIT- 396B23CD772DEC7617B4A2A5	<code>split_zero_moonshine_repair.tex</code> , SHA256 7e672f7f1c21ae8dca327a65d8395d1a8b201bbddebb4d0317a96ab9e2382538; complete 1–826	Character-coefficient positivity and retained negative polar data were read; no moonshine theorem is applied to infer Weil positivity.

These received manuscripts are identified as programme sources. The independently written complete receiving proofs are included here. No private correspondence is reproduced.

Human sources

- Alain Connes and Caterina Consani, *Weil positivity and Trace formula, the archimedean place*, arXiv:2006.13771v1; original author source archive, `weil-compo.tex`, Appendix A “Fourier versus Mellin transforms” (`appenmellinapp`) and Appendix B “Explicit formula” (`appendix2`), 2011–2070. This exact source block was read. Labels `bombieriexplicit`, `bombieriexplicit1`, `burnolexplicit1` fix the signs and conventions. The original archive is retained privately unchanged. WA1–WA30 proves the extension to the actual entire test functions, with convergence.
- Terence Tao, *A digestion of the Jacobian conjecture counterexample*, 21 July 2026, displayed original polynomial map; David Speyer, *The new counterexample to the Jacobian conjecture*, 20 July 2026, marked-factor discussion, as cited by the programme’s source. The first displayed map was checked live. Reading those posts does not claim a fresh audit of every original counterexample argument. EFI1–EFI3 and its exact check script verify the polynomial map and its inverse chart directly.
- NIST DLMF, §25.11, definitions, recurrence, special values and parameter derivative, equations 25.11.1, 25.11.3, 25.11.13, 25.11.14, 25.11.17, were checked in the primary HTML. Equation-TeX downloads returned HTTP 403 and the browser extractor rejected the TeX content type; no source download is claimed, and no PDF was substituted. SW3–SW8 supplies the needed continuation and derivative proofs independently.

The historical cotangent and prism foundations remain Luc Illusie’s regular-immersion and transitivity theorems and Bhatt–Scholze’s prism definitions, at the exact original-source locators retained in the linked OCF/OIF paper. This addition proves the specified polynomial and coefficient maps; it does not assert an unproved equality between complex Weil pairings and prismatic cohomology.

ES and endpoint heat source use

The ES inputs belong to the established Erdős–Straus collection. The source read is its original TeX, including the raw quarter projectors, forced metric and tetrahedral actions, original spatial affine map, simplex linking algebra and free-shell carrier. The new derivations reproduce every finite matrix and every connecting map they use. They do not attribute these older ES constructions to this addition.

The exact 488-page reading edition is pinned at 20709bee872ad10e4cc7d7b976d975e24532a803. The original TeX/build ZIP contains `reader/erdos_straus_project_reader.tex`, SHA256 678b3f70ee7d2a6d00f6c588d7778e390118ab4743a5759612dbba6430984c78. The intact local archive has SHA256 521582fa5d25eae3ab8771fc0668ca52374f21ff0f8871a72e652aa090849dd8. The local PDF was byte-matched to the pinned public Git blob.

Exact reader-source blocks read for this addition are 4981–5121 (raw quarter spectrum and projectors), 20151–20370 (the full simplex 1–2–3 linking map and its proof), and 23426–23960 (ordered shell blades,

simplex support, directed carrier and complete control-length/determinant proof). The corresponding labels are `cor:raw-blade-cells-quarter-three-plus-one`, `eq:raw-negative-cell-projector`, `thm:simplex-123-modular-intertwiner`, `cor:general-shell-simplex-Lorentz-pullback`, and `thm:general-shell-control-determinant-volume`. Figures 10 and 12 are the supplied images' linking and shell constructions. Further source reading in the current ES manuscript covered the exact forced metric, D3 action, regular tetrahedral completion and moving spatial map at the labels recorded in ESQ. This is bounded reading, not a claim to have reread the entire collection.

The endpoint convention is the same primary-source explicit formula already read in Connes–Consani's original author TeX, at the Appendix B locators above. HEB13–HEB17 prove the exact endpoint isomorphism; HEB26–HEB30 prove its rational completion correction. The original polynomial map and heat speed retain the public source-mechanism and incompressible-fibre citations above. No new claim is made for a global arithmetic identification of the ES operator or its infinite fibre trace.

The original eight-state source is the finite signed-root completion, FS1–FS31, read in full. Its source SHA256 is `eaf980a16e7d639e21953e5779e7f76abf18e8bdb80ae923c4160b456c22da6f`. The original polynomial and escaping arcs are FC36–FC47, read at public-source lines 590–769, SHA256 `10906ba3bd21c06645571560e4c7b0c3948c0ccaf65d8b9e525abf7cd732794c`. ESH preserves these original constructions and proves the new heat receivers and whole-fibre trace comparisons.

Supported-zero trace and signed-cover holonomy

The supported-zero derivation uses the original author TeX of Alain Connes, Trace formula in noncommutative geometry and the zeros of the Riemann zeta function, arXiv:math/9811068v1, Section III (two-endpoint function space), Section VI (formal orbital discussion), Appendix I (full Poisson formula), and Appendix II Theorem6 (unconditional explicit formula). SZW derives its stated arithmetic formula by a convergent contour argument and constructs its separate divisor-jet Hilbert receiver explicitly. It does not identify that receiver with an unproved global operator trace. The complete preceding TPS, TC and FL proofs are included in `supporting_proofs`, with original-source identity and exact scope preserved. Private correspondence is not included.

The actual order-192 monodromy was already proved in SM1–SM19, source lines2097–2513. The downloaded original has SHA256 `10906ba3bd21c06645571560e4c7b0c3948c0ccaf65d8b9e525abf7cd732794c`. Root read that complete section and the full new SZW, EHM and HWA proofs. Independent mathematical review checked the analytic trace construction and the complete finite averaging; symbolic checks supplement, rather than replace, those proofs. EHM derives the new explicit infinity loop, heat collision maps and nonzero residue in original coordinates. Figures31–33 display the corresponding proved maps and were rendered and inspected.

Class-field source and precise application

Alain Connes and Caterina Consani, Knots, primes and class field theory, arXiv:2501.06560v1, introductory Theorem1 and original source labels `coverdef`, `mappingtorus1`, `artinrec`, `main`. Root read the original author TeX body and bibliography in full, lines169 through end. The intact original archive remains preserved; its SHA256 is `5c8578f1296d2fc2c1cbf0b98f0692402b84f0b5907e090951ffe27c128cce82`, and `arxiv-pisa.tex` has SHA256 `249fed347469cfaf6da97915d547c2ca704a4ba21cc339613713e570f14d5cea`. The source identity is also recorded in `CONNES_CLASS_FIELD_SOURCE.json`.

The source proves a functor from finite abelian extensions to adelic covers, identifying unramified periodic monodromy with arithmetic Frobenius. PHW derives the correctly oriented periodic traces, inertia and fixed-support coefficient maps, and their convergent Euler-contour calculation. CBR computes the actual signed frame character and its conductor-23 arithmetic specialization, including the different integral model at2 and the heat collision residue. The paper is not represented as proving RH or positivity from nontrivial monodromy. PHW and CBR have root mathematical review; CBR also has72 exact finite and polynomial checks. The additional topological specialization-space calculation remains outside this edition. The integral collision sources are linked explicitly in DP11; DP, EC and SG now supply the complete distinguished-prism construction, extension classification and supported-spectrum maps.

Original supported-zero identity, heat source, and residue duality

The original programme singlet-algebra proof, theorem `thm:structure_A`, was reread at lines1146–1260; the operation table is at578–625. The intact source has an empty author field and is retained as an earlier programme source, not attributed to this edition. The globalization note, propositions `prop:projection-kernel`

and prop:units-irr and theorem thm:zero-prime-not-irr, retains both the singlet identity and ambient unit statements. ZH proves the complete semimodule and heat comparisons directly.

Brad Rodgers and Terence Tao, The De Bruijn–Newman constant is non-negative, arXiv:1801.05914v5, original fmp-template.tex: introduction definitions and theorem main, lines64–133, and heat discussion166–315, were read. Dave Platt and Tim Trudgian, The Riemann hypothesis is true up to $3 \cdot 10^{12}$, arXiv:2004.09765v1, original source131–159, gives the stated bound0.2. The source archives remain intact; no full reading of either verification proof is claimed. ZH uses their exact heat conventions and HZ proves the local persistence argument itself.

RD proves the integral Frobenius pairings and trace identities in full, with108 supplementary exact checks. Its actual packet calculation uses the retained WP multiplier and WA24 analytic formula, with the complete SZW label-resolved endpoints. Independent mathematical review checked every RD formula, including the cotangent presentation and signed residue defect. It does not supply a general positive Weil form or construct an off-critical zero.

Distinguished collision prism and extension sources

Bhargav Bhatt and Peter Scholze, Prisms and Prismatic Cohomology, arXiv:1905.08229v4, original author TeX, source labels DefPrismCat, distinguishedDivide, prismcrit and PrismMapTaut. The exact source identity, SHA256 and bounded reading coverage are recorded in DP11. DP proves all specialized prism axioms and the universal quotient directly. EC proves its extension group and connecting maps; SG proves every supported prime map. Original archives remain retained in the source library.

EC, DP and SG were read in full and checked against their generator formulas. SG received an independent mathematical review. The extension checker tests all 27 extension classes and their 243-element middle modules; the quotient checker tests all 81 ring elements and all ring pairs for its additive and Leibniz identities. These finite checks supplement the general proofs. The two diagrams preserve exact maps, signs and coefficients.

Actual heat, global divisor trace, and arithmetic variation

Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, arXiv:1801.05914v5, original author TeX, equations hoz, sas, phidef, htdef, supplies the original heat coordinates. The exact coordinate factor gives the coefficient $1/4$ in the arithmetic variable. The complete supported formula is SZW19–38 in this reader, with its human Connes source retained. The new proofs establish the global rational trace, its arithmetic derivative and higher mixed-prime coefficients, and the complete matrix criterion directly. Classical complex-analysis and Schur-transfer methods retain their standard attribution; no novelty claim for those methods or for an equivalent RH criterion is made.

The 102 symbolic arithmetic checks verify the coordinate factors, Cauchy/endpoints identities, Dirichlet convolution coefficients and multiplicity response. Three high-precision actual-zeta evaluations are consistency checks, not interval certificates or positivity proofs. The complete arguments and exact domains are in the four appended proofs. The reproducible formula diagram records the original constants and every support coordinate.

Classical transfer method and exact reflection-index continuation

Ball, Joseph A.; Biswas, Animikh; Fang, Quanlei; ter Horst, Sanne. *Multivariable generalizations of the Schur class: positive kernel characterization and transfer function realization*. arXiv:0705.2042v3, 2 November 2007. <https://arxiv.org/abs/0705.2042v3>. Original author TeX inspected, §1, Theorem 1.1, Proposition 1.2 and the proof discussion through (2) $\Rightarrow$ (3).

The inspected treatment attributes the scalar positive-kernel Hilbert-space construction to N. Aronszajn, *Theory of reproducing kernels*, Transactions of the American Mathematical Society 68 (1950), 337–404. It cites J. A. Ball, *Linear systems, operator model theory and scattering: multivariable generalizations*, in *Operator Theory and Its Applications* (Winnipeg, MB, 1998), Fields Institute Communications 25, American Mathematical Society, 2000, 151–178, for the lurking-isometry terminology. These two original works were not inspected for this repair; this historical attribution is through the inspected Ball–Biswas–Fang–ter Horst source. No historical novelty for these classical methods is claimed.

The complete CK proof now has the inspected primary-source attribution at its point of use. NI1–32 and FJ1–29 give all density, index, finite-jet and heat-coefficient arguments in full. The companion original

HR, HC and HF TeX files are retained unchanged in this source package; their coordinate and multiplicity conventions are preserved. The new finite symbolic checks concern the displayed algebra, not the sign of an infinite arithmetic form.

Complete primitive prime-window continuation

Juan Arias de Reyna, *An infinitely differentiable function with compact support: Definition and properties*, Rev. Real Acad. Ciencias Madrid **76** (1982), 21–38; author’s English translation arXiv:1702.05442v1, submitted 17 February 2017. The original author TeX `09-Function.tex` was inspected at lines 1–720 for this revision. Theorem 1 and equations (1), (4), (9) provide the classical bump, Fourier product and its integer zero multiplicities. UP includes the exact coordinate comparison $b_r(v) = r^{-1}\varphi(v/r)$, $G_r(z) = \widehat{\varphi}(rz/(2\pi i))$ at its point of use. The source file SHA256 is `a87d1eea98bb7c767a48bc2375f3beb2404e6479ea0be0203bbb6e99a95f5649`; the intact private source archive SHA256 is `3585f57d5be347e9e1c469857c23ad8a561c8991a5c17af07700579506817719`. The original source archive is retained for reading and is not republished in this edition.

The inspected author source credits Børge Jessen and Aurel Wintner, *Distribution functions and the Riemann zeta function*, Transactions of the American Mathematical Society **38** (1935), 48–88, for the Fourier construction. This historical credit is through Arias de Reyna; the 1935 original was not inspected in this revision. The complete bump and zero-set proofs remain in UP, with no historical novelty claim. The derivative formula in the same source, Theorem 4 and equations (21)–(23), supplies a concrete next calculation of the fixed primitive’s norms and autocorrelation, not an assumed arithmetic estimate.

The four new complete proofs are PM, PS, UP and PW, linked in the reading guide. The original programme Hurwitz operator proof, HCR1–42, and primitive boundary proof, BPZ1–23, accompany them unchanged. Their authorship and original citations remain in those sources. The arithmetic input retains the classical Möbius inversion identity and the original Weil formula. Connes and Consani’s original author source for Weil positivity at the archimedean place, Section sectsupport, gives the historical operator comparison in PS. No priority is claimed for classical local positivity, infinite convolution bumps, or translation criteria. The collection’s explicit support corrections, complete derivations and exact arithmetic maps are proved in the displayed sources. The unresolved prime-window bound is not stated as a proved inequality.